Solomonoff Axiom Induction:
Definitions, Basic Theorems, and What a Theory Is and Does

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Abstract

We study Solomonoff axiom induction (SAI), a Solomonoff-style prior over axiom systems. A universal machine reads a random tape and emits first-order sentences; the prior probability of sentences labelled true or false is the probability that the emitted theory is consistent and proves every datum. Its predictions form a belief function with three outcomes: proved, refuted, open. We develop its basic theory and relate it exactly to Solomonoff operator induction (SOI): version by version (coin-free, one-tape, Kolmogorov), SAI agrees with SOI over consistent labellers up to constant factors, and on labelled atoms of a fresh predicate it is SOI on strings. Logic adds a lottery: data on which the one-tape inducer has bounded loss although no consistent r.e. theory proves them; in arithmetic, adversarially chosen true data can concentrate the limit posterior on false theories. For Boolean laws and bounded reasoners, the least length of a usable written theory depends exponentially on the pair (reasoner, vocabulary), which program length cannot see. On literal data with charged deduction, the classical reply to Hempel's theoretician's dilemma becomes a theorem about Craig's construction, not about theoretical vocabulary. Under SAI's pricing an approximately true law is a law with a calibrated error model; Kepler's laws are worked through. The upshot: the idealised formalism cannot see the difference between a theory and a program; where shape matters, it is relative to a reasoner and a vocabulary.

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*Claude is an AI system made by Anthropic. Claude produced the proofs, the computations and the text, under Kaarel Hänni's direction and starting from his questions and working notes (Hänni, 2026). The results were checked by separate Claude instances acting as referees, which re-derived the proofs and re-implemented the computations with independent code. No human has yet independently checked all of the mathematics, and nothing was verified in a proof assistant.

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1 Introduction

What does a scientific theory do, and what is it? On one view, the view behind Solomonoff's theory of inductive inference (Solomonoff, 1964), a theory predicts observations and is a program: the ideal inductive reasoner is a Bayes mixture over all programs that predict the data, each with a weight exponentially small in its length. On another, a theory answers questions posed in some language, why-questions included (van Fraassen, 1980, ch. 5), and is a logical theory, a set of sentences closed under consequence from which the answers are derived (Suppe, 1977). Other answers have been given to both questions: theories explain (Hempel and Oppenheim, 1948) or solve problems (Laudan, 1977); they are classes of models (Suppes, 1960), or practices of exemplars and skills (Kuhn, 1970; Polanyi, 1958).[1] This paper puts the second view into the form of the first. We study Solomonoff axiom induction (SAI), a universal prior over axiom systems in the style of Solomonoff induction, and use it to make precise, and to test, claims about what theories do and what they are. The word "axiom" refers to the hypotheses, which are axiom systems, sets of first-order sentences produced by programs; it does not refer to the induction axiom of arithmetic.

The object. Fix a first-order language $\mathcal{L}$ and a decidable consistent background theory B. The data are finitely many sentences labelled true or false. A universal machine reads a random tape, runs the program it finds at its start on the rest of the tape, and emits sentences, so the emitted theory is random. The prior probability $M_A(D)$ of a data set D is the probability that this theory is consistent with B and proves every datum: each sentence labelled true, and the negation of each sentence labelled false (Definition 2.4). A hypothesis must explain each datum, not merely tolerate it. At a new sentence the posterior splits three ways: the random theory proves it, refutes it, or leaves it open. The open outcome is native, and with the probability Inc_0

[1] Several of the questions studied here first appeared in Hänni's working notes (Hänni, 2026), among them whether theories are better seen as answering questions than as predicting raw data, and how a written theory relates to the reasoner who uses it.

that the random theory is inconsistent it makes up the defect of the semimeasure M_A, which we keep rather than renormalise (Remark 2.6). The predictions form a belief function (Shafer, 1976), updated by geometric conditioning; requiring only that a hypothesis be consistent with the data gives Dempster's rule instead, under which data eliminate hypotheses but never change the odds between the survivors (Propositions 3.2, 3.3 and 3.19).

Raw data and questions. Solomonoff induction scores a hypothesis by how well it predicts a stream of observations, ideally raw. Science predicts much more particular things: where Mars is at a given time, to within two minutes of arc; the period of a planet, given the size of its orbit; whether an orbit is planar. These questions are posed with abstractions that are already available, such as positions, periods and orbits, which may themselves be highly theoretical. We order views of what a theory does on a ladder, each step further from predicting raw data: (1) predicting raw data, such as pixels or observation sequences; (2) answering questions posed with existing abstractions; (3) theories that extend the language with theoretical terms that pay for themselves, while the data stay in the base language; (4) theories that revise the base abstractions, as when a concept changes its meaning or a question is shown to be ill-posed. SAI over a fixed language sits on step 2. Its base language is whatever abstractions are available; it is not an observation language in the logical empiricists' sense, so the classical objections to a theory-neutral observation language do not apply to it in the same way, although theory-ladenness biases comparisons between theories that disagree about the base language (Section 6.1). We take step 2 to be a step in the right direction. We argue that it differs from step 1 in the target, the data type, logical coherence and partiality, not in the inference principle (Section 6.3 and Table 3): SAI is a Bayes mixture over a universal class, and version by version it agrees with Solomonoff operator induction restricted to consistent labellers (Section 4). We have partial results on step 3 (Sections 7 and 8); step 4 has no formal handle here.

What a theory is. On the second question the idealised formalism is neutral, and this is a theorem. Every consistent labeller of sentences, a program that answers some questions consistently, is an axiom system at a cost of $O(1)$ bits: coin-free SAI agrees with coin-free consistent SOI through Craig's construction inside the universal prior, and one-tape with one-tape, tape by tape (Theorems 4.2 and 4.3); new vocabulary costs $O(1)$ bits (Corollary 3.10 and Theorem 8.5); and program length sees no reasoner (Theorem 7.2). So SAI cannot see the difference between a theory and a program. We therefore keep two claims apart. That textbook presentations of fundamental theories are declarative, stating laws, definitions and models given by equations rather than procedures that print predictions, is an observation, with two qualifications: declarative is not sentential, and theories in use have procedural parts (Section 6.2). Why that shape is good is an open question. Craig's construction rules out an answer based only on what can be represented, so any answer must be relative to a reasoner with costs, or to a community of reasoners (Question 6.1); the best formal answer we have locates the large effects in the pair (reasoner, vocabulary), not in logical form (Section 7).

A preview: Kepler's third law. Our running example is Kepler's laws of planetary motion. The third law, K3, says that the squares of the periods of the planets are proportional to the cubes of the semi-major axes of their orbits. What it says about observations depends on the data language. In a language of heliocentric directions it adds nothing to Kepler's first two laws, since such a direction does not depend on the size of the orbit (Proposition 10.2). In a geocentric language, a modern idealisation of the data Kepler worked from, it has content: given the other

elements of the orbits of the Earth and Mars, it fixes the size of the orbit of Mars and with it every geocentric direction of Mars, which without it ranges over an arc (Propositions 10.3 and 10.4). A computation puts what K3 thus saves on the elements of Mars at 11.3 to 14.7 bits for precisions from $8'$ to $1'$, against a law text of 115 to 375 bits in the three codes we tried; so in none of these codes does compression alone select K3 at Kepler's precision (an extrapolation from Mars to the other planets, conditional on the code; Example 10.5). And Kepler's laws are false: against the plan94 ephemeris a fixed ellipse misses Mars by $1.6'$ to $1.9'$ over 12 to 30 years (Section 10.3). How a proof-requiring inducer should treat such a law is the subject of Section 9.

1.1 Contributions

The main results, roughly in order of importance; the sections state them with their hypotheses, and Section 12 says what is new and what is not.

(1) *SAI is consistent SOI, version by version:* coin-free with coin-free and one-tape with one-tape, within constant factors on every data set, and Kolmogorov with Kolmogorov within an additive constant; on labelled atoms of a fresh predicate SAI is SOI on strings, and the companion paper's results on binary labels that hold up to additive constants transfer (Theorems 4.2, 4.3 and 4.8 and Corollary 4.9).

(2) *Random theories that no r.e. theory can replace:* a stream on which the one-tape loss is bounded while no consistent r.e. theory proves the data and the limit posterior lives on random theories (Theorem 5.1); over a decidable true background containing Robinson arithmetic Q, the data can all be true while the limit posterior is concentrated on false theories, a possibility result with adversarial data (Corollary 5.6); on Π_1 data over such a background the mixture is, up to a constant factor, one random true theory (Theorem 5.7).

(3) *Theory shape is relative to a reasoner and a vocabulary:* for Boolean laws, program length sees neither, while the least written length of a usable theory depends exponentially on the pair (reasoner, vocabulary) (Theorem 7.2 and Corollary 7.10); procedural form costs a factor m, the number of query directions of one parity law, over the base variables or with one round of reasoning, and at most a factor 9 in written length with intermediate concepts and unbounded rounds (Theorem E.21).

(4) *The theoretician's dilemma inside an ideal inducer:* with free deduction, theoretical vocabulary changes observational predictions only by a constant factor; on literal data with charged deduction, padded (Craig-type) observational axiomatisations with polynomial-time membership fit exactly the polynomial-time regularities, vocabulary is eliminable at polynomial cost, and to fit more than the regular sets a finite theory needs a polyadic theoretical relation or objects unlike every named one (Theorem 8.4).

(5) *Approximate laws:* over deterministic theories, proof-requiring compatibility rewards a false exact law over its true tolerance form; on data from a law with independent errors, the best loss rate is that of the law together with its error distribution, attained by M_A when the distribution is computable (Propositions 9.3 and G.9); on Kepler's residuals a law with a fitted error histogram wins every cell computed among the Kepler-based machines compared.

(6) *The basic theory of M_A:* dominance; a loss at most the complexity up to deductive equivalence, plus a constant, of any consistent r.e. theory that proves the data (semimeasure convention); an inconsistency probability Inc_0 that is Solovay complete and random, so that M_A is not lower semicomputable; no coding theorem (Theorems 3.5, 3.6, 3.11 and 3.18).

(7) *Where SAI and SOI separate:* logical closure and an observed "independent" label separate them (Propositions 4.11 and 4.12); the separations under time bounds come from free deduction, not from axioms (Theorem 4.15 and Proposition 4.16); with deduction charged, SAI is certificate SOI (Theorem 4.18).

(8) *Gödel phenomena:* over a background containing Q, belief or independent mass on a Π_1 sentence certifies its truth; over one containing PA, the coin-free posterior never learns Σ_1-soundness; and the ω-gap is the ω-incompleteness of the background plus the instances (Propositions 5.11 and 5.16 and Theorem 5.12).

(9) *What a theory does:* answers cost no more description than raw modelling plus an extractor; in a toy world a conservation law's gain per raw step equals the raw no-law predictor's loss on the question about the conserved quantity, while its share of the raw loss vanishes (Propositions 6.4 and D.1); deductively equivalent formulations keep their prior odds, and explanatory direction is recovered from simplicity only under a code that lists theories as written (Propositions 6.5 and D.5).

(10) *Best systems and explanatory power:* Lewis's "kind nature" is a kink in the hull of length against fit, and explanatory power is a per-bit log-likelihood ratio whose growth with the number of systems singles out laws (Proposition 9.5 and Definition 9.6).

(11) *Kepler and Newton:* over five historical forms for the orbit of Mars, on noise-free data the data decide compatibility and the syntax of the law texts decides among compatible forms, while with graded fit the data decide every step but one within tens of observations (Table 6 and Example 10.6).

1.2 What we do not claim

We do not claim that SAI shows a theory to be a logical theory rather than a program: in the idealised setting it cannot see the difference (Section 6.4). That textbook theories are declarative is an observation, not a theorem, and why that shape is good is open (Question 6.1). The results of Section 7 hold in a stated model, with Boolean laws, named reasoners and written length as the cost; they are not theorems about textbooks, and whether they extend beyond Boolean laws is open (Question 7.11). We say nothing realist about the theories an inducer favours: its limit posterior can live on random theories and, with true but adversarially chosen data, on false ones (Theorem 5.1 and Corollary 5.6). The proof-requiring core of SAI is the noise-free special case of the confrontation of theory with data; graded fit enters through randomised theories, not natively (Section 11.1). SAI is an ideal of the evaluation of theories, not of their discovery, and it is not computable. Finally, our philosophical conclusions are arguments; Section 6.1 says which of them rest on theorems.

Organisation. Sections 2 and 3 set up the definitions and the basic theory, Section 4 relates SAI to operator induction, and Section 5 studies what is specific to logic. Section 6 turns to what a theory does and is; Sections 7 and 8 treat theory shape and theoretical terms, Section 9 approximate laws and best systems, and Section 10 the Kepler example. Sections 11 to 13 discuss limitations, related work and open problems. Proofs and computations are in the appendices; Appendix I collects the notation.

2 Setting

The hypotheses of Solomonoff axiom induction are axiom systems, sets of first-order sentences produced by programs; the word "axiom" refers to them, not to the induction axiom. The data are sentences labelled true or false. A hypothesis is compatible with the data when it is consistent and proves every datum: it must *explain* each datum, not merely tolerate it. Randomness enters as in the one-tape version of Solomonoff operator induction (Hänni and Claude (Anthropic), 2026c), which we call the companion paper: a universal machine reads one random tape and emits sentences, and the prior probability of a data set is the probability that the emitted theory is consistent and proves every datum. This section fixes the definitions; Section 3 develops their basic theory, and the proofs are in Appendix A.

2.1 Language, background and data

Definition 2.1 (Setting and data)**.** $\mathcal{L}$ is a countable first-order language with equality and computable syntax, $\mathrm{Sent} := \mathrm{Sent}_{\mathcal{L}}$, and $\vdash$ is derivability in a fixed sound and complete calculus. The *background* $B \subseteq \mathrm{Sent}$ is decidable and consistent; by default $B = \emptyset$. Put $\mathrm{Cn}(\Gamma) := \{\varphi : B \cup \Gamma \vdash \varphi\}$. "$\Gamma$ proves φ" means $B \cup \Gamma \vdash \varphi$; Γ is *consistent* if $B \cup \Gamma$ is; Γ *decides* φ if it proves φ or $\neg\varphi$; $\varphi^1 := \varphi$ and $\varphi^0 := \neg\varphi$.

A *data set* is a finite partial function $D : \mathrm{Sent} \rightharpoonup \{0,1\}$ (label 1 means true); a *three-valued* data set maps into $Y_3 := \{1, 0, \iota\}$, where ι means "independent". Put $D^{\pm} := \{\varphi^{D(\varphi)} : D(\varphi) \in \{0,1\}\}$, and let ψ_D be the conjunction of $D^{\pm}$ in a fixed canonical order ($\top$ if $D^{\pm} = \emptyset$); D is *consistent* if $B \cup D^{\pm}$ is. $D + (\varphi, y)$ is defined for $\varphi \notin \operatorname{dom} D$. A *reveal sequence* $(\varphi_t, y_t)_{t \le n}$ is consistent if $D_n := \{(\varphi_t, y_t) : t \le n\}$ is a partial function; D_t are its prefixes. An infinite stream D_∞ has $D_\infty^{\pm} := \bigcup_n D_n^{\pm}$. For a map ν from data sets to $[0,1]$ the *conditional* is $\nu(y \mid D, \varphi) := \nu(D + (\varphi, y))/\nu(D)$ for $\varphi \notin \operatorname{dom} D$ (0 if $\nu(D) = 0$), and $\nu(y \mid D, \varphi) := \mathbf{1}[y = D(\varphi)]$ for $\varphi \in \operatorname{dom} D$ (the repeated-input convention of the companion paper).

An r.e. background would serve as well, since by Craig's trick it has a decidable axiomatisation with the same consequences. The data $(\varphi, 1)$ and $(\neg\varphi, 0)$ are different data with the same content; weights will depend only on content (Proposition 3.1(c)).

Algorithmic notation. V is a fixed universal prefix machine, K prefix complexity, $\mathbf{m}(i) := \sum_{V(p)=i} 2^{-|p|} \ge 2^{-K(i)}$ and $\Omega_V := \sum_{p \in \operatorname{dom} V} 2^{-|p|}$; W_e is the e-th r.e. set of sentences. $\mathbb{U}$ is the fair-coin measure on tapes $\omega \in \{0,1\}^\infty$, $[w]$ the cylinder of a finite string w, and $\omega_{1:n}$ the first n bits of ω. Logarithms are binary and losses are in bits. The constants $c, c', \ldots$ depend only on the fixed universal machines, the coding of syntax and B; "lsc" means lower semicomputable. We use two standard facts (see Li and Vitányi, 2008): $\mathbf{m}$ dominates every lsc discrete semimeasure up to a constant factor; and the conditional coding theorem: if $P(a \mid z)$ is lsc uniformly in (a, z) and $\sum_a P(a \mid z) \le 1$, then $K(a \mid z) \le -\log P(a \mid z) + K(P) + c$.

Definition 2.2 (Literal setting; fresh atoms)**.** In the *literal setting*, $\mathcal{L}$ contains a unary relation symbol R that does not occur in B, and closed terms t_w ($w \in \{0,1\}^*$) that some model of B interprets as pairwise distinct elements. *Fresh atoms* are sentences $R(t_w)$ for pairwise distinct words w; so $R(t_w)$ and $\neg R(t_w)$ are never two fresh atoms. The standard instance is the *literal language*: a constant $\mathtt{e}$, unary function symbols $\mathtt{s}_0, \mathtt{s}_1$ and R, with $t_w := \mathtt{s}_{w_1}(\cdots \mathtt{s}_{w_k}(\mathtt{e}) \cdots)$ and $\varphi_w := R(t_w)$, of $|w| + 2$ symbols; its *named structure* $\mathfrak{T}$ is the free term algebra on $\mathtt{e}, \mathtt{s}_0, \mathtt{s}_1$. For a set of words $X \subseteq \{0,1\}^*$, D^X lists $(\varphi_w, \mathbf{1}[w \in X])$ in length-lexicographic order and f_X is the

literal labeller of X. The *fit class* of an inducer is the set of X on whose sequence D^X it has bounded loss.

Every result stated for fresh atoms or on literal data assumes this setting. Two facts carry all such uses (Lemma A.4): a set of literals of R with at most one literal per word is consistent with B, and a consistent such set proves exactly its own members among the literals.

2.2 Theory machines and their laws

A hypothesis of Solomonoff axiom induction is produced by a machine that reads random bits and writes sentences. A function machine of Solomonoff operator induction must give each input at most one label; a theory machine may write anything. Emitting more axioms is always allowed, and consistency is a property of the output, a random event.

Definition 2.3 (Theory machine)**.** A *theory machine* is a c.e. set $G \subseteq \{0,1\}^* \times \mathrm{Sent}$ ("after reading w from the tape, emit χ"). For finite w put $A^G_w := \{\chi : (v,\chi) \in G,\ v \sqsubseteq w\}$, and for a tape ω let $A^G_\omega := \bigcup_n A^G_{\omega_{1:n}}$ be the axioms emitted on ω. Let

$$O^G_\chi := \{\omega : B \cup A^G_\omega \vdash \chi\}, \qquad \mathrm{Cons}^G := \{0,1\}^\infty \setminus O^G_\bot, \qquad E^G_D := O^G_{\psi_D} \cap \mathrm{Cons}^G.$$

The *law* of G is $\nu_G(D) := \mathbb{U}(E^G_D)$, the probability that the random theory is consistent and proves every datum. The *three-valued law* is $\nu^3_G(D) := \mathbb{U}(E^{3,G}_D)$ with $E^{3,G}_D := E^G_D \cap \bigcap_{D(\varphi)=\iota} (\{0,1\}^\infty \setminus (O^G_\varphi \cup O^G_{\neg\varphi}))$: in addition, the theory decides no ι-labelled sentence. On Cons^G the *trichotomy function* $\mathrm{tri}^G_\omega : \mathrm{Sent} \to Y_3$ is 1 at the sentences $B \cup A^G_\omega$ proves, 0 at those it refutes and ι elsewhere. (G_i) is a fixed effective enumeration of the c.e. subsets of $\{0,1\}^* \times \mathrm{Sent}$; $K(G) := \min\{K(i) : G_i = G\}$ and, for a law ν, $K(\nu) := \min\{K(i) : \nu_{G_i} = \nu\}$.

These events are measurable: O^G_χ is effectively open uniformly in χ and an index of G, Cons^G is a Π^0_1 class, and $\nu_G(D) = \mathbb{U}(O^G_{\psi_D}) - \mathbb{U}(O^G_\bot)$ (Lemma A.1). *Examples.* (i) A deterministic r.e. theory W_e: $G = \{(\epsilon,\chi) : \chi \in \mathsf{W}_e\}$, with $\nu_G(D) = \mathbf{1}[B \cup \mathsf{W}_e \text{ consistent and } \vdash \psi_D]$. (ii) Fair-coin facts on a family $(\alpha_j)_{j\geq 1}$ of sentences: $G = \{(w, \alpha_j^{w_j}) : |w| = j\}$. (iii) A theory plus coin facts: the union of (i) and (ii). (iv) A finite mixture of finite theories $T_1,\dots,T_r$: $G = \{(w_j,\chi) : \chi \in T_j\}$ for a prefix-free set $\{w_1,\dots,w_r\}$, which gives T_j weight $2^{-|w_j|}$; a tape with no prefix among the w_j emits nothing, and its theory is the empty one, consistent since B is. More generally, a theory machine can complete a law at random inside what the law allows; such randomised theories are how approximate laws enter Solomonoff axiom induction (Proposition G.3).

Definition 2.4 (Solomonoff axiom induction)**.** The *universal theory machine* is $G_U := \{(pw,\chi) : p \in \mathrm{dom}\, V,\ (w,\chi) \in G_{V(p)}\}$: it reads a self-delimiting program and runs that machine on the rest of the tape. Write $A_\omega := A^{G_U}_\omega$, $O_\chi := O^{G_U}_\chi$, $\mathrm{Cons} := \mathrm{Cons}^{G_U}$, $E_D := E^{G_U}_D$, $E^3_D := E^{3,G_U}_D$ and $\mathrm{tri}_\omega := \mathrm{tri}^{G_U}_\omega$.

- *SAI* is the one-tape proof-requiring law $M_A := \nu_{G_U}$. Put $M^3_A := \nu^3_{G_U}$, $L_A := -\log M_A$ and $\mathrm{Inc}_0 := \mathbb{U}(O_\bot)$, the probability that the random theory is inconsistent.
- The *posterior* after D with $M_A(D) > 0$ is the law of A_ω under $\mathbb{U}(\cdot \mid E_D)$; three-valued, under $\mathbb{U}(\cdot \mid E^3_D)$.
- *Trichotomy.* $\mathrm{Bel}_D(\chi) := \mathbb{U}(E_D \cap O_\chi)/\mathbb{U}(E_D)$ is the posterior probability that the random theory proves χ; $\mathrm{Dis}_D(\chi) := \mathrm{Bel}_D(\neg\chi)$; $\mathrm{Indep}_D(\chi) := 1 - \mathrm{Bel}_D(\chi) - \mathrm{Dis}_D(\chi)$.

- The *prior probability of provability* $\mathrm{Bel}_0(\chi) := M_A(\{(\chi, 1)\}) = \mathbb{U}(O_\chi) - \mathrm{Inc}_0$ is not normalised: $\mathrm{Bel}_0(\top) = 1 - \mathrm{Inc}_0$. Its normalisation is $\widehat{\mathrm{Bel}}_0 := \mathrm{Bel}_0/(1 - \mathrm{Inc}_0) = \mathrm{Bel}_\emptyset$. The *plausibility* is $\mathrm{Pl}_0(\chi) := \mathrm{Bel}_0(\top) - \mathrm{Bel}_0(\neg\chi)$, and $\mathrm{Pl}_D(\chi) := 1 - \mathrm{Bel}_D(\neg\chi)$.
- *Class probabilities.* For a class $\mathcal{C}$ of deductively closed theories, $\mathbb{U}_A(\mathcal{C}; D) := \mathbb{U}\{\omega : B \cup A_\omega \text{ consistent}, \vdash \psi_D, \mathrm{Cn}(A_\omega) \in \mathcal{C}\}$ and $\mathbb{U}_A(\mathcal{C}) := \mathbb{U}_A(\mathcal{C}; \emptyset)$. For a theory T, $\mathbb{U}_A[T] := \mathbb{U}\{\omega : B \cup A_\omega \text{ consistent}, \mathrm{Cn}(A_\omega) = \mathrm{Cn}(T)\}$ and $\mathbb{U}_A(\mathrm{Ext}T) := \mathbb{U}\{\omega : B \cup A_\omega \text{ consistent}, \mathrm{Cn}(A_\omega) \supseteq \mathrm{Cn}(T)\}$.

If $\omega = p\omega'$ with $p \in \mathrm{dom}\, V$, then $A_\omega = A_{\omega'}^{G_{V(p)}}$, since $\mathrm{dom}\, V$ is prefix-free; a tape with no prefix in $\mathrm{dom}\, V$ emits nothing. So G_U reads one tape in two parts, a program and that program's coins. By Proposition 3.1, $M_A(D) = \mathrm{Bel}_0(\psi_D)$, $M_A(b \mid D, \varphi) = \mathrm{Bel}_D(\varphi^b)$ and $M_A^3(\iota \mid D, \varphi) = \mathrm{Indep}_D(\varphi)$ for $\varphi \notin \mathrm{dom}\, D$: the predictions of SAI are posterior probabilities that the random theory proves, refutes, or leaves open the queried sentence.

2.3 Conventions and the two defects

Definition 2.5 (Loss conventions)**.** • *Semimeasure convention* (primary): the cumulative loss of a consistent reveal sequence is $\sum_{t \le n} -\log M_A(y_t \mid D_{t-1}, \varphi_t)$, which equals $L_A(D_n) - L_A(\emptyset)$ (Proposition 3.1(d)); for the coin-free version it is $-\log W_A(D_n) + \log W_A(\emptyset)$.

- *Normalised convention*: each conditional is divided by $\mathrm{Bel}_D(\varphi) + \mathrm{Dis}_D(\varphi)$.
- *Three-valued convention*: labels in Y_3, scored by M_A^3, whose conditionals sum to 1 (Proposition 3.1(f)).
- The consistency-requiring predictor (Definition 2.10) has its own loss, $\sum_t -\log \mathrm{Bel}^{\mathrm{nc}}_{D_{t-1}}(\varphi_t^{b_t})$.

Upper bounds proved in the semimeasure convention hold in the normalised convention, since each normalised conditional is at least the raw one. Lower bounds do not transfer: a lower bound in the semimeasure convention can fail in the normalised convention by an unbounded amount (Proposition 5.10), and the normalised loss depends on the reveal order (Theorem A.20). Every lower bound below names its convention; statements about K_A, K_{fin} and the posterior need none.

The missing mass of M_A has two sources, and we keep both rather than renormalise.[2]

Remark 2.6 (Two defects). M_A has a *global* defect $1 - M_A(\emptyset) = \mathrm{Inc}_0 > 0$ and a *local* defect $\mathrm{Indep}_D(\varphi)$ at each query (Proposition 3.1(b), (e)). The one-tape law M_F of SOI has only the local one ($M_F(\emptyset) = 1$), because function machines are consistent by definition. The correspondence is: "proves φ" is a Σ_1 property of the tape, like "halts with output 1"; "independent" is Π_1, like "diverges"; "inconsistent" corresponds to "halts with both outputs", which a function machine cannot do. Removing the independent mass and renormalising, or splitting it equally between the two labels, gives incoherent predictions (Examples 2.15, 2.16 and Proposition 2.17 of Hänni and Claude (Anthropic) (2026a)); keeping it, the pair $(\mathrm{Bel}_D, \mathrm{Pl}_D)$ is a coherent lower and upper probability (Proposition 3.2).

2.4 Versions and complexities of theories

As for SOI, the one-tape law has a coin-free version, a mixture over deterministic programs, and a Kolmogorov version, the shortest compatible program.

[2]The proof-requiring and consistency-requiring variants, the three outcomes true, false and independent, and the choice to keep the defect of the semimeasure rather than renormalise first appeared in Hänni's working notes (Hänni, 2026).

Definition 2.7 (Versions)**.** For a data set D:

- the coin-free version, over enumerators: $W_A(D) := \sum_{p\in\operatorname{dom} V} 2^{-|p|}\mathbf{1}[B\cup\mathsf{W}_{V(p)}$ consistent and $\vdash \psi_D]$, which equals $\sum_e \mathbf{m}(e)\mathbf{1}[B\cup\mathsf{W}_e$ consistent and $\vdash \psi_D]$;
- $K_A(D) := \min\{|p| : p \in \operatorname{dom} V,\ B\cup\mathsf{W}_{V(p)}$ consistent and $\vdash \psi_D\}$ (∞ if there is none; the Kolmogorov version);
- W_A^{dec}, K_A^{dec}: the same over *deciders*, the $p \in \operatorname{dom} V$ such that $V(p)$ is an index of a total $\{0,1\}$-valued function d, read as the axiom set $\{\chi : d(\chi) = 1\}$ (Definition 6.2 of Hänni and Claude (Anthropic) (2026a));
- $W_A^{\mathrm{gen},t}$: the same over enumerators of the delay form (Definition E.10 of Hänni and Claude (Anthropic) (2026a)), which write each axiom a within $O(t(|a|))$ steps of the previous one;
- $K_{\mathrm{fin}}(D) := \min\{K(\psi) : \psi \in \mathrm{Sent},\ B\cup\{\psi\}$ consistent, $B\cup\{\psi\}\vdash\psi_D\}$ and $W_{\mathrm{fin}}(D) := \sum_\psi \mathbf{m}(\psi)\mathbf{1}[B\cup\{\psi\}$ consistent, $B\cup\{\psi\}\vdash\psi_D]$ (finite axiomatisations).

Remark 2.8 (Deciders, enumerators and polynomial-delay enumerators agree)*.* There is c such that for every D: (a) $2^{-c}W_A^{\mathrm{dec}}(D) \le W_A(D) \le 2^c W_A^{\mathrm{dec}}(D)$ and $|K_A(D) - K_A^{\mathrm{dec}}(D)| \le c$; (b) under Assumption 4.5, for t time-constructible, nondecreasing and superadditive with $t(m) \ge m$ and $t(m) \ge m^{a^*}$ (a^* fixed by the universal machine): $2^{-c}W_A(D) \le W_A^{\mathrm{gen},t}(D) \le 2^c W_A(D)$, and likewise for the Kolmogorov versions; (c) $K_A(D) \le K_{\mathrm{fin}}(D) + c \le K(\psi_D) + c$ if D is consistent; if D is inconsistent, $K_A(D) = K_{\mathrm{fin}}(D) = \infty$, while $K(\psi_D)$ is finite.

Part (a) is Craig's trick between two kinds of axiom hypotheses; (b) uses the padded Craig sets of Hänni and Claude (Anthropic) (2026a, Lemma E.12), proved there under the self-simulation assumption, which we do not re-derive (Appendix A.1). The inequality $K_A \le K_{\mathrm{fin}} + c$ can be far from tight: $K_{\mathrm{fin}} - K_A$ is unbounded (Remark F.24).

Definition 2.9 (Complexities of theories)**.** For an r.e. theory T: $[T] := \{e : \mathrm{Cn}(\mathsf{W}_e) = \mathrm{Cn}(T)\}$; $K_{\equiv}(T) := \min_{e\in[T]} K(e)$, the complexity of T up to deductive equivalence; $\mathbf{m}_{\equiv}(T) := \sum_{e\in[T]}\mathbf{m}(e)$; $\mathrm{Ext}(T) := \{e : B\cup\mathsf{W}_e$ consistent, $\mathrm{Cn}(\mathsf{W}_e)\supseteq\mathrm{Cn}(T)\}$ and $\mathbf{m}(\mathrm{Ext}T) := \sum_{e\in\mathrm{Ext}T}\mathbf{m}(e)$. If $B\cup T$ is consistent, then $[T]\subseteq\mathrm{Ext}T$, so $2^{-K_{\equiv}(T)} \le \mathbf{m}_{\equiv}(T) \le \mathbf{m}(\mathrm{Ext}T)$, and $\mathbb{U}_A[T] \ge 2^{-c}\mathbf{m}_{\equiv}(T)$, $\mathbb{U}_A(\mathrm{Ext}T) \ge 2^{-c}\mathbf{m}(\mathrm{Ext}T)$ (Theorem 3.6(i)). The hypothesis is needed: for $T = \{\bot\}$, $\mathbb{U}_A[T] = 0$ and $\mathrm{Ext}T = \emptyset$, while $\mathbf{m}_{\equiv}(T) \ge 2^{-K_{\equiv}(T)} > 0$.

2.5 Variants and questions

Proof-requiring compatibility asks the hypothesis to prove each datum. Two weaker requirements are natural: that the hypothesis be consistent with the data, or that it be graded by how many data it proves. A third variant replaces labelled data by sentences sampled from the theory. Section 3.6 explains why we take the proof-requiring law as primary.

Definition 2.10 (Variants)**.** Let $\varepsilon_\emptyset := \mathbb{U}\{\omega : A_\omega = \emptyset\}$, the mass of the empty theory; $\varepsilon_\emptyset \ge 2^{-|p_\emptyset|} > 0$ for a V-program $p_\emptyset$ of an index of the empty machine.

- *Consistency-requiring,* 0/1: $M_A^{\mathrm{nc}}(D) := \mathbb{U}\{\omega : B\cup A_\omega\cup D^{\pm}$ consistent$\}$. Its posterior is the law of A_ω given that event, and its predictor uses the hypothesis together with the data: $\mathrm{Bel}_D^{\mathrm{nc}}(\varphi) := \mathbb{U}\{\omega : B\cup A_\omega\cup D^{\pm}$ consistent and $\vdash\varphi\}/M_A^{\mathrm{nc}}(D)$ and $\mathrm{Dis}_D^{\mathrm{nc}}(\varphi) := \mathrm{Bel}_D^{\mathrm{nc}}(\neg\varphi)$.

- *Graded consistency-requiring, grade* $\frac{1}{2}$: for consistent A let $\mathrm{gr}_A(\varphi) := 1$ if $B \cup A \vdash \varphi^{D(\varphi)}$ and $\mathrm{gr}_A(\varphi) := \frac{1}{2}$ otherwise, and

$$\nu_A^{1/2}(D) := \mathbf{1}[B \cup A \cup D^{\pm} \text{ consistent}] \prod_{\varphi \in \mathrm{dom}\, D} \mathrm{gr}_A(\varphi), \qquad M_A^{1/2}(D) := \int_{\mathrm{Cons}} \nu_{A_\omega}^{1/2}(D)\, d\mathbb{U}(\omega).$$

- *Generative*: a theory A has a law P_A on sentences, given by a derivation grammar with citation weights on the axioms (the likelihood L_1 of Definition 2.9 of Hänni and Claude (Anthropic) (2026a)); the data are unlabelled sentences $\sigma_1, \sigma_2, \ldots$, and the universal law of data streams is $\Xi(\sigma_{1:n}) := \int \prod_{i \le n} P_{A_\omega}(\sigma_i)\, d\mathbb{U}(\omega)$, with the sampling coins read from separate parts of the tape (Appendix A.10).

The graded weight is the symmetric form of the graded score S_g of Hänni and Claude (Anthropic) (2026a, Definition 2.12) with $g \equiv 1$ on proved data and $g_\infty = \frac{1}{2}$: S_g grades only the data labelled true, $\nu^{1/2}$ grades the data labelled false too. The results below use only data labelled true, where the two agree.

Definition 2.11 (Questions)**.** A *question* is a finite set $\mathcal{Q} = \{\varphi_1, \ldots, \varphi_k\}$ of pairwise B-incompatible sentences whose disjunction B proves, its *direct answers*; a yes/no question is $\{\varphi, \neg\varphi\}$. A theory *answers* $\mathcal{Q}$ if it proves some φ_i. A *conditional question* is a pair (α, ℓ), read "given α, is ℓ the case?". Under the *encoding* (E3), a theory T is compatible with the conditional datum (α, ℓ) iff $B \cup T \cup \{\alpha\}$ is consistent and proves ℓ; equivalently, $T \vdash \alpha \to \ell$ and $T \nvdash \neg\alpha$.

This follows the tradition of erotetic logic (cf. Belnap and Steel, 1976); results that need only pairwise incompatibility say so. (E3) is not a single datum of M_A^3: it combines "the label of $\alpha \to \ell$ is 1" with "the label of $\neg\alpha$ is not 1", a union of two M_A^3-events. The two-valued encoding by the datum $(\alpha \to \ell, 1)$ is degenerate, since a single world satisfying the law fits all of its question data (Proposition E.3); it is never used for conditional data. (E3) makes "given α" hypothetical: a compatible theory may not refute an antecedent, so no single world fits question data with two incompatible antecedents.

3 Basic theorems

This section develops the theory of M_A: its structure and updating rules, computability, dominance and invariance, loss bounds, the failure of a coding theorem, the reasons to prefer proof-requiring compatibility, and the limit of the posterior. Proofs not given in place are in Appendix A.

3.1 Structure and updating

Recall from the companion paper that a map ν from data sets to $[0, 1]$ is a *weak semimeasure* if $\nu(\emptyset) \le 1$ and the masses of the one-datum extensions of D at an input sum to at most $\nu(D)$ (its Definition 3.2), and that the *containment functional* of a random partial function F is $D \mapsto \Pr[F \supseteq D]$ (its Definition 3.4).

Proposition 3.1 (Structure)**.** *Let G be a theory machine. (a) If $\nu_G(D) > 0$, then D is consistent; $M_A(D) > 0$ iff D is consistent, and $M_A^3(D) > 0$ iff moreover $B \cup D^{\pm}$ decides no ι-labelled sentence. (b) For two-valued D and $\varphi \notin \mathrm{dom}\, D$, $\nu_G(D{+}(\varphi, 1)) + \nu_G(D{+}(\varphi, 0)) + \nu_G^3(D{+}(\varphi, \iota)) = \nu_G(D)$; so ν_G is a weak semimeasure, and for M_A the defect at φ divided by $M_A(D)$ is $\mathrm{Indep}_D(\varphi)$. (c) $\nu_G(D)$ depends on D only through $\mathrm{Cn}(D^{\pm})$; if $B \cup D^{\pm} \vdash \varphi^b$ and $\varphi \notin \mathrm{dom}\, D$, then $\nu_G(D{+}(\varphi, b)) = \nu_G(D)$*

and $\nu_G(D + (\varphi, 1-b)) = 0$*: entailed data cost nothing and contradicting data are fatal. (d) For every consistent reveal sequence, in any order and with repeats,* $\sum_{t\le n} -\log M_A(y_t \mid D_{t-1}, \varphi_t) = L_A(D_n) - L_A(\emptyset)$*, which depends only on* $\mathrm{Cn}(D_n^{\pm})$*. (e)* $0 < M_A(\emptyset) = 1 - \mathrm{Inc}_0 < 1$*. (f) For three-valued* D *and* $\varphi \notin \operatorname{dom} D$*,* $\sum_{y\in Y_3} \nu_G^3(D + (\varphi, y)) = \nu_G^3(D)$*, and* $\nu_G^3(\emptyset) = \nu_G(\emptyset)$*; so the conditionals of* M_A^3 *are probability distributions on* Y_3*. (g)* $\nu_G^3(D) = \mathbb{U}\{\omega \in \mathrm{Cons}^G : \mathrm{tri}_\omega^G \supseteq D\}$*, also for two-valued* D *and* ν_G*. So, when* $\nu_G(\emptyset) > 0$*,* $\nu_G^3/\nu_G(\emptyset)$ *is the containment functional of the random total function* tri_ω^G *under* $\mathbb{U}(\cdot \mid \mathrm{Cons}^G)$*, and* $\nu_G/\nu_G(\emptyset)$ *is that of the random partial function obtained by reading* ι *as "undefined".*

On E_D^G the random theory is consistent, so at φ it proves φ, refutes it, or decides neither: these are the three events of (b). By (g), M_A is a (sub-probability) containment functional, which is strictly stronger than being a weak semimeasure (the companion paper's Example 3.5; Proposition A.9(e)).

Proposition 3.2 (Proof-requiring updating is geometric conditioning)**.** *(a) If* $M_A(D) > 0$*, then* $\mathrm{Bel}_D(\chi) = \mathrm{Bel}_0(\psi_D \wedge \chi)/\mathrm{Bel}_0(\psi_D) = \widehat{\mathrm{Bel}}_0(\psi_D \wedge \chi)/\widehat{\mathrm{Bel}}_0(\psi_D)$ *for every* χ*. (b)* Bel_0 *and every* Bel_D *are totally monotone on the Lindenbaum algebra of* B*:* $\mathrm{Bel}_D(\bigvee_{i\le k}\chi_i) \ge \sum_{\emptyset\ne I\subseteq[k]}(-1)^{|I|+1}\mathrm{Bel}_D(\bigwedge_{i\in I}\chi_i)$ *for all* $\chi_1, \dots, \chi_k$*. (c) There is a probability* P_D *on the complete consistent extensions of* B *(a Borel probability on the Stone space) with* $\mathrm{Bel}_D(\chi) \le P_D(\chi) \le \mathrm{Pl}_D(\chi)$ *for every* χ*.*

So Bel_D is a belief function in the sense of Shafer (1976), with random focal set $\mathrm{Mod}(B \cup A_\omega)$ and plausibility Pl_D, and Bel_0 is the probability of provability, as Pearl (1988) puts it. Part (a), which holds because a theory proves ψ_D and χ iff it proves $\psi_D \wedge \chi$, is the *geometric rule* of conditioning of $\widehat{\mathrm{Bel}}_0$, usually attributed to Suppes and Zanotti (1977): keep the focal sets inside the conditioning event and renormalise. Parts (b) and (c) are the one-tape forms of Propositions 2.18 and 2.19 of Hänni and Claude (Anthropic) (2026a), stated there for countable classes of theories.

Proposition 3.3 (Consistency-requiring updating is Dempster conditioning)**.** *Let* $\psi := \psi_D$ *and* $M_A^{\mathrm{nc}}(D) > 0$*. Then* $M_A^{\mathrm{nc}}(D) = \mathrm{Pl}_0(\psi)$ *and* $\mathrm{Bel}_D^{\mathrm{nc}}(\varphi) = \dfrac{\mathrm{Bel}_0(\varphi \vee \neg\psi) - \mathrm{Bel}_0(\neg\psi)}{\mathrm{Bel}_0(\top) - \mathrm{Bel}_0(\neg\psi)} = \dfrac{\widehat{\mathrm{Bel}}_0(\varphi \vee \neg\psi) - \widehat{\mathrm{Bel}}_0(\neg\psi)}{1 - \widehat{\mathrm{Bel}}_0(\neg\psi)}$.

The right-hand side is Dempster's rule of conditioning (Dempster, 1967; Shafer, 1976). With focal sets $F_\omega := \mathrm{Mod}(B \cup A_\omega)$, proof-requiring updating keeps the F_ω contained in $\mathrm{Mod}(\psi)$, while Dempster's rule replaces each F_ω by $F_\omega \cap \mathrm{Mod}(\psi)$ and drops the empty ones. Under proof-requiring updating a datum must be explained by the hypothesis; under consistency-requiring updating it is added to it.

3.2 Computability

Proposition 3.4. $M_A(D) = \mathbb{U}(O_{\psi_D}) - \mathrm{Inc}_0$*, where* $\mathbb{U}(O_{\psi_D})$ *is left-c.e. uniformly in* D *and* $\mathrm{Inc}_0 = \mathbb{U}(O_\perp)$ *is left-c.e.*

Proof. Lemma A.1(a), (c) for $G = G_U$. □

The lsc function $D \mapsto \mathbb{U}(O_{\psi_D})$ is not a semimeasure: it gives an inconsistent theory credit for every datum. The correction that turns it into M_A is one real, and that real is as complex as a left-c.e. real can be.

Theorem 3.5 (The inconsistency probability is an Ω-like real)**.** *(a)* Inc_0 *is not computable. Hence* $M_A(\emptyset) = 1 - \mathrm{Inc}_0$ *is right-c.e. and not left-c.e., and* $D \mapsto M_A(D)$ *is not lower semicomputable. (b) For every left-c.e.* $\alpha \in [0,1]$ *there are a V-program* p_α *and a left-c.e. real* γ_α *with* $\mathrm{Inc}_0 = 2^{-|p_\alpha|}\alpha + \gamma_\alpha$*; consequently* $\alpha \leq_S \mathrm{Inc}_0$ *for every left-c.e.* α*, and* Inc_0 *is Solovay complete. (c)* Inc_0 *is Martin-Löf random.*

A theory machine that emits $\perp$ on the tapes whose binary real lies below α has inconsistency probability α; splitting the tapes by their first program gives (b), the halting set gives (a), and $\alpha = \Omega_V$ (Chaitin, 1975) with a lemma of Demuth's (Lemma A.2) gives (c). The theorem is a new application of the standard argument that Ω_V is Solovay complete (for $\leq_S$ see Solovay, 1975; Downey and Hirschfeldt, 2010) to the event that the random theory is inconsistent; probabilities of other machine behaviours are studied by Becher et al. (2006) and Barmpalias et al. (2016). By Kučera and Slaman (2001) and Calude et al. (2001), Inc_0 is then an Ω-number; we do not use this. Every machine law of SOI is lsc, and M_F is lsc with $\emptyset'$-computable conditionals (Propositions 3.7 and A.10 of the companion paper): function machines are consistent by construction, while the consistency of a theory machine's output is a Π^0_1 property of the tape. M_A keeps the second property and loses the first by exactly one real: its values and conditionals are $\emptyset'$-computable and $M_A + \mathrm{Inc}_0$ is lsc (Proposition A.3).

3.3 Dominance and invariance

Theorem 3.6 (Mixture identity and dominance)**.** *For every two-valued* D*,* $M_A(D) = \sum_{p \in \mathrm{dom}\, V} 2^{-|p|}\nu_{G_{V(p)}}(D) + (1 - \Omega_V)\mathbf{1}[B \vdash \psi_D] = \sum_i \mathbf{m}(i)\nu_{G_i}(D) + (1 - \Omega_V)\mathbf{1}[B \vdash \psi_D]$*, and the same holds for* M^3_A*, the* $\nu^3_{G_i}$*, and the indicator that* B *proves* ψ_D *and decides no* ι*-labelled sentence. Hence* $M_A \geq 2^{-K(\nu)}\nu$ *for every theory-machine law* ν*, and* $M^3_A \geq 2^{-K(G)}\nu^3_G$ *for every theory machine* G*. In particular: (i)* $M_A(D) \geq 2^{-K(e)-c}\mathbf{1}[B \cup \mathsf{W}_e$ *consistent,* $\vdash \psi_D]$ *for every index* e*; consequently, for every r.e.* T *with* $B \cup T$ *consistent,* $\mathbb{U}_A[T] \geq 2^{-c}\mathbf{m}_{\equiv}(T) \geq 2^{-K_{\equiv}(T)-c}$ *and* $\mathbb{U}_A(\mathrm{Ext}T) \geq 2^{-c}\mathbf{m}(\mathrm{Ext}T)$*; (ii)* $M_A(D) \geq 2^{-K(f)-c}\mathbf{1}[f \supseteq D]$ *for every partial computable* $f : \mathrm{Sent} \rightharpoonup \{0,1\}$ *with* $B \cup \Gamma_f$ *consistent, where* $\Gamma_f := \{\varphi^{f(\varphi)} : \varphi \in \mathrm{dom}\, f\}$*; (iii)* $M_A(D) \geq 2^{-K(v)-c}\sum_e v(e)\mathbf{1}[B \cup \mathsf{W}_e$ *consistent,* $\vdash \psi_D]$ *for every lsc semimeasure* v *on indices; in particular* $M_A \geq 2^{-c}W_A$*; (iv) for an r.e.* T *and a computable family* (α_j)*, the law of* T *plus the fair-coin facts* $\alpha_j^{\omega_j}$ *(Example (iii) after Definition 2.3) is dominated with constant* $2^{-K(T,(\alpha_j))-c}$*.*

On the cylinder $[p]$ of a program $p \in \mathrm{dom}\, V$, G_U emits what $G_{V(p)}$ emits on the rest of the tape, so $\mathbb{U}(E_D \cap [p]) = 2^{-|p|}\nu_{G_{V(p)}}(D)$. The mixture and dominance arguments are the standard ones for universal mixtures (Zvonkin and Levin, 1970; the companion paper's Theorem 3.8); what is new is the object. In particular $L_A(D) = \min_G[K(G) - \log \nu_G(D)] \pm c$: SAI is the best two-part code over theory machines, randomised theories included (Proposition G.4). Neither the universal machine nor the background matters beyond a constant factor:

Proposition 3.7 (Choice of the universal machine)**.** *Call a theory machine* G' universal *if for every theory machine* G *there is a string* p_G *with* $A^{G'}_{p_G\omega} = A^G_\omega$ *for all* ω*. If* G' *is universal, then* $2^{-c}M_A \leq \nu_{G'} \leq 2^c M_A$*, and the same for the three-valued laws, with* c *depending only on* G_U *and* G'*.*

Proposition 3.8 (The background is a prior shift)**.** *Let* B, B' *be decidable and consistent with* $B \subseteq \mathrm{Cn}_\emptyset(B')$ *(plain logical consequence), and write* M^B_A*,* $M^{B'}_A$ *for SAI with these backgrounds. Then* $M^B_A(D) \geq 2^{-K(B')-c}M^{B'}_A(D)$ *for every* D*, with* $K(B')$ *the complexity of an index of* B'*; likewise* $W^B_A \geq 2^{-K(B')-c}W^{B'}_A$ *and* $K^B_A \leq K^{B'}_A + K(B') + c$*. The converse fails: a* D *consistent with* B *and inconsistent with* B' *has* $M^{B'}_A(D) = 0 < M^B_A(D)$*.*

So building knowledge into the background gains at most $K(B')+c$ bits on any data. A change of language costs a constant too, if the translation preserves what is proved:

Theorem 3.9 (Faithful translations)**.** *Let $(\mathcal{L},B)$, $(\mathcal{L}',B')$ be two settings and $\tau:\mathrm{Sent}_{\mathcal{L}}\to\mathrm{Sent}_{\mathcal{L}'}$ computable with (T1) $B\cup\Gamma\vdash\varphi$ iff $B'\cup\tau[\Gamma]\vdash\tau(\varphi)$, for all $\Gamma\subseteq\mathrm{Sent}_{\mathcal{L}}$ and φ; and (T2) $B'\vdash\tau(\neg\varphi)\leftrightarrow\neg\tau(\varphi)$ for all φ. For D on $\mathcal{L}$ let $M'_A(\tau D):=\mathbb{U}\{\omega:B'\cup A'_\omega$ consistent and $\vdash\tau(\varphi)^{D(\varphi)}$ for all $\varphi\in\operatorname{dom}D\}$, with A' the universal theory machine of $\mathcal{L}'$, and define $W'_A(\tau D)$, $K'_A(\tau D)$ likewise. Then $2^{-K(\tau)-c}M_A(D)\le M'_A(\tau D)\le 2^{K(\tau)+c}M_A(D)$ for every D, the same holds for W_A, and $|K'_A(\tau D)-K_A(D)|\le K(\tau)+c$, with c depending on both settings.*

Corollary 3.10 (Adding vocabulary)**.** *Let $\mathcal{L}\subseteq\mathcal{L}'$, keep B, and let τ be the inclusion. Then (T1) and (T2) hold, and for data in the old vocabulary $M_A^{\mathcal{L}'}(D)$ and $M_A^{\mathcal{L}}(D)$ agree up to a factor $2^{\pm c}$, W_A likewise, $K_A^{\mathcal{L}'}(D)=K_A^{\mathcal{L}}(D)\pm c$, and predictive probabilities of $\mathcal{L}$-sentences agree up to $2^{\pm 2c}$.*

One direction of Theorem 3.9 emits $\tau[A_\omega]$, the other pulls a theory A' back to $\{\varphi:B'\cup A'\vdash\tau(\varphi)\}$; (T1) for the inclusion holds because every $\mathcal{L}$-structure expands to an $\mathcal{L}'$-structure. So in the unbounded universal setting new vocabulary buys at most a constant: the pull-back of a theory, its consequences in the old vocabulary, is r.e., and the universal machine enumerates it. This is the effective content of Craig's theorem (Craig, 1953) inside the prior; Theorem 8.5 extends it class by class, and Section 8 studies where vocabulary does pay.

3.4 Loss bounds

Theorem 3.11 (Loss against a theory that proves the data)**.** *Let $(\varphi_t,b_t)_{t\le n}$ be any consistent reveal sequence, chosen in any way (in advance, adaptively, adversarially or non-computably; in any order; with repeats), and let T be an r.e. theory with $B\cup T$ consistent and $B\cup T\vdash\varphi_t^{b_t}$ for all $t\le n$. Then in the semimeasure convention*

$$\begin{aligned}L_A(D_n)-L_A(\emptyset)\ \le\ L_A(D_n)\ &\le\ -\log\mathbb{U}_A(\mathrm{Ext}T)\\ &\le\ -\log\mathbf{m}(\mathrm{Ext}T)+c\ \le\ -\log\mathbf{m}_{\equiv}(T)+c\ \le\ K_{\equiv}(T)+c,\end{aligned}$$

and also $L_A(D_n)\le-\log W_A(D_n)+c\le K_A(D_n)+c$.

Proof. Every consistent extension of T proves every datum, so E_{D_n} contains every tape ω with $B\cup A_\omega$ consistent and $\mathrm{Cn}(A_\omega)\supseteq\mathrm{Cn}(T)$. Then use Theorem 3.6(i), $[T]\subseteq\mathrm{Ext}T$, $\mathbf{m}_{\equiv}(T)\ge 2^{-K_{\equiv}(T)}$ and $L_A(\emptyset)\ge 0$; the left side is the cumulative loss by Proposition 3.1(d). The last chain is Theorem 3.6(iii) and $W_A\ge 2^{-K_A}$. □

The complexity that appears is the algorithmic probability of the class of all consistent extensions of T, bounded by $K_{\equiv}(T)$, the complexity of T up to deductive equivalence. So the bound favours the weakest theory that decides the data, and a theory such as PA has small $K_{\equiv}$ although every axiomatisation of it is infinite. Compared with the bound $K(f)+c$ of the companion paper (its Theorem 5.1(b)) for data from a partial computable f, a program is replaced by a theory up to equivalence, which need decide only the sentences asked.

Proposition 3.12 (Laws plus exceptions)**.** *(a) $L_A(D_n)\le-\log\nu(D_n)+K(\nu)$ for every theory-machine law ν. (b) For every r.e. T and every $E\subseteq D_n$ with $B\cup T\cup E^{\pm}$ consistent and $B\cup T\cup E^{\pm}\vdash\psi_{D_n}$: $L_A(D_n)\le K_{\equiv}(T)+K(E)+c$. (c) Let $(\alpha_j)_{j\ge1}$ be a computable family of sentences and T r.e. with $B\cup T\cup\{\alpha_j^{x_j}:j\ge1\}$ consistent for every $x\in\{0,1\}^{\mathbb{N}}$. If every datum of D_n is proved by T or is one of the α_j with some label, and m distinct j occur, then $L_A(D_n)\le K(T,(\alpha_j))+m+c$.*

In (b) a law pays for its listed exceptions, positions included; in (c) each exception costs one bit, without paying for which atoms were exceptional. Section 9 extends (c) from atoms a law is silent about to answers it only constrains.

Proposition 3.13 (Mistake bounds)**.** *(a) Under the hypotheses of Theorem 3.11, the predictor that outputs the most probable of* $1, 0, \iota$ *under* $(\mathrm{Bel}, \mathrm{Dis}, \mathrm{Indep})_{D_{t-1}}(\varphi_t)$ *(ties broken arbitrarily; the stored label at a repeat) makes at most* $-\log(\mathbb{U}_A(\mathrm{Ext}T)/M_A(\emptyset)) \le K_{\equiv}(T) + c$ *mistakes, a mistake being any output other than* b_t*,* ι *included. (b) Fix a well-order of* $\mathrm{dom}\, V$ *that refines length. The* Kolmogorov predictor *takes at step* t *the first program* p_t *with* $B \cup \mathsf{W}_{V(p_t)}$ *consistent and proving* $\psi_{D_{t-1}}$*, predicts* 1 *(resp.* 0*) if* $\mathsf{W}_{V(p_t)}$ *proves* φ_t *(resp.* $\neg\varphi_t$*), and abstains otherwise. If* $k := \sup_n K_A(D_n) < \infty$ *(for instance* $k \le K_{\equiv}(T) + c$ *as in Theorem 3.11), it errs or abstains at most* 2^k *times. (c) The order hypothesis in (b) is needed: for every* m *there are a well-order of* $\mathrm{dom}\, V$ *of type* ω *and a data set* D *with* $K_A(D) \le K(m) + c \le 2\log m + c'$ *on which the Kolmogorov predictor with that order errs or abstains* $2^m - 1$ *times.*

In (a) each mistake costs at least one bit (the halving argument); in (b) a failing program is never chosen again, and there are at most 2^k programs of length at most k. The bound of (b) is essentially attained even for length-refining orders, on data where the Bayes predictor makes at most $K_A(D) + c$ mistakes (Theorem A.22): it is the selection rule that fails, not the prior. With three-valued data the questions may be chosen by any rule:

Theorem 3.14 (Merging for three-valued data)**.** *Let* G *be a theory machine with* $\nu_G(\emptyset) = 1$*,* $\omega \sim \mathbb{U}$ *and* $A := A^G_\omega$*. Let* (u_t) *be random variables independent of* ω*, with any joint law, and let the questions be* $\varphi_t := s_t(h_{t-1}, u_t)$ *for arbitrary measurable rules* s_t*, where* $h_{t-1} = (\varphi_1, y_1, \ldots, \varphi_{t-1}, y_{t-1})$ *and* $y_t \in Y_3$ *is the trichotomy label of* φ_t *under* A*. Let* $D_t := \{(\varphi_s, y_s) : s \le t\}$ *and* $d_t := \mathrm{KL}(\nu^3_G(\cdot \mid D_{t-1}, \varphi_t) \,\|\, M^3_A(\cdot \mid D_{t-1}, \varphi_t))$ *in bits (*0 *at a repeat). Then for every* n*,* $\sum_{t \le n} \mathbb{E} d_t = \mathbb{E}\log \frac{\nu^3_G(D_n)}{M^3_A(D_n)} + \log M_A(\emptyset) \le K(G) + \log M_A(\emptyset) < K(G)$*. Hence* $d_t \to 0$ *almost surely, and* $\sum_t \mathbb{E}\|\nu^3_G(\cdot \mid D_{t-1}, \varphi_t) - M^3_A(\cdot \mid D_{t-1}, \varphi_t)\|_1^2 \le 2\ln 2 \cdot K(G)$*.*

Given the history and the side randomness, the next label has law $\nu^3_G(\cdot \mid D_{t-1}, \varphi_t)$, so the log-ratio telescopes and dominance bounds it. With two-valued data the question rule would have to see A for the two-valued conditionals to be the data law, and the expected bound can fail (as in the companion paper's remark after its Theorem 5.3); the pathwise bound of Proposition 3.12(a) still holds.

Corollary 3.15 (Three-valued loss)**.** *If* D_n *is three-valued and labelled by the trichotomy of an r.e.* T *with* $B \cup T$ *consistent, then in any reveal order the loss in the three-valued convention is* $-\log M^3_A(D_n) + \log M_A(\emptyset) \le -\log \mathbb{U}_A[T] \le K_{\equiv}(T) + c$*.*

Proposition 3.16 (Lower bound)**.** $L_A(D) \ge K(b_D \mid \mathrm{dom}\, D) - c$ *for every* D*, where* b_D *is the vector of two-valued labels in canonical order. So in the semimeasure convention, on fair-coin labels of any fixed* n *sentences,* $\mathbb{E}L_A \ge n - c$*.*

3.5 No coding theorem

Proposition 3.17 (Coding lower bound and the chain)**.** *(a) If* S *is a finite set of pairwise* B*-incompatible sentences and* $\sigma \in S$*, then* $-\log \mathrm{Bel}_0(\sigma) \ge K(\sigma \mid S) - c$*. (b)* $L_A(D) \ge K(b_D \mid \mathrm{dom}\, D) - c$ *for every* D*. (c) For every consistent* D *and every finite set* S *of pairwise* B*-incompatible sentences with* $\psi_D \in S$*,* $2^{-K_A(D)} \le W_A(D) \le 2^c M_A(D) \le 2^{K(S)+c} 2^{-K_A(D)}$*. With* S *the* $2^{|\mathrm{dom}\, D|}$ *label conjunctions on the two-valued part of* $\mathrm{dom}\, D$*,* $K(S) \le K(\mathrm{dom}\, D) + c$*.*

Since M_A is not lsc, the coding theorem in (a) is applied to an auxiliary lsc semimeasure on S, the probability that a dovetailed proof search finds a given member of S first. The slack $K(S)$, in place of the companion paper's $K(\operatorname{dom} D)$ (its Theorem 4.4), can be much smaller: for "which of m incompatible laws holds" it is $K(m)+O(1)$ when S is computable from m. It cannot be removed:

Theorem 3.18 (No coding theorem for axiom systems)**.** *In the literal setting, for every $d \geq 2$ there are distinct $x_1, x_2 \in \{0,1\}^d$ such that $D := \{(R(t_{x_1}),0),(R(t_{x_2}),1)\}$ satisfies (i) $K_A(D) \geq \ell_d > \log d - 2$, with $\ell_d := \lceil \log(d/\log 3)\rceil - 1$, and $W_A(D) \leq \Omega_V^{\geq \ell_d}$; (ii) $-\log W_A(D) \leq 2 + K(\lceil \log(d+1)\rceil) + c$; (iii) $-\log M_A(D) \leq 2 + c$. Parts (ii) and (iii) hold for every data set of two literals on distinct words of length at most d. Hence $K_A(D) + \log W_A(D) \geq \log d - K(\lceil \log(d+1)\rceil) - c - 4 \to \infty$ and $M_A(D)/W_A(D) \geq 2^{-c-2}/\Omega_V^{\geq \ell_d} \to \infty$: no two of 2^{-K_A}, W_A, M_A agree up to a constant factor.*

Fewer than 2^{ℓ_d} programs are shorter than ℓ_d, and they sort the words of length d into fewer than 2^d classes by what they prove about each word's literal, so two words are never separated by a short program; a coin per atom proves a given labelling of two atoms with probability $\frac{1}{4}$. This is the companion paper's Theorem 4.9 for axiom systems, proved directly with theory machines: deterministic axiom sets cannot separate atoms cheaply, while a coin can. Since D is the single sentence ψ_D, no coding theorem holds for single sentences. On literal data the companion paper's two-point rate transfers (Corollary 4.9); for whole theories $K_{\equiv}$ and $-\log \mathbf{m}_{\equiv}$ agree up to a factor 3 and logarithmic terms (Proposition A.5).

3.6 Why proof-requiring compatibility is primary

Proof-requiring compatibility has the guarantees above. The weaker requirements of Definition 2.10 fail as follows; let $H_\omega := B \cup A_\omega \cup D^{\pm}$.

Proposition 3.19 (The 0/1 consistency-requiring weight does not discriminate)**.** *(a) For $\varphi \notin \operatorname{dom} D$, $M_A^{\mathrm{nc}}(D+(\varphi,1)) + M_A^{\mathrm{nc}}(D+(\varphi,0)) = M_A^{\mathrm{nc}}(D) + \mathbb{U}(N_\varphi) \geq M_A^{\mathrm{nc}}(D)$, where N_φ is the set of tapes ω such that H_ω is consistent and decides neither φ nor $\neg\varphi$; so M_A^{nc} is not a semimeasure; it is the plausibility $\mathrm{Pl}_0(\psi_D)$, the dual of a belief function. (b) $M_A^{\mathrm{nc}}(D) \geq \varepsilon_\emptyset$ for every consistent D. The coin-free consistency-requiring weight is at least $2^{-|p_\emptyset|}$, and the Kolmogorov version $\min\{|p| : B \cup \mathsf{W}_{V(p)} \cup D^{\pm}$ consistent$\}$ is at most $|p_\emptyset|$, on all consistent D. (c) The consistency-requiring posterior after D is the prior restricted to the tapes whose theory is consistent with D: data eliminate hypotheses but never change the odds between survivors.*

Part (c) is the one-tape form of Proposition 2.22(b) of Hänni and Claude (Anthropic) (2026a). The inductive work is left to adding the data to the hypothesis, and the proper predictor of this weight, Dempster's rule, pays for it:

Proposition 3.20 (The consistency-requiring predictor)**.** *(a) For every consistent D and $\varphi \notin \operatorname{dom} D$ with $B \cup D^{\pm} \nvdash \varphi^b$, $\mathrm{Bel}_D^{\mathrm{nc}}(\varphi^b) \leq 1 - \varepsilon_\emptyset / M_A^{\mathrm{nc}}(D) \leq 1 - \varepsilon_\emptyset$. So on every consistent reveal sequence the cumulative loss $\sum_t -\log \mathrm{Bel}_{D_{t-1}}^{\mathrm{nc}}(\varphi_t^{b_t})$ is at least $\log \frac{1}{1-\varepsilon_\emptyset}$ times the number of steps whose datum is not entailed by the earlier data. (b) In the literal setting, let $a_i, s_{i,j}$ $(i,j \geq 1)$ be fresh atoms, computable in (i,j), $\zeta_{i,j} := \neg s_{i,1} \wedge \cdots \wedge \neg s_{i,j-1} \wedge s_{i,j}$, $b_{i,j} := a_i \vee \zeta_{i,j}$, and $c_R := |p_R|$ for a V-program p_R of an index of the deterministic machine emitting $\forall x\, R(x)$. (b1) For $D_j := \{(a_1,1),(b_{1,j},1)\}$ and all but at most $1/(\mathrm{Bel}_0(a_1)\varepsilon_\emptyset)$ indices j, the consistency-requiring probability of D_j is strictly larger in the order $(a_1, b_{1,j})$ than in the order $(b_{1,j}, a_1)$. (b2) For every n there are $j_1, \ldots, j_n \leq \lceil 2^{c_R+1}/\varepsilon_\emptyset \rceil$ such that, for $D_n := \{(a_i,1),(b_{i,j_i},1) : i \leq n\}$, the cumulative*

loss in the order $(b_{1,j_1}, a_1, b_{2,j_2}, a_2, \dots)$ *exceeds that in the order* $(a_1, b_{1,j_1}, a_2, b_{2,j_2}, \dots)$ *by at least* $n \log(1 + 2^{-c_R}\varepsilon_\emptyset/2)$ *bits.*

On a stream $(\alpha_j, 1)_{j\geq 1}$ of fresh atoms no datum is entailed, so (a) gives a loss linear in n, while $\{\forall x\, R(x)\}$ proves the stream and the proof-requiring loss is $O(1)$ (Theorem 3.11). In (b) revealing the stronger datum first is more probable; under normalised proof-requiring prediction the weaker datum first is (Theorem A.20).

The graded weight is no better a law of theories (Proposition A.9). It is a weak semimeasure with $M_A^{1/2} \geq M_A$, but it charges for entailed data: for fresh atoms p, q, adding the entailed datum $(p \vee q, 1)$ to $\{(p, 1)\}$ lowers it by at least $\varepsilon_\emptyset/4$, so it is the law of no theory machine. It depends on how the data are packaged: $\{(p \wedge q, 1)\}$ and $\{(p, 1), (q, 1)\}$ have the same weight under M_A but weights differing by at least $\varepsilon_\emptyset/4$ under $M_A^{1/2}$. And the Möbius sum of $M_A^{1/2}$ on the inputs $\varphi, \neg\varphi$ (the companion paper's Proposition A.2) is $-\frac{1}{2}$ times the mass of the consistent random theories that decide neither, which is negative when B does not decide φ, while for M_A the same sum is that mass itself; so $M_A^{1/2}/M_A^{1/2}(\emptyset)$ is not even a containment functional. Its advantage over M_A can come within $2|D| + O(\log)$ bits of $K(\operatorname{dom} D) + |D|$ (Proposition A.11), which measures packaging, not induction. The consistency-requiring predictor is a mixture of the proof-requiring one and a component of weak theories plus the data (Proposition A.10). Finally, the distinction exists only because the random theory is incomplete:

Proposition 3.21. *If $B \cup T$ is complete and consistent, then $B \cup T \vdash \psi_D$ iff $B \cup T \cup D^{\pm}$ is consistent, for every D. Hence for any prior concentrated on complete theories, proof-requiring and consistency-requiring compatibility coincide, the trichotomy has no independent outcome, and conditioning on D restricts to the complete theories that contain $D^{\pm}$.*

Proof. A complete consistent $B \cup T$ proves ψ_D or $\neg\psi_D$, and is consistent with ψ_D iff it does not prove $\neg\psi_D$. □

In short: under proof-requiring compatibility a theory must explain every datum; under consistency-requiring compatibility it need only tolerate it, and toleration is too weak to discriminate between theories. Priors of Demski's type complete their random theory, and so cannot see the difference (Proposition A.25).

3.7 Limits and identification

Let D_∞ be a consistent stream and $E_\infty := \bigcap_n E_{D_n}$. If $\mathbb{U}(E_\infty) > 0$, the posterior converges in total variation to $P_\infty := \mathbb{U}(\cdot \mid E_\infty)$, the prior restricted to the tapes whose theory is a consistent extension of the data, complete or not, and the loss is bounded; otherwise the loss diverges (Proposition A.13). What the limit identifies depends on the truth and on the data format (Proposition A.14). If the data present a consistent r.e. theory T^* with $B \cup T^*$ complete, the posterior concentrates on the class of T^*, $\mathrm{Bel}_{D_n}(\chi) \to \mathbf{1}[T^* \vdash \chi]$ uniformly in χ, independent mass vanishes, and the limit loss is at most $K_{\equiv}(T^*) + c$. If $B \cup T^*$ is incomplete and the data are two-valued, the limit is the prior restricted to the consistent extensions of T^*: every sentence independent of $B \cup T^*$ keeps limit belief and disbelief in $(0, 1)$ and positive independent mass, and the most probable limit class can be a strict extension of T^*. Three-valued data, which label every sentence by its T^*-trichotomy, identify the class of T^*.

This is the Bayesian shadow of Gold's distinction between learning from text and from an informant (cf. Gold, 1967): no learner, computable or not, identifies from two-valued data every theory of a simple family of finite and infinite sets of fresh atoms (Proposition A.15), so the failure

is not an artefact of the prior. Above arithmetic the random theory is almost never complete: if $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$ and $\mathsf{Q} \subseteq \mathrm{Cn}(B)$, the consistent random theory of M_A is almost surely incomplete, and random labellers almost surely fail to separate the theorems of Q from its refutable sentences (Proposition A.18, a new application of the fact that separating sets of effectively inseparable pairs have measure zero).

3.8 Order dependence and the Kolmogorov predictor

The semimeasure-convention loss is order independent (Proposition 3.1(d)); the normalised loss is not. Already for $D = \{(a, 1), (a \vee \chi, 1)\}$, with $B \cup \{a\}$ consistent and $B \cup \{\neg a\}$ consistent and deciding neither χ nor $\neg\chi$, the normalised probability of D is strictly larger in the order $(a \vee \chi, a)$, because a theory proving a always decides $a \vee \chi$ and one refuting a need not; and two orders of the same $2n$ fresh-atom data can differ by at least $2^{-c_1-1}n$ bits (Theorem A.20). The witnesses use entailment between data, so this does not answer the companion paper's question on logically independent data (Question A.21). The single shortest compatible theory can also be a poor predictor where the mixture is good: on data built from fresh atoms $a_1, \ldots, a_m$, the Kolmogorov predictor errs or abstains $2^m - 1$ times for any well-order, while $K_A(D) \leq m + K(m) + c$ and the Bayes predictor makes at most $K_A(D) + c$ mistakes (Theorem A.22). We close with the main open question about the object itself. Call a Borel random variable Θ with values in the consistent, deductively closed sets of sentences a *random consistent theory*, and $\beta(\chi) := \Pr[\chi \in \Theta]$ its *provability function*.

Question 3.22. Does M_A dominate every lower semicomputable provability function, that is, is $\mathrm{Bel}_0 \geq 2^{-c_\beta}\beta$ whenever β is lsc?

Domination holds with the slack $K(S)$ of Proposition 3.17, and for the closure of a theory machine's output; on random sets of literals of a fresh predicate the question is equivalent to the companion paper's Question 3.14 for binary labels (Proposition A.26).

4 Relation to Solomonoff operator induction

Solomonoff operator induction (Hänni and Claude (Anthropic), 2026c) puts the universal prior on programs that label inputs and conditions on the labelled inputs seen. With sentences as inputs its hypotheses are partial labellers of sentences, and a labeller whose labels are consistent with B is the program-shaped counterpart of a theory; so we compare SAI with SOI restricted to consistent labellers.[3] The two agree version by version, up to constant factors (Section 4.2); on positive literals of a fresh predicate SAI *is* SOI on strings (Section 4.3); logically structured data separate them (Section 4.4); and under time bounds the separations come from free deduction (Sections 4.5 to 4.7). Table 1 collects the relations; the proofs are in Appendix B. Losses are in the semimeasure convention unless stated, and $\asymp$ is equality within $2^{\pm c}$ on every finite data set.

4.1 The objects and four maps

Definition 4.1 (The operator-induction side)**.** Let $\mathcal{L}$ and B be as in Definition 2.1.

- A *function machine* is a c.e. set $F \subseteq \mathrm{Sent} \times \{0,1\}^* \times \{0,1\}$ such that $(\varphi, w, y), (\varphi, w', y') \in F$ with w, w' comparable implies $y = y'$ (Hänni and Claude (Anthropic), 2026c, Definition 3.6).

[3] The comparison of SAI with consistent SOI, the example in which plain SOI beats SAI (Remark 4.10), and the axiom schema of Remark B.3 first appeared in Hänni's working notes (Hänni, 2026).

On a tape ω it computes the partial labeller f^F_ω: $f^F_\omega(\varphi) := y$ if $(\varphi, w, y) \in F$ for a prefix w of ω, and $f^F_\omega(\varphi)$ is undefined ($\perp$) otherwise; its law is $\nu_F(D) := \mathbb{U}\{\omega : f^F_\omega \supseteq D\}$. F_U is the universal function machine (read $p \in \operatorname{dom} V$, then run $F_{V(p)}$ on the rest of the tape), $f_\omega := f^{F_U}_\omega$, and $M_F := \nu_{F_U}$.

- For a partial $f : \mathrm{Sent} \rightharpoonup \{0,1\}$ let $\Gamma_f := \{\varphi^{f(\varphi)} : \varphi \in \operatorname{dom} f\}$. The labeller f is *consistent* if $B \cup \Gamma_f$ is consistent, and *closed* if moreover $f(\varphi) = b$ iff $B \cup \Gamma_f \vdash \varphi^b$, for all φ and b.
- *One-tape consistent and closed SOI*: $M^{\mathrm{cons}}_F(D) := \mathbb{U}\{\omega : f_\omega \supseteq D,\ f_\omega \text{ consistent}\}$, and $M^{\mathrm{cl}}_F(D)$ likewise with "closed and consistent"; ν^{cons}_F, ν^{cl}_F are the same for one machine F.
- *Random deciders* are function machines F with f^F_ω total for every ω, read as theory machines with axiom set $A^{\mathrm{dec}}_F(\omega) := \{\chi : f^F_\omega(\chi) = 1\}$; $M^{\mathrm{dec}}_A(D)$ is the sum of $2^{-|p|}\,\mathbb{U}\{\omega : B \cup A^{\mathrm{dec}}_{F_{V(p)}}(\omega) \text{ consistent and } \vdash \psi_D\}$ over the $p \in \operatorname{dom} V$ for which $F_{V(p)}$ is a random decider.
- *Coin-free and Kolmogorov*: $W^{\mathrm{cons}}(D)$ is the sum of $2^{-|p|}$ over the $p \in \operatorname{dom} V$ for which the partial computable labeller $\{V(p)\}$ is consistent and extends D, and $K^{\mathrm{cons}}_F(D)$ is the least such $|p|$; W^{cl}, K^{cl}_F likewise with closed labellers; W^{all} drops the consistency condition (plain coin-free SOI on sentences). W and K_F are the companion paper's coin-free and Kolmogorov versions on strings.

Four maps. The equivalences rest on four computable maps between hypotheses (Lemma B.2), each a constant-size prefix in front of a program, hence injective and weight-preserving up to a constant factor. *Proof search* turns a theory machine G into a function machine F_G that labels φ by searching for a derivation of φ or $\neg\varphi$ from B and the axioms emitted so far; on every tape on which the theory is consistent, its labeller is consistent and closed and decides every sentence as the theory does. *Emission* turns a function machine F into the theory machine G_F whose axioms on ω are $\Gamma_{f^F_\omega}$. *Craig's map with the tape* turns F into a random decider whose axioms on ω are padded copies $\psi^{\wedge(k+1)}$ of the members ψ of $\Gamma_{f^F_\omega}$, k the number of steps F takes to certify ψ. *Closure* turns a consistent labeller f into the closed labeller that searches for proofs from $B \cup \Gamma_f$.

4.2 Equivalence, version by version

Theorem 4.2 (Coin-free equivalence)**.** *There is c, depending only on V, the coding of syntax and B, such that for every finite data set D:*

(a) $2^{-c}W_A(D) \le W^{\mathrm{cons}}(D) \le 2^c W_A(D)$, $2^{-c}W_A(D) \le W^{\mathrm{dec}}_A(D) \le 2^c W_A(D)$, *and* $2^{-c}W^{\mathrm{cons}}(D) \le W^{\mathrm{cl}}(D) \le W^{\mathrm{cons}}(D)$*;*

(b) $K_A(D)$, $K^{\mathrm{dec}}_A(D)$, $K^{\mathrm{cons}}_F(D)$ *and* $K^{\mathrm{cl}}_F(D)$ *agree up to $\pm c$, and all are finite iff D is consistent;*

(c) for every $\varphi \notin \operatorname{dom} D$ and $o \in \{1, 0, \perp\}$, $2^{-c}W_A(D; \varphi \mapsto o) \le W^{\mathrm{cl}}(D; \varphi \mapsto o) \le 2^c W_A(D; \varphi \mapsto o)$*, where $W_A(D; \varphi \mapsto o)$ is the weight of the enumerators compatible with D whose theory proves φ ($o = 1$), proves $\neg\varphi$ ($o = 0$) or decides neither ($o = \perp$), and $W^{\mathrm{cl}}(D; \varphi \mapsto o)$ is the weight of the closed labellers extending D with $f(\varphi) = o$.*

Hence the semimeasure losses of W_A, W^{dec}_A, W^{cons} and W^{cl} agree within $2c$ on every data set, in every order.

Proof search and emission give $W_A \asymp W^{\mathrm{cons}}$ with no Craig construction; Craig's map, needed only for deciders, halts on every input even when the labeller is partial, since each of its runs is bounded by the exponent of the candidate conjunction. For deciders, (a) is Theorem 6.3 of our

earlier report (Hänni and Claude (Anthropic), 2026a). The construction is Craig's; the theorem shows that it works inside the universal prior, weight by weight, and apart from that report we have not found the weight equivalence stated in the literature. A different route, through the axiom schema $\mathrm{Acc}_f(\ulcorner\varphi\urcorner) \to \varphi$, $\mathrm{Rej}_f(\ulcorner\varphi\urcorner) \to \neg\varphi$, works with two sorts or with bounded witnesses but fails in one sort over PA (Remark B.3); the theorem does not need it.

Theorem 4.3 (One-tape equivalence)**.** *There is c, depending only on V, the coding and B, such that for every finite D, the empty one included:*

(a) $2^{-c}M_A(D) \le M_F^{\mathrm{cons}}(D) \le 2^c M_A(D)$ *and* $2^{-c}M_A(D) \le M_A^{\mathrm{dec}}(D) \le 2^c M_A(D)$*;*

(b) tape by tape: for every theory machine G and tape ω, if $B \cup A_\omega^G$ is consistent and proves ψ_D, then $f_\omega^{F_G}$ is consistent and extends D; for every function machine F and tape ω, if f_ω^F is consistent and extends D, then G_F and Craig's random decider of F are compatible with D on ω.

Part (b) is the four maps applied tape by tape, and summing over programs gives (a); the tapes that carry no program have theory B and are matched by the deterministic labeller "proof search from B". For $M_A \asymp M_F^{\mathrm{cons}}$ no Craig construction is needed: a theory machine reads a tape, so it can emit Γ_{f_ω} itself.

Corollary 4.4 (Closed labellers and the trichotomy)**.** *For every finite D and $\varphi \notin \operatorname{dom} D$, the three outcome weights $M_A(D + (\varphi, 1))$, $M_A(D + (\varphi, 0))$, $M_A^3(D + (\varphi, \iota))$ (the theory proves φ, proves $\neg\varphi$, decides neither) agree within $2^{\pm c}$ with the measures of the tapes on which f_ω is closed, consistent, extends D and has $f_\omega(\varphi) = 1$, 0, $\perp$ respectively; and $2^{-c}M_A \le M_F^{\mathrm{cl}} \le M_F^{\mathrm{cons}}$.*

So the trichotomy of SAI (true, false or independent of the sampled theory) is the three-way split of SOI over *closed* labellers; over all consistent labellers it is not (Proposition 4.11).

Penalties on the axiom side. One may charge a hypothesis for the time it takes to check or to generate its axioms, with deduction still free. $M_A^{\mathrm{chk},t}$ is the one-tape mixture over random decider programs that answer every query χ within $O(t(|\chi|))$ steps on every tape, and $M_A^{\mathrm{gen},t}$ the mixture over random enumerator programs that write each axiom a_i within $O(t(|a_i|))$ steps of the previous one on every tape (the delay form; the strict and cumulative forms are in Definition B.4); $W_A^{\mathrm{chk},t}$ and $W_A^{\mathrm{gen},t}$ are their coin-free forms (Definition E.10 of Hänni and Claude (Anthropic), 2026a). These results rest on an assumption about the universal machine U whose programs are the hypotheses, taken from Appendix E.6 of the earlier report; it is standard for multitape machines, and we do not prove it for a particular U.

Assumption 4.5 ((S): efficient self-simulation)**.** Penalised hypotheses are programs of a fixed universal prefix machine U, read to their end before the input. There are constants $\alpha, a \ge 1$ such that the fixed wrappers used in the proofs read, copy and measure a program q within $\alpha(|q| + 1)$ steps; parse, compare or copy strings of total length L within $\alpha(L + 1)^2$ steps; and run the *dovetailing schedule* of q (at stage s, start q on the s-th sentence and advance every unfinished run by one simulated step) while keeping a *work counter* T (the simulated steps, plus the lengths of the sentences started, plus $|q|$), within $\alpha(T + 1)^a$ steps up to work T, writing a string of length L at cost at most αL. Put $a^* := \max(a, 3)$.

Assumption 4.6 ($(\mathrm{S}_{\mathrm{tape}})$: self-simulation with a tape)**.** There are constants $\alpha, a \ge 1$ with the properties of Assumption 4.5 when q also reads a one-way random tape: all runs of the dovetailing schedule read the same tape ω one-way from its start, the work counter also counts tape reads, and up to work T the schedule costs at most $\alpha(T + 1)^a$ steps and reads at most T tape bits.

Theorem 4.7 (Axiom-side penalties change nothing)**.** *Let $t(m) \geq m^{a^*}$ be time-constructible. There is c such that, for every finite D and in the strict, delay and cumulative forms: (a) under Assumption 4.6, $2^{-c}M_F^{\mathrm{cons}}(D) \leq M_A^{\mathrm{chk},t}(D) \leq 2^c M_A(D)$ and $2^{-c}M_F^{\mathrm{cons}}(D) \leq M_A^{\mathrm{gen},t}(D) \leq 2^c M_A(D)$, the empty D included; so $M_A^{\mathrm{chk},t} \asymp M_A^{\mathrm{gen},t} \asymp M_A \asymp M_F^{\mathrm{cons}}$; (b) under Assumption 4.5, $2^{-c}W^{\mathrm{cons}}(D) \leq W_A^{\mathrm{chk},t}(D) \leq W_A^{\mathrm{dec}}(D) \leq 2^c W^{\mathrm{cons}}(D)$ and $2^{-c}W^{\mathrm{cons}}(D) \leq W_A^{\mathrm{gen},t}(D) \leq 2^c W^{\mathrm{cons}}(D)$.*

The lower bounds pad Craig's set by the work counter, one copy $\psi^{\wedge(T+1)}$ per detection at work T, tape by tape. The equivalence also holds class by class: the posterior odds of any two classes of deductively closed theories, genuine axiom systems and disguised labellers alike, move by at most 2^{2c} (Proposition B.6, under Assumption 4.6; coin-free, Proposition B.20).

4.3 Positive literals: SAI is SOI

Theorem 4.8 (Transfer principle)**.** *Let $\mathcal{I} \subseteq \{0,1\}^*$ be the companion paper's (infinite, decidable) input set and $x \mapsto t_x$ a computable injective map from $\mathcal{I}$ to closed $\mathcal{L}$-terms, with decidable range and computable inverse. Let R be a unary relation symbol that does not occur in B, and suppose some model of B interprets the terms t_x as pairwise distinct elements. For D with $\operatorname{dom} D \subseteq \{R(t_x) : x \in \mathcal{I}\}$ put $D^{\mathrm{str}} := \{(x,b) : (R(t_x),b) \in D\}$. Then there is c with*

$$|K_A(D) - K_F(D^{\mathrm{str}})| \leq c, \qquad W_A(D) \asymp W(D^{\mathrm{str}}), \qquad M_A(D) \asymp M_F(D^{\mathrm{str}}) \ \ (D \neq \emptyset),$$

where K_F, W, M_F are the companion paper's objects over $\mathcal{I}$ with labels $\{0,1\}$; the same holds for K_A^{dec}, W_A^{dec}, M_A^{dec}, and for the three outcome weights of a literal query $R(t_y)$ (proves, refutes, decides neither; label 1, 0, undefined at y).

A labeller of strings, read on the literals, is consistent: in a model of B with distinct t_x the symbol R can be interpreted as the set of the t_x labelled 1. Without distinct terms the transfer fails: if $B \vdash t_x = t_y$, the data $(R(t_x),1),(R(t_y),0)$ are inconsistent for SAI, while D^{str} is fine for SOI.

Corollary 4.9 (What transfers)**.** *In the setting of Theorem 4.8, every result of the companion paper about binary labels that holds up to additive constants in K, $-\log W$ and $-\log M_F$ holds for SAI on literal data, with the same quantities. In particular:*

- *(a)* no coding theorem for axiom systems*: for every $d \geq 2$ there are distinct $x_1, x_2 \in \{0,1\}^d$ such that $D := \{(R(t_{x_1}),0),(R(t_{x_2}),1)\}$ has $K_A(D) \geq \log d - 2 - c$, $-\log W_A(D) \leq 2 + K(\lceil \log(d+1) \rceil) + c$ and $-\log M_A(D) \leq 2 + c$; and over two-point literal data $\log d - c \leq \max_D (K_A(D) - L_A(D)) \leq \log d + O(\log\log d)$ (Hänni and Claude (Anthropic), 2026c, Theorem 4.9, Corollary 4.10); Theorem 3.18 proves the first three bounds directly;*
- *(b)* random theories beat deterministic ones*: $\sup_D M_A(D)/W_A(D) = \infty$ over two-point literal data, and $W_A(D) \geq 2^{-|D| - K^{\mathrm{pad}}(2^{d+1}) - c}$ for every literal data set over words of length at most d (Hänni and Claude (Anthropic), 2026c, Propositions 4.5(a), 4.6);*
- *(c)* shared randomness is not a mixture of private coins*, read as a comparison of M_A on literal data D with the companion paper's product mixture M_{prod} on D^{str}: M_A/M_{prod} is unbounded (Hänni and Claude (Anthropic), 2026c, Proposition 4.7); no product law internal to SAI is defined here;*

(d) Solomonoff induction*: with unary numerals $t_i := S^i0$ and $B \in \{\emptyset, \mathsf{Q}\}$, $M_A(\{(R(S^i0), x_{i+1}) : i < n\})$ is within $2^{\pm c}$ of the universal semimeasure of $x \in \{0,1\}^n$ (Hänni and Claude (Anthropic), 2026c, Proposition 3.24).*

Cumulative statements in the normalised convention do not transfer (each normalised conditional transfers only within $2^{\pm 2c}$), nor do computability statements (M_A is not lower semicomputable, Theorem 3.5), statements about private randomness other than (c), or statements about weak semimeasures. So the content of SAI beyond SOI lies in logically structured data, where sentences entail, exclude or leave open one another.

Remark 4.10 (An encoding effect). Let χ_i be $R(t_{w_i})$ or $\neg R(t_{w_i})$ by independent fair coins, for distinct words w_i, with data $(\chi_i, 1)$, $i \le n$. Plain coin-free SOI on sentences, W^{all}, has bounded loss, through the labeller "always 1", while SAI and consistent SOI, coin-free or one-tape, lose at least n bits in expectation (Proposition E.1 of Hänni and Claude (Anthropic), 2026a). This is an encoding effect: the map $(\neg R(t_w), 1) \mapsto (R(t_w), 0)$ turns these data into fair-coin labels on positive literals and leaves M_A and W_A unchanged, since they depend on D only through $\mathrm{Cn}(D^{\pm})$ (Proposition 3.1); there SAI is SOI on strings, which loses at least $n - c$ bits in expectation on fair coins. The labeller "always 1" is inconsistent; consistency is what ties $\neg R(t_w)$ to $R(t_w)$.

4.4 What breaks

Proposition 4.11 (Logical closure)**.** *Let D be consistent and $\chi \notin \mathrm{dom}\, D$.*

(a) If $B \cup D^{\pm} \vdash \chi$, every hypothesis of SAI compatible with D proves χ: $M_A(1 \mid D, \chi) = 1$, $M_A(0 \mid D, \chi) = 0$ and $\mathrm{Indep}_D(\chi) = 0$; the same for W_A.

(b) If $B \cup D^{\pm} \vdash \chi$, then $W^{\mathrm{cons}}(0 \mid D, \chi) = 0$ and $W^{\mathrm{cons}}(1 \mid D, \chi) \ge 2^{-2c}$; the same for M_F^{cons}.

(c) For every $\chi \notin \mathrm{dom}\, D$, entailed or not, the consistent labellers extending D that are undefined at χ have weight at least $2^{-K(\chi)-c}\, W^{\mathrm{cons}}(D)$, and the tapes on which f_ω is consistent, extends D and is undefined at χ have measure at least $2^{-K(\chi)-c}\, M_F^{\mathrm{cons}}(D)$.

(d) Over closed labellers, the three predictive probabilities agree with those of SAI within $2^{\pm 2c}$.

So in the semimeasure convention consistent SOI never becomes certain of a logical consequence: for a fixed simple χ its undefined mass stays above $2^{-K(\chi)-c}$ whatever the data (in the normalised convention its prediction on an entailed χ is 1, by (b)). For (c), make a consistent labeller undefined at χ, at the cost of a description of χ. The equivalences control the losses, not the undefined mass, which in SOI mixes logical independence with laziness.

An observed "independent" label. With three-valued data the label ι is observed. A partial labeller f with values in Y_3 is *consistent* if $B \cup \Gamma_f$ is consistent (Γ_f formed from the labels 1, 0) and decides none of the sentences f labels ι. $W_A^3(D)$ is the weight of the enumerators whose theory is consistent, proves the 1/0 data of D and decides no ι-labelled sentence; $W^{3,\mathrm{cons}}(D)$ is the weight of the consistent three-valued labellers extending D; M_A^3 is as in Definition 2.4, and $M_F^{3,\mathrm{cons}}(D) := \mathbb{U}\{\omega : f_\omega \supseteq D,\ f_\omega \text{ consistent}\}$ for the universal function machine with labels in Y_3. Ind_B is the set of sentences that B decides neither way.

Proposition 4.12 (An observed independent label breaks the equivalence)**.** *Let $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$, and let B be decidable and consistent with $\mathsf{Q} \subseteq \mathrm{Cn}(B)$.*

(a) $2^{-c} W^{3,\mathrm{cons}}(D) \le W_A^3(D)$ for every three-valued D, and $M_F^{3,\mathrm{cons}}(D) \le 2^c M_A^3(D)$ for every three-valued $D \neq \emptyset$.

(b) For every n there is $\sigma_n \in \mathrm{Ind}_B$ with $W^{3,\mathrm{cons}}(\{(\sigma_n,\iota)\}) \le \Omega_V^{\ge n+1}$ while $W_A^3(\{(\sigma_n,\iota)\}) \ge 2^{-c}$; and there is $\sigma'_n \in \mathrm{Ind}_B$ with $M_F^{3,\mathrm{cons}}(\{(\sigma'_n,\iota)\}) \le 2^{1-n} + \Omega_V^{\ge n+1}$ while $M_A^3(\{(\sigma'_n,\iota)\}) \ge 2^{-c}$.

(c) For every $\sigma \in \mathrm{Ind}_B$, $W_A^3(\{(\sigma,\iota)\}) \le 2^{K(\sigma)+c} W^{3,\mathrm{cons}}(\{(\sigma,\iota)\})$ and $M_F^{3,\mathrm{cons}}(\{(\sigma,\iota)\}) \ge 2^{-K(\sigma)-c}$.

Proposition 4.13 (Large gaps have no r.e. witnesses)**.** *In the setting of Proposition 4.12: (a) for every r.e. $S \subseteq \mathrm{Ind}_B$ and every $\sigma \in S$, $W^{3,\mathrm{cons}}(\{(\sigma,\iota)\}) \ge 2^{-K(S)-c}$ and $M_F^{3,\mathrm{cons}}(\{(\sigma,\iota)\}) \ge 2^{-K(S)-c}$, with $K(S)$ the complexity of an index of S; so no computable sequence $\sigma_n \in \mathrm{Ind}_B$ has $W^{3,\mathrm{cons}}(\{(\sigma_n,\iota)\}) \to 0$. (b) If $n(\sigma)$ is the length of a shortest program of a consistent three-valued labeller that labels σ with ι, then $2^{-n(\sigma)} \le W^{3,\mathrm{cons}}(\{(\sigma,\iota)\}) \le \Omega_V^{\ge n(\sigma)}$.*

The mechanism: that a theory decides neither σ nor $\neg\sigma$ is a Π_1 property, which a theory has without committing to anything, while a labeller must output ι after finitely many steps. The sentences that the short consistent labellers label ι form an r.e. subset of Ind_B, which is productive by a classical effective-inseparability argument (Lemma B.10; cf. Smullyan, 1961). One-tape, thresholding the ι-mass at a Lebesgue density point (Lemma C.1), in the manner of the majority vote of de Leeuw et al. (1956), restores such an r.e. subset. An r.e. family of independent sentences, in contrast, is labelled ι by one consistent labeller (Proposition 4.13).

Consistency constraints of unbounded arity separate the versions of consistent SOI: in the setting of Theorem 5.1, on its lottery stream, one-tape consistent SOI has bounded loss while coin-free consistent SOI and W_A do not, and plain W^{all} has bounded loss (Corollary C.7). Finally, none of the versions is lower semicomputable when consistency of finite data sets is not r.e.; W_A, W^{cons}, M_A and M_F^{cons} are computable relative to $\emptyset'$, and the closed and decider versions relative to $\emptyset''$ (Proposition B.12).

4.5 Time bounds with free deduction

On the labeller side, a labeller is run for a bounded time. For a time-constructible τ and a string f, let $f^\tau(\varphi) := U_0(f,\varphi)$ if a fixed plain universal machine U_0 halts within $\tau(|\varphi|)$ steps with a label, and let f^τ abstain otherwise; f has weight $w(f) := 2^{-|E(f)|}$, with E a prefix code (Elias-δ code of $|f|+1$, then f) whose weights sum to 1 with computable tails. *Clocked consistent* SOI is $W^{\mathrm{cons},\tau}(D) := \sum_f w(f)\mathbf{1}[f^\tau \text{ consistent and } \supseteq D]$, and $W^{\mathrm{all},\tau}$ drops consistency; the one-tape forms $M_F^{\mathrm{cons},\tau}$, $M_F^{\mathrm{all},\tau}$ run f with a tape-reading instruction (Theorem B.21). Let $c_\perp := -\log w(f_\perp)$ for the always-abstaining string $f_\perp$. A *computable predictor* P maps a data set D and a sentence $\varphi \notin \mathrm{dom}\, D$ to a pair (q_0,q_1) with $q_0+q_1 \le 1$, computable to any precision; its loss on a reveal sequence is $\ell_P(D_n) := \sum_{j\le n} -\log q_{b_j}(D_{j-1},\varphi_j)$.

Corollary 4.14 (A one-sided equivalence)**.** *For every time-constructible τ there is c_τ such that, on every data set D, the losses of W_A and W_A^{dec} exceed that of $W^{\mathrm{cons},\tau}$ by at most c_τ, and the loss of M_A exceeds that of $M_F^{\mathrm{cons},\tau}$ by at most c_τ. Under Assumption 4.5 the same holds for $W_A^{\mathrm{chk},t}$ and $W_A^{\mathrm{gen},t}$, $t(m) \ge m^{a^*}$ time-constructible.*

A clocked consistent labeller is, unclocked, a consistent labeller computing the same partial map, so a domination lemma (Theorem B.17) bounds the regret by the cost of the clock wrapper and $c_\perp$. The reverse fails, on sentences that encode a diagonal argument. Let $\mathrm{Out}(e,x,y)$ be a Σ_1 formula without $<$ that holds in $\mathbb{N}$ iff program e halts on input x with output y (by the MRDP theorem, Matiyasevich, 1970; Davis, 1973; Lemma E.14 of Hänni and Claude (Anthropic), 2026a), and let DET be the true Π_1 sentence saying that no program has two outputs on one input.

Theorem 4.15 (Diagonal separation)**.** *Let $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$ and let B be decidable and true in an expansion of $\mathbb{N}$. For every computable predictor P there is a program e such that, with $s_n := \mathrm{Out}(\underline{e}, \underline{n}, S0)$ labelled by truth: e is total with values in $\{0,1\}$; $\mathsf{Q} \cup \{\mathrm{DET}\}$ decides every s_n; $\ell_P(D_n) \geq n - \sum_j \log(1 + 2^{-j-2}) \geq n - 1/(4 \ln 2) \geq n - 0.361$; and W_A, $W_A^{\mathrm{chk},t}$, $W_A^{\mathrm{gen},t}$ lose at most $\log(1/\omega_0)$, where $\omega_0 > 0$ is the least of the weights of a fixed decider and a fixed enumerator of $\mathsf{Q} \cup \{\mathrm{DET}\}$ in these mixtures, independent of P. Moreover $W^{\mathrm{all},\tau}$ loses at least $n - 1.37$ and $W^{\mathrm{cons},\tau}$ at least $n - 1.37 - c_\perp$ on such sequences.*

The program e is a fixed point that runs P on its own sentences and outputs the less probable label (Theorem 6.15 of Hänni and Claude (Anthropic), 2026a). The same holds one-tape: $M_F^{\mathrm{all},\tau}$ and $M_F^{\mathrm{cons},\tau}$ lose at least $n - 1.37$ and $n - 1.37 - c_\perp$ on a computable sequence on which M_A loses at most $\log(1/\omega_0)$ (Theorem B.21).

Conversely, on the data of Remark 4.10 the time-bounded $W^{\mathrm{all},\tau}$ has bounded loss while every consistent inducer loses n bits in expectation, so $W^{\mathrm{all},\tau}$ and SAI are incomparable (Proposition B.18): the natural strengthening, that SAI is strictly stronger than every time-bounded predictor, fails, and what holds is strict superiority over clocked *consistent* SOI. Nor is this gain one of axioms over programs:

Proposition 4.16 (The separation comes from free deduction)**.** *(a) Let u be the labeller that, on a sentence $\mathrm{Out}(\underline{e}, \underline{n}, S0)$, runs e on n and outputs 1 if the result is 1 and 0 if it is anything else (diverging if e diverges), and is undefined on all other sentences. Then $\Gamma_u \subseteq \mathrm{Th}(\mathbb{N})$, so u is consistent with every B true in an expansion of $\mathbb{N}$, and $|u|$ is a constant. (b) On every sequence of Theorem 4.15,* unpenalised W^{cons} *(and W^{all}) has loss at most $|u| + c$.*

What the time bounds show. Penalising the axiom side, by the time to check or to generate axioms, does not change which theories win: the equivalence survives, also class by class (Theorem 4.7). Penalising the labeller side makes the relation one-sided (Corollary 4.14 and Theorem 4.15), and the gap is the free deduction of SAI, which unpenalised consistent SOI also has, through one constant-size evaluator (Proposition 4.16). So we read the time-bounded separations as showing not that axioms are better hypotheses than programs, but that free deduction is free computation. (Under a time bound the defeating sequence is itself computable within a slightly larger bound, a time-hierarchy effect; we give only a proof sketch, Remark B.19.)

4.6 Charged deduction

The natural repair is to charge deduction. We use the setting of Definition E.18 of Hänni and Claude (Anthropic) (2026a): $\mathcal{L}$ is finite, B is decidable in polynomial time, derivations are in the Hilbert calculus K of that report, with sizes counting symbols, and a hypothesis is a program p that decides an axiom set A_p in time $O(t)$. Each datum φ must have a derivation of φ^b from $B \cup A_p$ whose size is at most $s(|\varphi|)$ ($W_A[t,s]$); in which every axiom instance, logical ones included, has at most $h(|\varphi|)$ symbols (reading (a1), $W_A^{a1}[t,h]$), or every nonlogical instance does (reading (a2), $W_A^{a2}[t,h]$); or whose *material*, the total size of the distinct instances used, is at most $g(|\varphi|)$ (reading (b), $W_A^{b}[t,g]$), or whose nonlogical material is (reading (b2), $W_A^{b2}[t,g]$). *Certificate* SOI, $W^{\mathrm{cert}}[t',s']$, weighs the consistent labellers $f_{\mathcal{V}}$ defined by verifiers $\mathcal{V}$ running in time $O(t')$: $f_{\mathcal{V}}(\varphi)$ is 1 (resp. 0) if some certificate of length at most $s'(|\varphi|)$ makes $\mathcal{V}$ accept (resp. reject). All charged results assume the renaming condition of that report; without it the index of a single parameter, which costs one symbol, could carry a certificate of any length (Remark B.24).

Assumption 4.17 ((F1): renaming closure)**.** For every hypothesis p of a charged reading, $B \cup A_p$ is closed under permutations of the parameters.

Theorem 4.18 (Charged deduction is certificate SOI)**.** *Under Assumption 4.17, with time-constructible nondecreasing budgets: (a) in a normal derivation of φ every line sits at its own node of an axiom instance used or of φ, up to abstracting parameters; so its size is at most $(M + |\varphi|)(M + |\varphi| + 1)/2$, with M its material, and the factor $\frac{1}{2}$ is attained; (b) with $G(m) := (g+m+1)(g+m+2)/2$, $t^+(m) := m^{k_0}(t(m)+1)$ and $t'^{\#}(m) := m^{k_0} + t'(m)$, for constants k_0, β and c: $W_A[t,g] \le W_A^b[t,g] \le W_A[t,G]$ hypothesis by hypothesis; $W_A^b[t,g] \le 2^c\, W^{\mathrm{cert}}[t^+, \beta G \log(G+2)]$; and $W^{\mathrm{cert}}[t',s'] \le 2^c$ times each of $W_A^b[t'^{\#}, 2s'+m+31]$, $W_A^{a1}[t'^{\#}, s'+m+16]$ and $W_A[t'^{\#}, 2s'+2m+31]$.*

This is Theorem 6.16 of the earlier report, with explicit constants; its classical core is the correspondence of propositional proof systems with NP (Cook and Reckhow, 1979). A derivation is a certificate; conversely, a certificate ζ becomes the axiom $\theta_\zeta \to \varphi^b$, where θ_ζ is a tautology of size $|\zeta| + 14$ that encodes ζ. On literal sequences D^X (Definition 2.2), at polynomial budgets whose degrees are charged in the prior and under Assumption 4.17, this fixes the fit classes (Corollary 6.17 of the earlier report; Corollary B.22): reading (b) fits exactly $\mathsf{NP} \cap \mathsf{coNP}$, so it beats polynomial-time consistent SOI on some sequence if $\mathsf{NP} \cap \mathsf{coNP} \neq \mathsf{P}$; reading (a1) fits exactly EXPTIME, using APSPACE = EXPTIME (Chandra et al., 1981); readings (a2) and (b2) fit every decidable set.

What is charged. Every reading bounds the size of a derivation that *exists*; none charges *finding* it. Derivation-charged SAI is certificate SOI, a notion of the type $\mathsf{NP} \cap \mathsf{coNP}$, not deterministic time-bounded SOI. So charged deduction cannot express the point that a written theory is far from sufficient for a simple reasoner; a deterministic bounded reasoner can (Section 7, in particular Theorem E.26).

4.7 A natural separation: Presburger arithmetic

The diagonal sequences of Theorem 4.15 are built against the predictor; a natural theory separates as well. Let $\mathcal{L}_{\mathrm{Pr}} := \{0, 1, +\}$ (with equality), $B = \emptyset$, and let D^{Pres} list all $\mathcal{L}_{\mathrm{Pr}}$-sentences in length-lexicographic order, labelled by truth in $(\mathbb{N}, 0, 1, +)$; $\mathrm{PrA} := \mathrm{Th}(\mathbb{N}, 0, 1, +)$. The length of a sentence is its symbol count, and clocks and budgets are functions of it. Presburger's axioms (successor, addition and the induction schema) form a decidable, consistent and complete axiomatisation of PrA (Presburger, 1929), with membership decidable in polynomial time. Fischer and Rabin (1974) show that some $c_{\mathrm{FR}} > 0$ has the following property: every decision procedure for PrA, deterministic or nondeterministic, takes more than $2^{2^{c_{\mathrm{FR}} n}}$ steps on some sentence of length n, for every large n. Under a proviso on their measure of length (Remark B.26) this gives a constant $c > 0$ such that, for all large m, every such procedure takes more than $2^{2^{cm}}$ steps on some sentence with exactly m symbols; c denotes this constant below.

Proposition 4.19 (Presburger arithmetic separates SAI from clocked SOI)**.** *(a) W_A, W_A^{dec} and M_A, and the penalised $W_A^{\mathrm{chk},t}$, $W_A^{\mathrm{gen},t}$ and $M_A^{\mathrm{chk},t}$ for a large enough polynomial t, have bounded loss on D^{Pres}. (b) If τ is time-constructible and $\tau(m) \le 2^{2^{c'm}}$ for large m, with $c' < c$, the losses of $W^{\mathrm{cons},\tau}$ and $W^{\mathrm{all},\tau}$ on D_n^{Pres} tend to infinity. (c) If $\tau(m) \le 2^{c'm}$ for large m, with $c' < c$, the losses of $M_F^{\mathrm{cons},\tau}$ and $M_F^{\mathrm{all},\tau}$ on D_n^{Pres} tend to infinity.*

The decider of Presburger's axioms is one deterministic hypothesis compatible with every prefix. Conversely, bounded loss of a clocked mixture yields one clocked labeller that labels every sentence correctly: a decision procedure for PrA below the Fischer–Rabin bound, deterministically in (b), and in (c) by a majority vote over the tape completions at a density point, which costs a factor $2^{\tau(m)}$. Under two further assumptions, a Σ_2-time lower bound for PrA and a time-bounded form of $\mathsf{BPP} \subseteq \Sigma_2^p$, one-tape clocked SOI is also unbounded for doubly exponential clocks (Proposition B.29). With deduction charged, SAI meets the same bound:

Proposition 4.20 (Charged deduction on Presburger data)**.** *Let p be a hypothesis of a charged reading over $\mathcal{L}_{\mathrm{Pr}}$ with $B = \emptyset$, deciding A_p in polynomial time and satisfying Assumption 4.17, and compatible with every prefix of D^{Pres}. Let $c'' < c$. (a) Size reading, budget s: $s(m) > 2^{2^{c''m}}$ for infinitely many m, and for all large m if s is time-constructible. (b) Material reading (b), budget g: the same with g in place of s. (c) Reading (a1), with h nondecreasing and time-constructible: $h(m) > 2^{c''m}$ for all large m.*

A mixture of such hypotheses has bounded loss on D^{Pres} only if one of them is compatible with every prefix. For (a) and (b), guessing a derivation within the budget and checking it is a nondeterministic decision procedure; for (c), closing the set of formulas within the instance budget under the rules is a deterministic one, in time $2^{O(h \log h)}$. Whether (a1) needs doubly exponential budgets, and what (a2) and (b2) need, is open (Question B.30).

Each charged reading matches one version of SOI. Once deduction is charged, the budget SAI needs on D^{Pres} is that of the matching version of SOI: under the size and material readings SAI is certificate SOI (Theorem 4.18), and the nondeterministic Fischer–Rabin bound forces doubly exponential budgets on both; under (a1) SAI behaves like deterministic SOI with an exponential clock (Proposition E.23 and Corollary 6.17(b) of the earlier report; the converse direction is proved there for literal data only, so for sentence data we offer this as a reading), and the deterministic bound forces exponential instance sizes, which the clock turns into doubly exponential time. So on a natural theory, as on the diagonal sequences, the separation from clocked SOI is the free deduction, for every reading in which the derivation is charged and the conclusion is proved; for (a2) and (b2) the question is open.

4.8 Summary

How to read the table. The equivalence is version by version: coin-free with coin-free (through Craig's construction), one-tape with one-tape (without it), Kolmogorov with Kolmogorov. It controls semimeasure losses, not the meaning of "undefined": the undefined mass of SOI mixes logical independence with laziness, and once "independent" is an observed label the relation is one-sided. Penalising the axiom side changes nothing; penalising the labeller side makes the relation one-sided, and the gap is the free deduction of SAI. Charging deduction turns SAI into certificate SOI, a nondeterministic notion. The natural separation is Presburger arithmetic, and once deduction is charged its strength depends on the reading. On literals SAI and SOI coincide; they separate only through logical closure, observed independence, consistency constraints of unbounded arity, or deduction that is free. In the idealised setting every consistent labeller is an axiom system at a cost of $O(1)$ bits, and every theory a closed labeller, so the formalism cannot see the difference in shape between a theory and a program; where shape matters, it is relative to a reasoner and a vocabulary (Section 7).

pair	relation (hypotheses)	where
Equivalences		
W_A, W_A^{dec} – $W^{\mathrm{cons}}, W^{\mathrm{cl}}$; K_A, K_A^{dec} – $K_F^{\mathrm{cons}}, K_F^{\mathrm{cl}}$	$\asymp$, also the three outcome weights against W^{cl}; $\pm c$, finite iff D consistent	Thm. 4.2
M_A, M_A^{dec} – $M_F^{\mathrm{cons}}, M_F^{\mathrm{cl}}$	$\asymp$ on every finite D, tape by tape; outcome weights against M_F^{cl}	Thm. 4.3 and Cor. 4.4
axiom-side penalties $M_A^{\mathrm{chk},t}, M_A^{\mathrm{gen},t}$; $W_A^{\mathrm{chk},t}, W_A^{\mathrm{gen},t}$; $W_A^{\mathrm{Kt},t}$	$\asymp$ unpenalised, also class by class; a Levin penalty costs a labeller $\log(\lvert f\rvert+1)+O(1)$ bits ($t \geq m^{a^*}$; ($\mathrm{S_{tape}}$), (S))	Thm. 4.7 and Props. B.6 and B.16
any version on positive literals – the same version on strings	$\asymp$, $\pm c$: SAI *is* SOI there (fresh predicate, distinct terms)	Thm. 4.8 and Cor. 4.9
all axioms – observational axioms	$\asymp$ in every version, class by class (free deduction)	Thm. 8.5
r.e. axioms in $\mathcal{L}$ – finite theories in a three-sorted $\mathcal{L}^+$	$K_{\mathrm{fin}}^{\mathcal{L}^+} = K_A^{\mathcal{L}} \pm c$, $W_{\mathrm{fin}}^{\mathcal{L}^+} \asymp W_A^{\mathcal{L}}$; not one-tape ($\mathcal{L}$ finite)	Thm. F.28
$W_A[t,s]$, $W_A^b[t,g]$ – W^{cert}	$\asymp$ up to $O(1)$ bits and polynomial changes of the budgets	Thm. 4.18
finite axiom sets – verifier circuits	polynomially related; Bayes over verifier circuits as in the companion paper (literal data)	Thm. B.31 and Cor. B.32
Gaps between versions		
2^{-K_A} – W_A – M_A	no coding theorem: $K_A \geq \log d - 2 - c$, $-\log M_A \leq 2 + c$ on two fresh atoms	Thm. 3.18 and Cor. 4.9
W_A – M_A on literal data	gap $\leq \delta_D + K^{\mathrm{pad}}(2^{d+1}) + c$, δ_D the randomness deficiency of the labels (words of length $\leq d$)	Prop. B.8
W^{cons} – M_F^{cons} (and W_A – M_A)	on the lottery M_F^{cons} is bounded, W^{cons} unbounded, W^{all} bounded	Cor. C.7
M_A – W_A on Π_1 data	a coding theorem holds (B true, $\supseteq \mathsf{Q}$)	Cor. 5.8
M_F^{cons} – machine-wise M_F^{mw}	$\asymp$ if consistency of finite sets is decidable; open in general	Prop. B.14 and Q. B.15
computability	none lsc if finite consistency is not r.e.; $\emptyset'$ resp. $\emptyset''$	Prop. B.12
Separations		
SAI– W^{all}	W^{all} wins by n bits on coin literals labelled 1: an encoding effect	Rem. 4.10
entailed query	SAI certain; consistent SOI keeps undefined mass $\geq 2^{-K(\chi)-c}$ (semimeasure convention)	Prop. 4.11
observed independence	one-sided; unbounded gap on one datum; bounded on r.e. families ($\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$, $\mathsf{Q} \subseteq \mathrm{Cn}(B)$)	Props. 4.12 and 4.13
SAI– clocked $W^{\mathrm{cons},\tau}$, $M_F^{\mathrm{cons},\tau}$	one-sided: regret $\leq c_\tau$; reverse $\geq n - 1.37 - c_\perp$ ($\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$, B true)	Cor. 4.14 and Thm. 4.15
SAI– $W^{\mathrm{all},\tau}$	incomparable	Prop. B.18
unpenalised W^{cons}, M_F^{cons} on diagonal sequences	loss $\leq \lvert u\rvert + c$: the gap is free deduction	Prop. 4.16 and Thm. B.21
SAI– clocked SOI on Presburger data	bounded against unbounded for $\tau \leq 2^{2^{c'm}}$ (coin-free), $\tau \leq 2^{c'm}$ (one-tape); doubly exponential one-tape conditional	Props. 4.19 and B.29
charged SAI on Presburger data	size, material: doubly exponential; (a1): exponential, doubly exponential open; (a2), (b2): open	Prop. 4.20 and Q. B.30
Fit classes under charged deduction (literal data, polynomial budgets)		
readings (b); (a1); (a2), (b2)	$\mathsf{NP} \cap \mathsf{coNP}$; $\mathsf{EXPTIME}$; every decidable set	Cor. B.22
run-padded (Craig-type) observational axiomatisations	exactly P, in every reading	Thm. 8.7 and Cor. F.4
observational hypotheses	the same classes as all hypotheses	Prop. F.7 and Thm. F.10
finite theories with unary theoretical predicates and constants	exactly the regular sets, in every reading (compatible with the named structure)	Thm. 8.8
finite theories with one binary theoretical relation	every decidable set ((a2), (b2)); $\mathsf{NP} \cap \mathsf{coNP}$ ((b)); $\mathsf{EXPTIME}$ ((a1)) (about the named objects)	Thm. F.10

Table 1: SAI and SOI: equivalences, gaps between versions, separations and fit classes. Semimeasure convention unless stated; $\asymp$ is equality within $2^{\pm c}$ on every finite data set; B is decidable and consistent throughout; every row on charged deduction assumes (F1). "Open" and "conditional" mark the only entries that are not theorems.

5 Logic: random theories, Π_1 truth and Gödel phenomena

Up to constant factors, every version of SAI is the corresponding version of consistent SOI (Section 4). This section studies what is specific to data that are sentences of a logic. Random theories can succeed where every r.e. theory fails (Sections 5.1 and 5.2); on Π_1 data the mixture is, up to a constant factor, one random true theory (Section 5.3); Gödel's theorems leave fixed marks on the posterior (Section 5.4); given all its instances, a universal sentence gets belief 1 only when the background and the instances prove it (Section 5.5); and the loss on true arithmetic grows at least polynomially (Section 5.6). Proofs are in Appendix C.

Conventions. $\mathcal{L}_{\mathrm{ar}} = \{0, S, +, \cdot, <\}$, with numerals $\underline{n}$; Δ_0, Σ_1, Π_1 are syntactic classes, and Π_1^* consists of the finite conjunctions of Π_1 sentences and negations of Σ_1 sentences; "Q-equivalent" means equivalent provably in Q. "B is true" means that B has a model whose $\mathcal{L}_{\mathrm{ar}}$-reduct is $\mathbb{N}$, and "$B \supseteq \mathsf{Q}$" means $\mathsf{Q} \subseteq \mathrm{Cn}(B)$ (likewise for PA and $B \cup T$). A theory is Σ_1-*sound* if it proves no false Σ_1 sentence; $\mathcal{U}$ is the class of Σ_1-unsound deductively closed theories, $\overline{\mathcal{U}}$ its complement, and $P_D(\mathcal{C}) := \mathbb{U}_A(\mathcal{C}; D)/M_A(D)$. Standard facts (G2: Gödel's second theorem for r.e. extensions of PA; Uniq: uniqueness of computation codes; Σ_1-completeness of Q; the diagonal lemma) are listed in Appendix C.1 (see Smoryński, 1977; Boolos et al., 2007). Losses are in the semimeasure convention unless stated otherwise.

5.1 The lottery

The one-tape loss on a stream is bounded iff $\mathbb{U}(E_\infty) > 0$ (Proposition A.13). For SOI on strings, bounded one-tape loss implies that some partial computable labeller extends the stream (Proposition 5.2(b)). Does bounded one-tape loss of SAI imply that some consistent r.e. theory proves every datum? It does not; the obstruction is a consistency constraint of unbounded arity.

Let $\mathcal{L}$ contain 0, S and a binary relation symbol Win, and let B be decidable and consistent, not mention Win, and have a model in which the numerals denote pairwise distinct elements (for example $B = \emptyset$, Q or PA). With $N_k := 2^{k+1} - 1$, the *lottery sentences*

$$\mathrm{Lot}_k := \exists x\, \mathrm{Win}(\underline{k}, x) \ \wedge\ \forall x \left(\mathrm{Win}(\underline{k}, x) \to x = \underline{0} \vee \cdots \vee x = \underline{N_k}\right)$$

say that lottery k has a winner among its tickets $0, \ldots, N_k$. The *random lottery theory* G_{lot} reads the tape in consecutive blocks of $k+1$ bits, reads block k as a number $J_k \in \{0, \ldots, N_k\}$, and emits Lot_k, $\mathrm{Win}(\underline{k}, \underline{J_k})$, and $\neg\mathrm{Win}(\underline{k}, \underline{j})$ for $j \le N_k$, $j \neq J_k$. Its theory is consistent with B on every tape.

Theorem 5.1 (The lottery)**.** *There is an infinite stream D_∞ of labelled sentences, computable from $\emptyset'$, such that:*

(a) for every n, $M_A(D_n) \ge 2^{-K(G_{\mathrm{lot}})-c} \prod_{k\ge 0}(1 - 2^{-(k+1)})$; so $\mathbb{U}(E_\infty) > 0$, the cumulative loss is at most $K(G_{\mathrm{lot}}) - \log \prod_{k\ge 0}(1 - 2^{-(k+1)}) + c$, and the posterior converges in total variation to P_∞;

(b) no consistent r.e. theory proves (together with B) every member of $D_\infty^{\pm}$;

(c) $W_A(D_n) \to 0$ and $K_A(D_n) \to \infty$, so the coin-free loss is unbounded; and $K_{\mathrm{fin}}^{\mathcal{L}'}(D_n) \to \infty$ for every language $\mathcal{L}' \supseteq \mathcal{L}$ with computable syntax;

(d) the posterior probability that the tape's program is a deterministic machine (one that emits only under the empty prefix) tends to 0, and no tape in E_∞ has a theory with r.e. deductive closure: the limit posterior lives on random theories whose closure is not r.e.

The product equals $0.288788\ldots$ (a computation; code in the accompanying repository). The stream is a diagonal: if $B \cup \mathsf{W}_e$ is consistent and proves Lot_e, it cannot refute every ticket of lottery e, and the stream reports as a loser the least ticket j_e^* that it does not refute. So every consistent r.e. theory fails at its own lottery, while a random lottery theory dies at lottery k only if its winner is the reported loser, with probability $2^{-(k+1)}$. Such a stream is neither r.e. nor, for the lottery classes, co-r.e., and the phenomenon needs proof-requiring compatibility (Remark C.2). The random lottery theories share the law $\{\mathrm{Lot}_k\}_k$ and differ only in random contingent facts, the winners: a forced form of the split between laws and random initial conditions (Section 9).

Where it cannot happen. The lottery's minimal inconsistent sets $\{\mathrm{Lot}_k\} \cup \{\neg\mathrm{Win}(\underline{k}, \underline{j}) : j \leq N_k\}$ have unbounded size, and at every threshold the majority-vote theory refutes every ticket of the late lotteries, as in Kyburg's lottery paradox (Kyburg, 1961). Bounded arity removes the obstruction.

Proposition 5.2 (Majority vote)**.** *(a) Let $Z \subseteq$ Sent be decidable and $r \geq 1$, and suppose that every B-inconsistent finite subset of Z contains a B-inconsistent subset with at most r elements. If $D_\infty^\pm \subseteq Z$ and $\mathbb{U}(E_\infty) > 0$, then some consistent r.e. $T \subseteq Z$ contains $D_\infty^\pm$; hence $\inf_n W_A(D_n) > 0$, and the coin-free loss is bounded too. (b) For the one-tape law M_F of SOI with inputs X and labels Y: if $D_\infty : X \rightharpoonup Y$ is infinite and $\lim_n M_F(D_n) > 0$, some partial computable f extends D_∞, so the coin-free version W also has bounded loss on D_∞.*

The proof takes the majority vote at threshold $1 - 1/(r+1)$ above a Lebesgue density point of E_∞ (Lemma C.1), as in de Leeuw et al. (1956); the threshold $1/(r+1)$ is exact for the argument (Appendix C.2).

Corollary 5.3 (Bounded loss and r.e. theories)**.** $\mathbb{U}(E_\infty) > 0$ *does not in general imply that a consistent r.e. theory proves every datum of D_∞ (Theorem 5.1; inside true arithmetic, Theorem 5.5). It does when the inconsistency arity of the data's sentence class is bounded (Proposition 5.2), for streams of Π_1 type over a true background $B \supseteq \mathsf{Q}$ (Corollary C.13), and for co-r.e. streams over block-decidable sentence classes, which include the lottery classes (Proposition C.4); and if a positive-measure set of true random theories proves the stream, a true r.e. theory does (Proposition C.3).*

We do not claim that these are the only cases; the last holds because truth, unlike consistency, is a constraint of arity one. Since consistent SOI is SAI version by version (Theorems 4.2 and 4.3), the lottery separates the versions of consistent SOI: on its stream one-tape consistent SOI has bounded loss and coin-free consistent SOI unbounded loss, while plain coin-free SOI on sentences has bounded loss, through an inconsistent labeller (Corollary C.7).

On the lottery stream the coin-free loss diverges more slowly than any computable unbounded nondecreasing function, and $K_A(D_n) \leq \lceil n/2 \rceil + O(\log n)$ (Corollary C.6, through the proof horizon of Proposition C.5).

Question 5.4. Is there a stream with bounded one-tape loss on which $-\log W_A(D_n) \geq f(n)$ for some computable unbounded nondecreasing f? And does bounded one-tape loss on a co-r.e. stream imply, in general, that a consistent r.e. theory proves it?

Relatives. The survival condition $\sum_k 2^{-(k+1)} < \infty$ is the convergence condition of bounded diagonally non-computable functions (cf. Bienvenu and Porter, 2016; Appendix C.2), and randomised learners can succeed where deterministic ones cannot (Pitt, 1989; Ambainis, 1999); here

the random objects are the hypotheses. Random axiom systems drawn by a universal machine are not new: the modified Demski prior samples them (Garrabrant, 2015; Garrabrant et al., 2016b), and it too gives the lottery stream positive probability. To our knowledge, the comparison of the one-tape law with its own coin-free and Kolmogorov versions, the identification of the succeeding hypotheses as random theories r.e. in the tape, the arity boundary and the arithmetic versions are new.

5.2 Lotteries in arithmetic

Inside arithmetic, self-referential formulas behave like independent atoms with respect to consistency with a true background, with prescribed truth values in $\mathbb{N}$ (Lemma C.8, a variant of the flexible formulas of Kripke, 1962). With them the lottery needs no fresh symbol, and survives even with true data. Let $\langle k, j\rangle$ be a Δ_0 pairing with $k, j \le \langle k, j\rangle$.

Theorem 5.5 (Lotteries in arithmetic)**.** *Let $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$, and let B be decidable and true with $B \supseteq \mathsf{Q}$. For a formula $\Upsilon(x)$ put $\mathrm{Tick}_{k,j} := \Upsilon(\langle k, j\rangle)$ and $\mathrm{Lot}^{\mathrm{ar}}_k := \bigvee_{j \le N_k} \mathrm{Tick}_{k,j}$. (a) There are a formula Υ_0 and a stream $((\mathrm{Lot}^{\mathrm{ar}}_k, 1), (\mathrm{Tick}_{k,j^*_k}, 0))_{k\ge 0}$ satisfying (a)–(d) of Theorem 5.1 in which every $\mathrm{Tick}_{k,j}$ is Q-equivalent to a false Σ_1 sentence; so the data are Q-equivalent to false Σ_1 sentences labelled true and to true Π^*_1 sentences. (b) There are a formula Υ_1 and a stream of the same form satisfying (a)–(d) of Theorem 5.1 in which every datum is true in $\mathbb{N}$ and Q-equivalent to a Boolean combination of Σ_1 sentences.*

In (b), Υ_1 is chosen by self-reference so that the true winner of lottery k is the least ticket that W_k refutes (0 if none); the diagonal reports a different, unrefuted ticket as the loser, and that report is true.

Corollary 5.6 (True data, false theories)**.** *On the stream of Theorem 5.5(b): (i) every datum is true, $L_A(D_n)$ is bounded, and $K_A(D_n) - L_A(D_n) \to \infty$; so no coding theorem of the form $K_A \le L_A + f(L_A)$ holds for true data beyond $\Sigma_1 \cup \Pi_1$; (ii) $P_\infty\{\omega : B \cup A_\omega \subseteq \mathrm{Th}(\mathbb{N})\} = 0$: the limit posterior is concentrated on false theories, although every datum is true; (iii) the random lottery theory is Σ_1-unsound on every tape on which, for some k, its winner J_k differs from the true winner of lottery k; conditionally on its survival this has probability* 1.

Part (ii) holds because a positive-measure set of true theories would give a true r.e. theory proving the stream (Proposition C.3). Fed this stream of true sentences, the one-tape inducer predicts with bounded loss and comes to believe, with probability one, false theories: a random theory that guesses winners is right where the data check it and wrong where they do not. The data are adversarial (a Δ_2 diagonal), so this is a possibility result about inferring truth from predictive success in an idealised setting, not evidence about actual science; it is one reason we draw no realist conclusion about the theories an inducer favours. Whether also $P_\infty(\overline{\mathcal{U}}) = 0$ is open (Question C.9).

5.3 Π_1 data: derandomisation, a coding theorem, Chaitin's theorem

On Π_1 data positive probability of provability certifies truth, and the mixture collapses, up to a constant factor, to one random true theory. Throughout, $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$ and B is decidable and true with $B \supseteq \mathsf{Q}$. A data set D is *of* Π_1 *type* if each literal in $D^{\pm}$ is B-equivalent to a Π^*_1 sentence or to a true Σ_1 sentence; then $B \vdash \psi_D \leftrightarrow \pi_D$ for some $\pi_D \in \Pi^*_1$, and $M_A(D) = \mathrm{Bel}_0(\pi_D)$ (Appendix C.4).

Theorem 5.7 (The threshold machine)**.** *Let G_{thr} be the theory machine that, after reading s tape bits u, emits every $\pi \in \Pi_1^*$ for which some lower approximation of $\mathbb{U}(O_\pi)$ exceeds $0.u + 2^{-s}$ ($0.u$ the dyadic rational with digits u). For a tape r that is not a dyadic rational, read as a real, $A_r^{G_{\mathrm{thr}}} = \{\pi \in \Pi_1^* : \mathbb{U}(O_\pi) > r\}$; this theory is inconsistent if $r < \mathrm{Inc}_0$ and true if $r > \mathrm{Inc}_0$. For every data set D of Π_1 type, $M_A(D) \le \nu_{G_{\mathrm{thr}}}(D) \le 2^{K(G_{\mathrm{thr}})+c} M_A(D)$.*

So on Π_1 data the universal mixture is, up to a constant factor, one random true theory: every Π_1^* sentence whose prior probability of being proved exceeds a uniformly random level. The tape helps only by choosing the level, and that costs only what the probability of the datum already charges. Fixing $k+2$ bits of the level just above Inc_0 gives a true r.e. theory T_k of complexity at most $k + K(k) + c$ that proves every Π_1^* sentence of prior probability of provability at least 2^{-k} (Theorem C.10).

Corollary 5.8 (A coding theorem for Π_1 data)**.** *For every D of Π_1 type, $L_A(D) \le -\log W_A(D) + c \le K_A(D) + c \le L_A(D) + K(\lceil L_A(D) \rceil) + c'$.*

Likewise, if every finite part of a stream is of Π_1 type and $\sup_n L_A(D_n) \le k$, the single true theory $B \cup T_k$ proves every datum (Corollary C.13). For literals no coding theorem holds (Theorem 3.18). For W_A the bound improves to $-\log W_A \le L_A + K^{\mathrm{pad}}(\lceil L_A \rceil + 1) + c$ (Remark C.11); whether $K_A \le L_A + O(1)$ is open (Question C.12). Now code strings by length-lexicographic rank, so that the strings of length at most m are the codes at most $\mathrm{rk}_m := 2^{m+1} - 2$, and let $\mathrm{Comp}(p, t, y)$ be Kleene's Δ_0 predicate "t codes the halting computation of V on p, with output y". The Π_1 sentence $\mathrm{Kgt}_m(x) := \forall t \, \forall p \, (p \le \underline{\mathrm{rk}_m} \to \neg \mathrm{Comp}(p, t, \underline{x}))$ is true iff $K(x) > m$.

Corollary 5.9 (Chaitin's theorem for random theories)**.** *For all m and x, in the semimeasure convention: (a) if $K(x) \le m$, then $L_A(\{(\mathrm{Kgt}_m(x), 1)\}) = \infty$; if $K(x) > m$, then $L_A(\{(\mathrm{Kgt}_m(x), 1)\}) \ge m - K(m) - K(\lceil L_A \rceil) - c \ge m - 2\log m - O(\log\log m)$; (b) if $K(x) > m$, then $L_A(\{(\mathrm{Kgt}_m(x), 1)\}) \le K_A(\{(\mathrm{Kgt}_m(x), 1)\}) + c \le m + K(m) + c'$, and one true theory, B plus a single Π_1 sentence of complexity at most $m + K(m) + c$, proves $\mathrm{Kgt}_m(x)$ for every such x at once. The same bounds, with other constants, hold in the three-valued convention, in which the cost of the label 1 on this single datum is $-\log \mathrm{Bel}_\emptyset(\mathrm{Kgt}_m(x)) = L_A(\{(\mathrm{Kgt}_m(x), 1)\}) + \log(1 - \mathrm{Inc}_0)$.*

The lower bound combines Corollary 5.8 with Chaitin's bound (Chaitin, 1974). One m-bit Π_1 law explains every true complexity lower bound at level m. Yet the datum carries at most one bit of label information: unconstrained SOI predicts it at cost $O(1)$ by always answering "true", and the lower bound of Proposition 3.16 is $O(1)$ here. In the normalised convention the m bits disappear:

Proposition 5.10 (The normalised convention: deciding, not predicting)**.** *Let B be decidable and true with $\mathsf{PA} \subseteq \mathrm{Cn}(B)$. There is c such that for all m, d and $j \ge 0$, all strings $x \in \{0,1\}^d$ except at most $2^{2m+K(m)+j+c} + 2^{m+1}$ of them satisfy $K(x) > m$ and $\mathrm{Dis}_\emptyset(\mathrm{Kgt}_m(x)) \le 2^{-j} \mathrm{Bel}_\emptyset(\mathrm{Kgt}_m(x))$; so the loss of $(\mathrm{Kgt}_m(x), 1)$ in the normalised convention is at most $\log(1 + 2^{-j}) \le 1.45 \cdot 2^{-j}$.*

The m bits are the price of deciding the sentence, not of predicting its truth value: every Π_1 affirmation must be backed by a true theory, while a consistent extension of PA refutes $\mathrm{Kgt}_m(x)$ for fewer than 2^{m+1} strings.

The boundary of the coding theorem is exact. With $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$ and B decidable and true with $B \supseteq \mathsf{Q}$, one false Σ_1 datum breaks it (Proposition C.15), and so do two true data one level up: for every $d \ge 2$ there are true data D_d, two sentences each Q-equivalent to a Boolean combination of Σ_1 sentences, with $K_A(D_d) > \log d - 2$ and $L_A(D_d) \le 2 + c$ (Proposition C.16); on a stream, the

gap is bounded against unbounded (Corollary 5.6). The threshold device is close to the averaging of de Leeuw et al. (1956) and to soundness arguments for random true axioms (Bienvenu et al., 2014; Shen, 2011), which show among other things that all true statements "$C(x) \geq n - c$" together prove every true Π_1 sentence. To our knowledge, the threshold just above the unknown Inc_0, the single random threshold theory and the coding theorem for an inductive inference rule are new; we have not found them stated, but cannot exclude that they are known.

5.4 Gödel phenomena

Proposition 5.11 (The arithmetical shape of the trichotomy)**.** *Let B be decidable and consistent with $\mathsf{Q} \subseteq \mathrm{Cn}(B)$, and let D be consistent. (a) If σ is a true Σ_1 sentence, then $\mathrm{Bel}_D(\sigma) = 1$. (b) If $\pi \in \Pi_1^*$ is false, then $(\mathrm{Bel}_D, \mathrm{Dis}_D, \mathrm{Indep}_D)(\pi) = (0, 1, 0)$; equivalently, $\mathrm{Bel}_D(\pi) > 0$ or $\mathrm{Indep}_D(\pi) > 0$ implies that π is true. (c) If $B \cup D^{\pm}$ is true, $\pi \in \Pi_1^*$ is true and $B \cup D^{\pm} \nvdash \pi$, then $\mathrm{Bel}_D(\pi)$, $\mathrm{Dis}_D(\pi)$, $\mathrm{Indep}_D(\pi)$ are all positive. (d) If $B \cup D^{\pm}$ is true, σ is a false Σ_1 sentence and $B \cup D^{\pm} \nvdash \neg\sigma$, then all three values of σ are positive.*

The inducer errs one-sidedly: on Σ_1 sentences it may believe a falsehood but never disbelieves a truth; on Π_1 sentences (consistency statements, Gödel sentences, $\mathrm{Kgt}_m(x)$) the reverse. So "independent" on a Π_1 question certifies truth, the inducer's form of "the Gödel sentence is true because it is unprovable"; the normalised convention discards exactly this share.

Theorem 5.12 (Soundness is never learned, coin-free)**.** *Let B be decidable and consistent with $\mathsf{PA} \subseteq \mathrm{Cn}(B)$. There is c such that $W_A(\mathcal{U}; D) \geq 2^{-c} W_A(\overline{\mathcal{U}}; D)$ for every consistent D. Hence the coin-free posterior probability of Σ_1-soundness, and a fortiori of truth, is at most $1/(1 + 2^{-c})$ for every D, in particular for every finite part of true arithmetic; and in the Kolmogorov version the shortest compatible Σ_1-unsound theory is at most c bits longer than the shortest compatible Σ_1-sound one.*

Every consistent W_e has a companion $\mathsf{W}_e \cup \{\neg\mathrm{Con}(B \cup \mathsf{W}_e)\}$, consistent by (G2), unsound, and only $O(1)$ bits longer. One-tape it gives $P_D(\mathcal{U}) \geq 2^{L_A(D) - K_A(D) - c}$ (Proposition C.17): confidence in soundness needs a large coding gap. On Π_1 data the bound is uniform:

Theorem 5.13 (Soundness is never learned from Π_1 data)**.** *Let B be decidable and true with $\mathsf{PA} \subseteq \mathrm{Cn}(B)$. There is c such that for every D of Π_1 type with $M_A(D) > 0$, $\mathbb{U}_A(\mathcal{U}; D) \geq 2^{-c} M_A(D)$; so $P_D(\overline{\mathcal{U}}) \leq 1 - 2^{-c}$.*

The companion of G_{thr} adds, for each prefix w of the tape, the negated consistency statement of what G_{thr} emits on extensions of w. The answer to the next question is yes on Π_1 data, with a bounded coding gap, and on the Win-lottery stream (Appendix C.5).

Question 5.14. With B true and $\mathsf{PA} \subseteq \mathrm{Cn}(B)$, is $P_D(\overline{\mathcal{U}})$ bounded away from 1 uniformly over all data D?

Given data presenting an r.e. theory T with $B \cup T$ true and $\mathsf{PA} \subseteq \mathrm{Cn}(B \cup T)$, the consistency statement $\mathrm{Con}(B \cup T)$ keeps positive limit shares on all three answers, its belief and independent shares certify its truth, and which of "true" and "false" is favoured is a choice of universal machine (Proposition C.18). The truth hypothesis is required: for $B = \mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$ and $T = \emptyset$, B refutes $\mathrm{Con}(B)$, and the limit of Dis is 1. Checking instances of a decidable property, the classic case of induction in mathematics,[4] confirms nothing when deduction is free:

[4]The question of how checking instances can confirm a universal mathematical statement first appeared in Hänni's working notes (Hänni, 2026).

Proposition 5.15 (Old evidence)**.** *If $B \vdash \varphi^b$ for every (φ, b) in D, then $M_A(D) = M_A(\emptyset)$ and the posterior equals the prior; likewise for W_A and K_A. In particular, with $\mathsf{Q} \subseteq \mathrm{Cn}(B)$ and a Δ_0 formula $P(x)$, the data $\{(P(\underline{n}), 1) : n \leq N\}$ (true, since otherwise they are inconsistent with B) leave $\mathrm{Bel}(\forall x\, P(x))$ at its prior value, for every N.*

The checks are entailed by Q and carry no information, the problem of old evidence and logical omniscience (cf. Hacking, 1967; Garber, 1983); instances of a fresh predicate do confirm, up to the ω-gap. Induction in mathematics needs a reasoner for whom computing $P(\underline{n})$ is costly, such as a logical inductor (Garrabrant et al., 2016a) or a derivation-charged inducer (Appendix C.5).

5.5 The ω-gap

Proposition 5.16 (Anatomy of the ω-gap)**.** *Let $\mathcal{L}$ have a nonempty set* CT *of closed terms, B be decidable and consistent, $\varphi(x)$ have one free variable, and $I := B \cup \{\varphi(t) : t \in \mathrm{CT}\}$ be consistent. Let the stream list $(\varphi(t), 1)$ for all $t \in \mathrm{CT}$, in any order. Then $\mathbb{U}(E_\infty) > 0$, the posterior converges in total variation to the prior restricted to the consistent extensions of I, and the limit trichotomy of $\forall x\, \varphi$ is (i) $(1, 0, 0)$ if $I \vdash \forall x\, \varphi$; (ii) $(0, 1, 0)$ if $I \vdash \neg\forall x\, \varphi$ (I is ω-inconsistent at φ); (iii) otherwise in $(0, 1)^3$, with $\lim \mathrm{Dis} = P_\infty$(the theory is ω-inconsistent at φ) and $\lim \mathrm{Indep} = P_\infty$(the theory proves every instance and decides $\forall x\, \varphi$ neither way).*

So the Bayesian ω-gap is the ω-incompleteness of the background plus the instances, weighted by prior shares. In particular Bel satisfies the Gaifman condition (Gaifman, 1964), a desideratum of Hutter et al. (2013), exactly when $I \vdash \forall x\, \varphi$; otherwise the gap $\lim_n \mathrm{Bel}_0(\bigwedge_{i \leq n} \varphi(t_i)) - \mathrm{Bel}_0(\forall x\, \varphi)$ is the mass of the consistent random theories that prove every instance but not $\forall x\, \varphi$, at least $2^{-K(I)-c}$ (Proposition C.19). This is the provability analogue of Proposition B.10 of Hänni and Claude (Anthropic) (2026a), where for probabilities on structures the gap is the mass of structures with non-standard elements in which every closed instance holds. The Gaifman condition is a desideratum for credences about truth; Bel is a probability of provability, and its failure is not a defect. The universal prior does not close the gap, even for true Π_1 sentences with trivially checkable instances:

Theorem 5.17 (A uniform Gödelian ω-gap)**.** *Let B be decidable and true with $\mathsf{PA} \subseteq \mathrm{Cn}(B)$. For every k there is a Δ_0 formula $\zeta_k(x)$ such that $\forall x\, \zeta_k$ is true, every instance $\zeta_k(\underline{n})$ is a true Δ_0 sentence (so Q-provable, and instance data are free by Proposition 5.15), and $2^{-k-K(k)-c} \leq \mathrm{Bel}_0(\forall x\, \zeta_k) < 2^{-k}$. Hence after any instance data the posterior belief is $\mathrm{Bel}_0(\forall x\, \zeta_k)/(1 - \mathrm{Inc}_0) < 2^{-k+c'}$, and $\mathrm{Dis} + \mathrm{Indep} > 1 - 2^{-k+c'}$.*

$\forall x\, \zeta_k$ is the consistency statement of $B \cup T_k$: were its belief at least 2^{-k}, T_k would contain it and $B \cup T_k$ would prove its own consistency, against (G2).

Remark 5.18 (Where the gap closes). (a) Under the ω-rule, with $B \supseteq \mathsf{Q}$, every random theory that is consistent under that rule has $\mathcal{L}_{\mathrm{ar}}$-content $\mathrm{Th}(\mathbb{N})$: the inducer is omniscient about arithmetic a priori. (b) Under full second-order consequence the loss on any presentation of $\mathrm{Th}(\mathbb{N})$ is at most $K(\mathsf{PA}^2) + c$: the unlearnability of $\mathrm{Th}(\mathbb{N})$ (Proposition A.17) rests on effective consequence. (c) The generative variant closes the gap only with the universal sentence among the data: by Theorem 3.9(c1) of Hänni and Claude (Anthropic) (2026a), with a faithful likelihood and $\forall x\, \varphi$ in the data space, the posterior probability of "proves $\forall x\, \varphi$" tends to $\mathbf{1}[T^* \vdash_{\mathsf{C}} \forall x\, \varphi]$, whether the generator T^* proves it, not to its truth value; for filtered instance data the gap stays open (its part (d1)).

A bound on the number of lines of derivations does not close it either: under a full-support prior the limit belief is then 1 iff a Kreisel-type property holds, and that fails already for a fresh predicate and nine lines (Proposition C.20 and Example C.21).

5.6 True arithmetic

Let $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$, $B \subseteq \mathrm{Th}(\mathbb{N})$ decidable, $(\varphi_i)_{i\geq 1}$ a computable enumeration of all sentences, D_n the first n of them labelled by truth, and $b_{1:n}$ the labels. The loss is unbounded and at most $n + K(n) + c$ (Proposition A.17). It grows at least polynomially, because with binary numerals the sentences "the k-th digit of Ω_V is 1" occur early.

Theorem 5.19 (A polynomial lower bound)**.** *Suppose the enumeration lists sentences by length, and within a length in a fixed order, and that (1) at most $C\kappa^L$ sentences have length at most L, for every L, and (2) there are terms $\underline{k}^{\mathrm{b}}$ with value k and length at most $c_1 \log k + c_2$. Put $\xi := 1/(c_1 \log \kappa)$ and $m(n) := \lfloor (n/C')^{\xi} \rfloor$, where C' depends only on the syntax and on a fixed formula for the binary digits of Ω_V. Then for all n, in the semimeasure convention: (a) $K(b_{1:n} \mid n) \geq m(n) - K(n) - c$; (b) $L_A(D_n) \geq C'^{-\xi} n^{\xi} - K(n) - c$; (c) $L_A(D_n) \geq (2C')^{-\xi} n^{\xi} - 2\log\log n - c$ for $n \geq 4$.*

For a stated Polish-notation syntax, $\xi \geq 0.0823$: a computation with exact rational arithmetic (code in the accompanying repository) certifies the bound on formula counts that this needs (Appendix C.7). The digit sentences among the first n carry about $m(n)$ incompressible bits (Chaitin, 1975), and a true theory decides correctly, so:

Corollary 5.20 (Quantitative incompleteness)**.** *Let T be r.e. with $B \cup T \subseteq \mathrm{Th}(\mathbb{N})$, and let $u_n(T)$ be the number of $i \leq n$ such that $B \cup T$ proves neither φ_i nor $\neg\varphi_i$. Then $u_n(T) \geq C'^{-\xi} n^{\xi}/(2\log n)$ for all large n.*

Conversely, $L_A(D_n) = o(n)$ if some true r.e. theory decides a set of sentences of asymptotic density one in the enumeration (Proposition C.23), a question of generic decidability that depends on the syntax.

Question 5.21. In such an enumeration of the sentences of $\mathcal{L}_{\mathrm{ar}}$ labelled by truth, is $L_A(D_n) = o(n)$?

6 What a theory does and what it is

What does a scientific theory do? It may predict observations, the view behind Solomonoff induction (Solomonoff, 1964); answer questions posed in some language, why-questions included (van Fraassen, 1980, ch. 5); explain (Hempel and Oppenheim, 1948); or solve problems (Laudan, 1977). And what is it? A program that predicts observations; a set of sentences closed under consequence (Suppe, 1977); a class of models (Suppes, 1960; van Fraassen, 1980); a core of models with intended applications (Sneed, 1971; Balzer et al., 1987); or a practice of exemplars and skills (Kuhn, 1970, 1977; Polanyi, 1958). The *axiom view* answers that a theory answers questions posed in a given language and is a logical theory; the *raw-data view*, that it predicts observations and is a program. We assess the axiom view against the theorems of this paper, and say which of our claims are theorems (proofs in Appendix D) and which are arguments.

claim	our assessment (an argument)	formal results it rests on
D answering	true in comparative form; not a new inference principle; survives the objection from statistical practice	Thms. 4.2 and 4.3 and Props. 4.11, 4.12, 6.3, 6.4, D.1 and D.3
S logical form	descriptive part true of declarative textbook presentations (an observation); explanatory part open, and relative to a reasoner	§ 6.4; Thms. 7.2, 8.4 and E.21 and Cor. 7.10
E explanation	true in the unification sense, with at least four gaps	Props. 6.5, D.5, G.32 and G.33
L the ladder	SAI is on step 2, with partial results on step 3; step 4 has no formal handle	Thms. 8.4 and 8.5 and Cor. 7.10
P progress	to be replaced by a list of formal differences; none concerns the shape of hypotheses in the idealised setting	Tab. 3

Table 2: The argument map: our assessments, which are arguments, and the results they rest on, which are theorems in their stated settings.

6.1 An argument map

We separate five claims (Table 2). **D**: a theory answers questions posed in a given language rather than predicting raw data, read comparatively, as a step away from raw-data prediction rather than the end point. **S**: a theory is a logical theory, not a program. **E**: theories explain propositions. **L**: the ladder of Section 1 (raw data; questions posed with existing abstractions; theories that extend the language; theories that revise the base abstractions). **P**: SAI is progress over raw-data prediction, both in what a theory does and in what it is.

What rests on theorems. Proved, in the stated settings: the idealised formalism cannot see the difference between a theory and a program (Section 6.4); answering questions differs from raw-data prediction in target, data type, coherence and partiality, not in the inference principle (Section 6.3); with a cost on the written theory, usefulness depends exponentially on the pair (reasoner, vocabulary), and on declarative against procedural form only by a constant factor once intermediate concepts are available and the reasoner can chain through them (in the model of Section 7; open beyond it); theoretical vocabulary is dispensable in the ideal mixture, while relations or entities pay for finite theories and under charged deduction (Section 8); and "Kepler's laws are only approximately right" is a limitation of proof-requiring compatibility with deterministic theories, not of the axiom view, since in M_A an approximately true law is a law with a calibrated error model (Section 9). Computed, conditional on the pool of forms and the codes: on noise-free interval data, simplicity among compatible forms is relative to the language, and with graded fit the data decide (Section 10). A theorem plus an argued identification: Hempel's inductive-systematisation reply fails in the idealised setting (Section 8.6). Arguments only: the descriptive observation; why declarative shape is good (open); that D survives the objection from statistical practice; the readings of the toys and the general claims about explanation; that the axiom view misses the procedural, statistical and dynamic sides of theories (Section 11); and our refusal, resting on a possibility result (Corollary 5.6), to say anything realist about induced theories.

The ladder. We accept L. The base language of step 2 may be highly theoretical (heliocentric longitudes presuppose Copernican geometry); no theory-neutral observation language is assumed, so the classical objections to one do not apply in the same way. But theory-ladenness (Hanson, 1958) biases comparisons between theories that disagree about the base language, since a theory

that revises it is scored in a language it rejects; this is why the ladder has a step 4.

6.2 The descriptive observation and the explanatory question

Claim S combines a description with an explanation, which we keep apart. *The observation:* textbook presentations of fundamental theories are declarative. They state laws, definitions, principles and models given by equations, not procedures that print predictions; this needs no theory of why it holds. Two qualifications. *Declarative is not sentential:* physics texts write something closer to a class of models specified by equations (Suppes, 1960, 1962) than to first-order axioms. We use sentences because questions and data are sentences and the prior of SAI is over texts; the question-answering view itself is neutral between sentences and classes of models, since in partition semantics a question is a partition of logical space (cf. Groenendijk and Stokhof, 1984). *Theories in use have procedural parts:* methods attached to concepts, numerical schemes, simulation codes; in quantum chemistry a "level of theory" is a computational method, and one position of Mars from Kepler's laws already needs root-finding for Kepler's equation. So the observation concerns textbook presentations, not theories in all their uses.

The question, why the declarative shape is good and whether an ideal inducer should produce it, is open. Craig's construction rules out an answer based only on what can be represented (Section 6.4), so an answer must be relative to a reasoner with costs, or to a community of reasoners. In the model of Section 7, the best formal answer we have, what matters is a reasoner that can do algebra or chain through intermediate concepts, together with those concepts, not logical form as such. The explanation may lie in communication and piecemeal checking, which have no formal setting yet.

Question 6.1 (Why declarative shape?)**.** Is there a formal setting for a community of bounded reasoners who communicate, check and extend theories piece by piece, in which theories of declarative shape provably do better than programs?

A natural argument for S is that a theory answers in many directions (Kepler's third law gives the period from the semi-major axis and conversely; Definition 10.1), while a program computes one direction cheaply. The second half holds of particular programs, not of programs as a class: a generic solver applied to a declarative law costs $O(1)$ extra length, a program with internal variables answers all m directions of a parity law on m variables with $O(m)$ operations, and under program length directions cost nothing. They matter only for a fixed weak reasoner and a cost on the written theory: for one parity law, forward chaining needs exactly m times as many rules as unit propagation needs clauses, over the base variables or with one round, and at most 9 times the written length of the least clausal theory with intermediate concepts and unbounded rounds (Theorems 7.2 and E.21).

6.3 What a theory does: questions against raw data

Questions, and the answers a theory proves, are defined in Definition 2.11, following the logic of questions (cf. Belnap and Steel, 1976; Hintikka, 1999); $[k] := \{1, \dots, k\}$.

Definition 6.2 (Answers)**.** A theory T *answers* a question $\mathcal{Q} = \{\varphi_1, \dots, \varphi_k\}$ if $B \cup T \vdash \varphi_i$ for some i, and *answers it partially* if $B \cup T \vdash \bigvee_{i \in J} \varphi_i$ for some J with $\emptyset \neq J \subsetneq [k]$.

In the Kepler example (Definition 10.1) a question is a cover by interval sentences ("where is Mars at time t, to within $2'$?") or structural ("is the orbit of Mars planar?"); without the

elements of the orbits, Kepler's laws answer structural questions but entail no direction datum with tolerance below π (Proposition H.1). A raw-data predictor answers one question, "what is the next observation?", and answers it fully. SAI answers every question the base language can pose, and reports how much of its posterior answers none:

Proposition 6.3 (The answer distribution)**.** *Let $M_A(D) > 0$, let $\mathcal{Q} = \{\varphi_1, \dots, \varphi_k\}$ be a question, and let $\mathrm{Un}_D(\mathcal{Q}) := \mathbb{U}(E_D \setminus \bigcup_i O_{\varphi_i})/\mathbb{U}(E_D)$, the posterior probability that the random theory proves no direct answer. (a) $\sum_{i \le k} \mathrm{Bel}_D(\varphi_i) + \mathrm{Un}_D(\mathcal{Q}) = 1$. (b) For $\emptyset \neq J \subsetneq [k]$, $\mathrm{Bel}_D(\bigvee_{i \in J} \varphi_i) \geq \sum_{i \in J} \mathrm{Bel}_D(\varphi_i)$, and the difference is the posterior probability that the random theory answers $\mathcal{Q}$ partially through J but proves no φ_i with $i \in J$; for $J = [k]$, $\mathrm{Bel}_D(\bigvee_i \varphi_i) = 1$ and the difference is $\mathrm{Un}_D(\mathcal{Q})$. (c) For $\mathcal{Q} = \{\varphi, \neg\varphi\}$, (a) is the trichotomy $\mathrm{Bel}_D(\varphi) + \mathrm{Dis}_D(\varphi) + \mathrm{Indep}_D(\varphi) = 1$ of Proposition 3.1.*

On E_D the random theory is consistent, so it proves at most one direct answer (Appendix D.1). In SOI the analogue of Un_D mixes logical independence with a labeller's silence (Proposition 4.11).

The conservation-law world. What is a law worth to each task? In a toy world (Proposition D.1 in Appendix D.2) a micro-state $x \in \{0,1\}^m$ moves by a uniformly random permutation of each weight class, so its weight is conserved; the raw data are the trajectory, and the questions are "what is $|x_t|$?". Knowing the law saves a Bayes predictor of the raw data between $\frac{1}{2}\log\frac{\pi m}{2} + o(1)$ bits and one bit more per step (for $t \le Z/2$, Z the size of the weight class), a share $O(\log m/m)$ of its step loss, until the trajectory returns to its start; on $t_{\max}+1$ questions it saves the share $t_{\max}/(t_{\max}+1)$ of the no-law loss. Yet by the chain rule the raw saving at each step equals the loss of the raw no-law predictor's own answer to "what is $|x_{t+1}|$?", and in each task a Bayes mixture of the two hypotheses pays at most one bit for learning the law. A computation (code in the accompanying repository) gives savings of 2.348, 3.331 and 5.326 bits at the start of a trajectory at $m = 16$, 64, 1024, against $\frac{1}{2}\log\frac{\pi m}{2} = 2.326$, 3.326, 5.326, and shares 0.15, 0.052, 0.0052. So the share of the raw loss that the law saves tends to 0, because of micro-detail that no theory addresses, while its absolute gain per step equals its value for the corresponding question, and both tasks learn the law at the same price: the theory is "in" the raw predictor. The tasks differ in their *target*, which losses are counted, not in what must be learned nor in what the law is worth per step; scoring questions is right when what we want is answers. (The saving on questions holds by construction: the toy asks only about the conserved quantity.)

Answers never cost more description than raw modelling plus an *extractor* that reads them off the raw data. Let $\mathbb{X}$ be a finite alphabet, μ a computable probability measure on $\mathbb{X}^{\mathbb{N}}$ (the raw world), $(\varphi_q)_{q \in \mathbb{N}}$ a fixed computable family of sentences (the questions), and η a partial computable function with $\eta(q,x) \in \{0,1\}$ computed from a finite prefix of x; put $f_x(\varphi_q) := \eta(q,x)$ and $\mu_\eta(D) := \mu\{x : f_x \supseteq D\}$.

Proposition 6.4 (Raw modelling plus an extractor)**.** *If f_x is consistent with B for μ-almost every x, then $M_A(D) \geq 2^{-K(\mu)-K(\eta)-c}\mu_\eta(D)$ for every finite data set D of question–answer pairs, so $L_A(D) \leq -\log\mu_\eta(D) + K(\mu) + K(\eta) + c$; here c depends also on the family (φ_q). For a computable sequence x with f_x consistent with B, $L_A(D) \leq K(x) + K(\eta) + c$ on every D with $f_x \supseteq D$.*

A theory machine samples the raw world and emits the extractor's labels (Appendix D.3). The extractor is the bridge from raw data to the question language, and its cost $K(\eta)$ is not small in real cases: it contains whatever recognises a planet in an image. Answering is also the stronger achievement (Proposition D.3 in Appendix D.4): a consistent r.e. theory that proves the data bounds the loss of M_A by its complexity, but on the lottery stream M_A predicts well

while no consistent r.e. theory proves the data; predicting data that present an incomplete theory leaves its open questions open; and the beliefs of SAI are monotone under entailment and closed under conjunction up to additive error, unlike, in the semimeasure convention, the predictions of consistent SOI (Proposition 4.11).

The objection from statistical practice. Much of physics confronts theory with near-raw data through generative models and information-theoretic losses: gravitational-wave searches match waveform templates against the strain series with a Gaussian noise likelihood, cosmology fits likelihoods of microwave-background data, particle physics simulates detectors event by event. D survives in comparative form, we think, because these pipelines exist to answer a few questions (a mass, a spin, a cosmological parameter, whether a particle exists). But the generative model is theory plus instrument plus noise, and the proof-requiring core of SAI, with no native noise model, sits far from this practice (Section 11.1). So D holds in comparative form; its formal content is a change of data type, logical coherence, partiality and the choice of target, not a new inference principle (one-tape SAI $\asymp$ one-tape consistent SOI, Theorem 4.3); and the real difficulty, the extractor, SAI takes as given.

Purpose and logarithmic loss. Grading a prediction for use and scoring credences are different jobs. For the first, the purpose decides what is graded and with which tolerance, as with the interval data of Section 10; Solomonoff axiom induction lets the purpose fix the propositions, and scores credences on those. For the second, the logarithmic score is, up to positive affine transformation, the only strictly proper local scoring rule on more than two outcomes (a result attributed to Bernardo, 1979), but which proper score is privileged is contested (accuracy-first arguments favour quadratic inaccuracy, Leitgeb and Pettigrew, 2010; Pettigrew, 2016), and for bounded or misspecified models different proper losses select different best models. We do not claim that the logarithmic score is privileged; even inside SAI the price of a datum can depend on the loss convention by an unbounded amount (Proposition 5.10).

6.4 What a theory is, after Craig

The idealised formalism cannot see shape. Every consistent assigner of truth values is an axiom set at $O(1)$ extra bits. Coin-free SAI $\asymp$ coin-free consistent SOI, by Craig's construction inside the universal prior (Theorem 4.2), and one-tape SAI $\asymp$ one-tape consistent SOI (Theorem 4.3). Adding vocabulary changes the weights of old-vocabulary data by $O(1)$ (Corollary 3.10 and Theorem 8.5); coin-free, r.e. axiom induction is finite-theory induction in a three-sorted extension, up to $O(1)$ (Theorem F.28). Program length sees neither the reasoner nor the vocabulary (Theorem 7.2); the posterior cannot move between deductively equivalent formulations, whose complexities in the flagpole toys below differ by $O(1)$ however many laws are added (Proposition D.5); and the diagonal separations from SOI under time bounds come from free deduction, not from axioms (Proposition 4.16). So the idealised formalism cannot see the difference between a theory and a program. Nor does D support S as a claim about what an ideal inducer should produce: the labellers of SOI carry no theory and are equivalent to theories, and Khardon and Roth (1997) give a learning-to-reason algorithm for a class of propositional languages that has no efficient reasoning algorithm when represented as a formula-based knowledge base.

Where shape matters, it is relative to a reasoner and a vocabulary, and only when the written theory is charged: in the model of Section 7 the exponential separations are between pairs (reasoner, vocabulary) (Corollary 7.10); in the settings of Section 8 what pays is theoretical re-

lations and entities, not vocabulary (Theorem 8.4). We still find it plausible that good theories are shaped like logical theories; what is missing is a formal setting in which the shape matters (Question 6.1), and nothing in the formal results favours sentences over classes of models. Since the limit posterior can live on random theories, and in true arithmetic with true but adversarially chosen data on false ones (Theorem 5.1 and Corollary 5.6; a possibility result), we say nothing realist about induced theories.

6.5 Explanation

In the axiom view T explains φ when $B \cup T \vdash \varphi$, the deductive-nomological pattern (Hempel and Oppenheim, 1948), and an explanation is better the more a short theory derives, as in unification accounts (Friedman, 1974; Kitcher, 1981); the explanatory power of Definition 9.6 makes this precise (Proposition G.31 relates it to Myrvold's account, Myrvold, 2003). But derivability is symmetric where explanation is not: a pole's height and the sun's elevation entail the shadow's length, and the shadow's length and the elevation entail the height (Bromberger's example, Bromberger, 1966, as attributed by Glymour, 1978). Over the theory B of real closed fields, with poles p_i, heights $\mathrm{ht}(p)$ and shadow lengths $\mathrm{sh}(p,u)$ when the tangent of the sun's elevation is $u > 0$, the shadow law has two formulations: Λ_h says $\mathrm{sh}(p,u) \cdot u = \mathrm{ht}(p)$; Λ_s says $\mathrm{sh}(p,u) \cdot u = \mathrm{sh}(p,1)$ and $\mathrm{ht}(p) = \mathrm{sh}(p,1)$. With initial conditions on the heights or on the reference shadows $\mathrm{sh}(p_i,1)$ they give T_H, T_S (with Λ_h) and T'_H, T'_S (with Λ_s). A derivation is charged by the number of closed instances of the law's universally quantified conjuncts that it uses; C_H, C_S extend T_H, T_S by intervention vocabulary with the height, resp. the reference shadow, exogenous (precise forms in Appendix D.5).

Proposition 6.5 (The flagpole)**.** *(a) T_H, T_S, T'_H, T'_S are deductively equivalent over B; so every data set is compatible with all four or with none, in the coin-free version the posterior odds of any two programs for them equal their prior odds on every data set compatible with them, and $|K(T_S) - K(T_H)| \leq c$. (b) Under derivation charges, every shadow datum $\mathrm{sh}(p_i,u) = v$ with $u \neq 1$ that they entail needs exactly one instance from T_H and T'_S, whose initial conditions sit at the hub of their formulation, and exactly two from T_S and T'_H: which quantity is the hub is a property of the formulation, not of the world. (c) C_H and C_S have the same consequences in the old vocabulary, and one interventional datum separates them.*

What the toys show. The explanation relation of SAI is direction-free, and its posterior cannot move between deductively equivalent theories. Kitcher's unificationist answer (Kitcher, 1989), in its usual summary, is that the best systematisation of all our beliefs derives the dimensions of objects from their origin and development; a height-from-shadow pattern would add a pattern, which fails for objects that cast no shadow. With k further laws in which the height figures (Proposition D.5 in Appendix D.6), the formulation with the shadow as hub is longer by an amount growing with k in a literal code without definitions (relative to the primitives, or because some objects cast no shadow), by a constant with one definition, and by $O(1)$ under the universal prior; under derivation charges only a mixed theory loses. So Kitcher's answer is an effect of whole-theory simplicity that SAI reproduces only under a listing code, like the textbook point of Section 7. Direction enters cleanly with interventional questions, once the base language can pose them; but the intervention semantics was written into C_H and C_S by hand, so the toy is consistent with the interventionist account (Woodward, 2003) and gives it no support over rivals such as the causal-process account (Salmon, 1984).

We conclude that derivation from a short theory is explanation in the unification sense, with at least four known gaps; we do not claim that derivation is explanation outright. The gaps are *asymmetry*, as above; *explanation by approximate correction*, since Newton's theory does not entail Kepler's laws but, in an exact toy, proves the honest old law on every window and refutes the exact one (Proposition G.33), so that "T explains φ" must include "T derives an honest approximation of φ with an error bound on a domain"; *irrelevance*, as in the standard examples of hexed salt and of a man taking birth-control pills, where the length prior penalises an irrelevant axiom in a whole theory but a derivation does not mark which axioms it used; and the *conjunction problem* (L is derived from $L \wedge L'$), which the length prior and the explanatory power address in part. Unification can register an explanatory gain without a gain in accuracy: against Ptolemy's system with the coincidences of retrogression and opposition stated as axioms, which has the same fit, Copernicus's wins at every steepness exactly when the heliocentric reorganisation costs fewer bits than the coincidences (Proposition G.32).

6.6 What separates SAI from raw-data prediction

We replace claim P by the formal differences between SAI and a predictor of an observation stream, sorted into advantages for claims D and E, neutral differences and liabilities (Table 3).What does not separate the two is the inference principle, a Bayes mixture over a universal class, and, in the idealised setting, the shape of hypotheses. This is the progress claim we believe: for what a theory does, SAI is a step in the right direction; for what a theory is, the idealised formalism is neutral.

7 Theory shape relative to a reasoner

The axioms of a mechanics textbook do not yield the solutions of its exercises by a simple mechanical procedure: the reader brings methods, lemmas and understanding that are not written down.[5] The inducers of the previous sections cannot express this: they charge a theory by the length of a program for it, a short program can contain a generic solver, and by Craig's construction every consistent assigner of truth values is an axiom set at constant cost (Theorem 4.2 and Section 6.2). A sense in which a written theory is well or badly shaped must be relative to a reasoner that uses it and to a cost that charges what is written. In the simplest model in which this can be stated, program length sees neither the reasoner nor the vocabulary; charged as written, the size of a usable theory depends exponentially on the pair (reasoner, vocabulary) (Table 4); and declarative against procedural form matters only by a constant factor once intermediate concepts are available and the reasoner can chain through them, and, without intermediate concepts or with a single round, by the number m of query directions of one parity law.

7.1 The model

Laws constrain the joint values of m observed quantities, and questions give some values and ask for another (the conditional questions of Definition 2.11, with literals as sentences).

Definition 7.1 (Theories, laws, reasoners, inducers)**.** Fix base variables $x_1, \ldots, x_m$ and auxiliary variables $y_1, y_2, \ldots$, the *intermediate concepts* (0-ary relation symbols of a first-order language $\mathcal{L}_m$ with equality). (a) An *explicit theory* Φ is a finite set of non-tautological clauses, a *linear theory*

[5] That written theories are far from sufficient for a simple reasoner, and are aids for a reasoner with much tacit understanding, is argued in Hänni's working notes (Hänni, 2026).

	difference	support
Advantages for claims D and E		
1	*Data type:* labelled sentences of a chosen language; questions are sets of sentences.	Defs. 2.11 and 6.2
2	*Logical coherence:* consequences of the data get posterior probability 1, while in the semimeasure convention consistent SOI is never certain of an unasked consequence; Bel is monotone under entailment and closed under conjunction up to additive error.	Props. 4.11 and D.3
3	*Partiality:* "independent" is a native third value, whose mass is the semimeasure defect; for a k-way question the answer masses and the unanswered mass sum to 1.	Props. 3.1 and 6.3
4	*An observed "independent" label* separates theories from labellers by an unbounded amount on one datum, coin-free and one-tape, when $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$ and $\mathsf{Q} \subseteq \mathrm{Cn}(B)$.	Prop. 4.12
5	*Laws are sentences with belief values:* $\mathrm{Bel}_D(\forall x\, \varphi)$ differs from the probability that all future data conform.	Prop. 5.16
6	*Answers cost no more description than raw modelling plus an extractor,* and can cost far less.	Props. 6.4 and D.1
Neutral differences		
7	*Piecemeal confirmation:* credence in single laws, though only whole theories meet the data: the standard Bayesian treatment of Duhem's problem (Dorling, 1979), which Strevens (2001) argues begs the question. SAI inherits the debate; its allocation of blame is fixed by the prior, hence by the syntax.	definitions; an argument
8	*Truth from unprovability:* if $\mathsf{Q} \subseteq \mathrm{Cn}(B)$, belief or independent mass on a Π_1 sentence certifies its truth; a curiosity of arithmetic data.	Prop. 5.11
9	*Explanatory direction needs questions the language can pose:* interventional data separate causal theories that the old vocabulary cannot.	Prop. 6.5(c); in general an argument
Liabilities		
10	*Random theories, false theories:* on the lottery stream bounded one-tape loss is achieved by random theories and by no consistent r.e. theory; in true arithmetic with true (adversarial) data the limit posterior can live on false theories, a possibility result.	Thm. 5.1 and Cor. 5.6
11	*Language-relative simplicity, within bounds:* faithful translations change weights by $2^{\pm(K(\tau)+c)}$; within that bound the base language decides real choices, such as the preferred orbit of Mars on noise-free interval data.	Thm. 3.9; § 10

Table 3: What formally separates SAI from raw-data prediction. Items 7 and 9 and the sorting are arguments; the other items are results.

a finite set of equations over GF(2), in these variables, up to renaming of the auxiliary ones; $\operatorname{proj}\Phi := \{x \in \{0,1\}^m : \exists y\ (x,y) \models \Phi\}$. The hypothesis classes are $\mathcal{E}_m^{\mathrm{base}}$ (no auxiliary variables) and $\mathcal{E}_m$ (any); likewise for linear theories. $|\Phi|$ counts clauses (equations), the *written length* $\|\Phi\|$ literal occurrences. (b) A *law* is a nonempty $\mathcal{S} \subseteq \{0,1\}^m$. $\mathcal{Q}(\mathcal{S})$ is the set of conditional questions (α, ℓ), α a partial assignment to the base variables consistent with $\mathcal{S}$ and ℓ a literal on a base variable outside $\operatorname{dom}\alpha$, with $\mathcal{S} \wedge \alpha \models \ell$; a *conditional data set* is a finite set of such pairs. (c) A *reasoner* $\mathcal{R}$ maps a theory and an antecedent α to a set of base literals or to a conflict $\perp$; it is *sound* if $\ell \in \mathcal{R}(\Phi, \alpha)$ implies $\Phi \cup \alpha \models \ell$, and $\perp$ occurs only if $\Phi \cup \alpha$ is unsatisfiable. The named reasoners are sound: UP_k, k *parallel* rounds of unit propagation (in each round every clause whose literals are all falsified but one, that one unassigned, assigns it; opposite derivations or a falsified clause give $\perp$); $\mathrm{UP} := \mathrm{UP}_\infty$; Gaussian elimination GE on linear theories and unbounded proof search PS_∞, which return exactly the entailed base literals. (d) Φ is $\mathcal{R}$*-compatible* with D if for every $(\alpha, \ell) \in D$, $\Phi \cup \alpha$ is satisfiable and $\ell \in \mathcal{R}(\Phi, \alpha)$; $\mathcal{R}$*-usable* for $\mathcal{S}$ if $\operatorname{proj}\Phi = \mathcal{S}$ and Φ is $\mathcal{R}$-compatible with $\mathcal{Q}(\mathcal{S})$. (e) The inducers range over the $\mathcal{R}$-compatible $\Phi \in \mathcal{E}_m$ (linear theories for GE); a superscript base restricts them to $\mathcal{E}_m^{\mathrm{base}}$. *Program length*: $K_A^{\mathcal{R}}(D) := \min|p|$ over the p with $V(p) = \langle\Phi\rangle$, and $W_A^{\mathcal{R}}(D) := \sum 2^{-|p|}$ over the same p. *Listing*: $K_{\mathrm{lst}}^{\mathcal{R}}(D) := \min \ell_{\mathrm{lst}}(\Phi)$ and

$W^{\mathcal{R}}_{\mathrm{lst}}(D) := \sum 2^{-\ell_{\mathrm{lst}}(\Phi)}$, for the prefix code $\ell_{\mathrm{lst}} \geq \|\cdot\|$ of Equation (4), which lists the clauses literal by literal. *Time-charged*: $Kt^{\mathcal{R}}_A(D) := \min(|p| + \log t_p)$ over the p of $K^{\mathcal{R}}_A$, t_p the running time of V on p.

Under the two-valued encoding a single point of the law would fit all its question data (Proposition E.3). Under (E3), with unbounded proof search, the program-length inducers are K_A and W_A on question data, up to a constant for r.e. hypotheses and up to $m + O(\log m)$ bits for finite explicit theories (Proposition E.4): they extend SAI. Call $\mathcal{S} = \{x : Ax = b\} \neq \emptyset$, with $A \in \mathrm{GF}(2)^{r\times m}$ of rank $r \geq 1$, an *affine law*. For affine laws and for the implication chain on $m \geq 3$ variables, compatibility with $\mathcal{Q}(\mathcal{S})$ already forces $\operatorname{proj}\Phi = \mathcal{S}$ (Lemma E.2).

7.2 Program length sees neither the reasoner nor the vocabulary

The prime implicates $\mathrm{PI}(\mathcal{S})$ of a law are its inclusion-minimal implicates among the clauses over the base variables; one round of unit propagation on them answers every question of $\mathcal{Q}(\mathcal{S})$ (Lemma E.5).

Theorem 7.2 (Reasoner invariance under program length)**.** *Let $\mathcal{R}'$ be a sound reasoner whose hypotheses are explicit or linear theories, of either class. For every conditional data set D, $K^{\mathrm{UP}_1,\mathrm{base}}_A(D) \leq K^{\mathcal{R}'}_A(D) + c$ and $W^{\mathrm{UP}_1,\mathrm{base}}_A(D) \geq 2^{-c} W^{\mathcal{R}'}_A(D)$; and $K^{\mathrm{UP}_1}_A = K^{\mathrm{UP}}_A = K^{\mathrm{PS}_\infty}_A \pm c$ for finite explicit theories of either class.*

Proof idea (Appendix E.2). If Φ' is $\mathcal{R}'$-compatible with D, soundness makes $\mathrm{PI}(\operatorname{proj}\Phi')$ UP_1-compatible with D, and a fixed prefix computes it from a program for Φ'.

This is Craig's collapse again, now for reasoners and vocabularies: the know-how a weak reasoner lacks can live in a generator of the written theory, at constant cost. It becomes visible when what is written, or the time to write it, is charged.

Theorem 7.3 (Where the reasoner and the vocabulary become visible)**.** *Let $\mathcal{S}$ be an affine law, $\mathcal{R}$ a sound reasoner, and take either class. (a) $K^{\mathcal{R}}_{\mathrm{lst}}(\mathcal{Q}(\mathcal{S}))$ is the least $\ell_{\mathrm{lst}}(\Phi)$ over the $\mathcal{R}$-usable Φ of the class. (b) $Kt^{\mathcal{R}}_A(\mathcal{Q}(\mathcal{S})) \geq \log\min\|\Phi\|$ over the same Φ; and $Kt^{\mathrm{UP}_1}_A(\mathcal{Q}(\mathcal{S})) \leq Kt(A,b) + \log|\mathrm{PI}(\mathcal{S})| + O(r + \log m)$, $Kt^{\mathrm{UP}}_A(\mathcal{Q}(\mathcal{S})) \leq Kt(A,b) + O(r + \log m)$, $Kt^{\mathrm{GE}}_A(\mathcal{Q}(\mathcal{S})) \leq Kt(A,b) + O(\log m)$. (c) $K^{\mathcal{R}}_A(\mathcal{Q}(\mathcal{S})) = K(\mathcal{S}) \pm c$ for $\mathcal{R} \in \{\mathrm{UP}_k\ (k \geq 1), \mathrm{UP}, \mathrm{PS}_\infty, \mathrm{GE}\}$.*

Proof idea (Appendix E.2). (a) is Lemma E.2. (b) Printing Φ takes $\|\Phi\|$ steps; a program can compute (A,b) and print the prime implicates, a library with intermediate concepts, or the equations. (c) $\mathcal{S}$ gives $\mathrm{PI}(\mathcal{S})$, and a compatible Φ gives $\operatorname{proj}\Phi = \mathcal{S}$. So the separations below concern written size, which by (a) is K_{lst} up to the code length per literal.

7.3 Over the base variables: the exact minimum

How large must a theory be for unit propagation if it may use only the observed quantities? A *derived equation* of an affine law is $u = yA$, with $u \cdot x = b_u := y \cdot b$ on $\mathcal{S}$; d_* is the least weight of a nonzero one; a *circuit* is a nonzero derived equation whose support contains the support of no other. Let $C_{Z,\beta}$ be the clause over the variable set Z falsified exactly by the assignment β; the prime implicates are the $C_{\mathrm{supp}(u),\beta}$ with u a circuit and $u \cdot \beta \neq b_u$ (Proposition E.7). A circuit u is *decomposable* if $u = u' + u''$ for nonzero derived equations whose supports meet in a single base variable outside $\mathrm{supp}(u)$: two derived equations chained through one intermediate base quantity. Let $\Phi_{\mathrm{ind}}(\mathcal{S}) := \{C_{\mathrm{supp}(u),\beta} : u \text{ an indecomposable circuit}, u \cdot \beta \neq b_u\}$.

Proposition 7.4 (What is forced)**.** *For every affine law, whatever d_*, the clauses contained in every* UP*-usable explicit theory over the base variables are exactly those of* $\Phi_{\mathrm{ind}}(\mathcal{S})$.

Theorem 7.5 (The unique minimum when $d_* \geq 3$)**.** *If $d_* \geq 3$, $\Phi_{\mathrm{ind}}(\mathcal{S})$ is* UP*-usable. Hence an explicit theory over the base variables equivalent to $\mathcal{S}$ is* UP*-usable if and only if it contains $\Phi_{\mathrm{ind}}(\mathcal{S})$, which is the unique minimum, with $\sum_{u \text{ indecomposable}} 2^{|u|-1}$ clauses.*

Proof idea (Appendix E.3). From an antecedent that fixes all of an indecomposable circuit but one variable, the first round must fire a clause supported on that circuit, since a clause through another variable would decompose it; any other implicate can be deleted from the set of all implicates. For sufficiency, induct on the weight: a decomposable circuit is answered in two steps through its intermediate variable, and $d_* \geq 3$ makes both parts lighter.

So a circuit must be written down if and only if it cannot be obtained by chaining two derived equations through one base quantity: base quantities already act as intermediate concepts (Corollary E.11 says when the given equations suffice; Remark E.12 treats $d_* \leq 2$).

Corollary 7.6 (The simplex family)**.** *Let $r \geq 3$, $m = 2^r - 1$, and let the columns of A be the nonzero vectors of* $\mathrm{GF}(2)^r$*. Every nonzero derived equation has weight $(m+1)/2$ and is an indecomposable circuit; the unique minimum* UP*-usable theory over the base variables is* $\mathrm{PI}(\mathcal{S})$*, with $m\,2^{(m-1)/2}$ clauses of width $(m+1)/2$, while* GE *uses the $r = \log(m+1)$ equations.*

The same holds for one round with any intermediate concepts (Proposition 7.7), but not for unbounded propagation with them: for $r \geq 4$, writing each circuit by its own chain of partial sums gives a UP-usable theory of $m(2m-4)$ clauses of width at most 3 (Remark E.15). So the simplex family separates the reasoners exponentially only over the base variables or for one round.

7.4 Intermediate concepts help only a reasoner that can chain through them

One parity law $\mathrm{Par}_m^b := \{x : x_1 \oplus \cdots \oplus x_m = b\}$, $m \geq 3$, needs all its 2^{m-1} clauses for unit propagation over the base variables (Corollary E.11). With the partial sums $y_1 = x_1 \oplus x_2$, $y_i = y_{i-1} \oplus x_{i+1}$ as intermediate concepts, $4m-6$ clauses suffice for unit propagation when $m \geq 4$ (Proposition E.13), but only for a reasoner that can chain.

Proposition 7.7 (One round cannot use intermediate concepts)**.** *For any law $\mathcal{S}$, every* UP_1*-usable explicit theory, with any auxiliary variables, contains* $\mathrm{PI}(\mathcal{S})$*: 2^{m-1} clauses for one parity law, $m2^{(m-1)/2}$ for the simplex family.*

Proof idea (Appendix E.4). In one round from base literals, a clause with an auxiliary literal can derive only that literal, so the answers come from the clauses over the base variables, which are implicates of $\mathcal{S}$; a prime implicate can then be answered only by itself.

Proposition 7.8 (Rounds and size for one parity law)**.** *(a) For $m \geq 4$ the chain encoding is* UP_{m-1}*-usable and not* UP_{m-2}*-usable. (b) The* balanced-tree encoding*, along a rooted binary tree of height $\lceil \log m \rceil$ whose leaves are the base variables, with $y_v = y_{v'} \oplus y_{v''}$ (four clauses) at each internal node v other than the root and $y_{v'} \oplus y_{v''} = b$ (two clauses) at the root, has $4m-6$ clauses ($m \geq 2$) and is* $\mathrm{UP}_{2\lceil \log m \rceil - 1}$*-usable. (c) There is an absolute constant $c_{\mathrm{H}} > 0$ such that for every $k \geq 1$ and every m large enough given k, every* UP_k*-usable Φ for* Par_m^b*, with any auxiliary variables, has $3\|\Phi\| + 2m \geq 2^{c_{\mathrm{H}}(m-1)^{1/(2k-1)}}/k$; likewise, with $w-1$ for $m-1$, for every affine law with a circuit of weight w. (d) Hence* UP_k*-usable theories of size polynomial in m for one parity law need $k = \Omega(\log m / \log\log m)$; $2\lceil \log m \rceil - 1$ rounds suffice for $4m-6$ clauses.*

Proof idea (Appendix E.4). (a), (b): follow the round in which each auxiliary value becomes known. (c): k rounds of unit propagation are a monotone circuit of depth $2k$ with at most $k(3\|\Phi\| + 2m)$ gates (Lemma E.18) that computes the parity of $m-1$ bits; apply Håstad's lower bound for parity (Håstad, 1986).

7.5 Systems of laws, and the regimes

For systems of laws, neither intermediate concepts nor depth need help. In the following law the observed quantities are fixed linear functionals of an unobserved state. Let Λ_q be the set of $v = (u, a) \in \mathrm{GF}(2)^{q+1}$ with $u \in \mathrm{GF}(2)^q$ of weight 1 or 3 and $a \in \{0, 1\}$, and $\mathbf{e}_\star := (0^q, 1)$. The *hidden-state law* on the $m = |\Lambda_q| + 1$ base variables $(x_v)_{v\in\Lambda_q}$, $x_\star$ is $\mathcal{H}_q := \{((z \cdot v)_{v\in\Lambda_q}, z \cdot \mathbf{e}_\star) : z \in \mathrm{GF}(2)^{q+1}\}$. With the state $z = (y, z_0)$ as intermediate concept it is written as the m equations $x_{(u,a)} = u \cdot y + a\, z_0$, $x_\star = z_0$, each of weight at most 5.

Theorem 7.9 (Systems of laws need exponential written length)**.** *The m equations above are* GE*-usable for $\mathcal{H}_q$. There are $\varepsilon > 0$ and q_0 such that for every $q \geq q_0$, every* UP*-usable explicit theory Φ for $\mathcal{H}_q$, with any auxiliary variables, clause widths and number of rounds, has $\|\Phi\| \geq 2^{q^\varepsilon/2 - O(1)} - m = 2^{\Omega(m^{\varepsilon/3})}$.*

Proof idea (Appendix E.4). Setting a set S of the x_v to 0 determines $x_\star$ exactly when the XOR system $\{u \cdot y = a : (u, a) \in S\}$ is unsatisfiable. So unit propagation on a usable Φ, read as a monotone circuit (Lemma E.18), decides this with $O((\|\Phi\| + m)^2)$ gates; the monotone 3-XOR-SAT function of Göös et al. (2019) is a projection of it and needs monotone circuits of size 2^{q^ε}.

The proof is a short combination of the monotone-circuit view of unit propagation (Bessiere et al., 2009) with the bound of Göös et al. (2019); it improves the superpolynomial bound of Gwynne and Kullmann (2014) for arc-consistent representations of XOR systems, which are our usable theories (Lemma E.17 and Remark E.19). We did not find it stated; it is probably known to specialists. Intermediate concepts private to one law add nothing to a system (Proposition E.16). A solver whose unit propagation does linear algebra over parity constraints (Beame and Sun, 2026) is a reasoner for which such laws are cheap.

Corollary 7.10 (The regimes)**.** *For one parity law ($m \geq 3$), the simplex family and the hidden-state law, the least sizes of theories usable by* UP_1*, by* UP_k *(k fixed, or $k \geq 2\lceil \log m \rceil - 1$), by* UP *and by* GE*, over the base variables or with any intermediate concepts, are as stated in Table 4.*

So the exponential separations are between pairs (reasoner, vocabulary). Intermediate quantities help a reasoner only if it can chain through them: not at all with one round, not enough with any fixed number of rounds, and fully, for one law, from about $2 \log m$ rounds. For the hidden-state law no vocabulary helps unit propagation at any depth, while elimination needs only the natural one.

7.6 Directionality

A natural guess is that logical form matters because a law can be used in every direction (Kepler's third law gives the period from the semi-major axis and conversely), while a program computes one. Take as the procedural counterpart a *rule base*: a set of sound directed rules $P \Rightarrow \ell$ over the base variables (a set P of literals consistent with $\mathcal{S}$ and a literal ℓ on another variable, with $\mathcal{S} \wedge P \models \ell$), used by forward chaining. For one parity law the only usable rule base is the set

reasoner $\mathcal{R}$	parity, base	parity, interm.	simplex, interm.	hidden state, interm.
UP_1	2^{m-1}	2^{m-1}	$m2^{(m-1)/2}$	$2^{\Omega(m^{\varepsilon/3})}$
UP_k, k fixed	2^{m-1}	$2^{\Omega(m^{1/(2k-1)})}$	$2^{\Omega(m^{1/(2k-1)})}$	$2^{\Omega(m^{\varepsilon/3})}$
UP_k, $k \geq 2\lceil \log m \rceil - 1$; UP	2^{m-1}	$\leq 4m-6$	$\leq m(2m-4)$	$2^{\Omega(m^{\varepsilon/3})}$
GE	1 equation	1 equation	$\log(m+1)$ equations	$\leq m$ equations

Table 4: The regimes (Cor. 7.10): the least size of a theory usable by the reasoner of the row, with the round budget the row states, in the class of the column (base: no auxiliary variables; interm.: any intermediate concepts). The inducer is $K^{\mathcal{R}}_{\mathrm{lst}}$ (superscript base in the first column), whose value on $\mathcal{Q}(\mathcal{S})$ is this size up to the code length per literal (Thm. 7.3(a)). Entries without a sign are exact and count clauses (equations for GE); $\leq$ marks the size of an explicit usable theory; $2^{\Omega(\cdot)}$ bounds the written length from below. Sources: first column Cor. E.11; UP_1 Prop. 7.7; fixed k and $4m-6$ Prop. 7.8; $m(2m-4)$ Rem. E.15; last column Thm. 7.9 (equations of weight ≤ 5). Under program length every entry is $K(\mathcal{S}) \pm c$ (Thm. 7.3(c)).

of all $m2^{m-1}$ sound rules, m times the 2^{m-1} clauses of the least UP-usable theory over the base variables: a factor m, the number of query directions, and the same factor holds for one round of reasoning with any vocabulary. But for every law and round budget the factor is at most the largest clause width of a smallest usable theory, with intermediate concepts and unbounded rounds it is at most 9 in written length, a program with internal variables answers all m directions of parity with $O(m)$ operations, and under program length the factor vanishes (Theorem E.21). So the dependence on directions is real but cheap: logical form does not get its support from directionality.

7.7 Where the know-how lives

The know-how to answer every question can be paid for in the reasoner (the r equations with GE), in written lemmas, or in a computation that the cost model does not charge: a generator under program length (Theorem 7.2), or a membership test with free search for derivations. For the latter, $\mathrm{PI}(\mathcal{S})$ has a decider of length $K^q(A,b)+c$ (q a suitable polynomial) running in time polynomial in m, and each $\bigwedge \alpha \to \ell$ with $(\alpha,\ell) \in \mathcal{Q}(\mathcal{S})$ follows from one of its members by propositional reasoning (Proposition E.24). Under the readings of the charged deduction of Theorem 4.18, extended to conditional data (Definition E.25), this decider is a compatible hypothesis of weight at least $2^{-K^q(A,b)-O(\log m)}$, given a polynomial bound on the derivations that we sketch, while every compatible hypothesis has weight at most $2^{-K(\mathcal{S})+c}$ (Theorem E.26). So these readings give the full lemma library ($m2^{(m-1)/2}$ clauses for the simplex family) a cost within $K^q(A,b) - K(\mathcal{S}) + O(\log m)$ bits of the least possible: the know-how lives in the decider, which runs elimination. This locates the fit classes of Corollary B.22. The readings bound the size of a derivation that exists and do not charge finding it (Section 4.6); they model a reasoner that is handed the derivation, and cannot express the textbook point, which concerns a reasoner that must find the answer. A deterministic bounded reasoner with a cost on the written theory can.

7.8 Learning, and depth against size

A weak reasoner must also acquire its know-how (Theorem E.29). For one parity law ($m \geq 3$), call the non-solution obtained by flipping the asked variable in the completion the *pattern* of a question; there are 2^{m-1} patterns. Over the base variables, every inclusion-minimal UP-compatible theory

fails on some question of every unseen pattern, so it is fully correct only after all have been seen, and on data made of whole patterns the listing inducer pays at least $\mathrm{len}_m \approx m \log m$ bits per pattern (semimeasure convention); with GE, or with intermediate concepts and unbounded propagation, it pays $O(m \log m)$ bits in all, and under program length there is no gap.

Depth trades against size even without new vocabulary (Theorem E.31). For the implication chain $x_1 \to \cdots \to x_m$, the least UP_k-usable theory over the base variables has as many clauses as a minimum k-hop transitive-closure spanner of the directed path, $\Theta(m\,\mathrm{ia}_k(m))$ (Alon and Schieber, 1987; Bhattacharyya et al., 2012), ia_k the k-th inverse-Ackermann row: $\binom{m}{2}$ at $k = 1$, $\Theta(m \log m)$ at $k = 2$, down to the $m - 1$ axioms of the law at $k \geq m - 1$. At depth 2, intermediate concepts and wide clauses save at most a factor of about 2 (Theorem E.34). In first-order logic the cut-free proofs from a lemma-free theory can be non-elementarily longer (Statman, 1979; Orevkov, 1982), and we sketch how valid lemma sentences restore short ones (Remark E.37).

7.9 What this shows and does not show

Question 7.11. Does the dependence of written size on the pair (reasoner, vocabulary) extend beyond Boolean laws: to real-valued laws with numerical solvers, to probabilistic programs with conditioning, to simulation codes?

This is the precise form of the open question why theories are shaped like logical theories (Question 6.1).

Reading. Whether a written theory suffices depends on the reader. Paired with a reasoner that can do algebra, a linear law's few equations suffice. Paired with one that only applies rules, the theory must spell out exponentially many derived laws, unless it is written with new intermediate quantities and the reader can chain many steps through them. Even then, some systems of laws need exponentially long written theories for such a reader, whatever quantities are introduced. The idealised inducer cannot see any of this, because it charges a theory by the length of a program that may compute the missing lemmas. The model does not show that logic, rather than probabilistic programs with conditioning or classes of models, is the right formal home for theories: the large separations are between pairs (reasoner, vocabulary), and a rule base with intermediate concepts is within a constant factor of a clausal theory. It does not touch the gap between a problem as posed and its formalisation, nor tacit skill (Section 11.2). Its laws are linear or implicational, its reasoners unit propagation, forward chaining and elimination, and its cost the written length. It is not a theorem about textbooks.

8 Theoretical terms: Craig, Ramsey and the theoretician's dilemma

Theories introduce vocabulary that their data do not contain: field, entropy, gene. Hempel's theoretician's dilemma (Hempel, 1958) makes this a puzzle. If theoretical terms serve their purpose of systematising the observable facts, they are dispensable: by Craig's theorem every recursively enumerable theory has a recursive axiomatisation in the observational vocabulary (Craig, 1953), and by Ramsey's method they can be quantified away (Ramsey, 1931). If they do not serve it, they are pointless. Solomonoff axiom induction gives the question a precise form. Data are labelled sentences of an observational language and hypotheses are axiom systems that may use further vocabulary, so one can ask in which hypothesis classes theoretical terms change the posterior,

and what it is that pays: vocabulary, relations or entities. This is the third step of the ladder of Section 6.1. Theorem 8.4 gives the answer; the rest of the section explains its parts and states the main ones precisely, and Appendix F has the proofs and the remaining statements.

8.1 Setting and the dilemma theorem

Definition 8.1 (Observational restriction)**.** Let $\mathcal{L} = \mathcal{L}_O \cup \mathcal{L}_T$ be a countable language with equality and computable syntax (finite where derivations are charged), with *observational* part $\mathcal{L}_O$; B may contain theoretical symbols. The *observational content* of $A \subseteq \mathrm{Sent}_{\mathcal{L}}$ is $T_O(A) := \mathrm{Cn}(A) \cap \mathrm{Sent}_{\mathcal{L}_O}$, and D is *observational* if $\mathrm{dom}\, D \subseteq \mathrm{Sent}_{\mathcal{L}_O}$. Let $G_U^O := \{(w,\chi) \in G_U : \chi \in \mathrm{Sent}_{\mathcal{L}_O}\}$, $M_A^{\mathcal{L}_O} := \nu_{G_U^O}$, $W_A^{\mathcal{L}_O}(D) := \sum_e \mathbf{m}(e)\mathbf{1}[B \cup (\mathsf{W}_e \cap \mathrm{Sent}_{\mathcal{L}_O})$ consistent and $\vdash \psi_D]$, and let $K_A^{\mathcal{L}_O}(D)$ be the least $K(e)$ over the same e. For a Borel set Θ of observational contents (subsets of $\mathrm{Sent}_{\mathcal{L}_O}$, product topology), $\mathcal{C}_\Theta := \{T : T \cap \mathrm{Sent}_{\mathcal{L}_O} \in \Theta\}$, and $\mathbb{U}_A(\mathcal{C}_\Theta; D)$, $M_A^{\mathcal{L}_O}(\mathcal{C}_\Theta; D)$, $W_A(\mathcal{C}_\Theta; D)$, $W_A^{\mathcal{L}_O}(\mathcal{C}_\Theta; D)$ are the class-restricted laws and weights (Definition 2.4). Observational restrictions of the penalised versions of Theorem 4.7 are their sub-sums over hypotheses whose axioms are $\mathcal{L}_O$-sentences on every tape.

Definition 8.2 (Literal data and charged deduction)**.** *Literal data* are those of Definition 2.2, in the fresh-atom setting, with $\mathcal{L}_O = \{\mathtt{e}, \mathtt{s}_0, \mathtt{s}_1, R\}$. *Charged deduction* is that of Section 4.6: a hypothesis decides an axiom set A_p within time t, satisfies Assumption 4.17, is consistent, and derives each datum in the calculus K with every axiom instance, logical ones included, of size at most h (reading (a1)), every nonlogical instance of size at most h ((a2)), material at most g ((b)), or nonlogical material at most g ((b2)); the mixtures are $W_A^{a1}[t,h]$, $W_A^{a2}[t,h]$, $W_A^{b}[t,g]$, $W_A^{b2}[t,g]$. At *polynomial budgets* the degrees of membership time and budget are charged in the prior (Definition E.18 of Hänni and Claude (Anthropic) (2026a)); then X is in the fit class iff one hypothesis is compatible with every D_n^X (Lemma E.11 there).

Definition 8.3 (The named objects)**.** $\mathrm{Th}_{\mathfrak{T}}$ is the first-order theory of the named structure $(\mathfrak{T}; \mathtt{e}, \mathtt{s}_0, \mathtt{s}_1)$. A theory in a language containing $\mathcal{L}_O$ is *about the named objects* if it has a model whose $\{\mathtt{e}, \mathtt{s}_0, \mathtt{s}_1\}$-reduct is $\mathfrak{T}$, and *compatible with the named structure* if it is consistent with $\mathrm{Th}_{\mathfrak{T}}$.

The second notion allows objects other than the named ones, but they obey every first-order law of the named objects (for instance, $\mathtt{s}_0, \mathtt{s}_1$ are injective with disjoint ranges that miss $\mathtt{e}$, and $\mathtt{s}_u x \neq x$ for every nonempty word u). "Named object" is our formal stand-in for "observable thing".

Theorem 8.4 (The theoretician's dilemma in Solomonoff axiom induction)**.** *Data are observational sentences; in (ii)–(iv), literal data under the readings of Definition 8.2 at polynomial budgets.*

(i) In principle. *With deduction free, restricting hypotheses to the observational vocabulary changes the prior weight of every Borel class of observational contents, the predictive probability of every observational sentence and the cumulative loss by at most a constant, in the one-tape, coin-free and Kolmogorov versions, and also when the time to check or to generate axioms is penalised (under Assumptions 4.5 and 4.6). Tape by tape, the observational projection of a random theory has exactly the same law on observational data.*

(ii) Craig's construction in practice. *When derivations are charged, Craig's axiomatisation of a theory whose observational consequences have derivations of size* ds *needs derivations of*

size at least 2^{ds}. *Every axiomatisation by padded observational literals with polynomial-time membership fits exactly the polynomial-time decidable regularities, under every reading. Axiom sets with theoretical symbols fit* $\mathsf{NP} \cap \mathsf{coNP}$ *(total material),* $\mathsf{EXPTIME}$ *(every instance short) and every decidable regularity (only the theory's own axioms short); so Craig's construction loses unconditionally under the last two readings and, under the first, iff* $\mathsf{NP} \cap \mathsf{coNP} \neq \mathsf{P}$.

(iii) Eliminating the vocabulary in practice. *Theoretical vocabulary can be eliminated at polynomial cost: per hypothesis, with* $O(1)$ *bits and a polynomial change of budget, under total material, by storing proofs instead of run times; and at the level of fit classes under (a1), (a2), (b2), by storing the eliminated relations in objects that no observational term names. Observational hypotheses, and already finite observational theories, fit what all hypotheses fit.*

(iv) What is indispensable. *A finite theory in the observational vocabulary with any unary theoretical predicates and constants, compatible with the first-order theory of the named objects, fits exactly the regular regularities, under every reading and also with free deduction. One binary theoretical relation over the named objects restores every decidable regularity ((a2), (b2)),* $\mathsf{NP} \cap \mathsf{coNP}$ *((b)) and* $\mathsf{EXPTIME}$ *((a1)). So does quantifying over unnamed objects; but a finite theory that does so without theoretical vocabulary, and fits a non-regular regularity, must prove some* $\{\mathsf{e}, \mathsf{s}_0, \mathsf{s}_1\}$*-sentence false in the named structure. For the theory built in Theorem F.10 that sentence is* $\exists z\, \mathsf{s}_1\mathsf{s}_1 z = z$*, true of no named object.*

(v) Free deduction, finite theories. *With deduction free, finite theories gain without bound from a theoretical function (infinity), from theoretical sorts (the theorems of* PA*) and from a binary relation over the named objects; and r.e. axiom induction is finite-theory induction with theoretical sorts up to* $O(1)$*, coin-free.*

Proof. (i) is Theorem 8.5. (ii) is Lemma F.2, Theorem 8.7, and Corollary F.4, given Corollary B.22, the time hierarchy theorem and $\mathsf{APSPACE} = \mathsf{EXPTIME}$ (Chandra et al., 1981). (iii) is Propositions F.5 and F.7 and Theorem F.10. (iv) is Theorems 8.8, F.10 and F.12 and Proposition F.15, given the Ehrenfeucht–Fraïssé theorem, the one-tape simulation of multitape machines and, for (a1), $\mathsf{APSPACE} = \mathsf{EXPTIME}$. (v) is Theorems F.23, F.26 and F.28 and Corollary F.14; the clause on PA is given Ryll-Nardzewski's theorem and Con(PA), the clause on sorts the Σ_1-completeness of Q. □

8.2 In principle

Corollary 3.10 showed that new vocabulary changes old-vocabulary weights by a constant factor. Part (i) needs more: classes of contents, a theoretical background, the penalised versions, and the identity tape by tape.

Theorem 8.5 (Theoretical vocabulary costs $O(1)$ bits, class by class)**.** *There is* c*, depending on* V*, the coding,* B *and* $\mathcal{L}_O \subseteq \mathcal{L}$*, such that for every finite observational* D *and every Borel set* Θ *of observational contents: (a)* $2^{-c}\mathbb{U}_A(\mathcal{C}_\Theta; D) \leq M_A^{\mathcal{L}_O}(\mathcal{C}_\Theta; D) \leq 2^c\mathbb{U}_A(\mathcal{C}_\Theta; D)$*; (b) the same holds for* $W_A(\mathcal{C}_\Theta; D)$ *and* $W_A^{\mathcal{L}_O}(\mathcal{C}_\Theta; D)$*, and* $|K_A^{\mathcal{L}_O}(D) - K_A(D)| \leq c$*; (c) for each time-constructible* $t(m) \geq m^{a^*}$ *there is* c_t *such that, class by class, the observational restrictions of* $W_A^{\mathrm{chk},t}$ *and* $W_A^{\mathrm{gen},t}$ *(all forms) agree with* W_A *within* $2^{\pm c_t}$ *under Assumption 4.5, and those of* $M_A^{\mathrm{chk},t}$ *and* $M_A^{\mathrm{gen},t}$ *agree with* M_A *within* $2^{\pm c_t}$ *under Assumption 4.6; (d) for every theory machine* G *in* $\mathcal{L}$*,* $G^O := \{(w, \varphi) : \varphi \in \mathrm{Sent}_{\mathcal{L}_O},\ B \cup A_w^G \vdash \varphi\}$ *is a theory machine emitting only* $\mathcal{L}_O$*-sentences, and*

on every tape ω: $A_\omega^{G^O} = T_O(A_\omega^G)$; $B \cup A_\omega^{G^O}$ is consistent iff $B \cup A_\omega^G$ is; and $B \cup A_\omega^{G^O} \vdash \varphi^b$ iff $B \cup A_\omega^G \vdash \varphi^b$, for all observational φ and $b \in \{0,1\}$. Hence, for every set Θ, the events defining $\nu_{G^O}(\mathcal{C}_\Theta; D)$ and $\nu_G(\mathcal{C}_\Theta; D)$ coincide tape by tape, and the two laws are equal.
Consequently, in each version, posterior odds between two classes of observational contents and predictive probabilities of observational sentences agree within $2^{\pm 2c}$, and cumulative semimeasure losses on observational reveal sequences within $2c$.

Proof idea (Appendix F.1). (d) is compactness. (a) applies (d) to G_U and the mixture identity of Theorem 3.6 both ways; (b) maps e to indices of $T_O(\mathsf{W}_e)$ and of $\mathsf{W}_e \cap \mathrm{Sent}_{\mathcal{L}_O}$; (c) pads the observational content as in Theorem 4.7, the only place where Craig's padding enters. G^O is the pull-back machine of Theorem 3.9 for the inclusion $\mathrm{Sent}_{\mathcal{L}_O} \subseteq \mathrm{Sent}_{\mathcal{L}}$. Projecting matters: on a propositional analogue, dropping the theoretical axioms instead changes the law of 37 of 40 random theory machines (a computation; code in the accompanying repository).

Remark 8.6 (Inductive systematisation in the mixture). By Theorem 8.5, the posterior predictive probability of every observational sentence, given observational data, is the same up to $2^{\pm 2c}$ whether or not hypotheses may use theoretical vocabulary. So in the universal mixture, with deduction free, theoretical terms are not needed for inductive inference among observational sentences. This is a statement about the mixture, not about a single theory (Section 8.6).

8.3 In practice: Craig's construction and better eliminations

Craig's construction makes an r.e. theory decidable by padding each consequence with the time at which it was found: the literal φ found after k steps becomes the axiom $\varphi^{\wedge(k+1)}$, a conjunction of $k+1$ copies. Under charged deduction this is useless for a precise reason. Canonical proof search tests all $2^{\mathrm{ds}} - 1$ strings shorter than the shortest derivation of φ^b, of size ds; and a set of sentences each equivalent to a literal proves a literal datum only through a member equivalent to it. So every derivation from the Craig set contains a line of size at least $2^{\mathrm{ds}}|\varphi^b|$ (Lemma F.2). For a seven-sentence Horn theory with two theoretical predicates for the parity of the number of 1s, in a small proof system for ground literals and at a word of length 24, a computation from closed forms gives a Craig axiom of more than $10^{204.6}$ symbols, against 795 for an axiom that carries a certificate (Example F.3). Call $A \subseteq \mathrm{Sent}_{\mathcal{L}_O}$ *run-padded* if each member is a power $\psi^{\wedge m}$ ($m \geq 1$) of a literal $\psi \in \{\varphi_w, \neg\varphi_w\}$.

Theorem 8.7 (Run-padded axiomatisations fit exactly P)**.** *On literal data: (a) if A is run-padded, with membership decidable in polynomial time, and is compatible with every D_n^X under reading (a1), (a2), (b) or (b2) with a polynomial budget h, or with derivations of size at most h, then $X \in \mathsf{P}$; (b) every $X \in \mathsf{P}$ is fitted under all these readings, at polynomial budgets with polynomial-time membership, by a run-padded set.*

Proof idea (Appendix F.2). The member that a derivation of φ_w^b cites and that is equivalent to φ_w^b has at most $h(|w|+2)$ symbols, hence at most that many factors; so $w \in X$ iff one of polynomially many powers of φ_w lies in A. A run-padded axiom carries $O(\log)$ bits, its length, while the axiom $\theta_\gamma \to \varphi^b$ of Theorem 4.18 carries the bits of a certificate γ. So the natural strengthening to all sets whose members are each equivalent to a literal fails: certificate sets are such sets and fit $\mathsf{NP} \cap \mathsf{coNP}$ under (b).

We did not find a complexity-theoretic form of the classical reply in the literature. Theorem 8.7 does not contradict the standard fact that every r.e. theory has an axiomatisation decidable in polynomial time, by a variant of Craig's construction (Krajíček, 2025): it concerns what such sets

fit when derivations are charged. Against all hypotheses (Corollary B.22), run-padded sets lose unconditionally under (a1), (a2), (b2), and under (b) iff $\mathsf{NP} \cap \mathsf{coNP} \neq \mathsf{P}$ (Corollary F.4).

The vocabulary itself can be eliminated at polynomial cost, by storing what it carries elsewhere. Under reading (b), replacing derivations by certificate axioms $\theta_\gamma \to \varphi^b$, which are observational because θ_γ uses only logical symbols, eliminates it hypothesis by hypothesis at $O(1)$ bits and a polynomial change of budget (Proposition F.5): what survives is the proofs. Under the other readings the content goes into objects that no observational term names. In the structure $\mathfrak{U}$ of words and formal pairs $\mathrm{pr}(a,b)$, with $\mathtt{s}_0\,\mathrm{pr}(a,b) := a$ and $\mathtt{s}_1\,\mathrm{pr}(a,b) := \mathrm{pr}(b,a)$, the observational formula $\theta_2(a,b) := \forall y\,(\mathtt{s}_1\mathtt{s}_1 y = y \to (\mathtt{s}_0 y = a \to (\mathtt{s}_0\mathtt{s}_1 y = b \to R(y))))$ says that R holds of $\mathrm{pr}(a,b)$ (Lemma F.6). Replacing each atom $\mathsf{C}(a,b)$ of a binary theoretical relation by $\theta_2(a,b)$, with quantifiers relativised to $\mathrm{Std}(y) := \neg\, \mathtt{s}_1\mathtt{s}_1 y = y$, translates hypotheses into observational ones (Proposition F.7), and finite theories into finite theories. For finite theories one binary relation suffices (Theorem F.10): for every *verifier* $\mathcal{V}$, a halting machine that accepts or rejects pairs (word, certificate) consistently, there are a finite theory in $\mathcal{L}_O \cup \{\mathsf{C}\}$ about the named objects and a finite set of $\mathcal{L}_O$-sentences that prove exactly the literals that $\mathcal{V}$ certifies, with derivations whose length and line size are linear in the time and space of a one-tape simulation of $\mathcal{V}$ plus the input length. So finite theories of either kind fit every decidable set under (a2) and (b2), $\mathsf{NP} \cap \mathsf{coNP}$ under (b) and, for alternating machines, $\mathsf{EXPTIME}$ under (a1): the fit classes of all hypotheses. The construction compiles $\mathcal{V}$ into a two-stack program on pairs of words and writes the theory backwards, $\mathsf{C}(r,l)$ saying that the configuration (r,l) leads to a halting configuration of the right kind (Definition F.8 and Lemma F.9). With deduction free the existence part is classical: as stated in Pakhomov and Visser (2019), every r.e. theory without finite models has a finitely axiomatised conservative extension in its language plus one binary relation symbol (Kleene, 1952), even one into which every model of the theory expands (Craig and Vaught, 1958), while additional unary predicates do not suffice in general. To our knowledge the construction with derivation bounds, hence the fit classes under charged deduction, and the observational translation are new; the comparison with unary vocabulary below is in the spirit of Craig and Vaught's example, in a different setting, and we have not established that it is new.

8.4 Vocabulary or ontology

Without objects outside the named structure, unary theoretical vocabulary does not help a finite theory. For $X \subseteq \{0,1\}^*$ and a word v, the *residual* $X_v := \{u : uv \in X\}$ is what the subtree below t_v sees, and X is regular iff it has finitely many residuals.

Theorem 8.8 (Unary theoretical vocabulary sees only regular regularities)**.** *Let T be a finite set of sentences in $\mathcal{L}_O$ plus unary predicate symbols $P_1, \dots, P_k$ and constants $\mathsf{c}_1, \dots, \mathsf{c}_j$ ($k, j \geq 0$), compatible with the named structure. If $T \vdash \varphi_w^{\mathbf{1}[w \in X]}$ for every word w, then X is regular. Quantitatively, X has at most $\mathrm{ncl}_q + |\mathrm{Suf}_\mathrm{c}|$ residuals, where q is the quantifier rank of the relational translation of $\bigwedge T$, ncl_q the number of classes of rank-q elementary equivalence in the relational signature of Appendix F.5, and Suf_c the set of suffixes of the words w_i with $\mathsf{c}_i = t_{w_i}$ in a fixed model of $T \cup \mathrm{Th}_\mathfrak{T}$, together with the empty word.*

Proof idea (Appendix F.5). By $\mathrm{Th}_\mathfrak{T}$ a single edge joins the subtree below t_v to the rest of a model, and unary symbols are labels; so by the composition lemma for Ehrenfeucht–Fraïssé games (Lemma F.11) the labels of one subtree can be replaced by those of another of the same rank-q type without changing the truth of T. As T proves the literals, the two residuals coincide.

Conversely, every regular X has a finite observational theory, true in $(\mathfrak{T}, X)$, with derivations of $O(|w|)$ lines of size $O(|w|)$ (Theorem F.12). So, under every reading, finite theories in $\mathcal{L}_O$ plus

unary predicates and constants, compatible with the named structure, fit exactly the regular sets, while with one binary relation over the named objects, or in $\mathcal{L}_O$ alone with quantification over objects outside the named structure, they fit what all finite theories fit (Corollary F.13); with deduction free the contrast holds for complexities too (Corollary F.14). The second way has a price. By Theorem 8.8 and compactness, a finite theory without theoretical vocabulary that fits a non-regular regularity must prove some $\{\mathtt{e}, \mathtt{s}_0, \mathtt{s}_1\}$-sentence false in the named structure; for the observational theory built above that sentence is $\exists z\, \mathtt{s}_1\mathtt{s}_1 z = z$, true of no named object (Proposition F.15). The theory with the binary relation, by contrast, is conservative over the theory of the named objects plus the data (Proposition F.16). Ramsey sentences eliminate theoretical predicates by quantifying over them in second order; the pairing gadget does so in first order by quantifying over objects, and then the observational theory must say that there are objects unlike every observable one. With only unary observables Theorem 8.8 survives, but one binary observable relation breaks it already for the words with as many 0s as 1s (Proposition F.17).

8.5 Finite theories with free deduction

With deduction free, r.e. hypotheses gain nothing from theoretical vocabulary; finite theories, the single sentences of K_{fin} and W_{fin} (Definition 2.7), can gain without bound. On the data "there are at least n things", the least complexity of a consistent sentence in pure equality plus unary predicates that proves the data is $K^{\text{pad}}(n) \pm c$, which tends to infinity, though more slowly than every computable unbounded function, while with one unary function symbol the sentence saying that it is injective and not surjective gives $O(1)$ (Theorem F.23); the same phenomenon separates K_{fin} from K_A with named objects (Remark F.24), and in literal symbols the costs are at least n in pure equality, $\Theta(\log n)$ with unary predicates and $O(1)$ with the function (Proposition F.25). Here the classical eliminations can be compared directly (Proposition F.30): the Ramsey sentence keeps the theory finite but leaves first-order logic; the observational axiomatisation $\{\exists x_1 \cdots \exists x_m \bigwedge_{i<j} x_i \neq x_j : m \geq 1\}$ stays first-order but is infinite; and no finite first-order theory without a function symbol has the same consequences. On the theorems of PA as data, finite $\mathcal{L}_{\text{ar}}$-theories diverge, by Ryll-Nardzewski's theorem that no consistent extension of PA in $\mathcal{L}_{\text{ar}}$ is finitely axiomatisable (Ryll-Nardzewski, 1952), while a finite theory with theoretical sorts and exactly the arithmetical consequences of PA has bounded complexity (Theorem F.26); ACA_0 is a classical instance (see Simpson, 2009). Sorts suffice for every r.e. theory: for finite $\mathcal{L}$, r.e. axiom induction in $\mathcal{L}$ is, up to a constant and coin-free, finite-theory induction in a three-sorted extension $\mathcal{L}^+$ with numbers, assignments and a satisfaction relation: $K_{\text{fin}}^{\mathcal{L}^+} = K_A^{\mathcal{L}} \pm c$ and $2^{-c} W_A^{\mathcal{L}} \leq W_{\text{fin}}^{\mathcal{L}^+} \leq 2^c W_A^{\mathcal{L}}$ (Theorem F.28). This is an effective, many-sorted form of the theorem of Kleene (1952) and Craig and Vaught (1958), of which only the Bayesian consequence is new: the r.e. prior already is a prior over finite theories with theoretical vocabulary and entities, which is why adding vocabulary costs nothing there. The equivalence fails for one tape: on the lottery stream of Theorem 5.1, random r.e. theories have bounded loss and random finite theories with any theoretical apparatus do not (Remark F.29).

A concept can also pay by factoring inferences.[6] If each of n_A unary predicates implies each of n_B others, a theory of atom-to-atom implications needs $n_A n_B$ members, and with a new intermediate predicate exactly $\min(n_A n_B, n_A + n_B)$; so the new predicate pays in this restricted format iff $(n_A - 1)(n_B - 1) > 1$, and only by a constant factor in full first-order logic (Proposition F.31). The natural claim that it always saves $n_A n_B - (n_A + n_B)$ members fails for small patterns. This is

[6] The idea of introducing a concept that factors a complete bipartite pattern of inferences first appeared in Hänni's working notes (Hänni, 2026).

the elementary instance, with the exact gain, of the compression argument for predicate invention in inductive logic programming (cf. Muggleton and Buntine, 1988).

8.6 What this does and does not show

The classical reply to Hempel, that Craigian axiomatisations are useless in practice, becomes a theorem here, but only about Craig's construction and every run-padded axiomatisation, which are fast labellers in disguise (Theorem 8.4(ii)); it is not a theorem about theoretical vocabulary, which can be eliminated at polynomial cost ((iii)). What bounded deduction and finiteness make indispensable is theoretical *relations* (polyadic) or theoretical *entities* (objects outside the named structure), either one ((iv)); theoretical *properties* of observable things, unary predicates, never help a finite theory in this setting. In the idealised setting nothing theoretical is indispensable ((i)), because the hypothesis class already contains the apparatus ((v)). So where theoretical terms pay is a property of the hypothesis class. The results concern one poor observational language and literal data, and say nothing about whether science should use theoretical terms; they locate, inside these inducers, what can and cannot be traded away.

Hempel's three replies. *Impracticality*: Craig's replacement axiomatisation is unmanageable. This is a theorem, but only about Craig's construction and run-padded axiomatisations under charged derivations; vocabulary as such is eliminable at polynomial cost (Theorem 8.4(ii), (iii)). *Inductive systematisation*: Hempel argued that theoretical terms may be needed for inductive, not deductive, systematisation (Hempel, 1958); the debate is surveyed by Niiniluoto (1972). Solomonoff axiom induction is an inductive systematisation, so the reply can be tested in it, and in the universal mixture with free deduction it fails: theoretical terms add no inductive power among observational sentences beyond a constant factor (Remark 8.6). This is a statement about the mixture; Hempel's own notion is relative to a single theory, and our statement is not a reconstruction of it. The reply survives, in corrected form, where something theoretical pays, for finite theories and under charged derivations, and there what pays is relations or entities, not vocabulary. In monadic inductive logic, by contrast, theoretical concepts can change inductive probabilities (Niiniluoto and Tuomela, 1973). *Aims beyond prediction*: Putnam (1965a) argued that the instrumentalist use of Craig's theorem rests on the view that science aims only at prediction. The question-answering view of Section 6 is a partial version of this reply: theories are scored on the questions they answer, which may concern unobservables only if the base language contains them.

9 Approximate laws and the best system

Kepler's laws are false. Against a modern planetary theory, a fixed ellipse misses Mars by one or two arcminutes over a few decades and the Moon by about a degree within a month (Table 9); yet the laws were, and are, good science. A proof-requiring inducer seems ill suited to such laws: a deterministic theory either proves a datum or does not, and an exact law that is slightly wrong is refuted by any datum precise enough to show its error. This section asks what SAI does with an approximately true law, and what becomes, inside it, of Lewis's best system and of explanatory power.[7] In brief, the difficulty is a limitation of proof-requiring compatibility with deterministic theories, not of the axiom view as such: M_A mixes randomised theories, and under SAI's pricing

[7]The questions of approximately true laws and of the domain of applicability of a model are discussed in Hänni's working notes (Hänni, 2026).

an approximately true law is a law with a calibrated error model (Proposition 9.3). These results carry the row on approximate truth of the argument map (Section 6.1). Appendix G states in full the results that this section states in words, and proves all of them.

9.1 Setting

The data are answers to questions (Definition 2.11), such as the cell of width Δ in which the longitude of Mars lies at time t (Definition 10.1).

Definition 9.1 (Answered data, exception codes, approximate laws)**.**

(a) *Answered data.* A datum is a pair $(\mathcal{Q}, a)$ of a question and its true answer; a data set D has pairwise distinct questions and is read as the two-valued data set $\{(a, 1) : (\mathcal{Q}, a) \in D\}$; $\operatorname{dom} D$ lists its questions, y_D their answers, and $n := |D|$. A theory T *proves* (*refutes*) the datum if $B \cup T \vdash a$ ($B \cup T \vdash \neg a$), and is *silent* otherwise; $A_T(\mathcal{Q}) := \{a' \in \mathcal{Q} : B \cup T \nvdash \neg a'\}$, and $\gamma(\mathcal{Q}) := \log|\mathcal{Q}|$ is the *literal cost*.

(b) *Codes.* For a prefix-free code c_S of sentences, $\ell(\chi) := 1 + |c_S(\chi)|$, and $\|\Gamma\| := \sum_{\chi\in\Gamma} \ell(\chi)$ for a finite set Γ of sentences (a theory or a *module* of one). If $B \cup T \cup D^{\pm}$ is consistent, the *exceptions* of T are the data it does not prove, priced by $\mathrm{Exc}^Q_D(T) := \sum \gamma(\mathcal{Q})$ (code Q), $\mathrm{Exc}^F_D(T) := \sum \ell(a)$ (code F) or $\mathrm{Exc}^R_D(T) := -\log P^R_T(D)$ (code R, below); otherwise $\mathrm{Exc}_D(T) := \infty$. Unmarked, Exc is code Q. $J_\lambda(T; D) := \lambda\|T\| + \mathrm{Exc}_D(T)$ for $\lambda > 0$ (and J^Q_λ, J^F_λ, J^R_λ); its minimisers are the *λ-MAPs* (Hänni and Claude (Anthropic), 2026d, Definition 3.6). T^{-Z} withdraws T at the finitely many times in Z, at about c_{time} bits per time.

(c) *Interval laws.* $B :=$ RCF (decidable and complete; standard, Tarski, 1951), with constants q_t for times t in a decidable set I of numerals, and a rational range $[0, \mathsf{R}]$. A *pointwise interval law* is $\{\underline{f}(t) \le q_t \le \overline{f}(t) : t \in I\}$ for total computable $\underline{f} \le \overline{f} : I \to \mathbb{Q} \cap [0, \mathsf{R}]$ named by a codeword of a fixed decidable prefix-free presentation language; $\|T\|$ is the codeword's length. The *exact law* T_0 has $\underline{f} = \overline{f} = f$, the *honest law* T^ε has $f \mp \varepsilon$ (clipped). *Interval data* answer questions that partition $[0, \mathsf{R})$ into finitely many cells with rational endpoints; with C_t the intersection of the cells that D assigns to t, $P^R_T(D) := \prod_t |C_t \cap [\underline{f}(t), \overline{f}(t)]|/(\overline{f}(t) - \underline{f}(t))$ (length; the factor is $\mathbf{1}[\underline{f}(t) \in C_t]$ if $\underline{f}(t) = \overline{f}(t)$).

(d) *Errors.* For a real quantity $q(t)$, $e(t) := f(t) - q(t)$, and $\varepsilon^* := \operatorname{ess\,sup}|e|$ under the law of the error at a data time. *Regime G:* datum j is "$q(t_j) \in [v_j - \delta, v_j + \delta]$" with $v_j = q(t_j) + \zeta_j$ rational and $|\zeta_j| \le \delta_0 \le \delta$. *Regime R: q* lives on a circle of circumference R; datum j is about a *fresh* time t_j (an injective computable sequence of numerals); the errors $e_j := e(t_j)$ are i.i.d.; and datum j is the cell of width Δ (R/Δ an integer) containing $q(t_j)$ in a grid shifted by a dither u_j, uniform and independent of everything else; $\gamma = \log(\mathsf{R}/\Delta)$, $p_\Delta := \mathbb{E}\min(|e|/\Delta, 1)$ and $s_\Delta := \min(2\varepsilon^*/\Delta, 1)$.

(e) *Peakedness.* $\mathrm{pk}(e) := \varepsilon^*/\mathbb{E}|e|$ and $\mathrm{pk}_\varepsilon(e) := \varepsilon/\mathbb{E}|e|$; every reported value names its fit (minimax or mean).

9.2 Deterministic theories and precision

Over deterministic theories, proof-requiring compatibility is monotone in logical strength, so data never move the odds towards a weaker theory: $T_0 \vdash T^\varepsilon$, every data set compatible with T^ε is compatible with T_0, and with prior weights $2^{-\|T\|}$ the posterior odds of T_0 against T^ε are at

least 2 on each, although T_0 is false when $e \not\equiv 0$ (Proposition G.9). Under guaranteed tolerances (regime G), T_0 is almost surely compatible with the data at every sample size when $\delta \geq \varepsilon^* + \delta_0$ and, under mild conditions on the measurement error, is eventually refuted when $\delta < \varepsilon^* + \delta_0$ (Proposition G.11). Rounding changes the picture.

Proposition 9.2 (Dithered rounding)**.** *In regime R, given $e(t_j)$, T_0 proves datum j with probability $(1 - |e(t_j)|/\Delta)^+$ and refutes it otherwise. So T_0 survives n data with probability $(1 - p_\Delta)^n$, its first refutation has a geometric index with mean $1/p_\Delta$, and it is almost surely refuted eventually iff $\mathbb{P}[e \neq 0] > 0$, for every Δ, however coarse. For $\Delta \geq \varepsilon^*$, $p_\Delta = \mathbb{E}|e|/\Delta$: the exact law survives about $\Delta/\mathbb{E}|e|$ data.*

This corrects the natural expectation that coarse data protect a false exact law: precision belongs to the data set, not to the datum, and an exact law with a sinusoidal error of amplitude $1'$ is refuted by positions rounded to a dithered grid of 1° after about 94 data (the proof and the comparison of several exact laws, Proposition G.12, are in Appendix G.3). We have not found these two regimes stated elsewhere; precision-dependent selection as such is familiar from accounts based on AIC, MDL and MML (cf. Forster and Sober, 1994; Rissanen, 1978; Wallace, 2005).

9.3 Randomised theories and laws with error models

A deterministic theory pays, in code F, for which questions are exceptions (Proposition G.2). A theory machine that reads a random tape answers every question at once, at random inside what the law allows: for a pointwise interval law T, $L_A(D) \leq \|T\| + \mathrm{Exc}_D^R(T) + c$ on interval data consistent with T (Proposition G.3). In fact $L_A(D) = \min_G[K(G) - \log \nu_G(D)] \pm c$ over theory machines G: SAI is the best two-part code over randomised theories, neither K_A nor literal-exception codes over finite theories are within an additive constant of it in general, and over all theory machines a steeper penalty collapses onto the universal machine, so best-system analyses use laws with private literal facts (Propositions G.4 and G.7). *SAI's pricing* of a form is the weight $2^{-(K(G) - \log \nu_G(D))}$ of a named theory machine G in M_A's mixture, not the whole posterior.

With exceptions named in the law at c_{time} bits each (a cost model valid only while times are that cheap to name; Proposition G.15), the honest law T^{ε^*} has a lower rate than the patched exact law T_0^{-Z}, for $\Delta \geq 2\varepsilon^*$, iff $\lambda c_{\mathrm{time}} > \gamma(2\mathrm{pk}(e) - 1)$: spread errors favour honesty and peaked ones exceptions (Propositions G.16 and G.17). Under random completion silence costs at most $\log \mathrm{e}$ bits per straddled cell boundary, e Euler's number (Lemma G.18), and the comparison moves inside the band. For $\varepsilon \geq \varepsilon^*$ and a computable distribution $\mathcal{G}$ on $[0, 1]$, the *band-confined completion* $\mathrm{BC}_\mathcal{G}$ emits T^ε and, for each time t_i, the value $f(t_i) - \varepsilon + 2\varepsilon\hat{w}_i$ with $\hat{w}_i \sim \mathcal{G}$ drawn from the tape. On data at fresh times its law is $\prod_j \mathcal{G}(C'_j)$, with C'_j the cell of datum j in band coordinates (the band mapped affinely onto $[0, 1]$), and $K(\mathrm{BC}_\mathcal{G}) \leq \|T^\varepsilon\| + K(\mathcal{G}) + c$. Uniform $\mathcal{G}$ gives the *uniform band*; $\mathrm{HY}(\rho)$, with $\mathcal{G} = (1 - \rho)\delta_{1/2} + \rho\,\mathrm{Unif}[0, 1]$, is "exact, except in a share ρ of cases, and then within ε".

Proposition 9.3 (Laws with error models)**.** *In regime R let $\varepsilon \geq \varepsilon^*$, and let $\mathcal{F}$ be the distribution function of $w_j := (\varepsilon - e_j)/(2\varepsilon)$, the position of the true value in the band.*

(a) If $\Delta \geq 2\varepsilon$, the rate of $\mathrm{BC}_\mathcal{G}$ is $(2\varepsilon/\Delta)\, c_\mathcal{G}$, with

$$c_\mathcal{G} := \mathbb{E}_w \int_0^1 \left(\mathbf{1}[v < w](-\log(1 - \mathcal{G}(v))) + \mathbf{1}[v > w](-\log \mathcal{G}(v))\right) dv.$$

So band-confined completions are ranked in the same way at every such Δ.

(b) $c_{\mathcal{G}} \geq c_{\mathcal{F}} := \int_0^1 H_2(\mathcal{F}(v))\,dv$, H_2 *the binary entropy, with equality iff* $\mathcal{G} = \mathcal{F}$ *almost everywhere on* $[0,1]$. *In particular the uniform band is the best band-confined completion iff* e *is uniform on* $[-\varepsilon, \varepsilon]$.

(c) *For every theory machine* G', G_U *included, almost surely* $\liminf_n \frac{1}{n}(-\log \nu_{G'}(D_n)) \geq r_{\mathcal{F}}(\Delta) := \mathbb{E}[-\log \mathcal{F}(C_1')]$, *the conditional entropy rate of the answers given the questions, while* $-\frac{1}{n}\log \nu_{\mathrm{BC}_{\mathcal{F}}}(D_n) \to r_{\mathcal{F}}(\Delta)$. *If* $\mathcal{F}$ *is computable, then* $L_A(D_n)/n \to r_{\mathcal{F}}(\Delta)$ *almost surely. For* $\Delta \geq 2\varepsilon$, $r_{\mathcal{F}}(\Delta) = (2\varepsilon/\Delta)\, c_{\mathcal{F}}$.

(d) $c_{\mathrm{HY}(\rho)}$ *is convex in* $\rho \in (0,1]$, *and some* $\mathrm{HY}(\rho)$ *with* $\rho < 1$ *beats the uniform band iff* $\mathbb{E}\ln(1 + |e|/\varepsilon) < 2\ln 2 - 1$. *This requires* $\mathrm{pk}_\varepsilon(e) > \ln 2/(2\ln 2 - 1) \approx 1.79$, *and holds whenever* $\mathrm{pk}_\varepsilon(e) > \mathrm{e}/(4 - \mathrm{e}) \approx 2.12$.

Proof idea. A cell boundary falls into the band with probability $2\varepsilon/\Delta$, at a uniform position v, and the datum then costs $-\log$ of the $\mathcal{G}$-mass on the true side; (b) is Gibbs' inequality given v; (c) uses that a theory proves at most one answer list, with Markov's inequality and the Borel–Cantelli lemma; (d) is convexity and the sign of the derivative at the uniform band (Appendix G.4).

So under SAI's pricing an approximately true law is a law with a calibrated error model, not a silent partial theory and not an exact law with listed exceptions; the dichotomy between the two survives inside the band, with peakedness thresholds near 2. That a model should include its noise distribution is the standard situation of MDL (cf. Rissanen, 1978; Grünwald, 2007); what is specific is the reading for a proof-requiring inducer, in which the honest partial theory is one completion among many. Selection among finitely many forms is almost sure (Theorem G.20 and Corollary G.21). For deterministic theories the trichotomy is the formal home of humility: on a finer question, the honest and the empty law put their posterior mass on "independent", and whether the inducer ends humble is decided not by the prior, which favours the exact law, but by the error's shape and the price of silence (Proposition G.22). For the randomised theories that M_A mixes, humility is calibration: by (c) the preferred spread over the cells is the true error distribution.

Kepler's residuals. For seven bodies and six windows each (three for the Moon) we kept the fixed Kepler ellipse with the least largest longitude error (the minimax fit) and the one with the least mean error (the mean fit), and took their errors at 8000 uniform times as error laws, against the plan94 and moon98 ephemerides (Table 8 in Appendix G.4). The errors are moderately spread: the peakedness is 1.60–3.95 for the minimax fit and 1.89–7.08 for the mean fit, against $\pi/2$ for a sinusoid and 2 for uniform $|e|$. A computation of the rates at eleven cell widths from $0.1'$ to 30° (code in the accompanying repository; details in Appendix G.4) gives the following. Among the Kepler-based theory machines compared, the law with a 40-bin error histogram, fitted on half of the times and scored on the other half, beats the uniform band and the exact law with exceptions in all 429 (body, window, cell width) cells, by 0.4% to 25.7% of the band's rate, although the uniform band wins every cell within the pool $\{\emptyset, T_0^{-Z}, \text{uniform band}\}$; and in the coarse regime the two-point form beats the band for 34 of the 39 body–window pairs, exactly where Proposition 9.3(d) says. The residuals are deterministic functions of time, so a better theory (Newton's, or the ephemeris itself) eventually beats every Kepler-based machine.

9.4 Sound reasoning with approximate models

If the background is true, true honest models of one quantity are jointly consistent, and every statement about the quantity derived from them is a set containing its true value. If the

background also decides the comparisons involved (as RCF does for semialgebraic models), exact approximations that differ anywhere on a common domain are jointly inconsistent and so prove everything, while joint consistency of their honest forms is a pairwise check of tolerances in one dimension and for boxes, and an $(m+1)$-wise check for Euclidean balls in $\mathbb{R}^m$, by Helly's theorem (Proposition G.25). An honest dynamical law with initial tolerance $\varepsilon^{(0)}$ and tolerance ε per step proves, after r steps of an L-Lipschitz map, the tolerance $L^r\varepsilon^{(0)} + \varepsilon(L^r - 1)/(L - 1)$ $(L \neq 1)$, tightly for linear maps in one dimension: it has a derivation horizon, and an exact approximation is guaranteed correct up to that horizon (for linear maps in one dimension, only up to it) (Propositions G.23 and G.24). The device is sound, but it does not say how to choose a model, a domain or a tolerance (Section 11.4).

Remark 9.4 (Typical against worst case). With $L = 1$ and $r = 100$ steps the propagated tolerance is 100ε, while with independent random-sign errors $\pm\varepsilon$ the mean deviation is $\varepsilon r\binom{r}{r/2}2^{-r} \approx 7.96\varepsilon$: a probabilistic error model answers much further out, with a stated confidence instead of a guarantee.

9.5 The best system as a steeper MAP

On Lewis's account the laws are the theorems of the deductive system that best balances simplicity against strength (cf. Lewis, 1983, 1994). Take strength to be fit to data in a fixed base language, the balance to be λ, and the candidates a class of laws with private literal facts; the relation to Lewis is an analogy, since best systems are a metaphysics of laws and SAI is inference from finite data. The points $(\|T\|, \mathrm{Exc}_D(T))$ have a convex hull with vertices v_i in order of length, joined by edges of slopes $-\sigma_i$, $\infty =: \sigma_0 > \sigma_1 > \cdots > 0$; for $\lambda \in (\sigma_{i+1}, \sigma_i)$ the λ-MAPs are the theories at v_i (Hänni and Claude (Anthropic), 2026d, Lemma 3.8). A point $(\alpha_{\mathrm{r}}, L_{\mathrm{r}})$ attained by some theory is a *zero-window* (σ, τ)*-kink* (Hänni and Claude (Anthropic), 2026d, Definition 5.2) if every theory shorter by $x > 0$ has fit at least $L_{\mathrm{r}} + \sigma x$ and every theory longer by $x > 0$ has fit at least $L_{\mathrm{r}} - \tau x$.

Proposition 9.5 (Kind nature is a kink)**.** *For a family of candidate theories satisfying the hypotheses of the hull lemma (Proposition G.7) and $0 \leq \tau < \sigma \leq \infty$, the following are equivalent: (a) the set of λ-MAPs is the same for every $\lambda \in (\tau, \sigma)$; (b) some hull vertex v_i has $\sigma_{i+1} \leq \tau$ and $\sigma \leq \sigma_i$; (c) the point family has a zero-window (σ, τ)-kink at a point that is a hull vertex.*

Proof idea. The λ-MAP set is fixed between consecutive edge slopes and changes at each; (b)⇔(c) is Hänni and Claude (Anthropic) (2026d, Corollary 5.4(c)) (Appendix G.6).

So nature is kind on an interval of balances exactly when the data's hull has a kink there. In a solar-system toy with eight planets of 655 to 3838 observations, sixty asteroids with a median of about five and thirty comets with one or two (Proposition G.29), the λ-MAP is "the law plus the elements of all eight planets" for every $\lambda \in (7.03, 135.37)$, while below $\lambda \approx 7$ it changes with every asteroid (code in the accompanying repository). Inside such an interval the worry that the best system depends on an arbitrary choice of balance (cf. Woodward, 2014) is moot, and kindness is a checkable property of a language, a family of candidates and a data set. Lewis's trivialisation worry, that a predicate true of exactly the actual things makes its universal closure maximally simple and strong (cf. Lewis, 1983), is blocked by the coding bound, up to the complexity of a language chosen after the data (Proposition G.27).

Definition 9.6 (Explanatory power)**.** For a theory T and a module $\mathcal{A} \subseteq T$, $\mathrm{EP}_D(\mathcal{A}; T) := (\mathrm{Exc}_D(T \setminus \mathcal{A}) - \mathrm{Exc}_D(T))/\|\mathcal{A}\|$: the bits of data the module explains per bit of its statement, in the context T.

Every module of a λ-MAP has EP $\geq \lambda$, no addition to it has EP $> \lambda$, and for $\lambda > 1$ a fact used once never enters (Propositions G.26 and G.28). So the model–randomness split of the companion paper separates facts reused across many data from facts used once, not laws from initial conditions: in the toy, the elements of a well-observed body explain hundreds of bits per bit and sit on the steep side of the hull with the law (Proposition G.29). What singles out the law is how its EP scales: with m_{b} bodies of n_0 observations each, all stated, a law of L bits has EP $= m_{\mathrm{b}} n_0 \gamma / L$, which grows with the number of systems, while each body's elements explain only that body's observations. Kepler's third law K3 is a law about initial conditions: it shortens every elements module by s bits, and when the set S of bodies whose elements are stated is the same with and without it, it lowers the objective iff $s|S|$ exceeds its length, whatever λ is; this compares law choices at exchange rate 1 and is not membership of the λ-MAP. In the question format, elements are question antecedents and never enter a λ-MAP (Proposition G.30). Section 10 computes what K3 is worth in Kepler's setting.

9.6 Explanation and correction

$\|\mathcal{A}\| \, \mathrm{EP}_D(\mathcal{A}; T)$ is the log-likelihood ratio of T against $T \setminus \mathcal{A}$ for $P_T(D) := 2^{-\mathrm{Exc}_D(T)}$. The support a hypothesis gets from joint data $D_1 \cup D_2$ splits into the supports from the parts plus a term U, the change in the mutual information between D_1 and D_2 that the hypothesis brings about, which has the form of Myrvold's measure of unification (Myrvold, 2003): with stated parameters, for laws that are products over questions, $U = 0$, and when a k-bit parameter is integrated out, U is k within one system, 0 across systems with independent parameters, and the bits of a shared constant across systems that share one (Proposition G.31). At $\lambda = 1$ a module pays iff its EP exceeds 1, the compression criterion of inductive logic programming (cf. Muggleton, 1995; Srinivasan et al., 1994). Copernicus against Ptolemy is unification without better prediction: against Ptolemy with the coincidence of each superior planet's retrograde motion and opposition stated as an axiom, at a_p bits each, which predicts the same, Copernicus, at H extra law bits, wins at every λ iff $H < \sum_p a_p$, and against Ptolemy with free phases it wins iff $\lambda < \sum_p a_p / H$ (Proposition G.32; the lengths are not measured; for the history, see Kuhn, 1957).

Newton's theory does not entail Kepler's laws; it contradicts them (cf. Duhem, 1954) and derives corrected statements, as in reduction to a corrected reduced theory (cf. Schaffner, 1967). In an exact rational toy, uniform circular motion perturbed to $x_{t+1} = Ax_t$ with A a rotation plus $\eta_{\mathrm{c}}\mathbf{P}$, the successor proves the honest old law on the window $t \leq \bar{t}$ with uniform tolerance ε iff $\varepsilon \geq \varepsilon_{\min}(\bar{t})$, where $\varepsilon_{\min}(\bar{t}) \leq ((1 + \eta_{\mathrm{c}}\|\mathbf{P}\|)^{\bar{t}} - 1)|a|$ for the initial state a, and, if $\eta_{\mathrm{c}}\mathbf{P}a \neq 0$, it refutes the exact old law; among uniform-tolerance forms, the strongest it proves is the strongest both theories prove, while time-dependent tolerances are stronger common consequences and the whole common content is the disjunction of the two theories (Proposition G.33). The old law's domain of applicability is derived, not stipulated. Deriving "Kepler within ε on a window" from Newton needs validated perturbation bounds (Section 10.5).

9.7 Domains of applicability and retention

A domain restriction that leaves one system's data literal pays asymptotically iff the law, even with an honest tolerance for that system, is worse than literal data, exact ties being decided by the constants (Proposition G.35). Under SAI's pricing that never happens, at cell widths below R/e, for a system that the law confines within about 66° of longitude (a sufficient bound): an honest band of half-width ε^* costs at most $\log(2\varepsilon^*/\Delta) + \log \mathrm{e}$ bits per datum, below the literal $\log(\mathsf{R}/\Delta)$ iff $\varepsilon^* < \mathsf{R}/(2\mathrm{e})$. So the Moon ($\varepsilon^* = 2.6°$ over a year) is kept in Kepler's law with its

own tolerance at every cell width computed, from $10'$ to $30°$, whereas under literal pricing at $10'$ and $60'$ a cheap domain clause excludes it. A superseded theory is retained where its derivation savings exceed its loss of fit, for all cell widths above a computable Δ^* within the range where its rate is monotone (Proposition G.36).

Domains are learned as tolerances. We conclude that under SAI's pricing a domain of applicability shows up as a tolerance and an error distribution that depend on the system and the window, not as a sharp boundary: for every planet and window computed, and for the Moon over a month, a year and 18.6 years, a law with its own tolerance has a lower rate than leaving the body's data literal. A sharp boundary is selected only when even an honest tolerance is worse than literal data, or under literal pricing at fine precision. The posterior over an explicit pool of domain predicates is not computed (Question G.37).

10 A worked example: Kepler and Newton

We work through one example: Kepler's laws of planetary motion, the theories of the motion of Mars that preceded the ellipse, and what Newton's theory adds. It shows that what a law says depends on the data language (Section 10.2); that in a proof-requiring posterior over the historical theories of Mars the data decide which theories are compatible, while among compatible theories, on noise-free data, the language of the law texts decides (Section 10.4); and that the gain of a successor theory need not lie in a better fit to the old data (Section 10.5). The data of every computation are the plan94 ephemeris of the ERFA library, so *every arcminute figure in this section is an error against plan94, not against the sky.* The code is in the accompanying repository; Appendix H gives the proofs, the law texts, the full tables and the elements that witness each fit.

10.1 The language and the data

Definition 10.1 (the Kepler languages and theories)**.**

(i) *Data languages.* Sorts bodies and reals; body constants Sun, Me, Ve, Ea, Ma, Ju, Sa, Mo; a predicate Planet. The *heliocentric language* $\mathcal{L}_{\mathrm{hel}}$ has the heliocentric ecliptic longitude, latitude and distance $\theta(b,t), \beta(b,t), r(b,t)$ of body b at time t (for the Moon, geocentric); its direction-only part $\mathcal{L}_{\mathrm{dir}}$ omits r. The *geocentric language* $\mathcal{L}_{\mathrm{geo}}$ has $\theta_g(b,t), \beta_g(b,t)$, giving the direction $\mathrm{dir}_g(b,t) \in S^2$: a modern idealisation of Tycho's data (our data are geometric plan94 positions, without aberration or light time).

(ii) *Background.* B: the axioms of real closed fields with sin, cos, π, and enough true analysis to prove the numerical bounds used (rational-function bounds of the kind MetiTarski proves (Akbarpour and Paulson, 2010)). The theory of the reals with sin is undecidable ($\sin(\pi x) = 0$ defines $\mathbb{Z}$), so B is a sound decidable fragment; each result names the true sentences it needs.

(iii) *Kepler's first two laws* K12, in the theoretical vocabulary of heliocentric positions $\mathbf{r}(b,t) \in \mathbb{R}^3$ and the elements $a, e, i, \Omega, \varpi, M_0, P$ (functions from bodies to reals: semi-major axis, eccentricity, inclination, node, longitude of perihelion, mean anomaly at the epoch, period): for every planet b and every t, with $M := M_0(b) + 2\pi t/P(b)$, there is ψ with $\psi - e(b)\sin\psi = M$ and $\mathbf{r}(b,t) = \mathrm{Rot}(b)\big(a(b)(\cos\psi - e(b)),\ a(b)\sqrt{1-e(b)^2}\sin\psi,\ 0\big)$, where $\mathrm{Rot}(b)$ is the rotation given by $i(b), \Omega(b), \varpi(b)$; moreover $0 \le e(b) < 1$, $a(b), P(b) > 0$ and $\mathbf{r}(\mathrm{Sun}, t) = 0$.

Over $\mathcal{L}_{\text{geo}}$, K12 also contains the frame convention $i(\text{Ea}) = 0$ and the *bridge*: $\text{dir}_g(b,t)$ is the unit vector of $\mathbf{r}(b,t) - \mathbf{r}(\text{Ea},t)$ if this is nonzero, and the north ecliptic pole $\mathbf{z}$ otherwise.

(iv) *Kepler's third law* K3: $P(b)^2 a(b')^3 = P(b')^2 a(b)^3$ for all planets b, b'.

(v) *Elements* (initial conditions): numerals, or intervals, for the elements of a body. *Data*: true interval sentences $\text{Near}(b,t,\mathbf{v},\delta)$, "the direction of b at t lies within angle δ of $\mathbf{v}$" (Near_g for geocentric directions), with numerals for $t, \mathbf{v}, \delta$, and, where available, $r(b,t) \in [r_1, r_2]$. *Questions* (Definition 2.11): covers by such sentences ("where is Mars at t, to within $2'$?"), and structural questions, which the laws answer without elements (is the orbit planar; does the direction return periodically; are equal areas swept in equal times).

Laws alone answer no position question;[8] laws with elements do. If the elements are put into the antecedents of conditional questions (Definition 2.11), the laws alone answer every such question, and in the solar-system model of Proposition G.30 no steeper MAP then contains elements. This is where the distinction between laws and initial conditions sits formally.

10.2 What the laws say, and in which language

Proposition 10.2 (no content in heliocentric directions)**.** *For every B that contains the axioms of real closed fields,* K12 *and* $\text{K12} \cup \text{K3}$ *have the same $\mathcal{L}_{\text{dir}}$-consequences.*

Proof idea. A Kepler direction does not depend on a. Replacing each $a(b)$ in a model of $B \cup \text{K12}$ by $(k^2 P(b)^2/4\pi^2)^{1/3}$ keeps every direction and makes K3 true (Appendix H.1).

Proposition 10.3 (content in geocentric directions)**.** *Let χ_{opp} be the $\mathcal{L}_{\text{geo}}$-sentence $\exists t\ \text{dir}_g(\text{Ma},t) \neq -\text{dir}_g(\text{Sun},t)$ ("Mars is not permanently in opposition").*

(a) χ_{opp} holds in every model of $\text{K12} \cup \text{K3}$ *whose real sort is* $\mathbb{R}$*. Hence $B \cup \text{K12} \cup \text{K3} \vdash \chi_{\text{opp}}$ for every B that contains the two true sentences ζ_{K3} and ζ_{pos} of the real field with* sin *and* cos *given in Appendix H.1.*

(b) For every B true in $\mathbb{R}$, $B \cup \text{K12} \nvdash \chi_{\text{opp}}$.

Proof idea. (b) Circular coplanar orbits with equal periods and phases, Mars at distance $\rho > 1$, satisfy K12 (which relates no bodies), violate K3 and keep Mars in opposition. (a) Over $\mathbb{R}$, a Mars permanently in opposition stays on the ray from the Sun through the Earth, and Kepler's second law forces $\mathbf{r}(\text{Ma},t) = \kappa\, \mathbf{r}(\text{Ea},t)$ with a constant $\kappa > 1$: equal periods, different semi-major axes, against K3. ζ_{K3} states this about the elements, and ζ_{pos} (the Earth is never at the Sun) lets the argument cover non-standard models.

The sentence χ_{opp} is odd (its negation describes a fantastic world); it only shows that the geocentric language can register K3. The natural form of K3's content is this:

Proposition 10.4 (what K3 does in geocentric directions)**.** *Let* Num^- *be numerals for all elements of the Earth and Mars except $a(\text{Ma})$, with $a(\text{Ea}) = 1$. Let $\mathbf{F}(t)$ be Mars's heliocentric position computed from* Num^- *with semi-major axis* 1*, so that $\mathbf{r}(\text{Ma},t) = \rho\, \mathbf{F}(t)$ with $\rho := a(\text{Ma})$, and let $\mathbf{u}(t)$ be the unit vector of $\mathbf{r}(\text{Ea},t)$.*

(a) $\text{K12} \cup \text{K3} \cup \text{Num}^-$ *fixes $\rho = (P(\text{Ma})/P(\text{Ea}))^{2/3}$ and hence every geocentric direction of Mars: it proves $\text{Near}_g(\text{Ma},t,\mathbf{v},\delta)$ whenever the computed direction lies within δ of $\mathbf{v}$ and B proves the numerical bound involved.*

[8] Even the Ramsey sentence of $\text{K12} \cup \text{K3}$, together with any background true in $\mathbb{R}$, entails (in second-order semantic consequence) no datum $\text{Near}(b,t,\mathbf{v},\delta)$ with $\delta < \pi$ (Proposition H.1).

(b) $\mathrm{K12}\cup\mathrm{Num}^-$ *leaves* ρ *free. At every* t *at which* $\mathbf{r}(\mathrm{Ea},t)$ *and* $\mathbf{F}(t)$ *are not parallel, the geocentric directions of Mars at* t *in the models of* $\mathrm{K12}\cup\mathrm{Num}^-$ *over* $\mathbb{R}$ *(with* ρ *ranging over* $(0,\infty)$*) form the open great-circle arc from* $-\mathbf{u}(t)$*, the direction of the Sun, to* $\mathbf{F}(t)/|\mathbf{F}(t)|$*; its length* $\alpha(t)\in(0,\pi)$ *is the angle between these two directions. So for every* B *true in* $\mathbb{R}$*,* $B\cup\mathrm{K12}\cup\mathrm{Num}^-$ *proves no datum* $\mathrm{Near}_g(\mathrm{Ma},t,\mathbf{v},\delta)$ *with* $\delta<\alpha(t)/2$*.*

Proof idea. As ρ runs over $(0,\infty)$, the vector $\rho\mathbf{F}(t)-\mathbf{r}(\mathrm{Ea},t)$ runs along an open ray that misses the origin, so its direction sweeps the open arc; an arc of length α lies in no cap of radius less than $\alpha/2$ (Appendix H.1).

Example 10.5 (K3's worth against its cost; a computation)**.** Given the other elements, K3's geocentric content is thus exactly ρ, and it is large: over 1580–1600 the median of $\alpha(t)$ is 90° (5th and 95th percentiles 9.5° and 170°). Take as data the daily geocentric longitudes of Mars over 1580.0–1600.0 at a common tolerance δ, and Kepler ellipses for the Earth and Mars with $\rho\in[0,4)$ free or given by K3. With fixed-length numerals for the elements, the posterior odds for K3 are $2^{s_{\mathrm{K3}}(\delta)-|\mathrm{K3}|}$, where $|\mathrm{K3}|$ is the length of its text and $s_{\mathrm{K3}}(\delta):=\log(4\,\mathrm{Vol}_8(\delta)/\mathrm{Vol}_9(\delta))$ ($\mathrm{Vol}_8,\mathrm{Vol}_9$ the volumes of the compatible element regions with and without K3) is what K3 saves (Appendix H.5). A computation gives $s_{\mathrm{K3}}=11.3$, 12.3, 13.4, 14.7 bits at $\delta=8'$, $4'$, $2'$, $1'$, about one more bit per halving of δ, as for any real-valued element; K3 costs $0.005'$ of fit. Its text costs 375 bits in the syntax S1 of Section 10.4 (brackets counted), 165 in Polish notation and 115 in a compact Polish code; it is paid once and saves $s_{\mathrm{K3}}(\delta)$ bits per body whose size it fixes. By Proposition G.29 (with the law in place, at exchange rate 1; a comparison of law choices, not membership of a steeper MAP), K3 pays once it fixes the sizes of about 25–33, 11–15 or 8–10 bodies in the three codes, while the five planets other than the Earth give about 55–75 bits if each saves about as much as Mars (an extrapolation; we computed Mars only). So in none of these codes does compression alone select K3 at Kepler's precision. This is conditional on the code: under the universal prior the relevant cost is the complexity of K3 given the rest of the theory, which these computations do not determine.

K3 is thus a law about initial conditions, worth what it saves on the elements of the bodies it relates. This depends on Kepler's kinematic parametrisation, with the period and the semi-major axis both free; in a state-space parametrisation K3 fixes the mean motion from the energy, a different question, and in Newton's theory the third law, with its mass correction, follows from the one law that does all the other work and costs nothing extra.

10.3 How wrong Kepler's laws are

Kepler's laws are approximately true, with an error that grows with the domain: one fixed ellipse, fitted by least squares, misses Mars by $1.6'$–$1.9'$ over 12–30 years, $3.61'$ over a century and $51.8'$ over 2000 years, and the Moon by 1.1° within a month (Table 9 in Appendix H.2). The third-law "constant" $q:=(2\pi/P)^2a^3/k^2$ (k Gauss's constant) differs between planets at the 10^{-3} level: Jupiter's fitted value matches the Newtonian two-body value $1+m/M$, while Saturn's needs Jupiter's pull. Yet at the precision of Tycho's data K3's own error is invisible: an exact K3 calibrated on the Earth misplaces Jupiter and Saturn geocentrically by at most $0.23'$ and $0.19'$ over 1580–1600. *The error budget:* by the accuracy that the documentation of ERFA's routine quotes, plan94's own error in the heliocentric longitude of Mars over 1580–1600 is at most about $0.43'$; for Jupiter and Saturn it is larger, and their 12-year residuals measure the ellipse against plan94 rather than the sky (Appendix H.2).

Table 5: Minimax errors of the Mars pool, 1580–1600 (arcminutes): against plan94 (upper bounds witnessed by elements, App. H.3), and against an exact ellipse.

form	k	plan94	exact
circle	2	640.79	–
eccentric	4	23.05	22.34
equant	4	8.17	7.43
vicarious	5	1.41	0.62
ellipse	4	1.07	0

Table 6: The proof-requiring posterior over the Mars pool: law-text lengths and Occam factors (bits; "–": δ below the minimax error; all values in App. H.3). A form's weight is $\propto 2^{-(\text{text+Occam factor})}$. K3's text: 375 bits in S1 and S3, 450 in S2.

	circle	eccentric	equant	vicarious	ellipse
text, S1	155	445	990	990	545
text, S2	186	342	450	468	654
text, S3	155	445	990	990	260
$\text{Occ}_{\mathsf{F}}(720')$	20.93	20.24	21.20	22.77	21.18
$\text{Occ}_{\mathsf{F}}(30')$	–	45.38	40.35	42.83	39.65
$\text{Occ}_{\mathsf{F}}(9')$	–	–	57.34	51.78	46.81
$\text{Occ}_{\mathsf{F}}(1.6')$	–	–	–	74.54	59.93

10.4 The proof-requiring posterior over the historical Mars pool

The pool and the data. We take a hand-picked pool of five forms of theory for the motion of Mars, as heliocentric reconstructions (the earlier historical models were geocentric and also represented the Earth's motion): the circle about the Sun (2 elements), the eccentric circle of Hipparchus (4), Ptolemy's bisected equant (4), Kepler's vicarious hypothesis (5) and the ellipse K12 (4) (Appendix H.3). The data are the 7306 daily plan94 heliocentric longitudes v_i of Mars over 1580.0–1600.0, in the best plane through the Sun, each stated as the interval sentence "$\theta(t_i)$ lies within δ of v_i": the guaranteed-tolerance regime of Proposition G.11 with zero measurement error, the noise-free special case. A form with suitable elements proves every datum iff δ is at least its *minimax error*, the least over elements of its largest error (Table 5); here this is an upper bound witnessed by elements and reproduced to four decimals by an independent search (only the circle's is proved optimal). The equant's $8'$ is intrinsic (it remains against an exact ellipse), close to the famous discrepancy of about eight minutes of arc in Kepler's *Astronomia Nova* (Kepler, 1992); more than half of the vicarious hypothesis's $1.41'$ is perturbation error shared with the ellipse, and the $0.34'$ between the two is of the size of plan94's error, so against the sky their ordering rests on the errors against an exact ellipse.

The posterior. A theory is a law text in one of three syntaxes plus fixed-length 64-bit numerals for its elements; the proof-requiring rule of M_A (Definition 2.4), with the universal machine replaced by this code, gives it weight $2^{-\text{length}}$ if it proves every datum (it is then consistent with the data) and 0 otherwise. Summing over numerals, form F has weight proportional to $2^{-(|\text{LAW}_{\mathsf{F}}|_S + \text{Occ}_{\mathsf{F}}(\delta))}$, with $|\text{LAW}_{\mathsf{F}}|_S$ the length of its law text in syntax S and $\text{Occ}_{\mathsf{F}}(\delta) := -\log(\text{Vol}_{\mathsf{F}}(\delta)/\prod \text{ranges})$ its *Occam factor*, $\text{Vol}_{\mathsf{F}}(\delta)$ being the volume of the elements with which F proves all the data (the term is MacKay's (MacKay, 1992); Jefferys and Berger, 1992 read Bayes factors this way for Mercury's perihelion). The syntaxes are *S1*, arithmetic $(+, -, \cdot, /, \sin, \cos, \text{atan2}, \sqrt{\ })$, the numerals $0, 1, 2$, quantifiers); *S2*, epicyclic (S1 with primitives for unit vectors, directions, and the point where a ray meets a circle); and *S3*, Keplerian (S1 with the true anomaly as a function of the eccentricity and the mean anomaly).

What the data decide and what the syntax decides (Table 6 and Figure 1). (i) *The data decide compatibility:* as δ falls the forms drop out in the historical order, circle, eccentric, equant,

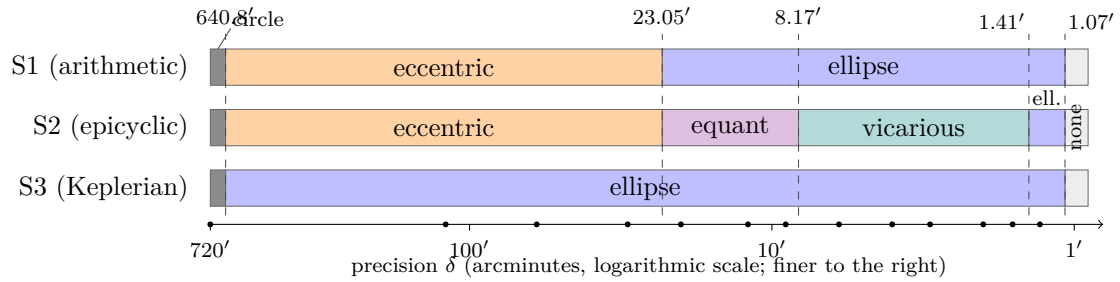


Figure 1: The MAP form of the proof-requiring posterior over the Mars pool against the precision δ of the data, in the three syntaxes, in the noise-free common-tolerance regime (tolerances against plan94). Dashed lines: the minimax errors of Table 5, below which a form is incompatible with the data; below $1.07'$ no form in the pool is compatible. Dots: the precisions at which the posterior was computed; at each of them the runner-up has at most $2^{-12.4}$ of the MAP's weight. Between dots each switch is drawn at the minimax error of the form that drops out; that form stops being the MAP somewhere between this line and the nearest dot to its left.

vicarious hypothesis, ellipse, in every syntax, as one expects, since each successor was adopted because it fitted better. (ii) *Among compatible forms the data contribute only the Occam factors, which on their own point weakly to the true form:* with a uniform prior over the five forms and no text prior, the MAP is the ellipse at every $\delta \leq 60'$, by 0.20 to 14.61 bits. (iii) *The text prior decides the rest:* law-text differences are 100–800 bits for every pair of forms except one, the epicyclic syntax's equant and vicarious texts, 18 bits apart, where the Occam factors move the margin by up to 5.6 bits, so the text still decides. (In S1 and S3 these two texts have equal length, but both lose to the ellipse by more than 400 bits.) So in the noise-free common-tolerance regime the syntax sets the sequence of preferred theories: with epicyclic primitives the posterior runs through every form in the historical order, switching only when a form becomes incompatible; with plain arithmetic it goes from the eccentric to the ellipse at $23'$; with a Keplerian primitive, from the circle to the ellipse at $641'$. A variable-length numeral code, Polish notation for the texts, or a natural form outside the pool changes no MAP form (Appendix H.3).

"The simplest theory compatible with the data" is thus, in this regime, relative to the language of the law texts. The universal prior bounds this relativity by the cost $K(\tau)$ of a faithful translation (Theorem 3.9), but that cost, here the definitions of the extra primitives in S1, is a few hundred bits (an estimate), as large as the text differences. Best-system accounts of laws face the same relativity (Lewis treats it as a problem and requires natural properties (Lewis, 1983); Cohen and Callender accept relativity to a chosen set of kinds (Cohen and Callender, 2009)); the relation is an analogy, since best systems are a metaphysics of laws and SAI is inference from finite data. And the noise-free regime is special: a form proves all the data or none, so the power of the data to discriminate among compatible forms does not grow with their number.

Example 10.6 (graded fit; a computation)**.** Under dithered rounding every form in the pool, the ellipse included, is eventually refuted (Proposition 9.2); but inside M_A a theory may be a sampler, the law with a Gaussian error model of stated σ (Proposition G.3). At $\sigma = 2'$, roughly Tycho's precision, the expected penalty per independent observation relative to the ellipse is 45.05 bits (eccentric), 4.96 (equant) and 0.035 (vicarious hypothesis), so the text advantages over the ellipse are overcome after 2 and 7 observations (eccentric, S1 and S2), 41 (equant, S2) and about 5,300 (vicarious hypothesis, S2); a simulation agrees. Honest uniform-tolerance theories give 23 to 71

and about 470 data, a conservative estimate, since SAI's mixture prefers a law with a calibrated error model (Proposition 9.3; Appendix H.4). So with graded fit the data, not the syntax, decide every step within tens of observations, except the epicyclic syntax's choice between the vicarious hypothesis and the ellipse on longitudes, which needs thousands of longitudes at Tycho-level noise.

Distances as data decide that last step. Add for every day the sentence "$r(t_i)$ lies within a factor $1 \pm \tau_{\mathrm{d}}$ of r_i" (r_i the plan94 distance), and give each form one more element, its scale. At $\tau_{\mathrm{d}} = 1\%$ the least longitude tolerance at which the vicarious hypothesis fits all the data is at most $5.12'$ (witnessed by elements; two independent searches found nothing lower), while the ellipse fits both kinds of data at $1.07'$ (Table 10 in Appendix H.6). So for δ below the former only the ellipse is compatible, in every syntax; with longitudes alone, only below $1.41'$. Compatibility alone thus eliminates the circle-based vicarious hypothesis once the data language contains distances: this reproduces the abandonment of circular orbits, not Kepler's last step from an oval to the ellipse, which is outside the pool. Lower bounds on the least tolerances and Occam factors with distance data were not computed.

10.5 What Newton adds

Newton's theory adds masses, the constant G and accelerations, with the law $m_b \ddot{\mathbf{r}}_b = \sum_{b' \neq b} G m_b m_{b'} (\mathbf{r}_{b'} - \mathbf{r}_b)/|\mathbf{r}_{b'} - \mathbf{r}_b|^3$; the data language is unchanged (step 3 of the ladder of Section 6.1), and masses are theoretical in Sneed's sense, measurable only by using the theory (Sneed, 1971; Balzer et al., 1987). Most of what follows is the achievement of Newtonian mechanics over about 150 years, from Newton to Laplace and Le Verrier, not of Newton alone (Newton, for instance, accounted for about half of the motion of the lunar apsides, and Clairaut for the rest in 1749 (Nauenberg, 2015; Bodenmann, 2010)). We see five points. (1) *No gain in parameters or in fit on the old data:* per body six initial conditions, as for Kepler, plus a mass, and a much longer law; on planetary longitudes at the precision of Kepler's data the extra parameters buy nothing measurable over 20 years (the ellipse fits Mars to $1.07'$), so we think that an SAI posterior on such data alone would not prefer Newton's theory (consistently with Example 10.6). (2) *Unification across domains,* where the gain lies: one law covers the planets, the Moon (the Moon test), falling bodies, tides, comets and satellite systems, and gives the third law the mass-dependent constant $q = 1 + m/M$ of the two-body form $(2\pi/P)^2 a^3 = G(M + m)$, which Jupiter's fitted q matches; Saturn's deviation is accounted for by the orbit-averaged pull of Jupiter, acting like an interior ring (a heuristic estimate gives $q = 1.00154$). Some of these domains need more base vocabulary. (3) *Corrections inside the old domain:* in the classical decomposition (Clemence, 1947), as given by secondary sources, Newtonian planetary perturbations account for about $531''$ of the fixed-frame advance of Mercury's perihelion, about $574''$ per century ($575.4''$ in our fits to plan94), and the remaining $43''$ or so is the effect of general relativity; Laplace explained the great inequality of Jupiter and Saturn (Wilson, 1985), of period about 900 years. (4) *New questions:* in the old language, where Neptune is (Le Verrier's prediction from the residuals of Uranus, also a standard example of reference fixed by a description (Kripke, 1980)); in a new language, the masses of the Sun and of the planets with satellites. (5) *No deductive entailment:* Newton's axioms do not entail K12 or K3, and with the planets' masses they contradict their exact form, as Duhem (1954) observed; with masses and initial conditions they entail honest tolerance forms, "K12 holds within $\varepsilon(b, W)$ on the window W" (Proposition G.33 is a provable toy version). A formal derivation of even one such bound needs an existence-and-uniqueness theorem and a validated integration or perturbation estimate, a very long derivation that no

textbook writes out: a concrete case of the computational gap between a written theory and the reasoner who uses it (Section 11.2).

11 Limitations

The axiom view, and SAI as its formal model, leave out much of what scientific theories are and do. For each limitation we say whether it is partially addressable, and by what, or a genuine limit of a declarative view. The judgements are arguments, not theorems; sentences that rest on a theorem cite it.

11.1 Statistical confrontation with data

In physics a theory meets data statistically: with models of the instruments and the noise it assigns likelihoods to data, often near-raw (Section 6.3), and statistical theories answer with probabilities, tested against frequencies. The proof-requiring core of SAI has no native noise model: a datum is proved or not. Graded fit enters only through randomised theories in M_A, such as laws with calibrated error models (Propositions 9.3 and G.3); through the generative variant, whose updating we prove given a support condition and whose identification we only sketch (Proposition A.23 and Remark A.24); or through sentences about probabilities, whose relation to finite frequencies is not deductive. It matters for our computations: on noise-free interval data the syntax of the law texts decides among the forms compatible with the Mars data, and with graded fit the data decide (Table 6 and Example 10.6). So the proof-requiring core is the noise-free special case; its natural extension inside M_A is a law with a calibrated error model; a native treatment is open.

11.2 Written theories and their readers

A written theory joined to a simple reasoner does not solve the problems the theory is used for. We think this is true, and see three gaps. *The bridge gap:* a problem posed in ordinary terms (a rope over a pulley) becomes formal only through idealisations and bridge principles that the theory does not contain (cf. Cartwright, 1983; Morgan and Morrison, 1999); Proposition 6.4 prices the bridge from raw data to questions as the complexity $K(\eta)$ of an extractor, large in real cases. *The computational gap:* for the formalised problem, theory and mathematics suffice logically, but finding the answer needs methods, lemmas and approximation schemes, as in deriving "Kepler's laws hold within ε on a window" from Newton's theory (Section 10.5). In the model of Section 7 this gap is a theorem, visible only when what is written is what is charged: program length sees neither reasoner nor vocabulary (Theorem 7.2), while charged as written the size of a usable theory depends exponentially on the pair (reasoner, vocabulary) (Corollary 7.10); charged deduction, which bounds a derivation that exists without charging its search, cannot express the gap (Theorem E.26, with the derivation sizes sketched in Remark E.27). So insufficiency for a simple reasoner is a property of the pair (theory, reasoner) and of the vocabulary, when what is written is what is charged. This is not a theorem about textbooks (Section 7.9); it shows that a declarative account of a weak reasoner's competence must include the lemmas, not that people carry exponentially many of them. *The tacit gap:* perceptual and manual skills (Polanyi, 1958, 1966) and exemplars, through which new problems are seen as like solved ones (Kuhn, 1970, 1977), are not sentences; in SAI exemplars are only data. This is a genuine limit.

11.3 Approximate truth

Kepler's laws as stated are false. The difficulty this seems to pose for a proof-requiring inducer is a limitation of proof-requiring compatibility with deterministic theories, which rewards the false exact law over its true weakening (Proposition G.9), not of the axiom view: inside M_A an approximately true law is a law with a calibrated error model (Proposition 9.3). Whether a deterministic theory holds a law honestly or with exceptions depends on how exceptions and silence are priced, a modelling choice (Proposition G.16), and humility is partiality for deterministic theories, with the trichotomy as its formal home, and calibration for randomised ones (Proposition G.22). Open are a canonical metric for grading approximate predictions, which we think does not exist, since purposes set the tolerances, and a general comparative notion of truthlikeness, which is known to be hard: on Popper's definition no false theory is closer to the truth than another (cf. Popper, 1963; Miller, 1974; Tichý, 1974), and the similarity approach needs a metric (cf. Niiniluoto, 1987). Our notion is deliberately local: a question family, a tolerance, a domain.

11.4 Domains, several models, and inconsistent theories in use

Practice uses mutually inconsistent models of one system, each in its domain (the Earth as flat, as a sphere, as an oblate spheroid), while proof-requiring SAI gives a globally inconsistent theory weight 0 (classical electrodynamics of point charges has been argued to be inconsistent, Frisch, 2004, which is disputed). A sound device exists: state each model in its honest form, "q lies within ε of f on the domain", and reason classically. Over a true background that decides the comparisons involved, exact models that give one length the values 100 and 99.9 prove everything, while honest forms of true models are jointly consistent and give true conclusions with propagated tolerances (Proposition G.25 and Remark 9.4). But the device neither explains practice nor reproduces it: physicists rarely state or propagate tolerances, worst-case propagation becomes vacuous over long chains and for chaotic dynamics, and the rigorous honest form of the Boltzmann equation, Lanford's theorem as described in expositions (Lanford, 1975), holds only for a fraction of the mean free time in the low-density limit, while practice uses the equation far beyond that, and rightly. The device blocks the catastrophe, but does not say how to choose a model, a domain or a tolerance, and would forbid much correct practice; we take that gap to be part of what written theories leave to tacit understanding. Under SAI's pricing a domain usually appears as a tolerance with an error distribution that varies with the system, not as a sharp boundary (Proposition G.35 and Section 9.7). Knowing when a model applies is a recognizer, a genuine limit.

11.5 Concepts

Theories are built from concepts. Table 7 lists features of human concepts that an account of concepts as pieces of code, combined to generate predictions, misses, and says whether the axiom view captures them.[9] We accept them all as features of human concepts, with two qualifications: that inferential relations are constitutive is one major account (cf. Sellars, 1953), disputed by referential accounts (cf. Kripke, 1980) and supported for theoretical terms by the Ramsey–Lewis treatment (Lewis, 1970); and natural types we find plausible. One feature supports the axiom view: if a notion can have different recognizers, its identity is fixed otherwise, by inferential role or reference.

For analogy the universal prior gives a bound. Let $K(T)$ be the least $K(e)$ with $\mathsf{W}_e = T$. For

[9] The list follows, with changes, an unpublished draft essay of Hänni's on concepts.

feature of concepts	axiom view	how captured, or what is missing
for questions and problems, not predictions	yes/no	yes for questions posed in a language (§ 6.3); no for problems needing search or action
inert alone; parts of a mind	a limit	worth depends on reasoner and vocabulary (Cor. 7.10); free deduction hides it
inferential role constitutive	yes	axioms are inferential relations (meaning postulates, Ramsey–Lewis)
compositional	yes/no	yes for contents, as formulas; no for recognizers, which are procedures
learning by new links	partly	a persisting symbol, but a global mixture over whole theories
soft; meaning change	partly	"within ε of a sphere" is a true partial theory; not meaning change
analogy	partly	partial interpretations, rewarded as they compress (1); not finding them
new concepts from old	partly	definitions: $O(1)$ universally (Cor. 3.10), useful to a chaining reasoner (Prop. 7.8)
methods attached	partly	lemmas for a bounded reasoner, or axioms via Craig at exponential cost (Lem. F.2); not motor methods
recognizers	no	a procedure on situations; the extractor of Prop. 6.4, taken as given
one notion, many recognizers	yes	identity is the symbol with its axioms; recognizers are bridges
built from pointers	yes	a constant fixed by one axiom, which gathers further axioms
purposes	no	axioms can state functions; their use in search is not modelled
clarification; facing action	no	not in the inducer
natural types	partly	sorts and arities; their emergence is not modelled

Table 7: Features of human concepts, and whether the axiom view captures them (an assessment).

every r.e. theory T_1 and computable map τ from sentences to sentences,

$$K\big(T_1 \cup \tau[T_1]\big) \;\leq\; K(T_1) + K(\tau) + c, \tag{1}$$

since an index of $T_1 \cup \tau[T_1]$ is computable from indices of T_1 and τ, and $K(x, y) \leq K(x) + K(y) + c$: a theory with its image under an analogy costs at most $K(\tau) + c$ bits more than the theory alone. Structure mapping (Gentner, 1983) and the models of science (cf. Hesse, 1963) are accounts of such partial interpretations. A name whose reference is fixed by a description, like Le Verrier's "Neptune" (cf. Kripke, 1980), is a pointer. Concepts that behave in patches (cf. Wilson, 2006) or stretch under counterexamples (cf. Lakatos, 1976), and interpretation as abduction (Hobbs et al., 1993), are outside the view.

11.6 Other limitations

Science predicts very particular things, in set-ups it builds. SAI captures half: its data are selected sentences and its theories partial. Which questions are worth asking, and the making of phenomena by experiment (cf. Hacking, 1983), are outside it; questions weighted by value would be a handle on the first. *Concept formation, discovery and understanding.* Step 3 of the ladder (Section 6.1) is partially addressable (Theorem 8.4 and Corollary 7.10); step 4, the revision of the base abstractions, is a genuine limit for now. SAI is an ideal of the evaluation of theories over a fixed space of hypotheses, not of discovery or understanding: in Section 10 the pool of forms is given, not found (on discovery systems, see Section 12). *Theory-laden and fallible data.* A base language may presuppose a theory (heliocentric longitudes presuppose Copernican geometry; Proposition 10.3), which biases comparisons between theories that disagree about it. And SAI takes labels as correct: a deterministic theory must prove even a mislabelled datum, by an exception axiom; a noise component in the likelihood, as in Hänni and Claude (Anthropic) (2026a), is a partial remedy. *Purposes and technology* (engines, circuits, lenses) are outside the view, a

genuine limit. *Science as a social structure,* in which theories are communicated, divided, checked piece by piece and extended, is a genuine limit of a model with one inducer, and may be where the question of Question 6.1 finds its answer. *The mathematics is free, and large* (in Section 10, ordered fields with trigonometry). Building a background B' into B gains at most $K(B') + c$ bits on any data (Proposition 3.8), so charging for it changes little in the universal setting; we expect it to matter much more under charged deduction. And with free deduction, data that the background entails carry no information: if $\mathsf{Q} \subseteq \mathrm{Cn}(B)$, checking instances $P(\underline{0}), \ldots, P(\underline{N})$ of a Δ_0 property leaves $\mathrm{Bel}(\forall x\, P(x))$ at its prior (Proposition 5.15), the problem of old evidence and logical omniscience (cf. Hacking, 1967; Garber, 1983). Induction in mathematics needs a reasoner for whom deduction is costly.

What the axiom view captures. It captures the static, declarative and inferential side of theories and concepts: contents, inferential relations, the composition of contents, learning by weighing axioms (globally, not concept by concept), pointers, partiality, and approximate truth in honest or error-model form. It misses the procedural side (recognizers, methods, the search for derivations, action), the statistical side of the contact with data in its proof-requiring core, and the dynamic side (purposes, softness, clarification, meaning change, discovery, the social process). That is less than a theory of concepts asks for, but we do not offer one. We offer a model of the declarative core of scientific theories, in which the procedural side reappears formally as the dependence of a written theory's worth on its reasoner and its vocabulary (Section 7).

12 Related work

Universal and operator induction. Solomonoff induction predicts sequences with a universal prior over programs (Solomonoff, 1964, 1978; Li and Vitányi, 2008). Operator induction predicts the answer to a new question from an unordered set of question–answer pairs (Solomonoff, 1999, 2008); the companion papers develop its one-tape universal law and its versions (Hänni and Claude (Anthropic), 2026c,d), and Section 4 relates SAI to them version by version: SAI is operator induction in which the questions are sentences and deduction ties the answers together. Whether a universal prior justifies a preference for simplicity is disputed (Sterkenburg, 2016; on universal learning machines, Sterkenburg, 2019); nothing here depends on it. The computable, finite-class precursors are two earlier reports: Hänni and Claude (Anthropic) (2026a) study a Bayesian axiom inducer with template priors and derivation likelihoods, and coin-free axiom induction under time penalties, which Section 4 extends; Hänni and Claude (Anthropic) (2026b) learn axioms and schemas from their instances.

Probabilities on sentences. Logical probabilities on the sentences of a language go back to Carnap (1950); coherent probabilities on first-order sentences were studied by Gaifman (1964), Scott and Krauss (1966) and Gaifman and Snir (1982); Hutter et al. (2013) extend given probabilities of some sentences to all sentences of a higher-order logic under desiderata that include the Gaifman condition; Christiano (2014) gives a tractable algorithm; and logical inductors (Garrabrant et al., 2016a) assign probabilities to the sentences of a given theory and refine them over time. These are probabilities of truth, and a logical inductor is given the axioms, while SAI is given labelled sentences and is uncertain about the axioms. Bel, the probability that a random theory proves a sentence, is a belief function (Dempster, 1967; Shafer, 1976) with an independent share, and it satisfies the Gaifman condition exactly when the instances prove the universal sentence (Proposition C.19). Sawin and Demski (2013) show that computable distributions that come to

believe the true Π_1 sentences must give some short true Π_2 sentences arbitrarily low limiting probability; Theorem 5.7 and Corollary 5.6 show a parallel divide for the ideal SAI posterior, not a consequence of theirs. Demski's prior (Demski, 2012), as it is usually described, draws random sentences and keeps each one consistent with those kept; the modification due to Fallenstein, Demski and Garrabrant (Garrabrant, 2015), used in Garrabrant et al. (2016b), draws theories as the outputs of a universal machine on random bits. So random axiom systems drawn by a universal machine are not new here: the sampling stage of M_A is that of the modified prior. Those processes iterate draws until the theory is complete; M_A uses a single draw, whose theory is incomplete with positive probability, and for complete theories proof-requiring and consistency-requiring compatibility coincide (Propositions 3.21 and A.25).

Learning theories from data. Formal learning theory studies identification in the limit (Gold, 1967; Putnam, 1965b), also of first-order theories (Osherson et al., 1991; Kelly, 1996; Martin and Osherson, 1998); Proposition A.15 is a Bayesian form of Gold's distinction between text and informant. Shapiro's model inference, which infers theories from facts by conjectures and refutations (Shapiro, 1981, 1991), is the nearest learning-theoretic precedent of proof-requiring compatibility; it outputs one conjecture at a time, without a prior. Inductive logic programming (cf. Muggleton and De Raedt, 1994; Cropper and Dumančić, 2022) asks a hypothesis to entail the positive examples and to be consistent with the negative ones, a mixture of the two compatibility notions of Definition 2.10, invents predicates (Muggleton and Buntine, 1988), and justifies theories by compression (Muggleton, 1995; Conklin and Witten, 1994). In cognitive science, intuitive theories are sets of logical laws generated by a probabilistic grammar and learned by Bayesian inference with stochastic search (Kemp et al., 2008, 2010; Ullman et al., 2012): models over finite spaces of hypotheses, of the same design as the template prior of Hänni and Claude (Anthropic) (2026a), where SAI is universal.

Random axioms and Gödel phenomena. What a probabilistic machine produces with positive probability can be produced deterministically (cf. de Leeuw et al., 1956); the lottery (Theorem 5.1) is where this averaging fails for random theories, since majority votes over consistency constraints of unbounded arity can be inconsistent (cf. the lottery paradox, Kyburg, 1961). Its closer relative is that randomness helps for mass problems: a class of positive measure, such as a class of bounded diagonally non-computable functions, can have no computable member (cf. Kučera, 1985; Bienvenu and Porter, 2016). That randomisation can succeed where determinism cannot, below a success-probability threshold, is familiar for learners (Pitt, 1989; Ambainis, 1999); here the random objects are the hypotheses themselves. Bienvenu et al. (2014) add randomly chosen true complexity axioms to PA and show that they do not help to prove new interesting theorems but can shorten proofs unless $\mathsf{NP} = \mathsf{PSPACE}$; the threshold theories of Theorem 5.7 use a soundness threshold of a related kind, placed just above the unknown inconsistency probability of the universal random theory. Section 5 also meets Chaitin's theorem (Chaitin, 1974; Raatikainen, 1998), flexible formulas (cf. Kripke, 1962; Woodin, 2011), Kreisel's conjecture (cf. Hrubeš, 2009) and the problem of old evidence (Garber, 1983).

Craig, Ramsey, Hempel. Craig's theorem (Craig, 1953, 1956) is the background of Hempel's theoretician's dilemma (Hempel, 1958) and of its replies: Craigian axiomatisations are unmanageable; theoretical terms serve inductive systematisation (cf. Niiniluoto, 1972); science does not aim only at prediction (cf. Putnam, 1965a). Section 8 sets them against our results. In monadic inductive logic theoretical concepts can change inductive probabilities (cf. Niiniluoto and Tuomela,

1973); in the universal mixture they cannot, beyond a constant factor (Theorem 8.5). Every r.e. theory has a polynomial-time axiomatisation by a variant of Craig's trick (Krajíček, 2025), and every r.e. theory without finite models has a finitely axiomatised conservative extension with one more binary relation symbol (Kleene, 1952; Craig and Vaught, 1958, as reported by Pakhomov and Visser, 2019); Theorem F.28 is an effective many-sorted form. Ramsey sentences and Lewis's definitions of theoretical terms (Ramsey, 1931; Lewis, 1970) appear in Proposition F.30. The complexity facts we use are standard (cf. Fischer and Rabin, 1974; Cook and Reckhow, 1979; Chandra et al., 1981; on lengths of proofs, Gödel, 1936; Pudlák, 1998).

Usable theories. The size of a theory that a weak reasoner can use is studied in knowledge compilation (Selman and Kautz, 1996; Darwiche and Marquis, 2002), in compilation for complete unit resolution and propagation (del Val, 1994; Babka et al., 2013), and for encodings of XOR constraints (Gwynne and Kullmann, 2014), with monotone-circuit methods (Bessiere et al., 2009; Göös et al., 2019). The declarative/procedural question has a history in AI: the advice taker (McCarthy, 1959), the controversy discussed by Winograd (1975), and "algorithm = logic + control" (Kowalski, 1979). Khardon and Roth (1997) give learning-to-reason algorithms for a propositional class without efficient reasoning from a formula-based knowledge base, and program synthesis can learn a library of reusable abstractions (Ellis et al., 2021).

Laws, approximation, explanation, and what a theory is. Lewis's best-system account (Lewis, 1983, 1994) and its language-relative version (Cohen and Callender, 2009) are the counterpart of Section 9.5, by analogy only. That a model includes its noise distribution, and that the precision of measurement enters the code length, are standard in MDL and MML (Rissanen, 1978; Wallace, 2005; Grünwald, 2007); the split into model and noise is the subject of algorithmic statistics (Gács et al., 2001; Vereshchagin and Vitányi, 2004); Akaike-based accounts give a statistical form of the dependence of selection on precision (Forster and Sober, 1994). Truthlikeness (cf. Miller, 1974; Niiniluoto, 1987) is a global notion where ours is local. The accounts of explanation (Hempel and Oppenheim, 1948; Kitcher, 1989; van Fraassen, 1980; Woodward, 2003) and of what a theory is (Suppe, 1977; Suppes, 1960; Sneed, 1971; Halvorson, 2012; Cartwright, 1983) are the background of Section 6. Discovery systems find laws in data: BACON rediscovers Kepler's third law (Langley et al., 1987), symbolic regression finds physical laws (Schmidt and Lipson, 2009; Udrescu and Tegmark, 2020), and the Apperception Engine constructs symbolic causal theories of sensory sequences (Evans et al., 2021). They search, where SAI evaluates.

What is new. To our knowledge the following are new: the single-draw proof-requiring law M_A and its basic theory (Section 3); the one-tape and Kolmogorov equivalences with consistent SOI and the transfer principle (Theorems 4.2, 4.3 and 4.8), while the coin-free weight equivalence is Craig's construction inside the universal prior, proved for deciders in Hänni and Claude (Anthropic) (2026a); the lottery, its comparison of versions and its arithmetic versions with true data (Theorem 5.1 and Corollary 5.6); the threshold theories on Π_1 data and the coding theorem for an inductive inference rule (Theorem 5.7 and Corollary 5.8); the reasoner-relative separations inside a universal inducer (Section 7), where the bound for systems of laws (Theorem 7.9) is a short combination of known results that we did not find stated and that is probably known to specialists; a complexity-theoretic form of the impracticality reply to Hempel, which we did not find in the literature, and the derivation-bounded constructions and fit classes of Section 8; and, for a proof-requiring inducer, the choice between honest laws, laws with exceptions and laws with error models (Propositions 9.3 and G.16). Not new are random axiom systems drawn by a universal

machine, the mixture and dominance arguments, Craig's construction, the rules of conditioning for belief functions, and the reading of a law with its noise distribution as a two-part code.

13 Discussion

What the formalism shows. Solomonoff axiom induction turns the view that theories answer questions posed in a given language into a precise object, and the view survives in comparative form. Its formal content is a change of target and of data type, logical coherence and partiality: consequences of the data are certain, "independent" is a native outcome, and a law is a sentence with a belief of its own (Section 6.3 and Table 3). The inference principle is unchanged: version by version, SAI is operator induction over consistent labellers (Section 4). What logic adds is a set of phenomena with no analogue for labellers: random theories that succeed where every r.e. theory fails, a single random true theory that is, up to a constant factor, the whole mixture on Π_1 data, and the fixed marks that Gödel's theorems leave on the posterior (Section 5). On what a theory is, the idealised formalism is neutral: it cannot see the difference between a theory and a program. Where shape matters, it is relative to a reasoner and a vocabulary, and it matters only when the written theory is charged or the class of hypotheses is restricted (Sections 7 and 8). We still find it plausible that good theories are shaped like logical theories, and that textbook theories are declarative is an observation we accept; what is missing is a formal setting in which that shape provably pays. Finally, that Kepler's laws are only approximately true is a limitation of proof-requiring compatibility with deterministic theories, not of the axiom view: under SAI's pricing an approximately true law is a law with a calibrated error model (Section 9).

What it cannot show. The axiom view models the declarative core of scientific theories: contents, inferential relations, partiality and approximate truth. It misses their procedural side (recognizers, methods, the search for derivations), the statistical confrontation of theory with data, of which its proof-requiring core is the noise-free special case, and their dynamic side (meaning change, discovery, the social process of science) (Section 11). Its limit posterior can live on random theories and, with adversarially chosen true data, on false ones, so it supports no realist conclusion about the theories it favours (Theorem 5.1 and Corollary 5.6). It evaluates theories and does not find them, and it is not computable.

13.1 Open problems

(1) Does M_A dominate every lower semicomputable provability function of a random consistent theory (Question 3.22)?

(2) Is there a stream with bounded one-tape loss on which the coin-free loss diverges computably fast, and does bounded one-tape loss on a co-r.e. stream imply that a consistent r.e. theory proves it (Question 5.4)?

(3) Over a true background containing PA, is the one-tape posterior probability of Σ_1-soundness bounded away from 1 uniformly over all data (Question 5.14)?

(4) Does the dependence of written size on the pair (reasoner, vocabulary) extend beyond Boolean laws, to real-valued laws with numerical solvers, probabilistic programs with conditioning and simulation codes (Question 7.11)?

(5) Is there a formal setting for communities of bounded reasoners who communicate, check and extend theories piece by piece, in which theories of declarative shape provably do better than programs (Question 6.1)?

Further questions concern the random tape and the order of the data (Questions A.8 and A.21); the computability and the versions of consistent operator induction (Questions B.13 and B.15); Presburger data under charged deduction (Question B.30); soundness on true data, a coding theorem with constant slack and the loss on true arithmetic (Questions 5.21, C.9 and C.12); a Kreisel-type property for finite theories (Question C.22); the fit classes of richer observational languages, of infinite hypotheses about the named objects and of unary theoretical functions (Questions F.18, F.20 and F.21); exact minima, learning and time bounds for usable theories (Questions E.23, E.30 and E.36); and restricted classes of approximate laws and domains of applicability (Questions G.6 and G.37).

Directions. Four directions seem to us most worth pursuing. The first is step 4 of the ladder, the revision of the base abstractions, for which we have no formal handle. The second is the setting of the fifth problem above, which may also be where the declarative shape of written theories finds its explanation. The third is a costed-deduction mixture over unknown axiom systems, combining SAI, which is uncertain about the axioms, with logical induction (Garrabrant et al., 2016a), which is uncertain about their consequences; such a reasoner is what induction in mathematics needs, since with free deduction checking instances confirms nothing (Proposition 5.15). The fourth is a native treatment of statistical confrontation inside the proof-requiring core, beyond randomised theories and the generative variant (Section 11.1). Smaller questions remain: whether explanatory asymmetry is interventional in general; a derivation of Kepler's laws within stated tolerances from Newton's theory in a prover for validated numerics, a concrete case of the computational gap of Section 11.2; and the Mars posterior with distances as data.

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Several of the questions and examples studied here first appeared in Hänni's working notes (Hänni, 2026). Independent instances of Claude checked each part: they re-derived the proofs and re-implemented the computations, and the results were revised in response. The results of the earlier report (Hänni and Claude (Anthropic), 2026a) and of the companion papers (Hänni and Claude (Anthropic), 2026c,d) that we use were prepared and checked in the same way. No human has yet independently checked all of the mathematics, and nothing was verified in a proof assistant; readers should weigh the results accordingly.

Code and data

All computations reported here were done with seeded or deterministic Python scripts (Python 3.11 with NumPy, SciPy and mpmath; the planetary computations use pyerfa, the Python interface to the ERFA library). The scripts, their saved outputs and independent re-implementations of the main computations are in the folder `solomonoff-axiom-induction/` of the repository `https://github.com/kaarelh/AI-works`; its README lists the script behind each reported number and how to re-run it. Re-running the scripts reproduces the saved outputs, apart from lines that report run times; the longest computations, on the Mars pool,

take more than half an hour each, and their figures were also reproduced by independent re-implementations.

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A Proofs for the setting and the basic theorems

This appendix proves the results of Sections 2 and 3 in their order, and states and proves the results that the main text only mentions. Throughout, G is a theory machine and the notation is that of Definitions 2.3 and 2.4.

A.1 The setting

Lemma A.1 (Measurability and the basic form)**.** *(a)* $O^G_\chi = \bigcup\{[w] : B \cup A^G_w \vdash \chi\}$ *is an effectively open class, uniformly in an index of* G *and in* χ*; hence* $\mathbb{U}(O^G_\chi)$ *is left-c.e., uniformly. (b)* Cons^G *is a* Π^0_1 *class, and* $O^G_\bot \subseteq O^G_\chi$ *for every* χ*. (c)* $\nu_G(D) = \mathbb{U}(O^G_{\psi_D}) - \mathbb{U}(O^G_\bot)$*, and* $E^{3,G}_D$ *is Borel.*

Proof. (a) If $B \cup A^G_\omega \vdash \chi$, a derivation uses finitely many axioms $\chi_1, \ldots, \chi_r \in A^G_\omega$, each emitted under some prefix of ω. The prefixes of ω are linearly ordered, so the longest of them, w, has $\chi_1, \ldots, \chi_r \in A^G_w$, and every $\omega' \sqsupseteq w$ has $B \cup A^G_{\omega'} \vdash \chi$. Conversely $[w] \subseteq O^G_\chi$ whenever $B \cup A^G_w \vdash \chi$. The set of pairs (w, χ) with $B \cup A^G_w \vdash \chi$ is c.e.: enumerate G and all derivations, using that B is decidable. (b) Cons^G is the complement of an effectively open class, and an inconsistent set proves everything. (c) $E^G_D = O^G_{\psi_D} \setminus O^G_\bot$ with $O^G_\bot \subseteq O^G_{\psi_D}$; and $E^{3,G}_D$ is E^G_D intersected with finitely many closed classes. □

Proof of Remark 2.8. (a) *Deciders to enumerators.* By universality of V there is a string r_1 such that, whenever p is a decider, $V(r_1 p)$ is an index of the r.e. set $\{\chi : \{V(p)\}(\chi) = 1\}$. The map $p \mapsto r_1 p$ is injective, adds $|r_1|$ to the length and preserves the axiom set. *Enumerators to deciders (Craig's trick).* Write $\chi^{\wedge 1} := \chi$ and $\chi^{\wedge(m+1)} := \chi \wedge \chi^{\wedge m}$. For an index e fix the step-by-step enumeration of W_e and let $A^{\mathrm{C}}_e := \{\chi^{\wedge(s+1)} : \chi \text{ is enumerated at step } s\}$. A sentence θ has at most two readings $\theta = \chi^{\wedge m}$: $(\chi, m) = (\theta, 1)$, and, if θ is a conjunction, χ its left conjunct with m the number of times χ can be stripped off. (If $\theta = \chi^{\wedge m}$ with $m \geq 2$, then χ is the left conjunct of θ, and stripping stops exactly at χ, because χ is not of the form $\chi \wedge X$.) So A^{C}_e is decided by the procedure: find the readings, and for each reading (χ, m) run the enumeration for exactly $m - 1$ steps and check whether χ appears at step $m - 1$. It halts on every input, also when W_e

is finite. Since $\chi^{\wedge m}$ is logically equivalent to χ, $\mathrm{Cn}(A_e^{\mathrm{C}}) = \mathrm{Cn}(\mathsf{W}_e)$. A fixed string r_2 makes $V(r_2 p)$ an index of this decider for $e = V(p)$; the map $p \mapsto r_2 p$ is injective and adds $|r_2|$. Both maps preserve consistency and what is proved, so summing over programs, and taking shortest programs, gives (a) with $c := \max(|r_1|, |r_2|)$. This is the argument of Theorem 6.3 of Hänni and Claude (Anthropic) (2026a), here between two kinds of axiom hypotheses rather than between axiom and labeller hypotheses.

(b) Under Assumption 4.5, Theorem 6.14(a) of Hänni and Claude (Anthropic) (2026a) shows that $W_A^{\mathrm{gen},t}$ is within a constant factor of the coin-free consistent mixture W^{cons} of labellers, and its Theorem 6.3 shows $W^{\mathrm{cons}} \asymp W_A^{\mathrm{dec}}$; combined with (a) this gives the claim for $W_A^{\mathrm{gen},t}$. For the Kolmogorov versions we use the map behind it. An enumerator p of A is sent to the padded enumerator of the semi-decider of A ("yes on χ iff χ is enumerated") of Lemma E.12 of Hänni and Claude (Anthropic) (2026a); that lemma, proved there under Assumption 4.5, gives an enumerator of the delay form with the same consequences as A, injectively and with length at most $|p| + c$. Conversely, an enumerator of the delay form is an enumerator. The translation between the universal machine of the earlier report and programs of V that print indices costs a constant. We do not re-derive Assumption 4.5.

(c) If ψ witnesses $K_{\mathrm{fin}}(D)$, a fixed prefix r_3 turns a shortest program for ψ into a program for an index of $\{\psi\}$, which is consistent with B and proves ψ_D; so $K_A \leq K_{\mathrm{fin}} + |r_3|$. For consistent D, ψ_D itself is a candidate for K_{fin}. If D is inconsistent, no consistent set proves ψ_D. □

A.2 Structure and updating

Proof of Proposition 3.1. (a) If D is inconsistent, every set proving ψ_D is inconsistent, so $E_D^G = \emptyset$. If D is consistent, the deterministic machine $G_D := \{(\epsilon, \psi_D)\}$ has $\nu_{G_D}(D) = 1$, so $M_A(D) \geq 2^{-K(G_D)} > 0$ by Theorem 3.6. Three-valued: $B \cup \{\psi_D\}$ decides exactly what $B \cup D^{\pm}$ decides, so if $B \cup D^{\pm}$ decides no ι-labelled sentence then $\nu^3_{G_D}(D) = 1$; conversely, if $B \cup D^{\pm}$ decides an ι-labelled sentence, so does every set proving ψ_D.

(b) O_χ^G depends only on the B-equivalence class of χ, and $\psi_{D+(\varphi,b)}$ is equivalent to $\psi_D \wedge \varphi^b$. On E_D^G the random theory is consistent, so exactly one of three things happens: it proves φ, it proves $\neg\varphi$, or it decides neither. The three events are $E^G_{D+(\varphi,1)}$, $E^G_{D+(\varphi,0)}$ and $E^{3,G}_{D+(\varphi,\iota)}$. For G_U, dividing by $\mathbb{U}(E_D)$ shows that the third term is $M_A(D)(1 - \mathrm{Bel}_D(\varphi) - \mathrm{Dis}_D(\varphi))$.

(c) $B \cup A \vdash \psi_D$ iff $\mathrm{Cn}(D^{\pm}) \subseteq \mathrm{Cn}(A)$. If $B \cup D^{\pm} \vdash \varphi^b$, then $\mathrm{Cn}(D^{\pm} \cup \{\varphi^b\}) = \mathrm{Cn}(D^{\pm})$, and $D + (\varphi, 1-b)$ is inconsistent, so (a) applies to it.

(d) At a step with $\varphi_t \notin \mathrm{dom}\, D_{t-1}$ the factor is $M_A(D_t)/M_A(D_{t-1})$; at a repeat (necessarily with the stored label, since the sequence is consistent) it is 1 and $D_t = D_{t-1}$. The product telescopes to $M_A(D_n)/M_A(\emptyset)$. If some $M_A(D_t) = 0$, the first such step has factor 0, and both sides are $+\infty$ by the monotonicity in (b). Dependence only on $\mathrm{Cn}(D_n^{\pm})$ is (c).

(e) $M_A(\emptyset) = \mathbb{U}(\mathrm{Cons})$. The tapes beginning with a V-program for an index of the empty machine have $A_\omega = \emptyset$, consistent since B is; the tapes beginning with a program for the machine $\{(\epsilon, \bot)\}$ are inconsistent. Both sets have positive measure.

(f) As (b), with $E_D^{3,G}$ in place of E_D^G; and $\nu_G^3(\emptyset) = \mathbb{U}(\mathrm{Cons}^G) = \nu_G(\emptyset)$.

(g) By definition $E_D^{3,G} = \{\omega \in \mathrm{Cons}^G : \mathrm{tri}_\omega^G \supseteq D\}$, and for two-valued D this is E_D^G. □

Proof of Proposition 3.2. (a) A theory proves ψ_D and χ iff it proves $\psi_D \wedge \chi$, so $E_D \cap O_\chi = O_{\psi_D \wedge \chi} \setminus O_\bot$, and both sides equal $\mathbb{U}(O_{\psi_D \wedge \chi} \setminus O_\bot)/\mathbb{U}(O_{\psi_D} \setminus O_\bot)$. The normalising constant $1 - \mathrm{Inc}_0$ cancels.

(b) Let $P := \mathbb{U}(\cdot \mid E_D)$ and $Q_i := O_{\chi_i}$. Then $\bigcap_{i\in I} Q_i = O_{\bigwedge_{i\in I}\chi_i}$, since on one tape a theory proves each χ_i iff it proves their conjunction, and $\bigcup_i Q_i \subseteq O_{\bigvee_i \chi_i}$. Inclusion–exclusion for $P(\bigcup_i Q_i)$ gives the inequality. Bel_0 is the unnormalised case, with P the restriction of $\mathbb{U}$ to Cons.

(c) Fix an enumeration (σ_k) of Sent. For $\omega \in \mathrm{Cons}$ let $T_0(\omega) := B \cup A_\omega$, and $T_{k+1} := T_k \cup \{\sigma_k\}$ if that set is consistent, $T_{k+1} := T_k \cup \{\neg\sigma_k\}$ otherwise. $T_\infty(\omega) := \bigcup_k T_k$ is a complete consistent extension of B: it is consistent by compactness, and complete and deductively closed because it is maximal consistent. Each choice depends on whether $B \cup A_\omega \cup F$ is consistent for a finite set F, that is, on membership of ω in the open class $O_{\neg\bigwedge F}$; so $\omega \mapsto T_\infty(\omega)$ is Borel. Let P_D be its law under $\mathbb{U}(\cdot \mid E_D)$. If $B \cup A_\omega \vdash \chi$ then $\chi \in T_\infty$, and if $\chi \in T_\infty$ then $B \cup A_\omega \nvdash \neg\chi$. So $\mathrm{Bel}_D \le P_D \le \mathrm{Pl}_D$. □

Proof of Proposition 3.3. For consistent $B \cup A$: $B \cup A \cup \{\psi\}$ is consistent iff $B \cup A \nvdash \neg\psi$, and $B \cup A \cup \{\psi\} \vdash \varphi$ iff $B \cup A \vdash \varphi \vee \neg\psi$ (the deduction theorem; ψ is a sentence). So $M_A^{\mathrm{nc}}(D) = \mathbb{U}(\mathrm{Cons}) - \mathbb{U}(\mathrm{Cons} \cap O_{\neg\psi}) = \mathrm{Bel}_0(\top) - \mathrm{Bel}_0(\neg\psi) = \mathrm{Pl}_0(\psi)$. Since $B \cup A \vdash \neg\psi$ implies $B \cup A \vdash \varphi \vee \neg\psi$, the numerator of $\mathrm{Bel}_D^{\mathrm{nc}}(\varphi)$ is $\mathbb{U}(\mathrm{Cons} \cap O_{\varphi\vee\neg\psi}) - \mathbb{U}(\mathrm{Cons} \cap O_{\neg\psi}) = \mathrm{Bel}_0(\varphi \vee \neg\psi) - \mathrm{Bel}_0(\neg\psi)$. Dividing numerator and denominator by $1 - \mathrm{Inc}_0$ gives the form with $\widehat{\mathrm{Bel}}_0$. □

A.3 Computability

The following lemma is a form of a theorem of Demuth (Demuth, 1975), as it is attributed in the literature: a left-c.e. real plus a random left-c.e. real is random (the converse is attributed to Downey et al., 2002). We prove the form we need, for completeness. For a real $x \in (0,1)$, $x_{1:n}$ denotes its first n binary digits; we use the Levin–Schnorr characterisation (standard; see Li and Vitányi, 2008): x is Martin-Löf random iff $K(x_{1:n}) \ge n - c_x$ for all n.

Lemma A.2 (Randomness passes upwards)**.** *Let α, γ be left-c.e. reals with α Martin-Löf random, let $k \ge 0$, and let $\beta := 2^{-k}\alpha + \gamma < 1$. Then $K(\alpha_{1:n}) \le K(\beta_{1:n}) + k + c$ for all n, where c depends on k and on fixed approximating programs for α and γ. Hence β is Martin-Löf random.*

Proof. Let $\alpha_s \uparrow \alpha$ and $\gamma_s \uparrow \gamma$ be computable rational approximations and $\beta_s := 2^{-k}\alpha_s + \gamma_s$. Given the string $u := \beta_{1:n}$ (which also gives n), put $q := 0.u - 2^{-n} < \beta$ and find the first s with $\beta_s > q$. Then $\beta - \beta_s < \beta - q \le 2^{1-n}$, hence $\alpha - \alpha_s \le 2^k(\beta - \beta_s) < 2^{k+1-n}$, because $\gamma - \gamma_s \ge 0$. Since α is random, it is irrational, so $\alpha_{1:n}$ is the index of the dyadic interval of length 2^{-n} that contains α, one of the at most $2^{k+1} + 1$ such intervals that meet $[\alpha_s, \alpha_s + 2^{k+1-n})$. So $\alpha_{1:n}$ is determined by u and a $(k+2)$-bit index, and $K(\alpha_{1:n}) \le K(\beta_{1:n}) + k + c$. By the Levin–Schnorr characterisation $K(\alpha_{1:n}) \ge n - c'$ for all n; hence $K(\beta_{1:n}) \ge n - k - c - c'$ for all n, and β is random by the same characterisation. □

Proof of Theorem 3.5. The machine G_α. Let $\alpha_s \uparrow \alpha$ be a computable nondecreasing sequence of rationals, and let $G_\alpha := \{(w, \bot) : \exists s\ 0.w + 2^{-|w|} \le \alpha_s\}$, where $0.w$ is the dyadic rational with binary digits w. Then $A_\omega^{G_\alpha} \subseteq \{\bot\}$, and $\bot \in A_\omega^{G_\alpha}$ iff some prefix w of ω has $[w] \subseteq [0, \alpha_s]$ for some s, read as reals. This holds for every ω with $0.\omega < \alpha$ (take s with $\alpha_s > 0.\omega$ and a long prefix) and fails for every ω with $0.\omega > \alpha$. So the inconsistency probability of G_α is α.

The decomposition. Let p_α be a V-program for an index of G_α. Splitting the tapes by their first block in $\mathrm{dom}\, V$ (tapes with no such block have $A_\omega = \emptyset$, which is consistent),

$$\mathrm{Inc}_0 = 2^{-|p_\alpha|}\alpha + \gamma_\alpha, \qquad \gamma_\alpha := \sum_{p \in \mathrm{dom}\, V \setminus \{p_\alpha\}} 2^{-|p|}\, \mathbb{U}(O_\bot^{G_{V(p)}}),$$

and γ_α is left-c.e., because $\operatorname{dom} V \setminus \{p_\alpha\}$ is c.e. and each term is left-c.e. uniformly (Lemma A.1(a)).

(a) Let $\mathcal{H}$ be the halting set and $\alpha := \sum_{n\in\mathcal{H}} 2^{-n-1}$, which is left-c.e. Its binary expansion is the characteristic sequence of $\mathcal{H}$ (unique, since $\mathcal{H}$ is infinite and co-infinite), so α is not computable. If Inc_0 were computable, $\alpha = 2^{|p_\alpha|}(\mathrm{Inc}_0 - \gamma_\alpha)$ would be right-c.e.; being also left-c.e., it would be computable. Since a real that is both left-c.e. and right-c.e. is computable, the right-c.e. real $1 - \mathrm{Inc}_0$ is not left-c.e.; as $M_A(\emptyset) = 1 - \mathrm{Inc}_0$ is a value of M_A, M_A is not lower semicomputable.

(b) *Solovay reducibility* (Solovay, 1975; we use the definition of Downey and Hirschfeldt, 2010): $\alpha \le_S \beta$ iff there are a constant k and a partial computable function f on the rationals such that for every rational $q < \beta$, $f(q)$ is defined, $f(q) < \alpha$, and $\alpha - f(q) \le 2^k(\beta - q)$. Let $\beta := \mathrm{Inc}_0$, with approximations $\beta_s := 2^{-|p_\alpha|}\alpha_s + \gamma_{\alpha,s} \uparrow \beta$. If α is computable the reduction is trivial: given $q < \beta$, find s with $\beta_s > q$ and output a rational in $(\alpha - (\beta_s - q), \alpha)$. Otherwise $\alpha_s < \alpha$ for all s; let $f(q) := \alpha_s$ for the first s with $\beta_s > q$. Then, since $\gamma_\alpha - \gamma_{\alpha,s} \ge 0$,

$$\alpha - \alpha_s = 2^{|p_\alpha|}\big[(\beta - \beta_s) - (\gamma_\alpha - \gamma_{\alpha,s})\big] \le 2^{|p_\alpha|}(\beta - \beta_s) < 2^{|p_\alpha|}(\beta - q).$$

(c) Ω_V is Martin-Löf random (Chaitin, 1975). Apply the decomposition with $\alpha := \Omega_V$, and Lemma A.2 with $k = |p_\alpha|$ and $\gamma = \gamma_\alpha$; here $\mathrm{Inc}_0 < 1$ by Proposition 3.1(e), and k and γ_α are fixed by V, so the constant of the lemma is a machine constant. □

Proposition A.3 (Computability relative to the halting problem)**.** *(a) $M_A(D)$ and the conditionals $M_A(b \mid D, \varphi)$, $M_A^3(y \mid D, \varphi)$ and $\mathrm{Bel}_D(\chi)$ are computable relative to $\emptyset'$, uniformly. (b) $D \mapsto M_A(D) + \mathrm{Inc}_0$ is lower semicomputable; so M_A is lower semicomputable relative to the single real Inc_0.*

Proof. (a) Every left-c.e. real is $\emptyset'$-computable (decide "$\exists s\ a_s > q$" for rationals q), and differences and ratios of $\emptyset'$-computable reals are $\emptyset'$-computable once the denominator is known to be positive. By Proposition 3.1(a), $M_A(D) > 0$ iff D is consistent, a Π^0_1 question, which $\emptyset'$ decides. The three-valued events are Boolean combinations of effectively open classes, with $O_a \cap O_b = O_{a\wedge b}$, so their measures are sums and differences of left-c.e. reals by inclusion–exclusion. (b) is Proposition 3.4. □

A.4 Dominance and invariance

Proof of Theorem 3.6. $\operatorname{dom} V$ is prefix-free, so the cylinders $[p]$, $p \in \operatorname{dom} V$, are disjoint, and for $\omega = p\omega'$ we have $A_\omega = A_{\omega'}^{G_{V(p)}}$ with ω' uniform given $[p]$. Hence $\mathbb{U}(E_D \cap [p]) = 2^{-|p|}\nu_{G_{V(p)}}(D)$. The tapes with no prefix in $\operatorname{dom} V$ have measure $1 - \Omega_V$ and $A_\omega = \emptyset$, which proves ψ_D iff B does. Grouping the p with $V(p) = i$ gives the weight $\mathbf{m}(i) \ge 2^{-K(i)}$, and taking i with $\nu_{G_i} = \nu$ and $K(i) = K(\nu)$ gives $M_A \ge 2^{-K(\nu)}\nu$. The three-valued case is identical.

(i) The deterministic machine $G'_e := \{(\epsilon, \chi) : \chi \in \mathsf{W}_e\}$ has an index $g(e)$ with g computable and injective, so $\mathbf{m}(g(e)) \ge 2^{-c}\mathbf{m}(e)$: the image of $\mathbf{m}$ under g is an lsc semimeasure, dominated by $\mathbf{m}$. For $e \in [T]$, the tapes beginning with a program for $g(e)$ have $A_\omega = \mathsf{W}_e$, with $\mathrm{Cn}(A_\omega) = \mathrm{Cn}(T)$, and $B \cup A_\omega$ consistent because $B \cup T$ is. These sets of tapes are disjoint for distinct e; summing gives $\mathbb{U}_A[T] \ge 2^{-c}\mathbf{m}_{\equiv}(T)$, and $\mathbf{m}_{\equiv}(T) \ge 2^{-K_{\equiv}(T)}$. The same argument over $e \in \mathrm{Ext}T$ gives $\mathbb{U}_A(\mathrm{Ext}T) \ge 2^{-c}\mathbf{m}(\mathrm{Ext}T)$.

(ii) $G_f := \{(\epsilon, \varphi^b) : f(\varphi) = b\}$ enumerates Γ_f, which is consistent with B and proves ψ_D when $f \supseteq D$; and $K(G_f) \le K(f) + c$.

(iii) The map $i \mapsto \sum_{g(e)=i} v(e)$ is an lsc semimeasure of complexity at most $K(v) + c$, so $\mathbf{m}(g(e)) \ge 2^{-K(v)-c}v(e)$. With $v = \mathbf{m}$ this gives $M_A \ge 2^{-c}W_A$.

(iv) The machine of Example (iii) after Definition 2.3 has an index computable from indices of T and of the family. □

Proof of Proposition 3.7. G_U is universal, with p_G a V-program for an index of G. $M_A \geq 2^{-K(G')}\nu_{G'}$ by Theorem 3.6. Conversely, the tapes $p_{G_U}\omega'$ give G' the axiom set $A_{\omega'}$, so $\nu_{G'}(D) \geq 2^{-|p_{G_U}|}M_A(D)$. The three-valued laws are handled in the same way. So posteriors change by at most a constant factor, and losses in the semimeasure convention by an additive constant. □

Proof of Proposition 3.8. The machine $G_{B'} := \{(\epsilon, \chi) : \chi \in B'\} \cup G_U$ emits $B' \cup A_\omega$ on ω. Under the background B, the set $B \cup B' \cup A_\omega$ is equivalent to $B' \cup A_\omega$, so the law of $G_{B'}$ under the background B is $M_A^{B'}$; apply Theorem 3.6. (Both laws use the same G_U, which does not depend on the background.) The same map on single programs gives the statements for W_A and K_A. The converse fails as stated, since D is consistent with B (Proposition 3.1(a)). □

Proof of Theorem 3.9. A preliminary fact: $B' \vdash \neg\tau(\bot)$. By (T1) with $\Gamma = \emptyset$, $B \vdash \neg\bot$ gives $B' \vdash \tau(\neg\bot)$, and (T2) converts this.

Push forward. The machine "run G_U and emit $\tau(\chi)$ for every emitted χ" emits $\tau[A_\omega]$. By (T1), $B' \cup \tau[A] \vdash \tau(\bot)$ iff $B \cup A \vdash \bot$; with the preliminary fact, $B' \cup \tau[A]$ is consistent iff $B \cup A$ is. By (T1) and (T2), $B' \cup \tau[A] \vdash \tau(\varphi)^b$ iff $B \cup A \vdash \varphi^b$ (for $b = 0$: $B' \cup \tau[A] \vdash \neg\tau(\varphi)$ iff it proves $\tau(\neg\varphi)$ iff $B \cup A \vdash \neg\varphi$). So the event of this machine for τD is exactly E_D, and Theorem 3.6 in $\mathcal{L}'$ gives $M'_A(\tau D) \geq 2^{-K(\tau)-c}M_A(D)$.

Pull back. For $A' \subseteq \mathrm{Sent}_{\mathcal{L}'}$ let $\tau^*(A') := \{\varphi \in \mathrm{Sent}_{\mathcal{L}} : B' \cup A' \vdash \tau(\varphi)\}$. The set $G^* := \{(w, \varphi) : B' \cup A'_w \vdash \tau(\varphi)\}$ is c.e., and by compactness, as in Lemma A.1, $A_\omega^{G^*} = \tau^*(A'_\omega)$. If $B' \cup A'$ is consistent, so is $B \cup \tau^*(A')$: if $B \cup \Gamma \vdash \bot$ for a finite $\Gamma \subseteq \tau^*(A')$, then $B' \cup \tau[\Gamma] \vdash \tau(\bot)$ by (T1), and $B' \cup A'$ proves every member of $\tau[\Gamma]$, so $B' \cup A' \vdash \tau(\bot) \wedge \neg\tau(\bot)$. If $B' \cup A' \vdash \tau(\varphi)^b$, then $\varphi^b \in \tau^*(A')$ (for $b = 0$ via (T2)). So the event of G^* for D contains the event for τD, and $M_A(D) \geq 2^{-K(\tau)-c}M'_A(\tau D)$.

The same two maps act on single enumerator programs, with a prefix of length $K(\tau) + c$; this gives the statements for W_A and K_A. □

Proof of Corollary 3.10. (T1) holds for the inclusion because every $\mathcal{L}$-structure expands to an $\mathcal{L}'$-structure, so by completeness $B \cup \Gamma \vdash_{\mathcal{L}'} \varphi$ iff $B \cup \Gamma \vdash_{\mathcal{L}} \varphi$ for $\mathcal{L}$-sentences; (T2) is trivial. Theorem 3.9 with $K(\tau) = O(1)$ gives the weights, and a predictive probability is a ratio of two weights, each changed by a factor $2^{\pm c}$. □

A.5 Loss bounds

Proof of Proposition 3.12. (a) is Theorem 3.6. (b) For $e \in [T]$, the set $\mathsf{W}_e \cup \{\psi_E\}$ has an index computable from (e, E); it is consistent with B, since $B \cup T \cup E^{\pm}$ is, and it proves ψ_{D_n}. As $K(e, E) \leq K(e) + K(E) + c$, Theorem 3.6(i) gives $L_A(D_n) \leq K(e) + K(E) + c'$; minimise over $e \in [T]$. (c) Consider the machine that emits T and $\alpha_j^{\omega_j}$ for every j. By the hypothesis its theory is consistent on every tape, and T decides no α_j. Its theory proves every datum proved by T, and proves the α_j-data whenever ω agrees with their m labels, which has probability 2^{-m}. The machine has an index computable from those of T and of the family; apply (a). □

Proof of Proposition 3.13(a), (b). (a) At a repeat the predictor outputs the stored label and makes no mistake. Otherwise the posterior trichotomy sums to 1, so if $M_A(b_t \mid D_{t-1}, \varphi_t) = \mathrm{Bel}_{D_{t-1}}(\varphi_t^{b_t}) > \frac{1}{2}$, then b_t is the unique maximiser. Each mistake therefore costs at least one bit of cumulative loss in the semimeasure convention, and by Theorem 3.11 the cumulative loss is

$L_A(D_n) - L_A(\emptyset) \le -\log \mathbb{U}_A(\mathrm{Ext}T) + \log M_A(\emptyset)$. This is the halving argument (the companion paper's Corollary 5.4).

(b) At an error or abstention, p_t does not prove $\varphi_t^{b_t}$, so it is incompatible with D_t and with every later $D_{t'}$, and it is never chosen again. (At a repeat p_t proves the stored label, since it is compatible with D_{t-1}.) Since the order refines length, each p_t is a shortest compatible program, of length $K_A(D_{t-1}) \le k$. Kraft's inequality for the prefix-free set $\mathrm{dom}\, V$ allows at most 2^k programs of length at most k. □

Part (c) is proved in Appendix A.9, after Theorem A.22.

Proof of Theorem 3.14. Step 1: the conditional law of the next label. Histories take countably many values. For a value h of the history and a value of $u_{<t}$, the event $\{h_{t-1} = h\}$ is the intersection of an event in $u_{<t}$ (the rules produce the questions of h from its labels) with the event $\{\mathrm{tri}_\omega^G \supseteq D_h\}$ (Proposition 3.1(g)), where D_h is the data set of h. Since $\nu_G(\emptyset) = 1$, the theory is consistent and tri_ω^G is defined on almost every tape. Since (u_t) is independent of ω, given $(h_{t-1}, u_{\le t})$ the tape has the law $\mathbb{U}(\cdot \mid \mathrm{tri}_\omega^G \supseteq D_{t-1})$, and $y_t = \mathrm{tri}_\omega^G(\varphi_t)$ has the law $\nu_G^3(\cdot \mid D_{t-1}, \varphi_t)$. (Almost surely $\nu_G^3(D_{t-1}) > 0$.)

Step 2: telescoping. Let $Z_t := \log \nu_G^3(y_t \mid D_{t-1}, \varphi_t) - \log M_A^3(y_t \mid D_{t-1}, \varphi_t)$ (0 at a repeat) and $S_n := \sum_{t \le n} Z_t$. Since $\nu_G^3(\emptyset) = 1$ and $M_A^3(\emptyset) = M_A(\emptyset)$, $S_n = \log \nu_G^3(D_n) - \log M_A^3(D_n) + \log M_A(\emptyset)$.

Step 3: integrability. By Theorem 3.6, $M_A^3 \ge 2^{-K(G)} \nu_G^3$, so $S_n \le K(G) + \log M_A(\emptyset)$ pointwise. Since $-\log M_A^3(D_n) \ge 0$, $S_n \ge \log \nu_G^3(D_n) + \log M_A(\emptyset)$. And $\mathbb{E}[-\log \nu_G^3(D_n)] = \sum_{t \le n} \mathbb{E}[H_t] \le n \log 3$, where H_t is the entropy of $\nu_G^3(\cdot \mid D_{t-1}, \varphi_t)$, a distribution on three points: the summands $-\log \nu_G^3(y_t \mid D_{t-1}, \varphi_t)$ are nonnegative, so the tower property applies by Step 1 (Tonelli). Hence S_n is integrable, and so is each $Z_t = S_t - S_{t-1}$.

Step 4. By Step 1, $\mathbb{E}[Z_t \mid h_{t-1}, u_{\le t}] = d_t$, which is nonnegative because the conditionals of M_A^3 are probability distributions (Proposition 3.1(f)). Taking expectations and summing gives $\sum_{t \le n} \mathbb{E} d_t = \mathbb{E} S_n$, which is the identity and the bound. Pinsker's inequality in bits, $\mathrm{KL} \ge \|P - Q\|_1^2 / (2 \ln 2)$, gives the last claim, and $\mathbb{E} \sum_t d_t < \infty$ gives $d_t \to 0$ almost surely. □

Proof of Corollary 3.15. The tapes with $B \cup A_\omega$ consistent and $\mathrm{Cn}(A_\omega) = \mathrm{Cn}(T)$ lie in $E_{D_n}^3$. Order independence holds for M_A^3 with the proof of Proposition 3.1(d), using (f) in place of (b); then apply Theorem 3.6(i). Here $[T]$ replaces Ext$T$: three-valued data penalise theories that decide more than T. □

Proof of Proposition 3.16. The inequality is Proposition 3.17(b). For the expectation, the weights $2^{-K(b|\mathrm{dom}\, D)}$ sum to at most 1 (Kraft's inequality), so Gibbs' inequality gives $\mathbb{E}_b K(b \mid \mathrm{dom}\, D) \ge n$ for uniform $b \in \{0,1\}^n$. With logically dependent sentences, inconsistent labellings have $L_A = \infty$, and the bound holds trivially for them. □

A.6 Coding bounds and the absence of a coding theorem

Lemma A.4 (Literals of a fresh predicate)**.** *In the literal setting: (L1) every set Λ of literals $R(t_w)^b$ with at most one literal per word is consistent with B; (L2) if $B \cup \Lambda$ is consistent, then $B \cup \Lambda \vdash R(t_y)^b$ only if $R(t_y)^b \in \Lambda$. The same holds for finitely many fresh predicates. Moreover a propositional combination of fresh atoms is consistent with B iff it is propositionally satisfiable, and B together with such combinations proves another such combination iff they imply it propositionally.*

Proof. (L1) Expand the model of B in which the t_w are pairwise distinct by interpreting R as the set of the elements t_w with $R(t_w) \in \Lambda$; B does not mention R, so it stays true. (L2) If $R(t_y)^{1-b} \in \Lambda$, a proof of $R(t_y)^b$ would make $B \cup \Lambda$ inconsistent. If neither literal of t_y is in Λ, then $\Lambda \cup \{R(t_y)^{1-b}\}$ has at most one literal per word and is consistent by (L1). For the last sentence: every valuation of finitely many fresh atoms is realised in an expansion of the model of B, so a propositional combination of fresh atoms is consistent with B iff it is propositionally satisfiable, and the implication statement follows by applying this to the premises together with the negated conclusion. □

Proof of Proposition 3.17. (a) Let $Q_S(\sigma)$ be the probability, over ω, that the following procedure outputs σ: at stage s, read $\omega_{1:s}$, run s steps of the enumeration of G_U on it, and test all derivations of size at most s from B and the axioms emitted so far, for every member of S in a fixed order; output the first member found. The output events are disjoint open sets, so $\sum_{\sigma \in S} Q_S(\sigma) \leq 1$, and $Q_S(\sigma)$ is lsc uniformly in (S, σ). If $B \cup A_\omega$ is consistent and proves σ, it proves no other member of S, so the procedure outputs σ; hence $\mathrm{Bel}_0(\sigma) \leq Q_S(\sigma)$. The conditional coding theorem (Section 2.1) gives $K(\sigma \mid S) \leq -\log Q_S(\sigma) + c$. (So the coding theorem is applied to Q_S, which is lsc, and not to M_A, which is not; Theorem 3.5.)

(b) Apply (a) with S the label conjunctions on $\operatorname{dom} D$, which are pairwise logically incompatible and computable from $\operatorname{dom} D$, and $\sigma := \psi_D$, so that $M_A(D) = \mathrm{Bel}_0(\psi_D)$ (Proposition 3.1(c)). From $\operatorname{dom} D$ and ψ_D one computes b_D, so $K(b_D \mid \operatorname{dom} D) \leq K(\psi_D \mid S) + c$.

(c) The first inequality is one term of W_A; the second is Theorem 3.6(iii). For the third, $K_A(D) \leq K(\psi_D) + c$ (Remark 2.8(c)) $\leq K(S) + K(\psi_D \mid S) + c'$, and (a) bounds $K(\psi_D \mid S)$ by $L_A(D) + c$. □

Proof of Theorem 3.18. (i) Let $\ell := \ell_d$, so that $2^\ell < d/\log 3$ and $3^{2^\ell} < 2^d$. There are fewer than 2^ℓ programs of length less than ℓ. Colour each $x \in \{0,1\}^d$ by the vector, over these programs p, of the following symbols: "$+$" if $B \cup \mathsf{W}_{V(p)}$ is consistent and proves $R(t_x)$; "$-$" if it is consistent and proves $\neg R(t_x)$; "$*$" otherwise. There are at most $3^{2^\ell - 1} < 2^d$ colours, so two distinct words x_1, x_2 share a colour. A program p with $|p| < \ell$ that is compatible with D would colour x_1 "$-$" and x_2 "$+$". So every compatible program has length at least ℓ, which gives both claims of (i); and $\ell \geq \log(d/\log 3) - 1 > \log d - 2$, since $\log\log 3 < 0.67$.

(ii) Let $j(x)$ be the position of x in the length-lexicographic order ($j(\epsilon) = 0$), so that $j(x) \leq 2^{d+1} - 2$ for $|x| \leq d$. Let $s := \lceil \log(d+1) \rceil$ and $L := 2^{2^s} \geq 2^{d+1}$. For $u \in \{0,1\}^L$ let $\Lambda_u := \{R(t_x)^{u_{j(x)+1}} : j(x) < L\}$; it is consistent by (L1) of Lemma A.4 and proves $R(t_x)^b$ iff $u_{j(x)+1} = b$, by (L2). A fixed prefix machine reads a V-program for s, computes L, reads L further bits u and outputs an index of Λ_u; by universality of V a fixed string r makes $r\,q\,u \in \operatorname{dom} V$ a program for that index, for q a shortest program for s. These 2^L programs are distinct, form a prefix-free family (since q fixes L), and have length $|r| + K(s) + L$. For a two-point D on distinct words of length at most d, exactly a quarter of the strings u give a compatible Λ_u. So $W_A(D) \geq 2^L \cdot \frac{1}{4} \cdot 2^{-|r| - K(s) - L}$.

(iii) The theory machine $G_{\mathrm{coin}} := \{(w, R(t_x)^{w_{j(x)+1}}) : |w| = j(x) + 1\}$ emits $\{R(t_x)^{\omega_{j(x)+1}} : x \in \{0,1\}^*\}$, which is consistent by (L1); its theory proves D iff ω agrees with the labels of D at two positions (L2), which has probability $\frac{1}{4}$. Apply Theorem 3.6.

The consequences: $K_A + \log W_A \geq \ell_d - 2 - K(s) - c$, and $\Omega_V^{\geq \ell}$ is the tail of a convergent series, with $\ell_d \to \infty$. □

The bound $K_{\equiv}(T) \leq K_A(D) + c$ of Theorem 3.11 is about single theories; for whole theories the question is how $K_{\equiv}$ compares with the mixture weight $\mathbf{m}_{\equiv}$. For an r.e. set $Y \subseteq$ Sent let $K_{\mathrm{enum}}(Y) := \min\{|p| : p \in \operatorname{dom} V,\ \mathsf{W}_{V(p)} = Y\}$ and $\mathbf{m}_{\mathrm{enum}}(Y) := \sum\{2^{-|p|} : p \in$

$\operatorname{dom} V,\ \mathsf{W}_{V(p)} = Y\}$, its enumeration complexity and the corresponding discrete semimeasure. A theorem of Solovay (Solovay, 1977), in the form stated by Vereshchagin (2007), says that $K_{\mathrm{enum}}(Y) \le 3I(Y) + O(\log I(Y))$ for every r.e. set Y, where $I(Y)$ is the negative logarithm of the probability that a machine with a random input tape enumerates exactly Y.

Proposition A.5 (Coding for whole theories)**.** *(a) For every r.e. T: $K_{\equiv}(T) = K_{\mathrm{enum}}(\mathrm{Cn}T) \pm c$ and $2^{-c}\mathbf{m}_{\mathrm{enum}}(\mathrm{Cn}T) \le \mathbf{m}_{\equiv}(T) \le 2^{c}\mathbf{m}_{\mathrm{enum}}(\mathrm{Cn}T)$. (b) For every r.e. T: $-\log \mathbf{m}_{\equiv}(T) \le K_{\equiv}(T) \le 3\,(-\log \mathbf{m}_{\equiv}(T)) + O\big(\log(-\log \mathbf{m}_{\equiv}(T))\big) + c$.*

Proof. (a) Let κ be computable and injective with $\mathsf{W}_{\kappa(e)} = \mathrm{Cn}(\mathsf{W}_e)$ (the s-m-n theorem with padding). Every e' with $\mathsf{W}_{e'} = \mathrm{Cn}T$ lies in $[T]$, so $K_{\equiv}(T) \le K_{\mathrm{enum}}(\mathrm{Cn}T) + c$ and $\mathbf{m}_{\equiv}(T) \ge 2^{-c}\mathbf{m}_{\mathrm{enum}}(\mathrm{Cn}T)$ (grouping programs by the index they print). Conversely κ maps $[T]$ into $\{e' : \mathsf{W}_{e'} = \mathrm{Cn}T\}$, with $K(\kappa(e)) \le K(e) + c$ and $\mathbf{m}(\kappa(e)) \ge 2^{-c}\mathbf{m}(e)$, since the image of $\mathbf{m}$ under κ is an lsc semimeasure. (b) The left inequality holds because $2^{-K_{\equiv}(T)}$ is one term of $\mathbf{m}_{\equiv}(T)$. For the right one, the probability that a machine with a random input tape enumerates exactly Y is at least $2^{-c}\mathbf{m}_{\mathrm{enum}}(Y)$, since a deterministic program is a randomised algorithm that ignores its tape; so Solovay's inequality also holds with $-\log \mathbf{m}_{\mathrm{enum}}(Y)$ in place of $I(Y)$. Apply it to $Y = \mathrm{Cn}T$ and use (a). □

Remark A.6 (Factor two). Vereshchagin (2007) replaces Solovay's constant 3 by 2 for finite sets, and Shekhovtsov and Zakharov (2023) show that a bound of this kind for finite sets transfers to all sets up to logarithmic terms. So the factor 3 in Proposition A.5(b) and in Proposition A.7(c) can be replaced by 2, up to logarithmic terms whose form we do not state.

Whether the random tape helps for whole theories is the question whether a probabilistic enumerator of a set can be much more likely than any short deterministic one. Let $\mathbb{U}_{\mathrm{enum}}(Y) := \mathbb{U}\{\omega : A_\omega = Y\}$, the probability that the universal theory machine emits exactly Y; for sets of strings, let $\mathbb{U}^{\mathrm{str}}_{\mathrm{enum}}$, $\mathbf{m}^{\mathrm{str}}_{\mathrm{enum}}$ and $K^{\mathrm{str}}_{\mathrm{enum}}$ be the same quantities for a universal probabilistic enumerator of sets of strings and for programs of V printing indices of r.e. sets of strings.

Proposition A.7 (The random tape and whole theories)**.** *(a) For every r.e. T with $B \cup T$ consistent: $2^{-c}\mathbb{U}_{\mathrm{enum}}(\mathrm{Cn}T) \le \mathbb{U}_A[T] \le 2^{c}\mathbb{U}_{\mathrm{enum}}(\mathrm{Cn}T)$, and $\mathbb{U}_A[T] \ge 2^{-c}\mathbf{m}_{\equiv}(T)$. (b) In the literal setting: $\mathbb{U}_A[T] \asymp \mathbf{m}_{\equiv}(T)$ for all consistent r.e. T iff $\mathbb{U}^{\mathrm{str}}_{\mathrm{enum}}(Y) \asymp \mathbf{m}^{\mathrm{str}}_{\mathrm{enum}}(Y)$ for all r.e. $Y \subseteq \{0,1\}^*$; and $K_{\equiv} \le -\log \mathbf{m}_{\equiv} + O(1)$ iff $K^{\mathrm{str}}_{\mathrm{enum}} \le -\log \mathbf{m}^{\mathrm{str}}_{\mathrm{enum}} + O(1)$. (c) $-\log \mathbb{U}_A[T] \le -\log \mathbf{m}_{\equiv}(T) + c \le K_{\equiv}(T) + c \le 3\,(-\log \mathbb{U}_A[T]) + O\big(\log(-\log \mathbb{U}_A[T])\big) + c$ for every r.e. T with $B \cup T$ consistent.*

Proof. (a) $\{\omega : A_\omega = \mathrm{Cn}T\} \subseteq \{\omega : \mathrm{Cn}(A_\omega) = \mathrm{Cn}T\}$, and both sets consist of consistent theories, so $\mathbb{U}_A[T] \ge \mathbb{U}_{\mathrm{enum}}(\mathrm{Cn}T)$. Conversely, the machine G_{cl} that runs G_U and emits $\mathrm{Cn}(A_\omega)$ emits exactly $\mathrm{Cn}T$ on $\{\omega : \mathrm{Cn}(A_\omega) = \mathrm{Cn}T\}$, and the tapes $p_{G_{\mathrm{cl}}}\omega'$, with $p_{G_{\mathrm{cl}}}$ a V-program for an index of G_{cl}, show $\mathbb{U}_{\mathrm{enum}}(\mathrm{Cn}T) \ge 2^{-|p_{G_{\mathrm{cl}}}|}\mathbb{U}_A[T]$. The last claim is Theorem 3.6(i).

(b) ($\Leftarrow$) By (a) and Proposition A.5(a), $\mathbb{U}_A[T] \asymp \mathbb{U}_{\mathrm{enum}}(\mathrm{Cn}T)$ and $\mathbf{m}_{\equiv}(T) \asymp \mathbf{m}_{\mathrm{enum}}(\mathrm{Cn}T)$, and coding sentences as strings changes the enumeration quantities by constant factors. ($\Rightarrow$) For $Y \subseteq \{0,1\}^*$ let $T_Y := \{R(t_x) : x \in Y\}$. By (L2) of Lemma A.4, $Y = \{x : R(t_x) \in \mathrm{Cn}(T_Y)\}$. A probabilistic enumerator of exactly Y yields one of exactly $\mathrm{Cn}(T_Y)$ (close under Cn) and conversely (extract the words x), with constant overhead, for probabilistic and for deterministic programs alike; and distinct Y give distinct $\mathrm{Cn}(T_Y)$. So $\mathbb{U}^{\mathrm{str}}_{\mathrm{enum}}(Y) \asymp \mathbb{U}_{\mathrm{enum}}(\mathrm{Cn}T_Y) \asymp \mathbb{U}_A[T_Y]$ and $\mathbf{m}^{\mathrm{str}}_{\mathrm{enum}}(Y) \asymp \mathbf{m}_{\equiv}(T_Y)$; likewise for the complexities.

(c) The first two inequalities are (a) and $\mathbf{m}_{\equiv}(T) \ge 2^{-K_{\equiv}(T)}$. For the last, Solovay's theorem bounds $K_{\mathrm{enum}}(Y)$ by $3I(Y) + O(\log I(Y))$, and $I(Y) = -\log \mathbb{U}_{\mathrm{enum}}(Y) \pm c$ for the universal probabilistic enumerator; apply it to $Y = \mathrm{Cn}T$, and use (a) and Proposition A.5(a). □

Question A.8. Does the random tape help for whole theories? That is, is $\mathbb{U}_A[T] \asymp \mathbf{m}_{\equiv}(T)$ for every consistent r.e. T, and is $K_{\equiv}(T) \leq -\log \mathbf{m}_{\equiv}(T) + O(1)$? By Proposition A.7(b) these are, in the literal setting, the questions whether randomness helps to enumerate a set of strings.

The question matters for identification: the limit loss on a complete presentation of T^* is $-\log \mathbb{U}_A[T^*] + \log M_A(\emptyset)$ (Proposition A.14(a)), and Question A.8 asks whether this is $-\log \mathbf{m}_{\equiv}(T^*)$ up to a constant.

A.7 The consistency-requiring variants

Throughout, $H_\omega := B \cup A_\omega \cup D^{\pm}$; when $M_A^{\mathrm{nc}}(D) > 0$, write $\Pr_D := \mathbb{U}(\cdot \mid H_\omega \text{ consistent})$, so that $\mathrm{Bel}_D^{\mathrm{nc}}(\varphi) = \Pr_D[H_\omega \vdash \varphi]$.

Proof of Proposition 3.19. (a) For consistent H, the set $H \cup \{\varphi^b\}$ is consistent iff $H \nvdash \varphi^{1-b}$; so both labels are possible iff H decides neither φ nor $\neg\varphi$, and exactly one is possible otherwise; for inconsistent H neither is. That $M_A^{\mathrm{nc}}(D) = \mathrm{Pl}_0(\psi_D)$ is Proposition 3.3. (b) If $A_\omega = \emptyset$, then $H_\omega = B \cup D^{\pm}$ is consistent; and these tapes include those beginning with $p_\emptyset$. For the coin-free and Kolmogorov versions, the program $p_\emptyset$ is a witness on every consistent D. (c) holds by definition. □

Proof of Proposition 3.20. (a) The tapes with $A_\omega = \emptyset$, of mass $\varepsilon_\emptyset$, lie in the event $\{H_\omega \text{ consistent}\}$ that normalises $\mathrm{Bel}_D^{\mathrm{nc}}$, and not in its numerator, since $B \cup D^{\pm} \nvdash \varphi^b$.

(b) Write $\zeta_{i,j}$ for the sentence $e_{i,j}$ of the statement, so that $b_{i,j} = a_i \vee \zeta_{i,j}$ (this avoids a clash with indices e).

Step 1: one pair, after any revealed set. Let D be consistent with $M_A^{\mathrm{nc}}(D) > 0$, let a, ζ be sentences and $b := a \vee \zeta$; write $\Pr := \Pr_D$ and $H := H_\omega$. Put

$$\kappa := \Pr[H \cup \{b\} \text{ consistent}], \qquad \theta := \Pr[H \cup \{b\} \text{ consistent},\ H \vdash b \to a],$$
$$r := \Pr[H \cup \{\zeta \wedge \neg a\} \text{ consistent}], \quad q := \Pr[H \vdash b,\ H \nvdash a].$$

The consistency-requiring probability of revealing $(a,1)$ and then $(b,1)$ after D is $P_{(a,b)} := \mathrm{Bel}_D^{\mathrm{nc}}(a)\mathrm{Bel}_{D+(a,1)}^{\mathrm{nc}}(b) = \Pr[H \vdash a]$, because $a \vdash b$, so that $H \cup \{a\}$ proves b whenever it is consistent. The probability of the other order is $P_{(b,a)} := \mathrm{Bel}_D^{\mathrm{nc}}(b)\mathrm{Bel}_{D+(b,1)}^{\mathrm{nc}}(a) = \Pr[H \vdash b] \cdot \theta/\kappa$: by the deduction theorem $H \cup \{b\} \vdash a$ iff $H \vdash b \to a$, and the ratio of the $\mathbb{U}$-measures of two events inside $\{H \text{ consistent}\}$ equals the ratio of their Pr-measures. Two identities hold. First, $\kappa - \theta = r$: $H \nvdash b \to a$ iff $H \cup \{b, \neg a\}$ is consistent, which implies that $H \cup \{b\}$ is consistent, and $\{b, \neg a\}$ is equivalent to $\{\zeta \wedge \neg a\}$. Second, $\Pr[H \vdash b] = \Pr[H \vdash a] + q$, since $H \vdash a$ implies $H \vdash b$. If $\Pr[H \vdash a] > 0$, then $\theta \geq \Pr[H \vdash a] > 0$ and $\Pr[H \vdash b] > 0$, and substituting,

$$\frac{P_{(a,b)}}{P_{(b,a)}} = \frac{\Pr[H \vdash a]\,(\theta + r)}{(\Pr[H \vdash a] + q)\,\theta} = 1 + \frac{z}{\Pr[H \vdash b]\,\theta}, \qquad z := \Pr[H \vdash a]\,r - q\,\theta \ \geq\ \Pr[H \vdash a]\,r - q.$$

Step 2: disjointness. For fixed i and D, the events $\{H \vdash b_{i,j},\ H \nvdash a_i\}$ are pairwise disjoint in j. Indeed $\zeta_{i,j}$ and $\zeta_{i,j'}$ $(j < j')$ are incompatible, since $\zeta_{i,j}$ asserts $s_{i,j}$ and $\zeta_{i,j'}$ denies it; so $H \vdash a_i \vee \zeta_{i,j}$ and $H \vdash a_i \vee \zeta_{i,j'}$ give $H \vdash a_i \vee (\zeta_{i,j} \wedge \zeta_{i,j'})$, hence $H \vdash a_i$. So $\sum_j q_{i,j} \leq 1$, where $q_{i,j}$ is q for the pair $(a_i, b_{i,j})$.

Step 3: lower bounds. Let D be $\emptyset$ (for (b1)) or $D_{i-1} := \{(a_{i'}, 1), (b_{i',j_{i'}}, 1) : i' < i\}$ (for (b2)); then $D^{\pm}$ is implied by $\{a_{i'} : i' < i\}$.

- $r_{i,j} \geq \varepsilon_\emptyset$. On the tapes with $A_\omega = \emptyset$ we have $H = B \cup D^\pm$, and $H \cup \{\zeta_{i,j} \wedge \neg a_i\}$ is consistent by (L1) of Lemma A.4: it is implied by the literal set $\{a_{i'} : i' < i\} \cup \{\neg a_i\} \cup \{\text{the literals of } \zeta_{i,j}\}$, which has one literal per word. These tapes have Pr-mass $\varepsilon_\emptyset / M_A^{\mathrm{nc}}(D) \geq \varepsilon_\emptyset$.
- $\Pr[H \vdash a_i] \geq 2^{-c_R}$ for (b2). On the tapes beginning with p_R, $H = B \cup \{\forall x\, R(x)\} \cup D^\pm$ is consistent (B does not mention R, so interpret R as the whole domain; $D^\pm$ consists of the $a_{i'}$ and disjunctions containing them) and proves a_i; their Pr-mass is $2^{-c_R} / M_A^{\mathrm{nc}}(D) \geq 2^{-c_R}$.
- For (b1), $D = \emptyset$, $\Pr = \mathbb{U}(\cdot \mid \mathrm{Cons})$ and $\Pr[H \vdash a_1] = \mathrm{Bel}_0(a_1)/(1 - \mathrm{Inc}_0) \geq \mathrm{Bel}_0(a_1) > 0$, by the one-axiom theory $\{a_1\}$.

Conclusion. (b1) By Steps 1 and 3, $z_j \geq \mathrm{Bel}_0(a_1)\varepsilon_\emptyset - q_{1,j}$, which is positive unless $q_{1,j} \geq \mathrm{Bel}_0(a_1)\varepsilon_\emptyset$; by Step 2 that happens for at most $1/(\mathrm{Bel}_0(a_1)\varepsilon_\emptyset)$ indices j. For the others $P_{(a,b)}/P_{(b,a)} > 1$. (b2) Choose $j_1, j_2, \ldots$ in turn. At step i, with $D = D_{i-1}$ and $J := \lceil 2^{c_R+1}/\varepsilon_\emptyset \rceil$, Step 2 gives some $j_i \leq J$ with $q_{i,j_i} \leq 1/J \leq 2^{-c_R}\varepsilon_\emptyset/2$, and then $z_i \geq 2^{-c_R}\varepsilon_\emptyset - 2^{-c_R}\varepsilon_\emptyset/2 = 2^{-c_R}\varepsilon_\emptyset/2$ by Step 3. Since $0 < \Pr[H \vdash b]\,\theta \leq 1$, $\log(P_{(a,b)}/P_{(b,a)}) \geq \log(1 + z_i) \geq \log(1 + 2^{-c_R}\varepsilon_\emptyset/2)$. The conditionals of the consistency-requiring predictor depend on the revealed data only as a set, and after $i-1$ completed pairs both orders have revealed D_{i-1}. So the loss of the order with b first minus that of the order with a first is $\sum_{i \leq n} \log(P^{(i)}_{(a,b)}/P^{(i)}_{(b,a)}) \geq n \log(1 + 2^{-c_R}\varepsilon_\emptyset/2)$. □

For a single pair the sign of z is not determined by universality; the counting over j in Step 2 is what makes (b) hold for M_A itself. The streams of (b2) are not computable (the j_i depend on M_A); they use at most J candidate atoms per block. The consistency-requiring posterior is order independent, since it depends on D as a set; its predictive probabilities do not chain, because $\mathrm{Bel}^{\mathrm{nc}}_D(\varphi) \neq M_A^{\mathrm{nc}}(D + (\varphi, 1))/M_A^{\mathrm{nc}}(D)$ in general.

An explicit law. For a single law the order effect is easy to see. Let a, χ be fresh atoms and $G := \{(1, a)\}$, whose theory is $\{a\}$ on the tapes beginning with 1 and $\emptyset$ otherwise, and let $D := \{(a, 1), (a \vee \chi, 1)\}$. In the order $(a, a \vee \chi)$ the consistency-requiring probability of D under G is $\frac{1}{2}$: only $\{a\}$ proves a, and then $a \vee \chi$ is entailed. In the order $(a \vee \chi, a)$ it is $\frac{1}{4}$: the first factor is $\frac{1}{2}$, and given $a \vee \chi$ both theories remain consistent and only $\{a\} \cup \{a \vee \chi\}$ proves a, so the second factor is $\frac{1}{2}$.

Proposition A.9 (Graded consistency-requiring weight)**.** *(a) For consistent A, $\nu_A^{1/2}$ is an order-independent weak semimeasure with $\nu_A^{1/2}(\emptyset) = 1$; hence $M_A^{1/2}$ is a weak semimeasure with $M_A^{1/2}(\emptyset) = 1 - \mathrm{Inc}_0$, and the conditionals of $M_A^{1/2}/M_A^{1/2}(\emptyset)$ form an order-independent predictor. (b) $M_A^{1/2} \geq M_A$. (c) For fresh atoms p, q and $D := \{(p, 1)\}$, $M_A^{1/2}(D + (p \vee q, 1)) \leq M_A^{1/2}(D) - \varepsilon_\emptyset/4$, although $B \cup D^\pm \vdash p \vee q$; so $M_A^{1/2}$ is not the law of any theory machine. (d) $M_A(\{(p \wedge q, 1)\}) = M_A(\{(p, 1), (q, 1)\})$, but $M_A^{1/2}(\{(p \wedge q, 1)\}) - M_A^{1/2}(\{(p, 1), (q, 1)\}) \geq \varepsilon_\emptyset/4$. (e) $M_A^{1/2}/M_A^{1/2}(\emptyset)$ is not a containment functional, not even of a random partial function on* Sent *that ignores logic: for every φ, its Möbius sum on the inputs $\varphi, \neg\varphi$ (the companion paper's Proposition A.2) equals $-\frac{1}{2}\mathbb{U}\{\omega \in \mathrm{Cons} : A_\omega$ decides neither φ nor $\neg\varphi\}$, which is at most $-\frac{1}{2}\varepsilon_\emptyset < 0$ when B does not decide φ; for M_A the same sum is $\mathbb{U}\{\omega \in \mathrm{Cons} : A_\omega$ decides neither$\} \geq 0$.*

Proof of Proposition A.9. (a) Let $B \cup A \cup D^\pm$ be consistent; otherwise every term is 0. If $B \cup A \vdash \varphi^{b^*}$, then $D + (\varphi, b^*)$ keeps the value (it is consistent with A and has factor 1), and $D + (\varphi, 1 - b^*)$ is inconsistent with A. If A decides neither φ nor $\neg\varphi$, each extension has factor at most $\frac{1}{2}$. Integrating over Cons preserves the inequality, and dividing by $M_A^{1/2}(\emptyset) = 1 - \mathrm{Inc}_0 > 0$ gives a

weak semimeasure with value 1 at $\emptyset$, to which the companion paper's Proposition 3.3 applies. (b) For a consistent A that proves ψ_D, $\nu_A^{1/2}(D) = 1$. (c) For each consistent A the new datum keeps consistency (it is entailed) and multiplies the value by 1 or $\frac{1}{2}$; for $A = \emptyset$ the value drops from $\frac{1}{2}$ to $\frac{1}{4}$. (d) Per consistent theory: if it proves $p \wedge q$, both values are 1; if it proves exactly one of p, q and is consistent with the other, both are $\frac{1}{2}$; if it proves neither and is consistent with $p \wedge q$, they are $\frac{1}{2}$ and $\frac{1}{4}$; otherwise both are 0. Integrate; the empty theory contributes at least $\varepsilon_\emptyset/4$ to the difference. (e) The Möbius sum on the inputs $\varphi, \neg\varphi$ at $\emptyset$ is $\nu(\emptyset) - \sum_b \nu(\{(\varphi, b)\}) - \sum_b \nu(\{(\neg\varphi, b)\}) + \sum_{b,b'} \nu(\{(\varphi, b), (\neg\varphi, b')\})$, which we compute per consistent theory A for $\nu = \nu_A^{1/2}$. If A decides φ, say $B \cup A \vdash \varphi^{b^*}$, the singletons consistent with A are (φ, b^*) and $(\neg\varphi, 1 - b^*)$, both proved (value 1), and the only pair consistent with A is $\{(\varphi, b^*), (\neg\varphi, 1 - b^*)\}$ (value 1); the sum is $1 - 1 - 1 + 1 = 0$. If A decides neither, all four singletons are consistent and unproved (value $\frac{1}{2}$ each); of the four pairs, $\{(\varphi, 1), (\neg\varphi, 0)\}$ and $\{(\varphi, 0), (\neg\varphi, 1)\}$ are consistent and unproved (value $\frac{1}{4}$ each), and the other two are inconsistent; the sum is $1 - 2 + \frac{1}{2} = -\frac{1}{2}$. Integrate over Cons; the empty-theory tapes decide nothing that B does not decide. For M_A the same computation gives 0 for a deciding theory and $1 - 0 - 0 + 0 = 1$ for an undeciding one, since an undeciding theory proves no singleton datum on these inputs; so its sum is the undecided mass. By the companion paper's Proposition A.2, a weak ν with $\nu(\emptyset) = 1$ is a containment functional iff all its Möbius sums are nonnegative. □

Proposition A.10 (The consistency-requiring predictor is a mixture)**.** *For every D with $M_A^{\mathrm{nc}}(D) > 0$ and every φ,*

$$M_A^{\mathrm{nc}}(D)\,\mathrm{Bel}_D^{\mathrm{nc}}(\varphi) = \mathbb{U}(E_D \cap O_\varphi) + \mathbb{U}\{\omega : H_\omega \text{ consistent},\ B \cup A_\omega \nvdash \psi_D,\ H_\omega \vdash \varphi\},$$

and $\mathbb{U}(E_D \cap O_\varphi) = M_A(D)\mathrm{Bel}_D(\varphi)$ when $M_A(D) > 0$. So the consistency-requiring predictor is the mixture, with weight $M_A(D)/M_A^{\mathrm{nc}}(D)$, of the proof-requiring predictor and a predictor that uses weak theories together with the data.

Proof. Split the event $\{H_\omega$ consistent, $H_\omega \vdash \varphi\}$ by whether $B \cup A_\omega \vdash \psi_D$. If it does, H_ω is equivalent to $B \cup A_\omega$: the consistency of H_ω is that of $B \cup A_\omega$, and $H_\omega \vdash \varphi$ iff $B \cup A_\omega \vdash \varphi$. So that part of the event is $E_D \cap O_\varphi$. □

The weak component contains the empty theory, of mass at least $\varepsilon_\emptyset$, which predicts from the data alone; it is the source of Proposition 3.20(a).

Proposition A.11 (How much the graded weight can exceed M_A)**.** *(a) For every consistent D: $0 \le \log\left(M_A^{1/2}(D)/M_A(D)\right) \le K(\mathrm{dom}\, D) + |D| + c$. (b) Let $m \ge 3$ and $k \ge 1$, let $a_{j,i}$ $(j \le k,\ i < m)$ be fresh atoms, and put $\chi_i^{(j)} := \neg a_{j,1} \wedge \cdots \wedge \neg a_{j,i-1} \wedge a_{j,i}$ for $1 \le i < m$ and $\chi_m^{(j)} := \neg a_{j,1} \wedge \cdots \wedge \neg a_{j,m-1}$. For $\vec{\imath} \in [m]^k$ let $D_{\vec{\imath}} := \{(\chi_{i_j}^{(j)}, 1) : j \le k\}$. Then some $\vec{\imath}$ has*

$$\log \frac{M_A^{1/2}(D_{\vec{\imath}})}{M_A(D_{\vec{\imath}})} \ \ge\ k \log \frac{m}{2} + \log \varepsilon_\emptyset, \qquad |D_{\vec{\imath}}| = k, \qquad K(\mathrm{dom}\, D_{\vec{\imath}}) \le k \log m + K(k, m) + c.$$

For $m = 3$ the gap is $k \log \frac{3}{2} - c$, linear in $|D|$; for $k = 1$ it is $\log m - 1 - c$ on a single datum. The lower bound of (b) is within $2|D| + K(k, m) + c$ of the upper bound of (a). The source is the per-datum grading of Proposition A.9(d): a datum whose content is a long conjunction costs one bit under the graded weight and its full content under M_A; m pairwise incompatible single data, each worth $\frac{1}{2}$ to the empty theory, have total graded mass $m/2 > 1$.

Proof. (a) The lower bound is Proposition A.9(b). For the upper bound, $M_A^{1/2} \le 1$, and $M_A(D) \ge 2^{-K(\psi_D)-c}$ by the one-axiom theory $\{\psi_D\}$, with $K(\psi_D) \le K(\operatorname{dom} D) + |D| + c$, since the labels are $|D|$ bits given $\operatorname{dom} D$. (b) Each $D_{\vec\imath}$ is consistent, since each $\chi_i^{(j)}$ is a consistent conjunction of literals and the blocks use disjoint atoms (L1), and B proves none of its data (L2 with $\Lambda = \emptyset$). The empty theory gives $\nu_\emptyset^{1/2}(D_{\vec\imath}) = 2^{-k}$, so $M_A^{1/2}(D_{\vec\imath}) \ge \varepsilon_\emptyset 2^{-k}$. Two different $\vec\imath$ differ in some block, where their sentences are incompatible; so the m^k conjunctions $\psi_{D_{\vec\imath}}$ are pairwise incompatible, $\sum_{\vec\imath} M_A(D_{\vec\imath}) \le M_A(\emptyset) \le 1$, and some $\vec\imath$ has $M_A(D_{\vec\imath}) \le m^{-k}$. Its gap is at least $\log(\varepsilon_\emptyset 2^{-k} m^k)$. Finally $\operatorname{dom} D_{\vec\imath}$ is computable from (k, m) and the position of $\vec\imath$ among the m^k values. □

A.8 Limits and identification

The following lemma is the averaging argument associated with de Leeuw, Moore, Shannon and Shapiro (cf. de Leeuw et al., 1956), in the form we need. It is used again in Section 5.

Lemma A.12 (Majority vote)**.** *Let Z be a decidable set of sentences, M an oracle machine that on tape ω and input $\sigma \in Z$ outputs $\mathsf{M}^\omega(\sigma) \in \{0,1\}$ or diverges, $S \subseteq Z$, $\kappa : S \to \{0,1\}$, and $\mathcal{G}$ a Borel set of tapes with $\mathbb{U}(\mathcal{G}) > 0$ such that for every $\omega \in \mathcal{G}$, $\mathsf{M}^\omega(\sigma)$ is defined for all $\sigma \in Z$ and equals $\kappa(\sigma)$ for $\sigma \in S$. Then some total computable $h : Z \to \{0,1\}$ extends κ.*

Proof. By outer regularity there is an open $U \supseteq \mathcal{G}$ with $\mathbb{U}(U) < \frac{4}{3}\mathbb{U}(\mathcal{G})$; writing U as a disjoint union of cylinders, some cylinder $[u]$ has $\mathbb{U}(\mathcal{G} \cap [u]) > \frac{3}{4}\, 2^{-|u|}$. For $b \in \{0,1\}$ let $Q_b(\sigma) := \{\omega \in [u] : \mathsf{M}^\omega(\sigma) = b\}$. These classes are effectively open uniformly in σ (a halting computation reads a finite prefix) and disjoint for $b = 0, 1$, so their measures are left-c.e., with computable approximations from below. Let $h(\sigma)$ be the b with the larger approximated measure at the first stage at which the two approximations sum to more than $\frac{1}{2}\, 2^{-|u|}$. That stage exists for every $\sigma \in Z$, because $\mathbb{U}(Q_0(\sigma) \cup Q_1(\sigma)) \ge \mathbb{U}(\mathcal{G} \cap [u]) > \frac{3}{4}\, 2^{-|u|}$. For $\sigma \in S$ the wrong class $Q_{1-\kappa(\sigma)}(\sigma)$ lies in $[u] \setminus \mathcal{G}$, of measure less than $\frac{1}{4}\, 2^{-|u|}$; so at that stage the approximation of the right class exceeds $\frac{1}{4}\, 2^{-|u|}$, more than that of the wrong one. □

Throughout the rest of this subsection D_∞ is a consistent infinite stream, $E_n := E_{D_n}$ (a decreasing sequence), $E_\infty := \bigcap_n E_n = \{\omega : B \cup A_\omega \text{ consistent, } \operatorname{Cn}(A_\omega) \supseteq D_\infty^{\pm}\}$, $a_n := \mathbb{U}(E_n)$, $a := \mathbb{U}(E_\infty)$ and $P_n := \mathbb{U}(\cdot \mid E_n)$.

Proposition A.13 (The limit posterior)**.** *(a) If $a > 0$, then $\sup_{\mathcal{X}} |P_n(\mathcal{X}) - P_\infty(\mathcal{X})| \le (a_n - a)/a_n \to 0$, the supremum over Borel sets of tapes, with $P_\infty := \mathbb{U}(\cdot \mid E_\infty)$. Hence $\sup_\chi |\mathrm{Bel}_{D_n}(\chi) - \mathrm{Bel}_\infty(\chi)| \to 0$ with $\mathrm{Bel}_\infty(\chi) := P_\infty(O_\chi)$, likewise for* Dis *and* Indep*, and the cumulative loss in the semimeasure convention converges to $-\log(a/M_A(\emptyset))$. (b) If $a = 0$, then $M_A(D_n) \to 0$ and the cumulative loss tends to ∞. (c) $a \ge \mathbb{U}_A(\mathrm{Ext}T) \ge 2^{-K_{\equiv}(T)-c}$ whenever some r.e. T with $B \cup T$ consistent proves every datum of D_∞. (d) For every set Γ of sentences with $\operatorname{Cn}(\Gamma) = \operatorname{Cn}(D_\infty^{\pm})$, complete or not, $E_\infty = \{\omega : B \cup A_\omega$ consistent, $\operatorname{Cn}(A_\omega) \supseteq \operatorname{Cn}(\Gamma)\}$; so, if $a > 0$, P_∞ is the prior restricted to the tapes whose theory is a consistent extension of Γ.*

Proof. (a) For a Borel $\mathcal{X}$ let $x := \mathbb{U}(\mathcal{X} \cap E_\infty) \le a$ and $y_n := \mathbb{U}(\mathcal{X} \cap (E_n \setminus E_\infty)) \le a_n - a$. Then

$$\Big|\frac{x + y_n}{a_n} - \frac{x}{a}\Big| = \frac{|y_n a - x(a_n - a)|}{a\, a_n} \le \frac{\max(y_n a,\, x(a_n - a))}{a\, a_n} \le \frac{a_n - a}{a_n},$$

and $a_n \downarrow a$ by continuity from above. The loss is $L_A(D_n) - L_A(\emptyset) = -\log(a_n/M_A(\emptyset))$ by Proposition 3.1(d). (b) Continuity from above. (c) As in the proof of Theorem 3.11. (d) A theory has $D_\infty^{\pm}$ among its consequences iff it contains $\operatorname{Cn}(D_\infty^{\pm}) = \operatorname{Cn}(\Gamma)$. □

Proposition A.14 (Limits)**.** *Let T^* be consistent and r.e. (a) If $B \cup T^*$ is complete (hence decidable) and $\mathrm{Cn}(D^{\pm}_{\infty}) = \mathrm{Cn}(T^*)$, then $P_n\{\omega : \mathrm{Cn}(A_\omega) = \mathrm{Cn}(T^*)\} \to 1$, $\sup_\chi |\mathrm{Bel}_{D_n}(\chi) - \mathbf{1}[T^* \vdash \chi]| \to 0$, $\sup_\chi \mathrm{Indep}_{D_n}(\chi) \to 0$, and the cumulative loss converges to $-\log(\mathbb{U}_A[T^*]/M_A(\emptyset)) \leq K_{\equiv}(T^*) + c$. If T^* is axiomatised by one sentence τ and $(\tau, 1) \in D_n$, the first probability is 1 from then on. (b) If $B \cup T^*$ is incomplete and $\mathrm{Cn}(D^{\pm}_{\infty}) = \mathrm{Cn}(T^*)$, then P_∞ is the prior restricted to the consistent extensions of T^*, and for every ψ independent of $B \cup T^*$:*

$$\lim_n \mathrm{Bel}_{D_n}(\psi) = \frac{\mathbb{U}_A(\mathrm{Ext}(T^* + \psi))}{\mathbb{U}_A(\mathrm{Ext}T^*)} \in (0,1), \qquad \lim_n \mathrm{Dis}_{D_n}(\psi) = \frac{\mathbb{U}_A(\mathrm{Ext}(T^* + \neg\psi))}{\mathbb{U}_A(\mathrm{Ext}T^*)} \in (0,1),$$
$$\lim_n \mathrm{Indep}_{D_n}(\psi) \geq \frac{\mathbb{U}_A[T^*]}{\mathbb{U}_A(\mathrm{Ext}T^*)} > 0.$$

The most probable limit class can be a strict extension: in the literal setting, with fresh atoms α_i on a computable sequence of distinct words, $U := \{\alpha_i : i \in \mathbb{N}\}$ and $T_j := U \setminus \{\alpha_j\}$, for all but finitely many j the limit posterior on data presenting T_j gives the class of U more probability than the class of T_j; a most probable class exists, and it strictly extends T_j. (c) If every sentence appears with its T^*-trichotomy label, then $\mathbb{U}(\{\omega : \mathrm{Cn}(A_\omega) = \mathrm{Cn}(T^*)\} \mid E^3_{D_n}) \to 1$.

Proof of Proposition A.14. (a) A complete consistent theory has no proper consistent extension, so $E_\infty = \{\omega : \mathrm{Cn}(A_\omega) = \mathrm{Cn}(T^*)\}$ by Proposition A.13(d), and this event has positive measure by Theorem 3.6(i). On it the trichotomy is $\mathbf{1}[T^* \vdash \chi]$ for all χ at once. Apply Proposition A.13(a); the limit loss is $-\log(\mathbb{U}_A[T^*]/M_A(\emptyset)) \leq K_{\equiv}(T^*) + c$ by Theorem 3.6(i). If T^* is axiomatised by τ and $(\tau, 1) \in D_n$, then every consistent theory proving ψ_{D_n} contains $\mathrm{Cn}(T^*)$, so $E_n = E_\infty$.

(b) E_∞ is as stated by Proposition A.13(d), and $\lim_n \mathrm{Bel}_{D_n}(\psi) = P_\infty(O_\psi)$ by Proposition A.13(a), likewise for Dis and Indep. Since $T^* + \psi$, $T^* + \neg\psi$ and T^* are consistent r.e. theories, all three ratios are positive by Theorem 3.6(i); each of the first two is less than 1 because the other is positive and the two events are disjoint; and the tapes with $\mathrm{Cn}(A_\omega) = \mathrm{Cn}(T^*)$ lie in the independent part. For the example: $T_j \nvdash \alpha_j$ by (L2) of Lemma A.4, while $T_{j'} \vdash \alpha_j$ for $j' \neq j$; so the classes $[T_j]$ are pairwise distinct and their events disjoint, and $\sum_j \mathbb{U}_A[T_j] \leq 1$, so $\mathbb{U}_A[T_j] \to 0$. U is r.e. with an index of complexity $O(1)$ and is consistent by (L1), so $\mathbb{U}_A[U] \geq 2^{-c}$. The limit posterior odds of $[U]$ against $[T_j]$ are $\mathbb{U}_A[U]/\mathbb{U}_A[T_j]$, which exceed 1 for all but finitely many j. *A most probable class exists:* under P_∞ the classes partition E_∞, so only finitely many classes have limit probability above the positive number $\mathbb{U}_A[U]/\mathbb{U}_A(\mathrm{Ext}T_j)$, which is the limit probability of $[U]$; a maximiser exists among them, lies in $\mathrm{Ext}(T_j)$, and is not $[T_j]$, which $[U]$ beats; so it is a strict extension of T_j.

(c) A consistent theory that proves every theorem of T^* and decides no sentence independent of $B \cup T^*$ has the same consequences as T^*: a theorem $\sigma \notin \mathrm{Cn}(T^*)$ of it would be refutable from T^* (then the theory is inconsistent) or independent of T^* (then σ is labelled ι and decided). So the three-valued analogue of E_∞ is $\{\omega : \mathrm{Cn}(A_\omega) = \mathrm{Cn}(T^*)\}$, of positive measure by Theorem 3.6(i), and the proof of Proposition A.13(a) applies to M^3_A unchanged. □

Since $\sum_j \mathbb{U}_A[T_j] \leq 1$, the same disjointness argument bounds $\mathbb{U}_A[T]$ for all but finitely many members of any infinite family of pairwise inequivalent theories with a common simple consistent extension. Two-valued data cannot do better than Proposition A.14(b), whatever the learner. Call a sequence (σ_t) of sentences a *text* for T if $\mathrm{Cn}(\{\sigma_t : t \geq 1\}) = \mathrm{Cn}(T)$; it presents T by the two-valued data $(\sigma_t, 1)$, a datum $(\varphi, 0)$ being the datum $(\neg\varphi, 1)$. A function F from finite sequences of sentences to deductive classes *identifies* $[T]$ on a text if its values on the prefixes of the text are eventually $[T]$.

Proposition A.15 (Text against informant)**.** *In the literal setting, let α_i be fresh atoms on a computable sequence of distinct words, $\Lambda_k := \{\alpha_i : i < k\}$ and $\Lambda_\infty := \{\alpha_i : i \in \mathbb{N}\}$. No function F from finite sequences of sentences to deductive classes, computable or not, identifies on every text every member of $\{[\Lambda_k] : k \geq 1\} \cup \{[\Lambda_\infty]\}$. In particular no learner identifies all consistent r.e. theories from two-valued data, while three-valued data identify every consistent r.e. T^* in the limit posterior (Proposition A.14(c)), non-computably.*

Proof. Build a text for Λ_∞ in stages. Stage $k \geq 1$ appends α_{k-1} repeatedly until F outputs $[\Lambda_k]$. This happens, because the sequence built so far, continued by α_{k-1} forever, is a text for Λ_k, on which F eventually outputs $[\Lambda_k]$. The resulting infinite sequence contains every α_i, so it is a text for Λ_∞; and F outputs $[\Lambda_k] \neq [\Lambda_\infty]$ for infinitely many k, since $\Lambda_k \nvdash \alpha_k$ by (L2) of Lemma A.4. □

This is Gold's distinction between learning from text and from an informant (cf. Gold, 1967): a class that contains all finite languages and one infinite language is not identifiable from text, by the same locking argument. So the failure of two-valued SAI to identify incomplete truths is not an artefact of its prior.

Proposition A.16 (Computable structures; the basic ω-gap)**.** *Let $\mathfrak{M}$ be a computable structure in which every element is the value of a closed term, let $\mathrm{Diag}(\mathfrak{M})$ be its atomic diagram (the closed atomic sentences and negations of closed atomic sentences true in $\mathfrak{M}$; a decidable set), and let the data list $\mathrm{Diag}(\mathfrak{M})$ in a computable order, labelled true. (a) If B is true in $\mathfrak{M}$, then $L_A(D_n) \leq K_{\equiv}(\mathrm{Diag}\,\mathfrak{M}) + c \leq K(\mathfrak{M}) + c$, with $K(\mathfrak{M})$ the complexity of an index of $\mathfrak{M}$. (b) P_∞ is the prior restricted to the consistent extensions of $\mathrm{Diag}(\mathfrak{M})$. For $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$, $\mathfrak{M} = (\mathbb{N}, 0, S, +, \cdot, <)$, $B = \emptyset$ (or any B true both in $\mathbb{N}$ and in the model $\mathbb{N} \sqcup \{\star\}$ of the proof) and $\vartheta := \forall x\,(x + 0 = x)$, the limits $\lim_n \mathrm{Bel}_{D_n}(\vartheta)$ and $\lim_n \mathrm{Dis}_{D_n}(\vartheta)$ both lie in $(0, 1)$: instances never settle the universal sentence. (If instead $\mathsf{Q} \subseteq \mathrm{Cn}(B)$, then $B \vdash \vartheta$ and $\lim_n \mathrm{Bel}_{D_n}(\vartheta) = 1$.) (c) If $\mathrm{Th}(\mathfrak{M})$ is decidable and the data present it, Proposition A.14(a) applies.*

Proof. (a) Theorem 3.11 with $T = \mathrm{Diag}(\mathfrak{M})$, which is consistent with B, proves every datum, and has an index computable from one of $\mathfrak{M}$. (b) Proposition A.14(b) with $\psi = \vartheta$; it needs $\mathrm{Diag}(\mathbb{N}) \cup \{\vartheta\}$ and $\mathrm{Diag}(\mathbb{N}) \cup \{\neg\vartheta\}$ to be consistent with B. The first holds in $\mathbb{N}$. For the second take $\mathbb{N} \sqcup \{\star\}$ with $S(\star) := \star$, $\star + y := 0$ for every y (including $y = \star$), $y + \star := \star$ for $y \in \mathbb{N}$, $\star \cdot y = y \cdot \star := \star$ for every y, and $\star$ incomparable under $<$ with everything. Closed terms take values in $\mathbb{N}$, which is closed under the operations, so the diagram holds; and $\star + 0 = 0 \neq \star$. □

This is the one-tape form of the coin-free ω-gap of Hänni and Claude (Anthropic) (2026a, Theorem 3.9 and Proposition B.8); Section 5 studies its anatomy.

Proposition A.17 (Complete truths that are not decidable)**.** *(a) If $\mathrm{Cn}(D^{\pm}_\infty)$ is complete and not decidable (for example $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$, $B \subseteq \mathrm{Th}(\mathbb{N})$ decidable, and every sentence labelled by its truth in $\mathbb{N}$), then $a = 0$: $M_A(D_n) \to 0$, and the cumulative loss in the semimeasure convention is unbounded. (b) If (σ_t) is a computable enumeration of all sentences and D_n labels $\sigma_1, \ldots, \sigma_n$ by a complete consistent theory, with label vector $b_{1:n}$, then $K(b_{1:n} \mid n) - c \leq L_A(D_n) \leq n + K(n) + c$.*

Proof. (a) On E_∞ the theory is consistent and contains the complete set $\mathrm{Cn}(D^{\pm}_\infty)$, so it has exactly these consequences. Let $\mathsf{M}^\omega(\sigma)$ be 1 or 0 according as σ or $\neg\sigma$ is found first to be provable from $B \cup A_\omega$. On E_∞ it is total and decides $\mathrm{Cn}(D^{\pm}_\infty)$. If $a > 0$, Lemma A.12, with $Z = S := \mathrm{Sent}$, $\mathcal{G} = E_\infty$ and κ the characteristic function of $\mathrm{Cn}(D^{\pm}_\infty)$, gives a decision procedure for it, a contradiction. ($\mathrm{Th}(\mathbb{N})$ is not decidable, for instance by the inseparability step in the proof

of Proposition A.18.) (b) The lower bound is Proposition 3.17(b), since $\operatorname{dom} D_n$ is computable from n. For the upper bound, $K_A(D_n) \le K(\psi_{D_n}) + c \le K(n) + n + c'$, since given n the labels are n bits; then use $L_A \le K_A + c$ (Theorem 3.11). □

The coin-free analogues are Proposition 7.11 and Lemma E.11 of Hänni and Claude (Anthropic) (2026a); Section 5 gives the growth rate on Th(ℕ).

For the next proposition, F is a function machine of the companion paper (its Definition 3.6) on the inputs $\mathrm{Sent}_{\mathcal{L}_{\mathrm{ar}}}$ with labels $\{0,1\}$, and f^F_ω its random partial labeller. We use two standard facts about Q (see Boolos et al., 2007): it strongly represents every computable set, and it satisfies the diagonal lemma.

Proposition A.18 (Random theories above Q are almost surely incomplete)**.** *Let $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$. (a) For every theory machine G, the set of tapes ω such that $B \cup A^G_\omega$ is consistent, proves every axiom of Q, and decides every $\mathcal{L}_{\mathrm{ar}}$-sentence has measure 0. In particular, if $\mathsf{Q} \subseteq \mathrm{Cn}(B)$, the consistent random theory of M_A is almost surely incomplete, and every posterior gives probability 0 to complete theories. (b) For every function machine F as above: $\mathbb{U}\{\omega : f^F_\omega$ is total, labels every Q-provable sentence 1 and every Q-refutable sentence $0\} = 0$; likewise with PA, or any consistent r.e. extension of Q, in place of Q.*

Proof. Inseparability. No total computable $h : \mathrm{Sent}_{\mathcal{L}_{\mathrm{ar}}} \to \{0,1\}$ is 1 on the theorems of Q and 0 on its refutable sentences. Otherwise the set $\{\ulcorner\sigma\urcorner : h(\sigma) = 1\}$ is strongly represented in Q by some formula $\rho_h(x)$, and the diagonal lemma gives σ with $\mathsf{Q} \vdash \sigma \leftrightarrow \neg\rho_h(\ulcorner\sigma\urcorner)$. If $h(\sigma) = 1$, then $\mathsf{Q} \vdash \rho_h(\ulcorner\sigma\urcorner)$, so $\mathsf{Q} \vdash \neg\sigma$, and h is 1 on a refutable sentence. If $h(\sigma) = 0$, then $\mathsf{Q} \vdash \neg\rho_h(\ulcorner\sigma\urcorner)$, so $\mathsf{Q} \vdash \sigma$, and h is 0 on a theorem. A separator for a consistent r.e. $T \supseteq \mathsf{Q}$ separates the pair of Q too.

(a) On the event, $\mathsf{M}^\omega(\sigma)$, the first of $1, 0$ found according as σ or $\neg\sigma$ is provable from $B \cup A^G_\omega$, is total on $\mathcal{L}_{\mathrm{ar}}$-sentences and separates the theorems of Q from its refutable sentences: a consistent theory that proves Q proves every theorem of Q and refutes every refutable sentence. If the event had positive measure, Lemma A.12 (with $Z := \mathrm{Sent}_{\mathcal{L}_{\mathrm{ar}}}$ and S the sentences Q decides) would give a computable separator. (b) The same, with $\mathsf{M}^\omega(\sigma) := f^F_\omega(\sigma)$, computed by running F on σ with tape ω. □

The proposition is a new application of the fact that separating sets of effectively inseparable pairs have measure zero (for effective inseparability, cf. Smullyan, 1961); Lemma A.12 makes the proof self-contained. Part (b) says that a random tape does not help a labeller to get every Q-decided sentence right.

Corollary A.19 (Local defects above Q)**.** *Let $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$ and let G be a theory machine with $\nu_G(\emptyset) > 0$ whose random theory proves Q almost surely on Cons^G (for instance, any G when $\mathsf{Q} \subseteq \mathrm{Cn}(B)$). Then ν_G has positive local defect at $D = \emptyset$ at some $\mathcal{L}_{\mathrm{ar}}$-sentence φ: $\nu_G(\{(\varphi,1)\}) + \nu_G(\{(\varphi,0)\}) < \nu_G(\emptyset)$. In particular, above Q there is no theory machine whose law is "a theory plus a fair coin on every sentence it leaves undecided" with zero local defect everywhere.*

Proof. By Proposition 3.1(b) at $D = \emptyset$, the defect at φ is $\nu^3_G(\{(\varphi,\iota)\})$, the measure of the set of $\omega \in \mathrm{Cons}^G$ such that $B \cup A^G_\omega$ decides neither φ nor $\neg\varphi$. If it were 0 for every $\mathcal{L}_{\mathrm{ar}}$-sentence, then, by a countable union, almost every $\omega \in \mathrm{Cons}^G$ would have a theory that decides every $\mathcal{L}_{\mathrm{ar}}$-sentence; with the hypothesis, the event of Proposition A.18(a) would have measure $\mathbb{U}(\mathrm{Cons}^G) = \nu_G(\emptyset) > 0$, a contradiction. □

A.9 Order dependence and the Kolmogorov predictor

In the normalised convention (Definition 2.5) each conditional of M_A is divided by $\mathrm{Bel}_D(\varphi) + \mathrm{Dis}_D(\varphi)$, so the loss of a reveal sequence is no longer a function of the final data set.

Theorem A.20 (Order dependence in the normalised convention)**.** *(a) Let a, χ be sentences such that $B \cup \{a\}$ is consistent, and $B \cup \{\neg a\}$ is consistent and decides neither χ nor $\neg\chi$. Let $D := \{(a,1), (a \vee \chi, 1)\}$, and for $x, y \in Y_3$ let $p_{xy} := \mathbb{U}\{\omega \in \mathrm{Cons} : \mathrm{tri}_\omega(a) = x,\ \mathrm{tri}_\omega(a \vee \chi) = y\}$. Then $p_{10} = p_{1\iota} = p_{\iota 0} = 0$, and the normalised probabilities q_1 and q_2 of D in the orders $(a, a \vee \chi)$ and $(a \vee \chi, a)$ satisfy*

$$\frac{q_2}{q_1} = 1 + \frac{(p_{11} + p_{01})\, p_{0\iota} + p_{\iota 1}\, (p_{00} + p_{0\iota})}{(p_{11} + p_{01} + p_{\iota 1} + p_{00})\, (p_{11} + p_{01})} \ > \ 1.$$

So the normalised cumulative loss of M_A on D is strictly smaller in the order $(a \vee \chi, a)$, while the loss in the semimeasure convention is the same in both orders. (b) Let a_i, a_i' $(i \geq 1)$ be fresh atoms on pairwise distinct words. For $y \in \{0,1\}^n$ put $\alpha_i := a_i^{y_i}$, $\beta_i := \alpha_i \vee a_i'$ and $D_n(y) := \{(\alpha_i, 1), (\beta_i, 1) : i \leq n\}$, and compare the orders $(\alpha_1, \beta_1, \alpha_2, \beta_2, \dots)$ and $(\beta_1, \alpha_1, \beta_2, \alpha_2, \dots)$. For every y the normalised loss of the first order exceeds that of the second by at least $\sum_{i \leq n} \log(1 + v_i(y)) \geq 0$, where $v_i(y)$ is the posterior probability, after the first $i-1$ pairs, that the random theory refutes α_i and does not decide a_i'. The average of this difference over $y \in \{0,1\}^n$ is at least $2^{-c_1 - 1} n$, where c_1 is the length of a V-program for the machine $G_{\mathrm{ca}} := \{(w, a_j^{w_j}) : |w| = j\}$ (fair coins on the atoms a_j). So for every n some data set of $2n$ data has two reveal orders whose normalised losses differ by at least $2^{-c_1 - 1} n$ bits.

Proof. (a) If the random theory proves a, it proves $a \vee \chi$; if it refutes $a \vee \chi$, it refutes a. This gives the three zero cells. In the order $(a, a \vee \chi)$ the first normalised factor is $\mathrm{Bel}_\emptyset(a)/(\mathrm{Bel}_\emptyset(a) + \mathrm{Dis}_\emptyset(a)) = p_{11}/(p_{11} + p_{01} + p_{00} + p_{0\iota})$, and the second is 1, since $a \vee \chi$ is then entailed (Proposition 3.1(c)). In the order $(a \vee \chi, a)$ the first factor is $(p_{11} + p_{01} + p_{\iota 1})/(p_{11} + p_{01} + p_{\iota 1} + p_{00})$; given $(a \vee \chi, 1)$ the posterior is the prior restricted to $\mathrm{tri}_\omega(a \vee \chi) = 1$, on which $\mathrm{Bel}(a) \propto p_{11}$ and $\mathrm{Dis}(a) \propto p_{01}$, so the second factor is $p_{11}/(p_{11} + p_{01})$. (The normalising constant $1 - \mathrm{Inc}_0$ cancels in each factor.) Dividing,

$$\frac{q_2}{q_1} = \frac{(p_{11} + p_{01} + p_{\iota 1})(p_{11} + p_{01} + p_{00} + p_{0\iota})}{(p_{11} + p_{01} + p_{\iota 1} + p_{00})(p_{11} + p_{01})},$$

and the numerator minus the denominator is $(p_{11} + p_{01})p_{0\iota} + p_{\iota 1}(p_{00} + p_{0\iota})$. For strictness: $p_{11} > 0$ (the tapes beginning with a program for $\{a\}$), and $p_{0\iota} > 0$ (the tapes beginning with a program for $\{\neg a\}$, which are consistent, refute a, and do not decide $a \vee \chi$, because $B \cup \{\neg a\} \vdash a \vee \chi$ iff $B \cup \{\neg a\} \vdash \chi$, and $B \cup \{\neg a\} \vdash \neg(a \vee \chi)$ iff $B \cup \{\neg a\} \vdash \neg\chi$).

(b) After the first $i-1$ pairs both orders have revealed the same set $D_{i-1}(y)$, and the normalised conditionals depend only on the current set and the queried sentence. So the difference is the sum over i of $\log(q_2^{(i)}/q_1^{(i)})$, computed as in (a) under the posterior $P_{i-1} := \mathbb{U}(\cdot \mid E_{D_{i-1}(y)})$ in place of the prior restricted to Cons; the zero cells of (a) hold for every consistent theory. Write $m_1 := p_{11} + p_{01}$, $m_2 := p_{\iota 1}$, $m_3 := p_{00}$ and $m_4 := p_{0\iota}$ for the cell masses under P_{i-1}; they sum to at most 1, and $m_1 > 0$ (the tapes of G_{ca} that agree with y up to i). The ratio is $1 + [m_1 m_4 + m_2(m_3 + m_4)]/[(m_1 + m_2 + m_3)m_1] \geq 1 + m_4/(m_1 + m_2 + m_3) \geq 1 + m_4 = 1 + v_i(y)$. Next, $\log(1+v) \geq v$ for $v \in [0,1]$. The tapes $p\,\omega'$, with p the program for G_{ca}, $\omega'_{<i} = y_{<i}$ and $\omega'_i \neq y_i$, lie in $E_{D_{i-1}(y)}$ (their theory proves α_j, hence β_j, for $j < i$), refute α_i, and do not decide a_i' (by (L2) of Lemma A.4);

they have measure 2^{-c_1-i}. So $v_i(y) \geq 2^{-c_1-i}/M_A(D_{i-1}(y))$. Over uniform y, Jensen's inequality gives $\mathbb{E}[1/M_A(D_{i-1})] \geq 1/\mathbb{E}[M_A(D_{i-1})]$. Moreover $M_A(D_{i-1}(y)) = \mathrm{Bel}_0(\bigwedge_{j<i} \alpha_j)$, and these conjunctions are pairwise incompatible for distinct $y_{<i}$, so $\sum_{y_{<i}} M_A(D_{i-1}(y)) \leq 1$ and $\mathbb{E}[M_A(D_{i-1})] \leq 2^{-(i-1)}$. Hence $\mathbb{E}\, v_i \geq 2^{-c_1-1}$; sum over $i \leq n$. □

The mechanism is the failure of the companion paper's two-point criterion (its Proposition A.7): whether the random theory decides $a \vee \chi$ depends on its answer to a, since a theory proving a always decides $a \vee \chi$ and one refuting a need not. The witnesses use logical structure: in one order the second datum is entailed. On logically independent data the question stays open.

Question A.21. Is the normalised loss of M_A order dependent on logically independent data, for instance on positive literals of a fresh predicate? This is the companion paper's Question A.8 in another guise: on such data $M_A \asymp M_F$ (Theorem 4.8), but constant factors do not control normalised conditionals.

The single shortest compatible theory can be a poor predictor where the mixture is good. Let $a_1, \ldots, a_m$ be fresh atoms, and for $w \in \{0,1\}^m$ let $\mathrm{desc}(w) := \bigwedge_{i \leq m} a_i^{w_i}$.

Theorem A.22 (The Kolmogorov predictor can be exponentially worse than Bayes)**.** *Fix any well-order $\prec$ of $\mathrm{dom}\, V$, and let the Kolmogorov predictor of Proposition 3.13(b) use the $\prec$-first compatible program. There are a data set D of $2^m - 1$ sentences, all of the form $\neg\mathrm{desc}(v)$ labelled true, and a reveal order of D, such that: (i) the Kolmogorov predictor errs or abstains at every one of the $2^m - 1$ steps; (ii) $K_A(D) \leq m + K(m) + c$, and the Bayes predictor of Proposition 3.13(a) makes at most $K_A(D) + c \leq m + K(m) + c'$ mistakes on the same sequence. If $\prec$ refines length, then $m \leq K_A(D) \leq m + K(m) + c$; so with $k := K_A(D)$ the number of errors or abstentions is at least $2^{k-K(m)-c} - 1 \geq 2^{k-2\log k - c'} - 1$, while the Bayes predictor makes at most $k + c$ mistakes. The upper bound of Proposition 3.13(b) is 2^k.*

Proof. For each w let p_w be the $\prec$-first $p \in \mathrm{dom}\, V$ with $B \cup \mathsf{W}_{V(p)}$ consistent and proving $\mathrm{desc}(w)$. It exists, since $\{\mathrm{desc}(w)\}$ is consistent (Lemma A.4) and $\prec$ is a well-order. A consistent theory proves at most one $\mathrm{desc}(w)$, so $w \mapsto p_w$ is injective. Let w^* be the w whose p_w is $\prec$-last among these 2^m programs. Starting from $D = \emptyset$, the adversary repeats the following as long as the $\prec$-first program p compatible with the current data does not prove $\mathrm{desc}(w^*)$.

- p is consistent and $B \vdash \bigvee_v \mathrm{desc}(v)$, so the set $\mathrm{Open}(p)$ of v with $B \cup \mathsf{W}_{V(p)} \cup \{\mathrm{desc}(v)\}$ consistent is nonempty. It contains some $v \neq w^*$: otherwise p proves $\neg\mathrm{desc}(v)$ for all $v \neq w^*$, hence $\mathrm{desc}(w^*)$.
- Present $(\neg\mathrm{desc}(v), 1)$ for such a v.

Then: (1) every presented sentence is true in the valuation w^*, so the data stay consistent and $\{\mathrm{desc}(w^*)\}$ proves them all. (2) p does not prove $\neg\mathrm{desc}(v)$, since v is open for p; so the predictor errs (if p proves $\mathrm{desc}(v)$) or abstains. (3) The sentence $\neg\mathrm{desc}(v)$ was not presented before, since p is compatible with the earlier data. (4) For $w \neq w^*$, p_w proves $\neg\mathrm{desc}(v)$ for every $v \neq w$, so it stays compatible until $\neg\mathrm{desc}(w)$ is presented. While some p_w with $w \neq w^*$ is compatible, the $\prec$-first compatible program is $\preceq p_w \prec p_{w^*}$, and, being consistent and $\prec$-before p_{w^*}, it does not prove $\mathrm{desc}(w^*)$; so the loop continues. By (3) the loop runs at most $2^m - 1$ times, and by (4) it does not stop before every $\neg\mathrm{desc}(w)$, $w \neq w^*$, has been presented. So it runs exactly $2^m - 1$ times, and by (2) every step is an error or an abstention.

(ii) $\bigwedge_{v \neq w^*} \neg\mathrm{desc}(v)$ is equivalent to $\mathrm{desc}(w^*)$, so $K_A(D) \leq K(\mathrm{desc}(w^*)) + c \leq m + K(m) + c'$. For the Bayes predictor, apply Proposition 3.13(a) with $T := \mathsf{W}_{V(p)}$ for a shortest program p compatible with D: $K_{\equiv}(T) \leq |p| + c = K_A(D) + c$, and T proves every datum.

If $\prec$ refines length, every program compatible with D proves $\mathrm{desc}(w^*)$, and the $\prec$-first of them is p_{w^*}, so $K_A(D) = |p_{w^*}|$. The 2^m distinct programs p_w all have length at most $|p_{w^*}|$, and Kraft's inequality gives $2^m \le 2^{|p_{w^*}|}$. Finally $K(m) \le 2\log m + c \le 2\log k + c$. □

The stream is not computable, since it depends on the programs p_w. The construction is logical: in SOI on strings the analogous decoys would be beaten by the constant labeller "true", which is not available here because $\{\neg\mathrm{desc}(v) : v \in \{0,1\}^m\}$ is inconsistent. As in the conjunction ladder of the companion paper (its Theorem 9.10), it is the selection rule that fails, not the prior.

Proof of Proposition 3.13(c). For each $w \neq 0^m$ fix a program $q_w \in \mathrm{dom}\, V$ for an index of $\{\mathrm{desc}(w)\}$. Let $\prec$ list the $2^m - 1$ programs q_w first, in any order, and then the rest of $\mathrm{dom}\, V$ by length; this is a well-order of type ω. A consistent theory proves at most one $\mathrm{desc}(w)$, so $p_w = q_w$ for $w \neq 0^m$ (notation of the proof of Theorem A.22), and p_{0^m} lies in the tail; hence $w^* = 0^m$. Theorem A.22, which holds for every well-order, gives $2^m - 1$ errors or abstentions on a data set D whose conjunction is equivalent to $\mathrm{desc}(0^m)$, so $K_A(D) \le K(\mathrm{desc}(0^m)) + c \le K(m) + c'$; and $K(m) \le 2\log m + c''$. □

A.10 The generative variant

We make the generative variant of Definition 2.10 precise. Take $B = \emptyset$, or include B among the axioms. Fix the likelihood of Definition 2.9 of Hänni and Claude (Anthropic) (2026a): a derivation grammar with positive rule probabilities, a full-support instantiation grammar, and admissible parameters. For a set A of axioms with an enumeration $(\chi_1, \chi_2, \dots)$, give χ_j the citation weight proportional to 2^{-j} (normalised; no axiom is cited if $A = \emptyset$), and let P_A be the resulting law on sentences. In the one-tape form, split the tape into $\omega^{(0)}, \omega^{(1)}, \dots$ by a computable interleaving; the theory is $A_\omega := A^{G_U}_{\omega^{(0)}}$, enumerated in the order of emission, and the i-th datum σ_i is sampled from P_{A_ω} with the coins $\omega^{(i)}$. The universal law of unlabelled data streams is $\Xi(\sigma_{1:n}) := \int \prod_{i \le n} P_{A_\omega}(\sigma_i)\, d\mathbb{U}(\omega)$, and the generative posterior ϱ_n after n data has density proportional to $\prod_{i \le n} P_{A_\omega}(\sigma_i)$ with respect to $\mathbb{U}$.

Consider the *support condition*: $\mathrm{supp}\, P_A = \mathrm{Cn}(A)$ for every set of axioms A. Lemma 2.10(c) of Hänni and Claude (Anthropic) (2026a) proves it for finite template theories; we use it here as a hypothesis.

Proposition A.23 (Generative updating, given the support condition)**.** *Assume the support condition. (a)* Reweighting. *On data $\sigma_1, \dots, \sigma_n$, a consistent A has positive likelihood iff $A \vdash \sigma_i$ for all i, which is proof-requiring compatibility with $D := \{(\sigma_i, 1) : i \le n\}$. So on consistent theories the generative posterior is the proof-requiring posterior reweighted by $\prod_i P_A(\sigma_i) \le 1$. Inconsistent theories have positive likelihood, since their support is every sentence. (b)* Question conditioning. *For a yes/no question $\mathcal{Q} = \{\varphi, \neg\varphi\}$ and a consistent A: $P_A(\mathcal{Q}) > 0$ iff A decides φ, and then $P_A(\varphi^b \mid \sigma \in \mathcal{Q}) = \mathbf{1}[A \vdash \varphi^b]$. Hence, on the consistent theories, the generative predictive probability that the next datum answers $\mathcal{Q}$ by φ^b, given that it answers $\mathcal{Q}$, is*

$$\frac{\int \varrho_n(d\omega)\, P_{A_\omega}(\mathcal{Q})\, \mathbf{1}[A_\omega \vdash \varphi^b]}{\int \varrho_n(d\omega)\, P_{A_\omega}(\mathcal{Q})},$$

the proof-requiring belief under a posterior reweighted by how likely each theory was to produce an answer to $\mathcal{Q}$; inconsistent theories contribute their own $P_A(\varphi^b \mid \sigma \in \mathcal{Q})$. The outcome "independent" gets probability 0.

Proof. (a) By the support condition, $P_A(\sigma_i) > 0$ iff $\sigma_i \in \mathrm{Cn}(A)$. (b) $\operatorname{supp} P_A = \mathrm{Cn}(A)$ contains at most one of $\varphi, \neg\varphi$, by consistency, and contains one of them iff A decides φ; then apply Bayes' rule. □

Inconsistent theories pay the size principle: by Lemma 2.21 of Hänni and Claude (Anthropic) (2026a), a theory with mass ε outside the support of the data law loses at least $\log \frac{1}{1-\varepsilon}$ bits per datum in expectation. The generative variant thus models which sentences get stated; the outcome "independent" disappears because the model presupposes that asked questions are answered.

Remark A.24 (Supports and identification). We expect the following two statements to hold, and give only proof sketches. (i) *Supports*: the support condition holds for every axiom set in which every axiom has positive citation weight. (ii) *Identification*: if $\sigma_1, \sigma_2, \ldots$ are independent draws from P_{T^*}, the law of the deterministic machine that emits a consistent r.e. T^*, then the generative posterior probability of $\{\omega : \mathrm{Cn}(A_\omega) = \mathrm{Cn}(T^*)\}$ tends to 1 almost surely; by contrast, the proof-requiring posterior on two-valued data presenting T^* reaches only the prior restricted to the consistent extensions of T^* (Proposition A.14(b)).

Proof sketch. We do not write out the support condition for r.e. axiom sets, nor the measurability of $\omega \mapsto P_{A_\omega}$ that (ii) needs. For (i), the argument of Lemma 2.10(c) of Hänni and Claude (Anthropic) (2026a) applies to such axiom sets: each derivation tree that cites finitely many axioms has positive probability. For (ii), $\operatorname{supp} P_{A_\omega} = \mathrm{Cn}(A_\omega)$ is almost surely a function of the infinite sample (the set of sentences that occur), so Doob's consistency theorem (Theorem 4.2 of Hänni and Claude (Anthropic), 2026a; Doob, 1949) gives posterior consistency for it at $\mathbb{U}$-almost every ω; the tapes whose theory has the law P_{T^*} form a set of positive measure, so the statement holds almost surely under P_{T^*}.

A.11 Priors that complete their random theory

We compare M_A with two processes that draw sentences or theories until the theory is complete. We define them explicitly and prove their properties for the processes as defined. Process 1 is Demski's construction (Demski, 2012) as it is usually described. Process 2 is, at the level of the published descriptions, the modification of Demski's prior due to Fallenstein, Demski and Garrabrant (Garrabrant, 2015), used in Garrabrant et al. (2016b), which samples machines on random bits and reads their outputs as sets of sentences; its sampling stage is that of G_U. We use nothing about either construction beyond the definitions below.

- *Process 1 (sentence draws).* Let μ be a probability on Sent with $\mu(\varphi) > 0$ for every φ. Draw sentences independently from μ, and keep a drawn sentence iff it is consistent with B and the sentences kept so far. Let S_t be the set kept after t draws, $T_\infty := \mathrm{Cn}(\bigcup_t S_t)$ and $P_\mu(\varphi) := \Pr[\varphi \in T_\infty]$.
- *Process 2 (theory draws).* At each step t draw an independent tape $\omega^{(t)}$ and the theory $A_{\omega^{(t)}}$; keep it iff it is consistent together with B and what has been kept. Let T'_∞ be the deductive closure of everything kept. (In a variant, the sentences of a drawn theory are kept one by one, as long as they are consistent with what has been kept.)

Proposition A.25 (Completing processes against M_A)**.** *(a) T_∞ is consistent and complete with probability* 1*. Hence P_μ is a probability on the Lindenbaum algebra of B ($P_\mu(\varphi) + P_\mu(\neg\varphi) = 1$), and $\Pr[\mathrm{Cn}(T_\infty) = \mathrm{Cn}(T)] = 0$ for every incomplete T. (b) T'_∞ is consistent and complete with probability* 1*, also in the variant; so Process 2 also gives a coherent probability over completions, and gives every incomplete theory class probability* 0*. (c) Under M_A, by contrast: for every φ*

undecided by B, the prior independent mass $\mathrm{Bel}_0(\top) - \mathrm{Bel}_0(\varphi) - \mathrm{Bel}_0(\neg\varphi)$ *is at least* $\varepsilon_\emptyset > 0$*; every r.e. T with $B \cup T$ consistent, complete or not, has* $\mathbb{U}_A[T] \geq 2^{-K_{\equiv}(T)-c} > 0$*; and if* $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$ *and* $\mathsf{Q} \subseteq \mathrm{Cn}(B)$*, the random theory is almost surely incomplete (Proposition A.18). (d) For any prior concentrated on complete theories, such as those of (a) and (b), proof-requiring and consistency-requiring compatibility coincide, and the trichotomy has no independent outcome (Proposition 3.21).*

Proof. (a) Fix φ. With probability 1 it is drawn at some step, since $\mu(\varphi) > 0$. At the first such step t, either $B \cup S_{t-1} \cup \{\varphi\}$ is consistent and φ is kept, or $B \cup S_{t-1} \vdash \neg\varphi$; either way T_∞ decides φ. There are countably many sentences. Every S_t is consistent with B, so their union is, by compactness. (b) Fix φ. At each step the drawn theory is $\{\varphi\}$ with probability at least $2^{-K(\varphi)-c} > 0$ (the tapes that begin with a program for the machine emitting φ), independently across steps; so with probability 1 some step draws $\{\varphi\}$. At the first such step, either $\{\varphi\}$ is consistent with what has been kept, and is kept, or what has been kept proves $\neg\varphi$. Either way T'_∞ decides φ. Take the countable union; consistency follows by compactness as in (a). In the variant, the same step decides φ. (c) The tapes with $A_\omega = \emptyset$ have mass $\varepsilon_\emptyset$ and decide only what B decides; the rest is Theorem 3.6(i) and Proposition A.18. (d) is Proposition 3.21. □

Running either process needs a consistency test at every step: already for $B = \emptyset$, a language with a binary relation symbol and one draw, it asks whether a sentence is satisfiable, which is undecidable by Church's theorem in Kalmár's form (Church, 1936; Kalmár, 1937); the processes are computable relative to $\emptyset'$, since consistency of a finite set is Π^0_1 (a single draw of Process 2 can output one sentence). A theory machine emits sentences unconditionally, and consistency enters M_A only once, as the event measured (Proposition 3.4). So Process 2 and M_A share the sampling stage, and with it the program-length credit for infinite schemas and dominance over the laws of machines; that stage is not new here. They differ in completion (iterated draws to a complete theory against a single draw, whose theory is incomplete with positive probability), in what a prediction is (a coherent probability over completions against a belief function of provability with an independent outcome), and in what is conditioned on (for complete theories the two compatibility notions coincide; for M_A they differ, and the proof-requiring one carries the learning guarantees of Section 3).

A.12 Domination of provability functions

Recall from Section 3.8 that a random consistent theory Θ has the provability function $\beta(\chi) = \Pr[\chi \in \Theta]$, and that Question 3.22 asks whether $\mathrm{Bel}_0 \geq 2^{-c_\beta}\beta$ whenever β is lsc. Here "consistent" means consistent with B.

Proposition A.26 (Partial domination)**.** *(a) For every lsc provability function β, every finite set S of pairwise B-incompatible sentences and every $\sigma \in S$:* $\mathrm{Bel}_0(\sigma) \geq 2^{-K(\beta)-K(S)-c}\beta(\sigma)$*. In particular* $M_A(D) \geq 2^{-K(\beta)-K(\mathrm{dom}\, D)-c}\beta(\psi_D)$*. (b) If Θ is the deductive closure of the output of a theory machine G whose output is almost surely consistent, then* $\mathrm{Bel}_0 \geq 2^{-K(G)}\beta$*. (c) In the literal setting, Question 3.22 restricted to random sets of literals of R with at most one literal per word (with a lower semicomputable containment functional on literal data, and domination required on literal data) is equivalent to the companion paper's Question 3.14 for labels $\{0,1\}$: whether its one-tape law M_F dominates every lower semicomputable law of a random partial function, up to a constant factor.*

Proof. (a) Θ contains at most one member of S, by consistency, so $\sum_{\sigma\in S}\beta(\sigma) \leq 1$, and β restricted to S is a semimeasure, lsc uniformly in S. The conditional coding theorem (Section 2.1)

gives $K(\sigma \mid S) \leq -\log \beta(\sigma) + K(\beta) + c$. If $\beta(\sigma) > 0$, then σ is consistent with B, and the one-axiom theory $\{\sigma\}$ gives $\mathrm{Bel}_0(\sigma) \geq 2^{-K(\sigma)-c} \geq 2^{-K(S)-K(\sigma|S)-c'}$. For D, take S to be the label conjunctions on $\operatorname{dom} D$. (b) Then $\beta(\chi) = \nu_G(\{(\chi, 1)\})$, and Theorem 3.6 gives $\mathrm{Bel}_0(\chi) = M_A(\{(\chi, 1)\}) \geq 2^{-K(G)}\nu_G(\{(\chi, 1)\})$. (c) A random set Λ of literals with at most one literal per word corresponds to the random partial function h_Λ with $h_\Lambda(x) = b$ iff $R(t_x)^b \in \Lambda$, and $\Pr[\Lambda \supseteq D^{\pm}] = \Pr[h_\Lambda \supseteq D^{\mathrm{str}}]$, with D^{str} the string data set of D; and $M_A(D) \asymp M_F(D^{\mathrm{str}})$ on literal data (Theorem 4.8). □

So Question 3.22 is at least as hard as the binary case of the companion paper's question, and for general sentences it is open. The obstacle is the same: a lower semicomputable law of a random object need not be realisable by a machine that reads one tape. Note that Bel_0 itself is not lower semicomputable (Theorem 3.5), while $\mathrm{Bel}_0 + \mathrm{Inc}_0$ is; and by Corollary A.19, a theory-machine law above Q always has a local defect somewhere, so the question is not trivial.

B Proofs for the relation to operator induction

This appendix proves the results of Section 4 and states those that the main text mentions in one sentence. Results of our earlier report (Hänni and Claude (Anthropic), 2026a) and of the companion paper (Hänni and Claude (Anthropic), 2026c) are cited by their printed numbers and re-proved only where the setting changes. Throughout, $c, c', \ldots$ are machine constants. For a computable map h on indices, universality of V gives a string r_h with $V(r_h s) = h(V(s))$ for all $s \in \operatorname{dom} V$; so $p \mapsto r_h p$ is injective, maps $\operatorname{dom} V$ into $\operatorname{dom} V$ and adds $|r_h|$ to the length. "Tape by tape" means: for every tape ω.

B.1 Decompositions and the four maps

Measurability. $\{\omega : f_\omega \supseteq D\}$ is open. "f_ω is inconsistent" is open: an inconsistency uses finitely many labels, each certified under a prefix of ω, and these prefixes are comparable. "f_ω is closed" is a countable intersection of Borel conditions. For theory machines $\{\omega : B \cup A_\omega \vdash \psi\}$ is open and consistency is closed (Lemma A.1). All events below are Borel.

Lemma B.1 (Mixture decompositions)**.** *For every finite $D \neq \emptyset$,*

$$M_F^{\mathrm{cons}}(D) = \sum_{p \in \operatorname{dom} V} 2^{-|p|} \nu^{\mathrm{cons}}_{F_{V(p)}}(D), \tag{2}$$

and the same holds for ν^{cl} (with M_F^{cl}) and for ν (with M_F). For $D = \emptyset$ add $1 - \Omega_V$ on the right for ν^{cons} and ν, and nothing for ν^{cl}. For every finite D, the empty one included,

$$M_A(D) = \sum_{p \in \operatorname{dom} V} 2^{-|p|} \nu_{G_{V(p)}}(D) + (1 - \Omega_V)\, \mathbf{1}[B \vdash \psi_D]. \tag{3}$$

Proof. $\operatorname{dom} V$ is prefix-free, so every tape has at most one prefix $p \in \operatorname{dom} V$. On a tape $p\omega'$, $f^{F_U}_{p\omega'} = f^{F_{V(p)}}_{\omega'}$ and $A^{G_U}_{p\omega'} = A^{G_{V(p)}}_{\omega'}$, with ω' uniform given p. On the tapes with no prefix in $\operatorname{dom} V$ (measure $1 - \Omega_V$), f_ω is nowhere defined, so it extends D only if $D = \emptyset$, and then it is consistent, B being consistent; it is never closed, because B proves every tautology and the empty labeller labels none. On these tapes $A_\omega = \emptyset$, so the theory is B, which is consistent and proves ψ_D iff $B \vdash \psi_D$. □

The weight of an index i is $\mathbf{m}(i) = \sum_{V(p)=i} 2^{-|p|}$, so (2) reads $M_F^{\mathrm{cons}} = \sum_i \mathbf{m}(i)\nu_{F_i}^{\mathrm{cons}}$ on nonempty D.

Lemma B.2 (The four maps)**.** *(i)* Proof search. *For a theory machine G let F_G answer a query φ as follows: in stages $s = 1, 2, \ldots$ read the tape up to bit s, run s steps of the enumeration of G restricted to pairs (w, χ) with w a prefix of the bits read, and test all derivations of size at most s from B and the sentences emitted so far; halt with* 1 *at the first derivation of φ and with* 0 *at the first derivation of $\neg\varphi$ (within a stage φ is tested first). Then F_G is a function machine, and on every tape ω with $B \cup A_\omega^G$ consistent, $f_\omega^{F_G}$ is consistent and closed and decides every sentence exactly as $B \cup A_\omega^G$ does.*

(ii) Emission. *For a function machine F let $G_F := \{(w, \varphi) : (\varphi, w, 1) \in F\} \cup \{(w, \neg\varphi) : (\varphi, w, 0) \in F\}$. Then $A_\omega^{G_F} = \Gamma_{f_\omega^F}$ for every ω. So if f_ω^F is consistent and extends D, then G_F is compatible with D on ω; if f_ω^F is closed and consistent, $B \cup A_\omega^{G_F}$ decides every sentence exactly as f_ω^F does.*

(iii) Craig's map with the tape. *For a function machine F, read as a machine that reads the tape one-way, let C_F answer a query χ as follows. Find the at most two decompositions $\chi = \psi^{\wedge m}$ (the right-nested m-fold conjunction: $m = 1$, or ψ the left conjunct of χ and m fixed by $|\chi|$). For each, test both readings: (1) run F on ψ for exactly $m-1$ steps, reading the tape as F does, and accept if F halts at step $m-1$ with label* 1*; (2) if $\psi = \neg\varphi'$, run F on φ' for exactly $m-1$ steps and accept if it halts at step $m-1$ with label* 0*. Answer* 1 *if some test accepts and* 0 *otherwise. Then C_F is a random decider, and $\mathrm{Cn}(A_{C_F}^{\mathrm{dec}}(\omega)) = \mathrm{Cn}(\Gamma_{f_\omega^F})$ for every ω.*

(iv) Closure. *For a partial computable labeller f let $\bar{f}$ search for proofs of φ and of $\neg\varphi$ from $B \cup \Gamma_f$, dovetailing the enumeration of Γ_f. If f is consistent, $\bar{f}$ extends f, is consistent and closed, and $\mathrm{Cn}(\Gamma_{\bar{f}}) = \mathrm{Cn}(\Gamma_f)$.*

Proof. (i) The run on a query is a deterministic function of the bits read, and a halting run that has read w runs identically on every tape extending w; so the triples (φ, w, y) of halting runs form a consistent c.e. set. Let $B \cup A_\omega^G$ be consistent. If it proves φ^b, a derivation uses finitely many axioms, each emitted under a prefix of ω; at a late enough stage all of them have been enumerated and the derivation has size at most s, so the run halts, and it cannot first find a derivation of φ^{1-b}, since none exists. If the theory decides neither, no derivation is found and the label is undefined. So $f_\omega^{F_G}(\varphi) = b$ iff $B \cup A_\omega^G \vdash \varphi^b$. Then $\Gamma_f \subseteq \mathrm{Cn}(A_\omega^G)$ is consistent with B, and f is closed: $B \cup \Gamma_f \vdash \varphi^b$ implies $B \cup A_\omega^G \vdash \varphi^b$, which implies $f(\varphi) = b$; the converse holds as $\varphi^b \in \Gamma_f$. (ii) G_F is c.e. and emits φ^y exactly under the prefixes that certify $f(\varphi) = y$, so $A_\omega^{G_F} = \Gamma_{f_\omega^F}$. If f is closed, $B \cup \Gamma_f \vdash \varphi^b$ iff $f(\varphi) = b$. (iii) A decomposition $\chi = \psi^{\wedge m}$ with $m \geq 2$ has ψ the left conjunct and m fixed by $|\chi|$, since $|\psi^{\wedge m}| = m|\psi| + 3(m-1)$ in primitive syntax (Lemma E.12(b) of Hänni and Claude (Anthropic), 2026a); the only other candidate is $m = 1$. Every run is bounded by $m-1$ simulated steps, so C_F halts on every query and every tape: it is a random decider. Its answer depends only on the tape bits read by the simulated runs, so its triples are consistent. Its accepted set on ω is $\{\varphi^{\wedge(k+1)} : F \text{ halts on } \varphi \text{ with 1 in exactly } k \text{ steps}\} \cup \{(\neg\varphi)^{\wedge(k+1)} : F \text{ halts on } \varphi \text{ with 0 in exactly } k \text{ steps}\}$: reading (1) gives the first set (also when φ is itself a negation), reading (2) the second. Each member is equivalent to a member of $\Gamma_{f_\omega^F}$, and each member of $\Gamma_{f_\omega^F}$ has one, so Cn is preserved. (Reading (1) alone would lose the members $(\neg\varphi')^{\wedge(k+1)}$ that come from $f(\neg\varphi') = 1$, and Cn would change.) (iv) As (i), with the r.e. set Γ_f in place of A_ω^G; $f(\varphi) = b$ puts $\varphi^b \in \Gamma_f$, so $\bar{f}(\varphi) = b$. □

Only inclusions of events are claimed. The inclusion given by proof search can be strict: on a tape on which the theory is inconsistent, the labeller may already have committed to a consistent set of labels before the inconsistency appeared. A computation on a finite propositional analogue (code in the accompanying repository) finds many such tapes.

B.2 The equivalences

Proof of Theorem 4.2. We use coin-free forms of the maps of Lemma B.2: deterministic machines ignore the tape. *Enumerators to labellers* (proof search): $p \mapsto r_{\mathrm{ps}}p$; if the enumerator p is compatible with D, the proof-search labeller is consistent and closed, extends D, and decides each φ as the theory does. *Labellers to enumerators* (emission): $p \mapsto r_{\mathrm{em}}p$, the enumerator of $\Gamma_{\{V(p)\}}$; if the labeller is consistent and extends D, the theory is consistent and proves $D^{\pm}$; if it is closed, it decides as the labeller does. *Labellers to deciders* (Craig's map without tape): the decider of Craig's set of the labeller, which halts on every input even when the labeller is partial, since every run is bounded by the exponent of the candidate conjunction. *Deciders to enumerators*: enumerate the decided set; the theory is the same. *Consistent to closed*: the closure map. All maps are injective on programs and add a constant to the length. Summing $2^{-|p|}$ over compatible programs gives (a) and (c), where (c) uses proof search and emission restricted to closed labellers, which preserve the decision of every sentence; taking shortest programs gives (b). Finiteness in (b): if D is consistent, the lookup table of D is a consistent labeller, its closure a closed one, and the decider of the finite set $D^{\pm}$ is compatible; if not, no consistent theory proves $D^{\pm}$ and no consistent labeller extends D. For the losses, each of the two terms of $-\log W(D) + \log W(\emptyset)$ changes by at most c. For deciders this is Theorem 6.3 of Hänni and Claude (Anthropic) (2026a), stated there for programs that read their input; the change of program format costs a constant. □

With weights $2^{-\lambda|p|}$, $\lambda \geq 1$, as in Hänni and Claude (Anthropic) (2026d), the same maps give the equivalence within $2^{\pm\lambda c}$.

Remark B.3 (The schema route). A different route to Theorem 4.2 turns a labeller f into the axiom schema $A_f := \{\mathrm{Acc}_f(\ulcorner\varphi\urcorner) \to \varphi\} \cup \{\mathrm{Rej}_f(\ulcorner\varphi\urcorner) \to \neg\varphi\}$, with Σ_1 formulas Acc_f, Rej_f, written without $<$, that express that f accepts or rejects.

(a) *Two sorts.* Let $B = B_A \cup B_L$, with arithmetic axioms $\mathsf{Q} \subseteq B_A \subseteq \mathrm{Th}(\mathbb{N})$ on one sort, axioms B_L on the other and no mixed axioms. If $B_L \cup \Gamma_f$ is consistent, so is $B \cup A_f$, and the sentences of the non-arithmetic sort's language that $B \cup A_f$ proves are exactly those that $B_L \cup \Gamma_f$ proves (Proposition 6.4 of Hänni and Claude (Anthropic), 2026a; it uses the Σ_1-completeness of Q).

(b) *One sort over* PA. Assuming $\mathrm{Con}(\mathsf{PA})$, the labeller f that accepts $\neg\mathrm{Con}(\mathsf{PA})$ and, on $\mathrm{Con}(\mathsf{PA})$, searches for a PA-proof of $\bot$ has $\Gamma_f = \{\neg\mathrm{Con}(\mathsf{PA})\}$, which is consistent with PA, while $\mathsf{PA} \cup A_f$ is inconsistent (Proposition 6.5 there). So the argument through the schema fails in one sort; the conclusion of Theorem 4.2 does not.

(c) *Bounded witnesses.* In one sort, for every decidable $B \supseteq \mathsf{Q}$, true or not, let A_f^{bd} use the forms $\mathrm{Acc}_f^{\leq k}$, $\mathrm{Rej}_f^{\leq k}$ whose witnesses are bounded by a binary numeral k. Then $\mathrm{Cn}(A_f^{\mathrm{bd}}) = \mathrm{Cn}(\Gamma_f)$ for every labeller f, and membership in A_f^{bd} is decidable in polynomial time without running f (Proposition E.17 there, given the MRDP theorem): each axiom is Q-equivalent to a member of Γ_f or is a Q-theorem. In substance this is Craig's idea, the bound playing the role of the padding.

(d) *Templates.* No non-ground template of that report's template class has all its instances in the acceptance part of A_f; but for every Σ_n-sound labeller f one ground reflection sentence

of length at most $a|f| + b_n$ reproduces f on Σ_n sentences over PA (Propositions 6.6 and 6.7 there, given partial truth predicates).

The one-tape equivalence avoids the schema altogether: a theory machine is an enumerator that reads a tape, so it may emit Γ_{f_ω} itself (emission); neither decidability, nor arithmetic, nor a schema is needed. The schema, like Craig's padding, matters only for the decider format.

Proof of Theorem 4.3. (b) is Lemma B.2(i)–(iii). (a), M_A against M_F^{cons}, for $D \neq \emptyset$: by (3), (b) and (2), with $p \mapsto r_{\mathrm{ps}}p$ (an index of $F_{G_{V(p)}}$) and $p \mapsto r_{\mathrm{em}}p$ (an index of $G_{F_{V(p)}}$),

$$\sum_p 2^{-|p|}\nu_{G_{V(p)}}(D) \leq \sum_p 2^{-|p|}\nu^{\mathrm{cons}}_{F_{V(r_{\mathrm{ps}}p)}}(D) \leq 2^{|r_{\mathrm{ps}}|}M_F^{\mathrm{cons}}(D),$$
$$M_F^{\mathrm{cons}}(D) \leq 2^{|r_{\mathrm{em}}|}\sum_p 2^{-|p|}\nu_{G_{V(p)}}(D) \leq 2^{|r_{\mathrm{em}}|}M_A(D).$$

The remaining term $(1-\Omega_V)\mathbf{1}[B \vdash \psi_D]$ of (3) is at most $\mathbf{1}[B \vdash \psi_D]$; if $B \vdash \psi_D$, the deterministic proof-search labeller of B is closed and consistent and extends D on every tape, so $M_F^{\mathrm{cons}}(D) \geq 2^{-c_B}$. Hence $M_A \leq (2^{|r_{\mathrm{ps}}|} + 2^{c_B})M_F^{\mathrm{cons}}$. For $D = \emptyset$ both $M_A(\emptyset)$ and $M_F^{\mathrm{cons}}(\emptyset)$ lie in $[1-\Omega_V, 1]$ (the tapes without a program give the consistent theory B and the empty labeller), so their ratio is at most $1/(1-\Omega_V)$. M_A against M_A^{dec}: a random decider is a theory machine (emit χ under the prefix that certifies the answer 1), so $M_A^{\mathrm{dec}} \leq 2^c M_A$; and Craig's map $C_{F_{V(p)}}$ gives $M_A^{\mathrm{dec}} \geq 2^{-c}M_F^{\mathrm{cons}} \geq 2^{-2c}M_A$ on $D \neq \emptyset$; for $D = \emptyset$ the decider of $\emptyset$ gives $M_A^{\mathrm{dec}}(\emptyset) \geq 2^{-c}$. □

Proof of Corollary 4.4. Proof search maps tapes with a consistent theory to tapes on which F_G is closed and decides as the theory does; emission maps tapes with a closed consistent labeller to tapes on which the theory decides as the labeller does. The program-less tapes (theory B) are covered by the closed deterministic labeller "proof search from B", of weight 2^{-c_B}, which decides as B does. Sum as in the proof of Theorem 4.3. The outcome weights of M_A are those of the corollary: $M_A(D + (\varphi, 1))$ is the measure of the tapes whose theory is consistent and proves ψ_D and φ, and likewise for the other two. □

Penalised versions. We use the format of Appendix E.6 of Hänni and Claude (Anthropic) (2026a): hypotheses are programs of the universal prefix machine U of Assumption 4.5, read to their end before the input, now with access to a one-way read-only random tape.

Definition B.4 (Penalised one-tape SAI)**.** A *random decider program* q halts on every pair (χ, ω) with yes or no; A^q_ω is its yes-set on ω, and $\mu_q^{\mathrm{dec}}(D) := \mathbb{U}\{\omega : B \cup A^q_\omega \text{ consistent and } \vdash \psi_D\}$. A *random enumerator program* q has no input and writes sentences $a_1(\omega), a_2(\omega), \ldots$, completing a_i at step $\mathrm{st}_i(\omega)$ ($\mathrm{st}_0 := 0$). For time-constructible t:

- $M_A^{\mathrm{chk},t}(D) := \sum_q 2^{-|q|}\mu_q^{\mathrm{dec}}(D)$, the sum over the random decider programs q whose running time on every query χ and every tape is at most a constant multiple of $t(|\chi|)$, that is, $\sup_{\chi,\omega} \mathrm{time}_q(\chi,\omega)/t(|\chi|) < \infty$;
- $M_A^{\mathrm{gen},t}(D) := \sum_q 2^{-|q|}\nu_q^{\mathrm{gen}}(D)$, with $\nu_q^{\mathrm{gen}}(D) := \mathbb{U}\{\omega : B \cup \{a_i(\omega)\}_i \text{ consistent and } \vdash \psi_D\}$, the sum over the random enumerator programs that meet the *delay form* $\sup_{\omega,i}(\mathrm{st}_i - \mathrm{st}_{i-1})/t(|a_i|) < \infty$; in the *strict form* the condition is on $\mathrm{st}_i/t(|a_i|)$, in the *cumulative form* on $\mathrm{st}_i/t(|a_1| + \cdots + |a_i|)$.

The coin-free $W_A^{\mathrm{chk},t}$, $W_A^{\mathrm{gen},t}$ and the Levin-type $W_A^{\mathrm{Kt},t}$ are those of Definition E.10 of Hänni and Claude (Anthropic) (2026a), which also assumes t nondecreasing and superadditive with $t(m) \geq m$.

Remark B.5 (On Assumption 4.6). *Proof sketch.* We do not write this out for a concrete U; like Assumption 4.5, the assumption is used, not proved. A tape-reading program can be simulated by keeping the bits read so far on a work tape and one pointer per run of the schedule; a simulated read copies a stored bit or reads the next tape bit, at cost $O(T)$ up to work T. So the exponent a of Assumption 4.5 should grow by at most 1.

In this format the unpenalised one-tape laws $M_A^U := \sum_q 2^{-|q|}\nu_{G_q}$, over all tape-reading programs read as theory machines, and the corresponding M_F^{cons} agree with M_A and M_F^{cons} within constant factors, since each universal mixture contains the other's universal machine behind a constant prefix.

Proof of Theorem 4.7. (a) *Lower bounds.* Let p be a tape-reading program for a function machine F. Let q_F say yes on χ if F labels χ with 1, or if $\chi = \neg\varphi$ and F labels φ with 0 (dovetailing the two runs on the same tape). On every tape its yes-set is $\Gamma_{f_\omega^F}$, and $|q_F| \le |p| + c$. The padded random decider d, a fixed wrapper followed by q_F, answers χ as in Lemma E.12(b) of Hänni and Claude (Anthropic) (2026a): it finds the at most two candidates $\chi = \psi^{\wedge m}$, rejects if $m - 1 < |q_F|$, and otherwise runs the dovetailing schedule of q_F on the tape, without writing, until the work counter exceeds $m - 1$, accepting iff q_F is first detected saying yes on ψ exactly at counter $m - 1$. Under Assumption 4.6 this takes at most $\alpha(|q_F| + 1) + 3\alpha(|\chi| + 1)^{a'}$ steps, $a' := \max(a, 2)$, on every tape, uniformly in ω (the argument of that lemma, verbatim with the tape); as $(|\chi| + 1)^{a'} \le 2^{a'}|\chi|^{a^*} \le 2^{a'}t(|\chi|)$, the membership bound holds. The answer depends only on the tape bits read by the schedule, so d is a random decider. On each tape its yes-set is $\mathrm{Pad}_\omega := \{\psi^{\wedge(T+1)} : q_F \text{ is first detected saying yes on } \psi \text{ at counter } T \text{ on } \omega\}$; distinct detections have distinct counters, so each $\psi \in \Gamma_{f_\omega^F}$ has exactly one padded copy and $\mathrm{Cn}(\mathrm{Pad}_\omega) = \mathrm{Cn}(\Gamma_{f_\omega^F})$ (Lemma E.12(a) there, per tape). So on every tape on which f_ω^F is consistent and extends D, d is compatible with D. The padded enumerator, a fixed wrapper followed by q_F, writes $\psi^{\wedge(T+1)}$ at each detection; Lemma E.12(c) there gives, on every tape, delay at most $2\alpha|a_i|^a$, strict completion at most $5\alpha|a_i|^{a^*}$ and cumulative completion at most $2\alpha(|a_1| + \cdots + |a_i|)^a$, uniformly in ω. Both maps are injective and add a constant to the length. Summing over p gives the lower bounds for $D \neq \emptyset$. For $D = \emptyset$, the decider that rejects every sentence runs in linear time with theory B, which is consistent, and the enumerator that writes nothing meets every delay condition vacuously; so $M_A^{\mathrm{chk},t}(\emptyset), M_A^{\mathrm{gen},t}(\emptyset) \ge 2^{-c} \ge 2^{-c}M_F^{\mathrm{cons}}(\emptyset)$. *Upper bounds.* A random decider program q is read as the theory machine that emits χ under the tape prefix read by a run of q on χ that answers yes; a random enumerator program as the theory machine that emits the i-th written sentence under the prefix read when it is completed. Each reading is a fixed computable map on programs, realised by a constant-size wrapper, which is injective and keeps the theory on every tape. So $M_A^{\mathrm{chk},t}, M_A^{\mathrm{gen},t} \le 2^c M_A^U \le 2^{c'} M_A$ for every D. With Theorem 4.3 this gives the chain of equivalences. (b) This is Theorem 6.14(a) of Hänni and Claude (Anthropic) (2026a): the coin-free version of the construction above, with the padded sets of Lemma E.12 there. The inequality $W_A^{\mathrm{chk},t} \le W_A^{\mathrm{dec}}$ holds because the penalised deciders are deciders with the same weights, and $W_A^{\mathrm{dec}} \le 2^c W^{\mathrm{cons}}$ by Theorem 4.2. □

Proposition B.6 (Class posteriors, one tape)**.** *Let $t(m) \ge m^{a^*}$ be time-constructible and X be $M_A^{\mathrm{chk},t}$ or $M_A^{\mathrm{gen},t}$ (any form). For a class $\mathcal{C}$ of deductively closed theories let $X(\mathcal{C}; D)$ be the measure, under X's mixture, of the tapes on which the hypothesis is compatible with D and the deductive closure of its axioms is in $\mathcal{C}$. Under Assumption 4.6 there is c with $2^{-c}\mathbb{U}_A(\mathcal{C}; D) \le X(\mathcal{C}; D) \le 2^c\mathbb{U}_A(\mathcal{C}; D)$ for all $\mathcal{C}$ and D. So the posterior odds of any two classes of theories (genuine axiom systems and disguised labellers alike) under X and under M_A differ by a factor at most 2^{2c}.*

Proof. Lower bound: for a theory machine G with program p, let q_G say yes on χ iff G emits χ on the tape, and pad as in the proof of Theorem 4.7. On every tape the padded hypothesis has the same deductive closure as $B \cup A^G_\omega$, so it lies in $\mathcal{C}$ and is compatible with D exactly when G's theory is. The program-less tapes of M_A (theory B) are matched by the decider of $\emptyset$. Upper bound: a penalised hypothesis, read as a theory machine, has the same theory. For the odds, divide by $X(\text{all}; D)$ and $\mathbb{U}_A(\text{all}; D) = M_A(D)$. □

Proposition B.7 (The chain for the consistent versions)**.** *For every consistent $D \neq \emptyset$,*

$$2^{-K^{\text{cons}}_F(D)} \leq W^{\text{cons}}(D) \leq 2^c M^{\text{cons}}_F(D) \leq 2^{K(\operatorname{dom} D)+c}\, 2^{-K^{\text{cons}}_F(D)},$$

and $\log(M^{\text{cons}}_F(D)/W^{\text{cons}}(D)) \leq K^{\text{pad}}(L^{\text{cons}}_{1/2}(D)) + c$, where $L^{\text{cons}}_{1/2}(D)$ is the least L with $\mathbb{U}\{\omega :$ every label of D is certified within $\omega_{1:L}$ and f_ω is consistent$\} \geq M^{\text{cons}}_F(D)/2$. By Theorems 4.2 and 4.3 the same holds for K_A, W_A, M_A with a larger c.

Proof. A shortest consistent program extending D is a term of W^{cons}. A consistent deterministic labeller, read as a function machine that ignores the tape, is consistent on every tape; with (2) and the wrapper "treat $\{V(p)\}$ as a machine", $W^{\text{cons}} \leq 2^c M^{\text{cons}}_F$. Next, $M^{\text{cons}}_F \leq M_F \leq 2^{-K(D|\operatorname{dom} D)+c}$ by Theorem 3.21(a) of Hänni and Claude (Anthropic) (2026c), which holds for the input set Sent; the lookup table of D is consistent, so $K^{\text{cons}}_F(D) \leq K(D) + c \leq K(\operatorname{dom} D) + K(D \mid \operatorname{dom} D) + c$. For the K^{pad} bound, take $L' \geq L$ with $K(L') = K^{\text{pad}}(L)$, and for $w \in \{0,1\}^{L'}$ the stored-prefix labeller $\varphi \mapsto y$ iff some $(\varphi, w', y) \in F$ has $w' \sqsubseteq w$. It is a sub-function of f^F_ω for every $\omega \sqsupseteq w$, so it is consistent whenever some extension of w is a consistent tape. The count in the proof of Proposition 4.6(a) of the companion paper, restricted to such w, gives, with r the constant-size wrapper, $W^{\text{cons}}(D) \geq 2^{-|r|-K(F)-K^{\text{pad}}(L)}\mathbb{U}\{\omega : D$ certified within $\omega_{1:L}$, f^F_ω consistent$\}$; take $F = F_U$ and $L = L^{\text{cons}}_{1/2}(D)$. □

B.3 Positive literals

Proof of Theorem 4.8. By Theorems 4.2 and 4.3 it suffices to compare consistent SOI on sentences with plain SOI on $\mathcal{I}$. *Strings to sentences:* for a partial computable g on $\mathcal{I}$ let $g^P(R(t_x)) := g(x)$, undefined off the literals, and likewise for a function machine on $\mathcal{I}$ (decode x from t_x). Γ_{g^P} is a set of literals $\pm R(t_x)$, at most one per x, and it is consistent with B: in a model of B in which the t_x are pairwise distinct, the symbol R, which B does not mention, can be interpreted as $\{t_x : g(x) = 1\}$. So g^P, and the machine version on every tape, is consistent, extends D iff g extends D^{str}, and costs a constant. *Sentences to strings:* $f^{\text{str}}(x) := f(R(t_x))$, at a constant cost; $f \supseteq D$ implies $f^{\text{str}} \supseteq D^{\text{str}}$, and consistency is not needed. Sum over the injective maps, and take shortest programs. For the three outcome weights of $R(t_y)$: a consistent set of literals of this kind proves exactly its members (for $R(t_y)^b$ not in the set, reinterpret R at t_y), so the theory $\text{Cn}(\Gamma_{g^P})$ decides $R(t_y)$ iff $g(y)$ is defined, with the same value; and the proof-search labeller of a theory, restricted to literals, labels y exactly as the theory decides $R(t_y)$. □

Proof of Corollary 4.9. Theorem 4.8 changes K, $-\log W$ and $-\log M_F$ by at most a constant, so every statement that holds up to additive constants in these quantities transfers. (a) is Theorem 4.9 and Corollary 4.10 of Hänni and Claude (Anthropic) (2026c) applied to D^{str}; (b) is its Propositions 4.5(a) and 4.6; (c) is its Proposition 4.7, read through $M_A(D) \asymp M_F(D^{\text{str}})$, so that it compares M_A with M_{prod} on D^{str}; (d) is its Proposition 3.24 with $t_i := S^i 0$, which are pairwise distinct in $\mathbb{N}$, a model of $B \in \{\emptyset, \mathsf{Q}\}$ (with R interpreted arbitrarily). The statements that do not transfer are of the following kinds: labels from a set larger than $\{0,1\}$ (a fresh unary predicate

carries one bit); cumulative statements in the normalised convention (each normalised conditional transfers only within $2^{\pm 2c}$, so n steps cost $2cn$; single-step quantities do transfer up to a constant factor); private randomness per input or per query, and repeated inputs with fresh noise (no mixture over theory machines with a private coin per atom is defined here, and conflicting labels on one sentence are inconsistent data); history-aware and weak-semimeasure variants, which have no analogue here; computability statements, which are not "up to constants"; and other objects (time-bounded mixtures, circuits), which we compare directly (Theorems B.21 and B.31). □

Proposition B.8 (The price of the coin on literal data)**.** *In the setting of Theorem 4.8, for every literal data set D over words of length at most d,*

$$\log \frac{M_A(D)}{W_A(D)} \le \delta_D + K^{\mathrm{pad}}(2^{d+1}) + c \le \delta_D + K(\lceil \log(d+1) \rceil) + c', \qquad \delta_D := |D| - K(D^{\mathrm{str}} \mid \operatorname{dom} D^{\mathrm{str}}).$$

Proof. $W_A(D) \ge 2^{-c} W(D^{\mathrm{str}}) \ge 2^{-|D| - K^{\mathrm{pad}}(2^{d+1}) - c'}$ by Theorem 4.8 and Proposition 4.6(b) of Hänni and Claude (Anthropic) (2026c), and $M_A(D) \le 2^c M_F(D^{\mathrm{str}}) \le 2^{-K(D^{\mathrm{str}} | \operatorname{dom} D^{\mathrm{str}}) + c''}$ by its Theorem 3.21(a). □

So on literal data whose labels are algorithmically random given the inputs ($\delta_D = O(1)$), the coin-free version loses at most $K^{\mathrm{pad}}(2^{d+1}) + O(1)$ bits against the one-tape version, however many labels are predicted.

Proof of Remark 4.10. The lower bound for SAI and consistent SOI is Proposition E.1 of Hänni and Claude (Anthropic) (2026a): on every tape the theory (or the consistent labeller) cannot accept both $R(t_w)$ and $\neg R(t_w)$, so the two predictive probabilities sum to at most 1, and convexity of $-\log$ gives at least one bit per step in expectation; the argument applies tape by tape to M_A and M_F^{cons}. The re-encoding $(\neg R(t_w), 1) \mapsto (R(t_w), 0)$ is a bijection of data sets that keeps $D^{\pm}$, so it keeps M_A and W_A (Proposition 3.1). On positive literals SAI is SOI on strings (Theorem 4.8), and on fair-coin labels the latter loses at least $n - c$ bits in expectation, by Theorem 3.21(a) of Hänni and Claude (Anthropic) (2026c) and Gibbs' inequality. □

B.4 What breaks

Proof of Proposition 4.11. (a) A consistent theory proving ψ_D proves χ; and M_A depends on D only through $\mathrm{Cn}(D^{\pm})$ (Proposition 3.1). (b) A labeller $f \supseteq D$ with $f(\chi) = 0$ has $D^{\pm} \cup \{\neg\chi\} \subseteq \Gamma_f$, which is inconsistent with B. By Theorem 4.2, $W^{\mathrm{cons}}(D + (\chi, 1)) \ge 2^{-c} W_A(D + (\chi, 1)) = 2^{-c} W_A(D) \ge 2^{-2c} W^{\mathrm{cons}}(D)$; for one tape use Theorem 4.3. (c) Let q_χ be a shortest V-program for χ. The prefix machine that reads q_χ, then $s \in \operatorname{dom} V$, and outputs an index of "$\{V(s)\}$, made undefined at $V(q_\chi)$" is simulated by V with a prefix r. The map $s \mapsto r q_\chi s$ is injective; removing one label preserves consistency, and the restricted labeller still extends D because $\chi \notin \operatorname{dom} D$. So the consistent labellers extending D and undefined at χ have weight at least $2^{-|r| - K(\chi)} W^{\mathrm{cons}}(D)$. For one tape, apply the same map to function machines and use (2); on the program-less tapes the empty labeller is already undefined at χ. (d) is Theorem 4.2(c) and Corollary 4.4. □

Remark B.9 (The three outcomes are a measure for SAI). For $\varphi \notin \operatorname{dom} D$, $M_A(D + (\varphi, 1)) + M_A(D + (\varphi, 0)) + M_A^3(D + (\varphi, \iota)) = M_A(D)$ (Proposition 3.1), and likewise for W_A: each consistent theory proving ψ_D falls in exactly one of the three classes. For SOI the corresponding third mass is the defect, which collects abstention and divergence alike.

Productivity of the independent sentences. Ref_B and Thm_B are the sets of B-refutable and B-provable sentences. We use three textbook facts about Q (standard; see Boolos et al., 2007): (R) every decidable relation on numbers is strongly represented in Q, by a formula $\theta(\bar{x})$ with $\mathsf{Q} \vdash \theta(\underline{\bar{n}})$ when the relation holds of $\bar{n}$ and $\mathsf{Q} \vdash \neg\theta(\underline{\bar{n}})$ when it does not; (O) for every n, $\mathsf{Q} \vdash \forall z\,(z \le \underline{n} \to z = \underline{0} \vee \cdots \vee z = \underline{n})$ and $\mathsf{Q} \vdash \forall y\,(y \le \underline{n} \vee \underline{n} \le y)$, with $\le$ the usual abbreviation; (Diag) for every formula $\vartheta(x)$ there is a sentence ρ, computable from ϑ, with $\mathsf{Q} \vdash \rho \leftrightarrow \vartheta(\ulcorner\rho\urcorner)$.

Lemma B.10 (Productivity of the independent sentences)**.** *Let $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$, and let B be decidable and consistent with $\mathsf{Q} \subseteq \mathrm{Cn}(B)$. From an index of an r.e. set $S \subseteq \mathrm{Ind}_B$ one computes an $\mathcal{L}_{\mathrm{ar}}$-sentence $\rho \in \mathrm{Ind}_B \setminus S$. In particular Ind_B is productive, hence not r.e., and Ref_B is r.e. and not decidable.*

The lemma is classical: it is a form of the effective inseparability of the provable and the refutable sentences of consistent r.e. extensions of Q (cf. Smullyan, 1961). We include the proof, which is Rosser's argument relative to S.

Proof. Let $\mathsf{Neg}(y,x)$ be the decidable relation "y codes a B-derivation of the negation of the sentence coded by x, or x is enumerated into S within y steps", and $\mathsf{Prf}(y,x)$ the decidable relation "y codes a B-derivation of the sentence coded by x" (B is decidable). Let $\theta_{\mathrm{neg}}, \theta_{\mathrm{prf}}$ strongly represent them (R), and let ρ satisfy $\mathsf{Q} \vdash \rho \leftrightarrow \exists y\,(\theta_{\mathrm{neg}}(y,\ulcorner\rho\urcorner) \wedge \forall z \le y\ \neg\theta_{\mathrm{prf}}(z,\ulcorner\rho\urcorner))$ (Diag); ρ is computed uniformly from the index of S. $B \nvdash \rho$. Suppose k codes a B-derivation of ρ. Then $\rho \notin \mathrm{Ind}_B \supseteq S$, and $B \nvdash \neg\rho$ by consistency, so $\mathsf{Neg}(n,\ulcorner\rho\urcorner)$ fails for every n and $\mathsf{Q} \vdash \neg\theta_{\mathrm{neg}}(\underline{n},\ulcorner\rho\urcorner)$ for every n (R); also $\mathsf{Q} \vdash \theta_{\mathrm{prf}}(\underline{k},\ulcorner\rho\urcorner)$. Argue in Q: given y with $\theta_{\mathrm{neg}}(y,\ulcorner\rho\urcorner) \wedge \forall z \le y\,\neg\theta_{\mathrm{prf}}(z,\ulcorner\rho\urcorner)$, by (O) either $\underline{k} \le y$, which contradicts $\theta_{\mathrm{prf}}(\underline{k},\ulcorner\rho\urcorner)$, or $y \le \underline{k}$, so that y is one of $\underline{0},\dots,\underline{k}$, which contradicts $\neg\theta_{\mathrm{neg}}(\underline{n},\ulcorner\rho\urcorner)$. So $\mathsf{Q} \vdash \neg\rho$, hence $B \vdash \neg\rho$, and B is inconsistent, a contradiction. *$B \nvdash \neg\rho$ and $\rho \notin S$.* Otherwise $\mathsf{Neg}(n,\ulcorner\rho\urcorner)$ holds for some n, while $\mathsf{Prf}(m,\ulcorner\rho\urcorner)$ fails for every m by the previous step. By (R) and (O), $\mathsf{Q} \vdash \theta_{\mathrm{neg}}(\underline{n},\ulcorner\rho\urcorner) \wedge \forall z \le \underline{n}\,\neg\theta_{\mathrm{prf}}(z,\ulcorner\rho\urcorner)$, so $\mathsf{Q} \vdash \rho$ and $B \vdash \rho$, against the previous step. So $\rho \in \mathrm{Ind}_B \setminus S$. A productive set is not r.e. (take S to be the set itself). Ref_B is r.e. since B is decidable; if it were decidable, so would be $\mathrm{Thm}_B = \{\varphi : \neg\varphi \in \mathrm{Ref}_B\}$, hence Ind_B, which is not even r.e. □

Proof of Proposition 4.12. (a) Coin-free: map a consistent three-valued labeller f to the enumerator of Γ_f (its 1/0 part); the theory $\mathrm{Cn}(\Gamma_f)$ proves the 1/0 data and, by the consistency of f, decides none of the sentences f labels ι, which include those of D. One tape: emission, tape by tape, on the tapes on which f^F_ω is consistent and extends D, and (2). (b) Coin-free: let $\mathcal{F}_n$ be the finite (not computable) set of the $p \in \mathrm{dom}\, V$ with $|p| \le n$ for which $\{V(p)\}$ is a consistent three-valued labeller, and I_n the union of their ι-sets. It is a finite union of r.e. sets, hence r.e., and $I_n \subseteq \mathrm{Ind}_B$, since $f(\sigma) = \iota$ for a consistent f implies $B \cup \Gamma_f \nvdash \sigma, \neg\sigma$. Let $\sigma_n \in \mathrm{Ind}_B \setminus I_n$ (Lemma B.10). Every consistent labeller extending $\{(\sigma_n,\iota)\}$ labels σ_n with ι, so its program is not in $\mathcal{F}_n$ and is longer than n; hence $W^{3,\mathrm{cons}}(\{(\sigma_n,\iota)\}) \le \Omega_V^{\ge n+1}$. The enumerator of $\emptyset$ (theory B) is compatible with $\{(\sigma_n,\iota)\}$, so $W^3_A(\{(\sigma_n,\iota)\}) \ge 2^{-c}$. One tape: fix n and $\varepsilon := 2^{-n}$. For $p \in \mathrm{dom}\, V$ with $|p| \le n$ let $F_p := F_{V(p)}$ and Cons_p the (Borel) set of tapes on which $f^{F_p}_\omega$ is consistent. If $\mathbb{U}(\mathrm{Cons}_p) = 0$, p contributes nothing. Otherwise, by the Lebesgue density theorem (Lemma C.1) almost every $\omega \in \mathrm{Cons}_p$ has a prefix w with $\mathbb{U}(\mathrm{Cons}_p \cap [w]) > (1-\varepsilon)2^{-|w|}$. Let $\mathcal{W}_p$ be the set of minimal such prefixes, a prefix-free set with $\mathbb{U}(\mathrm{Cons}_p \setminus \bigcup_{w\in\mathcal{W}_p}[w]) = 0$, and choose a finite $\mathcal{W}'_p \subseteq \mathcal{W}_p$ with $\sum_{w\in\mathcal{W}_p\setminus\mathcal{W}'_p} 2^{-|w|} \le \varepsilon$. For $w \in \mathcal{W}'_p$ put

$$J_w := \{\sigma : \mathbb{U}([w] \cap \{\omega : f^{F_p}_\omega(\sigma) = \iota\}) > \varepsilon 2^{-|w|}\}.$$

J_w is r.e., uniformly in (p, w) and the rational ε, because the measure of the open set $[w] \cap \{f(\sigma) = \iota\}$ is lower semicomputable. And $J_w \subseteq \mathrm{Ind}_B$: if $\sigma \in J_w$, the set $[w] \cap \{f(\sigma) = \iota\}$ has measure larger than $\varepsilon 2^{-|w|} > \mathbb{U}([w] \setminus \mathrm{Cons}_p)$, so it meets Cons_p; on such a tape f_ω is consistent and labels σ with ι, so $B \cup \Gamma_{f_\omega}$, and hence B, decides neither σ nor $\neg\sigma$. Let $I'_n := \bigcup_{|p|\le n} \bigcup_{w \in \mathcal{W}'_p} J_w$, a finite union of r.e. subsets of Ind_B, and $\sigma'_n \in \mathrm{Ind}_B \setminus I'_n$ (Lemma B.10). For $|p| \le n$, since $\sigma'_n \notin J_w$,

$$\begin{aligned}\nu^{3,\mathrm{cons}}_{F_p}(\{(\sigma'_n, \iota)\}) &\le \sum_{w \in \mathcal{W}'_p} \mathbb{U}([w] \cap \{f(\sigma'_n) = \iota\}) + \sum_{w \in \mathcal{W}_p \setminus \mathcal{W}'_p} 2^{-|w|} + \mathbb{U}\Big(\mathrm{Cons}_p \setminus \bigcup_{w \in \mathcal{W}_p} [w]\Big) \\ &\le \sum_{w \in \mathcal{W}'_p} \varepsilon 2^{-|w|} + \varepsilon \le 2\varepsilon.\end{aligned}$$

By (2), $M^{3,\mathrm{cons}}_F(\{(\sigma'_n, \iota)\}) \le \sum_{|p|\le n} 2^{-|p|} 2\varepsilon + \Omega^{\ge n+1}_V \le 2^{1-n} + \Omega^{\ge n+1}_V$. The deterministic theory machine that emits nothing (theory B) is compatible with $\{(\sigma'_n, \iota)\}$ on every tape, so $M^3_A(\{(\sigma'_n, \iota)\}) \ge 2^{-c}$. (c) For $\sigma \in \mathrm{Ind}_B$ the labeller "ι at σ, undefined elsewhere" is consistent and has length $K(\sigma) + c$; with $W^3_A \le 1$ this gives the coin-free bound, and as a function machine that ignores the tape it gives the one-tape bound. □

The coin-free part of (b) is the special case of machines that ignore the tape: then J_w is the labeller's ι-set and no density argument is needed. One tape, the obstacle is that the set of sentences a random machine labels ι on consistent tapes need not be r.e., since it depends on the positivity of the measure of a Π^0_1 class; the density point restores an r.e. subset of Ind_B that carries almost all of the consistent ι-mass. A computation on finite analogues (code in the accompanying repository), with machines consistent on most tapes and noise that labels B-decided sentences ι, confirms the density step and the summed bound in every case tested.

Proof of Proposition 4.13. (a) The labeller "ι on every member of S (enumerate S), undefined elsewhere" has $\Gamma = \emptyset$ and labels only B-independent sentences, so it is consistent; it has length $K(S) + c$. As a function machine that ignores the tape it gives the one-tape bound. (b) Every program counted has length at least $n(\sigma)$, and one has length $n(\sigma)$. □

Remark B.11 (Shared versus private randomness). If a product law (private coins per input, Hänni and Claude (Anthropic), 2026c, Proposition 3.9(a)) gives positive probability to labelling φ true and to labelling ψ true, with $B \cup \{\varphi, \psi\}$ inconsistent, the sampled labeller is inconsistent with probability at least the product of the two. The consistent part of a product law is not a product law: for a B-independent φ, on the two inputs φ, $\neg\varphi$ with fair independent labels, the consistent outcomes are $(1, 0)$ and $(0, 1)$, each of mass $\frac{1}{4}$, while a product would need $P(1,1)P(0,0) = P(1,0)P(0,1) \ne 0$. (If B decides φ, one of these two outcomes is inconsistent too.) The shared tape is what makes a random labeller a random *theory*; private randomness is harmless on logically independent inputs such as positive literals (Theorem 4.8). Product laws are machine laws, so M^{cons}_F dominates their consistent parts.

Proposition B.12 (Computability of the versions)**.** *(a) Suppose the set of finite data sets D with $B \cup D^{\pm}$ consistent is not r.e. (for instance if $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$ and B is decidable and consistent with $\mathsf{Q} \subseteq \mathrm{Cn}(B)$, since then $\{\varphi : \{(\varphi, 1)\}$ is consistent$\}$ is the complement of Ref_B, which is r.e. and not decidable by Lemma B.10; or if $B = \emptyset$ and $\mathcal{L}$ has a binary relation symbol, by Church's theorem, Church, 1936). Then none of W_A, W^{dec}_A, W^{cons}, W^{cl}, M_A, M^{dec}_A, M^{cons}_F, M^{cl}_F is lower semicomputable.*

(b) In every setting, $M_A(\emptyset)$ and $M_F^{\mathrm{cons}}(\emptyset)$ are not left-c.e.; so M_A and M_F^{cons} are not lower semicomputable.

(c) W_A, W^{cons}, M_A and M_F^{cons} are computable relative to $\emptyset'$, uniformly in D.

(c′) W_A^{dec}, M_A^{dec}, W^{cl} and M_F^{cl} are computable relative to $\emptyset''$, uniformly in D.

Whether the four versions of (c′) are computable relative to $\emptyset'$ is not settled here. Being closed and being a (random) decider are Π_2 conditions, and for closedness the condition is Π_2-complete: with $\chi_0, \chi_1, \ldots$ an injective enumeration of the B-decided sentences, the labeller that labels χ_n with B's decision once program e halts on n is consistent, and it is closed iff e is total.

Proof. (a) For a function ν that is lower semicomputable uniformly in D, the support $\{D : \nu(D) > 0\}$ is r.e. Each object listed has support exactly the consistent data sets: positive weight needs a consistent hypothesis that proves or extends D; conversely, for a consistent D the lookup table of D (as a labeller, a finite theory, a decider, or a machine that ignores the tape) has positive weight in W_A, W_A^{dec}, W^{cons}, M_A, M_A^{dec} and M_F^{cons}, and for the closed versions W^{cl}, M_F^{cl} its closure "proof search from $B \cup D^{\pm}$", which is closed, consistent and extends D, has positive weight (the lookup table itself is never closed). (b) For M_A this is Theorem 3.5. For M_F^{cons}: for a left-c.e. real ζ with approximations $\zeta_s \uparrow \zeta$, let F_ζ label the sentence $\exists x\, x \neq x$ with 1 under a prefix w as soon as some stage s has $0.w + 2^{-|w|} \leq \zeta_s$, and be undefined elsewhere; its labelling is inconsistent with probability $\mathbb{U}\{\omega : 0.\omega < \zeta\} = \zeta$. With p_ζ a V-program for F_ζ, the probability that f_ω is inconsistent is $2^{-|p_\zeta|}\zeta + \kappa$ with κ left-c.e. If $M_F^{\mathrm{cons}}(\emptyset)$, which is one minus that probability, were left-c.e., the probability would be computable and ζ right-c.e.; take $\zeta = \Omega_V$. (c) W_A and W^{cons} are sums $\sum_p 2^{-|p|}\mathbf{1}[\cdot]$ of Δ_2 indicators ("proves ψ_D" and "extends D" are Σ_1, consistency is Π_1), and the tails $\Omega_V^{\geq \ell}$ are $\emptyset'$-computable; so $\emptyset'$ computes each value to any precision. For one tape, $M_A(D) = \mathbb{U}(O_{\psi_D}) - \mathbb{U}(O_\perp)$ (Proposition 3.4), and $M_F^{\mathrm{cons}}(D) = \mathbb{U}(O) - \mathbb{U}(O \setminus C)$ with $O := \{\omega : f_\omega \supseteq D\}$ effectively open and $C := \{\omega : f_\omega \text{ consistent}\}$ effectively closed: differences of left-c.e. reals, uniformly in D. (c′) Coin-free: "$V(p)$ is a total $\{0,1\}$-valued program" and "$\{V(p)\}$ is closed" $(\forall \varphi, b : B \cup \Gamma \vdash \varphi^b \to \{V(p)\}(\varphi) = b)$ are Π_2, so $\emptyset''$ decides each indicator and computes the sums. One tape, deciders: "$F_{V(p)}$ halts on every query and every tape" is Π_2, because for each query, halting on all tapes is, by compactness of Cantor space, equivalent to the Σ_1 condition that some n and s make every tape prefix of length n halt within s steps. For a random decider $\mu_F^{\mathrm{dec}}(D)$ is a difference of left-c.e. reals uniformly in an index, as in (c); so $\emptyset''$ computes $M_A^{\mathrm{dec}}(D)$. One tape, closed: "f_ω is closed" is the countable intersection over (φ, b) of the sets $\{\omega : B \cup \Gamma_{f_\omega} \nvdash \varphi^b\} \cup \{\omega : f_\omega(\varphi) = b\}$, each the union of an effectively closed and an effectively open set, hence an effective G_δ. So the event in $M_F^{\mathrm{cl}}(D)$ is an effective Π_2^0 class $\bigcap_k U_k$ with U_k uniformly effectively open and decreasing; its measure $\inf_k \mathbb{U}(U_k)$ is an infimum of uniformly left-c.e., hence uniformly $\emptyset'$-computable, reals, so it is right-c.e. relative to $\emptyset'$ and computable from $\emptyset''$. □

Question B.13. Can W_A be lower semicomputable when consistency of finite sets of sentences with B is decidable (for instance for a propositional language, or a monadic language without function symbols and $B = \emptyset$)?

There the support argument of Proposition B.12(a) says nothing, and the argument of (b) does not carry over directly to the coin-free version, because the weight of the inconsistent programs is not separated from Ω_V by a computable amount.

Pathwise and machine-wise consistency. M_F^{cons} conditions on the event that the *sampled* labeller is consistent. The machine-wise alternative is

$$M_F^{\mathrm{mw}}(D) := \sum_{p \in \mathrm{dom}\, V} 2^{-|p|}\, \mathbf{1}[F_{V(p)} \text{ is consistent on every tape}]\, \nu_{F_{V(p)}}(D),$$

and M_A^{mw} is defined likewise over the theory machines whose theory is consistent on every tape.

Proposition B.14 (Pathwise against machine-wise consistency)**.** *(a)* $W^{\mathrm{cons}} \le 2^c M_F^{\mathrm{mw}} \le 2^c M_F^{\mathrm{cons}}$ *on nonempty* D*, and* $M_F^{\mathrm{mw}} \asymp M_A^{\mathrm{mw}}$*.*

(b) For every consistent $D \neq \emptyset$*,* $M_F^{\mathrm{cons}}(D) \le 2^{\min\{K^{\mathrm{pad}}(L_{1/2}^{\mathrm{cons}}(D)),\, K(\mathrm{dom}\, D)\}+c}\, M_F^{\mathrm{mw}}(D)$*, with* $L_{1/2}^{\mathrm{cons}}$ *as in Proposition B.7.*

(c) If consistency of finite sets with B *is decidable (an algorithm decides, for finite* $\Lambda \subseteq \mathrm{Sent}$*, whether* $B \cup \Lambda$ *is consistent; for example if* $\mathrm{Cn}(B)$ *is decidable, for propositional languages, and for monadic languages without function symbols with* $B = \emptyset$*, by the decidability of monadic first-order logic, Löwenheim, 1915; Behmann, 1922), then* $M_F^{\mathrm{cons}} \asymp M_F^{\mathrm{mw}}$ *and* $M_A \asymp M_A^{\mathrm{mw}}$*.*

Proof. (a) A deterministic consistent labeller is consistent on every tape, and a machine consistent on every tape has $\nu = \nu^{\mathrm{cons}}$. Proof search and emission preserve "consistent on every tape", since on each tape proof search of a consistent theory is consistent and emission of a consistent labelling is consistent. (b) Proposition B.7 with (a). (c) *The global-schedule filter.* For a function machine F, read as a machine that reads the tape one-way, let F' answer a query φ by running the *global schedule* of F: stage s reads tape bit s, starts F on the s-th sentence and advances every started run by one step; certification events (a run halting with a label) are ordered by stage, ties by sentence index. When φ's event occurs, F' checks whether the set of labels certified up to and including it is consistent with B (decidable by hypothesis); if so it outputs φ's label, otherwise it diverges. The answer is a deterministic function of the bits read, so F' is a function machine, and $K(F') \le K(F) + c$. On a tape ω let $l_1, l_2, \ldots$ be the labelled sentences in event order. F' outputs the label of l_i iff $\{l_1, \ldots, l_i\}$ is consistent; consistency of these prefixes is monotone, so the outputs form an initial segment all of whose finite subsets are consistent, hence a consistent set, by compactness. So F' is consistent on every tape, and $F' = F$ on every tape on which F is consistent. Hence $\nu_{F'}(D) \ge \nu_F^{\mathrm{cons}}(D)$ for every D, and summing with (2) gives $M_F^{\mathrm{mw}} \ge 2^{-c} M_F^{\mathrm{cons}}$. The theory-machine version filters emissions in the same way. □

With two unary function symbols, consistency of finite sets is not decidable, so in (c) "monadic" must exclude function symbols.

Question B.15. Is $M_F^{\mathrm{cons}}/M_F^{\mathrm{mw}}$ bounded in general, for instance for $B = \mathsf{Q}$, or for $B = \emptyset$ and a language with a binary relation symbol?

A machine consistent on every tape must avoid inconsistency, which is only semi-decidable, so its support is an r.e. family of consistent data sets; a pathwise-conditioned machine can exploit the Π_1^0 event that nothing inconsistent ever happens. The lottery does not separate the two: the lottery machine is consistent on every tape (Theorem 5.1).

B.5 Time bounds with free deduction

As in Appendix E.6 of Hänni and Claude (Anthropic) (2026a), the penalised coin-free hypotheses are programs of the machine U of Assumption 4.5, so $\mathrm{time}_p(x) \geq |p|$, and $a' := \max(a, 2)$. Clocked strings, their weights $w(f)$ and $c_\perp$ are as in Section 4.5; c_1 is the length of the fixed wrapper that runs a clocked string and diverges when it abstains. Simulations between U_0, U and multitape machines have polynomial overhead (standard; used in the same form as there).

Proposition B.16 (A Levin penalty costs a logarithm)**.** *Let $t(m) \geq m^{a^*}$ be time-constructible, m_0 the least length of a sentence, $W_A^{\mathrm{Kt},t}$ the mixture over all deciders p with weight $2^{-|p|}/c_p$, $c_p := \sup_\chi \mathrm{time}_p(\chi)/t(|\chi|)$ (Definition E.10 of Hänni and Claude (Anthropic), 2026a), and $W^1(D) := \sum_f 2^{-|f|}/(|f|+1)$ over the programs f of consistent labellers extending D. Under Assumption 4.5, $W_A^{\mathrm{Kt},t}(\emptyset) \leq t(m_0)$, and there is c with $2^{-c}W^1(D) \leq W_A^{\mathrm{Kt},t}(D) \leq 2^c t(m_0)W^1(D)$ for every D; so a labeller f pays $\log(|f|+1) + O(1)$ bits beyond its unpenalised weight.*

Proof. This is Proposition E.13 of Hänni and Claude (Anthropic) (2026a). Since $\mathrm{time}_p \geq |p|$, $c_p \geq |p|/t(m_0)$, so p has weight at most $t(m_0)2^{-|p|}/|p|$, and the total is at most $t(m_0)$ by Kraft's inequality. For the lower bound, the padded decider of a consistent labeller f (as in the proof of Theorem 4.7) has $c_d \leq \beta(|f|+1)$ for a constant β, by Lemma E.12(b) there and $(|\chi|+1)^{a'} \leq 2^{a'}t(|\chi|)$, so it has weight at least $2^{-|f|-2c}/(\beta(|f|+1))$; the map is injective. For the upper bound, map a compatible decider p to its proof-search labeller f_p, with $|p| \leq |f_p| \leq |p| + c$; the weight of p is at most $t(m_0)(c+2)2^c\, 2^{-|f_p|}/(|f_p|+1)$, and $p \mapsto f_p$ is injective. □

Theorem B.17 (Domination lemma)**.** *Let X be a coin-free mixture and v_X a weight on consistent labellers such that $W_X(D) \geq 2^{-c_X}\sum_g v_X(g)$, the sum over the consistent labellers g extending D, and $W_X(\emptyset) \leq \Omega_X$. Let Y be a mixture with $W_Y(D) = \sum_h v_Y(h, D)$, the sum over the hypotheses h compatible with D, with an injective map $h \mapsto g_h$ to consistent labellers that compute the same partial map as h, with $v_Y(h, D) \leq k_Y\, v_X(g_h)$, and with $W_Y(\emptyset) \geq \kappa_Y > 0$. Then $\ell_X(D) \leq \ell_Y(D) + c_X + \log(k_Y\Omega_X/\kappa_Y)$ for every D.*

Proof. A labeller g_h computes the same partial map as h, so it extends D iff h is compatible with D. Hence $W_X(D) \geq 2^{-c_X}\sum_h v_X(g_h) \geq 2^{-c_X}k_Y^{-1}W_Y(D)$, and $\ell_X(D) = -\log W_X(D) + \log W_X(\emptyset) \leq -\log W_Y(D) + c_X + \log(k_Y\Omega_X) = \ell_Y(D) - \log W_Y(\emptyset) + c_X + \log(k_Y\Omega_X)$. □

The weaker hypothesis $\Gamma_{g_h} = \Gamma_h$ does not suffice: a labeller that rejects ψ and one that accepts $\neg\psi$ and abstains on ψ have the same Γ but differ on the datum $(\psi, 0)$.

Proof of Corollary 4.14. Coin-free (Theorem 6.14(b) of Hänni and Claude (Anthropic), 2026a): apply Theorem B.17 with $X \in \{W_A, W_A^{\mathrm{dec}}, W_A^{\mathrm{chk},t}, W_A^{\mathrm{gen},t}\}$ and $Y = W^{\mathrm{cons},\tau}$. By Theorem 4.2 and, for the penalised versions under Assumption 4.5, Theorem 4.7(b), $W_X \geq 2^{-c}W^{\mathrm{cons}}$, and $W_X(\emptyset) \leq 1$ by Kraft's inequality. For a clocked string h let g_h be the wrapper that runs h^τ and diverges when it abstains, followed by $E(h)$; it computes the same partial map as h^τ, $|g_h| = |E(h)| + c_1$, and $h \mapsto g_h$ is injective. $W^{\mathrm{cons},\tau}(\emptyset) \geq 2^{-c_\perp}$, the weight of the always-abstaining string (B is consistent). So $c_\tau := c + c_1 + c_\perp$ works. One tape: the clocked function machine of a string f is a function machine whose V-program is a fixed wrapper followed by $E(f)$, injectively, so (2) gives $M_F^{\mathrm{cons}} \geq 2^{-c_1}M_F^{\mathrm{cons},\tau}$ on $D \neq \emptyset$; Theorem 4.3 gives $M_A \geq 2^{-c}M_F^{\mathrm{cons}}$; and $M_A(\emptyset) \leq 1$, $M_F^{\mathrm{cons},\tau}(\emptyset) \geq 2^{-c_\perp}$. □

Proof of Theorem 4.15. This is Theorem 6.15 of Hänni and Claude (Anthropic) (2026a); we recall the argument. By the MRDP theorem (Matiyasevich, 1970; Davis, 1973) there are polynomials

E, E' with natural-number coefficients such that program e halts on x with output y iff $\mathbb{N} \models \exists \bar{u}\,(E(e,x,y,\bar{u}) = E'(e,x,y,\bar{u}))$; $\mathrm{Out}(e,x,y)$ is this formula (Lemma E.14(a) there), and $\mathrm{DET} := \forall e\,\forall x\,\forall y\,\forall y'\,\forall \bar{u}\,\forall \bar{u}'\,(E(e,x,y,\bar{u}) = E'(e,x,y,\bar{u}) \wedge E(e,x,y',\bar{u}') = E'(e,x,y',\bar{u}') \to y = y')$, a true Π_1 sentence. Let $H_0 := \mathsf{Q} \cup \{\mathrm{DET}\}$. *Construction.* Let $\mathcal{G}(e',n)$ be the program that, for $j = 1,\dots,n$, forms $\sigma_j := \mathrm{Out}(\underline{e'}, \underline{j}, S0)$, asks P for approximations (a_0,a_1) within 2^{-j-3} of its pair on the history $((\sigma_1,\beta_1),\dots,(\sigma_{j-1},\beta_{j-1}))$ and the sentence σ_j, and sets $\beta_j := 0$ if $a_0 \le a_1$ and $\beta_j := 1$ otherwise; it outputs β_n. $\mathcal{G}$ is total, as P is. By Kleene's recursion theorem (standard) there is e with $e(n) = \mathcal{G}(e,n)$ for all n. The values β_j computed inside $\mathcal{G}(e,n)$ do not depend on n, so $e(j) = \beta_j \in \{0,1\}$ and $\sigma_j = s_j$; and $\mathbb{N} \models s_n$ iff $e(n) = 1$, so the label b_n is β_n, and P is queried exactly on the true histories D_{j-1}. H_0 *decides each* s_n. If $b_n = 1$, s_n is a true existential equation, which Q proves (Lemma E.14(c) there). If $b_n = 0$, $\mathrm{Out}(\underline{e}, \underline{n}, 0)$ is one, so Q proves it, and with s_n and DET it gives $S0 = 0$, which Q refutes; so $H_0 \vdash \neg s_n$. *The loss of* P. With $q_b := q_b(D_{j-1}, s_j)$ and $\varepsilon := 2^{-j-3}$, the tie rule gives $q_{b_j} \le a_{b_j} + \varepsilon \le a_{1-b_j} + \varepsilon \le q_{1-b_j} + 2\varepsilon$, and with $q_0 + q_1 \le 1$ this gives $q_{b_j} \le \frac{1}{2} + 2^{-j-3}$. So $-\log q_{b_j} \ge 1 - \log(1 + 2^{-j-2})$, and $\sum_j \log(1 + 2^{-j-2}) \le \sum_j 2^{-j-2}/\ln 2 = 1/(4\ln 2) < 0.361$. *Axiom induction.* H_0 is true in $\mathbb{N}$ and B in an expansion of $\mathbb{N}$, so $B \cup H_0$ is consistent, and H_0 is compatible with every D_n. It is finite, so it has a decider running in linear time and an enumerator writing finitely many axioms; let $\omega_0 > 0$ be the least of their weights in W_A, $W_A^{\mathrm{chk},t}$ and $W_A^{\mathrm{gen},t}$. Then $W(D_n) \ge \omega_0$, $W(\emptyset) \le 1$, and the loss is at most $\log(1/\omega_0)$, whatever P is. *Clocked SOI.* Put $P(D) := (W^{\mathrm{all},\tau}(D) + 2^{-|D|})/2$ and $q_b(D,\varphi) := P(D + (\varphi,b))/P(D)$. Then $q_0 + q_1 \le 1$, since $W^{\mathrm{all},\tau}$ is a semimeasure and the uniform part halves at each step; P is computable, since clocked compatibility is decidable and the weights have computable tails; $P(D) \ge 2^{-|D|-1} > 0$; and $W^{\mathrm{all},\tau}(\emptyset) = 1$, so $P(\emptyset) = 1$ and $\ell_P(D_n) = -\log P(D_n)$. On P's sequence $W^{\mathrm{cons},\tau}(D_n) \le W^{\mathrm{all},\tau}(D_n) \le 2P(D_n)$, so $\ell_{W^{\mathrm{all},\tau}}(D_n) \ge n - 1.37$ and $\ell_{W^{\mathrm{cons},\tau}}(D_n) \ge \ell_P(D_n) - 1 + \log W^{\mathrm{cons},\tau}(\emptyset) \ge n - 1.37 - c_\perp$. □

Proposition B.18 (Incomparability with time-bounded predictors that may be inconsistent)**.** *On the data of Remark 4.10,* $W^{\mathrm{all},\tau}$ *has bounded loss for every* τ *that covers the constant running time of "always* 1*", while every consistent inducer (SAI and consistent SOI, coin-free or one-tape) loses at least* n *bits in expectation; on the sequences of Theorem 4.15,* $W^{\mathrm{all},\tau}$ *loses at least* $n - 1.37$ *while SAI loses* $O(1)$*. So* $W^{\mathrm{all},\tau}$ *and SAI are incomparable.*

Proof. Once τ covers its running time, the clocked string "always 1" is compatible with every prefix of the data of Remark 4.10, so the loss of $W^{\mathrm{all},\tau}$ is at most $-\log w(\text{always 1})$. The other two claims are Remark 4.10 and Theorem 4.15. □

So the natural strengthening, that SAI is strictly stronger than every time-bounded predictor, fails; what holds is strict superiority over clocked *consistent* SOI (Corollary 4.14 and Theorem 4.15).

Proof of Proposition 4.16. This is Proposition E.15(a), (b) of Hänni and Claude (Anthropic) (2026a). (a) $\mathrm{Out}(\underline{e}, \underline{n}, S0)$ is true iff e outputs 1 on n; if e halts on n with another output, $\neg\mathrm{Out}(\underline{e}, \underline{n}, S0)$ is true, as a program has at most one output. So every label of u is true in $\mathbb{N}$. (b) In Theorem 4.15 e is total with values in $\{0,1\}$, so u labels every s_n correctly and extends every D_n; hence $W^{\mathrm{cons}}(D_n) \ge 2^{-|u|-c}$, where c pays for turning u into a V-program for a labeller index, and with $W^{\mathrm{cons}}(\emptyset) \le 1$ the loss is at most $|u| + c$. The same holds for $W^{\mathrm{all}} \ge W^{\mathrm{cons}}$. □

Remark B.19 (A time-hierarchy effect)*. Proof sketch.* We do not write out the version of the diagonal theorem that gives the compared predictors the same reuse of state, nor the dependence

on the machine model. The claim is that under a time bound the sequence that defeats a time-bounded predictor is itself computable within a slightly larger bound. For the predictors S_p of Proposition E.15(c) of Hänni and Claude (Anthropic) (2026a), which spend time $p(n)\log n$ per step and reuse their state, the labels that defeat S_p are produced by a predictor with per-step time $O(p(n)\log n)$ and state reuse that predicts its own labels with certainty; so $S_{p'}$, for a slightly larger p' (for example $np(n)$), has bounded loss on the sequence that defeats S_p.

Proposition B.20 (Disguised labellers keep their weight)**.** *Let $t(m) \geq m^{a^*}$ be time-constructible and X be $W_A^{\mathrm{chk},t}$ or $W_A^{\mathrm{gen},t}$ (any form). For a class $\mathcal{C}$ of deductively closed theories let $X(\mathcal{C};D)$ and $W_A^{\mathrm{dec}}(\mathcal{C};D)$ be the weights of the hypotheses compatible with D whose theory is in $\mathcal{C}$. Under Assumption 4.5 there is c with $2^{-c}W_A^{\mathrm{dec}}(\mathcal{C};D) \leq X(\mathcal{C};D) \leq 2^c W_A^{\mathrm{dec}}(\mathcal{C};D)$ for all $\mathcal{C}$ and D. So the posterior odds of any two classes of positive weight, genuine and disguised alike, move by at most 2^{2c}, and a slow consistent labeller f keeps prior weight at least $2^{-|f|-c}$, as a padded Craig set.*

Proof. This is Proposition E.16 of Hänni and Claude (Anthropic) (2026a): the padded sets of Lemma E.12 there keep the theory and meet the penalty, which gives the lower bound; for the upper bound the penalised deciders are deciders with the same weights, and an enumerator q maps to the padded decider of "yes iff q writes the input", which keeps the theory. Divide by the total weights. □

Theorem B.21 (One tape and the companion paper's time-bounded mixtures)**.** *Let $M_F^{\mathrm{cons},\tau} := \sum_f w(f)\nu_{F_f^\tau}^{\mathrm{cons}}$, where F_f^τ runs f on U_0 with a tape-reading instruction for at most $\tau(|\varphi|)$ steps and abstains otherwise, and let $M_F^{\mathrm{all},\tau}$ drop the consistency restriction. Parts (b)–(d) assume that $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$ and that B is decidable and true in an expansion of $\mathbb{N}$.*

- *(a) $\ell_{M_A}(D) \leq \ell_{M_F^{\mathrm{cons},\tau}}(D) + c_\tau$ for every D.*
- *(b) On a computable sequence of arithmetic sentences, $\ell_{M_F^{\mathrm{all},\tau}} \geq n - 1.37$ and $\ell_{M_F^{\mathrm{cons},\tau}} \geq n - 1.37 - c_\perp$, while $\ell_{M_A} \leq \log(1/\omega_0)$.*
- *(c) The length-activated and round-activated mixtures of Hänni and Claude (Anthropic) (2026c, Definition 9.1, Proposition 9.4), with computable weights whose tails are computable, are computable predictors, so each loses at least $n - 0.361$ on a computable sequence on which SAI loses $O(1)$. Conversely, on the coin literals of Remark 4.10, all labelled true, the round-activated mixture has bounded loss, while SAI and consistent SOI lose at least n bits in expectation. So the companion paper's time-bounded SOI and SAI are incomparable.*
- *(d) Unpenalised W^{cons} and M_F^{cons} have loss at most $|u| + c$ on the diagonal sequences of (b) and (c).*

Proof. (a) is the one-tape part of the proof of Corollary 4.14. (b) Each query of F_f^τ reads at most $\tau(|\varphi|)$ tape bits, so $\nu_{F_f^\tau}(D)$ is a finite union of cylinders, computable exactly; with computable tails, $M_F^{\mathrm{all},\tau}$ is computable, a semimeasure with value 1 at $\emptyset$, and $P(D) := (M_F^{\mathrm{all},\tau}(D) + 2^{-|D|})/2$ is a computable predictor; run Theorem 4.15 against P, as in its last step. The theory machine that emits H_0 is compatible with every D_n on every tape. (c) Theorem 4.15 needs only computability of the predictor. The converse is Proposition 9.4 of the companion paper with the constant comparator "label true with probability 1", which is in its class $\mathcal{C}_t$, and Proposition E.1 of Hänni and Claude (Anthropic) (2026a), applied tape by tape (Remark 4.10). (d) Proposition 4.16, with (2) for M_F^{cons}. On the coin literals of (c), by contrast, every consistent inducer, W^{cons} and M_F^{cons} included, loses at least n bits in expectation. □

B.6 Charged deduction

Proof of Theorem 4.18. This is Theorem 6.16 of Hänni and Claude (Anthropic) (2026a), with Lemma E.19 and Proposition E.21 there, whose budgets are time-constructible and nondecreasing. Part (a) is the subformula lemma (Lemma E.19(b), (d) there): in a normal derivation every line sits at its own formula node of an instance used or of the conclusion, up to abstracting the parameters of a chain of Gen steps, and a family with k vacuous Gen steps attains the factor $\frac{1}{2}$. Part (b): a derivation of size at most g has material at most g, and by (a) material at most g gives size at most G; this is the first chain, hypothesis by hypothesis. Under Assumption 4.17 the parameters of a derivation of size at most G can be renamed to $p_0, \dots, p_{G-1}$, so the derivation is a certificate of at most $\beta G \log(G+2)$ bits that a verifier checks in time t^+ (membership in A_p in time t per line); this is the second inequality. For the third, let Z be a closed tautology of size 4 and C_ζ a sentence of size $|\zeta|+4$ that encodes the certificate ζ, and put $\theta_\zeta := Z \to (C_\zeta \to Z)$, a tautology of size $|\zeta|+14$. The axiom set $\{\theta_\zeta \to \varphi^b : \zeta \text{ a certificate on which } \mathcal{V} \text{ accepts } (b=1) \text{ or rejects } (b=0)\ \varphi\}$ is decidable in time $t'^{\#}$, satisfies Assumption 4.17, and has the same consequences as $\Gamma_{f_\mathcal{V}}$. The derivation θ_ζ, $\theta_\zeta \to \varphi^b$, φ^b has material $2|\zeta|+29+|\varphi^b|$, largest instance $|\zeta|+15+|\varphi^b|$ and size $2|\zeta|+29+2|\varphi^b|$, with $|\varphi^b| \le |\varphi|+1$; these give the three stated budgets. All maps are injective and add a constant to the length. □

Corollary B.22 (Fit classes)**.** *Under Assumption 4.17, on literal sequences D^X, at polynomial budgets whose degrees are charged in the prior, with standard simulation overheads:*

(i) reading (b) fits exactly the $X \in \mathsf{NP} \cap \mathsf{coNP}$; it beats polynomial-time consistent SOI, $W^{\mathrm{cons,poly}}$, on some sequence if $\mathsf{NP} \cap \mathsf{coNP} \neq \mathsf{P}$, and fits the same sequences if $\mathsf{P} = \mathsf{NP}$; with degrees uncharged, $W^{\mathrm{cons,poly}}$ fails to dominate it, unconditionally; and, under Assumption 4.5, it beats $W^{\mathrm{cons},\tau}$ at a budget a hierarchy gap above τ;

(ii) reading (a1) fits exactly the $X \in \mathsf{EXPTIME}$; at any computable budgets some decidable X defeats it; and every predictor computable in time polynomial in the number of data loses at least $n - 0.361$ on a sequence on which (a1) at a linear budget loses $O(1)$;

(iii) readings (a2) and (b2) fit every decidable X, by a finite set of universally closed Horn sentences.

Proof. This is Corollary 6.17(a)–(c) of Hänni and Claude (Anthropic) (2026a), with its proofs there. (ii) uses $\mathsf{APSPACE} = \mathsf{EXPTIME}$ (Chandra et al., 1981) through Proposition E.23 there, and the comparisons use the time hierarchy theorems (standard; cf. Hartmanis and Stearns, 1965, and Seiferas et al., 1978 for nondeterministic time). □

Proposition B.23 (Soft charges)**.** *Let a hypothesis pay a factor $2^{-\kappa\ell}$ per datum, ℓ the bit length of a shortest derivation of the datum (soft SAI), and let a verifier pay likewise for its shortest certificate (soft certificate SOI). The two correspond only up to a constant factor in the rate κ; no pair of rates gives two-sided constant regret; and every soft loss diverges on every infinite sequence, so there are no soft fit classes.*

Proof. Proposition E.26 of Hänni and Claude (Anthropic) (2026a), which gives the explicit rates. □

Remark B.24 (The role of the renaming condition)*.* Without Assumption 4.17 the index of one parameter, which costs one symbol, can carry a certificate of any length (Proposition E.27 of Hänni and Claude (Anthropic), 2026a). The coarse statements (Theorem 6.9 there, and the inclusion of

the fit class of reading (b) in $\mathsf{NP}\cap\mathsf{coNP}$) survive in the reading that counts symbols, by truncating parameter indices (Lemma E.28 there); the fine bounds ($2^{O(h\log h)}$ for (a1), certificate lengths) are not claimed without it.

Remark B.25 (One tape). *Proof sketch.* We do not write out the bookkeeping of a verifier that reads a tape, which needs Assumption 4.6-style accounting. The maps of Theorem 4.18(b) act tape by tape: a random decider with time $O(t)$ on every tape gives a verifier that reads the tape to answer membership queries, and certificate axioms $\theta_\zeta \to \varphi^b$ can be emitted from a random verifier. So a one-tape form of the sandwich should hold with the same budgets.

B.7 Presburger arithmetic

Remark B.26 (Lengths). (i) The length $|\varphi|$ of a sentence is its symbol count: every occurrence of a connective, quantifier, nonlogical symbol, parenthesis or variable counts 1, whatever the variable's index. This is the size measure of the calculus K of Hänni and Claude (Anthropic) (2026a), in which a variable or parameter counts one symbol. Machines read φ as a bit string of length $b(\varphi) \le \kappa_0|\varphi|\log(|\varphi|+2)$, since a sentence with m symbols has at most m distinct variables, which can be renamed $x_1,\dots,x_m$ and written in binary. Clocks τ, derivation budgets s,g,h and the length m in the lower bounds are functions of $|\varphi|$. (ii) If a clock were a function of the bit length b, which can exceed the symbol count by a factor $\log m$, a clock $2^{2^{c'b}}$ could exceed every $2^{2^{cm}}$, and the contradictions below would fail. For clocks in bit length the results hold with $\tau(b) \le 2^{2^{c'b/\log(b+2)}}$ in place of $\tau(m) \le 2^{2^{c'm}}$, and likewise for the singly exponential clocks, since $b/\log(b+2) \le \kappa_0|\varphi|$. (iii) *Proviso:* Fischer and Rabin's length counts every symbol occurrence, variables included, at least once, and their nonlogical symbols are among $0,1,+,=,<$. Under the proviso every sentence of their length n has an equivalent $\mathcal{L}_{\mathrm{Pr}}$-sentence with at most κn symbols, computable in polynomial time (rewrite $x<y$ as $\exists z\,(x+z+1=y)$ with z fresh, and copy every other symbol). *Padding:* every $\mathcal{L}_{\mathrm{Pr}}$-sentence ψ has an equivalent sentence with exactly m symbols for every $m \ge |\psi|+c_0$, computable in polynomial time, because there are two equivalence-preserving paddings with coprime increments (in the size measure of K, $\psi\wedge 0=0$ adds 4 and $\psi\wedge\neg 0=1$ adds 5; every standard syntax has such a pair). *Consequence:* let Q be a decision procedure for PrA in our syntax, deterministic or nondeterministic. For $j=0,\dots,\kappa-1$ let P_j decide their sentences of length n by translating, padding to exactly $\kappa n+c_0+j$ symbols and running Q. Each P_j is a decision procedure in their syntax, with the same reading of nondeterministic time as Q; so by their theorem, for all large n, Q takes more than $2^{2^{c_{\mathrm{FR}}n}}-\mathrm{poly}(n)$ steps on some sentence with exactly $\kappa n+c_0+j$ symbols. As j runs through all residues, for all large m, Q takes more than $2^{2^{cm}}$ steps on some sentence with exactly m symbols, for every $c<c_{\mathrm{FR}}/\kappa$; c denotes such a constant. The exact-length form is what the "for all large m" arguments below use.

We use two known results. ($\mathrm{F_P}$) Presburger's axioms for successor and addition, with the induction schema for $\mathcal{L}_{\mathrm{Pr}}$-formulas, form a decidable, consistent and complete axiomatisation of PrA (Presburger, 1929), with membership decidable in polynomial time. ($\mathrm{F_{FR}}$) The bound of Fischer and Rabin (1974), in the form of Remark B.26(iii). Its statement does not say how the time of a nondeterministic procedure is measured. We distinguish two readings: (max) the time is the maximum over the computation paths, all of which halt, every output being correct and some path giving an output; (short) the time on a sentence is the length of the shortest computation that outputs the correct answer. (short) implies (max). Where only (max) is needed nothing is assumed; the one place that uses (short) says so.

Proof of Proposition 4.19. (a) The decider of Presburger's axioms is compatible with every pre-

fix of D^{Pres} (the axiomatisation is complete and consistent), runs in polynomial time and is a deterministic theory machine; with an enumerator of the same axioms it meets the axiom-side penalties for a large enough polynomial t. The losses are at most $-\log$ of its weight. (b) In a clocked mixture compatibility is inherited by prefixes, so the compatible strings form a decreasing family and bounded loss gives one τ-clocked f that labels every sentence correctly (Lemma E.11 of Hänni and Claude (Anthropic), 2026a). By the standard simulation overhead it yields a multi-tape decision procedure for PrA that runs in time $q(C_f\tau(m)+b)$, with q a fixed polynomial and $b \le \kappa_0 m\log(m+2)$ the input's bit length. When $c' < c$ this is at most $2^{2^{cm}}$ for large m, which contradicts $\mathrm{F_{FR}}$ in the deterministic case. (c) Write $M := \sum_f w(f)\nu_f$ for either one-tape mixture, each $\nu_f(D_n^{\mathrm{Pres}})$ being non-increasing in n. Bounded loss gives $\inf_n M(D_n^{\mathrm{Pres}}) > 0$; by dominated convergence some f has $\lim_n \nu_f(D_n^{\mathrm{Pres}}) > 0$, which by continuity from above is $\mathbb{U}(\mathcal{G}_f)$ for the set $\mathcal{G}_f$ of tapes on which F_f^τ labels every sentence correctly (and consistently, for the consistent version). By the Lebesgue density theorem (Lemma C.1) there is w_0 with $\mathbb{U}(\mathcal{G}_f \cap [w_0]) > \frac{3}{4}\,2^{-|w_0|}$, so for every sentence the tapes in $[w_0]$ on which F_f^τ answers correctly have conditional measure greater than $\frac{3}{4}$. A query of length m reads at most $\tau(m)$ tape bits, so the procedure "for every v of length $\max(0,\tau(m)-|w_0|)$ run F_f^τ with tape w_0v and output the majority answer" (w_0 hard-coded) decides PrA deterministically within $2^{\tau(m)}q(C_f\tau(m)+b) \le 2^{2^{cm}}$ steps for large m: a contradiction. The majority vote over the tape completions at a density point is the technique of de Leeuw et al. (1956); the argument applies to every mixture of machines whose tape use per query is bounded by τ. □

A computation with illustrative values of the constants (code in the accompanying repository; c is not known explicitly) gives the lengths from which these contradictions apply: for $c = 1$ and $c' = 0.9$, from about 11–26 symbols in (b) and 6–9 in (c). The separations are asymptotic: below these lengths a clocked hypothesis may label every short sentence correctly.

Assumption B.27 ((B*): a Σ_2-time lower bound for Presburger arithmetic)**.** There is $c_B > 0$ such that, for all large m, no alternating machine with at most two alternations (a Σ_2 machine) decides PrA within $2^{2^{c_B m}}$ steps on all sentences with m symbols.

Assumption B.28 ((SGL): time-bounded $\mathsf{BPP} \subseteq \Sigma_2$)**.** For time-constructible $T(m) \ge m$ and large T, $\mathsf{BPTIME}(T) \subseteq \Sigma_2\text{-}\mathsf{TIME}(T^3)$.

We do not prove either assumption. Berman's theorem (Berman, 1980), as reported in secondary sources, makes PrA complete for alternating doubly exponential time with linearly many alternations; (B*) would follow from its hardness part if the reductions used were of linear size, which we have not been able to check. If they are only of polynomial size, a bound $2^{2^{c_B m^\epsilon}}$ follows for some $\epsilon \in (0,1]$, and the clock condition below must be weakened to $\tau(m) \le 2^{2^{c' m^\epsilon}}$. (SGL) is the time-bounded form of $\mathsf{BPP} \subseteq \Sigma_2^p$ (Sipser, 1983; Lautemann, 1983): amplifying by $O(\log T)$ repetitions to $R = O(T\log T)$ random bits and error below $1/(2R)$, a Σ_2 machine guesses R shifts of R bits, universally chooses R bits, and runs the amplified machine R times, in time $O(T^2\log^2 T)$.

Proposition B.29 (One tape, doubly exponential clocks)**.** *Under Assumptions B.27 and B.28, if $\tau(m) \le 2^{2^{c' m}}$ for large m with $c' < c_B$, then the losses of $M_F^{\mathrm{cons},\tau}$ and $M_F^{\mathrm{all},\tau}$ on D_n^{Pres} tend to infinity.*

Proof. As in the proof of Proposition 4.19(c), obtain w_0 such that on $[w_0]$ the conditional probability of a correct answer exceeds $\frac{3}{4}$ for every sentence. The algorithm "draw v of length $\tau(m)$

at random, run F_f^τ with tape $w_0 v$, and output its answer" errs with probability below $\frac{1}{4}$ on every sentence and runs in time $T(m) := q(C_f\tau(m) + b)$, so PrA $\in \mathsf{BPTIME}(T) \subseteq \Sigma_2\text{-}\mathsf{TIME}(T^3)$ by Assumption B.28. Since $T(m)^3 \leq 2^{2^{c_B m}}$ for large m when $c' < c_B$, this contradicts Assumption B.27. □

A nondeterministic lower bound alone does not finish the doubly exponential case by this route: separating $\mathsf{NTIME}(2^{2^n})$ from Σ_2-time below it is not known in general.

Proof of Proposition 4.20. Since compatibility is inherited by prefixes, it suffices to treat one hypothesis p, deciding A_p in time $O(m^{k_p})$ and compatible with every prefix of D^{Pres}. Then $B \cup A_p = A_p$ is consistent and proves every true sentence, so it proves exactly PrA. (a) *Infinitely often.* Fix $c'' < c$ and suppose $s(m) \leq \tau''(m) := 2^{2^{c''m}}$ for all $m \geq m_1$; τ'' is time-constructible. The nondeterministic procedure N: on a sentence φ with $m < m_1$ symbols, look φ up in a table; otherwise compute $\tau''(m)$, guess a derivation of φ or of $\neg\varphi$ of size at most $\tau''(m)$ whose parameter numbers have at most $K + 1 + \lceil \log(\tau''(m) + 1) \rceil$ bits, $K = O(\tau''(m)^{k_p})$ (if a derivation of size at most $\tau''(m)$ exists, such a one exists, by Lemma E.28 of Hänni and Claude (Anthropic), 2026a), check it as in the proof of Theorem 6.9 there (each line a logical axiom, by matching, a member of A_p, in time $O(\tau''^{k_p})$, or a step of modus ponens or generalisation), and output the label it proves; a path whose guess is not a valid derivation outputs nothing. Every output is correct, every sentence has a path with an output, and every path halts within $\tau''(m)^{c_k}$ steps, c_k depending only on k_p and the checking overhead of K. Since $\tau''(m)^{c_k} = 2^{c_k 2^{c''m}} \leq 2^{2^{cm}}$ once $m \geq \log(c_k)/(c - c'')$, N contradicts $\mathrm{F_{FR}}$ (nondeterministic case, reading (max)). So $s(m) > 2^{2^{c''m}}$ for infinitely many m. *For all large m, s time-constructible.* Suppose $s(m) \leq \tau''(m)$ for infinitely many m, the *good* lengths. The procedure N'': on a sentence with m symbols, compute $s(m)$ with a clock of $\tau''(m)$ steps; if the computation finishes with $s(m) \leq \tau''(m)$, run N's guess-and-check with the budget $s(m)$; otherwise run a deterministic decision procedure for PrA, which exists by $\mathrm{F_P}$. Every output is correct and every path halts, and at each good length every path halts within $O(\tau''(m)) + \tau''(m)^{c_k} \leq 2^{2^{cm}}$ steps beyond the thresholds. By $\mathrm{F_{FR}}$ (reading (max)), for all large m some sentence with m symbols takes N'' more than $2^{2^{cm}}$ steps; a good length beyond the thresholds gives a contradiction. (For an arbitrary s the same conclusion follows if $\mathrm{F_{FR}}$ holds in reading (short): let a path whose guess is not a valid derivation run the deterministic procedure; at each good length every sentence then has a correct output within $\tau''(m)^{c_k} + O(1)$ steps.) (b) By the subformula lemma (Theorem 4.18(a)), material at most g gives a derivation of size at most $(g + m)(g + m + 1)/2$. If $g(m) \leq 2^{2^{c''m}}$ for all large m, this is at most $2^{2^{c'''m}}$ for large m, for any $c''' \in (c'', c)$, against (a) with c'''. The form for all large m transfers in the same way. (c) Proposition E.23(a) of Hänni and Claude (Anthropic) (2026a) gives the labeller g_p: with $H(\varphi) := \max(h(|\varphi|), |\varphi| + 1)$ and Der_H the formulas with a derivation from $B \cup A_p$ all of whose lines have size at most H, it accepts φ if $\varphi \in \mathrm{Der}_{H(\varphi)}$, rejects it if $\neg\varphi \in \mathrm{Der}_{H(\varphi)}$ and $\varphi \notin \mathrm{Der}_{H(\varphi)}$, and abstains otherwise. It runs in time $2^{O(H \log H)}(1 + t(H))^{O(1)}$, by closing the at most $2^{O(H \log H)}$ canonical formulas of size at most H under the rules, using Assumption 4.17; and it labels correctly every datum of every D with which p is (a1)-compatible, since a normal (a1)-derivation of φ^b has every line of size at most $\max(h(|\varphi|), |\varphi^b|) \leq H(\varphi)$ (the subformula lemma) and $B \cup A_p$ is consistent. Nothing in this argument uses the form of φ. On D^{Pres}, g_p therefore decides PrA deterministically (it never abstains, since every sentence is a datum) in time $T(m) \leq 2^{C_a H \log H}$ with $H = \max(h(m), m + 1)$ and C_a depending on k_p (computing $h(m)$ costs $O(h(m))$). By $\mathrm{F_{FR}}$, $T(m) > 2^{2^{cm}}$ for all large m, so $C_a H \log H > 2^{cm}$, hence $H > 2^{cm}/(C_a cm) > 2^{c''m}$ for all large m; as $m + 1 < 2^{c''m}$ for large m, $H = h(m)$. □

Question B.30. (i) Under reading (a1), does every hypothesis compatible with every prefix of D^{Pres} need doubly exponential instance budgets h? (ii) Does some hypothesis of reading (a1) in $\mathcal{L}_{\mathrm{Pr}}$ fit D^{Pres} at exponential budgets? (iii) What do readings (a2) and (b2) need on D^{Pres}, in either direction?

(i) would follow from a deterministic lower bound $\mathrm{PrA} \notin \mathsf{DTIME}(2^{2^{2^{c_3 m}}})$ for some $c_3 > 0$, by the argument of Proposition 4.20(c), which would then give $h(m) > 2^{2^{c_3' m}}$ infinitely often for every $c_3' < c_3$; no such bound is known to us. For (ii), Proposition E.23(b) of Hänni and Claude (Anthropic) (2026a) simulates alternating space-bounded machines, but it needs the literal language with two extra binary relation symbols, which $\mathcal{L}_{\mathrm{Pr}}$ with $B = \emptyset$ lacks. For (iii), logical instances are free under (a2) and (b2), so neither the guess-and-check route nor the enumeration route gives a bound, and Corollary B.22(iii) concerns literal data in the literal language, not D^{Pres}.

B.8 The circuit form of the sandwich

The sandwich of Theorem 4.18 has a routine circuit form, which connects finite axiom sets with the circuit Bayes theorem of the companion paper. Fix $d \geq 1$; the data are the literals $\varphi_x := R(t_x)$, $x \in \{0,1\}^d$, labelled by a partial $f : \{0,1\}^d \rightharpoonup \{0,1\}$.

A *verifier circuit* $\mathcal{V}$ of size $|\mathcal{V}|$ is a list of $|\mathcal{V}|$ two-input gates (any of the 16 Boolean functions) over the inputs $x_1, \ldots, x_d$, certificate bits $z_1, \ldots, z_k$ (without loss of generality $k \leq 2|\mathcal{V}|$), the constants and earlier gates; its last two gates are the outputs a (accept) and r (reject). It is *consistent* if no x has certificates ζ, ζ' with $\mathsf{a}(x,\zeta) = 1$ and $\mathsf{r}(x,\zeta') = 1$, and it defines $f_{\mathcal{V}}(x) := 1$ if some ζ has $\mathsf{a}(x,\zeta) = 1$, 0 if some ζ has $\mathsf{r}(x,\zeta) = 1$, and undefined otherwise. $\mathrm{vs}(f)$ is the least size of a consistent verifier circuit with $f_{\mathcal{V}} \supseteq f$; it is at most the circuit size of f in the sense of Hänni and Claude (Anthropic) (2026c, Definition 9.7) plus a constant. The *fixed language* is $\mathcal{L}_{\mathrm{circ}} := \{\mathsf{e}, \mathsf{s}_0, \mathsf{s}_1, \mathrm{Dig}, \mathrm{Gate}, R\}$, with Dig binary and Gate ternary. Index terms $\mathbf{n}_i$ and gate terms $\mathbf{g}_j$ are the word terms $t_{\beta(i)}$, $t_{\beta(j)}$, with $\beta(i)$ the binary expansion of $i+1$ without its leading 1; so $|\mathbf{n}_i| = O(\log i)$, the terms of each family are pairwise distinct, and the language does not grow with the circuit. $\mathrm{ax}_s(f)$ is the least total size of a finite set T of sentences of $\mathcal{L}_{\mathrm{circ}}$ (or of any fixed finite language containing $\mathsf{e}, \mathsf{s}_0, \mathsf{s}_1, R$) with $T \cup \Gamma_f$ consistent such that every datum $\varphi_x^{f(x)}$, $x \in \mathrm{dom}\, f$, has a K-derivation from T of size at most s.

The translation $T_{\mathcal{V}}$ (all sentences universally closed; $\leftrightarrow$ abbreviates $\neg((A \to B) \to \neg(B \to A))$). Digits: $\forall y\, \mathrm{Dig}(\mathbf{n}_0, \mathsf{s}_1 y)$, $\forall y\, \neg\mathrm{Dig}(\mathbf{n}_0, \mathsf{s}_0 y)$, and $\forall y\, (\mathrm{Dig}(\mathbf{n}_{i+1}, \mathsf{s}_a y) \leftrightarrow \mathrm{Dig}(\mathbf{n}_i, y))$ for $i < \max(d,k) - 1$ and $a \in \{0,1\}$; so $\mathrm{Dig}(\mathbf{n}_i, t_w)$ holds iff the $(i+1)$-st symbol of w is 1. Gates: a gate j that reads x_i (resp. z_i) has the axiom $\forall x\, \forall z\, (\mathrm{Gate}(\mathbf{g}_j, x, z) \leftrightarrow \mathrm{Dig}(\mathbf{n}_{i-1}, x))$ (resp. with z in place of x), and an internal gate j with operation op on gates a, b has $\forall x\, \forall z\, (\mathrm{Gate}(\mathbf{g}_j, x, z) \leftrightarrow \mathrm{op}(\mathrm{Gate}(\mathbf{g}_a, x, z), \mathrm{Gate}(\mathbf{g}_b, x, z)))$. Outputs: $\forall x\, \forall z\, (\mathrm{Gate}(\mathbf{g}_{\mathsf{a}}, x, z) \to R(x))$ and $\forall x\, \forall z\, (\mathrm{Gate}(\mathbf{g}_{\mathsf{r}}, x, z) \to \neg R(x))$.

Theorem B.31 (The circuit sandwich)**.** *(i) For every consistent verifier circuit $\mathcal{V}$ with k certificate bits, $T_{\mathcal{V}}$ has total size at most $C_1(|\mathcal{V}| + d + k)\log(|\mathcal{V}| + d + k + 2)$, $T_{\mathcal{V}} \cup \Gamma_{f_{\mathcal{V}}}$ is consistent, and every $x \in \mathrm{dom}\, f_{\mathcal{V}}$ has a derivation of $\varphi_x^{f_{\mathcal{V}}(x)}$ from $T_{\mathcal{V}}$ of size at most $s_{\mathcal{V}} := C_2(|\mathcal{V}|(\log(|\mathcal{V}| + 2) + d + k) + (d+k)^3)$, with absolute constants C_1, C_2. Hence, using $k \leq 2|\mathcal{V}|$, $\mathrm{ax}_s(f) \leq C_1'(\mathrm{vs}(f) + d)\log(\mathrm{vs}(f) + d + 2)$ for $s = C_2'(\mathrm{vs}(f)(\log \mathrm{vs}(f) + d) + (d + \mathrm{vs}(f))^3)$. (ii) Conversely, $\mathrm{vs}(f) \leq P(\mathrm{ax}_s(f), s, d)$ for a fixed polynomial P that depends only on the language and K.*

This is the routine circuit form of the sandwich, not a separate result of independent interest;

we include it because it carries the circuit Bayes theorem of the companion paper over to verifier circuits (Corollary B.32).

Proof. (i) *Size.* There are $O(d+k)$ digit axioms of size $O(\log(d+k))$, one axiom of size $O(\log(|\mathcal{V}|+d+k))$ per gate (two or three gate terms and at most two index terms), and two output axioms. *Consistency.* Take the free term algebra on $\mathsf{e}, \mathsf{s}_0, \mathsf{s}_1$, with equality the identity; every element is a word term. Read off the bits of an element y (the symbols of the word it spells, 0 beyond its length). Let $\mathrm{Dig}(t, y)$ hold iff $t = \mathbf{n}_i$ for some $i < \max(d, k)$ and bit i of y is 1; let $\mathrm{Gate}(t, x, z)$ hold iff $t = \mathbf{g}_j$ and gate j of $\mathcal{V}$ outputs 1 on the first d bits of x and the first k bits of z; and let $R(x)$ hold iff some ζ has $\mathsf{a}(\mathrm{bits}_d(x), \zeta) = 1$. The digit and gate axioms hold by construction. The first output axiom holds by the definition of R; the second holds because $\mathcal{V}$ is consistent: $\mathsf{r}(\mathrm{bits}_d(x), z) = 1$ excludes $\mathsf{a}(\mathrm{bits}_d(x), \zeta) = 1$ for every ζ. On word terms of length d, $R(t_x)$ holds iff $f_{\mathcal{V}}(x) = 1$, and $f_{\mathcal{V}}(x) = 0$ forces $\neg R(t_x)$; so the structure satisfies $T_{\mathcal{V}} \cup \Gamma_{f_{\mathcal{V}}}$. *Derivations.* For $x \in \mathrm{dom}\, f_{\mathcal{V}}$ take a certificate ζ for its label. Instantiate every gate axiom with $x := t_x$, $z := t_\zeta$ (two instances of the quantifier axiom and two modus ponens steps per axiom, each line of size $O(\log|\mathcal{V}| + d + k)$). Derive each needed digit, $\mathrm{Dig}(\mathbf{n}_i, t_w)$ or its negation, from an instance of the base axiom and i instances of the successor axioms (each of size $O(d+k)$), substituting at each step a fixed propositional derivation for biconditional elimination: $O((d+k)^2)$ per digit, $O((d+k)^3)$ in all. Then, gate by gate in order, derive $\mathrm{Gate}(\mathbf{g}_j, t_x, t_\zeta)$ or its negation from the ground biconditional and the literals of its inputs, by a substitution instance of a fixed propositional derivation (the propositional fragment of K is complete, and substitution instances of derivations are derivations of size linear in the substituted formulas, as in the proof of Theorem 6.16 of Hänni and Claude (Anthropic), 2026a). Finish with the output instance and modus ponens. The total is $O(|\mathcal{V}|(\log|\mathcal{V}|+d+k)+(d+k)^3)$. (ii) A certificate is the code of a derivation of size at most s. Since T is a set of sentences, Assumption 4.17 holds, and the parameters can be renamed to $p_0, \ldots, p_{s-1}$, so the code has $O(s \log s)$ bits (Proposition E.21(b) of Hänni and Claude (Anthropic), 2026a). The checker, with T hard-wired, verifies each line (logical axioms by matching, quantifier axioms by first-order matching, members of T by comparison, modus ponens, generalisation) and the last line against φ_x or $\neg\varphi_x$, in time polynomial in $s+|T|+d$; it accepts if the code is a derivation of φ_x and rejects if it is one of $\neg\varphi_x$. Simulating it by circuits (standard; size $O(T \log T)$ for time T, Pippenger and Fischer, 1979) gives a verifier circuit of size $P(|T|, s, d)$. It is consistent because T is consistent, and $f_{\mathcal{V}} \supseteq f$ because every datum has a derivation within s and T proves no wrong label, $T \cup \Gamma_f$ being consistent. □

A computation (code in the accompanying repository) checks the translation on random consistent verifier circuits with $d \in \{2, 3\}$: every axiom of $T_{\mathcal{V}}$ is true in the intended structure on a finite fragment, forward propagation through the ground instances derives the right label for every $x \in \mathrm{dom}\, f_{\mathcal{V}}$, every corruption of one gate that changes $f_{\mathcal{V}}$ is detected, and $|T_{\mathcal{V}}|$ stays between 7.9 and 9.0 times $(|\mathcal{V}| + d)\log(|\mathcal{V}| + d)$ for sizes from 33 to 1099 gates.

Corollary B.32 (Bayes over verifier circuits)**.** *Let W_{ver} be the mixture over consistent verifier circuits with prior $w_{\mathrm{ver}}(\mathcal{V}) := 2^{-|\mathcal{V}|}\,(2|\mathcal{V}|+1)^{-1}\,\Lambda^{-1}_{|\mathcal{V}|,k}$, where $k \le 2|\mathcal{V}|$ is uniform and $\Lambda_{j,k} := \prod_{i=1}^{j} 16(d+k+1+i)^2$ is the number of gate lists of size j. For a nonempty $J \subseteq \{0,1\}^d$, an ordering of J and the labels of f, let $L := -\log W_{\mathrm{ver}}(f|_J)$, $\ell_\emptyset := -\log W_{\mathrm{ver}}(\emptyset)$, and $\mathrm{Err} := \sum_i \epsilon_i$, with ϵ_i the posterior mass at step i that is wrong* or undefined *at x_i. Then: (a) some consistent verifier circuit of size at most $\lceil L \rceil$ agrees with f on J, and $L \le \mathrm{vs}_J + \log(2\mathrm{vs}_J+1) + \log \Lambda_{\mathrm{vs}_J, 2\mathrm{vs}_J} = O(\mathrm{vs}_J \log(\mathrm{vs}_J + d))$, with $\mathrm{vs}_J := \mathrm{vs}(f|_J)$; (b)* $\mathrm{Err} \le (\ln 2)(L - \ell_\emptyset)$ *and* $L - \ell_\emptyset \le 2\,\mathrm{Err}\,(\mathrm{gad}(\lceil L \rceil) + 3)$, *with* $\mathrm{gad}(j) := \mathrm{gg}\,(5 + 2\log(d + 3j + \mathrm{gg} + 1)) + \log(\mathrm{gg}+1)$ *and* $\mathrm{gg} := 2d + 2$.

So Bayes over nondeterministic verifier circuits makes few posterior mistakes on f exactly when f has a small consistent verifier circuit, and, by Theorem B.31, exactly when f has a small finite axiomatisation in a fixed language whose data have polynomially short derivations, up to polynomial changes.

Proof. (a) The prior has an exponential tail in the size (the total weight of sizes above S is at most 2^{-S}), and the verifier with no active output disagrees with f at every size, so the argument of Theorem 9.8(a) of Hänni and Claude (Anthropic) (2026c) applies; the bound on L is the weight of one smallest consistent verifier circuit for $f|_J$. (b) The first inequality is $\epsilon_i \le -\ln(1-\epsilon_i)$ with telescoping; now $W_{\mathrm{ver}}(\emptyset) < 1$, hence $\ell_\emptyset$. For the second we use an exception gadget for verifiers: given $\mathcal{V}$, a point a and a label b, append a circuit for $[x = a]$ (at most $2d-1$ gates), its negation, and new outputs $\mathsf{a}' := \mathsf{a} \vee [x = a]$, $\mathsf{r}' := \mathsf{r} \wedge \neg[x = a]$ if $b = 1$ (symmetrically if $b = 0$): at most $\mathrm{gg} = 2d + 2$ gates. The new circuit $\mathcal{V}'$ is consistent (at a only the b-output can fire, and elsewhere it is $\mathcal{V}$), is defined at a with label b, agrees with $\mathcal{V}$ off a, and $\mathcal{V} \mapsto \mathcal{V}'$ is injective, with $w_{\mathrm{ver}}(\mathcal{V}')/w_{\mathrm{ver}}(\mathcal{V}) \ge 2^{-\mathrm{gad}(|\mathcal{V}|)}$. It maps the hypotheses of size at most S that are wrong or undefined at x_i injectively to hypotheses that are right at x_i and still compatible with D_{i-1}, whose inputs differ from x_i. The rest of the proof of Theorem 9.8(b) of the companion paper is unchanged. □

Remark B.33 (The axiom-set prior itself). *Proof sketch.* We do not write out the converse direction for a prior over finite axiom sets; the obstacles are listed here. For a mixture over finite axiom sets with a derivation budget, the upper bounds $L = O(a \log a)$, with a the size of a smallest compatible axiom set, and $\mathrm{Err} \le (\ln 2)L$ hold directly (Theorem B.31(i) and telescoping). The converse needs an exception gadget for *every* hypothesis T: replace R by a fresh unary predicate R' in T and add $\forall x\,(x \ne t_a \to (R(x) \leftrightarrow R'(x)))$ and $R(t_a)^b$. This works if the language supplies fresh predicate symbols (so it is not a fixed finite language, and the symbol's index costs $O(\log)$ bits), if B contains the free-algebra axioms needed to derive $t_x \ne t_a$ (of size $O(d^2)$), and if the budget is charged in the prior, so that the gadget's extra derivation size $O(d^2)$ moves a hypothesis to a nearby budget at a cost of $O(\log)$ bits.

C Proofs for random theories, Π_1 data and Gödel phenomena

This appendix proves the results of Section 5 in their order and states and proves the results that the main text only mentions. The notation is that of Definitions 2.3, 2.4 and 2.7.

C.1 Standing facts and conventions

Facts from Section 3. We use four facts throughout.

- *(Dom)* If $\omega = p\omega'$ with $p \in \operatorname{dom} V$, then $A_\omega = A_{\omega'}^{G_{V(p)}}$ (Definition 2.4). So for every theory machine G, every event about the random theory of G is matched, at a cost of a factor $2^{-K(G)}$, by the same event for G_U: the tapes $p\omega'$ with p a shortest program for an index of G carry it. In particular $M_A \ge 2^{-K(\nu)}\nu$ for every theory-machine law ν, and $M_A(D) \ge 2^{-K(e)-c}$ whenever $B \cup \mathsf{W}_e$ is consistent and proves ψ_D (Theorem 3.6).
- *(Chain)* $2^{-K_A(D)} \le W_A(D) \le 2^c M_A(D)$ for consistent D (Proposition 3.17(c)).
- *(Lim)* If $\mathbb{U}(E_\infty) > 0$, the posterior converges in total variation to $P_\infty = \mathbb{U}(\cdot \mid E_\infty)$ and the cumulative loss is bounded; if $\mathbb{U}(E_\infty) = 0$, the loss tends to ∞. For every Γ with $\mathrm{Cn}(\Gamma) = \mathrm{Cn}(D^{\pm}_\infty)$, complete or not, $E_\infty = \{\omega : B \cup A_\omega \text{ consistent}, \ \mathrm{Cn}(A_\omega) \supseteq \mathrm{Cn}(\Gamma)\}$ (Proposition A.13).

- *(Old)* $M_A(D)$ depends on D only through $\mathrm{Cn}(D^{\pm})$, and data entailed by B change nothing (Proposition 3.1(c)). The same holds for W_A and K_A, whose definitions mention D only through "proves ψ_D".

$\mathbb{U}(O_\varphi \cap [w])$ is left-c.e. uniformly in φ and w (Lemma A.1). For a set $\mathcal{X}$ of tapes and a finite string w we write $\mathbb{U}(\mathcal{X} \mid [w]) = 2^{|w|}\mathbb{U}(\mathcal{X} \cap [w])$. For a class $\mathcal{C}$ of deductively closed theories, $P_D(\mathcal{C}) := \mathbb{U}_A(\mathcal{C}; D)/M_A(D)$ is the posterior probability that the random theory lies in $\mathcal{C}$, and $W_A(\mathcal{C}; D) := \sum_e \mathbf{m}(e)\mathbf{1}[B \cup \mathsf{W}_e \text{ consistent}, \vdash \psi_D, \ \mathrm{Cn}(\mathsf{W}_e) \in \mathcal{C}]$.

Arithmetic. $\mathcal{L}_{\mathrm{ar}} = \{0, S, +, \cdot, <\}$ with numerals $\underline{n} = S^n 0$. The classes Δ_0, Σ_1, Π_1 are syntactic; Π_1^* is the class of finite conjunctions of Π_1 sentences and negations of Σ_1 sentences; a sentence that is equivalent to a member of a class only provably in Q is called Q-*equivalent* to one. "B is true" means that B has a model whose $\mathcal{L}_{\mathrm{ar}}$-reduct is $\mathbb{N}$ (for $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$: $B \subseteq \mathrm{Th}(\mathbb{N})$), likewise "$B \cup T$ is true"; "$B \supseteq \mathsf{Q}$" means $\mathsf{Q} \subseteq \mathrm{Cn}(B)$, likewise for PA. A theory is Σ_1-*sound* if it proves no false Σ_1 sentence. Binary strings are coded by numbers through their length-lexicographic rank (the empty string is 0, then $0, 1, 00, 01, \ldots$); the coding is injective across lengths, the length of a string is computable from its code, and the strings of length at most m are exactly the codes at most $\mathrm{rk}_m := 2^{m+1} - 2$. We use the following standard facts (see Smoryński, 1977; Boolos et al., 2007).

- *(Σ_1 C)* Q proves every true Σ_1 sentence (in particular every true Δ_0 sentence) and refutes every false Δ_0 sentence.
- *(Π_1 T)* Hence, if $B \supseteq \mathsf{Q}$ and $B \cup A$ is consistent, every Π_1^* sentence that $B \cup A$ proves is true: if a conjunct were false, its negation would be a true Σ_1 sentence, provable in Q.
- *(O)* $\mathsf{Q} \vdash \forall z\,(z \le \underline{n} \to z = \underline{0} \vee \cdots \vee z = \underline{n})$ for every n; so Q unfolds bounded quantifiers with a numeral bound into finite conjunctions and disjunctions.
- *(G2)* For every consistent r.e. presentation Z with $\mathsf{PA} \subseteq \mathrm{Cn}(Z)$, $Z \nvdash \mathrm{Con}(Z)$, where $\mathrm{Con}(Z)$ is the canonical consistency statement of the presentation, a Π_1 sentence.
- *(Diag)* For every formula $\vartheta(z, x)$ there is a formula $\Upsilon(x)$ with $\mathsf{Q} \vdash \forall x\,(\Upsilon(x) \leftrightarrow \vartheta(\ulcorner \Upsilon \urcorner, x))$.
- *(Uniq)* Let $\mathrm{Comp}(p, t, y)$ be Kleene's Δ_0 predicate "t codes the halting computation of V on the program p, with output y". Each halting computation has a unique code, and $\mathsf{PA} \vdash \forall p, t, t', y, y'\,(\mathrm{Comp}(p, t, y) \wedge \mathrm{Comp}(p, t', y') \to t = t' \wedge y = y')$.

Loss conventions. As Definition 2.5 requires, every loss statement in Section 5 names its convention. Unless stated otherwise losses are in the semimeasure convention: the cumulative loss of D_n is $L_A(D_n) - L_A(\emptyset)$, and $-\log W_A(D_n) + \log W_A(\emptyset)$ for the coin-free version. The lower bounds Theorem 5.1(c), Corollary C.6, Corollary 5.9 and Theorem 5.19 are statements in this convention. Upper bounds transfer to the normalised convention, since each normalised conditional is at least the raw one; lower bounds do not, and Proposition 5.10 shows that the lower bound of Corollary 5.9 fails in the normalised convention for most strings. In the three-valued convention the conditional probability of the label 1 on a query φ after two-valued data D is $\mathrm{Bel}_D(\varphi)$, so a lower bound on $-\log \mathrm{Bel}_D(\varphi)$ is a lower bound on the three-valued cost of a true label. Statements about K_A, K_{fin} and the posterior (P_∞, Bel, Dis, Indep) depend on no convention.

C.2 The lottery

Lemma C.1 (Lebesgue density on Cantor space)**.** *Let $\mathcal{X}$ be a Borel set of tapes with $\mathbb{U}(\mathcal{X}) > 0$ and let $\varepsilon \in (0,1)$. Then there is a finite string w with $\mathbb{U}(\mathcal{X} \mid [w]) > 1 - \varepsilon$.*

Proof. By outer regularity there is an open $O \supseteq \mathcal{X}$ with $\mathbb{U}(O) < \mathbb{U}(\mathcal{X})/(1-\varepsilon)$. Write $O = \bigsqcup_i [w_i]$ with $\{w_i\}$ prefix-free. If $\mathbb{U}(\mathcal{X} \cap [w_i]) \le (1-\varepsilon)2^{-|w_i|}$ for every i, summing gives $\mathbb{U}(\mathcal{X}) \le (1-\varepsilon)\mathbb{U}(O) < \mathbb{U}(\mathcal{X})$, a contradiction. □

Proof of Theorem 5.1. The random lottery theory is consistent. Take a model $\mathfrak{M}$ of B in which the numerals denote pairwise distinct elements, and for a tape ω put $\mathrm{Win}^{\mathfrak{M}} := \{(\underline{k}^{\mathfrak{M}}, \underline{J_k}^{\mathfrak{M}}) : k \in \mathbb{N}\}$. Because the numerals are distinct, the only Win-partner of $\underline{k}^{\mathfrak{M}}$ is $\underline{J_k}^{\mathfrak{M}}$, so every sentence G_{lot} emits holds; B does not mention Win. So $B \cup A_\omega^{G_{\mathrm{lot}}}$ is consistent for every ω.

The stream. Let $\mathsf{W}_0, \mathsf{W}_1, \ldots$ enumerate the r.e. sets of $\mathcal{L}$-sentences. For each e define $j_e^* \in \{0, \ldots, N_e\}$: if $B \cup \mathsf{W}_e$ is consistent and proves Lot_e, let j_e^* be the least $j \le N_e$ with $B \cup \mathsf{W}_e \nvdash \neg\mathrm{Win}(\underline{e}, \underline{j})$; otherwise $j_e^* := 0$. In the first case such a j exists: if $B \cup \mathsf{W}_e$ proved $\neg\mathrm{Win}(\underline{e}, \underline{j})$ for all $j \le N_e$, then with the second conjunct of Lot_e and the equality axioms it would prove $\forall x\, \neg\mathrm{Win}(\underline{e}, x)$, contradicting the first conjunct and its consistency. Put

$$D_\infty := (\mathrm{Lot}_0, 1),\ (\mathrm{Win}(\underline{0}, \underline{j_0^*}), 0),\ (\mathrm{Lot}_1, 1),\ (\mathrm{Win}(\underline{1}, \underline{j_1^*}), 0),\ \ldots$$

"$B \cup \mathsf{W}_e$ is consistent" is Π_1, and "$B \cup \mathsf{W}_e \vdash \mathrm{Lot}_e$" and "$B \cup \mathsf{W}_e \vdash \neg\mathrm{Win}(\underline{e}, \underline{j})$" are Σ_1, uniformly; so $e \mapsto j_e^*$ and the stream are computable from $\emptyset'$.

(a) Let $\mathrm{Surv} := \{\omega : J_k(\omega) \neq j_k^* \text{ for all } k\}$. The J_k are independent and uniform on $\{0, \ldots, N_k\}$, so by continuity from above $\mathbb{U}(\mathrm{Surv}) = \prod_k (1 - 2^{-(k+1)})$, which is positive because $\sum_k 2^{-(k+1)} < \infty$. For $\omega \in \mathrm{Surv}$ the theory $B \cup A_\omega^{G_{\mathrm{lot}}}$ is consistent, emits every Lot_k, and emits $\neg\mathrm{Win}(\underline{k}, \underline{j_k^*})$ because $j_k^* \le N_k$ and $j_k^* \neq J_k$. So $\nu_{G_{\mathrm{lot}}}(D_n) \ge \mathbb{U}(\mathrm{Surv})$ for all n, and (Dom) gives the bound. The rest is (Lim), with $M_A(\emptyset) \le 1$ for the bound on the cumulative loss.

(b) If $B \cup \mathsf{W}_e$ is consistent and proves every member of $D_\infty^{\pm}$, it proves Lot_e, so j_e^* was chosen with $B \cup \mathsf{W}_e \nvdash \neg\mathrm{Win}(\underline{e}, \underline{j_e^*})$; but that sentence is in $D_\infty^{\pm}$.

(c) $W_A(D_n) = \sum_e \mathbf{m}(e)\mathbf{1}[B \cup \mathsf{W}_e \text{ consistent}, \vdash \psi_{D_n}]$. Each indicator is non-increasing in n and, by (b), eventually 0; dominated convergence ($\sum_e \mathbf{m}(e) \le 1$) gives $W_A(D_n) \to 0$, and $K_A \ge -\log W_A$ by (Chain). For a single sentence ψ of $\mathcal{L}'$ with $B \cup \{\psi\}$ consistent, its $\mathcal{L}$-content $\mathrm{Cont}(\psi) := \{\varphi \in \mathrm{Sent}_{\mathcal{L}} : B \cup \{\psi\} \vdash \varphi\}$ is r.e., because $\mathcal{L}'$ has computable syntax, and $B \cup \mathrm{Cont}(\psi)$ is consistent; so by (b) $\mathrm{Cont}(\psi) \not\supseteq D_\infty^{\pm}$, and every consistent ψ fails some datum. Only finitely many ψ have $K(\psi) \le j$, so $K_{\mathrm{fin}}^{\mathcal{L}'}(D_n) \to \infty$.

(d) Let P_{det} be the set of $p \in \mathrm{dom}\, V$ with $G_{V(p)} \subseteq \{\epsilon\} \times \mathrm{Sent}$, and $A^{(p)}$ the set that $G_{V(p)}$ emits. Then

$$\mathbb{U}\{\omega \in E_{D_n} : \omega \text{ begins with some } p \in P_{\mathrm{det}}\} = \sum_{p \in P_{\mathrm{det}}} 2^{-|p|}\mathbf{1}[B \cup A^{(p)} \text{ consistent}, \vdash \psi_{D_n}] \to 0$$

by (b) and dominated convergence; divide by $M_A(D_n)$, which is bounded below by (a). If $\omega \in E_\infty$ and $\mathrm{Cn}(B \cup A_\omega)$ were r.e., it would be a consistent r.e. theory proving every member of $D_\infty^{\pm}$, contrary to (b). □

A computation (code in the accompanying repository) confirms the survival probabilities $\prod_{k<K}(1 - 2^{-(k+1)})$ for $K \le 5$ by enumerating winner vectors, and runs the diagonal against random lists of theories and against an adversarial list in which each listed theory survives until its own lottery.

Remark C.2 (Which streams, which variant). (i) A stream with $\mathbb{U}(E_\infty) > 0$ that no consistent r.e. theory proves is not r.e.: otherwise $D^{\pm}_{\infty}$ itself would be such a theory. For the lottery classes it is not co-r.e. either (Proposition C.4). The streams constructed here are Δ_2; an arbitrary stream with these properties need not be arithmetical (for instance, a diagonal against a non-arithmetical list). (ii) The phenomenon needs proof-requiring compatibility: the r.e. theory $\{\mathrm{Lot}_k : k \in \mathbb{N}\}$ is consistent with every finite part of D_∞, so under the consistency-requiring variant it stays compatible forever. Proof-requiring compatibility demands that each reported loser be *derived*. (iii) On the stream of Theorem 5.1 every random *finite* theory, in any extended language with computable syntax, fails, because its content is r.e.; so random r.e. theories strictly beat random finite theories there.

Relatives of the lottery. The closest relative in computability theory is that randomness helps for mass problems: a class of positive measure can have no computable member. For a computable bound q, the q-bounded diagonally non-computable functions ($f(n) < q(n)$, and $f(n)$ differs from the value at n of a fixed partial computable diagonal function) form a Π^0_1 class with no computable member; under the uniform product measure on $\prod_n \{0, \dots, q(n)-1\}$ it has measure at least $\prod_n (1 - 1/q(n))$, which is positive exactly when $\sum_n 1/q(n) < \infty$. Every Martin-Löf random sequence computes a diagonally non-computable function (Kučera, 1985; see Bienvenu and Patey, 2017), and Bienvenu and Porter (2016) show that the bounded classes are not tt-negligible when $\sum_n 1/q(n) < \infty$ and are deep when the sum diverges. The lottery's survival condition $\sum_k 1/(N_k+1) = \sum_k 2^{-(k+1)} < \infty$ is this convergence condition, and the diagonal ticket j^*_k plays the role of the diagonal value. The differences: the lottery's diagonal is computed from $\emptyset'$, against r.e. theories rather than partial computable functions; what fails are consistent r.e. theories proving the stream, not computable members of a class; and the class of surviving tapes is the inducer's compatibility event, so its measure is the inducer's likelihood. The modified Demski prior (Garrabrant, 2015; Garrabrant et al., 2016b), which also draws random axiom systems from a universal machine, gives the lottery stream positive probability as well, for the process as described in those works: with probability at least $2^{-K(G_{\mathrm{lot}})-c} \prod_k (1 - 2^{-(k+1)})$ its first draw is a surviving lottery theory, every later draw is kept only if consistent with it, and the final theory extends it. Its completed theories containing the stream are not r.e., but completions of extensions of Q never are, so the contrast between random and r.e. hypotheses is not visible in that prior.

Proof of Proposition 5.2. (a) The argument is the majority vote of Lemma A.12, with a threshold adapted to the arity. Put $\varepsilon := 1/(r+1)$ and choose w by Lemma C.1 with $\mathbb{U}(E_\infty \mid [w]) > 1 - \varepsilon$. Let $T := \{\varphi \in Z : \mathbb{U}(O_\varphi \mid [w]) > 1 - \varepsilon\}$; it is r.e., because $\mathbb{U}(O_\varphi \mid [w])$ is left-c.e. uniformly in φ. For $\varphi \in D^{\pm}_{\infty}$ we have $O_\varphi \supseteq E_\infty$, so $\varphi \in T$. Suppose $B \cup T$ is inconsistent. By compactness and the hypothesis, some $\varphi_1, \dots, \varphi_j \in T$ with $j \le r$ are jointly B-inconsistent. Then $\mathbb{U}(\bigcap_i O_{\varphi_i} \mid [w]) > 1 - j\varepsilon \ge 1 - r\varepsilon = \varepsilon$, and every tape in that intersection has an inconsistent theory, so $\mathbb{U}(O_\perp \mid [w]) > \varepsilon$. But $O_\perp \cap E_\infty = \emptyset$ gives $\mathbb{U}(O_\perp \mid [w]) < \varepsilon$, a contradiction. So T is consistent, and $\inf_n W_A(D_n) \ge \mathbf{m}(e) > 0$ for an index e of T.

(b) $\mathbb{U}\{\omega : f_\omega \supseteq D_\infty\} = \lim_n M_F(D_n) > 0$ by continuity from above. Choose w with $\mathbb{U}(\{\omega : f_\omega \supseteq D_\infty\} \mid [w]) > \frac{1}{2}$ and let $f(x) := y$ iff $\mathbb{U}(\{\omega : f_\omega(x) = y\} \mid [w]) > \frac{1}{2}$. The events $\{f_\omega(x) = y\}$ are effectively open and disjoint in y, so f is a well-defined partial computable function; for $x \in \operatorname{dom} D_\infty$ the event $\{f_\omega(x) = D_\infty(x)\}$ contains $\{f_\omega \supseteq D_\infty\}$, so $f(x) = D_\infty(x)$.

The threshold is exact for the argument. Let $\{\varphi_1, \dots, \varphi_s\}$ be a minimal B-inconsistent set, and consider a law with inconsistent mass a and mass $(1-a)/s$ on each of s consistent types,

type i proving exactly the φ_j with $j \neq i$ (and their consequences). Then every φ_j is proved with probability $a + (s-1)(1-a)/s$, so the majority set at threshold $1-\varepsilon$ contains all of $\varphi_1, \ldots, \varphi_s$ iff $a > 1 - s\varepsilon$, while the consistent mass exceeds $1-\varepsilon$ iff $a < \varepsilon$. Both hold for some a iff $\varepsilon > 1/(s+1)$. So $1/(r+1)$ is the largest threshold at which the argument is guaranteed to return a consistent set, and a family of minimal inconsistent sets of unbounded size defeats every threshold. A computation (code in the accompanying repository) confirms the critical value for $s \leq 12$, and a linear program over all laws, for each of the 166 nonempty families of minimal inconsistent sets on four sentences, finds the same critical value. □

Instances. Literal data of a fresh predicate ($r = 2$, by Lemma A.4). Plain SOI, by (b). For the lottery class the sets $\{\mathrm{Lot}_k\} \cup \{\neg\mathrm{Win}(\underline{k}, \underline{j}) : j \leq N_k\}$ are minimal inconsistent, of size $N_k + 2$, so no r works; and for every threshold $1 - \varepsilon$ the majority set refutes every ticket of each lottery k with $2^{k+1} > 1/\varepsilon$, so it is inconsistent: this is the formal content of Kyburg's lottery paradox.

Proposition C.3 (Majority over true theories)**.** *Let $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$ and $B \subseteq \mathrm{Th}(\mathbb{N})$ decidable. If $\mathcal{T}_{\mathrm{true}} := \{\omega : B \cup A_\omega \subseteq \mathrm{Th}(\mathbb{N}),\ B \cup A_\omega \vdash D^{\pm}_{\infty}\}$ has positive measure, then some r.e. $T \subseteq \mathrm{Th}(\mathbb{N})$ has $B \cup T \vdash D^{\pm}_{\infty}$.*

Proof. $\mathcal{T}_{\mathrm{true}}$ is Borel: $\{\omega : A_\omega \subseteq \mathrm{Th}(\mathbb{N})\} = \bigcap_{\chi \text{ false}} \{\omega : \chi \notin A_\omega\}$ is closed, and the proving condition is a countable intersection of open sets. Choose w with $\mathbb{U}(\mathcal{T}_{\mathrm{true}} \mid [w]) > \frac{1}{2}$ by Lemma C.1 and let $T := \{\varphi : \mathbb{U}(O_\varphi \mid [w]) > \frac{1}{2}\}$, an r.e. set. It contains $D^{\pm}_{\infty}$, since $O_\varphi \supseteq \mathcal{T}_{\mathrm{true}}$ for $\varphi \in D^{\pm}_{\infty}$. For $\varphi \in T$, $\mathbb{U}(O_\varphi \cap \mathcal{T}_{\mathrm{true}} \mid [w]) > 0$, so a true theory proves φ and φ is true. Hence $T \subseteq \mathrm{Th}(\mathbb{N})$ and $B \cup T$ proves $D^{\pm}_{\infty}$. □

Truth is a constraint of arity one; this is why the majority argument transfers to it, while Σ_1-soundness, which is not a constraint of arity one, escapes it (Question C.9).

Proposition C.4 (Co-r.e. streams over block-decidable classes)**.** *Let $Z \subseteq \mathrm{Sent}$ be decidable and computably partitioned into finite blocks $Z_0, Z_1, \ldots$ (each given by a canonical index) such that (i) for every finite $X \subseteq Z$, $B \cup X$ is consistent iff $B \cup (X \cap Z_k)$ is consistent for every k, and (ii) consistency of $B \cup X$ for finite $X \subseteq Z_k$ is decidable, uniformly in k. If $D^{\pm}_{\infty} \subseteq Z$ is co-r.e. and $B \cup D^{\pm}_{\infty}$ is consistent, then some consistent decidable $T \subseteq Z$ contains $D^{\pm}_{\infty}$.*

Proof. $Z_k \setminus D^{\pm}_{\infty}$ is r.e. uniformly in k; let C^s_k be Z_k minus what has been enumerated into it by stage s. As Z_k is finite, $C^s_k = D^{\pm}_{\infty} \cap Z_k$ for large s, and that set is consistent with B; so by (ii) the least s_k with $B \cup C^{s_k}_k$ consistent exists and is computable from k. Put $T := \bigcup_k C^{s_k}_k$, a decidable set. $C^s_k \supseteq D^{\pm}_{\infty} \cap Z_k$ for all s, so $T \supseteq D^{\pm}_{\infty}$, and $B \cup T$ is consistent by compactness and (i). □

Instances. (1) The lottery class $Z = \{\mathrm{Lot}_k\} \cup \{\mathrm{Win}(\underline{k}, \underline{j})^b : j \leq N_k,\ b \in \{0,1\}\}$ with blocks by k: under the hypotheses of Theorem 5.1 a finite $X \subseteq Z$ is consistent with B iff no block contains a complementary pair, or Lot_k together with all $N_k + 1$ negative literals of lottery k (choose the Win-row of each $\underline{k}$ separately in a model with distinct numerals); so (i) and (ii) hold. A computation (code in the accompanying repository) checks this criterion against satisfiability in all structures with at most five elements, for $B = \emptyset$ and two blocks. (2) The arithmetic lottery classes of Theorem 5.5: by Lemma C.8(i) every propositionally satisfiable finite set of Boolean combinations of the tickets is consistent with B, so consistency is a propositional, decidable and blockwise property. Hence every stream of these kinds that answers the question of Corollary 5.3 negatively is neither r.e. (Remark C.2) nor co-r.e. Whether some co-r.e. stream over a sentence class on which consistency is not decidable has $\mathbb{U}(E_\infty) > 0$ and no consistent r.e. theory proving it is open (it is the second part of Question 5.4). The adversary in the proof waits, in each block,

for a stage at which the remaining data are certifiably consistent; a counterexample must deny it any such certificate at every finite stage.

Proof of Corollary 5.3. Theorem 5.1(a),(b) give the negative answer, and Theorem 5.5 gives one inside true arithmetic. The positive cases are Proposition 5.2(a), Corollary C.13 (for a stream of Π_1 type, bounded loss gives $\sup_n L_A(D_n) \le k$ for some k), Proposition C.4 (with $B \cup D_\infty^{\pm}$ consistent because $\mathbb{U}(E_\infty) > 0$), and Proposition C.3. □

For a theory machine G and a finite string u, A_u^G is the set of axioms emitted under prefixes of u (Definition 2.3).

Proposition C.5 (The proof horizon bounds the coin-free loss)**.** *Let G be a theory machine and D_∞ a stream with*

$$a_G := \mathbb{U}\{\omega : B \cup A_\omega^G \text{ consistent},\ B \cup A_\omega^G \vdash D_\infty^{\pm}\} > 0.$$

Let the proof horizon *$h_G(n)$ be the least m with $\mathbb{U}\{\omega : B \cup A_\omega^G$ consistent, $B \cup A_{\omega_{1:m}}^G \vdash \psi_{D_n}\} \ge a_G/2$. Then, in the semimeasure convention, for every n,*

$$-\log W_A(D_n) \ \le\ K^{\mathrm{pad}}(h_G(n)) + K(G) + \log(2/a_G) + c.$$

Proof. $h_G(n)$ exists: a derivation of ψ_{D_n} uses finitely many axioms, all emitted under some finite prefix, so the events in the definition increase in m to a set containing the one that defines a_G. Fix $m \ge h_G(n)$ and let Good_m be the set of $u \in \{0,1\}^m$ such that $B \cup A_u^G \vdash \psi_{D_n}$ and $[u]$ contains a tape ω with $B \cup A_\omega^G$ consistent. For $u \in \mathrm{Good}_m$, $A_u^G \subseteq A_\omega^G$ for such an ω, so $B \cup A_u^G$ is consistent and proves ψ_{D_n}; and $|\mathrm{Good}_m| 2^{-m} \ge a_G/2$. The set A_u^G is r.e. with an index $e(G,u)$ computable from (G,u) and injective in u (write u into the index). Hence, using $K(e(G,u)) \le K(G) + K(u) + c$ and $K(u) \le m + K(m) + c$ for $|u| = m$,

$$W_A(D_n) \ge \sum_{u \in \mathrm{Good}_m} \mathbf{m}(e(G,u)) \ge \sum_{u \in \mathrm{Good}_m} 2^{-K(G)-m-K(m)-c} \ge \tfrac{a_G}{2}\, 2^{-K(G)-K(m)-c}.$$

Minimise $K(m)$ over $m \ge h_G(n)$. □

Corollary C.6 (Rate on the lottery stream)**.** *On the stream of Theorem 5.1, in the semimeasure convention:*

(a) $-\log W_A(D_n) \le K^{\mathrm{pad}}(n) + c$ *for all n. Hence the coin-free loss diverges, but more slowly than any computable unbounded nondecreasing function: for every such f, $-\log W_A(D_n) \le \frac{2}{3} f(n) + c_f$.*

(b) $K_A(D_n) \le \lceil n/2 \rceil + O(\log n)$*, and $K_A(D_n) \to \infty$.*

(c) Batch lottery. *Give lottery k the 2^{2k+1} tickets $0, \dots, 2^{2k+1} - 1$ and diagonalise in it against every $e < 2^k$ at once. Then Theorem 5.1(a)–(d) hold with the same product, $W_A(D_n) \le \sum_{e \ge 2^k} \mathbf{m}(e)$ once lottery k is complete, and again $-\log W_A(D_n) \le K^{\mathrm{pad}}(n) + c$.*

The restriction to nondecreasing functions in (a) is needed: a computable unbounded function that vanishes on the odd numbers is not eventually above K^{pad}.

Proof. (a) For $G = G_{\mathrm{lot}}$, every tape gives a consistent theory, and $a_G = \mathbb{U}(\mathrm{Surv}) > 0$. For $\omega \in \mathrm{Surv}$ the axioms needed for D_n are emitted once the first $\lceil n/2 \rceil$ blocks are read, that is, by the prefix of length $m(n) := \sum_{k<\lceil n/2 \rceil}(k+1)$; so $h_{G_{\mathrm{lot}}}(n) \le m(n)$, which is computable from n, and $K^{\mathrm{pad}}(h_{G_{\mathrm{lot}}}(n)) \le K(m(n)) \le K(n)+c$. Proposition C.5 gives $-\log W_A(D_n) \le K(n)+c'$. As $D_n \subseteq D_{n'}$ for $n \le n'$, $W_A(D_n) \ge W_A(D_{n'})$, so $-\log W_A(D_n) \le \min_{n' \ge n} K(n')+c' = K^{\mathrm{pad}}(n)+c'$.
Slow growth. For a computable nondecreasing unbounded g let $\eta_g(j)$ be the least n' with $g(n') \ge j$, a computable function. Then $\eta_g(g(n)+1) > n$, so $K^{\mathrm{pad}}(n) \le K(\eta_g(g(n)+1)) \le 2\log(g(n)+1)+c$. For a computable unbounded nondecreasing f, apply this to $g := \lfloor 2^{f/3} \rfloor$, which gives $K^{\mathrm{pad}}(n) \le \frac{2}{3}f(n) + c_f$. Divergence is Theorem 5.1(c).

(b) Let $K := \lceil n/2 \rceil$. The tickets j^*_k, $k < K$, are determined by the truth values of the Σ_1 facts "$B \cup \mathsf{W}_k \vdash \mathrm{Lot}_k$" and "$B \cup \mathsf{W}_k \vdash \neg\mathrm{Win}(\underline{k}, \underline{j})$" ($j \le N_k$): j^*_k is the least unrefuted ticket if W_k proves Lot_k and some ticket is unrefuted, and 0 otherwise (an inconsistent W_k refutes every ticket). There are $2^{K+1}+K-2$ such facts. Given n and the number of true ones ($K+2$ bits, given K), enumerate proofs until that many facts are found true. So $K(\psi_{D_n}) \le K + O(\log n)$, and $K_A(D_n) \le K(\psi_{D_n}) + c$ by Remark 2.8(c). $K_A(D_n) \to \infty$ is Theorem 5.1(c).

(c) The data of lottery k are $(\mathrm{Lot}_k, 1)$ and $(\mathrm{Win}(\underline{k}, \underline{j^*_{k,e}}), 0)$ for $e < 2^k$, where Lot_k now names the tickets $0, \dots, 2^{2k+1}-1$ and $j^*_{k,e}$ is the least ticket that $B \cup \mathsf{W}_e$ does not refute if $B \cup \mathsf{W}_e$ is consistent and proves Lot_k, and 0 otherwise. The random theory reads a block of $2k+1$ bits for lottery k. At most 2^k tickets are reported, so it survives lottery k with probability at least $1 - 2^k/2^{2k+1} = 1 - 2^{-(k+1)}$, and the proof of Theorem 5.1 goes through with the same product. After lottery k every $e < 2^k$ has failed, which gives the bound on W_A. At most n lotteries are complete after n data, so the horizon is at most $\sum_{k<n}(2k+1) = n^2$, which is computable from n, and (a)'s argument applies. □

A computation (code in the accompanying repository) checks the bookkeeping of (b) and the survival probabilities of the batch lottery for three lotteries.

Corollary C.7 (The lottery separates the versions of consistent SOI)**.** *On the stream of Theorem 5.1: (a) $M^{\mathrm{cons}}_F(D_n)$ is bounded below, so one-tape consistent SOI has bounded loss; (b) $W^{\mathrm{cons}}(D_n) \to 0$ and $W_A(D_n) \to 0$, so coin-free consistent SOI has unbounded loss; (c) plain coin-free SOI on sentences has bounded loss: $W^{\mathrm{all}}(D_n) \ge \mathbf{m}(f) > 0$ for a partial computable f extending D_∞, and every such f is inconsistent.*

Proof. Theorem 4.3 gives $M^{\mathrm{cons}}_F \asymp M_A$, and Theorem 4.2 gives $W^{\mathrm{cons}} \asymp W_A$. (a) $M^{\mathrm{cons}}_F(D_n) \ge 2^{-c} M_A(D_n)$ is bounded below by Theorem 5.1(a). (b) $W^{\mathrm{cons}}(D_n) \le 2^c W_A(D_n) \to 0$ by Theorem 5.1(c). (c) $M_F(D_n) \ge M^{\mathrm{cons}}_F(D_n)$ is bounded below, so by Proposition 5.2(b), applied with inputs $\mathrm{Sent}_{\mathcal{L}}$ and labels $\{0,1\}$, some partial computable f extends D_∞, and $W^{\mathrm{all}}(D_n) \ge \mathbf{m}(f) > 0$. Such an f is inconsistent: a consistent one would give $W^{\mathrm{cons}}(D_n) \ge \mathbf{m}(f) > 0$, against (b). □

C.3 Lotteries in arithmetic

The lottery needs a family of sentences that behave, with respect to consistency with B, like independent atoms. Inside arithmetic with a true background, self-referential ("flexible") formulas provide them. Lemma C.8 is a self-contained proof of the weak form we need, in a version that also prescribes the truth values in $\mathbb{N}$. It is the standard self-referential construction behind Kripke's flexible formulas (Kripke, 1962) and Mostowski's independent sequences (Mostowski, 1961); Woodin's universal algorithm is a much stronger statement of the same kind (Woodin, 2011; Blanck and Enayat, 2017; Hamkins, 2018). We did not find the version with a prescribed

truth set stated in this form, but it is a routine variant, and we claim no novelty for it. For a code z let $\varphi_{[z]}$ be the formula with code z.

Lemma C.8 (Flexible formulas with prescribed truth)**.** *Let $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$ and let B be decidable and true with $B \supseteq \mathsf{Q}$. Write $\mathrm{In}(y, e)$ for a Δ_0 formula saying "y is in the finite set with canonical index e", and for a finite $F \subseteq \mathbb{N}$ with index e_F write $y \in F$ for $\mathrm{In}(y, \underline{e_F})$. For every formula $\Lambda(z, x)$ there is a formula $\Upsilon(x)$ such that:*

(i) $B + \forall y\,(\Upsilon(y) \leftrightarrow y \in F)$ *is consistent for every finite $F \subseteq \mathbb{N}$; hence every finite pattern $\{\Upsilon(\underline{n})^{b_n} : n \in F'\}$ is consistent with B;*

(ii) $\mathbb{N} \models \Upsilon(\underline{n})$ *iff* $\mathbb{N} \models \Lambda(\ulcorner\Upsilon\urcorner, \underline{n})$*, for every n;*

(iii) $\mathsf{Q} \vdash \forall x\,(\Upsilon(x) \leftrightarrow \sigma_\Upsilon(x) \vee (\pi_\Upsilon \wedge \Lambda(\ulcorner\Upsilon\urcorner, x)))$ *for a Σ_1 formula $\sigma_\Upsilon(x)$ that is false of every number and a true Π_1 sentence π_Υ.*

Proof. Fix a computable enumeration $\tau_0, \tau_1, \ldots$ of the theorems of B. Call a sentence *of form* (z, e) if it is $\neg\forall y\,(\varphi_{[z]}(y) \leftrightarrow \mathrm{In}(y, \underline{e}))$; whether τ_s is of form (z, e) for some e, and for which e, is decidable from (s, z). Let $\sigma_0(z, x)$ be the Σ_1 formula "there are s and e such that τ_s is of form (z, e), no $\tau_{s'}$ with $s' < s$ is of form (z, e') for any e', and $\mathrm{In}(x, e)$" (the computation of $\tau_0, \ldots, \tau_s$ is an existentially quantified witness, checked by a Δ_0 formula), and $\pi_0(z)$ the Π_1 formula "no τ_s is of form (z, e) for any e". By (Diag), applied to $\vartheta(z, x) := \sigma_0(z, x) \vee (\pi_0(z) \wedge \Lambda(z, x))$, take Υ with

$$\mathsf{Q} \vdash \forall x\,(\Upsilon(x) \leftrightarrow \sigma_0(\ulcorner\Upsilon\urcorner, x) \vee (\pi_0(\ulcorner\Upsilon\urcorner) \wedge \Lambda(\ulcorner\Upsilon\urcorner, x))).$$

As Q is true, the same equivalence holds in $\mathbb{N}$. Suppose some τ_s is of form $(\ulcorner\Upsilon\urcorner, e)$; let s_0 be the least such s, with index e_0 and set F_0. In $\mathbb{N}$, $\sigma_0(\ulcorner\Upsilon\urcorner, n)$ holds iff $n \in F_0$, and $\pi_0(\ulcorner\Upsilon\urcorner)$ fails; so $\mathbb{N} \models \forall y\,(\Upsilon(y) \leftrightarrow y \in F_0)$. But τ_{s_0}, the negation of this sentence, is a theorem of the true theory B, a contradiction. Hence no theorem of B is of form $(\ulcorner\Upsilon\urcorner, e)$, which is (i); the pattern follows from $\forall y\,(\Upsilon(y) \leftrightarrow y \in F)$ with $F := \{n \in F' : b_n = 1\}$, since Q decides each $\underline{n} \in F$ correctly by (Σ_1C). Moreover $\pi_0(\ulcorner\Upsilon\urcorner)$ is true and $\sigma_0(\ulcorner\Upsilon\urcorner, \underline{n})$ is false for every n, which gives (ii), and (iii) with $\sigma_\Upsilon(x) := \sigma_0(\ulcorner\Upsilon\urcorner, x)$ and $\pi_\Upsilon := \pi_0(\ulcorner\Upsilon\urcorner)$. □

Let $\langle k, j\rangle$ be a Δ_0 pairing with $k, j \le \langle k, j\rangle$.

Proof of Theorem 5.5. The tickets. In (a), Υ_0 is Lemma C.8's formula for $\Lambda(z, x) :\equiv (x \neq x)$. In (b), let $\mathrm{Ref}(z, k, j)$ be the Σ_1 formula "$B \cup \mathsf{W}_k \vdash \neg\varphi_{[z]}(\underline{\langle k, j\rangle})$", and let $\Lambda(z, x)$ be the formula, with all quantifiers outside Ref bounded,

$$\exists k \le x\, \exists t \le x \Big(x = \langle k, t\rangle \wedge t \le N_k \\ \wedge \big[(\mathrm{Ref}(z, k, t) \wedge \forall i < t\, \neg\mathrm{Ref}(z, k, i)) \vee (t = 0 \wedge \forall i \le N_k\, \neg\mathrm{Ref}(z, k, i))\big]\Big);$$

Υ_1 is the lemma's formula for this Λ. In both cases $\mathrm{Tick}_{k,j} := \Upsilon(\underline{\langle k, j\rangle})$ for the formula $\Upsilon \in \{\Upsilon_0, \Upsilon_1\}$, and $\mathrm{Lot}^{\mathrm{ar}}_k := \bigvee_{j \le N_k} \mathrm{Tick}_{k,j}$. In (a), j^*_e is defined as in Theorem 5.1, with $\mathrm{Tick}_{e,j}$ in place of $\mathrm{Win}(\underline{e}, \underline{j})$. In (b), let j°_k be the least $j \le N_k$ with $B \cup \mathsf{W}_k \vdash \neg\mathrm{Tick}_{k,j}$ ($j^\circ_k := 0$ if there is none), and let j^*_k be the least $j \le N_k$ with $j \neq j^\circ_k$ and $B \cup \mathsf{W}_k \nvdash \neg\mathrm{Tick}_{k,j}$ if $B \cup \mathsf{W}_k$ is consistent and proves $\mathrm{Lot}^{\mathrm{ar}}_k$, and the least $j \neq j^\circ_k$ otherwise. The stream is $((\mathrm{Lot}^{\mathrm{ar}}_k, 1), (\mathrm{Tick}_{k,j^*_k}, 0))_{k \ge 0}$.

The random theory. $G^{\mathrm{ar}}_{\mathrm{lot}}$ reads the tape in blocks as G_{lot} does and emits $\mathrm{Lot}^{\mathrm{ar}}_k$, Tick_{k,J_k} and $\neg\mathrm{Tick}_{k,j}$ ($j \le N_k$, $j \neq J_k$) after reading block k. A finite part of its output is implied by

$B + \forall y\,(\Upsilon(y) \leftrightarrow y \in F)$ with $F := \{\langle k, J_k\rangle : k \le K\}$ for some K, so by Lemma C.8(i) and compactness its theory is consistent with B on every tape.

The diagonal. If $B \cup \mathsf{W}_e$ is consistent and proves $\mathrm{Lot}_e^{\mathrm{ar}}$, it does not refute every $\mathrm{Tick}_{e,j}$, $j \le N_e$, since $\mathrm{Lot}_e^{\mathrm{ar}} \wedge \bigwedge_j \neg\mathrm{Tick}_{e,j}$ is propositionally contradictory. In (b), j_e^* exists in this case: if W_e refutes some ticket, then j_e° is refuted and every unrefuted ticket differs from j_e°; if it refutes none, then $j_e^\circ = 0$ and $j = 1$ is unrefuted ($N_e \ge 1$). Everything else is the proof of Theorem 5.1 verbatim, with Tick for Win: survival probability $\prod_k (1 - 2^{-(k+1)})$, the diagonal kills every consistent r.e. theory, dominated convergence, and the posterior on deterministic machines.

Truth and complexity. (a) By Lemma C.8(ii) with $\Lambda \equiv (x \neq x)$, every $\mathrm{Tick}_{k,j}$ is false, and by (iii) $\mathsf{Q} \vdash \mathrm{Tick}_{k,j} \leftrightarrow \sigma_{\Upsilon_0}(\underline{\langle k, j\rangle})$, since the second disjunct is Q-refutable: a Σ_1 sentence. So $\mathrm{Lot}_k^{\mathrm{ar}}$ is Q-equivalent to a false Σ_1 sentence and $\neg\mathrm{Tick}_{k,j_k^*}$ to a true Π_1^* sentence. (b) By Lemma C.8(ii), $\mathrm{Tick}_{k,j}$ is true iff $j = j_k^\circ$: for $z = \ulcorner \Upsilon_1 \urcorner$, $\mathrm{Ref}(z, k, j)$ holds iff $B \cup \mathsf{W}_k \vdash \neg\mathrm{Tick}_{k,j}$, and $\Lambda(z, \underline{\langle k, j\rangle})$ says exactly that j is the least refuted ticket, or 0 if none is refuted. Hence $\mathrm{Lot}_k^{\mathrm{ar}}$ is true, and Tick_{k,j_k^*} is false because $j_k^* \neq j_k^\circ$; every datum is true. By (iii), $\mathrm{Tick}_{k,j}$ is Q-equivalent to $\sigma_{\Upsilon_1}(\underline{\langle k, j\rangle}) \vee (\pi_{\Upsilon_1} \wedge \Lambda(\ulcorner \Upsilon_1 \urcorner, \underline{\langle k, j\rangle}))$. With a numeral argument, (O) and (Σ_1C) unfold the bounded quantifiers of Λ and decide its Δ_0 parts, provably in Q ($t \le N_k$ is Δ_0 because the graph of 2^x is); so $\Lambda(\ulcorner \Upsilon_1 \urcorner, \underline{\langle k, j\rangle})$ is Q-equivalent to a Boolean combination of the Σ_1 sentences $\mathrm{Ref}(\ulcorner \Upsilon_1 \urcorner, \underline{k}, \underline{i})$, $i \le N_k$. Hence every datum is Q-equivalent to a Boolean combination of Σ_1 sentences. □

Proof of Corollary 5.6. (i) Theorem 5.5(b) with Theorem 5.1(a),(c). (ii) If $\mathbb{U}\{\omega \in E_\infty : B \cup A_\omega \text{ true}\} > 0$, Proposition C.3 gives a true r.e. theory proving the stream, contrary to Theorem 5.1(b). (iii) Fix k with $J := J_k \neq j_k^\circ$. The random lottery theory proves $\mathrm{Tick}_{k,J}$, which is Q-equivalent to $\sigma_{k,J} \vee (\pi_{\Upsilon_1} \wedge \Lambda_{k,J})$ with $\sigma_{k,J} := \sigma_{\Upsilon_1}(\underline{\langle k, J\rangle})$ a false Σ_1 sentence and $\Lambda_{k,J} := \Lambda(\ulcorner \Upsilon_1 \urcorner, \underline{\langle k, J\rangle})$. Write Ref_i for $\mathrm{Ref}(\ulcorner \Upsilon_1 \urcorner, \underline{k}, \underline{i})$. If some ticket is refuted and $J > j_k^\circ$, then $\mathrm{Ref}_{j_k^\circ}$ is a true Σ_1 sentence, so $\mathsf{Q} \vdash \mathrm{Ref}_{j_k^\circ}$ and $\mathsf{Q} \vdash \neg\Lambda_{k,J}$, and the theory proves the false Σ_1 sentence $\sigma_{k,J}$. If $J < j_k^\circ$, or no ticket is refuted (then $j_k^\circ = 0 < J$), then Q proves $\Lambda_{k,J} \to \mathrm{Ref}_J$ (for $J = 0 < j_k^\circ$ through $\mathsf{Q} \vdash \mathrm{Ref}_{j_k^\circ}$, which refutes the "none refuted" disjunct), and Ref_J is false; so the theory proves $\sigma_{k,J} \vee \mathrm{Ref}_J$, a false Σ_1 sentence. Given survival ($J_k \neq j_k^*$ for all k), the events $J_k = j_k^\circ$ are independent with probabilities $1/N_k$, so all of them hold with probability $\prod_k 1/N_k = 0$. □

A computation (code in the accompanying repository) checks the bookkeeping of Theorem 5.5(b) and the three cases of (iii) exhaustively in a propositional model with up to eight tickets per lottery, for every behaviour of W_k and every winner.

Question C.9. On the stream of Theorem 5.5(b), is $P_\infty(\{\Sigma_1\text{-sound theories}\}) = 0$?

Corollary 5.6(ii) proves that the limit posterior lives on false theories, and (iii) that the random lottery theories among them are Σ_1-unsound; Proposition C.3's majority argument does not transfer to Σ_1-soundness, which is not a constraint of arity one.

C.4 Π_1 data

Throughout this subsection $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$ and B is decidable and true with $B \supseteq \mathsf{Q}$. Inc_0 is left-c.e. and not computable (Theorem 3.5(a)), hence irrational. Put $I'_k := \lfloor 2^{k+2}\mathrm{Inc}_0 \rfloor 2^{-(k+2)}$, so that $I'_k \le \mathrm{Inc}_0 < I'_k + 2^{-(k+2)}$; given k, I'_k is given by $k+2$ bits. If D is of Π_1 type, then ψ_D is B-equivalent to $\pi_D \wedge \sigma_D$ with $\pi_D \in \Pi_1^*$ (the class is closed under conjunction) and σ_D a conjunction of true Σ_1 sentences; $B \vdash \sigma_D$ by (Σ_1C); so $B \vdash \psi_D \leftrightarrow \pi_D$, $O_{\psi_D} = O_{\pi_D}$ and $M_A(D) = \mathrm{Bel}_0(\pi_D)$.

The definition is invariant under B-equivalence of literals, as it must be, since M_A, W_A and K_A depend on D only through $\mathrm{Cn}(D^{\pm})$ (Old).

Theorem C.10 (Threshold theories)**.** *There is c such that for every $k \geq 0$ the set*

$$T_k := \{\pi \in \Pi_1^* : \mathbb{U}(O_\pi) > I'_k + 2^{-(k+2)}\}$$

(i) is r.e., with an index e_k of complexity $K(e_k) \leq k + K(k) + c$; (ii) contains every $\pi \in \Pi_1^*$ with $\mathrm{Bel}_0(\pi) \geq 2^{-k}$; (iii) consists of true sentences of positive prior belief, so $B \cup T_k$ is true, consistent and Σ_1-sound.

Proof. (i) $\mathbb{U}(O_\pi)$ is left-c.e. uniformly in π, and Π_1^* is decidable, so T_k is r.e. uniformly in (k, I'_k); a program consists of a self-delimiting description of k, the $k+2$ bits of I'_k, and a fixed code. (ii) $\mathbb{U}(O_\pi) = \mathrm{Inc}_0 + \mathrm{Bel}_0(\pi) \geq I'_k + 2^{-k} > I'_k + 2^{-(k+2)}$. (iii) If $\mathbb{U}(O_\pi) > I'_k + 2^{-(k+2)} > \mathrm{Inc}_0$, then $\mathrm{Bel}_0(\pi) > 0$: some tape has $B \cup A_\omega$ consistent and proving π, and π is true by (Π_1T). □

Proof of Theorem 5.7. π is emitted on the tape r iff $\mathbb{U}(O_\pi) > 0.r_{1:s} + 2^{-s}$ for some s; as $0.r_{1:s} + 2^{-s}$ decreases strictly to r for non-dyadic r, this holds iff $\mathbb{U}(O_\pi) > r$. If $r < \mathrm{Inc}_0$, every $\pi \in \Pi_1^*$ has $\mathbb{U}(O_\pi) \geq \mathrm{Inc}_0 > r$, including false ones such as $\neg(0 = 0)$, so the theory is inconsistent. If $r > \mathrm{Inc}_0$, every emitted π has $\mathrm{Bel}_0(\pi) > 0$ and is true by (Π_1T); B is true; so $B \cup A_r^{G_{\mathrm{thr}}}$ is true. For D of Π_1 type: if $\mathrm{Inc}_0 < r < \mathbb{U}(O_{\pi_D})$, the theory is consistent and contains π_D, and $B \vdash \psi_D \leftrightarrow \pi_D$; the dyadic tapes form a null set, so $\nu_{G_{\mathrm{thr}}}(D) \geq \mathbb{U}(O_{\pi_D}) - \mathrm{Inc}_0 = \mathrm{Bel}_0(\pi_D) = M_A(D)$. The upper bound is (Dom), with $K(\nu_{G_{\mathrm{thr}}}) \leq K(G_{\mathrm{thr}}) + c$. □

A computation (code in the accompanying repository) checks Theorems 5.7 and C.10 in finite models of the prior.

Proof of Corollary 5.8. The first two inequalities are (Chain). For the last, let $k := \lceil L_A(D) \rceil$, so that $\mathrm{Bel}_0(\pi_D) = M_A(D) \geq 2^{-k}$. By Theorem C.10, $\pi_D \in T_k$, and $B \cup T_k$ is consistent and proves π_D, hence ψ_D. So $K_A(D) \leq K(e_k) + c \leq k + K(k) + c$. □

Remark C.11 (The coin-free version sums over levels). For every D of Π_1 type, $-\log W_A(D) \leq L_A(D) + K^{\mathrm{pad}}(\lceil L_A(D) \rceil + 1) + c$. This is a bound on $-\log W_A$; the proved bound on K_A is the weaker $L_A + K(\lceil L_A \rceil) + c'$ of Corollary 5.8.

Proof. For a pair (j, L) with $L \geq 0$ and $0 \leq j \leq 2^L$ let $Z_{j,L} := \{\pi \in \Pi_1^* : \mathbb{U}(O_\pi) > j2^{-L}\}$. It is r.e., with an index $e(j, L)$ computable from the *pair* and injective in the pair (write the pair into the index), and $K(e(j, L)) \leq K(j, L) + c \leq L + K(L) + c'$, since j has $L + 1$ bits given L. If $j2^{-L} \in (\mathrm{Inc}_0, \mathbb{U}(O_{\pi_D}))$, then every member of $Z_{j,L}$ has $\mathrm{Bel}_0 > 0$ and is true by (Π_1T), so $B \cup Z_{j,L}$ is true, contains π_D and proves ψ_D. With $k := \lceil L_A(D) \rceil$ the open interval $(\mathrm{Inc}_0, \mathbb{U}(O_{\pi_D}))$ has length $M_A(D) \geq 2^{-k}$, so for each $L \geq k + 1$ it contains at least $2^{L-k} - 1 \geq 2^{L-k-1}$ multiples of 2^{-L}. The indices $e(j, L)$ are distinct for distinct pairs, so

$$W_A(D) \;\geq\; \sum_{L \geq k+1} 2^{L-k-1}\, 2^{-L-K(L)-c'} \;=\; 2^{-k-1-c'} \sum_{L \geq k+1} 2^{-K(L)} \;\geq\; 2^{-k-1-c'-K^{\mathrm{pad}}(k+1)}.$$

Indexing the sets by the rational $j2^{-L}$ instead of by the pair would count $j2^{-L} = 2j2^{-(L+1)}$ once at every finer level, and the displayed sum would not bound W_A as written; a computation (code in the accompanying repository) confirms that the pair-indexed count meets the bound on all test intervals. □

Question C.12. On data of Π_1 type, is $K_A \le L_A + O(1)$?

The deterministic threshold needs the level written in full; the random threshold of Theorem 5.7 does not.

Corollary C.13 (Bounded loss on a Π_1 stream is one true theory)**.** *If every finite part of D_∞ is of Π_1 type and $\sup_n L_A(D_n) \le k$, then one true theory, $B \cup T_k$, of complexity at most $k + K(k) + c$, proves every datum; so $\sup_n -\log W_A(D_n) \le k + K(k) + c$.*

Proof. For each n, $\mathrm{Bel}_0(\pi_{D_n}) = M_A(D_n) \ge 2^{-k}$, so $\pi_{D_n} \in T_k$ by Theorem C.10(ii); T_k does not depend on n. By (iii), $B \cup T_k$ is true, and it proves every ψ_{D_n}; its index has complexity at most $k + K(k) + c$ by (i), which bounds $\sup_n -\log W_A(D_n)$ through Definition 2.7. □

Proof of Corollary 5.9. The deterministic bound (Chaitin). If $B \cup \mathsf{W}_e$ is consistent and proves $\mathrm{Kgt}_m(x)$, the program "given e and m, enumerate the theorems of $B \cup \mathsf{W}_e$ and output the first z such that $\mathrm{Kgt}_m(z)$ is among them" halts with some z, and $K(z) \le K(e) + K(m) + c$; $\mathrm{Kgt}_m(z)$ is true by (Π_1T), so $K(z) > m$, whence $K(e) > m - K(m) - c$. So $K_A(\{(\mathrm{Kgt}_m(x), 1)\}) \ge m - K(m) - c$.

(b) Let s_m be the largest code of a halting computation of V on a program of length at most m (0 if there is none); it exists because there are finitely many such programs and each halting computation has one code (Uniq). Let $\mathrm{Bd}_m := \forall t\, \forall p\, \forall y\, (p \le \underline{\mathrm{rk}_m} \wedge \mathrm{Comp}(p, t, y) \to t \le \underline{s_m})$, a true Π_1 sentence. For x with $K(x) > m$, the Δ_0 sentence $\forall p \le \underline{\mathrm{rk}_m}\, \forall t \le \underline{s_m}\, \neg\mathrm{Comp}(p, t, \underline{x})$ is true, hence Q-provable, and together with Bd_m it gives $\mathrm{Kgt}_m(x)$ by pure logic. s_m is computable from m and the number $H_m \le 2^m$ of halting programs of length at most m (Kraft's inequality), an $(m+1)$-bit number: run the programs until H_m of them have halted. So $K(\mathrm{Bd}_m) \le K(m) + m + c$. Since $B \cup \{\mathrm{Bd}_m\}$ is true, hence consistent, and proves $\mathrm{Kgt}_m(x)$, Remark 2.8(c) gives $K_A \le K_{\mathrm{fin}} + c \le m + K(m) + c'$; and $L_A \le K_A + c$ by (Chain). Without unique codes a halting program could have infinitely many codes, and Bd_m would be false.

(a) For $K(x) \le m$ the datum is a false Π_1 sentence, and $M_A = 0$ by (Π_1T). Otherwise Corollary 5.8 and the deterministic bound give $L_A \ge K_A - K(\lceil L_A \rceil) - c' \ge m - K(m) - K(\lceil L_A \rceil) - c''$. By (b), $\lceil L_A \rceil \le m + K(m) + c$, so $K(\lceil L_A \rceil) \le \log m + O(\log\log m)$, and $K(m) \le \log m + O(\log\log m)$. For the three-valued convention: for a single two-valued datum $M^3_A(\{(\varphi, 1)\}) = M_A(\{(\varphi, 1)\})$, and $M^3_A(\emptyset) = 1 - \mathrm{Inc}_0$, so the three-valued cost of the label 1 is $-\log \mathrm{Bel}_\emptyset(\mathrm{Kgt}_m(x)) = L_A(\{(\mathrm{Kgt}_m(x), 1)\}) + \log(1 - \mathrm{Inc}_0)$, which differs from L_A by a constant. □

The deterministic step is Chaitin's bound (Chaitin, 1974), used only in its upper-bound form (a consistent extension of Q that proves $\mathrm{Kgt}_m(x)$ has complexity greater than $m - K(m) - c$); it makes no claim about least constants, which depend on the coding of machines (Raatikainen, 1998).

Proof of Proposition 5.10. Step 1: few refutable strings per theory. Let $x_1, \ldots, x_{2^{m+1}}$ be distinct strings. By (O), $\mathsf{Q} \vdash p \le \underline{\mathrm{rk}_m} \to \bigvee_{q \le \mathrm{rk}_m} p = \underline{q}$, so $\neg\mathrm{Kgt}_m(x_i)$ is Q-equivalent to $\bigvee_{q \le \mathrm{rk}_m} \exists t\, \mathrm{Comp}(\underline{q}, t, \underline{x_i})$. Hence $\bigwedge_{i \le 2^{m+1}} \neg\mathrm{Kgt}_m(x_i)$ is Q-equivalent to the disjunction, over all maps $f : \{1, \ldots, 2^{m+1}\} \to \{0, \ldots, \mathrm{rk}_m\}$, of $\bigwedge_i \exists t\, \mathrm{Comp}(\underline{f(i)}, t, \underline{x_i})$. There are $\mathrm{rk}_m + 1 = 2^{m+1} - 1$ programs, fewer than the strings, so every f has $f(i) = f(i')$ for some $i \ne i'$, and then (Uniq) gives $x_i = x_{i'}$, which Q refutes. So $\mathsf{PA} \vdash \neg \bigwedge_{i \le 2^{m+1}} \neg\mathrm{Kgt}_m(x_i)$: every consistent extension of B refutes $\mathrm{Kgt}_m(x)$ for at most $2^{m+1} - 1$ strings x. Integrating over the consistent tapes (Tonelli), $\sum_x \mathrm{Bel}_0(\neg\mathrm{Kgt}_m(x)) \le 2^{m+1}$.

Step 2: a lower bound on belief. For $K(x) > m$, Corollary 5.9(b) and (Dom) give $\mathrm{Bel}_0(\mathrm{Kgt}_m(x)) \ge 2^{-m-K(m)-c}$.

Step 3: Markov's inequality. By Step 1, fewer than $2^{m+1} \cdot 2^{j+m+K(m)+c}$ strings have $\mathrm{Bel}_0(\neg \mathrm{Kgt}_m(x)) > 2^{-j}2^{-m-K(m)-c}$, and fewer than 2^{m+1} strings have $K(x) \le m$. For every other x, Step 2 gives $\mathrm{Bel}_0(\neg \mathrm{Kgt}_m(x)) \le 2^{-j}\mathrm{Bel}_0(\mathrm{Kgt}_m(x))$; and $\mathrm{Dis}_\emptyset$, $\mathrm{Bel}_\emptyset$ are $\mathrm{Bel}_0(\neg\cdot)$, $\mathrm{Bel}_0(\cdot)$ divided by the same $1-\mathrm{Inc}_0$. The normalised probability of the label 1 is $\mathrm{Bel}_\emptyset/(\mathrm{Bel}_\emptyset+\mathrm{Dis}_\emptyset) \ge 1/(1+2^{-j})$, and $\log(1+y) \le y/\ln 2$. □

So for $d \ge 2m + K(m) + j + c'$ almost every string of length d costs essentially nothing in the normalised convention, while every x with $K(x) > m$ costs $m \pm O(\log m)$ bits in the semimeasure and three-valued conventions (Corollary 5.9). In the three-valued convention most of the remaining mass goes to "independent": by Step 1, $\sum_x \mathrm{Dis}_\emptyset(\mathrm{Kgt}_m(x)) \le 2^{m+1}/(1-\mathrm{Inc}_0)$, so for most x the inducer predicts "independent" with probability $1 - 2^{-m+O(\log m)}$. One-tape consistent SOI behaves the same way: Theorem 4.3 transfers $M_A(\{(\varphi, b)\})$ to $M_F^{\mathrm{cons}}(\{(\varphi, b)\})$ up to factors $2^{\pm c}$, which leaves the semimeasure statement unchanged and shifts j by a constant in the normalised one. With $B = \mathsf{Q}$ alone, Step 1 needs Q to prove the uniqueness of outputs, which it may not; the normalised behaviour is then open. A computation (code in the accompanying repository) checks the finite bookkeeping of Steps 1 and 3 in finite models in which uniqueness of outputs is imposed.

Proposition C.14 (True Π_1^*-axiomatised theories)**.** *Let T be an r.e. set of true Π_1^* sentences and $k := \lceil -\log \mathbb{U}_A(\mathrm{Ext}T) \rceil$. Then*

$$-\log \mathbb{U}_A(\mathrm{Ext}T) - c \;\le\; -\log \mathbf{m}(\mathrm{Ext}T) \;\le\; -\log \mathbb{U}_A(\mathrm{Ext}T) + K(k) + c.$$

Proof. Right: each $\pi \in T$ has $\mathrm{Bel}_0(\pi) \ge \mathbb{U}_A(\mathrm{Ext}T) \ge 2^{-k}$, since every consistent theory extending $\mathrm{Cn}(T)$ proves π; so $T \subseteq T_k$. $B \cup T_k$ is consistent, so $e_k \in \mathrm{Ext}(T)$ and $\mathbf{m}(\mathrm{Ext}T) \ge \mathbf{m}(e_k) \ge 2^{-k-K(k)-c}$. Left: $\mathbb{U}_A(\mathrm{Ext}T) \ge 2^{-c}\mathbf{m}(\mathrm{Ext}T)$ (Definition 2.9; $B \cup T$ is consistent because it is true). □

The same comparison for the exact class $[T]$, and for general T, is part of Question A.8.

Proposition C.15 (One false Σ_1 datum)**.** *Let $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$, B decidable and true with $B \supseteq \mathsf{Q}$, and Υ_0 the formula of Theorem 5.5(a). For every $d \ge 2$ there are distinct $x_1, x_2 \in \{0,1\}^d$ such that $D := \{(\Upsilon_0(\underline{x_1}), 0), (\Upsilon_0(\underline{x_2}), 1)\}$ (strings coded by length-lexicographic rank) has $K_A(D) > \log d - 2$ and $L_A(D) \le 2 + c$. The literal $\neg\Upsilon_0(\underline{x_1})$ is Q-equivalent to a true Π_1^* sentence, and $\Upsilon_0(\underline{x_2})$ to a false Σ_1 sentence.*

Proof. Upper bound. The machine that, for every n, reads a fresh bit b_n and emits $\Upsilon_0(\underline{n})^{b_n}$ has a consistent theory on every tape (Lemma C.8(i) and compactness) and proves D with probability $\frac{1}{4}$; apply (Dom). *Lower bound* (the colouring argument of Theorem 3.18). Let $\ell := \lceil \log(d/\log 3) \rceil - 1$, so that $2^\ell < d/\log 3$ and $3^{2^\ell - 1} < 2^d$. Fewer than 2^ℓ programs have length less than ℓ. Colour each $x \in \{0,1\}^d$ by the vector, over these programs p, of "+" if $V(p)$ halts with an index e such that $B \cup \mathsf{W}_e$ is consistent and proves $\Upsilon_0(\underline{x})$, "−" if it is consistent and proves $\neg\Upsilon_0(\underline{x})$, and "∗" otherwise. Two distinct x_1, x_2 share a colour, and then no program shorter than ℓ gives a consistent theory proving $\neg\Upsilon_0(\underline{x_1}) \wedge \Upsilon_0(\underline{x_2})$. So $K_A(D) \ge \ell \ge \log d - \log\log 3 - 1 > \log d - 2$. The complexity claims are Theorem 5.5(a)'s. □

Proposition C.16 (Two true data)**.** *Let $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$, and let B be decidable and true with $B \supseteq \mathsf{Q}$. There is a formula Υ_2 such that for every $d \ge 2$ some distinct $x_1, x_2 \in \{0,1\}^d$ give true data $D_d := \{(\Upsilon_2(\underline{x_1}), 0), (\Upsilon_2(\underline{x_2}), 1)\}$ with $K_A(D_d) > \log d - 2$ and $L_A(D_d) \le 2 + c$. Each datum is Q-equivalent to a Boolean combination of Σ_1 sentences (hence, by prenexing, both to a Σ_2 and to a Π_2 sentence).*

Proof. The colour formulas. For a program p (a numeral) put

- $\mathrm{Halt}_p :\equiv \exists t\, \exists e\, \mathrm{Comp}(\underline{p}, t, e)$, a Σ_1 sentence ($V(p)$ halts);
- $\mathrm{Ok}_p :\equiv \forall t\, \forall e\, \big(\mathrm{Comp}(\underline{p}, t, e) \to \mathrm{Con}(B \cup \mathsf{W}_e)\big)$, a Π_1 sentence, with $\mathrm{Con}(B \cup \mathsf{W}_e)$ the canonical Π_1 formula in e;
- $\mathrm{Prv}_p^b(z, y) :\equiv \exists t\, \exists e\, \big(\mathrm{Comp}(\underline{p}, t, e) \wedge \mathrm{Prov}_{B \cup \mathsf{W}_e}(\ulcorner \varphi_{[z]}(\dot{y})^b \urcorner)\big)$, a Σ_1 formula, for $b \in \{0, 1\}$.

The colour entry of y at p is "$+$" iff $\mathrm{Halt}_p \wedge \mathrm{Ok}_p \wedge \mathrm{Prv}_p^1(z, y)$, "$-$" iff $\mathrm{Halt}_p \wedge \mathrm{Ok}_p \wedge \mathrm{Prv}_p^0(z, y)$, and "$*$" otherwise. In $\mathbb{N}$, where $V(p)$ has at most one output, these say exactly what the colouring in the proof of Proposition C.15 says (with $\varphi_{[z]}$ for Υ_0). The consistency condition is written with a universal quantifier because "$\exists e\, (\mathrm{Comp}(p, t, e) \wedge \mathrm{Con}(\dots))$" would be Σ_2. Let $\Lambda(z, x)$ say: "x is the code of a string of some length $d \geq 2$, and x is the second element of the lexicographically first pair of distinct strings of length d that have the same colour", colours taken over the programs of length less than $\ell_d := \lceil \log(d/\log 3) \rceil - 1$, with all quantifiers outside the colour formulas bounded by x. Let Υ_2 be Lemma C.8's formula for Λ, and (x_1, x_2) that pair for $z = \ulcorner \Upsilon_2 \urcorner$; it exists by the pigeonhole count of Proposition C.15. By Lemma C.8(ii), $\Upsilon_2(\underline{x_2})$ is true and $\Upsilon_2(\underline{x_1})$ is false, so both data are true. The fair-coin machine of Proposition C.15, with Υ_2 in place of Υ_0, gives $L_A(D_d) \leq 2 + c$ by Lemma C.8(i). The pair has the same colour for the actual Υ_2, which gives $K_A(D_d) \geq \ell_d > \log d - 2$ as in Proposition C.15. For a numeral argument, (O) and (Σ_1C) unfold the bounded quantifiers of $\Lambda(\ulcorner \Upsilon_2 \urcorner, \underline{x})$ and decide its Δ_0 parts in Q, leaving a Boolean combination of the Σ_1 sentences Halt_p, Prv_p^b and the Π_1 sentences Ok_p; with Lemma C.8(iii), each datum is Q-equivalent to a Boolean combination of Σ_1 sentences, hence, by prenexing, both to a Σ_2 and to a Π_2 sentence. □

C.5 Gödel phenomena

Proof of Proposition 5.11. (a) The posterior lives on consistent extensions of Q, and all of them prove σ by (Σ_1C). (b) Some conjunct of π is false; its negation is a true Σ_1 sentence, so every consistent extension of Q proves $\neg\pi$. (c) Consider the r.e. theories $D^{\pm} \cup \{\pi\}$, $D^{\pm} \cup \{\neg\pi\}$ and $D^{\pm}$. The first is true in a model of $B \cup D^{\pm}$ whose $\mathcal{L}_{\mathrm{ar}}$-reduct is $\mathbb{N}$, so it is consistent with B, and it proves π. The second is consistent with B because $B \cup D^{\pm} \nvdash \pi$ (by the deduction theorem for the finite set $D^{\pm}$), and it proves $\neg\pi$. The third proves neither: not π by hypothesis, and not $\neg\pi$ because $B \cup D^{\pm}$ is true and $\neg\pi$ is false. All three prove ψ_D and have positive weight by (Dom). (d) Apply (c) to the Π_1^* sentence $\neg\sigma$. □

An extreme case. If the data present $T := \mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$, a consistent and Σ_1-unsound theory, every consistent extension of T proves $\neg\mathrm{Con}(T)$, since $\mathsf{PA} \vdash \mathrm{Con}(T) \to \mathrm{Con}(\mathsf{PA})$; so $\mathrm{Dis}(\mathrm{Con}(T)) = 1$ although $\mathrm{Con}(T)$ is true. A computation (code in the accompanying repository) checks (a)–(d) and the one-sided error pattern on random instances of a finite model in which (Σ_1C) is the only arithmetical input.

Proof of Theorem 5.12. By the s-m-n theorem there is an injective computable g with $\mathsf{W}_{g(e)} = \mathsf{W}_e \cup \{\neg\mathrm{Con}(B \cup \mathsf{W}_e)\}$, the canonical consistency statement of the presentation e. If $B \cup \mathsf{W}_e$ is consistent, (G2) gives $B \cup \mathsf{W}_e \nvdash \mathrm{Con}(B \cup \mathsf{W}_e)$, so $B \cup \mathsf{W}_{g(e)}$ is consistent; its new axiom is a false Σ_1 sentence; and it proves everything W_e proves. As $\mathbf{m}(g(e)) \geq 2^{-c}\mathbf{m}(e)$ and $K(g(e)) \leq K(e) + c$, summing over the Σ_1-sound compatible e, whose images are distinct, gives the claim, and the posterior bound follows from $W_A(\mathcal{U}; D) \geq 2^{-c} W_A(\overline{\mathcal{U}}; D)$. A true theory is Σ_1-sound, which gives "a fortiori". □

The event that $B \cup A_\omega$ is Σ_1-unsound is $\bigcup_\sigma O_\sigma$ over the false Σ_1 sentences σ, an open set, so $\mathbb{U}_A(\mathcal{U}; D)$ and $P_D(\mathcal{U})$ are defined.

Proposition C.17 (One-tape soundness is controlled by the coding gap)**.** *Under the hypotheses of Theorem 5.12, for every consistent D,*

$$P_D(\mathcal{U}) \;\geq\; 2^{L_A(D)-K_A(D)-c}.$$

Proof. Let e witness $K_A(D)$. Its companion $g(e)$ from the proof of Theorem 5.12 is consistent, Σ_1-unsound, proves ψ_D, and has $K(g(e)) \leq K_A(D) + c$; by (Dom) the tapes that begin with a shortest program for it have measure at least $2^{-K_A(D)-c}$, and their theory is $\mathsf{W}_{g(e)}$. Divide by $M_A(D) = 2^{-L_A(D)}$. □

So one-tape SAI can become confident of soundness only on data where the coding theorem fails badly, which is where randomness dominates (Section 5.3). On data of Π_1 type, Proposition C.17 and Corollary 5.8 give $P_D(\mathcal{U}) \geq 2^{-K(\lceil L_A(D) \rceil)-c}$, a bound that degrades as the loss grows; the threshold machine removes the degradation.

Proof of Theorem 5.13. For a finite string w let U_w be the set of sentences that G_{thr} (Theorem 5.7) emits on some tape extending w; it is r.e. uniformly in w, and $U_w = \{\pi \in \Pi_1^* : \mathbb{U}(O_\pi) > 0.w\}$, since the emissions under prefixes $v \sqsupseteq w$ have thresholds $0.v + 2^{-|v|}$ with infimum $0.w$, not attained, and those under shorter prefixes $u \sqsubseteq w$ have thresholds at least $0.w + 2^{-|w|}$. Hence U_w is true if $0.w > \mathrm{Inc}_0$, and is all of Π_1^*, an inconsistent set, if $0.w < \mathrm{Inc}_0$ (equality is impossible, since Inc_0 is irrational). Moreover $U_{w'} \subseteq U_w$ for $w \sqsubseteq w'$, provably in PA from the definitions (every prefix of w' is comparable with w). Let $G^{\mathrm{u}}_{\mathrm{thr}}$ emit everything G_{thr} emits and, for every prefix w of the tape, $\chi_w := \neg\mathrm{Con}(B \cup U_w)$.

Consistency. Let $r > \mathrm{Inc}_0$ be a non-dyadic tape and F a finite subset of $A_r^{G^{\mathrm{u}}_{\mathrm{thr}}}$, with companion sentences $\chi_{w_1}, \ldots, \chi_{w_j}$; let w^* be the longest w_i. If $0.w^* < \mathrm{Inc}_0$, every $U_{w_i} \supseteq U_{w^*}$ is inconsistent, every χ_{w_i} is a true Σ_1 sentence, provable from B, and the rest of F lies in the true theory $A_r^{G_{\mathrm{thr}}}$. If $0.w^* > \mathrm{Inc}_0$, then $B \cup U_{w^*}$ is consistent, r.e. and contains PA, so $B \cup U_{w^*} \cup \{\chi_{w^*}\}$ is consistent by (G2); it contains $F \cap A_r^{G_{\mathrm{thr}}}$, because $A_r^{G_{\mathrm{thr}}} = \{\pi : \mathbb{U}(O_\pi) > r\} \subseteq U_{w^*}$ as $r > 0.w^*$; and it proves each χ_{w_i} with $w_i \sqsubseteq w^*$, because $U_{w^*} \subseteq U_{w_i}$ provably, so a proof of $\bot$ from $B \cup U_{w^*}$ is one from $B \cup U_{w_i}$.

Unsoundness. $r > \mathrm{Inc}_0$ has a prefix w with $0.w > \mathrm{Inc}_0$, since the truncations increase to r; for it χ_w is a false Σ_1 sentence.

Measure. For $\mathrm{Inc}_0 < r < \mathbb{U}(O_{\pi_D})$ the theory of $G^{\mathrm{u}}_{\mathrm{thr}}$ is consistent, Σ_1-unsound and contains π_D, and $B \vdash \psi_D \leftrightarrow \pi_D$. So $\nu_{G^{\mathrm{u}}_{\mathrm{thr}}}$ gives the event "consistent, unsound and proving ψ_D" probability at least $M_A(D)$, as in the proof of Theorem 5.7, and (Dom), tape by tape, gives $\mathbb{U}_A(\mathcal{U}; D) \geq 2^{-K(G^{\mathrm{u}}_{\mathrm{thr}})-c} M_A(D)$. □

A computation (code in the accompanying repository) checks the dyadic and prefix facts of this proof in finite models with non-dyadic weights.

What is known about Question 5.14. With B true and $\mathsf{PA} \subseteq \mathrm{Cn}(B)$, the answer is yes for data of Π_1 type (Theorem 5.13) and for data with a bounded coding gap $K_A - L_A$ (Proposition C.17). On the Win-lottery stream with $\mathcal{L} = \mathcal{L}_{\mathrm{ar}} \cup \{\mathrm{Win}\}$ it is yes as well: the machine "G_{lot} plus the axiom $\neg\mathrm{Con}(B)$" has the same survival event; its theories are consistent (a model of $B + \neg\mathrm{Con}(B)$ exists by (G2), has distinct numerals as every model of Q does, and is expanded by the winners) and Σ_1-unsound; so $P_{D_n}(\mathcal{U}) \geq 2^{-c} \prod_k (1 - 2^{-(k+1)})$ for every n. On the arithmetic

lotteries the random lottery theories are themselves unsound (Corollary 5.6(iii) for (b); in (a) they affirm the false Σ_1 sentences Tick_{k,J_k}). In general we can prove only the following. For a theory machine G let U_w^G be the set of sentences that G emits on some tape extending w, and $\mathcal{K}^G := \{\omega : B \cup U_w^G \text{ is consistent for some prefix } w \text{ of } \omega\}$. Under the hypotheses of Theorem 5.12, for every D,

$$\mathbb{U}_A(\mathcal{U}; D) \;\geq\; 2^{-K(G)-c}\,\mathbb{U}(\{\omega : B \cup A_\omega^G \text{ consistent}, \vdash \psi_D\} \cap \mathcal{K}^G).$$

Indeed, the companion G^{u} emits what G emits and $\neg\mathrm{Con}(B \cup U_w^G)$ for every prefix w of the tape, and $K(G^{\mathrm{u}}) \leq K(G) + c$; on a tape ω on which $B \cup A_\omega^G$ is consistent, the longest-prefix argument above shows that $B \cup A_\omega^{G^{\mathrm{u}}}$ is consistent ($A_\omega^G \subseteq U_{w^*}^G$ because w^* is a prefix of ω), and on $\mathcal{K}^G$ some added sentence is a false Σ_1 sentence; then apply (Dom) tape by tape. This answers the question for no class of data by itself: it fails for coin machines, all of whose sets U_w^G are inconsistent. The difficulty in general is that a companion must add a false Σ_1 sentence that the random theory A_ω does not refute, chosen from finitely much of the tape. $\neg\mathrm{Con}(A_\omega)$ is not a sentence, and A_ω may prove $\mathrm{Con}(Z)$ for every r.e. Z read off a finite prefix (a machine can emit, at each stage, the consistency of everything it has emitted so far). The natural companions need either a consistent U_w, which fails for coin machines, or a fixed false Σ_1 sentence that the random theory leaves open, which fails for such reflective machines.

Proposition C.18 ($\mathrm{Con}(B \cup T)$ given data from T)**.** *Let T be r.e., with $B \cup T$ true and $\mathsf{PA} \subseteq \mathrm{Cn}(B \cup T)$, and let the data present T: $\mathrm{Cn}(D_\infty^{\pm}) = \mathrm{Cn}(B \cup T)$. Write χ_T for $\mathrm{Con}(B \cup T)$ or the Gödel sentence of $B \cup T$. (a) $\lim_n \mathrm{Bel}_{D_n}(\chi_T)$, $\lim_n \mathrm{Dis}_{D_n}(\chi_T)$ and $\lim_n \mathrm{Indep}_{D_n}(\chi_T)$ exist and lie in $(0,1)$. (b) The belief and independent shares certify the truth of χ_T in the sense of Proposition 5.11(b). (c) Which of "true" and "false" gets more mass is a choice of universal machine: for every universal prefix machine V and every j there are universal prefix machines V', V'' under which the ratio of the limits of $\mathrm{Bel}(\chi_T)$ and $\mathrm{Dis}(\chi_T)$ is at least $2^j - 1$, respectively at most $1/(2^j - 1)$.*

Proof. (a) By (Lim) (Proposition A.13(d)), the limit posterior is the prior restricted to the consistent extensions of $B \cup T$, which is incomplete by (G2). Three deterministic theories have positive weight. $T + \mathrm{Con}(B \cup T)$ is true, because $B \cup T$ is true, hence consistent, and $\mathrm{Con}(B \cup T)$ is a true Π_1 sentence. $T + \neg\mathrm{Con}(B \cup T)$ is consistent with B by (G2) for the presentation $B \cup T \supseteq \mathsf{PA}$. T decides neither: not $\mathrm{Con}(B \cup T)$ by (G2), and not its negation because $B \cup T$ is true and $\neg\mathrm{Con}(B \cup T)$ is false. Each of the three limits is therefore positive, and each is less than 1 because the other two are positive. (b) is Proposition 5.11(b) applied to the Π_1 sentence $\mathrm{Con}(B \cup T)$. (c) Let $V'(0^i 1)$ be an index of the deterministic machine for $T + \mathrm{Con}(B \cup T)$ for $i < j$, and $V'(0^j p) := V(p)$; V' is a universal prefix machine. The tapes that begin with some $0^i 1$, $i < j$, have measure $1 - 2^{-j}$ and give a compatible theory proving $\mathrm{Con}(B \cup T)$; all tapes on which $\neg\mathrm{Con}(B \cup T)$ is proved begin with 0^j. So, within E_∞, the masses of the two events are at least $1 - 2^{-j}$ and at most 2^{-j}; dividing both by the common normaliser $\mathbb{U}(E_\infty)$ gives the ratio. V'' is symmetric, with $T + \neg\mathrm{Con}(B \cup T)$. The Gödel sentence of $B \cup T$ is PA-provably equivalent to $\mathrm{Con}(B \cup T)$ (Smoryński, 1977), so the same holds for it. □

The truth hypothesis is needed. With $B := \mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$, which is decidable and consistent by (G2), and $T := \emptyset$: since $\mathsf{PA} \vdash \mathrm{Con}(B) \to \mathrm{Con}(\mathsf{PA})$, $B \vdash \neg\mathrm{Con}(B)$, so every compatible theory refutes $\mathrm{Con}(B \cup T)$, $\lim \mathrm{Dis} = 1$, and (a) and (c) fail. A computation (code in the accompanying repository) checks the construction of (c) in a finite model.

Proof of Proposition 5.15. If $B \vdash \varphi^b$ for every $(\varphi, b) \in D$, then $\mathrm{Cn}(D^{\pm}) = \mathrm{Cn}(\emptyset)$, so $M_A(D) = M_A(\emptyset)$, $E_D = E_\emptyset$, and the same for W_A and K_A (Old). With $\mathsf{Q} \subseteq \mathrm{Cn}(B)$ and P a Δ_0 formula, every true instance $P(\underline{n})$ is Q-provable by (Σ_1C), and a false one is Q-refutable, so its datum would be inconsistent with B. □

Instances of a fresh predicate (empirical induction, $B = \emptyset$) do confirm $\forall x\, R(x)$, up to the ω-gap (Proposition 5.16). So the inference from checked instances needs a reasoner for whom computing $P(\underline{n})$ is costly: logical induction (Garrabrant et al., 2016a) is one, and a derivation-charged axiom inducer (Section 6 of Hänni and Claude (Anthropic), 2026a) is another. Whether a derivation-charged posterior then concentrates on provers of $\forall x\, P(x)$ depends on competitors that prove the instances cheaply without proving the universal (compare Proposition C.20); we do not settle this.

C.6 The ω-gap

Proof of Proposition 5.16. I is a consistent r.e. theory proving the stream, so $\mathbb{U}(E_\infty) \geq 2^{-K(I)-c} > 0$ (Proposition A.13(c)) and (Lim) applies; by Proposition A.13(d) with $\Gamma = I$, P_∞ is the prior restricted to the consistent extensions of I, whether or not I is complete. In (i) and (ii) every consistent extension of I proves what I proves. In (iii), I, $I + \forall x\, \varphi$ and $I + \neg\forall x\, \varphi$ are consistent r.e. theories with positive weight, which prove $\forall x\, \varphi$, refute it, and decide it neither way, respectively; and a consistent extension of I refutes $\forall x\, \varphi$ iff it is ω-inconsistent at φ, since it proves every instance. □

Two standard cases. With $B = \emptyset$ and the atomic diagram of a computable structure this is Proposition A.16. With $B \supseteq \mathsf{Q}$ and $\varphi \in \Delta_0$ the instances are free (Proposition 5.15) and the posterior does not move at all; case (iii) holds iff B decides $\forall x\, \varphi$ neither way. For Σ_1-sound B this is iff $B \nvdash \forall x\, \varphi$: with all instances true, $\neg\forall x\, \varphi$ is logically equivalent to the false Σ_1 sentence $\exists x\, \neg\varphi$, which a Σ_1-sound B does not prove. Without Σ_1-soundness this fails: for $B = \mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$ and $\varphi(x)$ the Δ_0 formula "x is not a PA-proof of $\bot$", every instance is true and Q-provable, so $\mathrm{Cn}(I) = \mathrm{Cn}(B)$, and $B \vdash \neg\forall x\, \varphi$: case (ii) holds although $B \nvdash \forall x\, \varphi$.

Proposition C.19 (The Gaifman condition)**.** *Under the hypotheses of Proposition 5.16, enumerate* $\mathrm{CT} = \{t_0, t_1, \dots\}$. *Then* $\lim_n \mathrm{Bel}_0(\bigwedge_{i \leq n} \varphi(t_i))$ *exists, and*

$$\lim_n \mathrm{Bel}_0\Big(\bigwedge_{i \leq n} \varphi(t_i)\Big) - \mathrm{Bel}_0(\forall x\, \varphi) = \mathbb{U}\{\omega : B \cup A_\omega \text{ consistent, proves every } \varphi(t_i),\ \nvdash \forall x\, \varphi\}.$$

This is 0 *iff* $I \vdash \forall x\, \varphi$, *and otherwise at least* $2^{-K(I)-c}$, *where* $K(I)$ *is the complexity of an index of* I. *The same holds for every posterior* Bel_D *with* D *proved by* I, *after dividing by* $M_A(D)$.

Proof. The events $\{\omega : B \cup A_\omega$ consistent and proves $\varphi(t_0), \dots, \varphi(t_n)\}$ decrease in n to the event of proving every instance; continuity from above gives the limit. A theory proving $\forall x\, \varphi$ proves every instance, so the difference is the stated measure. If $I \vdash \forall x\, \varphi$, every consistent theory proving all instances proves $\forall x\, \varphi$, and the difference is 0. Otherwise I itself lies in the event, with weight at least $2^{-K(I)-c}$ by (Dom). For Bel_D, divide by $M_A(D)$. □

"Prove", not "decide", is the right word: in the ω-inconsistent case (ii) of Proposition 5.16 the instances refute the universal, yet the gap equals the whole consistent mass of the extensions of I.

Proof of Theorem 5.17. Let $\forall x\,\zeta_k$ be the canonical consistency statement $\mathrm{Con}(B \cup T_k)$ of the presentation e_k (Theorem C.10), rewritten with one universal quantifier over a code of a pair (a derivation of $\bot$, an enumeration stage), with membership in B and in T_k certified inside the Δ_0 matrix by computation witnesses and the quantifiers contracted by Δ_0 pairing. For such a natural rewriting $\mathsf{PA} \vdash \forall x\,\zeta_k \leftrightarrow \mathrm{Con}(B \cup T_k)$ (standard). $B \cup T_k$ is true (Theorem C.10(iii)), so $\forall x\,\zeta_k$ is true and its instances are true Δ_0 sentences. If $\mathrm{Bel}_0(\forall x\,\zeta_k) \geq 2^{-k}$, Theorem C.10(ii) puts $\forall x\,\zeta_k$ into T_k, so $B \cup T_k \vdash \forall x\,\zeta_k$, hence, by the PA-provable equivalence, $B \cup T_k \vdash \mathrm{Con}(B \cup T_k)$, contradicting (G2) for the consistent r.e. presentation $B \cup T_k \supseteq \mathsf{PA}$. The lower bound is (Dom) for the true one-sentence theory $\{\forall x\,\zeta_k\}$, whose index is computable from e_k, of complexity at most $K(e_k) + c \leq k + K(k) + c'$. The posterior after B-entailed data is the prior normalised by $M_A(\emptyset) = 1 - \mathrm{Inc}_0$, that is $\widehat{\mathrm{Bel}}_0 = \mathrm{Bel}_\emptyset$, so the posterior belief is $\mathrm{Bel}_0(\forall x\,\zeta_k)/(1-\mathrm{Inc}_0) < 2^{-k+c'}$. □

The closures of Remark 5.18. (a) *The ω-rule.* Let $\vdash_\omega$ be closure under first-order deduction and the ω-rule (from $\varphi(\underline{n})$ for all n, infer $\forall x\,\varphi$). Then $\mathsf{Q} \vdash_\omega \chi$ for every true $\mathcal{L}_{\mathrm{ar}}$-sentence χ: write χ in prenex form with a Δ_0 matrix and argue by induction on the prefix. True Δ_0 sentences are Q-provable; a true $\exists x\,\psi(x)$ has a true instance $\psi(\underline{n})$, provable by induction, and $\exists$-introduction applies; a true $\forall x\,\psi(x)$ has all instances true and provable, and the ω-rule applies. So with $B \supseteq \mathsf{Q}$ every $\vdash_\omega$-consistent random theory has $\mathcal{L}_{\mathrm{ar}}$-content exactly $\mathrm{Th}(\mathbb{N})$: it proves every truth and, being consistent, no falsehood. The gap closes because the inducer is omniscient about arithmetic a priori. (b) *Full second-order consequence* (given Dedekind's categoricity theorem, standard). Let hypotheses be second-order sentences, and let compatibility mean "has a full model, and every full model satisfies ψ_D". Every full model of second-order Peano arithmetic PA^2 is isomorphic to $\mathbb{N}$, so $\{\mathsf{PA}^2\}$ is compatible with every finite correctly labelled data set. By the dominance argument (the tapes that begin with a program for $\{\mathsf{PA}^2\}$), the loss on any presentation of $\mathrm{Th}(\mathbb{N})$, in any order, is at most $K(\mathsf{PA}^2) + c$, and the limit posterior has $\mathrm{Bel}(\varphi) = \mathbf{1}[\mathbb{N} \models \varphi]$ for every $\mathcal{L}_{\mathrm{ar}}$-sentence, trivially, since every sentence is a datum; the content is the loss bound. This places the unlearnability of $\mathrm{Th}(\mathbb{N})$ (Proposition A.17) on effective consequence, not on the complexity of arithmetical truth. (c) *The generative variant.* Theorem 3.9 of Hänni and Claude (Anthropic) (2026a) concerns the generative inducer over a countable class with i.i.d. data from a generator T^*. Its parts (a), (b): the posterior probability of "proves $\forall x\,\varphi$" tends to the prior share of provers in the class of theories with the generator's law. Its part (c1): in the faithful regime, with $\forall x\,\varphi$ in the data space, the limit is $\mathbf{1}[T^* \vdash_{\mathsf{C}} \forall x\,\varphi]$, whether the generator proves the universal in the calculus C used there, not the truth value of $\forall x\,\varphi$. So this closes the gap only in the sense that the universal sentence is itself observed. For filtered data the gap stays open: for quantifier-free φ in the minimal class of that paper, the limit is 0 under its plain likelihood and the prior share under its selection-aware likelihood (its part (d1)), and with an unknown filter it is a filter-prior mixture (its part (e)). The generative variant is therefore not a closure of the ω-gap on instance data.

Line budgets. Let hypotheses be single sentences ψ, and call a datum χ *ℓ-compatible* with ψ if $B \cup \{\psi\}$ is consistent and proves χ in a fixed Hilbert calculus by a derivation of at most ℓ lines (lines, not symbols). The data are the closed instances $\varphi(S^n 0)$, $n \in \mathbb{N}$.

Proposition C.20 (Line budgets and a Kreisel-type property)**.** *Assume that some sentence is ℓ-compatible with every datum (for instance $\forall x\,\varphi$, when $B \cup \{\forall x\,\varphi\}$ is consistent and ℓ is at least the constant number of lines of instantiation). Then for any prior with full support on sentences, the limit posterior gives* $\mathrm{Bel}(\forall x\,\varphi) = 1$ *iff every ψ that is consistent with B and proves every $\varphi(S^n 0)$ within ℓ lines also proves $\forall x\,\varphi$ (a* Kreisel property *of (B, φ, ℓ)).*

Proof. The set of sentences compatible with the first n data decreases in n to the set of ψ that are consistent with B and ℓ-compatible with every datum, which is nonempty by hypothesis. If every such ψ proves $\forall x\,\varphi$, the limit belief is 1; a ψ in it that does not prove $\forall x\,\varphi$ has positive weight, and then the limit belief is less than 1. □

With $B = \emptyset$ and $\varphi(x) := x = 0 \vee \exists y\, x = Sy$, the property already fails at a constant budget, but that example is degenerate: every instance is logically valid, so the data are old evidence (Proposition 5.15), and the budget plays no role. The following example is not degenerate.

Example C.21 (A non-degenerate failure)**.** Let $B = \emptyset$, $\mathcal{L} = \{0, S, R\}$ with R a fresh unary predicate, and data $R(S^n0)$; they are not valid, since the one-point structure with R empty refutes $R(0)$. With $\varphi(x) := x = 0 \vee \exists y\, x = Sy$, let $\psi_1 := \forall x\,(\varphi(x) \to R(x))$ and $\psi_2 := \forall x\, R(x)$. Each instance $\varphi(S^n0)$ has a derivation of at most five lines (an equality axiom $S^n0 = S^n0$, the axiom $\chi(t) \to \exists y\,\chi(y)$ with $t = S^{n-1}0$ and modus ponens, the axiom $\beta \to \alpha \vee \beta$ and modus ponens; three lines for $n = 0$), so ψ_1 derives $R(S^n0)$ in nine lines (seven for $n = 0$): the hypothesis, its instance at S^n0 by $\forall$-elimination and modus ponens, the derivation of $\varphi(S^n0)$, and modus ponens. ψ_2 derives each datum in three lines. The structure $\{0, 1, 2, a\}$ with $S(0) = 1$, $S(1) = 2$, $S(2) = 2$, $S(a) = 0$ and $R := \{0, 1, 2\}$ satisfies ψ_1 (the only element outside R is a, and $\varphi(a)$ fails) and every $R(S^n0)$ (closed terms denote 0, 1 or 2), but not $\forall x\, R(x)$. So with $\ell = 9$ both ψ_1 and ψ_2 stay ℓ-compatible with every datum, ψ_1 is consistent and does not prove $\forall x\, R(x)$, and under any full-support prior the limit belief in $\forall x\, R(x)$ lies strictly between 0 and 1: the Kreisel property fails for $(\emptyset, R, 9)$, with instances that are not valid.

A computation (code in the accompanying repository) checks the derivations line by line for $n \le 60$ in a small Hilbert-style proof checker, and the countermodel.

Kreisel's conjecture says that if PA proves $\varphi(S^n0)$ in at most k lines for every n, then $\mathsf{PA} \vdash \forall x\,\varphi$; it is open for the usual Hilbert-style PA, with partial results and counterexamples for nearby theories (Parikh, 1973; Baaz and Pudlák, 1993; Hrubeš, 2007, 2009; Santos and Kahle, 2021). These concern infinite theories with schemata. The property of Proposition C.20 concerns finite extensions $B \cup \{\psi\}$, in which a k-line derivation can use only the finitely many axioms of ψ and logical axioms; we know of no result on it, so the link is a reformulation.

Question C.22. For $B = \mathsf{Q}$ and Δ_0 formulas φ, does every finite ψ that proves all instances of φ in boundedly many lines prove $\forall x\,\varphi$?

Over Q with $\varphi \in \Delta_0$ the instances are old evidence again; the question is which finite theories the line-budget inducer keeps, not what it learns from the instances.

C.7 True arithmetic

Proof of Theorem 5.19. Let $\Omega_s \uparrow \Omega_V$ be the computable approximations. Since Ω_V is irrational, its k-th binary digit is the eventual value of the k-th digit of Ω_s. Let $\mathrm{Bit}'(y)$ be a formula with $\mathbb{N} \models \mathrm{Bit}'(\underline{k})$ iff that digit is 1: "there is s such that for all $s' \ge s$ the y-th digit of $\Omega_{s'}$ is 1", with the digit predicate expressed by a Δ_0 formula about a computation witness. Put $\mathrm{Bit}(x) := \exists y\,(y = x \wedge \mathrm{Bit}'(y))$, so that x occurs once, and $\mathrm{Bit}_k := \mathrm{Bit}(\underline{k}^{\mathrm{b}})$. Then $|\mathrm{Bit}_k| \le |\mathrm{Bit}| + c_1 \log k + c_2$, and the position of Bit_k in the enumeration is at most $C\kappa^{|\mathrm{Bit}|+c_2}\kappa^{c_1 \log k} = C' k^{1/\xi}$. The first n sentences therefore contain $\mathrm{Bit}_1, \ldots, \mathrm{Bit}_{m(n)}$, at positions computable from k, and their labels are the digits $\Omega_{1:m(n)}$. So $\Omega_{1:m(n)}$ is computable from n and $b_{1:n}$, and $K(\Omega_{1:m(n)}) \le K(b_{1:n} \mid n) + K(n) + c$. By Chaitin's theorem $K(\Omega_{1:m}) \ge m - c$. This is (a). (b) is (a) with Proposition A.17(b) and

$m(n) \geq (n/C')^{\xi} - 1$. (c) $L_A(D_n)$ is nondecreasing in n, because $E_{D_{n+1}} \subseteq E_{D_n}$. For $n \geq 4$ let $n' := 2^{\lfloor \log n \rfloor} \in (n/2, n]$; then $K(n') \leq K(\lfloor \log n \rfloor) + c \leq 2 \log\log n + c'$, and by monotonicity and (b), $L_A(D_n) \geq L_A(D_{n'}) \geq (n/(2C'))^{\xi} - 2\log\log n - c''$. □

The exponent for a stated syntax. Take the Polish-notation syntax with fourteen symbols: terms 0, St, $+tt$, $\cdot tt$ and variables $x' \cdots'$ (x followed by $i \geq 0$ primes); formulas $=tt$, $<tt$, $\neg f$, $\wedge ff$, $\vee ff$, $\to ff$, $\forall vf$, $\exists vf$; and binary numerals $\underline{1}^{\mathrm{b}} = S0$, $\underline{2m}^{\mathrm{b}} = \cdot\, \underline{m}^{\mathrm{b}}\, SS0$, $\underline{2m+1}^{\mathrm{b}} = S\cdot\, \underline{m}^{\mathrm{b}}\, SS0$, so that $c_1 = 5$ and $c_2 = 2$. Counting all strings ($\kappa = 14$) gives $\xi \geq 1/(5 \log 14) > 0.0525$. Counting formulas gives more. With $V(x) = x/(1-x)$, the generating function of the variables, the generating functions of terms and formulas satisfy

$$T = x + V + xT + 2xT^2, \qquad F = 2xT^2 + xF + 3xF^2 + 2xVF,$$

and every term on the right carries a factor x. If rationals $x_0, T^*, F^* > 0$ satisfy both equations with "$\geq$" in place of "=", then the iterates of the right-hand sides from 0 have nonnegative coefficients, agree with the series up to degree k after k steps, and stay below (T^*, F^*) at x_0; so $\sum_L F_L x_0^L \leq F^*$, and the number of formulas, hence of sentences, of length at most L is at most $\frac{F^*}{1-x_0} x_0^{-L}$ for every L. A computation with exact rational arithmetic (code in the accompanying repository) verifies such a super-solution at $x_0 = 1000/5385$, with $T^* \approx 0.800042$ and $F^* \approx 0.609287$; this gives $C = 0.7482$ and $\kappa = 5.385$, hence $\xi \geq 1/(5 \log 5.385) > 0.0823$. The same computation finds no certificate at $\kappa = 5.38$; the growth rate of the formula counts is about 5.383.

Only the numeral length and the growth of the syntax enter ξ; the (large) size of Bit enters only the constant. With unary numerals the position of Bit_k is at most $C''\kappa^k$, so $m(n)$ is only about $\log_\kappa n$; then (b) is vacuous as written, since $K(n)$ can be about $\log n > \log_\kappa n$, but (c)'s monotonicity argument still gives $L_A(D_n) \geq \log_\kappa n - 2\log\log n - c$. The bound is weak in absolute terms (for $n = 10^6$, $n^{0.0823}$ is about 3.1); it is a statement about the shape of the growth, not a usable estimate.

Proof of Corollary 5.20. $b_{1:n}$ is computable from an index e_T of T, n, the set of undecided positions among the first n and their labels, since a true theory decides correctly. Using $\log \binom{n}{u} \leq u \log(en/u)$ and $K(u \mid n) \leq \log n + c$, this gives $K(b_{1:n} \mid n) \leq K(e_T) + u_n(T)(\log n + 3) + 2\log n + c$. Compare with Theorem 5.19(a), which is the inequality needed here (the statement (b) about L_A would not suffice): $u_n(T)(\log n + 3) \geq (n/C')^{\xi} - K(n) - K(e_T) - 2\log n - c'$, and the right-hand side grows like n^{ξ} while the subtracted terms are $O(\log n)$. □

Proposition C.23 (An upper bound by a reduction)**.** *For every r.e. T with $B \cup T \subseteq \mathrm{Th}(\mathbb{N})$ and every n,*

$$L_A(D_n) \leq K(e_T) + \log \tbinom{n}{u_n(T)} + u_n(T) + 2\log n + O(\log\log n).$$

Hence if some such T has $u_n(T)/n \to 0$, then $L_A(D_n) = o(n)$.

Proof. $B \cup T$ together with the correctly labelled undecided sentences among the first n is a consistent (true) r.e. theory proving D_n, with an index computable from e_T, n, u_n, the positions ($\log \binom{n}{u_n}$ bits, given n and u_n) and the u_n labels; then (Chain) gives $L_A \leq K_A + c$. If $u_n/n \to 0$, then $\log \binom{n}{u_n} \leq nH(u_n/n) = o(n)$, with H the binary entropy function. □

By Proposition C.23, a positive answer to Question 5.21 would follow if some true r.e. theory decided a set of sentences of asymptotic density one in the enumeration, that is, if the truth of arithmetic were generically decidable in the sense of generic-case complexity (Myasnikov and Rybalov, 2008). Such questions depend on the representation: the halting problem is decidable on

a set of asymptotic probability one for one model of Turing machines (Hamkins and Miasnikov, 2006). Calude and Jürgensen (2004) show that the probability that a true sentence of length n is provable tends to 0, in a probability space of statements with n-bit codes under their numbering. If a version of this held for the length enumeration used here, every true r.e. theory would leave a positive fraction of the sentences of each length undecided, and the sufficient condition of Proposition C.23 would fail for every T. It would not settle Question 5.21: $K(b_{1:n} \mid n)$ could still be $o(n)$, since truth values along the enumeration are highly redundant (for instance ψ and $\neg\neg\psi$). Corollary 5.20 gives only a polynomial lower bound on u_n.

D Proofs for what a theory does and is

This appendix proves the propositions of Section 6, and states in full the three that the section summarises in prose (Propositions D.1, D.3 and D.5). Appendices D.1 to D.4 concern questions against raw data; Appendices D.5 and D.6 give the precise setting of the flagpole toys, prove Proposition 6.5, and extend it to a system of laws.

D.1 The answer distribution

Proof of Proposition 6.3. On E_D the theory $B \cup A_\omega$ is consistent. If it proved φ_i and φ_j with $i \neq j$, it would prove $\varphi_i \wedge \varphi_j$, while $B \vdash \neg(\varphi_i \wedge \varphi_j)$; so it proves at most one direct answer. Hence the sets $E_D \cap O_{\varphi_i}$, $i \leq k$, are pairwise disjoint, and together with $E_D \setminus \bigcup_i O_{\varphi_i}$ they partition E_D. Dividing the measures by $\mathbb{U}(E_D) > 0$ gives (a), since $\mathrm{Bel}_D(\varphi_i) = \mathbb{U}(E_D \cap O_{\varphi_i})/\mathbb{U}(E_D)$.

(b) For $i \in J$, $O_{\varphi_i} \subseteq O_{\bigvee_{j\in J}\varphi_j}$, and the sets $E_D \cap O_{\varphi_i}$ are pairwise disjoint; so

$$\mathbb{U}\Big(E_D \cap O_{\bigvee_{j\in J}\varphi_j}\Big) = \sum_{i\in J} \mathbb{U}(E_D \cap O_{\varphi_i}) + \mathbb{U}\Big(E_D \cap O_{\bigvee_{j\in J}\varphi_j} \setminus \bigcup_{i\in J} O_{\varphi_i}\Big).$$

Divide by $\mathbb{U}(E_D)$: the last term is the posterior probability that the random theory proves $\bigvee_{j\in J}\varphi_j$, that is, answers $\mathcal{Q}$ partially through J, but proves no φ_i with $i \in J$. For $J = [k]$, $B \vdash \bigvee_i \varphi_i$, so $E_D \subseteq O_{\bigvee_i \varphi_i}$ and $\mathrm{Bel}_D(\bigvee_i \varphi_i) = 1$; by (a) the difference is $1 - \sum_i \mathrm{Bel}_D(\varphi_i) = \mathrm{Un}_D(\mathcal{Q})$.

(c) For $\mathcal{Q} = \{\varphi, \neg\varphi\}$ ($k = 2$), $\mathrm{Bel}_D(\neg\varphi) = \mathrm{Dis}_D(\varphi)$ and $\mathrm{Un}_D(\mathcal{Q})$ is the posterior probability that the random theory decides neither, which is $\mathrm{Indep}_D(\varphi)$. □

D.2 The conservation-law world

For even m, micro-states are $x \in \{0,1\}^m$ with weight $|x|$; the dynamics σ is a uniformly random permutation of each weight class, so the weight is conserved (the law); x_0 is uniform on the class of weight $m/2$, of size $Z := \binom{m}{m/2}$, and $x_{t+1} := \sigma(x_t)$. The raw predictors Raw^+, Raw^- are Bayes for "σ is a uniformly random permutation of each weight class", resp. "of $\{0,1\}^m$", and their step-t loss is that of predicting x_{t+1} from $x_0, \ldots, x_t$. The questions are "what is $|x_t|$?", $t = 0, 1, \ldots$; the question predictors Qa^+, Qa^-, $\mathsf{Qa}^{\mathrm{bin}}$ are Bayes for "$|x_t| = |x_0|$, with $|x_0|$ uniform on $\{0, \ldots, m\}$", "the $|x_t|$ are i.i.d. uniform on $\{0, \ldots, m\}$" and "the $|x_t|$ are i.i.d. binomial$(m, \frac{1}{2})$". In each task Mix gives prior $\frac{1}{2}$ to each of the task's two hypotheses ($\mathsf{Raw}^\pm$, resp. $\mathsf{Qa}^\pm$). Let t^* be the length of the cycle of σ through x_0.

Proposition D.1 (The conservation-law world)**.** *(a) For $t < t^*$ the step-t losses of Raw^+ and Raw^- are $\log(Z - t)$ and $\log(2^m - t)$; for $t \geq t^*$ both are 0. The law's gain $\kappa_t := \log \frac{2^m - t}{Z - t}$ ($t < t^*$) is nondecreasing in t; for $t \leq Z/2$, $\log \frac{2^m}{Z} \leq \kappa_t \leq \log \frac{2^m}{Z} + 1$, where $\log \frac{2^m}{Z} = \frac{1}{2}\log \frac{\pi m}{2} + o(1)$, and*

the share $\kappa_t/\log(2^m-t)$ of the step loss is $O(\log m/m)$. (b) Qa^+ pays $\log(m+1)$ for $|x_0|$ and 0 afterwards, Qa^- pays $\log(m+1)$ per question; over $t_{\max}+1$ questions the law's share of the no-law loss is $t_{\max}/(t_{\max}+1)$. (c) In each task the cumulative loss of Mix exceeds that of the better of its two hypotheses by at most one bit, on every trajectory. (d) For $t<t^*$, $\log(2^m-t)=\kappa_t+\log(Z-t)$ with $\kappa_t=-\log P_{\mathsf{Raw}^-}(|x_{t+1}|=m/2 \mid x_0,\dots,x_t)$: the law's raw gain is exactly the loss of Raw^-'s own answer to "what is $|x_{t+1}|$?". The loss of $\mathsf{Qa}^{\mathrm{bin}}$ on each question is κ_0.

Lemma D.2 (Next state of a random permutation)**.** *Let σ be a uniformly random permutation of a finite set X, and let $x_0,\dots,x_t\in X$ be distinct with $\sigma(x_s)=x_{s+1}$ for $s<t$. Conditionally on these t values of σ, $\sigma(x_t)$ is uniform on $(X\setminus\{x_0,\dots,x_t\})\cup\{x_0\}$, a set of $|X|-t$ elements.*

Proof. Conditionally on $\sigma(x_s)=x_{s+1}$ for $s<t$, the restriction of σ to $X\setminus\{x_0,\dots,x_{t-1}\}$ is a uniformly random bijection onto $X\setminus\{x_1,\dots,x_t\}$, since every such bijection extends the given values to exactly one permutation. So $\sigma(x_t)$ is uniform on $X\setminus\{x_1,\dots,x_t\}=(X\setminus\{x_0,\dots,x_t\})\cup\{x_0\}$, which has $|X|-t$ elements. □

Proof of Proposition D.1. (a) For $t<t^*$ the states $x_0,\dots,x_t$ are distinct, all of weight $m/2$, and x_{t+1} is either unvisited or x_0. Under the hypothesis of Raw^+ the permutations of the weight classes are independent, and the trajectory stays in the class of x_0; Lemma D.2 with X that class ($|X|=Z$) gives the predictive probability $1/(Z-t)$ of the true x_{t+1}, a step loss of $\log(Z-t)$. Under the hypothesis of Raw^- the lemma with $X=\{0,1\}^m$ gives $\log(2^m-t)$. For $t\ge t^*$ the path has returned to x_0, the cycle through x_0 is known to both predictors, and later states are determined: both losses are 0.

Since $2^m>Z\ge t^*>t$, the map $t\mapsto(2^m-t)/(Z-t)$ is increasing on $[0,Z)$ (its derivative is $(2^m-Z)/(Z-t)^2>0$), so κ_t is nondecreasing and $\kappa_t\ge\kappa_0=\log(2^m/Z)$. For $t\le Z/2$, $(2^m-t)/(Z-t)\le 2^m/(Z/2)$, so $\kappa_t\le\log(2^m/Z)+1$. By Stirling's formula, $Z=\binom{m}{m/2}=2^m\sqrt{2/(\pi m)}\,(1+o(1))$, so $\log(2^m/Z)=\frac12\log\frac{\pi m}{2}+o(1)$. Finally $\kappa_t\le\log(2^m/Z)+1=O(\log m)$, and $\log(2^m-t)\ge\log(2^m-Z/2)\ge m-1$ since $Z\le 2^m$; so the share is $O(\log m/m)$ for $t\le Z/2$.

(b) Qa^+ predicts $|x_0|$ uniformly on $\{0,\dots,m\}$, a loss of $\log(m+1)$, and then $|x_t|=|x_0|$ with probability 1, which is true since the weight is conserved: loss 0. Qa^- pays $\log(m+1)$ on each question. Over the questions $t=0,\dots,t_{\max}$ the losses are $\log(m+1)$ and $(t_{\max}+1)\log(m+1)$, and the law saves the share $t_{\max}/(t_{\max}+1)$ of the latter.

(c) In each task Mix is the Bayes mixture with prior $\frac12$ on the task's two hypotheses; x_0 has the same law under both raw hypotheses, so the posterior after x_0 is still $(\frac12,\frac12)$. By the chain rule, the cumulative loss of a Bayes mixture is $-\log$ of the mixture probability of the whole sequence. If the two hypotheses give the sequence probabilities 2^{-a} and 2^{-b}, the mixture's loss is $-\log(\frac12 2^{-a}+\frac12 2^{-b})\le-\log(\frac12 2^{-\min(a,b)})=\min(a,b)+1$, on every trajectory.

(d) Under Raw^-, by Lemma D.2, x_{t+1} is uniform on a set of 2^m-t states, of which exactly $Z-t$ have weight $m/2$: the $Z-(t+1)$ unvisited states of the class of x_0, and x_0. So $P_{\mathsf{Raw}^-}(|x_{t+1}|=m/2 \mid x_0,\dots,x_t)=(Z-t)/(2^m-t)=2^{-\kappa_t}$, and conditionally on that weight x_{t+1} is uniform on those $Z-t$ states, which is exactly the predictive distribution of Raw^+. The chain rule splits the step loss $\log(2^m-t)$ of Raw^- into κ_t, the loss of its prediction of the weight, and $\log(Z-t)$, the loss of Raw^+. Finally, every $|x_t|$ equals $m/2$, and the binomial probability of $m/2$ is $Z2^{-m}$, so $\mathsf{Qa}^{\mathrm{bin}}$ pays $\log(2^m/Z)=\kappa_0$ on each question. □

Computations. A computation (code in the accompanying repository, seeded) gives the following. At the start of a trajectory the law's gain per raw step is 1.871, 2.148, 2.348, 2.837, 3.331, 3.829, 5.326 and 10.292 bits at $m=8$, 12, 16, 32, 64, 128, 1024 and 10^6, against $\frac12\log\frac{\pi m}{2}=1.826$,

2.118, 2.326, 2.826, 3.326, 3.826, 5.326, 10.292; its share of the raw step loss is 0.23, 0.18, 0.15, 0.089, 0.052, 0.030, 0.0052 and 1.0×10^{-5}. The value $\kappa_{Z/2}$ is 2.659, 2.976, 3.199, 3.732, 4.258, 4.777 and 6.308 bits at $m = 8, \dots, 1024$; the loss of Qa^- per question is 3.170, 3.700, 4.088, 5.044, 6.022, 7.011 and 10.001 bits, about twice κ_0. At $m = 12$ ($Z = 924$), averaging exactly over the cycle length (uniform on $\{1, \dots, Z\}$), the expected raw loss over 300 steps is 2415.6 bits with the law and 3004.9 without (share 0.196), and the mixture's is 2416.6; over the 301 questions the losses are 3.70 and 1113.8 bits (share 0.997). Over a whole cycle the raw share is 0.232 and the mean gain 2.75 bits per step. In 2000 simulated runs the mean raw losses are within one standard error of these values, and the mixture never exceeds the better hypothesis by more than one bit. The identity of (d) holds at $m = 12$ for every $t < Z$ to within 1.8×10^{-15} bits in floating point.

D.3 Questions from raw data and an extractor

Proof of Proposition 6.4. Sampling. There is a partial computable map Ψ from tapes to $\mathbb{X}^{\mathbb{N}}$, computable from an index of μ, that is defined for $\mathbb{U}$-almost every tape and satisfies $\Psi_* \mathbb{U} = \mu$. This is the standard construction: read tape bits to refine a dyadic interval, and output the next symbol s as soon as the interval lies inside the μ-interval of s given the symbols already output; it fails only on the null set of tapes that hit interval boundaries. Each output symbol depends on a finite prefix of the tape.

The theory machine. Let G be the theory machine that, on a tape ω, computes $\Psi(\omega)$ as far as needed and, for each q, emits $\varphi_q^{\eta(q,\Psi(\omega))}$ once the tape prefix read so far fixes a prefix of $\Psi(\omega)$ on which the computation of $\eta(q, \cdot)$ halts. Each emission depends on a finite prefix of ω, so G is a c.e. set of pairs (prefix, sentence) as Definition 2.3 requires, and $K(G) \le K(\mu) + K(\eta) + c$, where c includes an index of the fixed family (φ_q). If $\Psi(\omega)$ is defined, then $A_\omega^G = \Gamma_{f_{\Psi(\omega)}}$, the set of labelled sentences of the labelling $f_{\Psi(\omega)}$.

The bound. Let H be the set of x for which f_x is consistent with B; $\mu(H) = 1$ by hypothesis. On every tape ω with $\Psi(\omega)$ defined, $\Psi(\omega) \in H$ and $f_{\Psi(\omega)} \supseteq D$, the theory $B \cup A_\omega^G$ is consistent and proves ψ_D, so $\omega \in E_D^G$. Hence $\nu_G(D) \ge \mathbb{U}\{\omega : \Psi(\omega) \in H,\ f_{\Psi(\omega)} \supseteq D\} = \mu\{x \in H : f_x \supseteq D\} = \mu_\eta(D)$. By Theorem 3.6, $M_A(D) \ge 2^{-K(\nu_G)} \nu_G(D) \ge 2^{-K(G)} \nu_G(D) \ge 2^{-K(\mu)-K(\eta)-c} \mu_\eta(D)$, and taking $-\log$ gives the bound on L_A. For a computable sequence x, take $\mu = \delta_x$, with $K(\delta_x) \le K(x) + c$; the hypothesis is that f_x is consistent with B, and $\mu_\eta(D) = 1$ exactly when $f_x \supseteq D$. □

The constant depends on the question family because the theory machine must write the sentences φ_q. The converse of the proposition fails badly, since answers do not determine raw data; in the conservation-law world the raw loss exceeds the question loss by the whole micro-entropy (Proposition D.1).

D.4 Answering and predicting

The following gives a formal vocabulary for "answering questions is more than predicting data": answering implies predicting, (a), but not conversely, (b), and success at predicting data that present a theory does not settle the questions the data leave open, (c). Part (d) is the coherence that a predictor of separate labels lacks.

Proposition D.3 (Answering and predicting)**.** *(a) If a consistent r.e. theory T proves every datum of a consistent reveal sequence, the cumulative loss of M_A is at most $-\log \mathbb{U}_A(\mathrm{Ext}T) \le K_{\equiv}(T) + c$ in the semimeasure convention, in any order. (b) On the lottery stream of Theorem 5.1,*

M_A has bounded loss in the semimeasure convention, while no consistent r.e. theory proves every datum and $W_A(D_n) \to 0$. (c) On two-valued data presenting an incomplete consistent r.e. theory T^*, the limit posterior is the prior restricted to the consistent extensions of T^*, and on every sentence independent of $B \cup T^*$ all three values Bel, Dis, Indep *have positive limits; three-valued data identify the class of T^*. (d) For all D with $M_A(D) > 0$ and all sentences, $\mathrm{Bel}_D(\bigwedge_{i\le k}\varphi_i) \ge 1 - \sum_{i\le k}(1 - \mathrm{Bel}_D(\varphi_i))$, and $\mathrm{Bel}_D(\psi) \ge \mathrm{Bel}_D(\varphi)$ whenever $B \cup \{\varphi\} \vdash \psi$. So the sentences with belief at least $1-\delta$ are closed under entailment and, with error $k\delta$, under k-fold conjunction.*

Proof of Proposition D.3. (a) By Proposition 3.1(d) the cumulative loss in the semimeasure convention is $L_A(D_n) - L_A(\emptyset)$, in any order, and by Theorem 3.11 it is at most $-\log \mathbb{U}_A(\mathrm{Ext}T) \le K_{\equiv}(T) + c$.

(b) In the setting of Theorem 5.1, part (a) of that theorem bounds $M_A(D_n)$ below by a positive constant independent of n, so the cumulative loss $L_A(D_n) - L_A(\emptyset)$ is bounded; part (b) says that no consistent r.e. theory proves every datum (with B); and part (c) gives $W_A(D_n) \to 0$.

(c) This is Proposition A.14(b), (c): if $B \cup T^*$ is incomplete and $\mathrm{Cn}(D^{\pm}_{\infty}) = \mathrm{Cn}(T^*)$, the limit posterior is the prior restricted to the consistent extensions of T^*, and for every ψ independent of $B \cup T^*$ the limits of $\mathrm{Bel}_{D_n}(\psi)$ and $\mathrm{Dis}_{D_n}(\psi)$ lie in $(0,1)$ while $\lim_n \mathrm{Indep}_{D_n}(\psi) \ge \mathbb{U}_A[T^*]/\mathbb{U}_A(\mathrm{Ext}T^*) > 0$; and if every sentence appears with its T^*-trichotomy label, the posterior probability of the class of T^* tends to 1.

(d) A theory that proves φ and ψ proves $\varphi \wedge \psi$, so $O_\varphi \cap O_\psi \subseteq O_{\varphi\wedge\psi}$. Within E_D,

$$\mathbb{U}(O_\varphi \cap O_\psi \cap E_D) \ge \mathbb{U}(O_\varphi \cap E_D) + \mathbb{U}(O_\psi \cap E_D) - \mathbb{U}(E_D);$$

dividing by $\mathbb{U}(E_D)$ gives $\mathrm{Bel}_D(\varphi \wedge \psi) \ge \mathrm{Bel}_D(\varphi) + \mathrm{Bel}_D(\psi) - 1$, and induction on k gives $\mathrm{Bel}_D(\bigwedge_{i\le k}\varphi_i) \ge 1 - \sum_{i\le k}(1 - \mathrm{Bel}_D(\varphi_i))$. If $B \cup \{\varphi\} \vdash \psi$, then $O_\varphi \subseteq O_\psi$, so $\mathrm{Bel}_D(\psi) \ge \mathrm{Bel}_D(\varphi)$. □

D.5 The flagpole

Setting. The language has two sorts, poles and reals; B is the theory of real closed fields on the real sort. There are pole constants $p_1, \dots, p_{N_\mathrm{p}}$, a function symbol ht from poles to reals (the height) and a function symbol sh from poles and reals to reals ($\mathrm{sh}(p,u)$ is the length of the shadow of p when the tangent of the sun's elevation is $u > 0$). Data are sentences $\mathrm{sh}(p_i,u) = v$ and $\mathrm{ht}(p_i) = v$ with rational numerals u, v; interval versions behave in the same way. The two formulations of the shadow law are

$$\Lambda_h := \forall p\,\forall u\,(u > 0 \to \mathrm{sh}(p,u)\cdot u = \mathrm{ht}(p)),$$
$$\Lambda_s := \forall p\,\forall u\,(u > 0 \to \mathrm{sh}(p,u)\cdot u = \mathrm{sh}(p,1)) \ \wedge\ \forall p\,\mathrm{ht}(p) = \mathrm{sh}(p,1),$$

and, with rational numerals a_i for the heights, the four theories are

$$T_H := \Lambda_h + \{\mathrm{ht}(p_i) = a_i\}_{i\le N_\mathrm{p}}, \qquad T_S := \Lambda_h + \{\mathrm{sh}(p_i,1) = a_i\}_{i\le N_\mathrm{p}},$$
$$T'_H := \Lambda_s + \{\mathrm{ht}(p_i) = a_i\}_{i\le N_\mathrm{p}}, \qquad T'_S := \Lambda_s + \{\mathrm{sh}(p_i,1) = a_i\}_{i\le N_\mathrm{p}}.$$

In T_S the shadow at 45° "explains" the height. The *intervention vocabulary* consists of $\mathrm{sh}_H(p,u,v)$, the shadow length after the height of p is set to v, and $\mathrm{ht}_S(p,v)$, the height after the reference shadow $\mathrm{sh}(p,1)$ is set to v by an intervention on the shadow (for instance, by an obstacle). The causal extensions, with the height exogenous and with the shadow exogenous, are

$$C_H := T_H + \{\forall p\,\forall u\,\forall v\,(u > 0 \to \mathrm{sh}_H(p,u,v)\cdot u = v),\ \forall p\,\forall v\,\mathrm{ht}_S(p,v) = \mathrm{ht}(p)\},$$
$$C_S := T_S + \{\forall p\,\forall u\,\forall v\,\mathrm{sh}_H(p,u,v) = \mathrm{sh}(p,u),\ \forall p\,\forall v\,\mathrm{ht}_S(p,v) = v\}.$$

Derivation charges. The cost of a derivation is the number of closed instances it uses of the universally quantified conjuncts of the law axioms (Λ_h has one such conjunct, Λ_s two); steps in B and uses of the initial conditions are free. The *least cost* of a datum for a theory is the least number of such instances that, together with B and the theory's initial conditions, entail it. By completeness this is the least cost of a derivation, so the count is semantic, as in Herbrand's theorem.

The precise form of Proposition 6.5(b) is as follows. For a shadow datum $\mathrm{sh}(p_i, u) = a_i/u$ with $u \neq 1$, the least costs for T_H, T_S, T'_H, T'_S are 1, 2, 2, 1; for a height datum $\mathrm{ht}(p_i) = a_i$ they are 0, 1, 0, 1; and for a reference-shadow datum $\mathrm{sh}(p_i, 1) = a_i$ they are 1, 0, 1, 0. So the two hub theories T_H and T'_S have equal costs on all shadow data with $u \neq 1$. Their law texts differ by the constant $|\Lambda_s| - |\Lambda_h|$, and in a literal code (Appendix D.6) their initial conditions differ by $N_\mathrm{p}\Delta_\mathrm{sh}$, where $\Delta_\mathrm{sh} := |\mathrm{sh}(x, 1)| - |\mathrm{ht}(x)|$ is the extra length of the reference-shadow term; this is 0 if the language has a primitive for the reference shadow, and then the two texts differ by $|\Lambda_s| - |\Lambda_h|$, independently of N_p and the data. The precise form of (c) is: C_H and C_S have the same consequences in the language without sh_H and ht_S; $C_H \vdash \mathrm{sh}_H(p_1, 1, 2a_1) = 2a_1$ ("doubling the pole doubles its 45° shadow"), while $C_S \vdash \mathrm{sh}_H(p_1, 1, 2a_1) = a_1$, which refutes it when $a_1 \neq 0$; and $C_H \vdash \mathrm{ht}_S(p_1, v) = a_1$ while $C_S \vdash \mathrm{ht}_S(p_1, v) = v$, for every numeral v.

Lemma D.4 (Least costs in a star)**.** *Let T be one of the theories of this appendix without the dark poles: law axioms whose universally quantified conjuncts have closed instances of the form $\pi \to s_1 = \rho \cdot s_2$, with π a condition on real numerals, s_1, s_2 terms of one pole and ρ a real term that B proves nonzero when π holds, and one initial condition $s(p_i) = a_i$ per pole. For a pole p_i, let the* instance graph *have as vertices the closed terms* $\mathrm{ht}(p_i)$, $\mathrm{sh}(p_i, u)$ *(terms whose real arguments B proves equal are identified) and, in Appendix D.6,* $\mathrm{qt}_j(p_i)$*, and an edge for each closed instance whose condition B proves, joining the two terms it relates (an instance that relates a term to itself is trivial and is ignored). If the instance graph is a tree, the least cost of a datum $s'(p_i) = v$ that T entails is the distance in the graph from $s'(p_i)$ to the vertex fixed by the initial condition.*

Proof. The instances along the path from the fixed vertex to $s'(p_i)$, together with B and the initial condition, determine the value of $s'(p_i)$; since T entails the datum, that value is v, so the distance is an upper bound. Conversely, let I be a set of fewer instances than the distance. As the graph is a tree, the edges of I do not connect $s'(p_i)$ to the fixed vertex. Fix a model of T over $\mathbb{R}$ (each theory of this appendix has one). Build a model of B, the initial conditions and I over $\mathbb{R}$ as follows: on the connected component (for the edges of I) of the fixed vertex, and on every term of another pole, keep the values of the fixed model; on the component of $s'(p_i)$, give $s'(p_i)$ a value other than v and propagate it along the edges of I, which is consistent because the component is a tree and each edge is an invertible linear relation; give all other terms arbitrary values. Instances with a false condition hold trivially. This model falsifies the datum, so I does not entail it. □

Proof of Proposition 6.5. (a) At $u = 1$, Λ_h gives $\mathrm{sh}(p, 1) = \mathrm{ht}(p)$; given this, $\mathrm{sh}(p, u) \cdot u = \mathrm{ht}(p)$ and $\mathrm{sh}(p, u) \cdot u = \mathrm{sh}(p, 1)$ are equivalent. So $B \vdash \Lambda_h \to \Lambda_s$; conversely Λ_s gives $\mathrm{ht}(p) = \mathrm{sh}(p, 1)$ and hence Λ_h. Given either law, the initial conditions $\mathrm{ht}(p_i) = a_i$ and $\mathrm{sh}(p_i, 1) = a_i$ are interderivable, so the four theories are deductively equivalent over B. Compatibility of a theory with a data set (consistent with B and proving every datum) depends only on its deductive closure, so every data set is compatible with all four or with none. In the coin-free version the posterior weight of a program is $2^{-|p|}$ times the indicator of compatibility, divided by the normaliser, so the odds of two compatible programs are their prior odds. The map on axiom lists that replaces each $\mathrm{ht}(p_i) = a_i$ by $\mathrm{sh}(p_i, 1) = a_i$, and its inverse, are fixed computable maps, so $|K(T_S) - K(T_H)| \leq c$.

(b) For each pole, the instances of Λ_h join $\mathrm{ht}(p_i)$ to each $\mathrm{sh}(p_i,u)$ (a star centred at the height), and the instances of Λ_s join $\mathrm{sh}(p_i,1)$ to each $\mathrm{sh}(p_i,u)$ with $u\neq 1$ and to $\mathrm{ht}(p_i)$ (a star centred at the reference shadow); instances with $u\leq 0$ have false conditions, and instances at other poles are not connected to p_i. Both graphs are trees, and Lemma D.4 gives the counts: in the height-centred star with the height fixed (T_H), shadows and the reference shadow are at distance 1; with the reference shadow fixed (T_S), a shadow with $u\neq 1$ is at distance 2 and the height at 1; in the shadow-centred star with the height fixed (T'_H), a shadow with $u\neq 1$ is at distance 2 and the reference shadow at 1; with the reference shadow fixed (T'_S), shadows and the height are at distance 1. The law texts of T_H and T'_S are Λ_h and Λ_s; their initial conditions differ only in the terms $\mathrm{ht}(p_i)$ and $\mathrm{sh}(p_i,1)$, once per pole.

(c) Without the intervention symbols, C_H and C_S are T_H and T_S plus axioms about fresh symbols. Every model of T_H expands to a model of C_H (put $\mathrm{sh}_H(p,u,v):=v/u$ for $u>0$, arbitrary otherwise, and $\mathrm{ht}_S(p,v):=\mathrm{ht}(p)$), and every model of T_S to a model of C_S (put $\mathrm{sh}_H(p,u,v):=\mathrm{sh}(p,u)$ and $\mathrm{ht}_S(p,v):=v$). So C_H and C_S are conservative over T_H and T_S, whose consequences coincide by (a). The interventional consequences are instances of the added axioms: in C_H, $\mathrm{sh}_H(p_1,1,2a_1)\cdot 1=2a_1$ and $\mathrm{ht}_S(p_1,v)=\mathrm{ht}(p_1)=a_1$; in C_S, $\mathrm{sh}_H(p_1,1,2a_1)=\mathrm{sh}(p_1,1)=a_1$ and $\mathrm{ht}_S(p_1,v)=v$. If $a_1\neq 0$ then $2a_1\neq a_1$, so C_S refutes the datum $\mathrm{sh}_H(p_1,1,2a_1)=2a_1$ that C_H proves. □

Computations. A computation (code in the accompanying repository) confirms the counts by exhaustive search over all subsets of finite instance pools, with exact linear algebra over the rationals, on 500 random data sets with one to four poles (shadows at $u=2$, and at $u=3$ in a second implementation); on 2000 random data the four theories prove, refute or leave open exactly the same data. In the literal syntax of that code (numerals counted as one token), $|\Lambda_s|-|\Lambda_h|=20$ tokens and $\Delta_{\mathrm{sh}}=1$ token, so T_S is longer than T_H by one token per pole.

D.6 The flagpole with a system of laws

Kitcher's answer to the asymmetry appeals to other laws in which the height figures, which the toy of Proposition 6.5 lacks. We add such laws.

Setting. As in Appendix D.5, with further function symbols $\mathrm{qt}_1,\dots,\mathrm{qt}_k$ from poles to reals (further quantities: weight, the height on the construction plan, and so on), nonzero rational numerals $\theta_1,\dots,\theta_k$, and data $\mathrm{qt}_j(p_i)=v$. The laws in height form and in shadow form are $\Lambda_j^{\mathrm{ht}}:=\forall p\,\mathrm{qt}_j(p)=\theta_j\cdot\mathrm{ht}(p)$ and $\Lambda_j^{\mathrm{sh}}:=\forall p\,\mathrm{qt}_j(p)=\theta_j\cdot\mathrm{sh}(p,1)$. The theories are the height hub $T_H^k:=\Lambda_h+\{\Lambda_j^{\mathrm{ht}}\}_{j\leq k}+\{\mathrm{ht}(p_i)=a_i\}_i$, the mixed theory $T_S^k:=\Lambda_h+\{\Lambda_j^{\mathrm{ht}}\}_j+\{\mathrm{sh}(p_i,1)=a_i\}_i$, and the shadow hub $T_{SS}^k:=\Lambda_s+\{\Lambda_j^{\mathrm{sh}}\}_j+\{\mathrm{sh}(p_i,1)=a_i\}_i$.

Dark variant. Add a predicate Sunlit on poles, restrict the shadow law to sunlit poles, $\Lambda_h^{\mathrm{D}}:=\forall p\,(\mathrm{Sunlit}(p)\to\forall u\,(u>0\to\mathrm{sh}(p,u)\cdot u=\mathrm{ht}(p)))$, and take sunlit poles $p_1,\dots,p_{N_{\mathrm{p}}}$ and dark poles $p'_1,\dots,p'_{N_{\mathrm{d}}}$, with the axioms $\mathrm{Sunlit}(p_i)$ and $\neg\mathrm{Sunlit}(p'_l)$ in every theory and rational numerals a'_l for the heights of the dark poles. Laws in shadow form must split cases:

$$\Lambda_j^{\mathrm{sh,D}}:=\forall p\,\Big((\mathrm{Sunlit}(p)\to\mathrm{qt}_j(p)=\theta_j\cdot\mathrm{sh}(p,1))\wedge(\neg\mathrm{Sunlit}(p)\to\mathrm{qt}_j(p)=\theta_j\cdot\mathrm{ht}(p))\Big).$$

The theories are $T_H^{k,\mathrm{D}}:=\Lambda_h^{\mathrm{D}}+\{\Lambda_j^{\mathrm{ht}}\}_j+\{\mathrm{ht}(p_i)=a_i\}_i+\{\mathrm{ht}(p'_l)=a'_l\}_l$ and $T_{SS}^{k,\mathrm{D}}:=\Lambda_h^{\mathrm{D}}+\{\Lambda_j^{\mathrm{sh,D}}\}_j+\{\mathrm{sh}(p_i,1)=a_i\}_i+\{\mathrm{ht}(p'_l)=a'_l\}_l$, each with the Sunlit axioms. A *literal code* charges a theory by the number of symbols it writes, each axiom separately, without definitions; $|\cdot|$ denotes this

length, a symbol qt_j counts as one token, and $\Delta_{\mathrm{sh}} := |\mathrm{sh}(x,1)| - |\mathrm{ht}(x)|$ is the extra length of the reference-shadow term ($\Delta_{\mathrm{sh}} = 0$ if the language has a primitive for the reference shadow).

Proposition D.5 (The flagpole with a system of laws)**.** *(a)* Class tie. *T_H^k, T_S^k and T_{SS}^k are deductively equivalent over B, and so are $T_H^{k,\mathrm{D}}$ and $T_{SS}^{k,\mathrm{D}}$; so for every k no data set separates them, and their deductive classes coincide. (b)* Universal prior. *$|K(T_{SS}^k) - K(T_H^k)| \le c$ and $|K(T_{SS}^{k,\mathrm{D}}) - K(T_H^{k,\mathrm{D}})| \le c$, with c independent of k, N_{p}, N_{d} and the numerals. (c)* Literal codes. *$|T_{SS}^k| - |T_H^k| = (|\Lambda_s| - |\Lambda_h|) + (k + N_{\mathrm{p}})\Delta_{\mathrm{sh}}$. In the dark variant, $|T_{SS}^{k,\mathrm{D}}| - |T_H^{k,\mathrm{D}}| = k\,\Delta_{\mathrm{split}} + \sum_{j\le k}|\theta_j| + N_{\mathrm{p}}\Delta_{\mathrm{sh}}$, where $\Delta_{\mathrm{split}} > 0$ is the length of the case-split frame and $|\theta_j|$ the length of the numeral θ_j, which the split law writes twice; so the shadow hub loses by at least $k\,\Delta_{\mathrm{split}}$, an amount growing with k, even when $\Delta_{\mathrm{sh}} = 0$. With one definition the overhead of the laws is a constant: define $\mathrm{z}(p)$ once as $\mathrm{sh}(p,1)$ on sunlit poles and $\mathrm{ht}(p)$ on dark ones, and write every law in* z*; in every model of the theory,* z *is coextensive with* ht*. (d)* Derivation charges *(instances of universally quantified conjuncts, as in Appendix D.5). A datum $\mathrm{qt}_j(p_i) = \theta_j a_i$ has least cost* 1 *for T_H^k and T_{SS}^k, and* 2 *for the mixed theory T_S^k. On shadow data with $u \ne 1$ the two hub theories have least cost* 1 *each; on heights T_H^k has* 0 *and T_{SS}^k has* 1*; on reference shadows the reverse.*

Proof. (a) Each theory contains a form of the shadow law, which gives $B \vdash \mathrm{ht}(p) = \mathrm{sh}(p,1)$ for every pole p (for every sunlit pole in the dark variant), as in the proof of Proposition 6.5(a). Substituting, each law in shadow form is equivalent to the law in height form, and each reference-shadow initial condition to the corresponding height condition. In the dark variant, the case split collapses to $\mathrm{qt}_j(p) = \theta_j \cdot \mathrm{ht}(p)$, because on sunlit poles $\mathrm{sh}(p,1) = \mathrm{ht}(p)$ by Λ_h^{D} at $u = 1$.

(b) The syntactic translations and their inverses are fixed computable maps on axiom lists: from T_H^k to T_{SS}^k, replace Λ_h by Λ_s, each Λ_j^{ht} by Λ_j^{sh} and each $\mathrm{ht}(p_i) = a_i$ by $\mathrm{sh}(p_i,1) = a_i$; from $T_H^{k,\mathrm{D}}$ to $T_{SS}^{k,\mathrm{D}}$, replace each Λ_j^{ht} by $\Lambda_j^{\mathrm{sh,D}}$ (reading θ_j and qt_j off the axiom) and the initial condition of each pole declared sunlit by its reference-shadow form. So a program for one list yields a program for the other at a constant extra length, independent of k, N_{p}, N_{d} and the numerals.

(c) Counting symbols: each Λ_j^{sh} differs from Λ_j^{ht} by one occurrence of the replaced term, and so does each sunlit initial condition; this gives $(k + N_{\mathrm{p}})\Delta_{\mathrm{sh}}$ besides the law texts. Each $\Lambda_j^{\mathrm{sh,D}}$ is longer than Λ_j^{ht} by a fixed frame of length $\Delta_{\mathrm{split}} > 0$ (the connectives, the two Sunlit atoms, a second copy of $\mathrm{qt}_j(p) = \cdot$, and the term $\mathrm{sh}(p,1)$) and by a second copy of the numeral θ_j; both dark theories contain Λ_h^{D}, the dark initial conditions and the Sunlit axioms. The definition of z is a single axiom of constant length, and laws written in z have the length of the laws written in ht. In every model of the theory, $\mathrm{z}(p) = \mathrm{sh}(p,1) = \mathrm{ht}(p)$ on sunlit poles by Λ_h^{D} at $u = 1$, and $\mathrm{z}(p) = \mathrm{ht}(p)$ on dark poles by the definition.

(d) For each pole the instance graphs are trees: for T_H^k and T_S^k, a star centred at $\mathrm{ht}(p_i)$ with leaves $\mathrm{sh}(p_i,u)$ and $\mathrm{qt}_j(p_i)$; for T_{SS}^k, a star centred at $\mathrm{sh}(p_i,1)$ with leaves $\mathrm{sh}(p_i,u)$ ($u \ne 1$), $\mathrm{ht}(p_i)$ and $\mathrm{qt}_j(p_i)$. Lemma D.4 gives the counts: $\mathrm{qt}_j(p_i)$ is at distance 1 from the fixed centre in T_H^k and T_{SS}^k, and at distance 2 from the fixed leaf $\mathrm{sh}(p_i,1)$ in T_S^k; the shadow, height and reference-shadow counts are those of Proposition 6.5(b) for T_H and T_S'. □

E Proofs for theory shape

Throughout, $\mathcal{S}$ is a law on the base variables $x_1, \dots, x_m$, Φ an explicit or linear theory with m_{aux} auxiliary variables and $\bar{m} := m + m_{\mathrm{aux}}$ variables in all, and $\mathbf{e}_j$ the j-th unit vector. A partial

assignment is identified with the set of literals it makes true, and $\hat{x}$ is the conjunction of the base literals true at $x \in \{0,1\}^m$. For a sentence or literal φ, $\varphi^1 := \varphi$ and $\varphi^0 := \neg\varphi$.

E.1 The model: the listing code, the two encodings, the bridge

Theories that differ only by a renaming of their auxiliary variables are identified, and each class has a fixed canonical representative whose auxiliary variables are $y_1, \dots, y_{m_{\mathrm{aux}}}$ and whose clauses are listed in a fixed canonical order (for instance the lexicographically least listing). Let $\ell_{\mathbb{N}}(n) := |\mathrm{El}(n+1)|$ for $n \geq 0$, with El the Elias delta code of the positive integers; $\ell_{\mathbb{N}}$ is nondecreasing and the lengths of a prefix code of $\{0,1,2,\dots\}$. The *listing length* of a canonical explicit theory is

$$\ell_{\mathrm{lst}}(\Phi) := \ell_{\mathbb{N}}(m_{\mathrm{aux}}) + \ell_{\mathbb{N}}(|\Phi|) + \sum_{C \in \Phi} \big(\ell_{\mathbb{N}}(|C|) + |C|(\lceil \log \bar{m} \rceil + 1)\big), \tag{4}$$

each literal being written as a variable index and a sign; a linear theory is listed likewise, each equation by its support and its constant, with $\ell_{\mathbb{N}}(|u|) + |u|\lceil \log \bar{m} \rceil + 1$ bits for an equation of support size $|u|$. Clearly $\ell_{\mathrm{lst}}(\Phi) \geq \|\Phi\|$.

Lemma E.1 (The listing code)**.** $\sum 2^{-\ell_{\mathrm{lst}}(\Phi)} \leq 1$ *over the canonical explicit theories, and* $\ell_{\mathrm{lst}}(\Phi \setminus \{C\}) < \ell_{\mathrm{lst}}(\Phi)$ *for every* $C \in \Phi$*. In particular an* ℓ_{lst}*-shortest* $\mathcal{R}$*-compatible theory is inclusion-minimal among the* $\mathcal{R}$*-compatible theories.*

Proof. Fix m_{aux}. The code of a single clause (its length by $\ell_{\mathbb{N}}$, then that many literals of $\lceil \log \bar{m} \rceil + 1$ bits each) is a prefix code on finite sequences of literals, so $\sum_C 2^{-\ell(C)} \leq 1$ over clauses written in canonical order, where $\ell(C)$ is the summand of (4). The sets of s distinct clauses have total weight at most $(\sum_C 2^{-\ell(C)})^s / s! \leq 1$. The codes $\ell_{\mathbb{N}}(m_{\mathrm{aux}})$ and $\ell_{\mathbb{N}}(s)$ contribute factors whose sums over m_{aux} and over s are at most 1, and canonical theories are a subset of all clause sets; this gives the Kraft inequality. Removing a clause C removes $\ell(C) > 0$ bits and replaces $\ell_{\mathbb{N}}(s)$ by $\ell_{\mathbb{N}}(s-1) \leq \ell_{\mathbb{N}}(s)$; if it removes the last occurrence of an auxiliary variable, then m_{aux} and $\lceil \log \bar{m} \rceil$ do not increase either (after renaming to canonical form, which does not change ℓ_{lst}). □

For two base variables the Kraft sum is 0.2525 (a direct computation over all 256 theories; code in the accompanying repository).

Lemma E.2 (Compatibility with all questions forces the projection)**.** *Let* $\mathcal{R}$ *be sound and* Φ $\mathcal{R}$*-compatible with* $\mathcal{Q}(\mathcal{S})$*. Then* $\mathrm{proj}\,\Phi \subseteq \mathcal{S}$*; and* $\mathrm{proj}\,\Phi = \mathcal{S}$ *if every* $x \in \mathcal{S}$ *has a coordinate that is determined on* $\mathcal{S}$ *by the others. This holds for every affine law and for the chain law* $\mathrm{Ch}_m := \{x : x_1 \leq \cdots \leq x_m\}$ *with* $m \geq 3$*; so for these laws,* $\mathcal{R}$*-compatibility with* $\mathcal{Q}(\mathcal{S})$ *and* $\mathcal{R}$*-usability coincide.*

Proof. Let $x \notin \mathcal{S}$. Choose $I \subsetneq [m]$ maximal such that $\alpha := x|_I$ is consistent with $\mathcal{S}$ ($I = \emptyset$ qualifies since $\mathcal{S} \neq \emptyset$, and $I \neq [m]$ since $x \notin \mathcal{S}$). For $j \notin I$, the extension of α by the value of x_j at x is inconsistent with $\mathcal{S}$, so $\mathcal{S} \wedge \alpha$ entails the literal ℓ on x_j that is false at x, and $(\alpha, \ell) \in \mathcal{Q}(\mathcal{S})$. Compatibility and soundness give $\Phi \cup \alpha \models \ell$; x extends α and falsifies ℓ, so $x \notin \mathrm{proj}\,\Phi$. Conversely, let $x \in \mathcal{S}$ have its coordinate j determined by $\alpha := x|_{[m]\setminus\{j\}}$. Then $(\alpha, \ell) \in \mathcal{Q}(\mathcal{S})$ for the literal ℓ on x_j true at x, so $\Phi \cup \alpha$ is satisfiable, by a point of $\mathrm{proj}\,\Phi \subseteq \mathcal{S}$ extending α, and the only such point of $\mathcal{S}$ is x. For an affine law, a coordinate j is determined by the others if and only if j lies in the support of a nonzero derived equation (Lemma E.6 with $I = [m] \setminus \{j\}$); such a j exists since $r \geq 1$, and the property does not depend on x. For Ch_m, the points are $0^a 1^{m-a}$; if $a \geq 2$, then $x_2 = 0$ forces $x_1 = 0$, and if $m - a \geq 2$, then $x_{m-1} = 1$ forces $x_m = 1$; one of the two holds as $m \geq 3$. □

The hypothesis $m \geq 3$ is needed: $\Phi = \{\neg x_1 \vee x_2,\ x_1 \vee \neg x_2\}$ is UP- and PS_∞-compatible with $\mathcal{Q}(\mathrm{Ch}_2)$, whose only questions are "$x_1 = 1$, so $x_2 = 1$" and "$x_2 = 0$, so $x_1 = 0$", but $\operatorname{proj}\Phi = \{00, 11\} \neq \mathrm{Ch}_2$; the point 01 has no determined coordinate.

Proposition E.3 (The two-valued encoding is degenerate)**.** *Encode a conditional datum (α, ℓ) by the sentence $\bigwedge\alpha \to \ell$ labelled true, with $B = \emptyset$. Then every consistent theory that proves all sentences proved by a compatible theory is compatible. In particular, for every law $\mathcal{S}$ and every $x^0 \in \mathcal{S}$, the theory $T_{x^0} := \{x_i^{x^0_i} : i \leq m\}$ proves $\bigwedge\alpha \to \ell$ for every $(\alpha, \ell) \in \mathcal{Q}(\mathcal{S})$, so $K_A(\{(\bigwedge\alpha \to \ell, 1) : (\alpha, \ell) \in \mathcal{Q}(\mathcal{S})\}) \leq K(x^0) + c$; for a homogeneous affine law ($b = 0$, $x^0 = 0$) this is at most $K(m) + c$.*

Proof. Compatibility with a datum labelled true asks only that the theory be consistent and prove the sentence, which passes to consistent theories proving more. If α agrees with x^0, then x^0 is a point of $\mathcal{S}$ extending α, so x^0 satisfies ℓ and $T_{x^0} \vdash \ell$; otherwise $T_{x^0} \vdash \neg\bigwedge\alpha$. Either way $T_{x^0} \vdash \bigwedge\alpha \to \ell$, and T_{x^0} is consistent and computable from x^0. □

Under this encoding the data cannot distinguish a law from a single world that satisfies it. Encoding (E3) of Definition 2.11 makes "given α" hypothetical: a compatible theory may not refute an antecedent, so no single world fits question data with two incompatible antecedents. (E3) is not upward closed, so it cannot be expressed by two-valued proof-requiring labels. For a conditional data set D we write $K_A(D)$ and $W_A(D)$ for the inducers of Definition 2.7 over the language $\mathcal{L}_m$ with $B = \emptyset$, in which an r.e. theory T is compatible with D if it is compatible with every datum of D under (E3).

Proposition E.4 (The bridge to SAI)**.** *(a) With $\mathcal{R} = \mathrm{PS}_\infty$ and as hypotheses the r.e. sets of $\mathcal{L}_m$-sentences, or the r.e. sets of clauses over the base variables, $K_A^{\mathrm{PS}_\infty}(D) = K_A(D) \pm c$ and $W_A^{\mathrm{PS}_\infty}(D) \asymp W_A(D)$ on every conditional data set D. (b) With finite explicit theories, of either class, $K_A(D) - c \leq K_A^{\mathrm{PS}_\infty}(D) \leq K_A(D) + \ell_{\mathbb{N}}(m) + \ell_{\mathbb{N}}(2^m) + c$, that is, up to $m + O(\log m)$ bits.*

Proof. (a) For sets of sentences, PS_∞-compatibility of an r.e. set T is (E3), by the completeness theorem: $T \cup \alpha$ is satisfiable if and only if it is consistent, and PS_∞ returns ℓ if and only if $T \cup \alpha \vdash \ell$. For sets of clauses, a fixed program turns an enumeration of T into an enumeration of $\mathrm{Cl}(T)$, the set of clauses over the base variables derivable from T. For every question, $T \cup \alpha$ is inconsistent iff $T \vdash \neg\bigwedge\alpha$ iff the clause $\neg\alpha$ is in $\mathrm{Cl}(T)$ iff $\mathrm{Cl}(T) \cup \alpha$ is inconsistent (as $\mathrm{Cl}(T) \subseteq \mathrm{Cn}(T)$); and $T \cup \alpha \vdash \ell$ iff the clause $\neg\alpha \vee \ell$ (non-tautological, as ℓ is on a variable outside $\operatorname{dom}\alpha$) is in $\mathrm{Cl}(T)$ iff $\mathrm{Cl}(T) \cup \alpha \vdash \ell$. So T and $\mathrm{Cl}(T)$ are compatible with the same conditional data. (A theory of $\mathcal{L}_m$ need not consist of Boolean combinations of the atoms, since $\mathcal{L}_m$ has equality and quantifiers; $\mathrm{Cl}(T)$ replaces a normal form.) Conversely, clauses are sentences. The weights compare in the same way, through the fixed maps on programs. (b) *Left.* A fixed prefix replaces a finite explicit theory Φ, whose auxiliary variables are not symbols of $\mathcal{L}_m$, by $\mathrm{PI}(\operatorname{proj}\Phi)$, a finite set of $\mathcal{L}_m$-sentences. It is compatible with the same data: $\Phi \cup \alpha$ is satisfiable iff $\operatorname{proj}\Phi$ meets α, and for a base literal ℓ, $\Phi \cup \alpha \models \ell$ iff every point of $\operatorname{proj}\Phi$ extending α satisfies ℓ. *Right.* Let p enumerate a consistent T that is compatible with D under (E3), and let $\mathcal{S}_T := \{x : T \cup \{\hat{x}\} \text{ consistent}\}$. Then $\mathcal{S}_T \neq \emptyset$, since otherwise T proves $\bigwedge_x \neg\hat{x}$, which is refutable. Since $x \notin \mathcal{S}_T$ iff $T \vdash \neg\hat{x}$, a program given p, m and $|\mathcal{S}_T|$ (self-delimiting, with $\ell_{\mathbb{N}}(m) + \ell_{\mathbb{N}}(|\mathcal{S}_T|) + c$ extra bits) enumerates refutations until $2^m - |\mathcal{S}_T|$ points are refuted, and outputs $\mathrm{PI}(\mathcal{S}_T)$. By completeness, $T \cup \alpha$ is consistent iff $\mathcal{S}_T$ meets α, and $T \cup \alpha \vdash \ell$ iff every point of $\mathcal{S}_T$ extending α satisfies ℓ, as ℓ is a base literal. So $\mathrm{PI}(\mathcal{S}_T)$ is PS_∞-compatible with D exactly when T is, and $\ell_{\mathbb{N}}(|\mathcal{S}_T|) \leq \ell_{\mathbb{N}}(2^m)$. □

So the reasoner-relative inducers extend SAI: with an ideal deducer and (E3) they are SAI on question data, up to the gap between finite and r.e. hypotheses in (b). We do not know whether that gap is attained.

E.2 Program length, listing and time

Lemma E.5 (Prime implicates and one round of propagation)**.** *Let $\mathcal{S}$ be any law. (a) If α is consistent with $\mathcal{S}$ and $\mathcal{S} \wedge \alpha \models \ell$, one round of unit propagation on* $\mathrm{PI}(\mathcal{S}) \cup \alpha$ *derives ℓ. (b) Every* UP_1*-usable $\Phi \in \mathcal{E}_m^{\mathrm{base}}$ contains* $\mathrm{PI}(\mathcal{S})$*.*

Part (a) is folklore; compiling a theory so that unit resolution becomes complete is studied by del Val (1994).

Proof. (a) The clause $\neg\alpha \vee \ell$ is an implicate of $\mathcal{S}$, so it contains a prime implicate C. If $\ell \notin C$, then $C \subseteq \neg\alpha$ is falsified by α, contradicting the consistency of α with $\mathcal{S}$. So $\ell \in C$ and all other literals of C are falsified by α: C fires in round 1. No conflict arises, since $\mathrm{PI}(\mathcal{S}) \cup \alpha$ is satisfiable. (b) Let $C = \neg\alpha' \vee \ell \in \mathrm{PI}(\mathcal{S})$. Since C is prime, $\neg\alpha'$ is not an implicate, so α' is consistent with $\mathcal{S}$ and $(\alpha', \ell) \in \mathcal{Q}(\mathcal{S})$. In one round from α', the clause of Φ that derives ℓ has all its other literals falsified by α', so it is some $C' \subseteq C$ with $\ell \in C'$; C' is an implicate of $\mathcal{S}$ (as Φ is a theory over the base variables with $\mathrm{proj}\,\Phi = \mathcal{S}$), so $C' = C$ by primality. □

Proof of Theorem 7.2. Let $\mathrm{comp}(\Phi') := \mathrm{PI}(\mathrm{proj}\,\Phi')$, computable from Φ' (a linear theory is first written as clauses). If Φ' is $\mathcal{R}'$-compatible with D then, by soundness, for each $(\alpha, \ell) \in D$ the set $\Phi' \cup \alpha$ is satisfiable and entails ℓ; so $\mathrm{proj}\,\Phi'$ meets α, and every point of it extending α satisfies ℓ, as ℓ is a base literal. By Lemma E.5(a) applied to the law $\mathrm{proj}\,\Phi'$, $\mathrm{comp}(\Phi') \in \mathcal{E}_m^{\mathrm{base}}$ is UP_1-compatible with D. A program for $\mathrm{comp}(\Phi')$ is a fixed prefix followed by a program for Φ', which gives both inequalities. For the equalities: UP_1, UP and PS_∞ are sound, so each is covered by the inequality, within either class; conversely a UP_1-compatible theory is UP- and PS_∞-compatible, since on a satisfiable $\Phi \cup \alpha$ these reasoners derive at least what UP_1 derives and never reach a conflict. □

On the question data of any law, a one-round unit-propagation reasoner over the base variables is therefore as good, under program length, as Gaussian elimination or unbounded proof search with any intermediate concepts, up to a machine constant.

Proof of Theorem 7.3. (a) By Lemma E.2, for an affine law and a sound reasoner, compatibility with $\mathcal{Q}(\mathcal{S})$ is usability. (b) The lower bound is the printing time: V needs at least $\|\Phi\|$ steps to print $\langle\Phi\rangle$. For the upper bounds, a program runs a program p for (A, b) (time t_p), enumerates the 2^r derived equations, keeps the circuits (time $\mathrm{poly}(m)4^r$), and prints either their clauses, which form $\mathrm{PI}(\mathcal{S})$ (Proposition E.7; time $\mathrm{poly}(m)|\mathrm{PI}(\mathcal{S})|$), or the library of Proposition E.14 with intermediate concepts (at most 2^r circuits, each with at most $4m$ clauses), or, for GE, just the r equations, without enumerating the derived equations. Then log of the total time is at most $\log t_p$ plus $\log|\mathrm{PI}(\mathcal{S})| + O(r + \log m)$, resp. $O(r + \log m)$, resp. $O(\log m)$, and $|p| + \log t_p = Kt(A, b)$ for an optimal p. (A program of length $K(A, b)$ for (A, b) need not run within any computable time bound, which is why Kt appears.) (c) Upper bounds: from $\mathcal{S}$ a fixed program computes $\mathrm{PI}(\mathcal{S}) \in \mathcal{E}_m^{\mathrm{base}}$, which is usable for every UP_k, UP and PS_∞ (Lemma E.5(a)), and a canonical system of equations for $\mathcal{S}$, which is usable for GE. Lower bound: from a program for any compatible Φ a fixed program computes $\mathrm{proj}\,\Phi = \mathcal{S}$ (Lemma E.2). □

E.3 Linear laws over the base variables

Throughout this subsection $\mathcal{S} = \{x : Ax = b\} \neq \emptyset$ with $A \in \mathrm{GF}(2)^{r\times m}$ of rank $r \geq 1$, $\mathrm{Row}(A)$ is the set of derived equations, $b_u := y \cdot b$ for $u = yA$ (well defined as A has full row rank, and linear in u), and a_* is the number of derived equations of weight d_*. Over GF(2) a vector is determined by its support, so a circuit is the unique nonzero derived equation supported in its support.

Lemma E.6 (Determination)**.** *For $I \subseteq [m]$ and $j \notin I$ the following are equivalent: (i) x_j is a function of x_I on $\mathcal{S}$; (ii) some $u \in \mathrm{Row}(A)$ has $j \in \mathrm{supp}(u) \subseteq I \cup \{j\}$; (iii) some circuit has this property.*

Proof. (ii)⇒(i): $x_j = b_u + \sum_{i\in\mathrm{supp}(u)\setminus\{j\}} x_i$ on $\mathcal{S}$. (i)⇒(ii): if no such u exists, then $\mathbf{e}_j \notin \mathrm{Row}(A) + \mathrm{span}\{\mathbf{e}_i : i \in I\}$, so some z is orthogonal to that space and has $z_j = 1$; then $z \in \ker A$ and $z_i = 0$ on I, and $x, x + z \in \mathcal{S}$ agree on I and differ at j. (ii)⇒(iii): take u of least support among the witnesses of (ii). If a nonzero derived equation u' had $\mathrm{supp}(u') \subsetneq \mathrm{supp}(u)$, then either $j \in \mathrm{supp}(u')$, contradicting minimality, or $u + u'$, with support $\mathrm{supp}(u) \setminus \mathrm{supp}(u') \ni j$, is a smaller witness. So u is a circuit. □

Proposition E.7 (Prime implicates of an affine law)**.** *For $Z \subseteq [m]$, the projection of $\mathcal{S}$ to Z is the affine subspace cut out by the derived equations supported in Z; so $C_{Z,\beta}$ is an implicate of $\mathcal{S}$ if and only if β violates some derived equation supported in Z. The prime implicates of $\mathcal{S}$ are exactly the clauses $C_{\mathrm{supp}(u),\beta}$ with u a circuit and $u \cdot \beta \neq b_u$; there are $\sum_{\text{circuits } u} 2^{|u|-1}$ of them, and $\mathrm{PI}(\mathcal{S})$ is UP_1-usable.*

Proof. An affine subspace is the common solution set of the affine functionals constant on it. The functionals on $\{0,1\}^Z$ that are constant on the projection of $\mathcal{S}$ are the restrictions of the functionals constant on $\mathcal{S}$, that is of $\mathrm{Row}(A)$ (as $\mathcal{S} = x^0 + \ker A$ and $(\ker A)^\perp = \mathrm{Row}(A)$), that vanish off Z. So $C_{Z,\beta}$ is an implicate iff β is outside the projection iff β violates some derived equation u with $\mathrm{supp}(u) \subseteq Z$. Every such u is a sum of circuits supported in $\mathrm{supp}(u)$: if u is not a circuit, some circuit u' has $\mathrm{supp}(u') \subsetneq \mathrm{supp}(u)$, and $u + u'$ has the smaller support $\mathrm{supp}(u) \setminus \mathrm{supp}(u')$; induct. Since $u \mapsto b_u$ is linear, β violates one of these circuits, u_0, and $C_{\mathrm{supp}(u_0),\beta|_{\mathrm{supp}(u_0)}} \subseteq C_{Z,\beta}$ is an implicate. A clause $C_{\mathrm{supp}(u),\beta}$ with u a circuit and $u \cdot \beta \neq b_u$ is prime, since a strictly smaller implicate would be violated on a smaller support, by a derived equation supported there, against the minimality of circuits. For each circuit there are $2^{|u|-1}$ such β, and distinct circuits have distinct supports. $\mathrm{PI}(\mathcal{S})$ defines $\mathcal{S}$, and is UP_1-compatible with $\mathcal{Q}(\mathcal{S})$ by Lemma E.5(a). □

Proposition E.8 (Any CNF)**.** *Every implicate of $\mathcal{S}$ has width at least d_*, so every CNF over the base variables equivalent to $\mathcal{S}$ has at least $(1 - 2^{-r})2^{d_*}$ clauses.*

Proof. For $|Z| < d_*$ no nonzero derived equation is supported in Z, so the projection of $\mathcal{S}$ to Z is $\{0,1\}^Z$ (Proposition E.7) and no clause over Z is an implicate. A clause of width $w \geq d_*$ excludes $2^{m-w} \leq 2^{m-d_*}$ points, and the $2^m - 2^{m-r}$ non-solutions must all be excluded. □

Lemma E.9 (Subsumption)**.** *Let $\mathcal{S}$ be any law, Φ a set of implicates of $\mathcal{S}$ over the base variables, and Φ' obtained by replacing each clause C by any prime implicate $C' \subseteq C$. For every α consistent with $\mathcal{S}$ and every k, every literal derived by UP_k on $\Phi \cup \alpha$ is derived by UP_k on $\Phi' \cup \alpha$, and neither run reaches a conflict. So minimum usable theories over the base variables may be sought among subsets of $\mathrm{PI}(\mathcal{S})$, round for round.*

Proof. Both runs derive only literals entailed by $\mathcal{S} \wedge \alpha \neq \emptyset$, so neither conflicts. Induct on rounds. If C fires in round t of the Φ-run, deriving L, all other literals of C are falsified after round $t-1$, hence, by induction, in the Φ'-run. If $L \notin C'$, every literal of $C' \subseteq C$ would be falsified by the state of the Φ-run after round $t-1$; that state holds at every point of $\mathcal{S} \wedge \alpha$, and such a point would falsify the implicate C', which is impossible. So $L \in C'$, and C' fires by round t. □

Only one direction holds: Φ' derives at least what Φ derives, and the converse fails (see the counterexamples after Theorem E.31).

Decomposable circuits. Recall that a circuit u is decomposable if $u = u' + u''$ with $u', u'' \in \mathrm{Row}(A)$ nonzero and $\mathrm{supp}(u') \cap \mathrm{supp}(u'') = \{i\}$ for a single $i \notin \mathrm{supp}(u)$. Equivalently:

$$\text{for some, equivalently every, } j \in \mathrm{supp}(u) \text{ there are } u' \in \mathrm{Row}(A) \text{ and } i \notin \mathrm{supp}(u) \text{ with } i \in \mathrm{supp}(u') \subseteq (\mathrm{supp}(u) \setminus \{j\}) \cup \{i\}. \tag{5}$$

Indeed, a decomposition through i gives such a u' for every j, taking u' or u'', whichever does not contain j (they partition $\mathrm{supp}(u)$); conversely such a u' gives the decomposition $u = u' + (u + u')$, whose supports meet in $\{i\}$.

Proposition E.10 (Indecomposable circuits must be written)**.** *For every d_*, every* UP*-usable explicit theory over the base variables contains all $2^{|u|-1}$ clauses $C_{\mathrm{supp}(u),\beta}$, $u \cdot \beta \neq b_u$, of every indecomposable circuit u.*

Proof. Let Φ be usable (so its clauses are implicates), u indecomposable, $j \in \mathrm{supp}(u)$, $I := \mathrm{supp}(u) \setminus \{j\}$, and α any assignment to I. No nonzero derived equation is supported in I, as u is a circuit, so α is consistent and determines x_j (Proposition E.7 and Lemma E.6). Unit propagation never conflicts on the satisfiable $\Phi \cup \alpha$, and by usability it derives a literal outside I; so round 1 derives one. A literal derived in round 1 on x_i, $i \notin I$, comes from a clause $C \in \Phi$ whose other literals are on I and falsified by α, so $\mathrm{supp}(C) \subseteq I \cup \{i\}$, and C is an implicate. By Proposition E.7 some derived equation u' with $\mathrm{supp}(u') \subseteq \mathrm{supp}(C)$ is violated by the assignment that falsifies C, and $i \in \mathrm{supp}(u')$, since otherwise u' would be supported in I. If $i \neq j$, then $i \notin \mathrm{supp}(u)$ and u' contradicts indecomposability by (5). So $i = j$, $\mathrm{supp}(u') \subseteq \mathrm{supp}(u)$, hence $u' = u$ by the minimality of circuits, and $C = C_{\mathrm{supp}(u),\beta}$ for the assignment β that extends α by the value of x_j opposite to the derived one. The $2^{|u|-1}$ choices of α give all the clauses of u. □

Proof of Proposition 7.4. By Proposition E.10 it remains to show that for every clause $C \notin \Phi_{\mathrm{ind}}(\mathcal{S})$ some UP-usable theory over the base variables omits C. Let $\mathrm{Imp}(\mathcal{S})$ be the set of all non-tautological implicates over the base variables; we show that $\mathrm{Imp}(\mathcal{S}) \setminus \{C\}$ is usable. It consists of implicates, so unit propagation on it from a consistent antecedent never conflicts, and once it is compatible with $\mathcal{Q}(\mathcal{S})$ its projection is $\mathcal{S}$ (Lemma E.2). If C is not a prime implicate, the set contains $\mathrm{PI}(\mathcal{S})$ and is usable by Lemma E.5(a) and the monotonicity of unit propagation in the clause set. Otherwise $C = C_{\mathrm{supp}(u),\beta}$ for a decomposable circuit u. Let $(\alpha, \ell) \in \mathcal{Q}(\mathcal{S})$. In the one-round run on $\mathrm{PI}(\mathcal{S}) \cup \alpha$ some prime implicate $C' \subseteq \neg\alpha \vee \ell$ fires. If $C' \neq C$, it is in our set. If $C' = C$, then ℓ is the literal of C on some $j \in \mathrm{supp}(u)$, and α contains $\beta|_{\mathrm{supp}(u)\setminus\{j\}}$. By (5) there are u' and $i \notin \mathrm{supp}(u)$ with $i \in \mathrm{supp}(u') \subseteq (\mathrm{supp}(u) \setminus \{j\}) \cup \{i\}$; put $u'' := u + u'$, so that $j \in \mathrm{supp}(u'') \subseteq (\mathrm{supp}(u) \setminus \mathrm{supp}(u')) \cup \{i\}$. Let v_i be the value of x_i that u' forces given α. The clause on $\mathrm{supp}(u')$ falsified by $\alpha|_{\mathrm{supp}(u')\setminus\{i\}} \cup \{x_i = 1 - v_i\}$ is an implicate (that assignment violates u'), differs from C (its support contains $i \notin \mathrm{supp}(u)$) and fires in round 1, deriving $x_i = v_i$. The clause on $\mathrm{supp}(u'')$ falsified by the restriction of $\alpha \cup \{x_i = v_i\}$ to $\mathrm{supp}(u'') \setminus \{j\}$, extended by the

value of x_j opposite to ℓ, is likewise an implicate different from C, and fires in round 2, deriving ℓ ($\alpha \cup \{x_i = v_i\}$ is consistent and entails ℓ). So every question is answered. □

Proof of Theorem 7.5. We show by induction on $|w|$: for every nonzero $w \in \mathrm{Row}(A)$, every $j \in \mathrm{supp}(w)$ and every assignment α to $\mathrm{supp}(w) \setminus \{j\}$ consistent with $\mathcal{S}$, unit propagation on $\Phi_{\mathrm{ind}} \cup \alpha$ derives the value of x_j, and it never conflicts, as $\Phi_{\mathrm{ind}} \cup \alpha$ is satisfiable. By Lemma E.6 and the monotonicity of unit propagation in the antecedent (a larger consistent antecedent derives at least as much), this is usability; the projection is $\mathcal{S}$ by Lemma E.2.

- *w is not a circuit.* By Lemma E.6(iii) with $I = \mathrm{supp}(w) \setminus \{j\}$ there is a circuit $c' \neq w$ with $j \in \mathrm{supp}(c') \subseteq \mathrm{supp}(w)$, hence $|c'| < |w|$; apply the induction hypothesis to c' and the restriction of α.
- *$w = u$ is an indecomposable circuit.* Its clauses are in Φ_{ind}, and one of them fires.
- *$w = u$ is a decomposable circuit.* By (5) write $u = u' + u''$ through $i \notin \mathrm{supp}(u)$ with $j \in \mathrm{supp}(u'')$. If $\mathrm{supp}(u') = (\mathrm{supp}(u) \setminus \{j\}) \cup \{i\}$, then $u'' = \mathbf{e}_i + \mathbf{e}_j$ has weight $2 < d_*$, which is impossible; so $|u'| \leq |u| - 1$. Also $|\mathrm{supp}(u') \cap \mathrm{supp}(u)| = |u'| - 1 \geq d_* - 1 \geq 2$, so $|u''| = |\mathrm{supp}(u) \setminus \mathrm{supp}(u')| + 1 \leq |u| - 1$. By induction on u' (target i, antecedent the restriction of α to $\mathrm{supp}(u') \setminus \{i\} \subseteq \mathrm{supp}(u) \setminus \{j\}$), unit propagation derives the value v_i of x_i determined by α. The assignment $\alpha \cup \{x_i = v_i\}$ is consistent, and its restriction to $\mathrm{supp}(u'') \setminus \{j\} \subseteq (\mathrm{supp}(u) \setminus \{j\}) \cup \{i\}$ is an antecedent for u'' and target j; by induction unit propagation derives x_j.

The base case is contained in the induction: a circuit of weight $d_* \geq 3$ is indecomposable, since a decomposition would give, by the two bounds above, $|u'|, |u''| \leq d_* - 1$, against the minimality of d_*. Necessity: every usable theory over the base variables contains Φ_{ind} (Proposition E.10). Sufficiency: an explicit theory over the base variables equivalent to $\mathcal{S}$ consists of implicates, so adding its clauses to Φ_{ind} keeps every derivation and causes no conflict on consistent antecedents. □

Corollary E.11 (Counting, and when the given equations suffice)**.** *Let $d_* \geq 3$. (a) Every circuit of weight d_* is indecomposable, so every* UP*-usable theory over the base variables has at least $a_* 2^{d_* - 1}$ clauses. (b) The* core*, the r given equations each written as all of its $2^{|row|-1}$ clauses, is* UP*-usable if and only if every indecomposable circuit is one of the given rows. So a circuit must be written down if and only if it cannot be obtained by chaining two derived equations through one intermediate base quantity, and the core suffices for unit propagation exactly when it already contains every such circuit. (c) One parity law on $m \geq 3$ variables needs exactly its 2^{m-1} clauses over the base variables.*

Proof. (a) As in the base case of the proof of Theorem 7.5. (b) The core is equivalent to $\mathcal{S}$, so by Theorem 7.5 it is usable iff it contains Φ_{ind}. A clause on the support of a circuit u belongs to the core iff some row has that support, that is, as a vector is determined by its support, iff u is a row. (c) $\mathrm{Row}(A) = \{0, 1^m\}$, and 1^m is an indecomposable circuit of weight $m = d_* \geq 3$. □

A derived equation that is not a circuit, for example the sum of two circuits with disjoint supports, never needs to be written.

Proof of Corollary 7.6. Every nonzero $u = yA$ ($y \neq 0$) has $u_c = y \cdot c$ at the column c, which is 1 for exactly half of the 2^r vectors c, none of them 0: so u has weight $2^{r-1} = (m+1)/2$. Equal weights make every nonzero derived equation a circuit, and $d_* = 2^{r-1} \geq 4$, so all m of them are indecomposable by Corollary E.11(a). By Theorem 7.5 the unique minimum is $\Phi_{\mathrm{ind}} = \mathrm{PI}(\mathcal{S})$, with $m 2^{(m-1)/2}$ clauses of width $(m+1)/2$. □

For $r = 3, \ldots, 7$ the minimum is 56, 1,920, 1,015,808, about $1.35 \cdot 10^{11}$ and about $1.2 \cdot 10^{21}$ clauses, against r equations; for $r = 3$ a direct computation (code in the accompanying repository) confirms that $\mathrm{PI}(\mathcal{S})$ answers all 2,352 entailed literals and that deleting any one of its 56 clauses breaks usability.

Remark E.12 (The hypothesis $d_* \geq 3$). The natural extension of Theorem 7.5 and Corollary E.11(a) to $d_* = 2$ fails. The law $x_1 = x_2 = x_3 = x_4$ has $a_* = 6$ derived equations of weight 2, so the bound of Corollary E.11(a) would be 12 clauses, but 4 suffice (a directed cycle; see below); likewise the simplex family with $r = 2$ is the law $x_1 = x_2 = x_3$, with $|\mathrm{PI}| = 6 = a_* 2^{d_*-1}$ and minimum 3. For every d_*, Φ_{ind} is necessary (Proposition E.10) and nothing else is forced (Proposition 7.4), but for $d_* = 2$ it need not be usable: a direct computation (code in the accompanying repository) finds, for example, a law with $m = 6$, $r = 3$ and $d_* = 2$ for which Φ_{ind} has 4 clauses and the least usable theory over the base variables has 8. For $d_* \leq 2$ the difficulty is the choice among alternative clauses, not which clauses are forced.

One case is exact. Call $\mathcal{S}$ an *equivalence law* if $\mathrm{Row}(A)$ is spanned by derived equations of weight 2. Then $i \sim j :\iff \mathbf{e}_i + \mathbf{e}_j \in \mathrm{Row}(A)$ partitions the variables that occur in derived equations into classes $J_1, J_2, \ldots$ of sizes at least 2, and *the least* UP*-usable theory over the base variables has exactly* $|J_1| + |J_2| + \cdots$ *clauses: a directed cycle of implications on each class.*

Proof. Every derived equation is a sum of vectors $\mathbf{e}_i + \mathbf{e}_j$ with $i \sim j$, so its restriction to each class has even weight; hence a derived equation of weight at least 3 contains two variables of one class in its support, the circuits are exactly the weight-2 equations, which lie within classes, and there is no derived equation of weight 1. *Lower bound.* Fix a class J, $i \in J$ and $v \in \{0, 1\}$. The antecedent $\{x_i = v\}$ is consistent and entails a literal on each $x_{i'}$, $i' \in J \setminus \{i\}$, so unit propagation derives something in round 1, by a clause whose other literals are falsified by $\{x_i = v\}$: a unit clause, which is not an implicate (no variable is fixed), or a 2-clause containing the literal x_i^{1-v}. A 2-clause implicate has as support the support of a derived equation (Proposition E.7), a pair within one class. So each of the $2|J|$ literals on J has its negation in some 2-clause within J; a 2-clause contains two literals, so at least $|J|$ clauses lie within J, and different classes share no clause. *Upper bound.* Fix $x^0 \in \mathcal{S}$; on $\mathcal{S}$, $x_i^{x_i^0} \leftrightarrow x_{i'}^{x_{i'}^0}$ for $i \sim i'$. For $J = \{i_1, \ldots, i_{|J|}\}$ take the $|J|$ implicates $\neg x_{i_t}^{x_{i_t}^0} \vee x_{i_{t+1}}^{x_{i_{t+1}}^0}$ (indices mod $|J|$). A question (α, ℓ) with ℓ on x_j, $j \in J$, has α assigning some x_i with $i \in J$ (Lemma E.6(iii): the circuits are the pairs); if α makes $x_i^{x_i^0}$ true, propagation runs forward around the cycle to x_j, and if false, backward. No conflict arises. □

A direct computation (exhaustive over subsets of PI; code in the accompanying repository) confirms that the minimum is the size of the class for one class of $2, \ldots, 5$ variables, and is 3 for $x_1 = x_2 \oplus 1 = x_3$, and 5 and 6 for two classes of sizes $(2, 3)$ and $(3, 3)$.

E.4 Intermediate concepts, systems of laws and depth

Proposition E.13 (The chain encoding)**.** *Let $m \geq 4$. Encode $x_1 \oplus \cdots \oplus x_m = b$ by the constraints $y_1 = x_1 \oplus x_2$, $y_i = y_{i-1} \oplus x_{i+1}$ for $2 \leq i \leq m-2$, and $y_{m-2} \oplus x_m = b$, each three-variable constraint written as its 4 clauses and the last as its 2 clauses: $4m-6$ clauses and $m-2$ auxiliary variables. The encoding is* propagation complete on all variables*: for every partial assignment α to the base and auxiliary variables, unit propagation derives every literal entailed by the encoding and α, and reaches a conflict if and only if α is inconsistent with the encoding. Without auxiliary variables 2^{m-1} clauses are necessary (Corollary E.11(c)) and sufficient.*

Proof. Each constraint's clause set is the set of all prime implicates of that constraint, so (Lemma E.5(a)) it derives its last variable when the others are assigned, and gives a conflict

when all are assigned inconsistently. Consider a fixpoint of unit propagation without conflict. Every constraint is then either fully assigned, and satisfied, or has at least two unassigned variables; call the latter *active*. Each variable occurs in at most two constraints, and consecutive constraints share exactly one auxiliary variable, so the graph on the active constraints, joined when they share an unassigned variable, is a disjoint union of paths. Fix an unassigned variable z and a value for it. Process each component breadth-first from a root (in the component of z, a constraint containing z, with z set first). When a constraint is processed, at most one of its variables is already set: the one it shares with its parent, or z at the root. It has at least two unassigned variables, so at least one of them is still free: give the other free ones arbitrary values and solve for that one. This yields a solution extending the fixpoint with either value of z, so no unassigned variable is entailed. If α were inconsistent and no conflict arose, the same construction (without z) would give a solution extending α; so unit propagation reaches a conflict on every inconsistent α, and by soundness only on those. □

The chain encoding is standard. Every CNF encoding of parity with auxiliary variables has at least $3m$ clauses (Emdin et al., 2022), so the chain is within a factor $4/3$ of the least clause count even among encodings that need not be usable.

Proof of Proposition 7.7. Let $\Phi \in \mathcal{E}_m$ be UP_1-usable and α the antecedent of a question; it assigns base variables only. In round 1 a clause fires only if all its literals but one are falsified; auxiliary literals are unassigned, so a clause containing an auxiliary literal fires only if that literal is its only unfalsified one, and then it derives that auxiliary literal. Hence the base literals derived in round 1 are derived by the clauses of Φ over the base variables, Φ_{base}; no conflict occurs, as $\Phi \cup \alpha$ is satisfiable. Each $C \in \Phi_{\mathrm{base}}$ is an implicate of $\mathcal{S}$, as $\Phi \models C$ and $\operatorname{proj} \Phi = \mathcal{S}$. The proof of Lemma E.5(b), which uses only that the firing clause is an implicate of $\mathcal{S}$, gives $\Phi_{\mathrm{base}} \supseteq \mathrm{PI}(\mathcal{S})$. For one parity law PI consists of the 2^{m-1} clauses of its single circuit; for the simplex family, of $2^{(m-1)/2}$ clauses for each of its m circuits (Corollary 7.6). □

A direct computation (code in the accompanying repository) confirms on twelve random laws that deleting any clause of $\mathrm{PI}(\mathcal{S})$, in all its copies, from $\mathrm{PI}(\mathcal{S})$ together with a library with intermediate concepts destroys UP_1-usability, and that the chain-encoded libraries of the simplex family for $r = 3, 4$ are UP-usable but not UP_1-usable.

Proposition E.14 (Libraries with intermediate concepts)**.** *Let* $\mathrm{Circ}_{\mathrm{aux}}(\mathcal{S})$ *write each circuit* u *of an affine law by the cheaper of its full clause set and its chain encoding with its own auxiliary variables: directly if* $|u| \le 4$*, and by the chain if* $|u| \ge 5$*. It is* UP*-usable, of size* $\sum_{\text{circuits } u} \mathrm{sz}(|u|)$*, where* $\mathrm{sz}(w) := 2^{w-1}$ *for* $w \le 4$ *and* $\mathrm{sz}(w) := 4w - 6$ *for* $w \ge 5$*. The same holds for the library over all* $2^r - 1$ *nonzero derived equations, of size at most* $(2^r - 1)(4m - 6)$ *for* $m \ge 4$*.*

Proof. A question (α, ℓ), ℓ on x_j, has a circuit u with $j \in \mathrm{supp}(u) \subseteq \operatorname{dom} \alpha \cup \{j\}$ (Lemma E.6). The encoding of u is propagation complete (Proposition E.13, or Lemma E.5(a) for the full clause set of the one-equation law), and α fixes all of $\mathrm{supp}(u)$ but j, so unit propagation on that encoding alone derives ℓ; unit propagation is monotone in the clause set. Each auxiliary variable is a partial sum of base variables, so the union is satisfied exactly by the extensions of the points satisfying every circuit, and its projection is $\mathcal{S}$, since the circuits span $\mathrm{Row}(A)$ (proof of Proposition E.7); a consistent α therefore gives no conflict. For the second library, a derived equation of weight w costs at most $\max(8, 4w - 6) \le 4m - 6$ clauses when $m \ge 4$. □

The second form has size $O(m2^r)$, in line with the fixed-parameter tractability in the number of equations shown by Gwynne and Kullmann (2014).

Remark E.15 (The simplex family with intermediate concepts). For the simplex family with $r \geq 4$, $\mathrm{Circ}_{\mathrm{aux}}$ has $m(2m-4)$ clauses of width at most 3 and $m(m-3)/2$ auxiliary variables, and is UP-usable: a theory of polynomial size. Writing each circuit by the balanced-tree encoding of Proposition 7.8(b) instead of the chain gives the same size and a $\mathrm{UP}_{2\lceil\log((m+1)/2)\rceil-1}$-usable theory. So with intermediate concepts and enough rounds the gap to GE is polynomial in written length, and $O(\log m)$ bits under time charging.

Proof. Every circuit has weight $w = 2^{r-1} = (m+1)/2 \geq 8$, so each is chain-encoded with $4w-6 = 2m-4$ clauses and $w-2 = (m-3)/2$ auxiliary variables; usability is Proposition E.14. With balanced trees, each question is answered inside the encoding of one circuit, as in the proof of Proposition E.14, within $2\lceil\log w\rceil - 1$ rounds by Proposition 7.8(b), and unit propagation with a fixed number of rounds is monotone in the clause set on satisfiable inputs; the auxiliary variables are again partial sums, so the projection is $\mathcal{S}$. The simplex family is computable from m in polynomial time, so $Kt(A,b) = O(\log m)$, and Theorem 7.3(b) with $r = \log(m+1)$ gives $Kt_A^{\mathrm{UP}}(\mathcal{Q}(\mathcal{S})) = O(\log m)$. □

A direct computation (code in the accompanying repository) of the listing lengths (4) gives ratios of about 65, 137, 256 and 461 between this library and the r given equations for $r = 4, 5, 6, 7$: polynomial growth, not $2^{(m-1)/2}$; with chains the library at $r = 4$ needs 7 rounds, with balanced trees 5.

Proposition E.16 (Private intermediate concepts add nothing to a system). *Let $\Phi = \bigcup_t \Phi_t$, where Φ_t encodes a constraint c_t on base variables X_t with auxiliary variables Y_t private to it ($Y_t \cap Y_s = \emptyset$ for $t \neq s$, Y_t occurs only in Φ_t, and $\exists Y_t\, \Phi_t \equiv c_t$), and each Φ_t is propagation complete on $X_t \cup Y_t$. For a partial assignment α to the base variables, let its* arc-consistency closure $\mathrm{AC}(\alpha)$ *be the least set of base literals that contains α and is closed under adding, for every t, every base literal on X_t entailed by c_t and the current literals on X_t; call it* contradictory *if it contains two opposite literals or its literals on some X_t are inconsistent with c_t. Then unit propagation on $\Phi \cup \alpha$ reaches a conflict if and only if $\mathrm{AC}(\alpha)$ is contradictory, and otherwise the base literals at its fixpoint are exactly $\mathrm{AC}(\alpha)$. In particular the given equations written directly and the given equations each chain-encoded derive the same base literals and conflict on the same antecedents.*

Proof. Invariant. At every point of the run, the literals on Y_t are entailed by Φ_t and the literals on X_t: they are derived only by clauses of Φ_t, and each derivation step preserves this. Hence a literal derived by a clause of Φ_t is entailed by Φ_t and the current literals on X_t; if it is a base literal, it is entailed by c_t and the current literals on X_t, as $\exists Y_t\, \Phi_t \equiv c_t$. By induction on rounds, every base literal derived before a conflict lies in $\mathrm{AC}(\alpha)$. A conflict arises either from a clause of some Φ_t with all literals falsified, so that the current literals on $X_t \cup Y_t$ are inconsistent with Φ_t and, by the invariant, those on X_t are inconsistent with c_t; or from two opposite derivations, both from one Φ_t if the variable is auxiliary (with the same conclusion), or both base literals of $\mathrm{AC}(\alpha)$. Either way $\mathrm{AC}(\alpha)$ is contradictory. *Conversely,* suppose the run reaches a conflict-free fixpoint γ' with base part β. No clause of Φ_t fires at γ' and Φ_t is propagation complete, so the restriction of γ' to $X_t \cup Y_t$ is consistent with Φ_t and contains every literal it entails with Φ_t. So $\beta|_{X_t}$ is consistent with c_t, and if $c_t \wedge \beta|_{X_t} \models \ell$ then Φ_t and the literals of γ' on $X_t \cup Y_t$ entail ℓ, so $\ell \in \gamma'$. Thus $\beta \supseteq \alpha$ is closed and not contradictory; hence $\beta \supseteq \mathrm{AC}(\alpha)$ and $\mathrm{AC}(\alpha)$ is not contradictory, and with the first part $\beta = \mathrm{AC}(\alpha)$. The direct encoding of one equation (all its clauses) is propagation complete on its own variables by Lemma E.5(a), and the chain encoding by Proposition E.13. □

So intermediate concepts that each belong to one law cannot help a weak reasoner combine laws; the useful intermediate concepts are shared across laws, such as derived equations and their partial sums. On sparse random systems a direct computation (code in the accompanying repository) finds that unit propagation on the given equations fully answers only a part of the sampled questions, between about 0.62 and 0.89 of them depending on the system, while the circuit library answers all of them.

Lemma E.17 (Usability is arc consistency)**.** *If Φ is* UP*-usable for $\mathcal{S}$, then for every partial assignment α to the base variables that is inconsistent with $\mathcal{S}$, unit propagation on $\Phi \cup \alpha$ reaches a conflict. So* UP*-usability is* arc consistency *of a representation in the sense of Gwynne and Kullmann (2014): unit propagation detects every forced base literal and every inconsistency.*

Proof. Take $\alpha' \subseteq \alpha$ minimal inconsistent and a literal x_j^v of α'. Then $\alpha'' := \alpha' \setminus \{x_j^v\}$ is consistent and entails x_j^{1-v}, so unit propagation on $\Phi \cup \alpha''$ derives it. Unit propagation is monotone up to conflict: in the run on the larger $\Phi \cup \alpha$, every clause that fired in the smaller run either fires again or meets a conflict. So the larger run derives x_j^{1-v} against $x_j^v \in \alpha$, or conflicts earlier. □

Lemma E.18 (Unit propagation as a monotone circuit)**.** *Let Φ be an explicit theory with $\bar{m}$ variables, written length $\|\Phi\|$ and clause width at most w, and let $\mathcal{I}$ be a set of literals (the inputs). (a) There is a monotone circuit of $O(\bar{m}(\|\Phi\| + \bar{m}))$ fan-in-2 AND and OR gates, with one input per literal of $\mathcal{I}$, that outputs, for every literal L, whether unit propagation on $\Phi \cup \alpha$ derives L, on every input $\alpha \subseteq \mathcal{I}$ for which $\Phi \cup \alpha$ is satisfiable. (b) For every k there is a monotone circuit of unbounded fan-in AND and OR gates, of depth $2k$, with at most $k(\|\Phi\| + 2\bar{m})$ gates and at most $k(w\|\Phi\| + 2\bar{m})$ wires, with the same inputs, that outputs whether* UP_k *derives L, on the same inputs.*

The direction "small usable theory gives small monotone circuit" of the equivalence of Bessiere et al. (2009) is (a).

Proof. Define $o_0(L)$ as the input bit if $L \in \mathcal{I}$ and the constant 0 otherwise, and $o_{t+1}(L) := o_t(L) \vee \bigvee_{C \ni L} \bigwedge_{L' \in C,\, L' \neq L} o_t(\neg L')$, where the empty conjunction, for a unit clause $C = (L)$, is the constant 1. On a satisfiable $\Phi \cup \alpha$ unit propagation never conflicts, so a clause whose other literals are all falsified fires (its last literal is unassigned or already true), exactly as in the formula; by induction $o_t(L)$ says whether L is known after t rounds. (a) Prefix and suffix products give the $|C|$ conjunctions of a clause with $3|C|$ fan-in-2 gates, so a round costs $O(\|\Phi\| + \bar{m})$ gates; each round before the fixpoint assigns at least one new variable and no variable is assigned twice, so $\bar{m}$ rounds suffice. (b) Per round, one AND gate for each pair (clause, literal of it), of fan-in $|C| - 1 \leq w - 1$, and one OR gate for each literal, of fan-in one plus the number of clauses containing it: at most $\|\Phi\| + 2\bar{m}$ gates and $w\|\Phi\| + 2\bar{m}$ wires per round. AND layers and OR layers alternate (an OR gate feeding an OR gate of the next round is merged into it), so the depth is $2k$. Constants are removed by simplification without increasing size or depth. □

Proof of Theorem 7.9. The law. $|\Lambda_q| = 2\binom{q}{3} + 2q$, so $m = 2\binom{q}{3} + 2q + 1 = \Theta(q^3)$. The vectors of Λ_q span $\mathrm{GF}(2)^{q+1}$ (already $(\mathbf{e}_v, 0)$ and $(\mathbf{e}_v, 1)$, $v \leq q$, do), so $\dim \mathcal{H}_q = q + 1$ and $\mathcal{H}_q$ is an affine law of rank $r = m - q - 1 \geq 1$. The m equations $x_{(u,a)} = u \cdot y + a\, z_0$, $x_\star = z_0$, with $z = (y, z_0)$ as auxiliary variables, have projection $\mathcal{H}_q$, and GE returns exactly the entailed base literals, so they are GE-usable. *The lower bound.* For $S \subseteq \Lambda_q$ the antecedent $\alpha_S := \{\neg x_v : v \in S\}$ is consistent, as $0 \in \mathcal{H}_q$. On $\mathcal{H}_q$, $x_\star$ is determined by x_S iff $\mathbf{e}_\star \in \mathrm{span}(S)$ (if not, some z has $z \cdot v = 0$ for $v \in S$ and $z \cdot \mathbf{e}_\star = 1$, and z and 0 give two points that agree on S and differ at $x_\star$; if so, $x_\star$ is a sum of the

x_v, $v \in S$), iff the XOR system $\{u \cdot y = a : (u,a) \in S\}$ over $y \in \mathrm{GF}(2)^q$ is unsatisfiable (a subset of the equations sums to $0 = 1$, that is, a subset of S sums to $\mathbf{e}_\star$). If $x_\star$ is determined, its value is that at the point 0, so usability gives "unit propagation on $\Phi \cup \alpha_S$ derives $\neg x_\star$"; if not, soundness ($\mathrm{proj}\,\Phi = \mathcal{H}_q$) excludes any literal on $x_\star$. So the output for $\neg x_\star$ of the circuit of Lemma E.18(a), with inputs $\mathcal{I} = \{\neg x_v : v \in \Lambda_q\}$, computes the monotone function $\mathrm{Unsat}_q(S) :=$ [the system S is unsatisfiable], with $O((\|\Phi\| + m)^2)$ gates (auxiliary variables that do not occur in Φ are dropped, so $\bar{m} \le \|\Phi\| + m$). Göös et al. (2019) show that a monotone variant of XOR-SAT has exponential monotone circuit complexity. In the form we use, their function $\mathrm{3XorSat}_q$ has one input for each ordered triple $(v_1, v_2, v_3) \in [q]^3$ and each constant $a \in \{0,1\}$ ($2q^3$ inputs), accepts iff the selected constraints $y_{v_1} \oplus y_{v_2} \oplus y_{v_3} = a$ are unsatisfiable, and requires monotone circuits of size 2^{q^ε} for some $\varepsilon > 0$. Each input (v_1, v_2, v_3, a) is the constraint $u \cdot y = a$ with $u = \mathbf{e}_{v_1} + \mathbf{e}_{v_2} + \mathbf{e}_{v_3}$, of weight 1 (if two indices coincide) or 3, so $(u,a) \in \Lambda_q$; hence $\mathrm{3XorSat}_q(Z) = \mathrm{Unsat}_q(\mathrm{pr}(Z))$, where $\mathrm{pr}(Z)_v$ is the OR of the inputs that map to v. A monotone circuit for Unsat_q together with at most $2q^3$ OR gates is a monotone circuit for $\mathrm{3XorSat}_q$. (If the inputs are only the triples of distinct indices, $\mathrm{3XorSat}_q$ is Unsat_q with the weight-1 inputs set to 0, and a circuit for Unsat_q again gives one for it.) Hence $O((\|\Phi\| + m)^2) + 2q^3 \ge 2^{q^\varepsilon}$, so $\|\Phi\| \ge 2^{q^\varepsilon/2 - O(1)} - m$ for $q \ge q_0$, and $q^\varepsilon = \Theta(m^{\varepsilon/3})$. □

The weight-1 functionals matter: with weight-3 functionals only, the function computed would be the restriction of $\mathrm{3XorSat}_q$ that sets the inputs with repeated indices to 0, and a monotone lower bound does not pass to a restriction. A direct computation (code in the accompanying repository) confirms for $q = 3, 4$ that "$x_\star$ determined", "system unsatisfiable" and "$\mathbf{e}_\star \in \mathrm{span}(S)$" agree on all $2^{|\Lambda_q|}$ sets S, and that unit propagation on $\mathrm{PI}(\mathcal{H}_3)$ derives $\neg x_\star$ from α_S exactly when the system is unsatisfiable.

Remark E.19 (XOR systems without auxiliary-variable restrictions). Gwynne and Kullmann (2014) show that systems of XOR constraints have no polynomial-size arc-consistent representation in general, even with auxiliary variables, and that the standard translation is propagation complete for one equation and fails for two, the counterpart with auxiliary variables of Corollary E.11(b). By Lemma E.17 their notion is our UP-usability, so their superpolynomial bound applies to usable theories; Theorem 7.9 gives an explicit family with an exponential bound.

Proof of Proposition 7.8. For one parity law every question has $|\mathrm{dom}\,\alpha| = m-1$, as the only circuit is 1^m. (a) Write $C_1 = (x_1, x_2, y_1)$, $C_i = (y_{i-1}, x_{i+1}, y_i)$ for $2 \le i \le m-2$, and $C_{m-1} = (y_{m-2}, x_m)$ for the constraints. Each constraint's clauses mention only its own variables and fire only when all but one of them are assigned. *Upper bound:* let x_j be the unknown base variable and $C_{j'}$ its constraint ($j' = \max(1, j-1)$). The constraints to the left of $C_{j'}$ resolve from the left end (C_1 derives y_1 in round 1, C_2 derives y_2 in round 2, and so on), and those to the right from the right end (C_{m-1} derives y_{m-2} in round 1, and so on); $C_{j'}$ then has all its variables but x_j assigned by round $\max(j'-1, m-1-j')$, and derives x_j one round later, by round $m-1$. *Lower bound:* for the question with x_1 unknown, after round $t \le m-2$ exactly the auxiliary variables $y_{m-2}, \dots, y_{m-1-t}$ are assigned (induction on t: C_1 has the two unknowns x_1, y_1, and each C_i, $2 \le i \le m-2$, the two unknowns y_{i-1}, y_i, until the boundary reaches it, so in each round exactly one constraint has a single unknown); so x_1 is derived in round $m-1$ and not before. (b) Let $H := \lceil \log m \rceil$ be the height of the tree. Each constraint's clause set is its set of prime implicates, so it derives its third variable in the round after the other two are known (Lemma E.5(a)). Let x_j be the unknown leaf and $v_0, \dots, v_s = x_j$ the path from the root, $s \le H$; for $1 \le i \le s$ let w_i be the sibling of v_i. Every node v not on the path has only known leaves below it, and by induction on its height $\mathrm{ht}(v)$ its value is derived by round $\mathrm{ht}(v)$; w_i is at depth i, so $\mathrm{ht}(w_i) \le H - i$. The root constraint gives y_{v_1} by round $\mathrm{ht}(w_1) + 1 \le H$. If y_{v_i} is known by round $H + i - 1$, the constraint at v_i gives $y_{v_{i+1}}$ by

round $\max(H+i-1, H-i-1)+1 = H+i$. So x_j is derived by round $H+s-1 \le 2H-1$. No conflict arises, as the antecedent is consistent. Counting: there are $m-1$ internal nodes, $m-2$ of them non-root with 4 clauses each, and the root with 2. (c) Let Φ be UP_k-usable, and drop auxiliary variables that do not occur, so that $\bar{m} \le m + \|\Phi\|$. Apply Lemma E.18(b) with inputs the literals on $x_2, \dots, x_m$ (for a general affine law: on $\mathrm{supp}(u) \setminus \{j\}$ for a circuit u of weight w and some $j \in \mathrm{supp}(u)$). Every assignment to these variables is consistent (no nonzero derived equation is supported among them) and determines x_1 (resp. x_j) as a constant plus their parity (Lemma E.6); so by usability and soundness the output gate for the literal x_1 computes, on the inputs that make exactly one literal of each variable true, the parity of $m-1$ bits or its negation. This is a circuit of depth $2k$ with unbounded fan-in AND and OR gates over the literals of $m-1$ variables, with at most $k(\|\Phi\|+2\bar{m}) \le k(3\|\Phi\|+2m)$ gates. By Håstad's theorem (Håstad, 1986), in its standard form, every such circuit of depth $d \ge 2$ computing the parity of n variables has at least $2^{c_{\mathrm{H}} n^{1/(d-1)}}$ gates, for an absolute constant $c_{\mathrm{H}} > 0$ and n large enough given d. With $n = m-1$ (resp. $w-1$) and $d = 2k$ the claim follows. (For $k = 1$, Proposition 7.7 gives the exact value 2^{m-1}.) (d) By (c), for fixed k the written length is $2^{\Omega(m^{1/(2k-1)})}$, superpolynomial in m; and comparing a polynomial size $2^{O(\log m)}$ with $2^{c_{\mathrm{H}}(m-1)^{1/(2k-1)}}/k$ gives $(m-1)^{1/(2k-1)} = O(\log m)$, that is, $k = \Omega(\log m / \log\log m)$. This comparison applies (c) with k growing with m, that is, Håstad's bound for depths up to this order. By (b), $2\lceil \log m \rceil - 1$ rounds suffice for $4m-6$ clauses. □

A direct computation over all $m2^{m-1}$ questions, $m = 3, \dots, 10$ (code in the accompanying repository), confirms that the chain needs exactly $m-1$ rounds for $m \ge 4$ and that the balanced tree needs $1, 3, 4, 5, 5, 5, 6, 7$ rounds for $m = 3, \dots, 10$, within the bound $2\lceil \log m \rceil - 1$. Part (c) is a direct application of Håstad's theorem and very probably folklore.

Proof of Corollary 7.10. First column. One parity law has the single circuit 1^m, indecomposable, so every UP-usable theory over the base variables, and a fortiori every UP_k-usable one, contains its 2^{m-1} clauses (Proposition E.10), and these clauses are UP_1-usable (Corollary E.11(c), Proposition E.7). *Row* UP_1. Proposition 7.7, with $\mathrm{PI}(\mathcal{S})$ itself UP_1-usable. *Row* UP_k*, k fixed.* Proposition 7.8(c), for parity with $w = m$ and for the simplex family with $w = (m+1)/2$. *Row* UP_k*, $k \ge 2\lceil \log m \rceil - 1$, and* UP. Proposition 7.8(b) and Remark E.15 (for $r = 3$, $\mathrm{PI}(\mathcal{S})$ has $56 \le m(2m-4)$ clauses); more rounds never hurt. *Last column.* Theorem 7.9, for every number of rounds. *Row* GE. A linear theory with e equations in $m + m_{\mathrm{aux}}$ variables has a solution set of dimension at least $m + m_{\mathrm{aux}} - e$, whose projection has dimension at least $m - e$; so a theory for a law of rank r has at least r equations, and the given equations (one for parity, $r = \log(m+1)$ for the simplex family) or the m equations of Theorem 7.9 are usable. □

E.5 Directionality

A *directed rule* is $P \Rightarrow \ell$, with P a set of literals (the premise) and ℓ a literal on a variable that does not occur in P; a "rule" whose conclusion's variable occurs in its premise corresponds to a tautological clause and is excluded, as tautological clauses are excluded from explicit theories. Its written length is $|P| + 1$. *Forward chaining* FC_k applies, in each of k parallel rounds, every rule whose premise is satisfied by the current assignment; a conclusion whose opposite is assigned, or two opposite conclusions, give a conflict; $\mathrm{FC} := \mathrm{FC}_\infty$. A rule over the base variables is *sound* for $\mathcal{S}$ if P is consistent with $\mathcal{S}$ and $\mathcal{S} \wedge P \models \ell$. The projection of a rule set, possibly with auxiliary variables, is that of its clauses $\neg P \vee \ell$; a rule set is FC_k*-usable* for $\mathcal{S}$ if its projection is $\mathcal{S}$ and FC_k

derives every entailed base literal from every consistent antecedent. A *rule base* is a set of sound rules over the base variables.

Proposition E.20 (Directed rules against clauses)**.** *(a) For a clause C let* $\mathrm{rules}(C) := \{\neg(C \setminus \{L\}) \Rightarrow L : L \in C\}$*, where $\neg(C \setminus \{L\})$ is the set of negations of the other literals. For every theory Φ, every k and every α,* FC_k *with* $\mathrm{rules}(\Phi)$ *and* UP_k *with Φ derive the same literals and conflict together. Conversely a rule $P \Rightarrow \ell$ gives the clause $\neg P \vee \ell$, and unit propagation with the clause derives at least what forward chaining with the rule derives. Hence, for every law and every round budget k, with or without auxiliary variables, the smallest usable rule set has at least as many rules as the smallest usable clausal theory has clauses, and at most w times as many, w the largest clause width of a smallest usable clausal theory; in written length the factor is likewise at most w. (b)* One parity law on $m \geq 2$ variables, no auxiliary variables. *A sound rule concluding a literal on x_j has all other $m-1$ variables in its premise. So there are exactly $m2^{m-1}$ sound rules, each question needs its own rule, and the only usable rule base is the set of all of them: $m2^{m-1}$ rules of length m, against the unique minimum of 2^{m-1} clauses of length m; a factor m, the number of query directions, in count and in written length. (c)* With auxiliary variables, for every law. *Splitting each clause $(l_1 \vee \cdots \vee l_w)$ with $w \geq 4$ into $(l_1 \vee l_2 \vee z_1), (\neg z_1 \vee l_3 \vee z_2), \ldots, (\neg z_{w-3} \vee l_{w-1} \vee l_w)$ with fresh variables $z_1, \ldots, z_{w-3}$ preserves the projection and* UP*-usability. So every law has an* FC*-usable rule set of written length at most* 9 *times the least written length of a* UP*-usable clausal theory with auxiliary variables. For the chain encoding of parity: $4m-6$ clauses against $3(4m-6)-2$ rules. (d)* One round, any vocabulary. *For one parity law on $m \geq 2$ variables, every* FC_1*-usable set of sound rules, with any auxiliary variables, contains all $m2^{m-1}$ sound rules over the base variables, while every* UP_1*-usable theory with any auxiliary variables must contain the 2^{m-1} clauses of* $\mathrm{PI}(\mathcal{S})$ *(Proposition 7.7).*

Proof. (a) Induction on rounds: a clause C fires in round t deriving L exactly when the rule $\neg(C \setminus \{L\}) \Rightarrow L$ has its premise satisfied after round $t-1$ and L is not yet assigned; forward chaining signals a conflict when a conclusion's opposite is assigned, as unit propagation does when a clause is falsified. A rule $P \Rightarrow \ell$ whose premise is satisfied leaves its clause with all literals but ℓ falsified. The bounds follow by translating a smallest usable theory in either direction: $\mathrm{rules}(\Phi)$ has at most $w|\Phi|$ rules and written length $\sum_{C \in \Phi} |C|^2 \leq w\|\Phi\|$, and the clauses of a rule set are at most as many as its rules and are usable for the same round budget. (b) If a sound rule $P \Rightarrow x_j^v$ had a premise missing some x_i, $i \neq j$, then by Lemma E.6 (the only circuit is 1^m) x_j would not be determined by the consistent premise P, and the rule would be unsound. So each sound rule is $\beta|_{[m]\setminus\{j\}} \Rightarrow x_j^v$ with v the forced value, one for each j and each of the 2^{m-1} assignments β to the other variables, and the question with that antecedent is answered only by that rule. The minimum of 2^{m-1} clauses is Corollary E.11(c) for $m \geq 3$, and for $m = 2$ the equivalence-law case of Remark E.12 with one class of two variables. (c) The pieces, with the z's quantified out, are equivalent to the clause. If all literals of the clause but l_t are falsified, propagation from the left derives $z_1, \ldots, z_{t-2}$, from the right $\neg z_{w-3}, \ldots, \neg z_{t-1}$, and the piece containing l_t fires (the end cases are the same with one side empty). Every derivation of the original run is reproduced, in more rounds, and soundness is unchanged. A clause of length $w \geq 4$ becomes $w-2$ clauses of length 3, of total length at most $3w$, and each 3-clause becomes 3 rules of length 3; clauses of width at most 3 become rules of total length at most 3 times their own. The chain encoding has $4m-8$ clauses of width 3 and 2 of width 2, so rules gives $3(4m-8)+4 = 3(4m-6)-2$ rules. (d) From an antecedent α on base variables, a rule fires in round 1 only if its premise is satisfied by α, so its premise consists of base literals of α. A rule answering the question with antecedent $\beta|_{[m]\setminus\{j\}}$ concludes a literal on x_j, so it is a sound rule over the base variables, and by (b) its

premise is that whole antecedent. □

A direct computation (code in the accompanying repository) confirms for $m = 2, \dots, 7$ that the sound rules of one parity law number exactly $m2^{m-1}$, all with premises of size $m-1$, and that for $m \le 5$ deleting any one of them breaks usability; on random theories, forward chaining with rules(Φ) and unit propagation with Φ agree in every run.

Theorem E.21 (Directionality is real but cheap)**.** *(1) For one parity law on $m \ge 2$ variables, the only* FC*-usable rule base is the set of all $m2^{m-1}$ sound rules, of written length m each, while the least* UP*-usable theory over the base variables has 2^{m-1} clauses of length m: a factor m, the number of query directions. With any auxiliary variables, every* FC_1*-usable set of sound rules contains these $m2^{m-1}$ rules, while every* UP_1*-usable theory must contain 2^{m-1} clauses. (2) For every law and round budget k, with or without auxiliary variables, the least* FC_k*-usable rule set has at least as many rules as the least* UP_k*-usable theory has clauses and at most w times as many, w the largest clause width of a smallest such theory; in written length the factor is likewise at most w. With auxiliary variables and unbounded rounds, every law has an* FC*-usable rule set of written length at most 9 times the least written length of a* UP*-usable theory (at most 3 times for the chain encoding of parity). A program with internal variables answers all m directions of parity with $O(m)$ operations. (3) $K_A^{\mathrm{FC}_k} = K_A^{\mathrm{UP}_1} \pm c$ on every conditional data set.*

Proof. (1) is Proposition E.20(b), (d). (2) is Proposition E.20(a), (c); the chain encoding has width at most 3, so its rules have written length at most 3 times its own; and a program answers the question with x_j unknown by outputting b plus the sum of the given bits. (3) A fixed program maps an FC_k-compatible rule set R to its clauses and then to PI of their projection, which is UP_1-compatible with the same data by Lemma E.5(a), since forward chaining derives only literals entailed by the clauses; conversely it maps a UP_1-compatible Φ to rules(Φ), which is FC_1-compatible by Proposition E.20(a), hence FC_k-compatible for every $k \ge 1$. □

Remark E.22 (Programs, rule bases and time bounds)*.* A program may consist of a generic solver applied to a declarative law, at $O(1)$ extra length, so "programs lose" cannot be the statement; Theorem 7.2 is the precise form for reasoners: under program length the weakest reasoner together with a compiler is as good as any. The separations above concern what a *fixed* weak reasoner needs written. The translation of Proposition E.20(a) is round for round, so the directionality factor is the same under every round budget. A per-question time bound below that of elimination ($O(r^2 m)$ bit operations for GE on r equations) does not separate directed rules from clauses either: PI($\mathcal{S}$) and its rule translation both answer every question of the simplex family in one round. What a time bound changes is the comparison between reasoners: a reasoner that may not run elimination must be handed a lemma library, which over the base variables is exponentially larger (Appendix E.3); if its time is charged at least linearly in the size of the theory it reads, it cannot even read that library.

Question E.23. In a cell-probe-type model of a reasoner, is there a theory of size polynomial in m that answers every question of the simplex family in time below that of elimination?

E.6 Where the know-how lives

Proposition E.24 (Three homes of know-how)**.** *For an affine law $\mathcal{S} = \{x : Ax = b\}$, the know-how to answer every question of $\mathcal{Q}(\mathcal{S})$ can be placed in any of the following. (a)* In the reasoner: *the r equations with* GE*, of listing length $O(rm \log m)$ bits. (b)* In written lemmas: *for* UP *over*

explicit theories, exactly $\Phi_{\mathrm{ind}}(\mathcal{S})$ *over the base variables when* $d_* \geq 3$ *(Theorem 7.5; exponential for the simplex family), and a library such as* $\mathrm{Circ}_{\mathrm{aux}}$ *with intermediate concepts (Proposition E.14; polynomial for the simplex family by Remark E.15, exponential for some systems whatever the intermediate concepts by Theorem 7.9, and for one parity law at any fixed depth by Proposition 7.8(c)). (c)* In a computation that the cost model does not charge: *either a membership test with free search for derivations: for a suitable polynomial* q*, the axiom set* $\mathrm{PI}(\mathcal{S})$ *has a decider of length* $K^q(A,b)+c$ *that runs in time polynomial in* m *and in the length of its input, and for every* $(\alpha,\ell) \in \mathcal{Q}(\mathcal{S})$ *the sentence* $\bigwedge \alpha \to \ell$ *follows by propositional reasoning from a single member of* $\mathrm{PI}(\mathcal{S})$*; or a generator of the explicit theory under program length (Theorem 7.2: the compiler* $\Phi' \mapsto \mathrm{PI}(\mathrm{proj}\,\Phi')$ *costs* $O(1)$ *bits). Here* $K^q(A,b) \leq rm + r + O(\log m)$*, and* $K^q(A,b) = O(\log m)$ *for one parity law and for the simplex family.*

Proof. (a) and (b) collect the results cited. (c) A clause $C_{Z,\beta}$ is in $\mathrm{PI}(\mathcal{S})$ iff the derived equations supported in Z are exactly 0 and the all-ones vector 1_Z on Z, and β violates 1_Z (Proposition E.7; a circuit is the unique nonzero derived equation supported in its support). The derived equations supported in Z are the yA with y in the kernel of the columns of A outside Z, computable by Gaussian elimination in time polynomial in m once (A,b) is available, and a program of length $K^q(A,b)$ produces (A,b) within $q(|(A,b)|)$ steps. A question (α,ℓ) has a prime implicate $C = \neg\alpha' \vee \ell$ with $\alpha' \subseteq \alpha$ (Lemma E.5(a)), and $\bigwedge \alpha \to \ell$ follows from the single instance C by propositional reasoning. Printing the literal matrix and vector, with r and m coded self-delimitingly, takes polynomial time and gives $K^q(A,b) \leq rm + r + O(\log m)$; one parity law and the simplex family are computed from m in polynomial time. □

A direct computation (code in the accompanying repository) confirms that the decider agrees with prime-implicate membership computed by brute force on 9,852 clauses over 30 random laws.

Definition E.25 (Charged deduction with conditional data)**.** We use the readings of charged deduction of Theorem 4.18 (derivation size; every instance short, (a1), or every nonlogical instance short, (a2); total material, (b), or nonlogical material, (b2)) at polynomial budgets with the degrees charged, as in Definition E.18 of Hänni and Claude (Anthropic) (2026a). A hypothesis is a triple (p,i,j); $A_{p,i}$ is the set of sentences χ that p accepts within $(|\chi|+2)^i$ steps; the budget for a datum φ is $(|\varphi|+2)^j$; and the weight is $2^{-|\mathrm{El}(p)|-2\lceil\log(i+1)\rceil-2\lceil\log(j+1)\rceil-2}$, with $\mathrm{El}(p)$ a self-delimiting code of p of length $|p|+O(\log|p|)$. The *conditional* version: (p,i,j) is compatible with a conditional data set D if for each $(\alpha,\ell) \in D$ the set $B \cup A_{p,i} \cup \{\alpha\}$ is consistent (this is not charged) and the sentence $\bigwedge \alpha \to \ell$ has a derivation in the calculus K of that report from $B \cup A_{p,i}$ within the budget under the reading. In our setting $\mathcal{L}_m$ is finite, $B = \emptyset$ is decidable in polynomial time, and Assumption 4.17 holds vacuously, as the axioms are sentences of $\mathcal{L}_m$, which contain no parameters.

The readings are defined for two-valued data; question data reach them only through the two-valued encoding, which is degenerate (Proposition E.3), or through this definition.

Theorem E.26 (Charged deduction does not charge for the lemma library)**.** *Let* $\mathcal{S}$ *be an affine law and take any reading of charged deduction in Definition E.25, at polynomial budgets with the degrees charged. (i) If each derivation of Proposition E.24(c) has size at most* $(|\bigwedge \alpha \to \ell|+2)^{j_0}$*, the decider of* $\mathrm{PI}(\mathcal{S})$ *with derivation degree* j_0 *is a hypothesis compatible with* $\mathcal{Q}(\mathcal{S})$ *of weight at least* $2^{-K^q(A,b)-O(\log m)-2\lceil\log(j_0+1)\rceil}$*. (ii) Every hypothesis compatible with* $\mathcal{Q}(\mathcal{S})$ *has weight at most* $2^{-K(\mathcal{S})+c}$*.*

Proof. (i) By Proposition E.24(c) the decider runs in time $\mathrm{poly}(m)\cdot\mathrm{poly}(|\chi|)\le(|\chi|+2)^i$ with $i=O(\log m)$, as $|\chi|+2\ge 3$, so the degree charge is $O(\log\log m)$; it has length $K^q(A,b)+c$, so $|\mathrm{El}(p)|=K^q(A,b)+O(\log m)$. $\mathrm{PI}(\mathcal{S})\cup\alpha$ is consistent for every antecedent of $\mathcal{Q}(\mathcal{S})$, and by hypothesis each datum has a derivation from a single member of $\mathrm{PI}(\mathcal{S})$ of size at most $(|\bigwedge\alpha\to\ell|+2)^{j_0}$, which bounds its instances and its material as well, so it is within the budget under every reading. (ii) Let (p,i,j) be compatible with $\mathcal{Q}(\mathcal{S})$ and $\Gamma:=A_{p,i}$. *Claim:* $x\notin\mathcal{S}$ iff there are $I\subsetneq[m]$ and $j'\notin I$ such that $\bigwedge x|_I\to\ell$ has a derivation from Γ within the budget, where ℓ is the literal on $x_{j'}$ false at x. If $x\notin\mathcal{S}$, the proof of Lemma E.2 gives a question $(x|_I,\ell)\in\mathcal{Q}(\mathcal{S})$ with ℓ false at x, and compatibility gives the derivation. If $x\in\mathcal{S}$, then x has a determined coordinate j'' (Lemma E.2), and the datum $(x|_{[m]\setminus\{j''\}},\ell'')$, with ℓ'' the literal on $x_{j''}$ true at x, makes $\Gamma\cup\{\hat{x}\}$ consistent; a derivation of $\bigwedge x|_I\to\ell$ with ℓ false at x would refute $\Gamma\cup\{\hat{x}\}$. The condition is decidable from (p,i,j): there are finitely many derivations within the budget, and membership in Γ is decided by running p with its clock. So $\mathcal{S}$ is computable from (p,i,j), and $K(\mathcal{S})\le|\mathrm{El}(p)|+2\lceil\log(i+1)\rceil+2\lceil\log(j+1)\rceil+c$, which is the claim. □

Remark E.27 (Derivation sizes, and the two-valued encoding under charged deduction). We do not count the nodes of the derivations below. (1) In the calculus K (connectives $\neg$ and $\to$, with $\vee$ and $\wedge$ defined), the tautology $(\neg\alpha'\vee\ell)\to(\bigwedge\alpha\to\ell)$ has a derivation of size polynomial in $|\alpha|+|\ell|=O(m)$, assembled from the standard derivations of the deduction theorem, $\wedge$-elimination and the double-negation laws, used $O(|\alpha|)$ times; one modus ponens with the axiom $C=\neg\alpha'\vee\ell$ finishes. So the hypothesis of Theorem E.26(i) holds with $j_0=O(1)$, the decider has weight at least $2^{-K^q(A,b)-O(\log m)}$, and by Theorem E.26(ii) these readings give the full lemma library, whose explicit form has $m2^{(m-1)/2}$ clauses for the simplex family, a cost within $K^q(A,b)-K(\mathcal{S})+O(\log m)$ bits of the least possible. (2) Under the two-valued encoding these readings are degenerate as SAI is: the point theory T_{x^0} of Proposition E.3 has a polynomial-time decider and is consistent, and the case distinction in the proof of Proposition E.3 gives each $\bigwedge\alpha\to\ell$ from one literal of T_{x^0} by propositional steps of polynomial size.

Charged deduction thus measures the length of a membership test, not the length of what a reader must be handed; the membership test for the full library runs Gaussian elimination, so the know-how lives in the decider. On literal sequences, under Assumption 4.17, at polynomial budgets with the degrees charged, charged-deduction SAI corresponds to certificate SOI (Theorem 4.18 and Corollary B.22), and the nondeterminism is the free search for derivations.

E.7 Learning know-how pattern by pattern

Let $m\ge 3$ and $\mathrm{Par}_m:=\mathrm{Par}_m^0$ (the constant 1 is symmetric). Every question of $\mathcal{Q}(\mathrm{Par}_m)$ has an antecedent on $m-1$ variables; write it as (j,ζ) with $\zeta\in\mathrm{Par}_m$, its *positive point*, antecedent $\zeta|_{[m]\setminus\{j\}}$ and answer $x_j^{\zeta_j}$. Its *pattern* is the violating point $\mathrm{pat}(j,\zeta):=\zeta\oplus\mathbf{e}_j\in\mathcal{P}_m:=\{0,1\}^m\setminus\mathrm{Par}_m$. So $|\mathcal{Q}(\mathrm{Par}_m)|=m2^{m-1}$ and $|\mathcal{P}_m|=2^{m-1}$, and every pattern and every positive point carries exactly m questions. For a conditional data set $D\subseteq\mathcal{Q}(\mathrm{Par}_m)$, $\mathrm{pat}(D)$ is the set of *seen* patterns. For $\beta\in\{0,1\}^m$, C_β is the full-width clause falsified exactly by β; in the listing code it costs $\mathrm{len}_m:=\ell_{\mathbb{N}}(m)+m(\lceil\log m\rceil+1)$ bits. A theory *fails* on a question if it errs, abstains, or is inconsistent with the antecedent.

Lemma E.28 (Answers of a theory over the base variables)**.** *For an explicit theory Φ over the base variables and a question (j,ζ) the following are equivalent: (i) Φ is* UP*-compatible with (j,ζ); (ii) Φ answers it correctly (unit propagation derives exactly $x_j^{\zeta_j}$, without conflict); (iii) ζ satisfies Φ, and some clause of Φ with a literal on x_j is falsified by* $\mathrm{pat}(j,\zeta)$.

Proof. Only x_j is unassigned, so unit propagation either finds a clause falsified by the antecedent (a conflict), or derives values of x_j from clauses whose other literals are falsified by it. A clause with a literal on x_j and all other literals falsified by the antecedent derives $x_j^{\zeta_j}$ iff it is falsified by $\mathrm{pat}(j,\zeta)$, and derives the opposite iff it is falsified by ζ; a clause without x_j that is falsified by the antecedent is falsified by ζ. So (iii) gives a derivation of the answer and no conflict, that is (ii), and ζ witnesses satisfiability, so (i) follows. If (i), the satisfying completion of the antecedent is ζ, since the clause that derives $x_j^{\zeta_j}$ excludes $\mathrm{pat}(j,\zeta)$; this gives (iii). (ii)⇒(iii): a correct derivation needs a clause falsified by $\mathrm{pat}(j,\zeta)$, and a clause falsified by ζ would cause a conflict or a wrong derivation. □

Theorem E.29 (A weak reasoner learns its know-how pattern by pattern)**.** *(a)* Full correctness. *An explicit theory over the base variables answers every question of $\mathcal{Q}(\mathrm{Par}_m)$ correctly iff it is* $\mathrm{PI}(\mathrm{Par}_m) = \{C_\beta : \beta \in \mathcal{P}_m\}$. *(b)* Minimal theories, any order. *Let $D \subseteq \mathcal{Q}(\mathrm{Par}_m)$ and let Φ be an inclusion-minimal explicit theory over the base variables that is* UP*-compatible with D (for instance any ℓ_{lst}-shortest one, Lemma E.1). Then for every unseen pattern $\beta \notin \mathrm{pat}(D)$, Φ fails on some question of pattern β. Hence Φ fails on at least $|\mathcal{P}_m \setminus \mathrm{pat}(D)|$ questions, and a shortest compatible theory is fully correct only after all 2^{m-1} patterns have been seen. (c)* Random order. *If the questions are drawn independently and uniformly from $\mathcal{Q}(\mathrm{Par}_m)$, the patterns are independent and uniform on $\mathcal{P}_m$; after n questions the expected number of unseen patterns is $2^{m-1}(1-2^{1-m})^n$, and the expected time until all have been seen is $2^{m-1}\sum_{i=1}^{2^{m-1}} 1/i = 2^{m-1}\big((m-1)\ln 2 + 0.5772\ldots\big) + \frac{1}{2} + O(2^{-m})$, where $0.5772\ldots$ is Euler's constant. (d)* Bayes, block-complete data (semimeasure convention). *If D contains all m questions of each seen pattern, every* UP*-compatible explicit theory over the base variables contains $\{C_\beta : \beta \in \mathrm{pat}(D)\}$, and $-\log W_{\mathrm{lst}}^{\mathrm{UP,base}}(D) \geq |\mathrm{pat}(D)|\,\mathrm{len}_m$. So the cumulative loss is at least $|\mathrm{pat}(D)|\,\mathrm{len}_m - c$ bits: about $m \log m$ bits per pattern, and $2^{m-1}\mathrm{len}_m$ in all. (e)* With the reasoner or with intermediate concepts. *For every $D \subseteq \mathcal{Q}(\mathrm{Par}_m)$, $-\log W_{\mathrm{lst}}^{\mathrm{GE}}(D) \leq \ell_{\mathrm{lst}}$(the one equation) $= m\lceil \log m \rceil + O(\log m)$, and $-\log W_{\mathrm{lst}}^{\mathrm{UP}}(D) \leq \ell_{\mathrm{lst}}$(the chain encoding) $= O(m \log m)$ (Proposition E.13). So these inducers pay $O(m \log m)$ bits in total on every question sequence. (f)* Other measures. *Under program length there is no gap: $K_A^{\mathrm{UP,base}}(\mathcal{Q}(\mathrm{Par}_m)) = K(m) \pm c$. Under time charging, $Kt_A^{\mathrm{UP,base}}(\mathcal{Q}(\mathrm{Par}_m)) \geq \log(m2^{m-1}) = m - 1 + \log m$, while $Kt_A^{\mathrm{UP}}(\mathcal{Q}(\mathrm{Par}_m))$ and $Kt_A^{\mathrm{GE}}(\mathcal{Q}(\mathrm{Par}_m))$ are $O(\log m)$.*

Proof. (a) By Lemma E.28(iii) for all questions: every point of Par_m satisfies Φ, so every clause is an implicate, of width m (Proposition E.8, $d_* = m$), that is, some C_β with $\beta \in \mathcal{P}_m$; and every pattern β needs a clause falsified by it, which among full-width clauses is C_β itself. (b) C_β is falsified only by β, so by Lemma E.28 it serves only questions of pattern β; for unseen β, removing it keeps compatibility with D, so by minimality $C_\beta \notin \Phi$. If for some j no clause of Φ with a literal on x_j is falsified by β, then Φ fails on $(j, \beta \oplus \mathbf{e}_j)$, of pattern β. Otherwise take a clause $C \in \Phi$ falsified by β; $C \neq C_\beta$, so $|C| < m$; take $i \notin \mathrm{supp}(C)$. The question $(i, \beta \oplus \mathbf{e}_i)$ has pattern β, and its positive point $\beta \oplus \mathbf{e}_i$ agrees with β on $\mathrm{supp}(C)$, so it falsifies C, and Φ fails on it (Lemma E.28). Distinct patterns give distinct questions. (c) Each pattern carries m of the $m2^{m-1}$ questions; the rest is the coupon collector's problem, with $\sum_{i \leq N} 1/i = \ln N + 0.5772\ldots + \frac{1}{2N} + O(N^{-2})$. (d) For a seen pattern β, every $\beta \oplus \mathbf{e}_i$ is a seen positive point, so every clause of a compatible Φ is satisfied by all of them (Lemma E.28). A clause falsified by β of width less than m is falsified by $\beta \oplus \mathbf{e}_i$ for i outside its support; so the only admissible clause falsified by β is C_β, and Lemma E.28 requires one. With $R := \{C_\beta : \beta \in \mathrm{pat}(D)\}$ and $\sigma_D := |\mathrm{pat}(D)|$, summing the listing weights of all theories over the base variables that contain R (here $m_{\mathrm{aux}} = 0$) gives, as in the proof of Lemma E.1, at most

$\sum_s 2^{-\ell_{\mathbb{N}}(s)} 2^{-\sigma_D \mathrm{len}_m} \sum_{|\Phi'|=s-\sigma_D} \prod_{C\in\Phi'} 2^{-\ell(C)} \le 2^{-\sigma_D \mathrm{len}_m} \sum_s 2^{-\ell_{\mathbb{N}}(s)}/(s-\sigma_D)! \le 2^{-\sigma_D \mathrm{len}_m}$. The cumulative loss in the semimeasure convention is $-\log W_{\mathrm{lst}}(D) + \log W_{\mathrm{lst}}(\emptyset)$, and $W_{\mathrm{lst}}(\emptyset) \ge 2^{-c}$. (e) The single equation (respectively the chain encoding) is compatible with every $D \subseteq \mathcal{Q}(\mathrm{Par}_m)$, so $W_{\mathrm{lst}} \ge 2^{-\ell_{\mathrm{lst}}}$ of it; the listing lengths are direct from (4). (f) Theorem 7.3(b), (c): the unique usable theory over the base variables has 2^{m-1} clauses of width m, and the chain encoding and the equation are printed from m in polynomial time. □

A simulation (code in the accompanying repository) with uniformly random question sequences for $m = 4, 5, 6$, which computes an exactly ℓ_{lst}-shortest compatible theory at checkpoints, found in each of 330 checkpoints at least as many failures as unseen patterns, as (b) requires, and typically about m times as many; the exact expected times until all patterns are seen are 21.74, 54.09 and 129.87.

Question E.30. Is there a learning theorem of this kind for systems of laws, and for a fixed number of rounds with intermediate concepts? Is there a Bayes lower bound like (d) for arbitrary question sequences that are not block-complete?

E.8 Depth against size for the implication chain

Let $\mathrm{Ch}_m = \{x : x_1 \le \cdots \le x_m\}$, $m \ge 2$. A *k-hop transitive-closure spanner* of the directed path on $[m]$ is a set E of pairs (i, j), $i < j$, such that for all $i < j$ there is a path from i to j in E with at most k edges; $\mathrm{sp}_k(m)$ is the least size of one.

Theorem E.31 (Depth against size: the reduction to spanners)**.** *(i) The prime implicates of Ch_m are exactly the clauses $\neg x_i \vee x_j$, $i < j$. (ii) A 2-CNF of prime implicates, with edge set E, is UP_k-usable iff E is a k-hop transitive-closure spanner of the path. (iii) If an explicit theory Φ over the base variables, with clauses of any width, is UP_k-usable, then for every choice of a prime implicate contained in each clause, the chosen edges form a k-hop transitive-closure spanner of the path. (iv) Hence the least number of clauses of a UP_k-usable theory over the base variables is exactly $\mathrm{sp}_k(m)$, for every k.*

Proof. (i) A partial assignment is consistent with Ch_m iff it sets no $x_i = 1$, $x_j = 0$ with $i < j$; so a clause is an implicate iff it contains some $\neg x_i \vee x_j$ with $i < j$, and these 2-clauses are the prime implicates. (ii) For an edge set E, UP_k from a consistent α derives x_j iff some x_i true in α has a path to j of at most k edges, and $\neg x_i$ iff some x_j false in α is reached from i within k edges (the same clauses read contrapositively); and an entailed literal is entailed by a single literal of α. So usability is the spanner property (take $\alpha = \{x_i\}$ for necessity). (iii) Lemma E.9 holds for every law: the replaced theory derives, round for round, at least what Φ derives, so it is UP_k-usable, and it is a 2-CNF of prime implicates, so (ii) applies (two clauses may give the same edge). (iv) By (iii) a usable Φ has $|\Phi| \ge |E| \ge \mathrm{sp}_k(m)$, and a minimum spanner written as a 2-CNF is usable by (ii). □

The converse of (iii) fails for wide clauses. For $m = 3$, $k = 1$, the theory $\{\neg x_1 \vee x_2,\ \neg x_2 \vee x_3,\ \neg x_1 \vee \neg x_2 \vee x_3\}$ consists of implicates, and choosing $\neg x_1 \vee x_3$ inside the wide clause gives the 1-hop spanner $\{12, 23, 13\}$, but from $x_1 = 1$ one round does not derive x_3. The failure persists when the choice is forced: in $\{\neg x_1 \vee x_2,\ \neg x_1 \vee x_3,\ \neg x_2 \vee x_1 \vee x_3\}$ the wide clause contains only $\neg x_2 \vee x_3$, the edges $\{12, 13, 23\}$ form a 1-hop spanner, but from $x_3 = 0$ one round does not derive $x_2 = 0$. A wide clause propagates in fewer situations than the edge it contains.

The minimum size of a k-hop transitive-closure spanner of the directed path is $\Theta(m\,\mathrm{ia}_k(m))$, with ia_k the k-th row of the inverse Ackermann function: $\mathrm{ia}_2(m) = \log m$, $\mathrm{ia}_3(m) = \log\log m$,

$\mathrm{ia}_4(m) = \log^* m$. This is the bound of Alon and Schieber (1987) for answering product queries in at most k steps, in the language of transitive-closure spanners (Bhattacharyya et al., 2012; Berman et al., 2014); Yao (1982) gave the first inverse-Ackermann trade-off for range queries. A direct computation (integer programming, confirmed by exhaustive search for $m \le 7$; code in the accompanying repository) gives the exact values:

m	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
$\mathrm{sp}_1(m) = \binom{m}{2}$	1	3	6	10	15	21	28	36	45	55	66	78	91	105	120	136	153
$\mathrm{sp}_2(m)$	1	2	4	6	8	10	13	16	19	22	25	28	31	34	38	42	46
$\mathrm{sp}_3(m)$	1	2	3	5	6	8	10	12	14	16	18	20					
$\mathrm{sp}_4(m)$	1	2	3	4	6	7	9	10	12	14	16						

At these sizes $\mathrm{sp}_3(m) = 2m - 6$ for $6 \le m \le 13$; the $\log\log$ growth is not yet visible.

Proposition E.32 (The midpoint construction)**.** *The midpoint recursion, $S(1) = 0$ and $S(n) = n-1+S(\lceil \frac{n-1}{2} \rceil)+S(\lfloor \frac{n-1}{2} \rfloor)$, gives 2-hop spanners of the path, so $\mathrm{sp}_2(m) \le S(m) = \sum_{i=1}^{m} \lfloor \log i \rfloor = (m+1)L - 2^{L+1} + 2$, where $L := \lfloor \log m \rfloor$.*

Proof. Join a hub vertex h to all other vertices ($i \to h$ for $i < h$, $h \to j$ for $j > h$), with $\lceil \frac{n-1}{2} \rceil$ vertices on one side and $\lfloor \frac{n-1}{2} \rfloor$ on the other, and recurse on both sides: pairs on opposite sides of h, or containing h, are joined within 2 hops, and pairs on one side by the recursion. $\Sigma(m) := \sum_{i \le m} \lfloor \log i \rfloor$ satisfies the same recursion: splitting $i = 2i'$ and $i = 2i' + 1$ ($i \ge 2$), $\lfloor \log i \rfloor = 1 + \lfloor \log i' \rfloor$, so $\Sigma(m) = (m-1) + \Sigma(\lfloor m/2 \rfloor) + \Sigma(\lfloor (m-1)/2 \rfloor)$, and $\lfloor m/2 \rfloor = \lceil (m-1)/2 \rceil$. The closed form is standard. □

The construction attains $\mathrm{sp}_2(m)$ for every $m \le 18$ in the table above.

Conjecture E.33. $\mathrm{sp}_2(m) = \sum_{i=1}^{m} \lfloor \log i \rfloor$ *for every $m \ge 2$.*

We have neither a proof nor a counterexample; a computation confirms it for $m \le 18$. The best lower bound we prove is that of Theorem E.34, about $\frac{1}{2} m \log m$, which holds even with intermediate concepts.

Theorem E.34 (Intermediate concepts and wide clauses gain at most a factor about 2 at depth 2)**.** *Let $m \ge 4$. Every UP_2-usable explicit theory Φ for Ch_m, with any auxiliary variables and clauses of any width, has*

$$|\Phi| \;\ge\; \tfrac{1}{2}\Big((m-2)\log \tfrac{m-2}{2} - m\Big),$$

while $\mathrm{sp}_2(m) \le m \log m$. So at depth 2 the least usable theory has $\Theta(m \log m)$ clauses with or without intermediate concepts, and they save at most a factor of about 2.

Proof. Upper bound. In the midpoint recursion $S(n) = n - 1 + S(a) + S(b)$ with $a + b = n - 1$ and $a, b \le n/2$, so by induction $S(n) \le n - 1 + (n-1)\log(n/2) \le n \log n$. *Simplification.* Let U_0 be the set of literals derived by unit propagation from Φ alone; they are auxiliary, as Ch_m fixes no base variable. Simplifying Φ by U_0 (deleting the clauses satisfied by U_0 and the literals falsified by it) gives Φ' with $|\Phi'| \le |\Phi|$ and the same projection, which does not mention the variables of U_0 and has no unit clauses, and in which every UP_2 derivation of Φ outside U_0 is made no later (induction on rounds: a clause of Φ that fires in a run from a base antecedent contains no literal of U_0, since such a literal would be falsified, that is, its negation derived, against soundness; so its simplification is in Φ' and fires). So we may assume that Φ has no unit clauses. *Single sources.*

For a base index v let $\mathrm{Nb}(v) := \{x_v\} \cup \{L : (\neg x_v \vee L) \in \Phi\}$, the literals known after one round from $\alpha = \{x_v\}$. For a clause C containing x_j let $P(C, j) := \{\neg L' : L' \in C \setminus \{x_j\}\}$, its premises for deriving x_j. From $\{x_u\}$, UP_2 derives x_j iff some $C \ni x_j$ has $P(C, j) \subseteq \mathrm{Nb}(u)$ (round 1 is the case $C = \neg x_u \vee x_j$); say that C *serves* u for j. By soundness, if C serves u for j then $u \leq j$. A clause serving any source for j contains no other positive base literal $x_{w'}$, since its premise $\neg x_{w'}$ would follow from a single x_u, which is unsound. Let $\mathrm{cnt}(j)$ be the number of clauses containing x_j that serve some $u < j$; then $\sum_j \mathrm{cnt}(j) \leq |\Phi|$, and $\sum_v |\mathrm{Nb}(v)| \leq m + |\Phi|$, since a 2-clause $\neg x_v \vee L$ is counted only for v (a 2-clause with two negative base literals is not an implicate). *Random experiment.* Give every literal an independent fair coin, *left* or *right*. Call v *source-alive* if all of $\mathrm{Nb}(v)$ is right, and j *target-alive* if every clause containing x_j that serves some $u < j$ has a left premise. No $u < j$ has u source-alive and j target-alive: by usability some C serves u for j, and all its premises lie in $\mathrm{Nb}(u)$ and are right. *Pairing.* For $a = 1, \ldots, M' := \lfloor m/2 \rfloor$ let $\mathcal{B}_a$ be the event that $2a$ is source-alive and $2a - 1$ target-alive. For $a < a'$ we have $2a < 2a' - 1$, so at most one $\mathcal{B}_a$ occurs and $\sum_a \Pr[\mathcal{B}_a] \leq 1$. Given that $\mathrm{Nb}(2a)$ is all right, a clause serving some $u < 2a - 1$ for $2a - 1$ cannot have all its premises in $\mathrm{Nb}(2a)$ (it would serve $2a$ for $2a - 1$, which is unsound), so it is blocked by a left coin outside $\mathrm{Nb}(2a)$ with probability at least $\frac{1}{2}$, independently of the coins of $\mathrm{Nb}(2a)$; these blocking events are increasing in the set of left coins, so by Harris's inequality they occur together with probability at least $2^{-\mathrm{cnt}(2a-1)}$. Hence $\Pr[\mathcal{B}_a] \geq 2^{-|\mathrm{Nb}(2a)| - \mathrm{cnt}(2a-1)}$, and by the convexity of $x \mapsto 2^{-x}$, $\sum_a (|\mathrm{Nb}(2a)| + \mathrm{cnt}(2a - 1)) \geq M' \log M'$. The shifted pairing ($2a + 1$ source-alive and $2a$ target-alive, for $a \leq M'' := \lfloor (m - 1)/2 \rfloor$; $a < a'$ gives $2a + 1 < 2a'$) gives $\sum_a (|\mathrm{Nb}(2a + 1)| + \mathrm{cnt}(2a)) \geq M'' \log M''$. Adding, and using $M', M'' \geq (m - 2)/2 \geq 1$, $m + 2|\Phi| \geq \sum_v |\mathrm{Nb}(v)| + \sum_j \mathrm{cnt}(j) \geq (m - 2) \log \frac{m-2}{2}$. □

The bound is asymptotic: it is positive only from $m = 8$ on, and gives 11.7 at $m = 16$ against $\mathrm{sp}_2(16) = 38$. The argument adapts Hansel's lemma, that covering the complete graph on n vertices by bicliques needs total size at least $n \log n$ (Hansel, 1964; Katona and Szemerédi, 1967), to ordered pairs and to conjunctive premises: the pairing keeps the diagonal out, and Harris's inequality handles clauses with several premises. For 2-CNFs, after the simplification step the implication graph of a UP_2-usable theory (edges $\neg a \to b$ and $\neg b \to a$ for each clause $a \vee b$) is a Steiner 2-hop transitive-closure spanner of the path, with the positive base literals as its vertices and at most $2|\Phi|$ edges, so lower bounds for Steiner 2-transitive-closure spanners of directed hypergrids (Berman et al., 2014), of which the path is the one-dimensional case, bear on that case; what Theorem E.34 adds is the case of wide clauses and an explicit constant.

Proposition E.35 (Every fixed depth: superlinear with intermediate concepts)**.** *Suppose that for every fixed depth d, every monotone circuit of depth d with unbounded fan-in AND and OR gates that computes all the prefix ORs x_1, $x_1 \vee x_2$, …, $x_1 \vee \cdots \vee x_m$ has $m \cdot \omega(1)$ wires. Then for every fixed k and every bound w on clause width, every UP_k-usable theory Φ for Ch_m with clauses of width at most w, with any auxiliary variables, has $\|\Phi\| \geq m \cdot \omega(1)$.*

Lower bounds of this form for constant-depth prefix circuits are due to Chandra, Fortune and Lipton, in a companion paper to Chandra et al. (1985), which gives the matching constant-depth constructions; we state the property as a hypothesis. Size must be counted in wires: counting gates, prefix OR has depth-1 circuits with m gates.

Proof. On the antecedents $\alpha_{A'} = \{x_i : i \in A'\}$, $A' \subseteq [m]$, all consistent, soundness and usability give: UP_k derives x_j iff $A' \cap [1, j] \neq \emptyset$. So the circuit of Lemma E.18(b), with inputs the literals x_i, is a monotone circuit of depth $2k$ with at most $k(w\|\Phi\| + 2\bar{m})$ wires that computes all prefix ORs, and $\bar{m} \leq m + \|\Phi\|$. Hence $(w + 2)k\|\Phi\| + 2km \geq m \cdot \omega(1)$. □

Question E.36. (a) What is the least UP-usable theory over the base variables when $d_* = 2$, beyond equivalence laws? (b) What is the constant at depth 2 with intermediate concepts, between $\frac{1}{2} m \log m$ (Theorem E.34) and $\sum_{i \le m} \lfloor \log i \rfloor \approx m \log m$ (Conjecture E.33)? What is the effect of intermediate concepts at depth $k \ge 3$ beyond Proposition E.35? (c) How many rounds does one parity law need for usable theories of polynomial size: between $\Omega(\log m / \log \log m)$ and $2\lceil \log m \rceil - 1$ (Proposition 7.8(d))?

E.9 Lemmas in first-order logic

In first-order logic the cost of a lemma-free theory for an analytic reasoner can be non-elementary: there are sentences with proofs of size linear in k in Gentzen's sequent calculus LK with cuts whose cut-free proofs have size not bounded by any Kalmár-elementary function of k (Orevkov, 1982); see also the non-elementary lower bounds of Statman (1979). Cut formulas are lemmas, and cut-free proofs are lemma-free reasoning.

Remark E.37 (Valid lemma sentences restore short cut-free proofs). Let $\mathcal{D}$ be an LK proof of $T_0 \vdash \varphi$, with $T_0 \subseteq T$ finite and cuts numbered $1, \dots, n_{\mathrm{cut}}$, the i-th with left premise $\Gamma_i \vdash \Delta_i, A_i$. Let $L_i := \forall \vec{v} \, (\bigwedge \Gamma_i \to \bigvee \Delta_i \vee A_i)$, the universal closure over the free variables. Then (a) each L_i is valid, since the sequent $\Gamma_i \vdash \Delta_i, A_i$ is derivable and LK is sound, so $T \cup \{L_1, \dots, L_{n_{\mathrm{cut}}}\}$ is deductively equivalent to T; and (b) $L_1, \dots, L_{n_{\mathrm{cut}}}, T_0 \vdash \varphi$ has a cut-free proof of size polynomial in the size of $\mathcal{D}$.

Proof sketch. We do not write out the bookkeeping of contexts in the chosen sequent calculus. Replace the i-th cut, with premises $\Gamma \vdash \Delta, A$ and $A, \Gamma' \vdash \Delta'$, as follows. From the right premise and identity sequents $\delta \vdash \delta$ ($\delta \in \Delta$), $\vee$-left gives $\bigvee \Delta \vee A, \Gamma' \vdash \Delta, \Delta'$; from identity sequents $\gamma \vdash \gamma$, $\wedge$-right gives $\Gamma \vdash \bigwedge \Gamma$; $\to$-left gives $\bigwedge \Gamma \to \bigvee \Delta \vee A, \Gamma, \Gamma' \vdash \Delta, \Delta'$; and $\forall$-left steps, instantiating the bound variables by the free variables of the cut's premises, give $L_i, \Gamma, \Gamma' \vdash \Delta, \Delta'$. The subproof of the left premise is dropped. Carry L_i down to the end sequent, by weakening where branches meet and contraction for duplicates; L_i is closed, so no eigenvariable condition below the former cut is affected. Identity sequents for composite formulas have cut-free proofs of size polynomial in the formula. □

The added sentences are valid: the theory's deductive closure, and so everything the idealised inducer sees, is unchanged, and only its written form changes. If every cut's left premise has the form $T_i \vdash A_i$ with $T_i \subseteq T$, the closures $\forall \vec{v} \, A_i$ of the cut formulas are theorems of T and serve directly; in general they are not, since a cut formula can be arbitrary.

Proposition E.38 (When a lemma pays). *Let* $\mathrm{Obj}(T) := \mathrm{lh}(T) + \sum_{\varphi \in D} \rho_T(\varphi)$, *with* lh *additive over axioms and* $\rho_T(\varphi)$ *the cost of the cheapest derivation of the datum* φ *from* T, *non-increasing when axioms are added. Adding a theorem* $L \notin T$ *of* T *as an axiom lowers* Obj *if and only if* $\sum_{\varphi \in D} (\rho_T(\varphi) - \rho_{T \cup \{L\}}(\varphi)) > \mathrm{lh}(L)$*: a lemma enters iff the derivation cost it saves on the data exceeds its own length.*

Proof. By additivity, $\mathrm{Obj}(T \cup \{L\}) - \mathrm{Obj}(T) = \mathrm{lh}(L) - \sum_{\varphi \in D} (\rho_T(\varphi) - \rho_{T \cup \{L\}}(\varphi))$. □

This is the general form of the observation in Remark E.9(iv) of Hänni and Claude (Anthropic) (2026a) that a time-bounded likelihood favours adopting theorems with long proofs as axioms.

F Proofs for theoretical terms

Throughout, B is decidable and consistent, "consistent" means consistent with B, and Cn has B built in (Section 2). For a sentence ψ, $\psi^{\wedge 1} := \psi$ and $\psi^{\wedge (m+1)} := \psi^{\wedge m} \wedge \psi$, where $\alpha \wedge \beta$ abbreviates

$\neg(\alpha \to \neg\beta)$ in the syntax of K; so $|\psi^{\wedge m}| \geq m|\psi|$. On literal data (Definition 8.2) the literal setting of Definition 2.2 is assumed: R does not occur in B, and some model of B interprets the terms t_w as pairwise distinct elements. Sentence length is the symbol count. The computations quoted below check constructions and counts on finite fragments; the theorems are proved in the text, and the code is in the accompanying repository.

F.1 In principle

Proof of Theorem 8.5. (d) G^O is c.e. because G is and derivability from the decidable B is c.e. By compactness, and because A^G_w grows with w along the prefixes of ω (Lemma A.1(a)), $A^{G^O}_\omega = \bigcup_{w \sqsubset \omega}\{\varphi \in \mathrm{Sent}_{\mathcal{L}_O} : B \cup A^G_w \vdash \varphi\} = T_O(A^G_\omega)$. Since $T_O(A) \subseteq \mathrm{Cn}(A)$, consistency of $B \cup A$ passes to $B \cup T_O(A)$; if $B \cup A$ is inconsistent, $T_O(A)$ contains the observational sentence $\neg\forall x\, x = x$. Again from $T_O(A) \subseteq \mathrm{Cn}(A)$, $\mathrm{Cn}(T_O(A)) \cap \mathrm{Sent}_{\mathcal{L}_O} = T_O(A)$, so an observational φ^b follows from $B \cup T_O(A)$ iff it lies in $T_O(A)$ iff it follows from $B \cup A$. The observational content of G^O on ω is $T_O(A^G_\omega)$ itself, so membership in $\mathcal{C}_\Theta$ is preserved, and the two events coincide tape by tape.

(a) By the mixture identity of Theorem 3.6, on the cylinder of a program p with $V(p)$ an index of a machine G, the universal machine emits $A^G_{\omega'}$ on the rest ω' of the tape, hence G^O_U emits $A^G_{\omega'} \cap \mathrm{Sent}_{\mathcal{L}_O}$, and $\mathbb{U}(\,\cdot \cap [p]) = 2^{-|p|}\mathbb{U}(\cdot)$ on the shifted event; here the measurability of $\mathcal{C}_\Theta$ is used (the map $\omega \mapsto T_O(A_\omega)$ is Borel, since each coordinate "$\varphi \in T_O(A_\omega)$" is the effectively open event O_φ). *Lower bound.* Apply (d) to $G := G_U$. $(G_U)^O$ is a fixed machine emitting only $\mathcal{L}_O$-sentences, so on the tapes $p\omega'$ with $V(p)$ an index of $(G_U)^O$ the filtered machine G^O_U emits exactly $A^{(G_U)^O}_{\omega'}$; hence $M^{\mathcal{L}_O}_A(\mathcal{C}_\Theta; D) \geq 2^{-K((G_U)^O)}\nu_{(G_U)^O}(\mathcal{C}_\Theta; D) = 2^{-c}\mathbb{U}_A(\mathcal{C}_\Theta; D)$. *Upper bound.* G^O_U is a theory machine of $\mathcal{L}$, so the same identity in $\mathcal{L}$ gives $\mathbb{U}_A(\mathcal{C}_\Theta; D) \geq 2^{-K(G^O_U)}\nu_{G^O_U}(\mathcal{C}_\Theta; D) = 2^{-c}M^{\mathcal{L}_O}_A(\mathcal{C}_\Theta; D)$. The program-less tapes contribute $(1 - \Omega_V)\mathbf{1}[B \vdash \psi_D,\ T_O(\emptyset) \in \Theta]$ to both sides.

(b) Let $o(e)$ be an index of $\{\varphi \in \mathrm{Sent}_{\mathcal{L}_O} : B \cup \mathsf{W}_e \vdash \varphi\}$ and $f(e)$ one of $\mathsf{W}_e \cap \mathrm{Sent}_{\mathcal{L}_O}$; both maps are computable. By the argument of (d), $o(e)$, read in $W^{\mathcal{L}_O}_A$, is compatible with D and has content in Θ iff e, read in W_A, is; and $f(e)$ read in W_A behaves as e read in $W^{\mathcal{L}_O}_A$. The pushforward of $\mathbf{m}$ under a computable map is a lower semicomputable semimeasure, hence at most $2^c\mathbf{m}$. So, with P the compatibility-and-class predicate, $W_A(\mathcal{C}_\Theta; D) = \sum_e \mathbf{m}(e)\mathbf{1}[P(o(e))] \leq 2^c \sum_j \mathbf{m}(j)\mathbf{1}[P(j)] = 2^c W^{\mathcal{L}_O}_A(\mathcal{C}_\Theta; D)$, and symmetrically with f. Since $K(o(e)) \leq K(e) + c$ and $K(f(e)) \leq K(e) + c$, the statement on K_A follows.

(c) *Coin-free.* For an index e, let f^O_e be proof search from $B \cup \mathsf{W}_e$ restricted to observational inputs: accept φ if $B \cup \mathsf{W}_e \vdash \varphi$, reject if $B \cup \mathsf{W}_e \vdash \neg\varphi$, abstain on non-observational inputs. Then $\Gamma_{f^O_e} = T_O(\mathsf{W}_e)$, which is consistent iff W_e is. Theorem 6.14 of Hänni and Claude (Anthropic) (2026a) (Theorem 4.7(b)) maps every consistent labeller f, injectively and at length cost c_t, to the decider (respectively enumerator) of the padded set of Lemma E.12 there, whose members are conjunction-powers of members of Γ_f, with the same consequences as Γ_f and membership (respectively generation) within t, under Assumption 4.5. For $f = f^O_e$ the padded set is observational and has the observational content of W_e. Summing over e with the pushforward bound gives the lower bound. The upper bound holds because the observational restriction is a sub-sum of $W^{\mathrm{chk},t}_A$ (respectively $W^{\mathrm{gen},t}_A$), and a penalised hypothesis is an ordinary one with the same theory (Proposition B.20 for the class form). *One tape.* The padded random decider in the proof of Theorem 4.7(a), applied to the random decider program that says yes on χ iff χ is observational and $B \cup A^G_\omega \vdash \chi$ (dovetailed proof search over the tape), has on every tape the padded copy of $T_O(A^G_\omega)$ as its yes-set; this set is observational, has the observational content of A^G_ω, and meets the time bound uniformly in ω under Assumption 4.6. The program-less tapes are matched by

the decider of $\emptyset$, as in Proposition B.6. Sum as in (a). For the upper bound, a random decider or enumerator is read as a theory machine through a constant-size wrapper, and Proposition B.6 bounds the penalised mixture by M_A class by class.

The consequences. Divide the class bounds for Θ_1 and Θ_2, and the bounds for $D + (\varphi, b)$ and D; the semimeasure loss is $-\log M(D_n) + \log M(\emptyset)$ for each version M. □

A direct computation on a propositional analogue (three observational atoms, two theoretical atoms, the theoretical background $(t_0 \to o_0) \wedge (t_1 \to \neg o_2)$, 40 random theory machines on 16 tapes) finds the law of G^O equal to that of G on all 24480 pairs of machine and observational data set and on 21830 class checks, while dropping the theoretical axioms instead of projecting changes the law of 37 of the 40 machines; an independent implementation with three backgrounds agrees.

F.2 Craig's construction

Lemma F.1 (Re-checking a literal)**.** *On literal data, let A be a set of sentences each logically equivalent to a literal $\pm\varphi_{w'}$, with $B \cup A$ consistent. If $B \cup F \vdash \varphi_w^b$ for a finite $F \subseteq A$, then F contains a member equivalent to φ_w^b.*

Proof. Suppose not. The members of F are equivalent to literals; no two of them are complementary, since $B \cup F$ is consistent; and those about w, if any, are equivalent to φ_w^{1-b}. Take a model of B in which the $t_{w'}$ are pairwise distinct and reinterpret R, which does not occur in B: put $t_{w'}$ into R iff $R(t_{w'})$ is among the literals of F, and in addition put t_w into R iff $b = 0$. This model satisfies B, every member of F and $\neg\varphi_w^b$. □

Lemma F.2 (Craig's padding writes a run time in unary)**.** *Let $B \cup T$ be consistent.* Canonical proof search f_T^{can} *tests, on input φ, the binary strings in length-lexicographic order as codes of* K*-derivations from $B \cup T$, in a code in which a derivation of symbol size s has code length at least s, spending at least one step per string, and stops at the first derivation of φ or of $\neg\varphi$. Let* $\mathrm{ds}(\varphi^b)$ *be the least symbol size of a derivation of whichever of $\varphi, \neg\varphi$ is provable from $B \cup T$. (a) The running time of f_T^{can} on φ is at least $2^{\mathrm{ds}(\varphi^b)} - 1$. (b) On literal data, let $A^{\mathrm{Cr}} := \{(\varphi_w^b)^{\wedge(k+1)} :$ f_T^{can} decides φ_w with b in exactly k steps$\}$ be Craig's set of f_T^{can} restricted to literal inputs, or its version padded by a work counter (Lemma E.12 of Hänni and Claude (Anthropic) (2026a)). Every* K*-derivation of φ_w^b from $B \cup A^{\mathrm{Cr}}$ contains a line of size at least $2^{\mathrm{ds}(\varphi_w^b)}|\varphi_w^b|$; so its size, its material and its largest axiom instance are all at least that. (c) With Craig's original padding by the Gödel number* gn *of a derivation (the number whose binary expansion is* 1 *followed by the code), $A^{\mathrm{gn}} := \{(\varphi_w^b)^{\wedge \mathrm{gn}}\}$, the same bound holds.*

Proof. (a) Every derivation of φ^b has size at least ds, hence code length at least ds, and no derivation of φ^{1-b} exists. So all $2^0 + 2^1 + \cdots + 2^{\mathrm{ds}-1} = 2^{\mathrm{ds}} - 1$ strings of length less than ds are tested and fail. (b) By Lemma F.1 the derivation cites a member of A^{Cr} equivalent to φ_w^b. The only such member is $(\varphi_w^b)^{\wedge(k+1)}$, with $k \geq 2^{\mathrm{ds}} - 1$ by (a); for the padded version the counter is at least the simulated time. That member is a line of the derivation, of size at least $(k+1)|\varphi_w^b|$. (c) Every member of A^{gn} equivalent to φ_w^b is a power with $\mathrm{gn} \geq 2^\ell$ factors, where $\ell \geq \mathrm{ds}$ is the code length; apply Lemma F.1. □

Proof of Theorem 8.7. (a) Under each reading, the member of A that a derivation of φ_w^b cites and that is equivalent to φ_w^b (Lemma F.1) has size at most $h(|\varphi_w|)$: it is an axiom line, a nonlogical instance, a part of the material, and a line of a derivation of size at most h. Its

form is $(\varphi_w^b)^{\wedge m}$ with $m \le h(|\varphi_w|)$. Decide w as follows: for $m = 1, \dots, h(|w|+2)$ test whether $(\varphi_w)^{\wedge m} \in A$, and accept iff one is. These are polynomially many tests of polynomial time. If a power of φ_w is in A, then $B \cup A \vdash \varphi_w$, so $w \in X$ (otherwise $B \cup A$ would also prove the datum $\neg\varphi_w$ and be inconsistent). If $w \in X$, the derivation within budget cites such a power with $m \le h$. (b) Clock a polynomial-time decider f of X at $\tau(m) := (m+2)^k$. Its Craig set $\{(\varphi_w^b)^{\wedge(j+1)} : f \text{ decides } \varphi_w \text{ with } b \text{ in exactly } j \text{ steps}\}$ is parameter-free, has polynomial-time membership (read the word off a candidate with m factors and run f for $m-1$ steps), and derives each datum by citing $(\varphi_w^b)^{\wedge(j+1)}$, one substitution instance of a fixed propositional derivation of $(\beta \wedge \gamma) \to \gamma$, and modus ponens; size, material and largest instance are $O((\tau(m)+1)(m+4))$. This is Lemma E.22 of Hänni and Claude (Anthropic) (2026a). □

Example F.3 (The size of a Craig axiom)**.** Take the seven-sentence Horn theory with theoretical predicates Ev, Od (an even, respectively odd, number of 1s): $\mathrm{Ev}(\mathsf{e})$; $\mathrm{Ev}(x) \to \mathrm{Ev}(\mathsf{s}_0 x)$; $\mathrm{Ev}(x) \to \mathrm{Od}(\mathsf{s}_1 x)$; $\mathrm{Od}(x) \to \mathrm{Od}(\mathsf{s}_0 x)$; $\mathrm{Od}(x) \to \mathrm{Ev}(\mathsf{s}_1 x)$; $\mathrm{Ev}(x) \to R(x)$; $\mathrm{Od}(x) \to \neg R(x)$. In a small proof system, a derivation is a sequence of ground literals over Ev, Od, R with terms t_u, $|u| \le |w|$, each justified by a rule whose premise occurs earlier; a literal has $|u|+2$ symbols, one more if negated; canonical search tests sequences by number of lines, then lexicographically. For a word w of length n there are $N_{\mathrm{cand}} = 4(2^{n+1}-1)$ candidate lines; a shortest derivation has $L^* = n+2$ lines ($\mathrm{Ev}(\mathsf{e})$, the parity atoms of the n nonempty suffixes of w, and the literal), and none is shorter, since each rule application extends the term by one letter and needs its premise earlier; its size is $\mathrm{ds}_T = \sum_{j=0}^{n}(j+2) + (n+2) + \mathbf{1}[\#_1 w \text{ odd}]$; and the number k of sequences tested is at least $\sum_{L=1}^{L^*-1} N_{\mathrm{cand}}^L$, since every sequence with fewer lines is tested and fails. The Craig axiom has at least $(k+1)|\varphi^b|$ symbols; the certificate axiom $\theta_\gamma \to \varphi^b$, with γ coding the shortest derivation in $L^*(2+\lceil \log N_{\mathrm{cand}} \rceil)$ bits, has $|\gamma|+15+|\varphi^b|$; the Craig axiom of a fast parity labeller has $(n+3)|\varphi^b|$.

At $n = 24$ these closed forms give $N_{\mathrm{cand}} = 134217724$, $L^* = 26$, $\mathrm{ds}_T = 376$, a Craig axiom of more than $10^{204.6}$ symbols, and 795 and 702 symbols for the certificate axiom and for the fast labeller's Craig axiom (a direct computation, which also reproduces, for $|w| \le 4$, an exact count by exhaustive search; code in the accompanying repository).

Corollary F.4 (The practical horn)**.** *Under Assumption 4.17, on literal sequences, at polynomial budgets with polynomial membership and degrees charged, the fit classes are:*

hypotheses	*(b)*	*(a1)*	*(a2), (b2)*
run-padded observational sets (Craig type)	P	P	P
observational certificate sets	$\supseteq \mathsf{NP} \cap \mathsf{coNP}$	$\supseteq \mathsf{NP} \cap \mathsf{coNP}$	$\supseteq \mathsf{NP} \cap \mathsf{coNP}$
all observational hypotheses	$\mathsf{NP} \cap \mathsf{coNP}$	$\mathsf{EXPTIME}$	*all decidable*
*finite theories in $\mathcal{L}_O$, unary predicates, constants**	*regular*	*regular*	*regular*
all hypotheses	$\mathsf{NP} \cap \mathsf{coNP}$	$\mathsf{EXPTIME}$	*all decidable*

*(*compatible with the named structure.) Hence under (a1) the run-padded sets fit strictly less than the axiom sets with theoretical symbols, or than the observational hypotheses that are not run-padded, unconditionally; under (a2) and (b2) also unconditionally; and under (b) iff* $\mathsf{NP} \cap \mathsf{coNP} \neq \mathsf{P}$.

Proof. Row 1 is Theorem 8.7. Row 2: the certificate sentence $\theta_\gamma := Z \to (C_\gamma \to Z)$ of Theorem 4.18, with Z a closed tautology and C_γ a valid sentence coding γ, uses only logical symbols, so the certificate sets $\{\theta_\gamma \to \varphi_w^b\}$ are observational; they fit $\mathsf{NP} \cap \mathsf{coNP}$ under every reading by the third inequality of Theorem 4.18(b) (material $2|\gamma| + 29 + |\varphi^b|$, largest instance $|\gamma| + 15 + |\varphi^b|$,

every nonlogical instance a member). Row 3 is Proposition F.5 for (b), Proposition F.7 for (a1) and Theorem F.10 for (a2), (b2), each bounded above by row 5. Row 4 is Theorems 8.8 and F.12. Row 5 is Corollary B.22. The separations: $\mathsf{P} \subsetneq \mathsf{EXPTIME}$ by the time hierarchy theorem; some decidable set is not in P; and under (b) the classes are P and $\mathsf{NP} \cap \mathsf{coNP}$. □

F.3 Eliminating vocabulary

Proposition F.5 (Elimination by proof padding)**.** *Let $\mathcal{L}$ be finite and B decidable in polynomial time and closed under permutations of the parameters; assume Assumption 4.17. With the constants k_0, β of Theorem 4.18, put $\bar{g}(m) := (g(m)+m+1)(g(m)+m+2)/2$, $\bar{g}'(m) := \beta\bar{g}(m)\log(\bar{g}(m)+2)$ and $t^{+\#}(m) := m^{k_0} + m^{k_0}(t(m)+1)$. For every hypothesis p of $W_A^b[t,g]$ there is a hypothesis p' whose axiom set is observational, with $|p'| \le |p| + c$ and $p \mapsto p'$ injective, such that p' is compatible with every observational D with which p is, under (b) with budget $2\bar{g}'(m)+m+31$ and under (a1) with budget $\bar{g}'(m)+m+16$, with membership time $t^{+\#}$. Hence, on observational data,*

$$W_A^b[t,g](D) \le 2^c\, W_A^{\mathcal{L}_O,b}[t^{+\#}, 2\bar{g}'+m+31](D), \qquad W_A^b[t,g](D) \le 2^c\, W_A^{\mathcal{L}_O,a1}[t^{+\#}, \bar{g}'+m+16](D),$$

where $W_A^{\mathcal{L}_O,b}$ and $W_A^{\mathcal{L}_O,a1}$ are the sub-sums over hypotheses with observational axiom sets.

Proof. By the second inequality of Theorem 4.18(b) (Proposition E.21 of Hänni and Claude (Anthropic) (2026a)), p maps to the verifier $\mathcal{V}_p(\varphi,\gamma)$ that outputs acc if γ codes a derivation of φ from $B \cup A_p$ of material at most $g(|\varphi|)$, rej if it codes one of $\neg\varphi$, and $\bot$ otherwise; it runs in time $O(t^+)$, and every datum that p derives has a certificate of length at most $\bar{g}'(|\varphi|)$ (this step uses Assumption 4.17). Restrict $\mathcal{V}_p$ to observational φ, with output $\bot$ elsewhere (a linear-time syntax test). The third inequality maps a verifier $\mathcal{V}$ to $A_\mathcal{V} := \{\theta_\gamma \to \varphi : \mathcal{V}(\varphi,\gamma) = \text{acc}\} \cup \{\theta_\gamma \to \neg\varphi : \mathcal{V}(\varphi,\gamma) = \text{rej}\}$ (certificates of length at most $\bar{g}'$), a parameter-free set with membership time $t^{+\#}$. Each member is observational and logically equivalent to an observational theorem of $B \cup A_p$, so $B \cup A_\mathcal{V} \subseteq \mathrm{Cn}(A_p)$ is consistent. Each observational datum derived by p within material g has the derivation θ_γ, $\theta_\gamma \to \varphi^b$, φ^b, of material $2|\gamma| + 29 + |\varphi^b|$ and largest instance $|\gamma| + 15 + |\varphi^b|$. Lengths add constants, and the maps are injective. □

Under (a1) we compare only fit classes, through Proposition F.7 below; a hypothesis-by-hypothesis translation is not claimed. Under (a2) and (b2), at a fixed budget, an observational substitute may need a table of the data below the size of its own members, so its description length need not be $|p| + O(1)$; again only fit classes are compared (Theorem F.10).

The pairing structure. $\mathfrak{U}$ is the least set that contains the words (the elements t_w of $\mathfrak{T}$) and is closed under a formal pairing $(a,b) \mapsto \mathrm{pr}(a,b)$, where pr is injective and its values are new (not words). $\mathtt{e}, \mathtt{s}_0, \mathtt{s}_1$ act on words as in $\mathfrak{T}$, and on pairs by $\mathtt{s}_0\,\mathrm{pr}(a,b) := a$ and $\mathtt{s}_1\,\mathrm{pr}(a,b) := \mathrm{pr}(b,a)$; R is left free. In $\mathcal{L}_O$ let $\mathrm{Pair}(z) := \mathtt{s}_1\mathtt{s}_1 z = z$, $\mathrm{Std}(y) := \neg\mathrm{Pair}(y)$ and

$$\theta_2(a,b) := \forall y_\theta\,(\mathrm{Pair}(y_\theta) \to (\mathtt{s}_0 y_\theta = a \to (\mathtt{s}_0\mathtt{s}_1 y_\theta = b \to R(y_\theta)))),$$

where the bound variable y_θ occurs in no other formula below. It must be fresh: $\theta_2(a,b)$ is substituted for atoms inside formulas with their own quantified variables, and a bound variable named like one of them would capture it.

Lemma F.6 (Pairing)**.** *In $\mathfrak{U}$, for every interpretation of R: (i)* Pair *holds exactly on the pairs, so* Std *holds exactly on the words, and $\mathfrak{U} \models \mathrm{Std}(\mathsf{e})$ and $\mathfrak{U} \models \forall y\,(\mathrm{Std}(y) \to \mathrm{Std}(\mathsf{s}_a y))$ for $a = 0, 1$; (ii) for all $a, b \in \mathfrak{U}$, $\mathrm{Pair}(z) \wedge \mathsf{s}_0 z = a \wedge \mathsf{s}_0\mathsf{s}_1 z = b$ holds iff $z = \mathrm{pr}(a, b)$; (iii) for all $a, b \in \mathfrak{U}$, $\theta_2(a, b)$ holds iff $R(\mathrm{pr}(a, b))$.*

Proof. (i) On words, $\mathsf{s}_1\mathsf{s}_1 t_v = t_{11v} \neq t_v$; on pairs, $\mathsf{s}_1\mathsf{s}_1\,\mathrm{pr}(a, b) = \mathsf{s}_1\,\mathrm{pr}(b, a) = \mathrm{pr}(a, b)$; and words are closed under $\mathsf{s}_0, \mathsf{s}_1$. (ii) A pair $z = \mathrm{pr}(c, d)$ has $\mathsf{s}_0 z = c$ and $\mathsf{s}_0\mathsf{s}_1 z = \mathsf{s}_0\,\mathrm{pr}(d, c) = d$; by injectivity of pr it equals $\mathrm{pr}(a, b)$ iff $(c, d) = (a, b)$. (iii) By (ii) the premise singles out $y_\theta = \mathrm{pr}(a, b)$. □

A computation on a fragment of $\mathfrak{U}$ with 800 elements (583200 triples), and independently on a fragment of 552 elements including pairs of pairs, finds no exception.

Proposition F.7 (Ground pairing translation, reading (a1))**.** *Let $\mathcal{M}$ be an alternating machine with branching at most 2 that decides X in space $s(m) \geq m$ and halts on every path, and let $A_{\mathcal{M}}$ be the ground axioms of Proposition E.23(b) of Hänni and Claude (Anthropic) (2026a) in $\{\mathsf{e}, \mathsf{s}_0, \mathsf{s}_1, R, \mathsf{C}, \mathsf{C}'\}$, where $\mathsf{C}(t_w, t_\varsigma)$ says that the configuration ς of $\mathcal{M}$ on w accepts and $\mathsf{C}'(t_w, t_\varsigma)$ that it rejects. Let $A^O_{\mathcal{M}}$ replace every atom $\mathsf{C}(t_w, t_\varsigma)$ by $\theta_2(t_w, \mathsf{s}_0 t_\varsigma)$ and every $\mathsf{C}'(t_w, t_\varsigma)$ by $\theta_2(t_w, \mathsf{s}_1 t_\varsigma)$. Then $A^O_{\mathcal{M}}$ is observational and parameter-free with polynomial-time membership (by a decider of length $|\mathcal{M}| + c$); $A^O_{\mathcal{M}} \cup \Gamma_{f_X}$ is consistent; and every datum has a derivation from $A^O_{\mathcal{M}}$ by modus ponens alone with $2^{O(s(|w|)+1)}$ lines, each of size $O(s(|w|) + 1)$. Hence the fit class of the observational hypotheses of W^{a1}_A at polynomial budgets is* EXPTIME*, as for all hypotheses.*

Proof. Membership: parse the fixed shape of the θ_2-atoms, read off t_w, the tag letter and t_ς, and run the check of Proposition E.23(b) there. *Derivations:* the derivations there use modus ponens only, between ground atoms and ground members; modus ponens is syntactic in the major premise, so translating atom by atom gives derivations from $A^O_{\mathcal{M}}$, and each atom grows by $O(1)$ symbols. *Consistency:* in $\mathfrak{U}$ put $R(t_w) := \mathbf{1}[w \in X]$, $R(\mathrm{pr}(t_w, t_{0\varsigma})) := \mathbf{1}[\varsigma \text{ accepts}]$, $R(\mathrm{pr}(t_w, t_{1\varsigma})) := \mathbf{1}[\varsigma \text{ rejects}]$, and R false on all other pairs. By Lemma F.6(iii) each translated atom has the truth value of its original in the model of Proposition E.23(b) (the term algebra, with C, C' the least fixed points), so $A^O_{\mathcal{M}}$ and Γ_{f_X} are true in $\mathfrak{U}$. *Fit class:* $\supseteq$ EXPTIME by Corollary 6.17(b) there, with $A^O_{\mathcal{M}}$ in place of $A_{\mathcal{M}}$ and EXPTIME = APSPACE (Chandra et al., 1981); $\subseteq$ by Proposition E.23(a) there, which holds for every hypothesis satisfying (F1). □

A computation on an alternating machine for an interval game (positions are intervals of the word, the players alternately delete the first or last letter, and the existential player wins iff the last letter is 1) finds valid derivations, against an independent parsing decider, for all 2046 data with $1 \leq |w| \leq 10$, and all 14322 members with $|w| \leq 7$ recognised and true in $\mathfrak{U}$.

F.4 One binary relation: two-stack programs stated backwards

For a word $u = u_1 \cdots u_k$ and a term τ write $u\tau$ for $\mathsf{s}_{u_1}(\cdots \mathsf{s}_{u_k}(\tau) \cdots)$; so $t_w = w\mathsf{e}$, and a pair of words is a pair of closed terms.

Definition F.8 (Two-stack programs and their backward theories)**.** A *pattern* is $\pi = (r_\pi, \epsilon_r; l_\pi, \epsilon_l)$ with words r_π, l_π and $\epsilon_r, \epsilon_l \in \{\mathrm{var}, \mathrm{exact}\}$. A pair of words (r, l) *matches* π with values (Z_r, Z_l) if $r = r_\pi Z_r$ when $\epsilon_r = \mathrm{var}$, and $r = r_\pi$ (with Z_r absent) when $\epsilon_r = \mathrm{exact}$, and likewise $l = l_\pi Z_l$ or $l = l_\pi$. A *rule* $\pi \Rightarrow \pi'$ is a pair of patterns with the same ϵ's; it is *linear* (each variable occurs once on each side), and applied to $(r_\pi Z_r, l_\pi Z_l)$ it gives $(r_{\pi'} Z_r, l_{\pi'} Z_l)$. A *two-stack program* $\mathcal{P}$ is a finite set of rules; it is *deterministic* if no pair of words matches the current patterns of two rules.

Programs carry a *tag* letter $\mathrm{tg} \in \{0,1\}$ at the front of l (0 for an acceptance run, 1 for a rejection run), which every rule preserves, followed by a fixed-length *state code* $\mathrm{st}(q)$; the *halting prefixes* are $H_{\mathrm{acc}} := 0\,\mathrm{st}(\mathrm{ACC})$ and $H_{\mathrm{rej}} := 1\,\mathrm{st}(\mathrm{REJ})$, and the *start prefixes* are $\mathrm{tg}\,\mathrm{st}(\mathrm{start})$.

The *backward theory* of $\mathcal{P}$ in $\mathcal{L}_O \cup \{\mathsf{C}\}$, C a binary relation symbol, is

$$T_{\mathcal{P}}^{\mathsf{C}} := \{\forall Z_r \forall Z_l\,(\mathsf{C}(\pi') \to \mathsf{C}(\pi)) : (\pi \Rightarrow \pi') \in \mathcal{P}\} \cup \{\forall Z_r \forall Z_l\,\mathsf{C}(Z_r, H_{\mathrm{acc}} Z_l),\ \forall Z_r \forall Z_l\,\mathsf{C}(Z_r, H_{\mathrm{rej}} Z_l)\}$$
$$\cup\, \{\forall x \forall z\,(\mathsf{C}(x, 0\,\mathrm{st}(\mathrm{start})\,z) \to R(x)),\ \forall x \forall z\,(\mathsf{C}(x, 1\,\mathrm{st}(\mathrm{start})\,z) \to \neg R(x))\},$$

where $\mathsf{C}(\pi)$ is $\mathsf{C}(r_\pi Z_r, l_\pi Z_l)$ (without a variable where the pattern is exact) and the quantifiers range over the variables present. The two *output rules* carry a free certificate variable z; they are used only for programs with a certificate phase that consumes z (Lemma F.9), never for the alternating programs in the proof of Theorem F.10(iv), which use exact output rules. The *observational translation* $T_{\mathcal{P}}^{O}$ replaces each atom $\mathsf{C}(a,b)$ by $\theta_2(a,b)$, prefixes the matrix of every member by $\mathrm{Std}(v) \to \cdots$ for each of its universally quantified variables v, and adds $\mathrm{Std}(\mathsf{e})$ and $\forall y\,(\mathrm{Std}(y) \to \mathrm{Std}(\mathsf{s}_a y))$ for $a = 0, 1$; it is a finite set of $\mathcal{L}_O$-sentences.

A forward theory (a start axiom, rules $\mathsf{C}(\pi) \to \mathsf{C}(\pi')$, an output rule reading the input from a component) must keep the input word in the configuration until the end, which needs a relation of higher arity. Stated backwards, C means "this configuration leads to a halting configuration of the right kind", and the input is read only by the output rules, at the start configuration.

Lemma F.9 (Every verifier is a deterministic two-stack program)**.** *Let $\mathcal{V}$ be a verifier: a deterministic multitape machine that halts on every pair (x,z) of words with output acc, rej or $\bot$, such that no x has both an accepting and a rejecting certificate ($f_{\mathcal{V}}$ is consistent); let $X_{\mathrm{acc}} := \{x : \exists z\ \mathcal{V}(x,z) = \mathrm{acc}\}$ and X_{rej} likewise. There is a deterministic two-stack program $\mathcal{P}_{\mathcal{V}}$ such that for all words x, z and both tags: (i) every rule is linear, the current patterns are pairwise disjoint, and no rule applies to a pair whose l begins with a halting prefix or with the code of the state $\bot$; (ii) the run from the* start pair $(x, \mathrm{tg}\,\mathrm{st}(\mathrm{start})\,z)$ *reaches a pair with prefix H_{acc} (for* $\mathrm{tg} = 0$*) iff $\mathcal{V}(x,z) = \mathrm{acc}$, and one with prefix H_{rej} (for* $\mathrm{tg} = 1$*) iff $\mathcal{V}(x,z) = \mathrm{rej}$; (iii) the run has $N_{\mathrm{run}} = O(T_1 + |x| + |z|)$ steps, every pair on it has length $O(S_1 + |x| + |z|)$, and each step changes $O(1)$ letters, where T_1 and S_1 are the time and space of a one-tape simulation of $\mathcal{V}$ (so T_1 is polynomial in the time of $\mathcal{V}$ and $S_1 = O(S_{\mathcal{V}} + |x| + |z|)$).*

Proof. (1) The one-tape machine. By the one-tape simulation of multitape machines (standard; cf. Hartmanis and Stearns, 1965) there is a deterministic one-tape machine $\mathcal{M}_1$ that, started on the tape $\mathtt{Z}_{z_m} \cdots \mathtt{Z}_{z_1} \# x_1 \cdots x_{|x|}$ (the certificate reversed, in letters $\mathtt{Z}_0, \mathtt{Z}_1$, then a separator, then the input) with the head on cell 0, halts in state ACC, REJ or $\bot$ according to $\mathcal{V}(x,z)$; rearranging the input format costs $O((|x| + |z|)^2)$ steps, and the simulation is polynomial in the time of $\mathcal{V}$ and linear in its space. Bring $\mathcal{M}_1$ to a normal form: its tape alphabet $\Sigma_{\mathcal{M}}$ (without the blank $\square$) contains $\#, 0, 1, \mathtt{Z}_0, \mathtt{Z}_1$ and a symbol $\mathtt{B}$ on which $\mathcal{M}_1$ has exactly the transitions it has on $\square$, and wherever $\mathcal{M}_1$ would write $\square$ it writes $\mathtt{B}$; so every transition writes a symbol of $\Sigma_{\mathcal{M}}$. Moving left on cell 0 leaves the head there. Then on every run from a start tape the non-blank cells form an initial segment $[0, \mathrm{nb})$ and the head position hp satisfies $\mathrm{hp} \le \mathrm{nb}$: a step at $\mathrm{hp} < \mathrm{nb}$ keeps both properties, and a step at $\mathrm{hp} = \mathrm{nb}$ writes a symbol of $\Sigma_{\mathcal{M}}$ there, so nb grows by one.

(2) Coding. Fix injective codes $\mathrm{st} : Q \to \{0,1\}^{k_Q}$ (the states of $\mathcal{M}_1$ and the phase states $\mathrm{start}, \mathrm{cert}, \mathrm{sep}, \mathrm{inp}, \mathrm{rew}$) and $\mathrm{blk} : \Sigma_{\mathcal{M}} \to \{0,1\}^{k_B}$. A configuration of $\mathcal{M}_1$ with state q, tape $a_0 \cdots a_{\mathrm{nb}-1} \square\square \cdots$ and head $\mathrm{hp} \le \mathrm{nb}$ is coded, with tag tg, as

$$[\![q, a, \mathrm{hp}]\!]_{\mathrm{tg}} := \big(\mathrm{blk}(a_{\mathrm{hp}}) \cdots \mathrm{blk}(a_{\mathrm{nb}-1}),\quad \mathrm{tg}\,\mathrm{st}(q)\,\mathrm{blk}(a_{\mathrm{hp}-1}) \cdots \mathrm{blk}(a_0)\big) :$$

r holds the cells from the head on, l the tag, the state and the cells to the left of the head, nearest first; $r = \varepsilon$ means that the head reads $\square$.

(3) Rules, for each tag, with $\mathrm{cd}(q) := \mathrm{tg}\,\mathrm{st}(q)$; a pattern is written $(\rho; \lambda)$, and "$\varepsilon \,|\, \mathrm{exact}$" or "$\mathrm{cd}(q) \,|\, \mathrm{exact}$" marks an exact component:

- *start:* $(Z_r;\, \mathrm{cd}(\mathrm{start})Z_l) \Rightarrow (\mathrm{blk}(\#)Z_r;\, \mathrm{cd}(\mathrm{cert})Z_l)$, which pushes the separator onto r, above the raw input;
- *phase cert:* $(Z_r;\, \mathrm{cd}(\mathrm{cert})aZ_l) \Rightarrow (\mathrm{blk}(\mathtt{Z}_a)Z_r;\, \mathrm{cd}(\mathrm{cert})Z_l)$ for $a \in \{0,1\}$ (a raw certificate letter below the state code becomes a block on r), and $(Z_r;\, \mathrm{cd}(\mathrm{cert}) \,|\, \mathrm{exact}) \Rightarrow (Z_r;\, \mathrm{cd}(\mathrm{sep}) \,|\, \mathrm{exact})$;
- *phase sep:* $(\mathrm{blk}(\mathtt{Z}_a)Z_r;\, \mathrm{cd}(\mathrm{sep})Z_l) \Rightarrow (Z_r;\, \mathrm{cd}(\mathrm{sep})\mathrm{blk}(\mathtt{Z}_a)Z_l)$, and $(\mathrm{blk}(\#)Z_r;\, \mathrm{cd}(\mathrm{sep})Z_l) \Rightarrow (Z_r;\, \mathrm{cd}(\mathrm{inp})\mathrm{blk}(\#)Z_l)$;
- *phase inp:* $(aZ_r;\, \mathrm{cd}(\mathrm{inp})Z_l) \Rightarrow (Z_r;\, \mathrm{cd}(\mathrm{inp})\mathrm{blk}(a)Z_l)$ for $a \in \{0,1\}$, and $(\varepsilon \,|\, \mathrm{exact};\, \mathrm{cd}(\mathrm{inp})Z_l) \Rightarrow (\varepsilon \,|\, \mathrm{exact};\, \mathrm{cd}(\mathrm{rew})Z_l)$;
- *phase rew:* $(Z_r;\, \mathrm{cd}(\mathrm{rew})\mathrm{blk}(e)Z_l) \Rightarrow (\mathrm{blk}(e)Z_r;\, \mathrm{cd}(\mathrm{rew})Z_l)$ for $e \in \{\mathtt{Z}_0, \mathtt{Z}_1, \#, 0, 1\}$, and $(Z_r;\, \mathrm{cd}(\mathrm{rew}) \,|\, \mathrm{exact}) \Rightarrow (Z_r;\, \mathrm{cd}(q_0) \,|\, \mathrm{exact})$;
- *the machine:* for each transition $\delta(q,a) = (q', b, \mathrm{dir})$, with read pattern $\rho := \mathrm{blk}(a)Z_r$ if $a \in \Sigma_{\mathcal{M}}$ and $\rho := \varepsilon \,|\, \mathrm{exact}$ if $a = \square$, and ρ' the remainder (Z_r, respectively $\varepsilon \,|\, \mathrm{exact}$): for $\mathrm{dir} = R$, $(\rho;\, \mathrm{cd}(q)Z_l) \Rightarrow (\rho';\, \mathrm{cd}(q')\mathrm{blk}(b)Z_l)$; for $\mathrm{dir} = S$, $(\rho;\, \mathrm{cd}(q)Z_l) \Rightarrow (\mathrm{blk}(b)\rho';\, \mathrm{cd}(q')Z_l)$; for $\mathrm{dir} = L$, for each $e \in \Sigma_{\mathcal{M}}$, $(\rho;\, \mathrm{cd}(q)\mathrm{blk}(e)Z_l) \Rightarrow (\mathrm{blk}(e)\mathrm{blk}(b)\rho';\, \mathrm{cd}(q')Z_l)$, and at the left end $(\rho;\, \mathrm{cd}(q) \,|\, \mathrm{exact}) \Rightarrow (\mathrm{blk}(b)\rho';\, \mathrm{cd}(q') \,|\, \mathrm{exact})$.

(i) Linearity is visible in the list. Rules with different tags or state codes differ in a fixed-length prefix of l. Within one state, the read patterns $\mathrm{blk}(a)Z_r$ (fixed length, injective code) and $\varepsilon \,|\, \mathrm{exact}$ are pairwise disjoint, as are, within one left-moving transition, the neighbour patterns $\mathrm{blk}(e)Z_l$ and the exact left-end pattern; in phase cert the raw-letter patterns $\mathrm{cd}(\mathrm{cert})aZ_l$ and the exact $\mathrm{cd}(\mathrm{cert})$ are disjoint, and in phase inp the raw-letter patterns aZ_r and the exact ε. There are no rules for ACC, REJ and $\bot$, and fixed-length state codes prevent a spurious halting prefix. (ii) *Phases.* From $(x, \mathrm{cd}(\mathrm{start})z)$ the start rule gives $(\mathrm{blk}(\#)x, \mathrm{cd}(\mathrm{cert})z)$; phase cert moves the letters of z, first letter first, onto r, ending in $(\mathrm{blk}(\mathtt{Z}_{z_m}) \cdots \mathrm{blk}(\mathtt{Z}_{z_1})\mathrm{blk}(\#)x,\ \mathrm{cd}(\mathrm{sep}))$; phase sep moves these blocks to l, the separator last, ending in $(x,\ \mathrm{cd}(\mathrm{inp})\mathrm{blk}(\#)\mathrm{blk}(\mathtt{Z}_{z_1}) \cdots \mathrm{blk}(\mathtt{Z}_{z_m}))$; phase inp converts the raw letters of x; and phase rew moves all blocks back to r and ends exactly at the left end in $[\![q_0,\ \mathtt{Z}_{z_m} \cdots \mathtt{Z}_{z_1}\#x,\ 0]\!]_{\mathrm{tg}}$, the code of the start configuration of $\mathcal{M}_1$. *Steps.* For a configuration with $\mathrm{hp} \le \mathrm{nb}$ and a non-blank initial segment, exactly one rule applies to its code, namely the rule of the transition taken, with the neighbour case given by the cell $\mathrm{hp} - 1$ or the left end, and its result is the code of the successor: for $\mathrm{dir} = R$ the written b moves to the top of l; for S it replaces the top of r; for L the left neighbour e moves from l to r above b; and at $a = \square$ ($r = \varepsilon$) the written b becomes the new last cell. Reading $\mathtt{B}$ is reading the block $\mathrm{blk}(\mathtt{B})$, with the transitions of $\square$. By (1) the invariant is preserved. So the run of $\mathcal{P}_{\mathcal{V}}$ from the start pair passes through the codes of the run of $\mathcal{M}_1$ and stops exactly when $\mathcal{M}_1$ halts. The tag is preserved, so a 0-tagged run can reach H_{acc} only through ACC, and a 1-tagged run H_{rej} only through REJ. (iii) The phases take $2|z| + |x| + (|z| + |x| + 1) + O(1)$ steps, the machine T_1; every rule changes $O(k_Q + k_B)$ letters. $\square$

Theorem F.10 (One binary theoretical relation suffices under charged deduction)**.** *Let $\mathcal{V}$ be a verifier as in Lemma F.9, $\mathcal{P} := \mathcal{P}_{\mathcal{V}}$, and let $\mathsf{C}^{\mathfrak{T}} \subseteq \mathfrak{T}^2$ be the least set of pairs of words that contains all pairs with l-prefix H_{acc} or H_{rej} and is closed under the rules read backwards (if the result of a rule instance is in the set, so is its current pair). Write $T^{\mathsf{C}}_{\mathcal{V}} := T^{\mathsf{C}}_{\mathcal{P}}$ and $T^{O}_{\mathcal{V}} := T^{O}_{\mathcal{P}}$.*

(i) Relation version. $(\mathfrak{T}, \mathsf{C}^{\mathfrak{T}}, R := X_{\mathrm{acc}}) \models T_{\mathcal{V}}^{\mathsf{C}} \cup \Gamma_{f_{\mathcal{V}}}$ *and* $(\mathfrak{T}, \mathsf{C}^{\mathfrak{T}}, R := \{w : w \notin X_{\mathrm{rej}}\}) \models T_{\mathcal{V}}^{\mathsf{C}}$. *So* $T_{\mathcal{V}}^{\mathsf{C}}$ *is a finite theory in* $\mathcal{L}_O \cup \{\mathsf{C}\}$ *about the named objects, consistent with* $\Gamma_{f_{\mathcal{V}}}$, *and* $T_{\mathcal{V}}^{\mathsf{C}} \vdash R(t_w)$ *iff* $w \in X_{\mathrm{acc}}$, $T_{\mathcal{V}}^{\mathsf{C}} \vdash \neg R(t_w)$ *iff* $w \in X_{\mathrm{rej}}$.

(ii) Observational version. *In* $\mathfrak{U}$ *put* $R(t_w) := \mathbf{1}[w \in X_{\mathrm{acc}}]$, $R(\mathrm{pr}(a,b)) := \mathbf{1}[(a,b) \in \mathsf{C}^{\mathfrak{T}}]$ *for words* a, b, *and* R *false on all other pairs. Then* $\mathfrak{U} \models T_{\mathcal{V}}^{O} \cup \Gamma_{f_{\mathcal{V}}}$, *and with* $R(t_w) := \mathbf{1}[w \notin X_{\mathrm{rej}}]$ *instead,* $\mathfrak{U} \models T_{\mathcal{V}}^{O}$. *So* $T_{\mathcal{V}}^{O}$ *is a finite set of* $\mathcal{L}_O$*-sentences, consistent with* $\Gamma_{f_{\mathcal{V}}}$, *proving exactly the literals* φ_w *with* $w \in X_{\mathrm{acc}}$ *and* $\neg\varphi_w$ *with* $w \in X_{\mathrm{rej}}$.

(iii) Derivations. *If* $\mathcal{V}(w,z) \in \{\mathrm{acc}, \mathrm{rej}\}$, *the corresponding literal has a* K*-derivation from* $T_{\mathcal{V}}^{\mathsf{C}}$ *(respectively* $T_{\mathcal{V}}^{O}$*) by instances of the quantifier axiom A4 and modus ponens only, with* $O(N_{\mathrm{run}})$ *(respectively* $O(N_{\mathrm{run}} + \ell)$*) lines,* N_{run} *the number of steps of Lemma F.9(iii) and* ℓ *the largest length of a pair on the run, each line of size* $O(\ell + 1)$*; its nonlogical instances are members of the theory, each of size at most* $\mathrm{mx}_{\mathcal{V}}$ *and together at most* $\mathrm{tot}_{\mathcal{V}}$.

(iv) Fit classes. *Hence, on literal sequences, finite theories of either kind are compatible with all of* D^X *for every decidable* X *under (a2) and (b2), at every nondecreasing unbounded budget with* $h(m) \geq m+1$ *(with a finite table of the data below the budget threshold added as literals; with a budget of at least* $\mathrm{mx}_{\mathcal{V}}$*, respectively* $\mathrm{tot}_{\mathcal{V}}$*, everywhere no table is needed); and, at polynomial budgets, for every* $X \in \mathsf{NP} \cap \mathsf{coNP}$ *under (b) (material* $O(N_{\mathrm{run}}\ell)$*, polynomial for polynomial-time verifiers), the fit class of all hypotheses (Corollary B.22). Under (a1): for every* $X \in \mathsf{EXPTIME}$ *there are a finite theory in* $\mathcal{L}_O \cup \{\mathsf{C}\}$ *about the named objects and a finite set of* $\mathcal{L}_O$*-sentences, each consistent with* Γ_{f_X}, *from which every datum* $\varphi_w^{\mathbf{1}[w \in X]}$ *has a* K*-derivation whose every line, logical axiom instances included, has size* $O(s(|w|) + |w|)$ *for a polynomial* s*; so at polynomial budgets the (a1) fit class of both kinds of finite theory is* $\mathsf{EXPTIME}$*, the class of all hypotheses. Membership is linear-time (finite sets of sentences) and Assumption 4.17 holds (sentences).*

Proof. (i) The rules hold in $(\mathfrak{T}, \mathsf{C}^{\mathfrak{T}})$ by the closure of $\mathsf{C}^{\mathfrak{T}}$, the halting axioms by its base. For the output rules we show: *a start pair* $\sigma = (x, \mathrm{tg}\,\mathrm{st}(\mathrm{start})\,z)$, z *any word, lies in* $\mathsf{C}^{\mathfrak{T}}$ *iff its run reaches a pair with prefix* H_{acc} *(*$\mathrm{tg} = 0$*) or* H_{rej} *(*$\mathrm{tg} = 1$*).* Let S be the set of pairs whose deterministic run reaches a halting prefix. S contains the base and is closed under the backward rules: if the result $\pi'(Z_r, Z_l)$ of a rule instance is in S, then $\pi(Z_r, Z_l)$ matches that rule and, by disjointness (Lemma F.9(i), on all pairs of words), no other, so its run moves to $\pi'(Z_r, Z_l)$ and continues into the base; so $\mathsf{C}^{\mathfrak{T}} \subseteq S$. Conversely $S \subseteq \mathsf{C}^{\mathfrak{T}}$ by induction on the number of steps to the base. With Lemma F.9(ii): $(x, 0\,\mathrm{st}(\mathrm{start})z) \in \mathsf{C}^{\mathfrak{T}}$ iff $\mathcal{V}(x,z) = \mathrm{acc}$, which implies $x \in X_{\mathrm{acc}}$, so the first output rule holds with $R = X_{\mathrm{acc}}$; and $(x, 1\,\mathrm{st}(\mathrm{start})z) \in \mathsf{C}^{\mathfrak{T}}$ iff $\mathcal{V}(x,z) = \mathrm{rej}$, which implies $x \in X_{\mathrm{rej}}$, hence $x \notin X_{\mathrm{acc}}$ by the consistency of $f_{\mathcal{V}}$, so the second holds. $\Gamma_{f_{\mathcal{V}}}$ holds since $X_{\mathrm{acc}} \cap X_{\mathrm{rej}} = \emptyset$. The second model is checked in the same way. The two models give that $T_{\mathcal{V}}^{\mathsf{C}} \vdash R(t_w)$ only if $w \in X_{\mathrm{acc}}$ and $T_{\mathcal{V}}^{\mathsf{C}} \vdash \neg R(t_w)$ only if $w \in X_{\mathrm{rej}}$; the converses are (iii).

(ii) For values of its variables in $\mathrm{Std}^{\mathfrak{U}} = \mathfrak{T}$ (Lemma F.6(i)), each member of $T_{\mathcal{V}}^{O}$ says what the corresponding member of $T_{\mathcal{V}}^{\mathsf{C}}$ says with $\mathsf{C}(a,b)$ read as $R(\mathrm{pr}(a,b))$ (Lemma F.6(iii)); terms built from words by $\mathtt{s}_0, \mathtt{s}_1$ stay words and are evaluated as in $\mathfrak{T}$. For values outside $\mathfrak{T}$ a Std premise fails. The Std axioms hold by Lemma F.6(i).

(iii) Let $\sigma_0, \ldots, \sigma_{N_{\mathrm{run}}}$ be the run from $\sigma_0 = (t_w, \mathrm{tg}\,\mathrm{st}(\mathrm{start})\,t_z)$. Instantiate the halting axiom at $\sigma_{N_{\mathrm{run}}}$ (two A4 instances and two modus ponens steps) to get $\mathsf{C}(\sigma_{N_{\mathrm{run}}})$; for $i = N_{\mathrm{run}} - 1, \ldots, 0$, instantiate the rule that applies to σ_i at its values (one A4 instance and one modus ponens step per variable) and apply modus ponens to $\mathsf{C}(\sigma_{i+1})$, getting $\mathsf{C}(\sigma_i)$; finally instantiate the output rule at $x := t_w$, $z := t_z$ and apply modus ponens. In $T_{\mathcal{V}}^{O}$ the instantiated members carry Std premises

for the values, all words: derive $\mathrm{Std}(t_v)$ for every suffix v of every component from $\mathrm{Std}(\mathsf{e})$ and the propagation axioms (one A4 instance and two modus ponens steps per letter, keeping the set of words with a derived Std fact closed under suffixes, each step adding $O(1)$ letters), and discharge them by modus ponens. Every line is a member, an instance of a member at the values of one pair on the run, or an atom about such a pair, so it has size $O(\ell+1)$.

(iv) For a decidable X take $\mathcal{V}(x,z) :=$ the decision of X on x (ignoring z). Under (a2) and (b2) every nonlogical instance is a member of the finite theory, so the budget needed is a constant; data below it are listed as table literals, of size $|\varphi_w|$ or $|\varphi_w|+1$, which keeps the theory finite and true in the models above. Under (b) the material is at most the number of distinct instances, $O(N_{\mathrm{run}})$, times their size, $O(\ell)$; for $X \in \mathsf{NP} \cap \mathsf{coNP}$ combine polynomial-time verifiers for X and its complement into one consistent verifier, and the upper bound is that of all hypotheses (Corollary B.22).

Reading (a1). Step 1 (the machine). An alternating one-tape machine has states partitioned into existential and universal states and the halting states ACC and REJ, and a transition relation giving, for each non-halting state q and symbol $a \in \Sigma_{\mathcal{M}} \cup \{\Box\}$, a list $\delta(q,a)$ of one or two moves $(q_i, b_i, \mathrm{dir}_i)$. A configuration *accepts* if its state is ACC, or it is existential and some successor accepts, or it is universal and all successors accept (least fixed point); when every computation path halts, the computation tree from a configuration is finite (König's lemma), acceptance is decided by induction on it, and a configuration that does not accept *rejects*. Normal form: (1) as in Lemma F.9(1) ($\mathtt{B}$ has the transitions of $\Box$, only symbols of $\Sigma_{\mathcal{M}}$ are written, a left move at cell 0 stays), so $\mathrm{hp} \le \mathrm{nb}$ on every path from a start tape; (2) $\delta(q,a)$ is nonempty for every non-halting q and every a (add a move to REJ at an existential state, and to ACC at a universal state, where none exists; this does not change acceptance); (3) every computation path from a start configuration (tape $\#x$, head on cell 0, state q_0) halts. Such a machine exists for every $X \in \mathsf{EXPTIME}$. By $\mathsf{EXPTIME} = \mathsf{APSPACE}$ (Chandra et al., 1981), X is decided by a multitape alternating machine $\mathcal{M}$ in polynomial space $S(m)$. *Clock.* $\mathcal{M}$ has at most $N_{\mathrm{cf}}(m) = 2^{c_{\mathcal{M}}(S(m)+\log m)}$ configurations on inputs of length m. Let A_j be the set of configurations with an accepting computation tree of height at most j; $A_j \subseteq A_{j+1}$, the accepting configurations are $\bigcup_j A_j$, and the sequence stabilises after at most $N_{\mathrm{cf}}(m)$ strict increases, so every accepting configuration has an accepting tree of height at most $N_{\mathrm{cf}}(m)$. Add a binary step counter of $c_{\mathcal{M}}(S(m)+\log m)+1$ bits on an extra tape, and go to REJ when it overflows. If $\mathcal{M}$ accepts x, the clocked machine follows an accepting tree of height at most $N_{\mathrm{cf}}(m)$ and accepts; if the clocked machine accepts, its accepting tree contains no overflow (overflow leads to REJ, which an accepting tree does not contain) and is an accepting tree of $\mathcal{M}$. Every path of the clocked machine halts. *One tape.* Simulate the tapes on tracks of one tape, each step of the clocked machine by a deterministic sequence of steps (existential states with one successor) that ends with the branching step; the computation trees correspond, so acceptance is preserved and all paths halt, in space $O(S+\log m)$. The normal form (1)–(2) is a local change. Call the result $\mathcal{M}_1$, and split moves so that $|\delta(q,a)| \le 2$.

Step 2 (the compile). Fix codes st (states of $\mathcal{M}_1$ and the phase states $\mathrm{start}, \mathrm{inp}, \mathrm{rew}$) and blk as in Lemma F.9(2). For each tag, with $\mathrm{cd}(q) := \mathrm{tg}\,\mathrm{st}(q)$:

- *start:* $(Z_r;\ \mathrm{cd}(\mathrm{start})Z_l) \Rightarrow (Z_r;\ \mathrm{cd}(\mathrm{inp})\,\mathrm{blk}(\#)Z_l)$: the separator goes onto l, and there is no certificate phase;
- *phase inp:* $(aZ_r;\ \mathrm{cd}(\mathrm{inp})Z_l) \Rightarrow (Z_r;\ \mathrm{cd}(\mathrm{inp})\mathrm{blk}(a)Z_l)$ for $a \in \{0,1\}$, and $(\varepsilon \mid \mathrm{exact};\ \mathrm{cd}(\mathrm{inp})Z_l) \Rightarrow (\varepsilon \mid \mathrm{exact};\ \mathrm{cd}(\mathrm{rew})Z_l)$;
- *phase rew:* $(Z_r;\ \mathrm{cd}(\mathrm{rew})\mathrm{blk}(e)Z_l) \Rightarrow (\mathrm{blk}(e)Z_r;\ \mathrm{cd}(\mathrm{rew})Z_l)$ for $e \in \{\#,0,1\}$, and

$(Z_r;\,\mathrm{cd}(\mathrm{rew})\,|\,\mathrm{exact}) \Rightarrow (Z_r;\,\mathrm{cd}(q_0)\,|\,\mathrm{exact})$;

- *the machine:* for each non-halting q and $a \in \Sigma_{\mathcal{M}} \cup \{\Box\}$, with read pattern ρ and remainder ρ' as in Lemma F.9(3), the *neighbour cases* are $\nu \in \{\mathrm{blk}(e)Z_l : e \in \Sigma_{\mathcal{M}}\} \cup \{|\,\mathrm{exact}\}$ if some move in $\delta(q,a)$ goes left, and the single case $\nu = Z_l$ otherwise. For each case the current pattern is $\pi := (\rho;\,\mathrm{cd}(q)\nu)$, and each move $(q_i, b_i, \mathrm{dir}_i)$ gives a next pattern π_i': $(\rho';\,\mathrm{cd}(q_i)\mathrm{blk}(b_i)\nu)$ for R; $(\mathrm{blk}(b_i)\rho';\,\mathrm{cd}(q_i)\nu)$ for S; $(\mathrm{blk}(e)\mathrm{blk}(b_i)\rho';\,\mathrm{cd}(q_i)Z_l)$ for L when $\nu = \mathrm{blk}(e)Z_l$; and $(\mathrm{blk}(b_i)\rho';\,\mathrm{cd}(q_i)\,|\,\mathrm{exact})$ for L at the left end. The rule is *conjunctive*, $\pi \Rightarrow \{\pi_1', \pi_2'\}$ with backward form $\forall Z_r \forall Z_l\,(\mathsf{C}(\pi_1') \to (\mathsf{C}(\pi_2') \to \mathsf{C}(\pi)))$, when all successors are needed: at universal states for tag 0 and at existential states for tag 1. Otherwise there is one rule $\pi \Rightarrow \pi_i'$ per move, with backward form $\forall Z_r \forall Z_l\,(\mathsf{C}(\pi_i') \to \mathsf{C}(\pi))$; a single move gives a single rule;
- *halting axioms:* $\forall Z_r \forall Z_l\,\mathsf{C}(Z_r, H_{\mathrm{acc}} Z_l)$ and $\forall Z_r \forall Z_l\,\mathsf{C}(Z_r, H_{\mathrm{rej}} Z_l)$;
- *exact output rules:* $\forall x\,(\mathsf{C}(x,\, 0\,\mathrm{st}(\mathrm{start})) \to R(x))$ and $\forall x\,(\mathsf{C}(x,\, 1\,\mathrm{st}(\mathrm{start})) \to \neg R(x))$, where $\mathrm{tg\,st}(\mathrm{start})$ stands for the closed term $t_{\mathrm{tg\,st}(\mathrm{start})}$. In the observational version they are $\forall x\,(\mathrm{Std}(x) \to (\theta_2(x, t_{0\,\mathrm{st}(\mathrm{start})}) \to R(x)))$ and its rejecting twin.

Call the result $\mathcal{A}$, $T_{\mathcal{A}}^{\mathsf{C}}$ its backward theory (rules, halting axioms, exact output rules) and $T_{\mathcal{A}}^{O}$ its observational translation (Definition F.8). The current patterns of distinct cases (state, read symbol, neighbour case, tag) are pairwise disjoint, by the argument of Lemma F.9(i); the two rules of a non-conjunctive branching share one current pattern.

Step 3 (the meaning of C *at reachable pairs).* Let $\mathsf{C}^{\mathfrak{T}}$ be the least set of pairs of words that contains the pairs with a halting prefix and is closed under the backward rules (a conjunctive rule adds π when all its next pairs are in, the others when their one next pair is in). Let Reach be the set of pairs reachable by the rules from the *exact* start pairs $(x,\, \mathrm{tg\,st}(\mathrm{start}))$, x a word. Every pair in Reach is a phase pair of the run from a start pair, or the code $[\![q, a, \mathrm{hp}]\!]_{\mathrm{tg}}$ of a configuration of $\mathcal{M}_1$ reachable from $\#x$: the start rule gives $(x,\, \mathrm{cd}(\mathrm{inp})\mathrm{blk}(\#))$, phase inp gives $(\varepsilon,\, \mathrm{cd}(\mathrm{rew})\mathrm{blk}(x_{|x|}) \cdots \mathrm{blk}(x_1)\mathrm{blk}(\#))$, phase rew gives $(\mathrm{blk}(\#)\mathrm{blk}(x_1) \cdots \mathrm{blk}(x_{|x|}),\, \mathrm{cd}(q_0)) = [\![q_0, \#x, 0]\!]_{\mathrm{tg}}$, and from a configuration code the patterns that match are exactly those of its case, whose next pairs are the codes of its successors (the step argument of Lemma F.9). For $\sigma \in$ Reach let $\mathrm{acc}(\sigma)$ mean that the configuration of σ accepts (for a phase pair: the start configuration it leads to accepts). *Claim: for* $\sigma \in$ Reach *with tag* 0, $\sigma \in \mathsf{C}^{\mathfrak{T}}$ *iff* $\mathrm{acc}(\sigma)$*; with tag* 1, $\sigma \in \mathsf{C}^{\mathfrak{T}}$ *iff not* $\mathrm{acc}(\sigma)$. ($\Leftarrow$, tag 0.) By induction on the height of the finite computation tree of σ, phase steps counting as deterministic steps: a halting accepting configuration has prefix H_{acc}, so it is in the base; at an existential configuration some successor accepts and the rule for that move puts σ in; at a universal one all successors accept and the conjunctive rule puts σ in; a deterministic or phase step uses its single rule. ($\Rightarrow$, tag 0.) The set $S_0 := \{\sigma \in \text{Reach} : \text{tag } 0,\ \mathrm{acc}(\sigma)\} \cup \{0\text{-tagged pairs outside Reach}\} \cup \{\text{all 1-tagged pairs}\}$ contains the base (a reachable pair with prefix H_{acc} is a configuration in state ACC) and is closed under the backward rules. Take a rule instance with current pair σ whose next pairs (all of them for a conjunctive rule, its one next pair otherwise) lie in S_0. If σ is 1-tagged or outside Reach, $\sigma \in S_0$. Otherwise $\sigma \in$ Reach is 0-tagged, the rule belongs to the case of σ, and its next pairs are codes of successors of σ, hence in Reach and 0-tagged, hence accepting. A conjunctive rule at tag 0 sits at a universal or deterministic configuration, all of whose successors are then accepting; a non-conjunctive one sits at an existential configuration with an accepting successor. Either way $\mathrm{acc}(\sigma)$, so $\mathsf{C}^{\mathfrak{T}} \subseteq S_0$. (Tag 1.) The same, with acceptance and rejection, existential and universal exchanged: at tag 1 the conjunctive rules sit at existential states. So for the exact start pairs, $(x, 0\,\mathrm{st}(\mathrm{start})) \in \mathsf{C}^{\mathfrak{T}}$ iff $\mathcal{M}_1$ accepts $\#x$ iff $x \in X$, and $(x, 1\,\mathrm{st}(\mathrm{start})) \in \mathsf{C}^{\mathfrak{T}}$ iff $x \notin X$.

Step 4 (theories, truth, derivations). By Step 3, $(\mathfrak{T}, \mathsf{C}^{\mathfrak{T}}, R := X) \models T^{\mathsf{C}}_{\mathcal{A}} \cup \Gamma_{f_X}$: the rules by closure, the halting axioms by the base, the exact output rules by the Claim. So $T^{\mathsf{C}}_{\mathcal{A}}$ is about the named objects, and it proves φ^b_w only if $b = \mathbf{1}[w \in X]$. In $\mathfrak{U}$ put $R(t_w) := \mathbf{1}[w \in X]$, $R(\mathrm{pr}(a,b)) := \mathbf{1}[(a,b) \in \mathsf{C}^{\mathfrak{T}}]$ for words a, b, and R false on other pairs; as in (ii), $\mathfrak{U} \models T^{Q}_{\mathcal{A}} \cup \Gamma_{f_X}$ (the ground term $t_{\mathrm{tg\,st(start)}}$ of the exact output rule is a word, so $\theta_2(x, t_{\mathrm{tg\,st(start)}})$ holds iff $(x, \mathrm{tg\,st(start)}) \in \mathsf{C}^{\mathfrak{T}}$). For $w \in X$ fix an accepting computation tree of $\mathcal{M}_1$ on $\#w$ (all successors at universal nodes, one accepting successor at existential nodes) and the phase pairs before it; derive $\mathsf{C}(\sigma)$ for its nodes bottom-up, a leaf by the halting axiom (two A4 instances, two modus ponens steps), an inner node by the instance of its rule at its values and one modus ponens step per premise; then apply the exact output rule at $x := t_w$ and modus ponens. Rejection is the same with a rejecting tree and tag 1. In $T^{Q}_{\mathcal{A}}$ the Std premises are discharged as in (iii). The number of lines may be exponential, but every line is a member, an instance of a member at the values of one reachable pair, or an atom about one, so every line, logical instances included, has size $O(S_1 + |w|)$, a polynomial. That is reading (a1) at a polynomial budget; the upper bound $\subseteq$ EXPTIME is Corollary B.22. □

Why the output rules must be exact for alternating programs. Without a certificate phase nothing consumes the variable z of the output rules of Definition F.8. Phase rew rewinds every block-aligned z onto the tape, so $\mathcal{M}_1$ runs on $z'\#x$, a tape about which the normal form says nothing. If $\mathcal{M}_1$ has, at q_0, transitions on the letters 0 and 1 (legal, since cell 0 of a start tape always holds #), the theory with free-z output rules proves both $R(t_x)$ and $\neg R(t_x)$ for every x. A computation confirms this on three machines with such transitions (checked derivations of $R(t_{0110})$ and $\neg R(t_{0110})$), and confirms the exact rules on the same machines. In Lemma F.9 phase cert consumes z, so the free-z output rules are correct there.

Relativisation to Std **is needed.** In $\mathfrak{U}$ every element is in the range of $\mathtt{s}_0$ and $\mathtt{s}_0$ is not injective, so unrelativised patterns uZ_l match pairs, and $u\,\mathrm{pr}(a,b)$ can be a word with the wrong prefix. A computation on the backward theory of a decider for the words with as many 0s as 1s, with $\mathsf{C}(a,b)$ read as $R(\mathrm{pr}(a,b))$ in $\mathfrak{U}$, finds all 3017 instances at word values true and 318 of 12519 instances at pair values false. So without Std the structure $\mathfrak{U}$ is not a model; whether the unrelativised theory is consistent we have not examined.

Computations. For that decider (140 rules) and a verifier whose accepting certificate is a position (274 rules), the compilation has no non-linear rule, no overlapping current patterns (checked symbolically, on all pairs of words) and no rule in a halting state, and the runs from all start pairs checked ($|x| \le 10$, $|z| \le 3$, respectively $|x| \le 6$, $|z| \le 5$) end as specified. All 511 data with $|w| \le 8$ have derivations from the relation version (144 sentences) and the observational version (147 sentences), checked by an independent derivation checker; corrupted derivations and a corrupted machine are rejected. For an alternating machine for the words in which every 1 has a 0 to its right, and three further machines with unreachable transitions out of q_0 (one universal and one existential with left moves, one deterministic that writes and reads B), all data with $|x| \le 6$ have checked derivations in both versions, and the largest line grows linearly in $|x|$, as reading (a1) requires. An independent implementation agrees.

F.5 Vocabulary or ontology

Residuals and the relational form. For $X \subseteq \{0,1\}^*$ and a word v, $X_v := \{u : uv \in X\}$. In the term model $R(t_{uv}) = R(u\,t_v)$, so X_v is what the subtree below t_v sees. X is regular iff it

has finitely many residuals: the X_v are the right quotients of X, that is, the reversals of the left quotients of the reversal of X, and the Myhill–Nerode theorem applies to the reversal (standard). Let $\mathcal{L}^{\mathrm{rel}}$ have a unary Root, binary $\mathrm{Gr}_0, \mathrm{Gr}_1$, unary $R, P_1, \ldots, P_k$ and unary $\mathrm{Cst}_1, \ldots, \mathrm{Cst}_j$. A structure $\mathfrak{A}$ for $\mathcal{L}_O \cup \{P_i\} \cup \{\mathtt{c}_i\}$ gives $\mathfrak{A}^{\mathrm{rel}}$ with Root $= \{\mathtt{e}\}$, Gr_a the graph of $\mathtt{s}_a$ ($\mathrm{Gr}_a(x, y)$ iff $\mathtt{s}_a x = y$) and $\mathrm{Cst}_i = \{\mathtt{c}_i\}$; every sentence ψ has a relational translation ψ^{rel}, obtained by unnesting terms, with $\mathfrak{A} \models \psi$ iff $\mathfrak{A}^{\mathrm{rel}} \models \psi^{\mathrm{rel}}$ (standard). $\equiv_q$ is agreement on all sentences of quantifier rank at most q. Over a finite relational signature there are finitely many $\equiv_q$-classes, and $\equiv_q$ holds iff the duplicator wins the q-round Ehrenfeucht–Fraïssé game (Ehrenfeucht, 1961; Fraïssé, 1954); see Ebbinghaus and Flum (1999) for the textbook form. In Theorem 8.8, ncl_q is the number of $\equiv_q$-classes of $\mathcal{L}^{\mathrm{rel}} \cup \{\mathrm{Rt}, \mathrm{Att}\}$-structures, Rt and Att unary.

Lemma F.11 (Composition)**.** *(a) For relational structures in one finite signature, $\mathfrak{A}_1 \equiv_q \mathfrak{B}_1$ and $\mathfrak{A}_2 \equiv_q \mathfrak{B}_2$ imply $\mathfrak{A}_1 \sqcup \mathfrak{A}_2 \equiv_q \mathfrak{B}_1 \sqcup \mathfrak{B}_2$, where $\sqcup$ is the disjoint union with each relation interpreted as the union. (b) If $\mathfrak{A}, \mathfrak{B}$ arise from $\mathfrak{A}', \mathfrak{B}'$ (on the same domains) by defining each relation by a quantifier-free formula, then $\mathfrak{A}' \equiv_q \mathfrak{B}'$ implies $\mathfrak{A} \equiv_q \mathfrak{B}$.*

Proof. (a) The duplicator answers a move in a component with her winning strategy for that component's game. Within a component the chosen tuples form a partial isomorphism; no relation holds across components on either side, and equality across components fails on both sides. (b) Substituting the defining formulas turns a rank-q sentence about $\mathfrak{A}$ into a rank-q sentence about $\mathfrak{A}'$. □

Proof of Theorem 8.8. Take $\mathfrak{A} \models T \cup \mathrm{Th}_{\mathfrak{T}}$. The closed terms denote pairwise distinct elements, and $R^{\mathfrak{A}}(t_w) = \mathbf{1}[w \in X]$ because T proves the literals. For a word $v \neq \varepsilon$, $v = a\hat{v}$ with a a letter, let $\mathrm{Sub}_v := \{t_{uv} : u \in \{0,1\}^*\}$ (standard elements), let $\mathfrak{S}_v$ be the restriction of $\mathfrak{A}^{\mathrm{rel}}$ to Sub_v with a new unary $\mathrm{Rt} := \{t_v\}$ (and $\mathrm{Att} := \emptyset$), and let $\mathfrak{C}_v$ be the restriction to the rest of the domain with a new unary $\mathrm{Att} := \{t_{\hat{v}}\}$ (and $\mathrm{Rt} := \emptyset$).

One edge between the parts. Sub_v is closed under $\mathtt{s}_0, \mathtt{s}_1$, and $\mathtt{e} \notin \mathrm{Sub}_v$, as $\mathtt{e}$ is not in the range of $\mathtt{s}_0, \mathtt{s}_1$. If $\mathtt{s}_b y = t_{uv}$ with $y \notin \mathrm{Sub}_v$, then $u = \varepsilon$: otherwise $u = b'u'$ and $t_{uv} = \mathtt{s}_{b'} t_{u'v}$, so $b = b'$ by disjoint ranges and $y = t_{u'v} \in \mathrm{Sub}_v$ by injectivity, both in $\mathrm{Th}_{\mathfrak{T}}$. So $\mathtt{s}_b y = t_v = \mathtt{s}_a t_{\hat{v}}$, hence $b = a$ and $y = t_{\hat{v}}$. Hence $\mathfrak{A}^{\mathrm{rel}}$ is obtained from $\mathfrak{C}_v \sqcup \mathfrak{S}_v$ by redefining $\mathrm{Gr}_a(x, y)$ as $\mathrm{Gr}_a(x, y) \vee (\mathrm{Att}(x) \wedge \mathrm{Rt}(y))$ and forgetting Att and Rt, a quantifier-free redefinition.

The subtrees are alike up to labels. For $v, v' \notin \mathrm{Suf}_{\mathrm{c}}$ the map $t_{uv} \mapsto t_{uv'}$ is an isomorphism of the $\{\mathrm{Root}, \mathrm{Gr}_0, \mathrm{Gr}_1, \mathrm{Rt}, \mathrm{Cst}_1, \ldots, \mathrm{Cst}_j\}$-reducts of $\mathfrak{S}_v$ and $\mathfrak{S}_{v'}$: Root and the Cst_i are empty on both, because a constant lies in Sub_v only if it is some t_{w_i} with v a suffix of w_i, and nonstandard denotations lie in no Sub_v.

Transplanting. Suppose X has more than $\mathrm{ncl}_q + |\mathrm{Suf}_{\mathrm{c}}|$ residuals. Choose one word per residual; more than ncl_q of them lie outside $\mathrm{Suf}_{\mathrm{c}}$, so two of them, $v \neq v'$, have $X_v \neq X_{v'}$ and $\mathfrak{S}_v \equiv_q \mathfrak{S}_{v'}$. Let $\mathfrak{A}^*$ be $\mathfrak{A}$ with the labels on Sub_v replaced by those copied from $\mathrm{Sub}_{v'}$: $R^*(t_{uv}) := R^{\mathfrak{A}}(t_{uv'})$ and $P_i^*(t_{uv}) := P_i^{\mathfrak{A}}(t_{uv'})$, everything else unchanged (in particular the $\{\mathtt{e}, \mathtt{s}_0, \mathtt{s}_1\}$-reduct and the constants). Then $\mathfrak{S}_v^* \cong \mathfrak{S}_{v'} \equiv_q \mathfrak{S}_v$ and $\mathfrak{C}_v^* = \mathfrak{C}_v$, so by Lemma F.11, $(\mathfrak{A}^*)^{\mathrm{rel}} \equiv_q \mathfrak{A}^{\mathrm{rel}}$ and $\mathfrak{A}^* \models T$. Its reduct still satisfies $\mathrm{Th}_{\mathfrak{T}}$, so, as T proves the literals, $R^*(t_{uv}) = \mathbf{1}[uv \in X]$ for all u. But $R^*(t_{uv}) = \mathbf{1}[uv' \in X]$. Hence $X_v = X_{v'}$, a contradiction. Overlapping subtrees, $\mathrm{Sub}_v \supset \mathrm{Sub}_{v'}$, are harmless, since the labels are copied from the original structure. □

Unary predicates and constants are labels: they do not connect a subtree to its context, so the composition lemma applies. A binary theoretical relation can connect any two subtrees (Theorem F.10(i)); objects that violate $\mathrm{Th}_{\mathfrak{T}}$ (pairs, with $\mathtt{s}_1 \mathtt{s}_1 z = z$) can carry information between subtrees through $\mathtt{s}_0$, which is not injective on them (Theorem F.10(ii)); and a finite set is essential,

since infinite sets of padded literals are true of the named objects and fit P (Theorem 8.7). The proof uses no tree automata and allows nonstandard elements and constants. Theorem 8.8 is in the spirit of the reported example of Craig and Vaught that unary additional predicates do not suffice in general, in a different setting; we have not established that it is new.

Theorem F.12 (Every regular regularity has a short observational theory)**.** *For every regular X there is a finite $T \subseteq \mathrm{Sent}_{\mathcal{L}_O}$, true in $(\mathfrak{T}, X)$, such that every datum has a K-derivation from T with $O(|w|)$ lines, each of size $O(|w|)$, the constants depending on X. So the budgets needed are linear under (a1); constant under (a2) and (b2) (at least the largest member, respectively the total size of T; with smaller budgets, a finite table of short data as in Theorem F.10(iv)); and quadratic under (b).*

Proof. Let k_X be the number of residuals of X, and write $A/a := \{u : ua \in A\}$, so $X_v/a = X_{av}$. Let $\approx_j$ be the equivalence "agree on all words of length less than j" on the finite set $\{X_v\}$. Then $A \approx_{j+1} A'$ iff $(\varepsilon \in A \Leftrightarrow \varepsilon \in A')$ and $A/a \approx_j A'/a$ for $a = 0, 1$, so $\approx_{j+1}$ is a fixed function of $\approx_j$; once $\approx_j = \approx_{j+1}$ the sequence is constant; $\approx_0$ has one class and each strict refinement adds a class; so $\approx_{k_X - 1}$ is equality. Hence the window $\{0,1\}^{<\mathrm{wd}_X}$ with $\mathrm{wd}_X := \max(k_X - 1, 1)$, or any smaller window that contains ε and separates the residuals, determines X_v. Put $\mathrm{sig}(v) := X_v \cap \{0,1\}^{<\mathrm{wd}_X}$; then $\mathrm{sig}(av)$ is a function $\sigma \mapsto \sigma_a$ of $\mathrm{sig}(v)$. With $\mathrm{Sg}_\sigma(x) := \bigwedge_{|u|<\mathrm{wd}_X} R(ux)^{\sigma(u)}$ let

$$T := \{\mathrm{Sg}_{\mathrm{sig}(\varepsilon)}(\mathsf{e})\} \cup \{\forall x\, (\mathrm{Sg}_\sigma(x) \to \mathrm{Sg}_{\sigma_a}(\mathtt{s}_a x)) : \sigma \text{ realised},\ a \in \{0,1\}\}.$$

$\mathrm{Sg}_\sigma(t_v)$ holds in $(\mathfrak{T}, X)$ iff $\mathrm{sig}(v) = \sigma$, so T is true there. Derive $\mathrm{Sg}_{\mathrm{sig}(v)}(t_v)$ for the suffixes v of w from the inside out (one A4 instance and modus ponens steps per letter) and read off the conjunct $u = \varepsilon$, which is $R(t_w)^{\mathbf{1}[w \in X]}$. In a curried form with one literal per conclusion only A4 and modus ponens occur. □

The bound $\max(k_X - 1, 1)$ on the window is attained: on random automata the least separating window equals $k_X - 1$ in many cases (90 of 1500 random automata with $k_X \geq 3$, and never above $\max(k_X - 1, 1)$; a computation, which also checks the theories of five regular languages on all data with $|w| \leq 9$).

Corollary F.13 (Vocabulary or ontology)**.** *On literal sequences with $B = \emptyset$ (membership is linear-time for finite sets), the fit classes of finite theories are:*

finite theories	*(a2), (b2)*†	*(b): polynomial*	*(a1): polynomial*
$\mathcal{L}_O$ + unary predicates + constants, compatible with the named structure	*regular*	*regular*	*regular*
$\mathcal{L}_O$ + one binary relation, about the named objects	*all decidable*	$\mathsf{NP} \cap \mathsf{coNP}$	EXPTIME
$\mathcal{L}_O$ only, quantifying over unnamed objects	*all decidable*	$\mathsf{NP} \cap \mathsf{coNP}$	EXPTIME
all finite theories, any vocabulary	*all decidable*	$\mathsf{NP} \cap \mathsf{coNP}$	EXPTIME

(†at every nondecreasing unbounded budget with $h(m) \geq m + 1$.) The theories of row 3 are inconsistent with $\mathrm{Th}_{\mathfrak{T}}$ *(Theorem 8.8).*

Proof. Row 1: Theorem 8.8 gives the upper bound (at any budget, and also with free deduction) and Theorem F.12 the lower bound. Rows 2 and 3: Theorem F.10. Row 4: the upper bounds are those of all hypotheses (Corollary B.22), the lower bounds rows 2 and 3. □

Corollary F.14 (With free deduction too)**.** *For a decidable non-regular X, let $K^{\mathrm{named}}_{\mathrm{fin}}(D)$ be the least $K(\psi)$ over single sentences ψ in $\mathcal{L}_O$ plus any finitely many unary predicates and constants that are compatible with the named structure and prove ψ_D. Then $K^{\mathrm{named}}_{\mathrm{fin}}(D^X_n) \to \infty$, while $K^{\mathcal{L}_O\cup\{\mathsf{C}\}}_{\mathrm{fin}}(D^X_n) \le K(\mathcal{V}) + c$ and $K^{\mathcal{L}_O}_{\mathrm{fin}}(D^X_n) \le K(\mathcal{V}) + c$ for a decider $\mathcal{V}$ of X.*

Proof. By Theorem 8.8 no admissible sentence proves all of D^X; each fails some datum, and only finitely many sentences have $K \le j$. The upper bounds are given by the conjunctions of $T^{\mathsf{C}}_{\mathcal{V}}$ and $T^{O}_{\mathcal{V}}$ (Theorem F.10), computable from $\mathcal{V}$. □

The upper bound with C is an effective instance of the theorem of Kleene and of Craig and Vaught (Kleene, 1952; Craig and Vaught, 1958), as stated in Pakhomov and Visser (2019): the axioms of $\mathrm{Th}_{\mathfrak{T}}$ listed after Definition 8.3 together with Γ_{f_X} form an r.e. theory without finite models, which therefore has a finitely axiomatised extension with one additional binary relation symbol into which every model of it, $(\mathfrak{T}, X)$ included, expands.

Proposition F.15 (A forced ontological commitment)**.** *(a) A finite theory in $\mathcal{L}_O$ plus unary predicates and constants that proves every literal datum of a non-regular X proves some $\{\mathtt{e}, \mathtt{s}_0, \mathtt{s}_1\}$-sentence false in the named structure. (b) For a verifier $\mathcal{V}$, $T^{O}_{\mathcal{V}} \vdash \exists z\, \mathtt{s}_1\mathtt{s}_1 z = z$, and $T^{O}_{\mathcal{V}} \vdash \neg\, \mathtt{s}_1\mathtt{s}_1 t_w = t_w$ for every word w. So $T^{O}_{\mathcal{V}} \cup \mathrm{Th}_{\mathfrak{T}}$ is inconsistent: $T^{O}_{\mathcal{V}}$ is consistent with all the data but proves an observational sentence that is false of every named object and refutable from the observational theory of the named structure. In particular $T^{O}_{\mathcal{V}}$ is not conservative over $\Gamma_{f_{\mathcal{V}}}$, although $\Gamma_{f_{\mathcal{V}}} \cup \mathrm{Th}_{\mathfrak{T}}$ is consistent.*

Proof. (a) By Theorem 8.8 the theory is inconsistent with $\mathrm{Th}_{\mathfrak{T}}$, so by compactness it proves the negation of some $\{\mathtt{e}, \mathtt{s}_0, \mathtt{s}_1\}$-sentence true in $\mathfrak{T}$. (b) In a model of $T^{O}_{\mathcal{V}} \cup \{\forall z \neg\, \mathtt{s}_1\mathtt{s}_1 z = z\}$, Std holds everywhere and every $\theta_2(a, b)$ holds vacuously, so the two output rules give $\forall x\, R(x)$ and $\forall x \neg R(x)$; no such model exists. $\mathrm{Std}(t_w)$ follows from $\mathrm{Std}(\mathtt{e})$ and the propagation axioms, and $\forall z \neg\, \mathtt{s}_1\mathtt{s}_1 z = z \in \mathrm{Th}_{\mathfrak{T}}$. Finally $(\mathfrak{T}, X_{\mathrm{acc}}) \models \mathrm{Th}_{\mathfrak{T}} \cup \Gamma_{f_{\mathcal{V}}}$. □

A computation confirms that without pairs the two output rules of the observational theory cannot both hold (for 200 of 200 random interpretations of R).

Proposition F.16 (Conservativity of the relation version)**.** *For a verifier $\mathcal{V}$, $T^{\mathsf{C}}_{\mathcal{V}}$ is conservative over $\mathrm{Th}_{\mathfrak{T}} \cup \Gamma_{f_{\mathcal{V}}}$: for every $\mathcal{L}_O$-sentence χ, $T^{\mathsf{C}}_{\mathcal{V}} \cup \mathrm{Th}_{\mathfrak{T}} \vdash \chi$ iff $\mathrm{Th}_{\mathfrak{T}} \cup \Gamma_{f_{\mathcal{V}}} \vdash \chi$.*

Proof. ($\Leftarrow$) $T^{\mathsf{C}}_{\mathcal{V}} \vdash \Gamma_{f_{\mathcal{V}}}$ by Theorem F.10. ($\Rightarrow$) Every model $\mathfrak{M}$ of $\mathrm{Th}_{\mathfrak{T}} \cup \Gamma_{f_{\mathcal{V}}}$ expands to a model of $T^{\mathsf{C}}_{\mathcal{V}}$. Let $\mathsf{C}^{\mathfrak{M}}$ be the least subset of M^2 that contains the pairs with a halting prefix and is closed under the backward rules, interpreted in $\mathfrak{M}$. The rules and halting axioms hold by construction. In $\mathfrak{M}$ the current patterns are still pairwise disjoint and the values (Z_r, Z_l) of a match are unique, since disjointness and injectivity are first-order consequences of $\mathrm{Th}_{\mathfrak{T}}$ (for instance $p\,Z \neq p'\,Z'$ for incomparable words p, p'); so, as in Theorem F.10(i), $\mathsf{C}^{\mathfrak{M}}$ is the set of pairs whose deterministic run in $\mathfrak{M}$ reaches a halting prefix. Consider a start pair $(x, \mathrm{tg}\,\mathrm{st}(\mathrm{start})\,z)$. If z is nonstandard, phase cert never ends: a nonstandard $z \neq \mathtt{e}$ is $\mathtt{s}_a z'$ for some letter a (a sentence of $\mathrm{Th}_{\mathfrak{T}}$) with z' nonstandard, so each step leaves a nonstandard remainder. If z is standard and x nonstandard, phase inp never ends, for the same reason. If both are standard, the run stays in the standard part, a copy of $\mathfrak{T}$, and is the run of Lemma F.9. So $(x, 0\,\mathrm{st}(\mathrm{start})z) \in \mathsf{C}^{\mathfrak{M}}$ implies $x = t_w$ with $\mathcal{V}(w, z') = \mathrm{acc}$ for some z', hence $\mathfrak{M} \models R(t_w)$ by $\Gamma_{f_{\mathcal{V}}}$; likewise for rejection. Completeness gives the claim. □

The design of Lemma F.9, which reads both raw words to their ends before the machine starts, is what makes this work. Proposition F.16 is the conservative-extension form of the theorem of Kleene and of Craig and Vaught for this particular theory, with a derivation-bounded extension.

Proposition F.17 (Richer observational languages)**.** *(a) Let the observational language have constants* $\mathtt{e}_1, \ldots, \mathtt{e}_{r_0}$*, unary function symbols* $\mathtt{f}_1, \ldots, \mathtt{f}_{r_1}$ *and unary observable predicates* $R_1, \ldots, R_{r_2}$*, the named structure being the free term algebra* $\mathfrak{T}'$ *on these constants and function symbols and the data all literals* $R_i(\tau)^{\pm}$*,* τ *closed. If a finite theory in this language plus unary theoretical predicates and constants is compatible with* $\mathrm{Th}(\mathfrak{T}')$ *and proves every datum, then each* R_i *has finitely many residuals* $\{u \in \{1, \ldots, r_1\}^* : R_i(\mathtt{f}_u\tau)\}$ *(*τ *closed); in particular* $\{u : R_i(\mathtt{f}_u\mathtt{e}_l)\}$ *is regular for each root* $\mathtt{e}_l$*. (b) Let the observational language be* $\{\mathtt{e}, \mathtt{s}_0, \mathtt{s}_1, R, J\}$ *with* J *a binary observable relation, the data being all literals of* R *and* J *on closed terms, and let the world be* $J := \{(t_u, t_v) : \#_0(u) = \#_1(v)\}$*,* $R := \mathrm{EQ} := \{w : \#_0(w) = \#_1(w)\}$*, which is not regular. The 13 observational sentences* $J(\mathtt{e}, \mathtt{e})$*;* $\forall y\, \neg J(\mathtt{e}, \mathtt{s}_1 y)$*;* $\forall x\, \neg J(\mathtt{s}_0 x, \mathtt{e})$*; the four sentences* $\forall y\, (J(\mathtt{e}, y) \to J(\mathtt{e}, \mathtt{s}_0 y))$*,* $\forall x \forall y\, (J(x, y) \to J(\mathtt{s}_0 x, \mathtt{s}_1 y))$*,* $\forall x \forall y\, (J(\mathtt{s}_0 x, y) \to J(\mathtt{s}_0 x, \mathtt{s}_0 y))$*,* $\forall x \forall y\, (J(x, y) \to J(\mathtt{s}_1 x, y))$*, each also with both* J*-atoms negated; and* $\forall x\, (J(x, x) \to R(x))$*,* $\forall x\, (\neg J(x, x) \to \neg R(x))$ *are true in* $(\mathfrak{T}, R, J)$*, so they form a theory about the named objects with no theoretical vocabulary, and they derive every datum with* $O(|u| + |v|)$ *lines. So with a binary observable, finite observational theories about the named objects fit a non-regular target.*

Proof. (a) Use, from $\mathrm{Th}(\mathfrak{T}')$: each $\mathtt{f}_i$ is injective, the ranges of distinct $\mathtt{f}_i$ are disjoint, no root is in any range, and distinct closed terms are distinct. For a non-root closed term $\tau = \mathtt{f}_a\tau'$ put $\mathrm{Sub}_\tau := \{\mathtt{f}_u\tau : u \in \{1, \ldots, r_1\}^*\}$. By these facts an element $y \notin \mathrm{Sub}_\tau$ with $\mathtt{f}_b y \in \mathrm{Sub}_\tau$ must be τ' with $b = a$ (the one-edge argument in the proof of Theorem 8.8, with $\{0, 1\}$ replaced by $\{1, \ldots, r_1\}$), so $\mathfrak{A}^{\mathrm{rel}}$ is $\mathfrak{C}_\tau \sqcup \mathfrak{S}_\tau$ with one quantifier-free redefined edge. The exceptional set is the set of closed subterms of the denotations of the constants, together with the roots (roots are never the top of a Sub_τ); for τ, τ' outside it, $\mathfrak{S}_\tau$ and $\mathfrak{S}_{\tau'}$ are isomorphic up to labels, possibly under different roots. The transplant, which copies all labels $R_1, \ldots, R_{r_2}, P_1, \ldots$ at once, then bounds the number of distinct residuals over all closed terms, for each R_i, by ncl_q plus the size of the exceptional set. (b) Truth: $\#_0$ and $\#_1$ change as the letters say. Every pair (u, v) is $(\varepsilon, \varepsilon)$, $(\varepsilon, 1v')$, $(\varepsilon, 0v')$, $(0u', \varepsilon)$, $(1u', v)$ (v arbitrary, including ε), $(0u', 1v')$ or $(0u', 0v')$, and each case is an instance of a sentence or reduces by one sentence, in its positive or negative form, to a pair of smaller total length (for $(1u', v)$ the sentence $\forall x \forall y\, (J(x, y) \to J(\mathtt{s}_1 x, y))$ at $y := t_v$). The two R-sentences finish. □

A computation finds the 13 sentences true for $|x|, |y| \le 6$, and checked derivations of all 3969 J-literals with $|u|, |v| \le 5$ and all 1023 R-literals with $|w| \le 9$; the windowed residual count of J grows $(5, 7, 9$ at depths $2, 3, 4)$.

Naming composites without observing them. If the observational language has a binary function symbol $\mathtt{p}$ but the data concern only the words t_w, then $T_{\mathcal{V}}^{\mathsf{C}}$ with every atom $\mathsf{C}(a, b)$ replaced by $R(\mathtt{p}(a, b))$, with no relativisation, is about the named objects (the term algebra on $\mathtt{e}, \mathtt{s}_0, \mathtt{s}_1, \mathtt{p}$, with $R(\mathtt{p}(a, b)) := \mathbf{1}[(a, b) \in \mathsf{C}^{\mathfrak{T}}]$ for words a, b, false for other composites, and $R := X_{\mathrm{acc}}$ on words), and fits every decidable X under (a2) and (b2). In the free term algebra matching is syntactic, and a start pair with a non-word component gets stuck when phase cert or phase inp reaches the $\mathtt{p}$-term, since no rule matches a term whose head is $\mathtt{p}$; so the output rules hold as in Theorem F.10. Theorem 8.8 thus depends on the composites being unnamed, or, if named, observed.

Question F.18 (Binary observables)**.** What is the fit class of finite observational theories about the named objects when the observational language contains a binary observable relation J, as in Proposition F.17(b)? Is it every decidable target? The answer must depend on the observable: with $J = \emptyset$ it is the class of regular sets, by the proof of Theorem 8.8.

Proposition F.19 (Reading (a1) for infinite hypotheses about the named objects)**.** *Let $X \in$ EXPTIME be* EXPTIME*-hard under polynomial-time many-one reductions. There is a parameter-free set $A \subseteq \mathrm{Sent}_{\mathcal{L}_O}$, true in $(\mathfrak{T}, X)$, with polynomial-time membership, such that every datum $\varphi_w^{\mathbf{1}[w\in X]}$ has a derivation from A by modus ponens and instances of the axiom scheme A1 of* K *in which every line has size polynomial in $|w|$. Hence, at polynomial budgets under (a1), the fit class of observational hypotheses about the named objects contains every* EXPTIME*-hard set in* EXPTIME*, besides* NP ∩ coNP*.*

Proof. Let $\mathcal{M}$ be an alternating polynomial-space machine for X with all paths halting (Step 1 of the proof of Theorem F.10(iv)), and let Acc $\in$ EXPTIME be the set of pairs (w, ς) with ς an accepting configuration of $\mathcal{M}$ on w. Let red be a polynomial-time reduction of Acc (pairs coded as words) to X; since every configuration accepts or rejects, ς rejects iff $\mathrm{red}(w, \varsigma) \notin X$. For a word u let θ_u be the valid certificate sentence of Theorem 4.18 encoding u, and $\langle\cdot\rangle$ a pairing of words. A consists of, for all w and configurations ς of $\mathcal{M}$ on w: $\theta_{\langle w,\varsigma\rangle} \to R(t_{\mathrm{red}(w,\varsigma)})$ for halting accepting ς and $\theta_{\langle w,\varsigma\rangle} \to \neg R(t_{\mathrm{red}(w,\varsigma)})$ for halting rejecting ς; at an existential ς with successors ς_1, ς_2, $\theta_{\langle w,\varsigma,\varsigma_i\rangle} \to (R(t_{\mathrm{red}(w,\varsigma_i)}) \to R(t_{\mathrm{red}(w,\varsigma)}))$ for $i = 1, 2$ and $\theta_{\langle w,\varsigma,\varsigma_1,\varsigma_2\rangle} \to (\neg R(t_{\mathrm{red}(w,\varsigma_1)}) \to (\neg R(t_{\mathrm{red}(w,\varsigma_2)}) \to \neg R(t_{\mathrm{red}(w,\varsigma)})))$, and at a universal ς the duals; and $\theta_{\langle w\rangle} \to (R(t_{\mathrm{red}(w,\mathrm{init}(w))}) \to R(t_w))$ and $\theta_{\langle w\rangle} \to (\neg R(t_{\mathrm{red}(w,\mathrm{init}(w))}) \to \neg R(t_w))$. Every member is true in $(\mathfrak{T}, X)$, since θ is valid and $R(t_{\mathrm{red}(w,\varsigma)})$ says that ς accepts. Membership: read the coded tuple off the θ-part, recompute the red-values and check the transition relation, in polynomial time. A datum is derived along an accepting (rejecting) computation tree: each step cites a member, the valid sentence θ_u (an A1 instance), and modus ponens; every line has polynomial size, as configurations and red-values do. □

For infinite observational axiom sets true of the named objects, with polynomial membership, the fit class at polynomial budgets is exactly NP ∩ coNP under (b): certificate sets are true of the named objects, since θ_γ is valid, and the class of all hypotheses bounds it (Corollary B.22). Under (a1) it lies between the classes of Proposition F.19 and EXPTIME.

Question F.20 (Reading (a1) for infinite hypotheses)**.** Is the (a1) fit class of infinite observational hypotheses about the named objects, at polynomial budgets, all of EXPTIME? For an X that is neither easy nor hard, the facts about the named objects that a derivation can use are the values of X itself and sentences about $(\mathfrak{T}, X)$, and it is not clear that they can store an exponential computation. If they cannot, theoretical vocabulary or entities would be needed under (a1) even for infinite hypotheses.

Question F.21 (A unary theoretical function; soft elimination)**.** (a) Does one unary theoretical function symbol over the named objects suffice for every decidable X? A function is not a label, so the proof of Theorem 8.8 does not apply; a two-stack configuration cannot be read off a single word by prefix patterns, so the construction of Theorem F.10 does not transfer. (b) Is there a soft version of Proposition F.5, with rate factor 1? Soft axiom induction and soft certificate induction admit no two-sided constant-regret correspondence at any pair of rates (Proposition B.23), which suggests that the answer is no.

F.6 Finite theories with free deduction

In this subsection $B = \emptyset$, and $K^{\mathcal{L}}_{\mathrm{fin}}$, $W^{\mathcal{L}}_{\mathrm{fin}}$ are the finite-axiomatisation versions of Definition 2.7 in the language $\mathcal{L}$. Recall $K^{\mathrm{pad}}(n) = \min_{m \geq n} K(m)$, which is nondecreasing and unbounded. $\mathcal{L}_{\mathrm{mon}}$, $\mathcal{L}_f$, Inf_n, D_n and Dinf_f are as in Section 8.5: $\mathcal{L}_{\mathrm{mon}} := \{=\} \cup \{P_i : i \in \mathbb{N}\}$, $\mathcal{L}_f := \mathcal{L}_{\mathrm{mon}} \cup \{f\}$ with f unary, $\mathrm{Inf}_n := \exists x_1 \cdots \exists x_n \bigwedge_{i<j} x_i \neq x_j$, $D_n := \{(\mathrm{Inf}_m, 1) : m \leq n\}$, so that $\psi_{D_n} \equiv \mathrm{Inf}_n$, and $\mathrm{Dinf}_f := \forall x \forall y\, (fx = fy \to x = y) \wedge \exists y \forall x\, fx \neq y$.

Lemma F.22 (A quantitative finite model property of monadic logic)**.** *If $\psi \in \mathrm{Sent}_{\mathcal{L}_{\mathrm{mon}}}$ uses k predicate symbols, has quantifier rank q and has a model, then it has a model with at most $2^k \max(q, 1)$ elements. For $k = 0$, all sets of size at least $\max(q, 1)$ satisfy the same rank-q sentences.*

Proof. Let $\mathfrak{M} \models \psi$ and $q' := \max(q, 1)$. Partition $\mathfrak{M}$ into the 2^k cells Cell_ϵ, $\epsilon \in \{0,1\}^k$, by the values of the k predicates, and let $\mathfrak{M}' \subseteq \mathfrak{M}$ keep $\min(|\mathrm{Cell}_\epsilon|, q')$ elements of each cell; $\mathfrak{M}' \neq \emptyset$. The duplicator wins the q-round game on $(\mathfrak{M}, \mathfrak{M}')$, keeping the invariant "chosen pairs lie in the same cell, and $a_i = a_j \iff b_i = b_j$": a repeated element is answered by its partner, and a fresh element in cell ϵ by a fresh element of that cell on the other side, which exists. If $|\mathrm{Cell}_\epsilon| \geq q'$, both sides have at least $q' \geq q$ elements of the cell and fewer than q are chosen; if $|\mathrm{Cell}_\epsilon| < q'$, both sides have exactly $|\mathrm{Cell}_\epsilon|$ elements of the cell, and the chosen ones are matched bijectively. The invariant gives a partial isomorphism for $=$ and the P_i, so $\mathfrak{M}' \equiv_q \mathfrak{M}$ and $\mathfrak{M}' \models \psi$. □

This is a quantitative form of the finite model property of monadic first-order logic, whose decidability goes back to Löwenheim (1915) and Behmann (1922). A computation by exact Ehrenfeucht–Fraïssé games confirms that truncation at $\max(q, 1)$ preserves $\equiv_q$ in all 185 structures tried, that truncation at $q - 1$ breaks it in all 40 tried, and that the bound $2^k q$ is attained.

Theorem F.23 (Infinity needs a theoretical function)**.** *Let $\mathrm{mb}(\psi) := 2^{k(\psi)} \max(q(\psi), 1)$ for $\psi \in \mathrm{Sent}_{\mathcal{L}_{\mathrm{mon}}}$, with $k(\psi)$ the number of predicate symbols in ψ and $q(\psi)$ its quantifier rank. (a) If $\psi \in \mathrm{Sent}_{\mathcal{L}_{\mathrm{mon}}}$ is consistent and $\psi \vdash \mathrm{Inf}_n$, then $n \leq \mathrm{mb}(\psi)$. No consistent finite monadic theory proves all of D_∞. (b) $K^{\mathcal{L}_{\mathrm{mon}}}_{\mathrm{fin}}(D_n) = K^{\mathrm{pad}}(n) \pm c$. So $K^{\mathcal{L}_{\mathrm{mon}}}_{\mathrm{fin}}(D_n) \to \infty$, and $K^{\mathcal{L}_{\mathrm{mon}}}_{\mathrm{fin}}(D_n) = o(u(n))$ for every computable unbounded nondecreasing $u : \mathbb{N} \to \mathbb{N}$. (c) Dinf_f is consistent and proves every Inf_n, so $K^{\mathcal{L}_f}_{\mathrm{fin}}(D_n) \leq K(\mathrm{Dinf}_f) + c$. (d) Under the prior $\mathbf{m}$ on $\mathrm{Sent}_{\mathcal{L}_f}$, the posterior probability, given D_n, that the finite hypothesis does not use f is at most $2^c \sum_{m \geq n} \mathbf{m}(m) / \mathbf{m}(\mathrm{Dinf}_f)$, which tends to 0. (e) $K^{\{=\}}_A(D_n) \leq c$, by the r.e. theory $\{\mathrm{Inf}_m : m \geq 1\}$.*

Proof. (a) By Lemma F.22, ψ has a model with at most $\mathrm{mb}(\psi)$ elements. (b) $\mathrm{mb}(\psi) \geq n$ is computable from ψ, so $K(\psi) \geq K(\mathrm{mb}(\psi)) - c \geq K^{\mathrm{pad}}(n) - c$. Conversely, Inf_m for $m \geq n$ is consistent, proves Inf_n and has $K(\mathrm{Inf}_m) \leq K(m) + c$. Divergence: K^{pad} is unbounded because only finitely many m have $K(m) \leq j$. Slow growth: for a computable unbounded nondecreasing u_1 let $h_{u_1}(j)$ be the least n' with $u_1(n') \geq j$; then $h_{u_1}(u_1(n)+1) > n$, so $K^{\mathrm{pad}}(n) \leq K(h_{u_1}(u_1(n)+1)) + c \leq 2\log(u_1(n)+1) + c'$. With $u_1 := \lfloor 2^{u/r} \rfloor$ this is at most $2u(n)/r + c''$ for every r, hence $o(u)$. (c) $(\mathbb{N}, \mathrm{successor}) \models \mathrm{Dinf}_f$. In any model, with y_0 outside the range of f, the elements $y_0, fy_0, \ldots, f^{n-1}y_0$ are distinct: $f^i y_0 = f^j y_0$ with $i < j$ gives $y_0 = f^{j-i} y_0$ by injectivity, which lies in the range. (d) The pushforward of $\mathbf{m}$ under the computable partial map $\psi \mapsto \mathrm{mb}(\psi)$ (on $\mathcal{L}_{\mathrm{mon}}$) is at most $2^c \mathbf{m}$; compatible monadic sentences have $\mathrm{mb}(\psi) \geq n$; and the compatible mass with f is at least $\mathbf{m}(\mathrm{Dinf}_f)$. (e) is immediate. □

A computation confirms that Dinf_f has no model with at most 7 elements.

Remark F.24 (The same with named objects). In the language $\{R\} \cup \{c_j : j \in \mathbb{N}\}$ (R unary, c_j constants), with $B = \emptyset$, the data sets $D_n := \{(R(c_j), 1) : j < n\} \cup \{(\exists x\, \neg R(x), 1)\}$ have $K_A(D_n) \le c$ and $K_{\mathrm{fin}}(D_n) = K^{\mathrm{pad}}(n) \pm c$. So $K_{\mathrm{fin}} - K_A$ is unbounded, and finite axiomatisability is a different simplicity measure.

Proof. $T := \{R(c_j) : j \in \mathbb{N}\} \cup \{\exists x\, \neg R(x)\}$ is r.e. with an index of complexity $O(1)$ and is consistent (domain $\mathbb{N} \cup \{*\}$, $c_j \mapsto j$, $R := \mathbb{N}$), so $K_A(D_n) \le c$. *Upper bound on* K_{fin}: for $N \ge n$ with $K(N) = K^{\mathrm{pad}}(n)$, the sentence $\bigwedge_{j<N} R(c_j) \wedge \exists x\, \neg R(x)$ is consistent, proves ψ_{D_n}, and has complexity at most $K(N) + c$. *Lower bound*: let ψ be consistent with $\psi \vdash \psi_{D_n}$. If some c_j, $j < n$, does not occur in ψ, then $\psi \vdash R(c_j)$ and the theorem on constants give $\psi \vdash \forall x\, R(x)$, which contradicts $\psi \vdash \exists x\, \neg R(x)$ and the consistency of ψ. So ψ mentions $c_0, \dots, c_{n-1}$, and $1 + \max\{j : c_j \text{ occurs in } \psi\} \ge n$ is computable from ψ; hence $K(\psi) \ge K^{\mathrm{pad}}(n) - c$. □

Proposition F.25 (The cost of "at least n things" in symbols and in bits)**.** *Let $|\psi|$ be the number of symbols of ψ (each P_i one symbol), and let ψ range over* consistent *sentences with $\psi \vdash \mathrm{Inf}_n$. (a) In pure equality, $q(\psi) \ge n$ for $n \ge 2$, so $|\psi| \ge n$; Inf_n itself has $\Theta(n^2)$ symbols. (b) For monadic ψ, $|\psi| \ge \log n - \log\log n - O(1)$, and $|\psi| = O(\log n)$ is attained by the* carry counter

$$\mathrm{Cnt}'_k := \forall x\, \mathrm{Cr}_0 x \;\wedge\; \bigwedge_{i<k} \forall x\, (\mathrm{Cr}_{i+1} x \leftrightarrow (\mathrm{Cr}_i x \wedge P_i x)) \;\wedge\; \exists x \bigwedge_{i<k} \neg P_i x \;\wedge\; \forall x \Big(\neg \mathrm{Cr}_k x \to \exists y \bigwedge_{i<k} (P_i y \leftrightarrow (P_i x \oplus \mathrm{Cr}_i x)) \Big),$$

with $\mathrm{Cr}_0, \dots, \mathrm{Cr}_k$ further unary predicates, which holds exactly in the monadic structures in which the Cr_i are the carries of the P-vector and all 2^k cells of $P_0, \dots, P_{k-1}$ are nonempty; so $\mathrm{Cnt}'_{\lceil \log n \rceil} \vdash \mathrm{Inf}_n$. (c) With f, $O(1)$ symbols suffice (Dinf_f). (d) Code sentences in Polish notation, symbol by symbol, with a prefix-free code on the symbols in which each of the finitely many logical symbols and variables used costs between 1 and c_0 bits and P_i costs $\ell(i)$ bits, with ℓ the length function of a prefix-free code on $\mathbb{N}$ with $\ell(i) \le 2\log(i+1) + O(1)$. Then the codes of sentences form a prefix-free set, so $\sum_\psi 2^{-\mathrm{bits}(\psi)} \le 1$ and $2^{-\alpha\,\mathrm{bits}(\psi)}$, $\alpha \ge 1$, is a sub-probability on sentences. In this code pure equality costs at least n bits, f costs $O(1)$ bits, and monadic sentences cost $\Theta(\log n \cdot \log\log n)$ bits.

Proof. (a) Let $\mathfrak{M} \models \psi$; then $|\mathfrak{M}| \ge n$. If $|\mathfrak{M}| \ge \max(q, 1)$, Lemma F.22 with $k = 0$ makes every set of size $\max(q, 1)$ a model of ψ, so $\max(q, 1) \ge n$; otherwise $\max(q, 1) > |\mathfrak{M}| \ge n$. Either way $q \ge n$ when $n \ge 2$, and each quantifier is a symbol. (b) *Lower bound*: Theorem F.23(a) gives $n \le 2^k \max(q, 1)$ with $k, q \le |\psi|$. *Correctness of* Cnt'_k: the definitional conjuncts force $\mathrm{Cr}_i x \leftrightarrow \bigwedge_{j<i} P_j x$, the carry into bit i of the binary increment. If all P-vectors are realised, the remaining conjuncts hold, since every non-maximal vector has its increment. Conversely, the vector 0 is realised, and if the vector of value $v < 2^k - 1$ is realised, so is that of $v + 1$ ($\neg \mathrm{Cr}_k x$ says that the vector of x is not all ones, and bit i of $v + 1$ is $P_i \oplus \mathrm{Cr}_i$); by induction all 2^k vectors are realised, so there are at least 2^k elements. Each conjunct indexed by i has constant size. (c) Theorem F.23(c). (d) *Prefix-freeness*: a sentence in Polish notation is determined, symbol by symbol, by its own beginning, so no sentence code is a proper prefix of another. *Lower bound, monadic*: a consistent $\psi \vdash \mathrm{Inf}_n$ with k distinct predicate symbols and rank q contains at least q quantifier symbols, of at least one bit each, and k distinct predicate codewords. Sort their lengths $\ell_{(1)} \le \dots \le \ell_{(k)}$; by Kraft's inequality $j\, 2^{-\ell_{(j)}} \le \sum_{i \le j} 2^{-\ell_{(i)}} \le 1$, so $\ell_{(j)} \ge \log j$ and the predicate codewords cost at least $\log k! \ge k \log k - k \log e$ bits. With $n \le 2^k \max(q, 1)$: if $k \ge \frac{1}{2} \log n$, the cost is at least $\frac{1}{2} \log n\, (\log\log n - 1) - \frac{\log e}{2} \log n$; if $k < \frac{1}{2} \log n$, then $q \ge n / 2^k > \sqrt{n}$. Either

way $\mathrm{bits}(\psi) = \Omega(\log n \log\log n)$. *Upper bound*: Cnt'_k with $k = \lceil \log n \rceil$, its $2k+1$ predicates numbered $1, \ldots, 2k+1$, has $O(k)$ symbols of at most $\max(c_0, 2\log(2k+2) + O(1))$ bits each, so $O(\log n \log\log n)$ bits. *Pure equality*: at least $q \geq n$ symbols of at least one bit. *With f*: Dinf_f is a fixed sentence. □

So the single best theory with f outweighs the best monadic theory by a factor $n^{\Theta(\alpha)}$ and the best pure-equality theory by $2^{\Omega(\alpha n)}$ under the weight $2^{-\alpha|\psi|}$, and by $2^{\Theta(\alpha \log n \log\log n)}$ and $2^{\Omega(\alpha n)}$ under the normalisable weight of (d); with each P_i one symbol, the weight $2^{-\alpha|\psi|}$ is not normalisable, so the first comparison is between single theories, not under a prior. A computation gives $64k + 13$ symbols for Cnt'_k in one convention (linear in k in three conventions), checks that Cnt'_k holds iff all cells are nonempty on all 6560 carry-labelled structures with $k = 3$ and cell counts at most 2, and finds the ratios of the lower and upper bit bounds to $\log n \log\log n$ bounded for $n = 2^8, \ldots, 2^{1024}$. The counter is elementary, and we make no novelty claim for it.

Theorem F.26 (The theorems of PA as data)**.** *Let $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$ and let D_∞ list the theorems of* PA*, labelled true, in the order of an effective enumeration of* PA*-derivations, so that every axiom of* PA *occurs. (a) $K_{\mathrm{fin}}^{\mathcal{L}_{\mathrm{ar}}}(D_n) \to \infty$ and $W_{\mathrm{fin}}^{\mathcal{L}_{\mathrm{ar}}}(D_n) \to 0$; and $K_{\mathrm{fin}}^{\mathcal{L}_{\mathrm{ar}}}(D_n) \leq K(n) + c$. (b) $K_A^{\mathcal{L}_{\mathrm{ar}}}(D_n) \leq K(\mathsf{PA}) + c = O(1)$. (c) In the three-sorted extension $\mathcal{L}_{\mathrm{ar}}^+$ of Theorem F.28, $K_{\mathrm{fin}}^{\mathcal{L}_{\mathrm{ar}}^+}(D_n) \leq c$, and the posterior over finite $\mathcal{L}_{\mathrm{ar}}^+$-theories puts mass tending to 1 on theories that use the new sorts. The winning theory adds no $\mathcal{L}_{\mathrm{ar}}$-content: its $\mathcal{L}_{\mathrm{ar}}$-consequences are exactly those of* PA*.*

Proof. We use the theorem of Ryll-Nardzewski (1952): no consistent extension of PA in $\mathcal{L}_{\mathrm{ar}}$ is finitely axiomatisable; equivalently, no consistent $\mathcal{L}_{\mathrm{ar}}$-sentence proves every axiom of PA. (a) So every consistent ψ fails to prove some axiom α_ψ of PA, which occurs in D_∞, and ψ is incompatible with D_n for large n. Only finitely many ψ have $K(\psi) \leq j$, which gives $K_{\mathrm{fin}} \to \infty$, and dominated convergence ($\sum \mathbf{m} \leq 1$) gives $W_{\mathrm{fin}} \to 0$. The conjunction of the first n data is consistent, given $\mathrm{Con}(\mathsf{PA})$, and computable from n. (b) PA is r.e. (c) Theorem F.28 with $\mathsf{W}_e = \mathsf{PA}$ gives a single $\mathcal{L}_{\mathrm{ar}}^+$-sentence Fin_e whose $\mathcal{L}_{\mathrm{ar}}$-consequences are those of PA; with (a), the posterior mass of the $\mathcal{L}_{\mathrm{ar}}$-sentences tends to 0 while that of Fin_e is a fixed positive number. □

A classical instance of (c) is ACA_0, in the two-sorted language of second-order arithmetic, which is finitely axiomatisable and has the same first-order consequences as PA (see Simpson, 2009).

Remark F.27 (A truth predicate for PA). A theoretical *predicate* on the same domain also appears to give $O(1)$: $\mathrm{I}\Sigma_1$ plus compositional truth clauses for a unary predicate Tr plus $\forall\varphi\,(\text{PA-Ax}(\varphi) \to \mathrm{Tr}(\varphi))$ is a finite theory, true in $\mathbb{N}$, that proves every axiom of PA.

Proof sketch. We do not write out the derivation of the Tarski biconditionals from the clauses over $\mathrm{I}\Sigma_1$. For each standard sentence α the theory proves $\mathrm{Tr}(\ulcorner\alpha\urcorner) \leftrightarrow \alpha$ by external induction on α from the compositional clauses, using that $\mathrm{I}\Sigma_1$ proves the needed facts about the coding (that numerals evaluate correctly needs Σ_1-induction); with $\mathrm{Tr}(\ulcorner\alpha\urcorner)$ for every axiom α of PA this proves every axiom. $\mathrm{I}\Sigma_1$ is finitely axiomatisable (standard; see Hájek and Pudlák, 1993). So in arithmetic a unary theoretical predicate appears to suffice, because the domain already carries $+$ and $\cdot$, which code syntax; in pure equality unary predicates never help (Theorem F.23(a)), nor over observation sequences (Theorem 8.8). We do not claim that unary predicates help only where the domain codes syntax.

Theorem F.28 (r.e. axiom induction is finite-theory induction with theoretical sorts)**.** *Let $\mathcal{L}$ be a finite language (relation, function and constant symbols, with equality) and $B \subseteq \mathrm{Sent}_{\mathcal{L}}$ decidable. Let $\mathcal{L}^+$ be the three-sorted extension (many-sorted first-order logic, every sort with a nonempty*

domain) with a sort o *carrying* $\mathcal{L}$*; a sort* n *with* $0, S, +, \cdot, <$*; and a sort* a *of assignments, with* $\mathrm{val} : \mathsf{a} \times \mathsf{n} \to \mathsf{o}$, $\mathrm{upd} : \mathsf{a} \times \mathsf{n} \times \mathsf{o} \to \mathsf{a}$, $\mathrm{den} : \mathsf{n} \times \mathsf{a} \to \mathsf{o}$ *and a relation* $\mathrm{Sat} \subseteq \mathsf{n} \times \mathsf{a}$*. There is a computable injective map* $e \mapsto \mathrm{Fin}_e$ *from indices to single* $\mathcal{L}^+$*-sentences such that for every* e

$$\{\chi \in \mathrm{Sent}_{\mathcal{L}} : B \cup \{\mathrm{Fin}_e\} \vdash \chi\} = \mathrm{Cn}(\mathsf{W}_e),$$

both sides being inconsistent together. Consequently, for every finite data set D *in* $\mathcal{L}$*,* $K^{\mathcal{L}^+}_{\mathrm{fin}}(D) = K^{\mathcal{L}}_A(D) \pm c$ *and* $2^{-c} W^{\mathcal{L}}_A(D) \le W^{\mathcal{L}^+}_{\mathrm{fin}}(D) \le 2^c W^{\mathcal{L}}_A(D)$*.*

Proof. Construction. Fix a Gödel numbering of the $\mathcal{L}$-terms and $\mathcal{L}$-formulas (variables $x_0, x_1, \dots$) and Σ_1 formulas of $\mathcal{L}_{\mathrm{ar}}$ that define in $\mathbb{N}$ the syntactic relations: $\mathrm{Var}(c, v)$, "c codes x_v"; for each function symbol g of arity k (constants: $k = 0$), $\mathrm{App}_{\mathsf{g}}(c, d_1, \dots, d_k)$, "the d_i code terms τ_i and c codes $\mathsf{g}(\tau_1, \dots, \tau_k)$"; for each relation symbol r, $\mathrm{At}_{\mathsf{r}}(c, \vec{d})$, and $\mathrm{Eq}(c, d, d')$, for atomic formulas over coded terms; $\mathrm{Neg}(c, d)$, "d codes a formula χ and c codes $\neg\chi$"; $\mathrm{Imp}(c, d, d')$; $\mathrm{All}(c, v, d)$, "d codes χ and c codes $\forall x_v \chi$"; and $\mathrm{Enum}_e(c)$, "$c \in \mathsf{W}_e$ and c codes an $\mathcal{L}$-sentence". The bounded quantifiers in these formulas use $<$. Fin_e is the conjunction of:

- (Ar) Robinson's Q on the sort n, including the defining axiom $\forall x \forall y\, (x < y \leftrightarrow \exists z\, (Sz + x = y))$, so that Q proves every true Σ_1 sentence whose bounded quantifiers use $<$ (standard; see Boolos et al., 2007);
- (Up) $\forall \alpha \forall v \forall x\, \mathrm{val}(\mathrm{upd}(\alpha, v, x), v) = x$ and $\forall \alpha \forall v \forall v' \forall x\, (v \neq v' \to \mathrm{val}(\mathrm{upd}(\alpha, v, x), v') = \mathrm{val}(\alpha, v'))$;
- (Den) $\forall c \forall v\, (\mathrm{Var}(c, v) \to \forall \alpha\, \mathrm{den}(c, \alpha) = \mathrm{val}(\alpha, v))$ and, for each g, $\forall c \forall \vec{d}\, (\mathrm{App}_{\mathsf{g}}(c, \vec{d}) \to \forall \alpha\, \mathrm{den}(c, \alpha) = \mathsf{g}(\mathrm{den}(d_1, \alpha), \dots, \mathrm{den}(d_k, \alpha)))$;
- (Sat) for each r, $\forall c \forall \vec{d}\, (\mathrm{At}_{\mathsf{r}}(c, \vec{d}) \to \forall \alpha\, (\mathrm{Sat}(c, \alpha) \leftrightarrow \mathsf{r}(\mathrm{den}(d_1, \alpha), \dots)))$; the same for Eq with $=$; $\forall c \forall d\, (\mathrm{Neg}(c, d) \to \forall \alpha\, (\mathrm{Sat}(c, \alpha) \leftrightarrow \neg \mathrm{Sat}(d, \alpha)))$; the same for Imp with $\to$; and $\forall c \forall v \forall d\, (\mathrm{All}(c, v, d) \to \forall \alpha\, (\mathrm{Sat}(c, \alpha) \leftrightarrow \forall x\, \mathrm{Sat}(d, \mathrm{upd}(\alpha, v, x))))$, with x of sort o;
- (Ax) $\forall c\, (\mathrm{Enum}_e(c) \to \forall \alpha\, \mathrm{Sat}(c, \alpha))$, with e written as a numeral, so that $e \mapsto \mathrm{Fin}_e$ is injective.

($\supseteq$) Let $\varphi \in \mathsf{W}_e$. Q proves the true Σ_1 sentence $\mathrm{Enum}_e(\underline{\ulcorner \varphi \urcorner})$, so $\mathrm{Fin}_e \vdash \forall \alpha\, \mathrm{Sat}(\underline{\ulcorner \varphi \urcorner}, \alpha)$. By external induction on terms τ and formulas χ whose variables are among $x_{v_1}, \dots, x_{v_j}$: $\mathrm{Fin}_e \vdash \forall \alpha\, \mathrm{den}(\underline{\ulcorner \tau \urcorner}, \alpha) = \tau^\alpha$ and $\mathrm{Fin}_e \vdash \forall \alpha\, (\mathrm{Sat}(\underline{\ulcorner \chi \urcorner}, \alpha) \leftrightarrow \chi^\alpha)$, where $^\alpha$ replaces each x_v by $\mathrm{val}(\alpha, \underline{v})$. Each step instantiates one clause at the numerals of the immediate subterms or subformulas, whose guard is a true Σ_1 sentence and hence provable in Q. The quantifier step uses (Up) and $\mathsf{Q} \vdash \underline{v} \neq \underline{v'}$ for $v \neq v'$, which give $\mathrm{val}(\mathrm{upd}(\alpha, \underline{v}, x), \underline{v}) = x$ and $\mathrm{val}(\mathrm{upd}(\alpha, \underline{v}, x), \underline{v'}) = \mathrm{val}(\alpha, \underline{v'})$, so the induction hypothesis at $\mathrm{upd}(\alpha, \underline{v}, x)$ turns $\forall x\, \mathrm{Sat}(\underline{\ulcorner \chi \urcorner}, \mathrm{upd}(\alpha, \underline{v}, x))$ into $(\forall x_v \chi)^\alpha$. Extra, nonstandard witnesses of the guards in other models only add constraints and cannot block these derivations. For the sentence φ, $\varphi^\alpha = \varphi$, and the sort a is nonempty, so $\mathrm{Fin}_e \vdash \varphi$. ($\subseteq$) Let $\mathfrak{M}$ be an $\mathcal{L}$-structure with $\mathfrak{M} \models B \cup \mathsf{W}_e$. Expand it: the sort n is $\mathbb{N}$, with the true $<$; the sort a is $M^{\mathbb{N}}$, with $\mathrm{val}(\alpha, v) := \alpha(v)$ and $\mathrm{upd}(\alpha, v, x) := \alpha[v \mapsto x]$; $\mathrm{den}(c, \alpha)$ is the value under α of the term coded by c (a fixed element if c codes no term); and $\mathrm{Sat}(c, \alpha)$ holds iff c codes a formula χ with $\mathfrak{M} \models \chi[\alpha]$. The guards are evaluated in $\mathbb{N}$, where the Σ_1 formulas mean what they say, so each clause holds (the guards of Neg, Imp and All require the components to code formulas, so no clause compares a non-formula with a formula), and (Ax) holds because $\mathfrak{M} \models \mathsf{W}_e$. Thus every model of $B \cup \mathsf{W}_e$ expands to a model of $B \cup \{\mathrm{Fin}_e\}$, and by completeness every $\mathcal{L}$-consequence of $B \cup \{\mathrm{Fin}_e\}$ is in $\mathrm{Cn}(\mathsf{W}_e)$. If $B \cup \mathsf{W}_e$ is inconsistent, so is $B \cup \{\mathrm{Fin}_e\}$, by ($\supseteq$); if it is consistent, a model expands, so $B \cup \{\mathrm{Fin}_e\}$ is consistent. *Complexities.* $K(\mathrm{Fin}_e) \le K(e) + c$, and Fin_e is compatible with D

iff e is. Conversely, the map $\chi \mapsto \mathrm{ct}(\chi)$, an index of $\{\varphi \in \mathrm{Sent}_{\mathcal{L}} : B \cup \{\chi\} \vdash \varphi\}$, is computable; $\mathrm{ct}(\chi)$ is compatible with D iff χ is (the set contains $\neg\forall x\, x = x$ iff $B \cup \{\chi\}$ is inconsistent); and $K(\mathrm{ct}(\chi)) \le K(\chi) + c$. For the weights, $\mathbf{m}(\mathrm{Fin}_e) \ge 2^{-c}\mathbf{m}(e)$ because $e \mapsto \mathrm{Fin}_e$ is computable and injective, and $\sum_{\chi:\mathrm{ct}(\chi)=j} \mathbf{m}(\chi) \le 2^c \mathbf{m}(j)$ because pushforwards of $\mathbf{m}$ under computable maps are dominated by $\mathbf{m}$. □

Theorem F.28 is an effective, many-sorted form of the theorem of Kleene (1952) and Craig and Vaught (1958) (as stated in Pakhomov and Visser, 2019). Using extra sorts to remove the condition that the theory have no finite models is routine; what we add is the Bayesian consequence, which needs an effective construction with an injective map and two-sided complexity bounds. We do not treat the one-sorted version.

Remark F.29 (Not one tape). Let $\mathcal{L}$ be finite and satisfy the hypotheses of Theorem 5.1, and for the universal theory machine of $\mathcal{L}^+$ let $M^{\mathrm{fin}}(D) := \mathbb{U}\{\omega : A_\omega \text{ is finite, } B \cup A_\omega \text{ is consistent and } \vdash \psi_D\}$. On the lottery stream D_∞ of Theorem 5.1, $M^{\mathrm{fin}}(D_n) \to 0$, while $M_A(D_n)$ is bounded below by a positive constant. So the equivalence of Theorem F.28 is not one-tape: random r.e. theories are strictly stronger than random finite theories with any amount of theoretical apparatus.

Proof. $\mathcal{L}^+$ is finite, so it has computable syntax, and for finite A_ω the $\mathcal{L}$-content $\{\chi \in \mathrm{Sent}_{\mathcal{L}} : B \cup A_\omega \vdash \chi\}$ is r.e. and, if $B \cup A_\omega$ is consistent, consistent. By Theorem 5.1(b) it does not prove every datum of D_∞. So the events $\{\omega : A_\omega$ finite, consistent with B, $\vdash \psi_{D_n}\}$ decrease in n to the empty set, and by continuity from above their measure tends to 0. The lower bound on $M_A(D_n)$ is Theorem 5.1(a). □

Proposition F.30 (Infinity: Ramsey, Craig and a theoretical function)**.** *In pure equality: (a) the Ramsey sentence $\exists f\, \mathrm{Dinf}_f$ of the theory $\{\mathrm{Dinf}_f\}$, a single existential second-order sentence with no theoretical constant, has in standard semantics exactly the infinite sets as models; (b) $\{\mathrm{Inf}_m : m \ge 1\}$ has exactly the infinite sets as models, and its consequences form a complete theory; (c) so $\exists f\, \mathrm{Dinf}_f$ and $\{\mathrm{Inf}_m : m \ge 1\}$ (a first-order observational axiomatisation, necessarily infinite by Theorem F.23(a)) have the same first-order $\{=\}$-consequences, and no finite first-order theory in $\{=\}$, nor in any monadic extension, has them; (d) the first-order theory $\{\mathrm{Dinf}_f\}$ in the extended language $\{=, f\}$ is finite and has them as its $\{=\}$-consequences.*

Proof. (a) If f is injective and not surjective on A, then A is infinite (Theorem F.23(c)); conversely an infinite A has a countable subset $\{a_0, a_1, \dots\}$ (countable choice), and the map $a_i \mapsto a_{i+1}$, the identity elsewhere, is injective and misses a_0. (b) The first claim is clear. Completeness: by Lemma F.22 with $k = 0$, all sets of size at least q satisfy the same rank-q sentences, so all infinite sets are elementarily equivalent. (c), (d) The same models give the same first-order consequences; the rest is Theorem F.23(a), (c). □

A computation on 1200 random pure-equality sentences of rank 1 to 4 finds their truth constant on sets of sizes q to $q + 2$.

Proposition F.31 (Factoring an inference graph)**.** *Let $\mathcal{L}$ contain unary predicates $\mathrm{A}_1, \dots, \mathrm{A}_{n_A}$ and $\mathrm{B}_1, \dots, \mathrm{B}_{n_B}$; let $T_{\mathrm{dir}} := \{\forall x\, (\mathrm{A}_i x \to \mathrm{B}_j x)\}_{i,j}$ and, in $\mathcal{L} \cup \{\mathrm{Mid}\}$, $T_{\mathrm{Mid}} := \{\forall x\, (\mathrm{A}_i x \to \mathrm{Mid}\, x)\}_i \cup \{\forall x\, (\mathrm{Mid}\, x \to \mathrm{B}_j x)\}_j$. An* atom-to-atom implication *is a sentence $\forall x\, (\mathrm{P}x \to \mathrm{P}'x)$ with P, P' distinct unary predicate symbols. (a) $\mathrm{Cn}(T_{\mathrm{Mid}}) \cap \mathrm{Sent}_{\mathcal{L}} = \mathrm{Cn}(T_{\mathrm{dir}})$. (b) Every theory of atom-to-atom implications over $\{\mathrm{A}_i, \mathrm{B}_j\}$ equivalent to T_{dir} has at least $n_A n_B$ members. (c) In full first-order logic, T_{dir} is equivalent to the single sentence $\forall x\, ((\mathrm{A}_1 x \vee \cdots \vee \mathrm{A}_{n_A} x) \to (\mathrm{B}_1 x \wedge \cdots \wedge \mathrm{B}_{n_B} x))$, of length $O(n_A + n_B)$, so* Mid *saves only a constant factor there. (d) Every theory*

S of atom-to-atom implications over $\{\mathrm{A}_i, \mathrm{B}_j, \mathrm{Mid}\}$ with $\mathrm{Cn}(S) \cap \mathrm{Sent}_{\mathcal{L}} = \mathrm{Cn}(T_{\mathrm{dir}})$ has at least $\min(n_A n_B, n_A + n_B)$ members, and T_{dir} or T_{Mid} attains the bound. So the new predicate shortens the theory iff $(n_A - 1)(n_B - 1) > 1$, by $\max(0, n_A n_B - n_A - n_B)$ members.

Proof. (a) $T_{\mathrm{Mid}} \vdash \mathrm{A}_i \to \mathrm{Mid} \to \mathrm{B}_j$, and every model of T_{dir} expands to one of T_{Mid} with $\mathrm{Mid} := \bigcup_i \mathrm{A}_i \subseteq \bigcap_j \mathrm{B}_j$. (b) For a set S of atom-to-atom implications over any set of unary predicates, $S \vdash \forall x\,(\mathrm{P}x \to \mathrm{P}'x)$ iff P' is reachable from P in the graph S: if not, the one-element structure in which exactly the atoms reachable from P hold satisfies S, P and $\neg\mathrm{P}'$. In models of T_{dir} no $\mathrm{A}_i \to \mathrm{A}_{i'}$, $\mathrm{B}_j \to \mathrm{B}_{j'}$ $(i \neq i',\, j \neq j')$ or $\mathrm{B}_j \to \mathrm{A}_i$ holds (take the one-element structures with only A_i true together with all B's, respectively only B_j true). So an equivalent S over $\{\mathrm{A}_i, \mathrm{B}_j\}$ contains only edges $\mathrm{A}_i \to \mathrm{B}_j$, and reachability from A_i to B_j then needs the edge itself: all $n_A n_B$ of them. (c) Propositional logic. (d) S proves every $\mathrm{A}_i \to \mathrm{B}_j$ and, by the equality of $\mathcal{L}$-consequences, none of the implications excluded in (b). Take a shortest path from A_i to B_j in the graph S; it has no repeated vertex. An interior vertex $\mathrm{A}_{i'}$ would give $S \vdash \mathrm{A}_i \to \mathrm{A}_{i'}$ $(i' \neq i)$, and an interior $\mathrm{B}_{j'}$ would give $S \vdash \mathrm{B}_{j'} \to \mathrm{B}_j$ $(j' \neq j)$. So the path is the edge $\mathrm{A}_i \to \mathrm{B}_j$ or the path $\mathrm{A}_i \to \mathrm{Mid} \to \mathrm{B}_j$. With $I_A := \{i : (\mathrm{A}_i \to \mathrm{Mid}) \in S\}$ and $I_B := \{j : (\mathrm{Mid} \to \mathrm{B}_j) \in S\}$, every pair outside $I_A \times I_B$ needs its own edge, so $|S| \geq |I_A| + |I_B| + n_A n_B - |I_A||I_B|$. This function is affine in each of $|I_A|$ and $|I_B|$, so its minimum over $[0, n_A] \times [0, n_B]$ is at a corner: $n_A n_B$, $n_A n_B + n_A$, $n_A n_B + n_B$ or $n_A + n_B$. Finally $n_A n_B - (n_A + n_B) = (n_A - 1)(n_B - 1) - 1$. □

A computation that decides the consequence relation exactly (a set of atom-to-atom implications says that the type of every element lies in a set of types, and the $\mathcal{L}$-reducts of its models are the structures whose types lie in the projection of that set) confirms the least sizes $n_A n_B$ without Mid and $\min(n_A n_B, n_A + n_B)$ with it for all ten pairs up to $(3, 3)$; Mid saves nothing except for $(2, 3)$, $(3, 2)$ (one member) and $(3, 3)$ (three members).

G Proofs for approximate laws and the best system

This appendix states in full the results of Section 9 that the main text states in words, and proves every result of that section. The notation is that of Definition 9.1. We measure a finite theory by $\|T\|$, without the stop bit of a prefix-free code of theories; using the prefix-free length $\|T\| + 1$ instead changes every statement below by an additive constant at most, and shifts J_λ by the same amount λ on every theory, so that it leaves the λ-MAPs unchanged. Computed numbers come from seeded scripts in the accompanying repository; they are never part of a statement.

G.1 The setting in full

Technical points. (i) *Rational cells.* In regime R the dither u_j is uniform on a grid $\{k/N_{\mathrm{d}} : 0 \leq k < N_{\mathrm{d}}\}$, so that cell endpoints are rational. Every event used below has the form "u_j lies in an interval of length ℓ'", whose grid probability is within $1/N_{\mathrm{d}}$ of ℓ'; the formulas are the limit $N_{\mathrm{d}} \to \infty$ (equivalently, endpoints may be real algebraic numbers). A cell that crosses the zero point of the circle is a disjunction of two interval sentences. (ii) *Fresh times.* Data are sentences, so they mention numerals, a countable set; no atomless law lives on it. Times drawn i.i.d. from a law on numerals repeat: the number of distinct times among n draws is $o(n)$ almost surely (take a finite set of mass at least $1 - \epsilon$ and apply the strong law of large numbers). A repeat changes the cost of every form except an exact law: a patched law withdraws the law at a whole time, so every later datum at a refuted time is an exception, and the cells of an honest law at one time intersect, which narrows its band. A computation with times drawn uniformly from the 4383 days

of a twelve-year window (sinusoidal error, $\varepsilon^* = 1'$, $\Delta = 60'$, $\lambda = 1$, $c_{\text{time}} = \log 4383 + 2$) gives a patched law's cost per datum of 0.40, 1.96 and 3.23 bits at $n = 10^3$, 10^5 and 2×10^5, against the rate 0.240 at fresh times, and a code-R cost of the honest law falling from 0.037 to 0.022, against 0.030. This is why regime R takes fresh times: an injective computable sequence of numerals, with i.i.d. errors and dithers. Fresh times have names that grow, and Proposition G.15 says what that costs. (iii) *Presentations.* No decidable set of programs covers all total computable maps, so the class of pointwise interval laws depends on the presentation language (for example, primitive recursive definitions, or programs paired with an explicit clock); every statement holds for each fixed choice. (iv) *A model of the data.* In regime R the error at a fresh time is a draw from a fixed law; for the Kepler residuals this law is the law of $e(t)$ for t uniform in the window. Answered data with a question $\mathcal{Q}$ are read through Definition 2.11: for consistent $B \cup T$, T proves $(\mathcal{Q}, a)$ iff $A_T(\mathcal{Q}) = \{a\}$, since $T \vdash a$ refutes every other answer by incompatibility, and refuting every other answer proves a because B proves $\bigvee \mathcal{Q}$.

Lemma G.1 (Decidability)**.** *Let T be a pointwise interval law and C a cell, a finite union of intervals with rational endpoints. Then T proves the datum "$q_t \in C$" iff $[\underline{f}(t), \overline{f}(t)] \subseteq C$, and refutes it iff $[\underline{f}(t), \overline{f}(t)] \cap C = \emptyset$; and $B \cup T \cup D^{\pm}$ is consistent iff, for every time t of D, $[\underline{f}(t), \overline{f}(t)]$ meets the intersection C_t of the cells that D assigns to t. All three are decidable from the presentation of T.*

Proof. Every choice of reals $q_t \in [\underline{f}(t), \overline{f}(t)]$ gives a model of $B \cup T$ in $\mathbb{R}$, and if the q_t also lie in the assigned cells, a model of $B \cup T \cup D^{\pm}$. This gives the "only if" of the first two claims (choose q_t outside the cell, respectively inside it) and the "if" of the third. Inclusion and disjointness of finite unions of rational intervals are Boolean combinations of inequalities between rational numerals, which RCF proves or refutes; this gives the "if" of the first two claims. If for some t the interval misses C_t, the finitely many inequalities involved are jointly refutable in RCF, which gives the "only if" of the third. The values $\underline{f}(t), \overline{f}(t)$ are computable from the presentation, so all three conditions are decidable. □

Proposition G.2 (Laws plus listed exceptions are deterministic hypotheses)**.**

(a) For every finite T with $B \cup T \cup D^{\pm}$ consistent, $L_A(D) - c \le K_A(D) \le \|T\| + \mathrm{Exc}_D^F(T) + c$.

(b) For an r.e. T and $E \subseteq D$ with $B \cup T \cup E^{\pm}$ consistent and $B \cup T \cup E^{\pm} \vdash \psi_D$, $L_A(D) \le K_{\equiv}(T) + K(E) + c$.

Proof. (a) The finite set $T' := T \cup \{a : (\mathcal{Q}, a)$ a datum of D that T does not prove$\}$ is contained in the consistent set $T \cup D^{\pm}$, so $B \cup T'$ is consistent, and T' proves every datum. Concatenating the codewords of the sentences of T, each after a continuation bit, the codewords $c_S(a)$ of the exceptions, each after a continuation bit, and a stop bit gives a prefix-free code of T' of length $\|T\| + \mathrm{Exc}_D^F(T) + 1$, and a fixed program turns it into an index of T'. So $K_A(D) \le \|T\| + \mathrm{Exc}_D^F(T) + c$. The instance form of dominance (Theorem 3.6) gives $M_A(D) \ge 2^{-K(e)-c}$ for every index e of a theory consistent with B that proves ψ_D, hence $L_A(D) \le K_A(D) + c$. (b) is Proposition 3.12(b). □

So code F pays for *which* questions are exceptions, since the answer sentence names its question; a deterministic theory cannot avoid this, because it does not see the questions.

G.2 Randomised theories and the best two-part code

Proposition G.3 (Randomised theories pay question-relative costs)**.** *Let T be a pointwise interval law and D a set of interval data consistent with T.*

(a) $L_A(D) \le K(T) + \mathrm{Exc}_D^R(T) + c \le \|T\| + \mathrm{Exc}_D^R(T) + c$.

(b) *If D comes from a computable question family with one question $\mathcal{Q}_t$ per time, and $k_t(T) := |A_T(\mathcal{Q}_t)| \le |\mathcal{Q}_t|$, then $L_A(D) \le K(T, (\mathcal{Q}_t)_t) + \sum \log k_t(T) + c$, the sum over the data that T does not prove; in particular $L_A(D) \le K(T, (\mathcal{Q}_t)_t) + \mathrm{Exc}_D^Q(T) + c$.*

(c) *Code R charges an exception of a patched law, at a time where $\underline{f} = 0$ and $\overline{f} = \mathsf{R}$, exactly $\log(\mathsf{R}/|C_t|)$, which is the code-Q cost for equal cells; it charges the honest law $-\log(|C_t \cap [f(t) - \varepsilon, f(t) + \varepsilon]|/2\varepsilon)$, which is 0 when the band lies inside the cell.*

Proof. (a) *The machine.* Enumerate I as $t_1, t_2, \ldots$ and fix a computable injection $(i, k) \mapsto \langle i, k\rangle$ into tape positions, increasing in k. On a tape ω put $U_i := \sum_{k \ge 1} \omega_{\langle i,k\rangle} 2^{-k}$; under $\mathbb{U}$ the U_i are independent and uniform on $[0, 1]$. Let G_T be the theory machine that emits the sentences of T at the empty prefix and, once the prefix read contains the positions $\langle i, 1\rangle, \ldots, \langle i, k\rangle$, whose bits form the binary numeral of an integer m, also emits

$$\underline{f}(t_i) + (\overline{f} - \underline{f})(t_i)\, m 2^{-k} \;\le\; q_{t_i} \;\le\; \underline{f}(t_i) + (\overline{f} - \underline{f})(t_i)\, (m+1) 2^{-k}.$$

This set of (prefix, sentence) pairs is c.e., and an index of it is computable from the presentation of T, so $K(G_T) \le K(T) + c$.

The random theory is always consistent. Put $q_{t_i} := v_i := \underline{f}(t_i) + (\overline{f} - \underline{f})(t_i) U_i$. Each emitted closed interval contains v_i, so these values satisfy every emitted sentence, and $\mathbb{R}$ with them is a model.

It proves a datum "$q_{t_i} \in C$" whenever v_i lies in the interior of C. The emitted intervals contain v_i and have lengths $(\overline{f} - \underline{f})(t_i) 2^{-k} \to 0$, so one of them lies in the interior of C, and RCF proves the inclusion. (If $\underline{f}(t_i) = \overline{f}(t_i)$, T alone decides the datum.) The times of D are distinct indices i, the U_i are independent, and endpoints have probability 0, so

$$\nu_{G_T}(D) \;\ge\; \mathbb{U}\{\omega : v_t \in \operatorname{int} C_t \text{ for every time } t \text{ of } D\} \;=\; P_T^R(D).$$

Dominance (Theorem 3.6), $M_A \ge 2^{-K(\nu)} \nu$ for every theory-machine law ν, gives $M_A(D) \ge 2^{-K(G_T) - c} \nu_{G_T}(D)$, which is (a); and $K(T) \le \|T\| + c$.

(b) Let the machine use U_i to pick the $\lfloor k_{t_i}(T) U_i \rfloor$-th element a of $A_T(\mathcal{Q}_{t_i})$, which is computable by Lemma G.1, and emit a as soon as enough bits have been read (almost surely after finitely many). The cell a meets $[\underline{f}(t_i), \overline{f}(t_i)]$, so the random theory has a model in $\mathbb{R}$; it proves the true answer with probability $1/k_{t_i}(T)$, independently across times, and $k_t(T) = 1$ when T proves the datum. Since D is consistent with T, T refutes no datum. The machine's index is computable from T and the question family, and dominance gives the bound; the last claim follows from $\log k_t(T) \le \log |\mathcal{Q}_t| = \gamma(\mathcal{Q}_t)$. (c) is the definition of P_T^R. □

Proposition G.4 (SAI is the best two-part code over theory machines)**.**

(a) *For every data set D, $L_A(D) = \min_G [K(G) - \log \nu_G(D)] \pm c$, the minimum over theory machines G; it is attained, up to c, by G_U.*

(b) *Restricting G gives upper bounds: $L_A(D) \le \min_T J_1^R(T; D) + c$ over pointwise interval laws, $L_A(D) \le \min_T J_1^F(T; D) + c$ over finite theories, and $L_A(D) \le K_A(D) + c$.*

(c) *Neither K_A nor $\min_T J_1^Q(T; \cdot)$ over finite theories is within an additive constant of L_A in general; nor, therefore, is $\min_T J_1^F(T; \cdot)$ when $\gamma \le \ell$.*

(d) *Over all theory machines steeper selection collapses. Take the point family of programs $p \in \operatorname{dom} V$ for theory machines, with length $|p|$ and fit $-\log \nu_{G_{V(p)}}(D)$, and let ℓ_U be the length of a program for G_U. For $\lambda > 1$, every λ-MAP p has $|p| \le \ell_U + \ell_U/(\lambda - 1)$.*

Proof. (a) Dominance gives $L_A(D) \le K(G) - \log \nu_G(D) + c$ for every theory machine G, and $M_A = \nu_{G_U}$ gives $K(G_U) - \log \nu_{G_U}(D) = L_A(D) + O(1)$. (b) is Propositions G.2 and G.3 with $K_{\equiv}$ and K of a finite theory at most its length plus a constant.

(c) That $K_A - L_A$ is unbounded is the absence of a coding theorem (Theorem 3.18). For $\min_T J_1^Q(T;\cdot)$, take the language $\{R\} \cup \{c_j : j \in \mathbb{N}\}$ with $B = \emptyset$ and the yes/no data $D_n := \{(R(c_j), 1) : j < n\} \cup \{(\exists x\, \neg R(x), 1)\}$, so that every datum has literal cost $\gamma = 1$. By Remark F.24, applied to the r.e. theory $\{R(c_j) : j \in \mathbb{N}\} \cup \{\exists x\, \neg R(x)\}$, $K_A(D_n) \le c$, hence $L_A(D_n) \le K_A(D_n) + c \le c'$. Let T be finite with $B \cup T \cup D_n^{\pm}$ consistent. If c_j does not occur in T and $T \vdash R(c_j)$, then $T \vdash \forall x\, R(x)$ by the theorem on constants, which contradicts consistency with $\exists x\, \neg R(x)$. So T proves at most the data $R(c_j)$ with c_j occurring in T, together with $\exists x\, \neg R(x)$. In a syntax code in which each symbol occurrence costs at least one bit, each constant occurring in T costs at least one bit of $\|T\|$, and each unproved datum one bit of Exc^Q; so $J_1^Q(T; D_n) \ge n$. Finally $\mathrm{Exc}^F \ge \mathrm{Exc}^Q$ when $\gamma \le \ell$, since then each exception costs at least as much in code F.

(d) Hänni and Claude (Anthropic) (2026d, Proposition 5.7(ii)) states: if a point family contains a member π_U, of length ℓ_U, with fit $L(\pi_U) \le L(\pi) + \ell(\pi) + \kappa_0$ for every member π, then for every $\lambda > 1$ every λ-MAP has length at most $\ell_U + (\ell_U + \kappa_0)/(\lambda - 1)$. The mixture identity of Theorem 3.6 gives $M_A(D) \ge \sum_{p \in \mathrm{dom}\, V} 2^{-|p|} \nu_{G_{V(p)}}(D) \ge 2^{-|p|} \nu_{G_{V(p)}}(D)$ for each p, so a program p_U for G_U has fit $-\log \nu_{G_U}(D) = L_A(D) \le -\log \nu_{G_{V(p)}}(D) + |p|$ for every p. This is the hypothesis with $\kappa_0 = 0$. □

Remark G.5 (Pointwise interval laws: the separation is open). For every set D of interval data, both $L_A(D)$ and $\min_T J_1^R(T; D)$ over pointwise interval laws are at least $K(y_D \mid \mathrm{dom}\, D) - c$. For L_A: the answer conjunctions of the questions in $\mathrm{dom}\, D$ form a finite set S of pairwise B-incompatible sentences, computable from $\mathrm{dom}\, D$, and $M_A(D) = M_A(\{(\psi_D, 1)\})$ (Proposition 3.1); so the coding lower bound of Proposition 3.17 gives $L_A(D) \ge K(\psi_D \mid S) - c \ge K(y_D \mid \mathrm{dom}\, D) - c'$. For J_1^R: Proposition G.8(b). So neither bound separates the two. The natural candidate for a separation, data in which pairs of times share a random cell, has $L_A \approx n\gamma/2$; but it does not separate them either, because a presentation may store values: the program that stores the $n/2$ shared cells and returns each for both times of its pair has length $n\gamma/2 + O(\log n)$ and $\mathrm{Exc}^R = 0$, so $\min_T J_1^R \le n\gamma/2 + O(\log n)$ as well.

Question G.6. Fix a presentation language for pointwise interval laws. Is $\min_T J_1^R(T; D) - L_A(D)$ bounded over all sets D of interval data? (For K_A and for $\min_T J_1^Q$ over finite theories the answer is no, by Proposition G.4(c).)

Proposition G.7 (Which hypotheses of the companion paper on steeper priors hold here)**.**

(a) Pointwise interval laws in code R *(a fixed presentation language; one question family with finitely many cells). The codewords form a decidable prefix-free set; P_T^R is computable uniformly in T, is the law ν_{G_T} of a theory machine (a shared hypothesis in the sense of Hänni and Claude (Anthropic), 2026d, Definition 3.4) and, on data at distinct times, a product hypothesis (Hänni and Claude (Anthropic), 2026d, Definition 3.3); Kraft's inequality holds, lengths are integers, and finitely many laws have each length. So Lemmas A.1, 3.8, 3.9 and 5.3, Corollaries 5.4 and 5.5 and Theorem 5.6 of Hänni and Claude (Anthropic) (2026d) apply, with $J_\lambda = J_\lambda^R$.*

(b) Finite theories over a decidable consistent B, in code Q, *with $P_T(D) := 2^{-\mathrm{Exc}_D^Q(T)} \mathbf{1}[B \cup T \cup D^{\pm}$ consistent$]$. P_T satisfies the semimeasure inequalities of Hänni and Claude (Anthropic) (2026d, Definition 3.4), and it is a product over questions when consistency can*

be checked question by question (pointwise laws at distinct times). It is not lower semicomputable in general (for $B = \emptyset$ and a language with a binary relation symbol), so this class is not a hypothesis class of the companion paper. Its hull lemma (Lemma 3.8), kink lemma (Lemma 5.3), Corollary 5.4, Theorem 5.6 and Lemma A.1(i) use only the point family $(\|T\|, \mathrm{Exc}^Q_D(T))_T$ and apply. Its coding bound (Lemma 3.9) is replaced by Proposition G.8, with which Corollary 5.5 holds with offset $b + \kappa + n$ in general, and $b + \kappa$ when provability of the answers from the theories is decidable (κ the constant of Proposition G.8).

Proof. Which hypotheses are used. Each listed property is checked in the statement: for (a), $P^R_T = \nu_{G_T}$ on interval data by the proof of Proposition G.3(a) (the random theory proves "$q_t \in C$" iff $v_t \in \operatorname{int} C$, up to null events), and the product form holds because the U_i are independent. That the hull, kink and converse results need only the point family can be read off their statements, which concern countable point families, and their proofs, which use only lengths, fits, and the finiteness of the set of programs of each length.

The semimeasure inequalities in (b). $P_T(\emptyset) = \mathbf{1}[B \cup T \text{ consistent}] \le 1$. Let $\mathcal{Q} \notin \operatorname{dom} D$ and $P_T(D) > 0$. If T proves an answer a_0, every other answer is refuted by T and contributes 0, while a_0 adds no cost and keeps consistency; so the sum is $P_T(D)$. If T proves no answer, each of the at most $|\mathcal{Q}|$ answers consistent with $B \cup T \cup D^{\pm}$ contributes at most $2^{-\gamma(\mathcal{Q})} P_T(D) = P_T(D)/|\mathcal{Q}|$. If $P_T(D) = 0$, every term is 0.

Not lower semicomputable. Take $B = \emptyset$ and a language with a binary relation symbol. If $T \mapsto P_T(\emptyset)$ were lower semicomputable uniformly in T, the set of finite theories with $P_T(\emptyset) > \frac{1}{2}$, that is the consistent ones, would be r.e. The inconsistent ones are r.e. as well, so consistency of finite theories would be decidable, contrary to Church's theorem (standard; Church, 1936). □

Proposition G.8 (Coding bound for laws with literal exceptions)**.** *Let $n := |D|$.*

(a) For every finite T with $B \cup T \cup D^{\pm}$ consistent, $\|T\| + n + \mathrm{Exc}^Q_D(T) \ge K(y_D \mid \operatorname{dom} D) - c$.

(b) Suppose that provability of answers from T is decidable uniformly in T, for T ranging over a class of presented theories with $K(T) \le \|T\| + c$ (finite theories over a decidable complete B, or pointwise interval laws). Then $\|T\| + \mathrm{Exc}^Q_D(T) \ge K(y_D \mid \operatorname{dom} D) - c$ and $\|T\| + \mathrm{Exc}^R_D(T) \ge K(y_D \mid \operatorname{dom} D) - c$ for every member T consistent with D.

Proof. (a) For a question $\mathcal{Q}$ put

$$w_T(a \mid \mathcal{Q}) := \tfrac{1}{2}\mathbf{1}[a \text{ is the first answer of } \mathcal{Q} \text{ found provable from } B \cup T] + \tfrac{1}{2}|\mathcal{Q}|^{-1},$$

where "first" refers to a fixed dovetailed proof search. Then $\sum_a w_T(a \mid \mathcal{Q}) \le 1$, and w_T is lower semicomputable uniformly in $(T, \mathcal{Q})$, because the first answer found never changes. So $P'_T(y) := \prod_j w_T(y_j \mid \mathcal{Q}_j)$ is a semimeasure on the answer lists for $\operatorname{dom} D$, lower semicomputable uniformly in $(T, \operatorname{dom} D)$. If $B \cup T$ is consistent and proves y_j, it proves no other answer of $\mathcal{Q}_j$, so y_j is the first found and $w_T(y_j \mid \mathcal{Q}_j) \ge \frac{1}{2}$; otherwise $w_T(y_j \mid \mathcal{Q}_j) \ge \frac{1}{2}|\mathcal{Q}_j|^{-1}$. Hence $-\log P'_T(y_D) \le n + \mathrm{Exc}^Q_D(T)$. The conditional coding theorem ($K(y \mid z) \le -\log P(y \mid z) + c_P$ for a semimeasure $P(\cdot \mid z)$ lower semicomputable uniformly in z; standard, see Li and Vitányi, 2008) gives $K(y_D \mid T, \operatorname{dom} D) \le n + \mathrm{Exc}^Q_D(T) + c$, and $K(y_D \mid \operatorname{dom} D) \le K(T) + K(y_D \mid T, \operatorname{dom} D) + c \le \|T\| + n + \mathrm{Exc}^Q_D(T) + c'$.

(b) With decidable provability put $w_T(a \mid \mathcal{Q}) := 1$ if a is the first answer, in a fixed order of $\mathcal{Q}$, that T proves; $:= 0$ for the other answers that T proves; and $:= |\mathcal{Q}|^{-1}$ for every answer if T proves none. This sums to at most 1 for every T, consistent or not, and it is computable. For T consistent with D the cost bound of (a) holds without the flag bit, and the same coding

argument gives the first inequality. For code R, $P_T^R(\cdot \mid \operatorname{dom} D)$ is itself a semimeasure on answer lists, computable uniformly in $(T, \operatorname{dom} D)$ (Proposition G.7(a)), and the same argument gives the second. □

So the right line of slope 1 around a sufficient law (Hänni and Claude (Anthropic), 2026d, Corollary 5.5(a)) holds in both classes; in code Q without decidability it carries the flag cost n, which makes it useless, and for that reason the analyses of Section 9 use laws whose provability on the data is decidable.

G.3 Deterministic theories and precision

Proposition G.9 (Compatibility is monotone in logical strength)**.**

(a) Let $B \cup T$ be consistent and $\mathrm{Cn}(T) \supseteq \mathrm{Cn}(T')$. On every data set with which T' is compatible ($B \cup T'$ consistent and proving every datum), T is compatible. So among the deterministic hypotheses of W_A and M_A the posterior odds of T against T' equal their prior odds on every data set with which T' is compatible, and T' is incompatible whenever T is: data never move the odds towards the weaker theory.

(b) $T_0 \vdash T^\varepsilon$ for every $\varepsilon \geq 0$, so every data set compatible with the honest law T^ε is compatible with the exact law T_0, at every sample size and every precision. With prior weights $2^{-\|T\|}$ in a code in which the tolerance clause costs at least one bit, the posterior odds of T_0 against T^ε are at least $2^{\|T^\varepsilon\| - \|T_0\|} \geq 2$ on every such data set.

(c) If $\sup_t |e(t)| > 0$, then T_0 is false and, on a question at precision $\delta' < \sup_t |e(t)|$, proves a false answer at some times; T^ε with $\varepsilon \geq \sup_t |e(t)|$ is true and is silent wherever its band straddles a cell boundary.

Proof. A consistent theory that proves everything T' proves proves every datum that T' proves; this is (a), and (b) is (a) with $T = T_0$, $T' = T^\varepsilon$. (c) is immediate from the definitions. □

Proposition G.10 (Precision monotonicity)**.** *If $B \cup T$ proves "$q(t) \in [v - \delta', v + \delta']$", then it proves "$q(t) \in [v - \delta, v + \delta]$" for every $\delta \geq \delta'$. So compatibility with data at a fixed set of centres is monotone in the precision.*

Proof. The first interval is contained in the second. □

Proposition G.11 (Guaranteed tolerances)**.** *In regime G:*

(a) T_0 proves datum j iff $|e(t_j) - \zeta_j| \leq \delta$, and refutes it otherwise.

(b) If $\delta \geq \varepsilon^* + \delta_0$, then T_0 is compatible, at every n, with every such data set whose errors lie in the essential range (so almost surely).

(c) Suppose the pairs $(e(t_j), \zeta_j)$ are i.i.d., $e(t_j)$ is independent of ζ_j, both $\pm\delta_0$ lie in the support of the law of ζ_j, and $\delta < \varepsilon^* + \delta_0$. Then the per-datum refutation probability $r(\delta) := \mathbb{P}[|e - \zeta| > \delta]$ is positive, T_0 survives n data with probability $(1 - r(\delta))^n$, and it is almost surely refuted eventually.

So the switch point $\delta = \varepsilon^* + \delta_0$ is sharp.

Proof. (a) T_0 proves $q(t_j) = f(t_j) = v_j + e(t_j) - \zeta_j$. This point lies in $[v_j - \delta, v_j + \delta]$ iff $|e(t_j) - \zeta_j| \leq \delta$, and if it does not, T_0 proves the negation of the datum. (b) $|e - \zeta| \leq |e| + |\zeta| \leq \varepsilon^* + \delta_0$ almost surely. (c) Pick $\varepsilon' < \varepsilon^*$ and $s > 0$ with $\varepsilon' + \delta_0 - s > \delta$. Since ε^* is the essential supremum of $|e|$, $\mathbb{P}[e > \varepsilon'] > 0$ or $\mathbb{P}[e < -\varepsilon'] > 0$. In the first case $\mathbb{P}[\zeta < -\delta_0 + s] > 0$ (support), and independence

gives $\mathbb{P}[e - \zeta > \delta] > 0$; the second case is symmetric, with $\zeta > \delta_0 - s$. Refutations of distinct data are independent, so survival has probability $(1 - r(\delta))^n \to 0$, and a refuted theory stays refuted. □

Proof of Proposition 9.2. T_0 proves "$q(t_j)$ lies in the cell of $f(t_j)$" and so refutes every other cell. Two points at distance $|e| < \Delta$ fall into different cells of a uniformly shifted grid of width Δ with probability $|e|/\Delta$, and at distance at least Δ they always do; this gives the conditional proof probability $(1 - |e(t_j)|/\Delta)^+$. The refutation indicators of distinct data are independent (they depend only on each datum's own error and dither), which gives the product $(1 - p_\Delta)^n$ and the geometric law of the first refutation; and $p_\Delta > 0$ iff $\mathbb{P}[e \neq 0] > 0$. For $\Delta \geq \varepsilon^*$, $\min(|e|/\Delta, 1) = |e|/\Delta$ almost surely. □

Proposition G.12 (Which exact theory a proof-requiring inducer settles on)**.** *Let $T_1, \ldots, T_k$ be exact pointwise laws for the same quantity, with lengths $\|T_1\| < \cdots < \|T_k\|$ and errors $e_1, \ldots, e_k$, and let the inducer choose the shortest compatible theory in this pool (the MAP of any prior that decreases with length, or K_A restricted to the pool). Let the data be i.i.d.*

(G) Under the hypotheses of Proposition G.11(c) at precision δ, almost surely the choice is eventually $T_{i^*(\delta)}$, $i^*(\delta) := \min\{i : \varepsilon_i^* + \delta_0 \leq \delta\}$; if no i qualifies, every theory is eventually refuted. As δ decreases past the thresholds $\varepsilon_i^* + \delta_0$, the eventual choice moves to longer and more exact theories.

(R) In regime R, for every $\Delta > 0$, almost surely the choice is eventually the shortest T_i with $\mathbb{P}[e_i \neq 0] = 0$, whatever Δ is; if there is none, every theory is eventually refuted. At sample size n, T_i survives with probability $(1 - p_{\Delta,i})^n$, and for $\Delta \geq \varepsilon_i^*$ it is abandoned after about $\Delta/\mathbb{E}|e_i|$ data.

Proof. Apply Proposition G.11 and Proposition 9.2 to each T_i. The choice at n is the first survivor in order of length, and a finite pool has a last refutation time almost surely. □

Computations. In regime G, with $\varepsilon^* = 1'$, $\delta_0 = 0.5'$ and ζ uniform on $[-\delta_0, \delta_0]$, the refutation probability is exactly $r(\delta) = 0.218$ at $\delta = 1'$, 6.0×10^{-4} at $\delta = 1.49'$, and 0 for $\delta \geq 1.5'$, for a sinusoidal error and for an error equal to $\pm\varepsilon^*$ on 5% of the times and 0 otherwise. In regime R, simulated survival after 1, 10 and 100 data matches $(1 - p_\Delta)^n$ at every Δ from $0.5'$ to $600'$; for the sinusoid with $\varepsilon^* = 1'$ the mean first-refutation index is 94.1 against the exact $1/p_\Delta = \pi\Delta/2 = 94.25$ at $\Delta = 60'$, and 942.7 against 942.48 at $\Delta = 600'$. For the pool T_1 (length 100, $\varepsilon^* = 10'$), T_2 (length 150, $\varepsilon^* = 1'$), T_3 (length 400, exact), with sinusoidal errors and 2000 runs, every run chooses the predicted theory at $n = 10^4$ in regime G with $\delta_0 = 0.5'$ at each of $\delta = 20'$, $10.5'$, $5'$, $1.5'$, $1.2'$ (predictions T_1, T_1, T_2, T_2, T_3); in regime R every run ends at T_3, T_1 being abandoned after about 94 data at $\Delta = 10^\circ$ and 9.4 at $\Delta = 1^\circ$, and T_2 after about 942 and 94.

G.4 Honest laws, laws with exceptions, laws with error models

Forms and pricings. Work in regime R. The forms are: the empty law $\emptyset$; the *patched tolerance law* $T^{\varepsilon,-Z}$, the tolerance law T^ε withdrawn at the set Z of times of the data it refutes (fewer patches are inconsistent with the data, more only add cost); for $\varepsilon = 0$ the exact law, written T_0 while it has no patch and T_0^{-Z} after; for $\varepsilon \geq \varepsilon^*$ the honest law T^ε, which is true and never patched; and a true exact theory T_{tr}. Their lengths, relative to a common part, are $\|\emptyset\| = 0$, $\|T_0\| = \ell_0$,

$\|T^\varepsilon\| = \ell_0 + h$ (h the length of the tolerance clause and its numeral) and $\|T_{\mathrm{tr}}\| = \ell_0 + h_{\mathrm{tr}}$. A pricing fixes the price γ_{si} of a datum on which a form is silent and the price γ_{ex} of an exception:

- *Q/det*: code Q, each patch naming its time inside the law at c_{time} bits; $\gamma_{\mathrm{si}} = \gamma$ and $\gamma_{\mathrm{ex}} = \lambda c_{\mathrm{time}} + \gamma$. This is a finite-horizon cost model (Proposition G.15).
- *Q/rand*: code Q with randomised exceptions: a theory machine withdraws the law at each time independently with probability ρ, and the fit is charged $-\log(1-\rho)$ per kept datum and $-\log\rho + \gamma$ per exception. The share ρ can be stated in $O(\log n)$ bits or integrated out: drawing ρ uniformly from the tape and then withdrawing independently is a single theory machine, whose cost for k exceptions among n data is $\log\left((n+1)\binom{n}{k}\right) + k\gamma$. With $\rho = p_\Delta$, or integrated out, the rate is $H_2(p_\Delta) + p_\Delta\gamma$, where H_2 is the binary entropy, so $\gamma_{\mathrm{ex}} = \gamma + H_2(p_\Delta)/p_\Delta$. These are exact theory-machine costs for every n.
- *R/det* and *R/rand*: as above, with a silent datum priced by random completion inside the band (code R); an exception of a patched law is priced γ in code R too (Proposition G.3(c)).

SAI's pricing is R/rand: the honest law with random completion and the exact law with random exceptions are theory machines, and R/rand compares their weights in M_A's mixture.

Lemma G.13 (What a tolerance law does with one datum)**.** *In regime R, fix a datum with error e and let $\varepsilon \geq 0$. Over the dither, T^ε refutes the datum with probability* $\min\left((|e| - \varepsilon)^+/\Delta, 1\right)$*, proves it with probability* $\left(1 - (\varepsilon + \max(|e|, \varepsilon))/\Delta\right)^+$*, and is silent otherwise. In the* coarse regime $\Delta \geq \varepsilon + \max(\varepsilon, \varepsilon^*)$ *it is silent with probability exactly $2\varepsilon/\Delta$, whatever e is, and refutes with probability $(|e| - \varepsilon)^+/\Delta$. For $\varepsilon = 0$ it refutes with probability* $\min(|e|/\Delta, 1)$ *and is never silent.*

Proof. Place q at 0. The cell containing it is $[-u\Delta, (1-u)\Delta)$ with u uniform on $[0,1)$, and the band is $[e - \varepsilon, e + \varepsilon]$, since $f = q + e$; by symmetry let $e \geq 0$. *Refutation* means that the band misses the cell. If $e \leq \varepsilon$, the band contains 0, so this is impossible. If $e > \varepsilon$, the band lies in $(0, \infty)$, and it misses the cell iff $(1-u)\Delta \leq e - \varepsilon$, of probability $\min((e-\varepsilon)/\Delta, 1)$. *Proof of the datum* means that the band lies in the cell: $-u\Delta \leq e - \varepsilon$ and $e + \varepsilon < (1-u)\Delta$. If $e \leq \varepsilon$, this is $u \in [(\varepsilon - e)/\Delta, (\Delta - \varepsilon - e)/\Delta)$, of probability $(1 - 2\varepsilon/\Delta)^+$; if $e > \varepsilon$, the first condition holds automatically and the second is $u < (\Delta - e - \varepsilon)/\Delta$. Both cases equal $(1 - (\varepsilon + \max(e, \varepsilon))/\Delta)^+$. When $\Delta \geq \varepsilon + \max(\varepsilon, \varepsilon^*)$ the positive parts are inactive for every e in the essential range, and silence has probability $1 - \left(1 - (\varepsilon + \max(e, \varepsilon))/\Delta\right) - (\max(e, \varepsilon) - \varepsilon)/\Delta = 2\varepsilon/\Delta$. □

The coarse-regime condition cannot be weakened to $\Delta \geq \varepsilon + \varepsilon^*$ when $\varepsilon > \varepsilon^*$: with $|e| = \varepsilon^* = 1$, $\varepsilon = 2$ and $\Delta = 3$ the formula $2\varepsilon/\Delta$ would give a silence probability of $4/3$.

Proposition G.14 (Rates)**.** *In regime R, write $z_j := (e_j, u_j)$. Each form X has $J_\lambda(X; D_n) = a_X + \sum_{j \leq n} c_X(z_j)$ with a constant a_X and i.i.d. per-datum costs $c_X(z_j)$ (under Q/rand and R/rand with ρ stated; integrating ρ out adds $O(\log n)$). In the coarse regime $\Delta \geq 2\varepsilon^*$ the rates $m_X := \mathbb{E}c_X$ are: $m_\emptyset = \gamma$; $m_{T_{\mathrm{tr}}} = 0$; $m_{T_0} = \infty$ if $\mathbb{P}[e \neq 0] > 0$ (an unpatched exact law is eventually refuted); $m_{T_0^- z} = p_\Delta\gamma_{\mathrm{ex}}$, that is $p_\Delta(\lambda c_{\mathrm{time}} + \gamma)$ under the det pricings and $H_2(p_\Delta) + p_\Delta\gamma$ under the rand pricings, with $p_\Delta = \mathbb{E}|e|/\Delta$; and $m_{T^{\varepsilon^*}} = s_\Delta\gamma_{\mathrm{si}}$ with $s_\Delta = 2\varepsilon^*/\Delta$, that is $s_\Delta\gamma$ in code Q and $s_\Delta\beta_e$ in code R, β_e as in Lemma G.18.*

Proof. For the patched exact law at its cheapest patch set, a datum costs γ_{ex} if T_0 refutes it and 0 otherwise: under det the patch names the time ($\lambda c_{\mathrm{time}}$) and the datum is then literal (γ); under rand it is the flag code. Lemma G.13 with $\varepsilon = 0$ gives the probability p_Δ. The honest law costs γ_{si} on a datum iff it is silent, which in the coarse regime has probability $2\varepsilon^*/\Delta$ independently of e (Lemma G.13); in code R the expected cost given silence, averaged over the errors, is β_e (Lemma G.18(i)). □

Proposition G.15 (The price of naming exceptions deterministically)**.** *In regime R, let $Z_n \subseteq \{1,\dots,n\}$ be the indices of the data that the exact law refutes, so that the events $j \in Z_n$ are independent with probability $p_\Delta \in (0,1)$.*

(a) For every prefix-free code of the finite subsets of $\{1,\dots,n\}$, even one that depends on n, the expected code length of Z_n is at least $nH_2(p_\Delta)$; so the expected cost per exception is at least $H_2(p_\Delta)/p_\Delta$ bits, the cost of randomised exceptions.

(b) If the patched law names each patched time t_j by a numeral in a prefix-free code ℓ_{num} of numerals, then $\mathbb{E}\sum_{j\in Z_n}\ell_{\mathrm{num}}(t_j) \geq p_\Delta n\log n$, so the cost per exception is at least $\log n$ on average. If $\ell_{\mathrm{num}}(t_j) \geq \log j - c$ for all j (any standard numeral code), the cost per datum $\frac{1}{n}\sum_{j\in Z_n}\ell_{\mathrm{num}}(t_j)$ tends to ∞ almost surely.

(c) Hence Q/det and R/det with a constant c_{time} are cost models valid only while the times used cost about c_{time} bits to name, that is for $n \lesssim 2^{c_{\mathrm{time}}}$. Asymptotically, at fresh times, deterministic patching costs at least as much per exception as randomised exceptions, and literal patching costs unboundedly more; so the patched exact law under det is eventually beaten by the exact law with random exceptions.

Proof. (a) Let P_Z be the law of Z_n and ℓ' a prefix-free code. By Kraft's inequality $\sum_Z 2^{-\ell'(Z)} \leq 1$, and by Jensen's inequality, with H the entropy,

$$\mathbb{E}\ell'(Z_n) - H(Z_n) = -\sum_Z P_Z(Z)\log\frac{2^{-\ell'(Z)}}{P_Z(Z)} \geq -\log\sum_Z 2^{-\ell'(Z)} \geq 0,$$

where $H(Z_n) = nH_2(p_\Delta)$. The expected number of exceptions is $p_\Delta n$. (b) The numerals $t_1,\dots,t_n$ are distinct, so Kraft gives $\sum_{j\leq n} 2^{-\ell_{\mathrm{num}}(t_j)} \leq 1$, and Jensen gives $2^{-\frac{1}{n}\sum_j \ell_{\mathrm{num}}(t_j)} \leq \frac{1}{n}\sum_j 2^{-\ell_{\mathrm{num}}(t_j)} \leq \frac{1}{n}$, that is $\sum_{j\leq n}\ell_{\mathrm{num}}(t_j) \geq n\log n$. Since $\mathbb{P}[j \in Z_n] = p_\Delta$ for every j, $\mathbb{E}\sum_{j\in Z_n}\ell_{\mathrm{num}}(t_j) \geq p_\Delta n\log n$. If $\ell_{\mathrm{num}}(t_j) \geq \log j - c$, then $\frac{1}{n}\sum_{j\in Z_n}\ell_{\mathrm{num}}(t_j) \geq \frac{1}{n}\#\{j \in Z_n : j > n/2\}(\log n - 1 - c)$, and the count over n tends to $p_\Delta/2$ almost surely by the strong law of large numbers. (c) follows from (a), (b) and Proposition G.14. □

A computation (code in the accompanying repository) gives, at $p_\Delta = 0.01$, 15.0, 23.0 and 30.3 bits per exception for literal listing by Elias delta numerals at $n = 10^3$, 10^5 and 10^7, and 9.02, 8.13 and 8.08 bits for an enumerative code of the set, against $H_2(p_\Delta)/p_\Delta = 8.08$.

Proposition G.16 (The peakedness criterion)**.** *Let $\Delta \geq 2\varepsilon^*$, under any of the four pricings (the deterministic ones as a cost model).*

(a) The honest law T^{ε^*} has a lower rate than the exact law with exceptions T_0^{-Z} iff $\mathrm{pk}(e) < \gamma_{\mathrm{ex}}/(2\gamma_{\mathrm{si}})$.

(b) If the exact law with exceptions comes from a law f_1 with error e_1, and the honest law from a law f_2 of the same length up to $O(1)$, with tolerance $\varepsilon_2 \geq \operatorname{ess\,sup}|e_2|$ and $\Delta \geq 2\varepsilon_2$, then the honest law has the lower rate iff $\varepsilon_2/\mathbb{E}|e_1| < \gamma_{\mathrm{ex}}/(2\gamma_{\mathrm{si}})$.

Proof. By Proposition G.14, $s_\Delta\gamma_{\mathrm{si}} < p_\Delta\gamma_{\mathrm{ex}}$ iff $2\varepsilon^*\gamma_{\mathrm{si}} < \mathbb{E}|e|\,\gamma_{\mathrm{ex}}$. In (b), $s_\Delta = 2\varepsilon_2/\Delta$ (Lemma G.13) and $p_\Delta = \mathbb{E}|e_1|/\Delta$. □

Under Q/det the criterion reads $\lambda c_{\mathrm{time}} > \gamma(2\mathrm{pk}(e) - 1)$. Simulated rates for a sinusoid ($\mathrm{pk} = \pi/2$) with $c_{\mathrm{time}} = 22$ pick the exact law with exceptions at $\lambda = 1$ and the honest law at $\lambda = 2$ for $\Delta = 2'$, $3'$ and $10'$, and the honest law at both for $\Delta = 60'$, as the threshold

$(\lambda c_{\text{time}}+\gamma)/(2\gamma)$ predicts; for an error with pk $=20$ they pick the exact law with exceptions everywhere. Under Q/rand it reads $\text{pk}(e)<\frac{1}{2}+H_2(p_\Delta)/(2p_\Delta\gamma)$; since $H_2(p)/p\approx\log(\text{e}/p)$ for small p, the threshold grows only logarithmically in $1/p_\Delta$: at $\gamma=\log 360$ it is 1.17 at $p_\Delta=10^{-3}$, 1.76 at 10^{-6} and 2.93 at 10^{-12}. For the sinusoid it is 0.637, 0.742 and 0.971 at $\Delta=3'$, $10'$ and $60'$, so the exact law with random exceptions wins although the error is spread. Within the pool $\{\emptyset,T_0^{-Z},T^{\varepsilon^*}\}$ under R/rand the threshold is $(\gamma+H_2(p_\Delta)/p_\Delta)/(2\beta_e)$, roughly 5 to 30 for typical cell widths (9.2–9.3 for the sinusoid, 20.4–20.5 for the error with pk $=20$). This comparison is relative to its pool: the exact law with random exceptions completes its exceptions over the whole circle, while the equally short two-point completion HY(ρ) completes them inside the band, and against it the error with pk $=20$ is held as "exact except in about 10% of cases" at a third of the uniform band's rate (Proposition 9.3(d) and the computation after its proof).

Proposition G.17 (Optimal tolerance)**.** *Under Q/det in the regime $\Delta\geq 2\varepsilon^*$, the patched tolerance law $T^{\varepsilon,-Z}$ with $0\leq\varepsilon\leq\varepsilon^*$ has rate*

$$\bar m(\varepsilon)=\frac{2\varepsilon\gamma+(\lambda c_{\text{time}}+\gamma)\,\mathbb{E}(|e|-\varepsilon)^+}{\Delta}.$$

$\bar m$ is convex, with right derivative $\big(2\gamma-(\lambda c_{\text{time}}+\gamma)\mathbb{P}[|e|>\varepsilon]\big)/\Delta$, and its minimisers are the ε with $\mathbb{P}[|e|>\varepsilon]\leq\theta_{\text{ex}}\leq\mathbb{P}[|e|\geq\varepsilon]$, where $\theta_{\text{ex}}:=2\gamma/(\lambda c_{\text{time}}+\gamma)$. So the optimal law states the $(1-\theta_{\text{ex}})$-quantile of $|e|$ as its tolerance and lists the remaining share θ_{ex} of cases as exceptions. If $\theta_{\text{ex}}\geq 1$ ($\lambda c_{\text{time}}\leq\gamma$), the exact law with patches is optimal. The pure honest law ($\varepsilon=\varepsilon^*$, no exceptions) is optimal iff $\mathbb{P}[|e|\geq\varepsilon^*]\geq\theta_{\text{ex}}$, which never holds for an error law without an atom at its maximum.

Proof. A datum costs γ if it is not proved, and in addition λc_{time} if it is refuted (the patch). By Lemma G.13, with $\Delta\geq 2\varepsilon^*\geq\varepsilon+\varepsilon^*$, $\mathbb{P}[\text{not proved}]=(2\varepsilon+\mathbb{E}(|e|-\varepsilon)^+)/\Delta$ and $\mathbb{P}[\text{refuted}]=\mathbb{E}(|e|-\varepsilon)^+/\Delta$, which gives $\bar m$. The function $\varepsilon\mapsto\mathbb{E}(|e|-\varepsilon)^+$ is convex, with right derivative $-\mathbb{P}[|e|>\varepsilon]$ and left derivative $-\mathbb{P}[|e|\geq\varepsilon]$. This gives convexity and the derivatives of $\bar m$, and a convex function is minimised exactly where its right derivative is non-negative and its left derivative non-positive, which is the stated condition. The last two claims are the cases $\varepsilon=0$ and $\varepsilon=\varepsilon^*$ of it. □

Minimising $\theta\varepsilon+\mathbb{E}(|e|-\varepsilon)^+$ over ε is the quantile-loss fact, and in ν-support-vector regression the parameter ν likewise bounds the share of points outside an ε-tube whose width is a quantile of the residuals (cf. Schölkopf et al., 2000). What is specific here is the reading for axiom inducers: the tube is a tolerance clause, the points outside it are patched exceptions, and θ_{ex} is a ratio of literal costs. The steeper the prior, or the dearer the naming of an exception, the smaller θ_{ex}: the law states a wider tolerance and lists fewer exceptions. A computation at $\Delta=10'$, $c_{\text{time}}=22$ finds the grid minimiser at 0.52, 0.80 and 0.94 for a sinusoid of amplitude 1 at $\lambda=1,2,4$, against the predicted quantiles 0.496, 0.807 and 0.939.

Lemma G.18 (The code-R cost of the honest law, exactly)**.** *Let $\varepsilon\geq\varepsilon^*$ and let $w:=(\varepsilon-e)/(2\varepsilon)\in[0,1]$ be the position of the true value in the band. Put $\vartheta(z):=1-z+z\ln z$ for $0\leq z<1$ ($\vartheta(0)=1$) and $\vartheta(z):=0$ for $z\geq 1$. The expected code-R cost of a datum for T^ε, over the dither, is*

$$\bar c_\Delta(w)=\begin{cases}\dfrac{2\varepsilon}{\Delta}\big(\log\text{e}-H_2(w)\big) & \text{if } \Delta\geq 2\varepsilon,\\[2ex] \log\dfrac{2\varepsilon}{\Delta}+\log\text{e}\,\Big[\vartheta\Big(\dfrac{2\varepsilon w}{\Delta}\Big)+\vartheta\Big(\dfrac{2\varepsilon(1-w)}{\Delta}\Big)\Big] & \text{if } \Delta\leq 2\varepsilon.\end{cases}$$

(i) In the coarse regime the cost given silence is $\log \mathrm{e} - H_2(w)$. *Averaged over the errors,* $\gamma_{\mathrm{si}} = \beta_e := \log \mathrm{e} - \mathbb{E}H_2(w) \in [\log \mathrm{e} - 1, \log \mathrm{e}] = [0.443, 1.443]$ *bits.*

(ii) In the fine regime T^ε *is silent on every datum, and* $\log(2\varepsilon/\Delta) \le \bar{c}_\Delta(w) \le \log(2\varepsilon/\Delta) + \log \mathrm{e}$*; the upper bound is approached as* $w \to 0$ *at* $\Delta = 2\varepsilon$.

(iii) For every w, $\bar{c}_\Delta(w)$ *is continuous and strictly decreasing in* $\Delta \in (0, \infty)$.

Proof. Coarse regime. The band straddles a cell boundary with probability $2\varepsilon/\Delta$ (Lemma G.13, since $\varepsilon \ge \varepsilon^*$); otherwise it lies in a cell and the datum costs 0. Given a straddle, exactly one boundary lies in the band, at a relative position v uniform on $[0, 1]$ (a uniformly shifted lattice of spacing at least the band's length). The true cell contains the part of the band on the true value's side, of relative length v if $w < v$ and $1 - v$ otherwise, and random completion inside the band puts that much probability on it. So the expected cost given a straddle is, in nats,

$$\int_w^1 -\ln v \, dv + \int_0^w -\ln(1 - v) \, dv = 1 + w \ln w + (1 - w) \ln(1 - w),$$

which is $\log \mathrm{e} - H_2(w)$ in bits.

Fine regime. Let y be the length of the part of the true cell inside the band; the cost is $\log(2\varepsilon/y) = \log(2\varepsilon/\Delta) - \log(y/\Delta)$. With $a := 2\varepsilon w/\Delta$ and $b := 2\varepsilon(1 - w)/\Delta$, so that $a + b = 2\varepsilon/\Delta \ge 1$, the band is $[-a\Delta, b\Delta]$ around the true value, and with dither u the true cell is $[-u\Delta, (1 - u)\Delta)$, so $y/\Delta = \min(u, a) + \min(1 - u, b)$. If $a, b \ge 1$, then $y = \Delta$ and the excess $-\ln(y/\Delta)$ is 0. If $a < 1 \le b$, then $y/\Delta = 1$ for $u < a$ and $1 + a - u$ for $u \ge a$, so the expected excess is $\int_a^1 -\ln(1 + a - u) \, du = \int_a^1 -\ln s \, ds = \vartheta(a)$; the case $b < 1 \le a$ is symmetric. If $a, b < 1$, then $1 - b \le a$, and $y/\Delta = u + b$ for $u < 1 - b$, 1 on $[1 - b, a]$, and $a + 1 - u$ for $u > a$; the expected excess is $\int_b^1 -\ln s \, ds + \int_a^1 -\ln s \, ds = \vartheta(a) + \vartheta(b)$.

(i) is the coarse case, and $\mathbb{E}H_2(w) \in [0, 1]$. (ii) In the fine regime $2\varepsilon \ge \Delta$, so T^ε is silent on every datum by Lemma G.13. $\vartheta \ge 0$ and ϑ is decreasing on $[0, 1]$ ($\vartheta'(z) = \ln z$). In the last case $\vartheta(a) + \vartheta(b)$ is largest on $a + b = 1$, where it equals $1 - H_2(a) \ln 2 \le 1$; in the middle case $\vartheta(a) \le \vartheta(0) = 1$. The value 1 is approached as $a \to 0$, $b \to 1$. (iii) Write $r := 2\varepsilon/\Delta$ and the cost in nats. In the coarse regime ($r \le 1$) it is $r(1 - H_2(w) \ln 2)$, increasing in r because $1 - H_2(w) \ln 2 \ge 1 - \ln 2 > 0$. In the fine regime ($r \ge 1$) it is $\ln r + \vartheta(wr) + \vartheta((1 - w)r)$, and on each piece its derivative in r is $\frac{1}{r}(1 + \sum z \ln z)$, the sum over $z \in \{wr, (1 - w)r\}$ with $z < 1$; since $z \ln z \ge -1/\mathrm{e}$, this is at least $\frac{1}{r}(1 - 2/\mathrm{e}) > 0$. The cost is continuous in r (ϑ is continuous at 1), and at $r = 1$ the two forms agree, since $\vartheta(w) + \vartheta(1 - w) = 1 - H_2(w) \ln 2$. So it is strictly increasing in r, that is strictly decreasing in Δ. □

A computation (code in the accompanying repository) confirms the closed form against a 5000-point quadrature over the dither to 10^{-4} bits on 36090 cells, with largest fine-regime excess $1.442695 = \log \mathrm{e}$; the coarse averages are $\beta_e = 0.885$ for a sinusoid and 0.493 for an error equal to $\pm\varepsilon^*$ on 5% of the times.

Band-confined completions. The machine $\mathrm{BC}_\mathcal{G}$ of Section 9 draws $\hat{w}_i \sim \mathcal{G}$ by the inverse distribution function from the uniform variables U_i of the proof of Proposition G.3, and emits the value $f(t_i) - \varepsilon + 2\varepsilon\hat{w}_i$ by narrowing intervals for the continuous part of $\mathcal{G}$ and exactly at an atom. A computable $\mathcal{G}$ is, for instance, piecewise linear with rational knots plus finitely many atoms at rational points. The argument of Proposition G.3(a) gives the claims made in Section 9: $K(\mathrm{BC}_\mathcal{G}) \le \|T^\varepsilon\| + K(\mathcal{G}) + c$, the random theory is always consistent, and on data at fresh times $\nu_{\mathrm{BC}_\mathcal{G}}(D) = \prod_j \mathcal{G}(C'_j)$ (cell endpoints are null events for continuous dithers).

Proof of Proposition 9.3. (a) As in Lemma G.18: a cell boundary falls into the band with probability $2\varepsilon/\Delta$, at a relative position v uniform on $[0,1]$ and independent of w. With no boundary in the band the cell contains the band and the datum costs 0. With a boundary at v, the true cell contains $[0,v)$ if $w<v$, of $\mathcal{G}$-mass $\mathcal{G}(v)$, and $[v,1]$ if $w>v$, of mass $1-\mathcal{G}(v)$; this gives $c_{\mathcal{G}}$, which does not depend on Δ.

(b) Given v, the event $w<v$ has probability $\mathcal{F}(v)$, so the conditional expected cost is $\mathcal{F}(v)(-\log\mathcal{G}(v))+(1-\mathcal{F}(v))(-\log(1-\mathcal{G}(v)))\geq H_2(\mathcal{F}(v))$, with equality iff $\mathcal{G}(v)=\mathcal{F}(v)$ (Gibbs' inequality). Integrate over v. The uniform band has $\mathcal{G}(v)=v$, which equals $\mathcal{F}$ on $[0,1]$ iff w is uniform, that is iff e is uniform on $[-\varepsilon,\varepsilon]$.

(c) Let $P^\star$ be the true conditional law of the answer list given the question list. The errors are independent of the questions, and $q=f-e$, so the conditional probability of the observed cell of $q(t_j)$ is $\mathcal{F}(C_j')$; hence $P^\star(D_n)=\nu_{\mathrm{BC}_{\mathcal{F}}}(D_n)=\prod_{j\leq n}\mathcal{F}(C_j')$, and $-\log\nu_{\mathrm{BC}_{\mathcal{F}}}(D_n)$ is a sum of i.i.d. terms with finite mean $r_{\mathcal{F}}(\Delta)$; the strong law of large numbers gives the limit. Let G' be any theory machine. Answers to one question are pairwise B-incompatible, so for a fixed question list the events "the random theory is consistent and proves the answer list y" are disjoint across y; hence $\sum_y\nu_{G'}(y)\leq 1$ and $\mathbb{E}_{P^\star}[\nu_{G'}(D_n)/P^\star(D_n)\mid\text{questions}]\leq 1$. By Markov's inequality $\mathbb{P}[-\log\nu_{G'}(D_n)<-\log P^\star(D_n)-2\log n]\leq n^{-2}$, and by the Borel–Cantelli lemma, almost surely $-\log\nu_{G'}(D_n)\geq-\log P^\star(D_n)-2\log n$ for all large n. Dividing by n gives the lim inf bound, and $G'=G_U$ gives it for L_A. If $\mathcal{F}$ is computable, dominance gives $L_A(D_n)\leq K(\mathrm{BC}_{\mathcal{F}})+c-\log\nu_{\mathrm{BC}_{\mathcal{F}}}(D_n)$, hence the limit. In the coarse regime $r_{\mathcal{F}}(\Delta)=(2\varepsilon/\Delta)c_{\mathcal{F}}$ by (a) and the equality case of (b).

(d) Write $\sigma:=1-\rho$ for the weight of the atom at $\frac12$. The family is symmetric, $\mathcal{G}(1-v)=1-\mathcal{G}(v)$, so the cost given w equals the cost given $1-w$, and we may take $w\leq\frac12$. *Convexity.* For fixed (w,v) the probability assigned to the true side is affine in σ and positive for $\sigma\in[0,1)$, and $-\ln$ of a positive affine function is convex; so $c_{\mathrm{HY}(\rho)}$, an expectation of convex functions, is convex in σ, hence in $\rho\in(0,1]$. *Right derivative at $\sigma=0$, in nats.* For $v<w$ the true side has probability $1-(1-\sigma)v$, and the derivative of $-\ln$ of it at $\sigma=0$ is $-v/(1-v)\in[-1,0]$; for $w<v<\frac12$ the probability is $(1-\sigma)v$, with derivative $+1$; for $v>\frac12$ it is $\sigma+(1-\sigma)v$, with derivative $-(1-v)/v\in[-1,0]$. Difference quotients of convex functions decrease monotonically as $\sigma\downarrow 0$, so by monotone convergence the derivative of the expectation is the expectation of the derivatives:

$$c'(0^+)=\mathbb{E}\Big[-\int_0^w\tfrac{v}{1-v}\,dv+(\tfrac12-w)-\int_{1/2}^1\tfrac{1-v}{v}\,dv\Big]=\mathbb{E}\big[\ln(1-w)+1-\ln 2\big],$$

using $\int_0^w\frac{v}{1-v}dv=-\ln(1-w)-w$ and $\int_{1/2}^1\frac{1-v}{v}dv=\ln 2-\frac12$. For $w\leq\frac12$, $1-w=(1+|e|/\varepsilon)/2$, so $c'(0^+)=\mathbb{E}\ln(1+|e|/\varepsilon)+1-2\ln 2$. *The criterion.* If $c'(0^+)<0$, some $\sigma>0$ beats $\sigma=0$, which is the uniform band; if $c'(0^+)\geq 0$, convexity gives $c(\sigma)\geq c(0)$ for every σ. *Necessity.* $\ln(1+z)\geq z\ln 2$ on $[0,1]$ (a concave function lies above its chord), so $\mathbb{E}\ln(1+|e|/\varepsilon)\geq\ln 2/\mathrm{pk}_\varepsilon(e)$, and $c'(0^+)<0$ forces $\mathrm{pk}_\varepsilon(e)>\ln 2/(2\ln 2-1)$. *Sufficiency.* Jensen's inequality gives $\mathbb{E}\ln(1+|e|/\varepsilon)\leq\ln(1+1/\mathrm{pk}_\varepsilon(e))$, which is below $2\ln 2-1$ iff $\mathrm{pk}_\varepsilon(e)>\mathrm{e}/(4-\mathrm{e})$. *Boundary.* $\int_0^1\ln(1+z)\,dz=2\ln 2-1$. □

The following computation (synthetic errors, 2×10^5 samples, band $\varepsilon=\varepsilon^*$; code in the accompanying repository) gives per-straddle costs in bits in the coarse regime. SH is a 40-bin smoothed histogram fitted on a random half of the samples and evaluated on the other half, $c_{\mathcal{F}}$ is the bound of Proposition 9.3(b) on that half, and $c'(0^+)$ (in nats) is the derivative that decides whether some $\mathrm{HY}(\rho)$ with $\rho<1$ beats the uniform band.

error	pk	uniform	best HY	ρ	SH	$c_{\mathcal{F}}$	$c'(0^+)$
sinusoid	1.57	0.8854	0.8854	1	0.8514	0.8510	$+0.087$
$\pm\varepsilon^*$ on 5%	20.1	0.4924	0.1790	0.099	0.2006	0.1704	-0.352
$\lvert e\rvert$ uniform	2.00	0.7213	0.7213	1	0.7227	0.7227	≈ 0
0.25 or 1 (90/10%)	3.08	0.5832	0.5543	0.68	0.4797	0.4640	-0.116

So the two-point form beats the uniform band exactly when $c'(0^+) < 0$, and the fitted histogram comes within about 0.02 bits of the Gibbs bound of the half it did not see.

Proposition G.19 (Fine data)**.** *Let $\Delta \le 2\varepsilon^*$. The honest law T^{ε^*} is silent on every datum.*

(a) In code Q its rate is $\gamma = m_\emptyset$, so it never beats the empty law, and T_0^{-Z} beats the empty law iff $p_\Delta \gamma_{\mathrm{ex}} < \gamma$.

(b) In code R its rate is at most $\log(2\varepsilon^*/\Delta) + \log \mathrm{e}$, which is below $m_\emptyset = \gamma = \log(\mathsf{R}/\Delta)$ by at least $\log(\mathsf{R}/2\varepsilon^*) - \log \mathrm{e}$ bits per datum; this is positive iff $\varepsilon^* < \mathsf{R}/(2\mathrm{e})$, about 66.2° for a longitude.

(c) If moreover $\varepsilon^* < \mathsf{R}/(2\mathrm{e})$ and $\mathbb{P}[\lvert e\rvert < \Delta]\log(1/\Delta) \to 0$ as $\Delta \to 0$ (for instance, if $\lvert e\rvert$ has a bounded density near 0), then in code R the honest law has the least rate among $\emptyset$, T_0^{-Z} and T^{ε^*} for all small Δ.

Proof. Silence and (a) are Lemma G.13 and Proposition G.14's argument with silence probability 1. (b) is Lemma G.18(ii). (c) $m_{T_0^{-}z} \ge p_\Delta \gamma$, and $1 - p_\Delta = \mathbb{E}(1 - \lvert e\rvert/\Delta)^+ \le \mathbb{P}[\lvert e\rvert < \Delta]$, so $m_{T_0^{-}z} - m_{T^{\varepsilon^*}} \ge \log(\mathsf{R}/2\varepsilon^*) - \log \mathrm{e} - \mathbb{P}[\lvert e\rvert < \Delta]\log(\mathsf{R}/\Delta)$, which is positive for small Δ; and (b) handles the empty law. □

So under SAI's pricing an approximate law keeps its value at precisions finer than its error: it narrows each answer to its band, which is worth $\log(\mathsf{R}/2\varepsilon^*)$ bits per datum (about 13 bits for a $1'$ error on a full circle of longitude), and by Proposition 9.3 a completion shaped like the error narrows it further.

Theorem G.20 (Almost-sure selection)**.** *Let $\mathcal{X}$ be a finite set of forms with $J_\lambda(X; D_n) = a_X + \sum_{j \le n} c_X(z_j) + r_X(n)$, where the z_j are i.i.d., $c_X(z) \in [0, \infty]$, and $r_X(n) = o(n)$ almost surely ($r_X = 0$ for deterministic forms and for stated ρ; $O(\log n)$ when ρ is integrated out). Let $m_X := \mathbb{E}c_X \in [0, \infty]$, $m_{\min} := \min_X m_X < \infty$ and $\mathcal{X}^* := \{X : m_X = m_{\min}\}$.*

(i) Almost surely, for all large n every λ-MAP within $\mathcal{X}$ lies in $\mathcal{X}^*$. If $\mathcal{X}^* = \{X^*\}$, the λ-MAP is eventually X^* alone.

(ii) Let $X, Y \in \mathcal{X}^*$ with $r_X = r_Y = 0$. If $c_X - c_Y$ is almost surely finite, with finite variance, and not almost surely constant, then almost surely $J_\lambda(X; D_n) < J_\lambda(Y; D_n)$ for infinitely many n and $J_\lambda(Y; D_n) < J_\lambda(X; D_n)$ for infinitely many n: there is no eventual winner. If $c_X - c_Y = 0$ almost surely, a_X against a_Y decides at every n.

(iii) Let $r_X = r_Y = 0$, $m_Y < m_X < \infty$, $\mathbb{E}\lvert c_X - c_Y\rvert < \infty$ and $x := a_Y - a_X > 0$. Let τ_x be the first n with $J_\lambda(Y; D_n) < J_\lambda(X; D_n)$, ς_x the last n with $J_\lambda(Y; D_n) \ge J_\lambda(X; D_n)$, and $\mu := m_X - m_Y$. Then $\mathbb{E}\tau_x > x/\mu$; if moreover $c_X - c_Y \le b$ almost surely, then $\mathbb{E}\tau_x \le (x + b)/\mu$; and on almost every sample path, $\tau_x/x \to 1/\mu$ and $\varsigma_x/x \to 1/\mu$ as $x \to \infty$.

Proof. The probabilistic facts used are standard (see Durrett, 2019). (i) Let $X^* \in \mathcal{X}^*$; then $\frac{1}{n} J_\lambda(X^*; D_n) \to m_{\min}$ almost surely. Let $X \notin \mathcal{X}^*$. If $\mathbb{P}[c_X = \infty] > 0$, almost surely some datum has infinite cost, after which $J_\lambda(X; D_n) = \infty$. If $c_X < \infty$ almost surely and $m_X = \infty$, then

for each K, $\frac{1}{n}\sum_{j\le n}\min(c_X(z_j),K)\to\mathbb{E}\min(c_X,K)$, and $\mathbb{E}\min(c_X,K)\to\infty$ as $K\to\infty$, so $\frac{1}{n}J_\lambda(X;D_n)\to\infty$. If $m_X<\infty$, then $\frac{1}{n}(J_\lambda(X;D_n)-J_\lambda(X^*;D_n))\to m_X-m_{\min}>0$. Since $\mathcal{X}$ is finite, the claim follows. (ii) $S_n:=\sum_{j\le n}(c_X-c_Y)(z_j)$ is a random walk with mean-zero, non-degenerate increments of finite variance. For each K, $\mathbb{P}[S_n>K]\to\frac{1}{2}$ by the central limit theorem, so $\mathbb{P}[S_n>K \text{ infinitely often}]\ge\limsup_n\mathbb{P}[S_n>K]=\frac{1}{2}$. This event is invariant under finite permutations of the increments, so by the Hewitt–Savage 0–1 law its probability is 1. Intersecting over $K\in\mathbb{N}$ gives $\limsup_n S_n=+\infty$ almost surely, and likewise $\liminf_n S_n=-\infty$; so $J_\lambda(X;D_n)-J_\lambda(Y;D_n)=a_X-a_Y+S_n$ changes sign infinitely often. The degenerate case is immediate. (iii) $S_n:=\sum_{j\le n}(c_X-c_Y)(z_j)$ has i.i.d. integrable increments with mean $\mu>0$, and $J_\lambda(Y;D_n)<J_\lambda(X;D_n)$ iff $S_n>x$. So $\tau_x=\min\{n:S_n>x\}$ is a stopping time with $\mathbb{E}\tau_x<\infty$ (a random walk with positive drift), and Wald's identity gives $\mathbb{E}S_{\tau_x}=\mu\mathbb{E}\tau_x$ with $S_{\tau_x}>x$; if the increments are at most b, also $S_{\tau_x}\le x+b$. For the path statement fix $\epsilon\in(0,\mu)$; by the strong law there is $N(\omega)$ with $|S_n-n\mu|\le\epsilon n$ for $n\ge N$. For $x>\max_{n\le N}S_n$ we have $\tau_x-1\ge N$, and $S_{\tau_x}>x\ge S_{\tau_x-1}$ gives $x/(\mu+\epsilon)<\tau_x\le x/(\mu-\epsilon)+1$. Similarly $\varsigma_x\ge\tau_x-1\ge N$ and $S_{\varsigma_x}\le x$ give $\varsigma_x\le x/(\mu-\epsilon)$. Divide by x, let $x\to\infty$ and then $\epsilon\to0$. □

The code-R cost of a silent datum is $-\log$ of a share that is uniform near 0 (the proof of Lemma G.18), so it has all moments, and (i)–(iii) cover every application in this section. The bounded-increment bound of (iii) applies to the race between T_0^{-Z} and T^{ε^*}, whose cost differences are bounded above by the largest cost of T_0^{-Z}; for the race between T^{ε^*} and T_{tr} in code R the increments are unbounded, and only the lower bound and the path limits apply.

Corollary G.21 (The phases of selection, coarse data)**.** *Take a pricing with i.i.d. per-datum costs as in Proposition G.14: exactly Q/rand and R/rand (with ρ stated), or Q/det and R/det as a constant-c_{time} cost model (Proposition G.15). Assume $\Delta\ge2\varepsilon^*$, $\mathbb{P}[e\ne0]>0$, $\varepsilon^*>0$ and $h_{\mathrm{tr}}>h$. The pool is $\{\emptyset,T_0 \text{ (and } T_0^{-Z}),T^{\varepsilon^*},T_{\mathrm{tr}}\}$, with $a_\emptyset=0$, $a_{T_0}=\lambda\ell_0$, $a_{T^{\varepsilon^*}}=\lambda(\ell_0+h)$ and $a_{T_{\mathrm{tr}}}=\lambda(\ell_0+h_{\mathrm{tr}})$. Almost surely:*

- *(a) the exact law is refuted for the first time at a geometric index of mean $1/p_\Delta$, and until then T_0 and T_0^{-Z} coincide;*
- *(b) T_{tr} is eventually the unique λ-MAP; the first time it beats T^{ε^*} has mean above $\lambda(h_{\mathrm{tr}}-h)/m_{T^{\varepsilon^*}}$, and at most $(\lambda(h_{\mathrm{tr}}-h)+\gamma)/m_{T^{\varepsilon^*}}$ under code Q, where the costs of T^{ε^*} are bounded by γ;*
- *(c) without T_{tr}, the eventual λ-MAP is the form of least rate among $m_\emptyset$, $m_{T_0^{-}z}$ and $m_{T^{\varepsilon^*}}$ (T^{ε^*} against T_0^{-Z} by Proposition G.16, ties by Theorem G.20(ii));*
- *(d) the expected switch points are: $\emptyset\to T_0$ at $\lambda\ell_0/\gamma$ (deterministic, if T_0 is not yet refuted); $T_0\to T_0^{-Z}$ at the first refutation, of mean $1/p_\Delta$; $T_0^{-Z}\to T^{\varepsilon^*}$, if $m_{T_0^{-}z}>m_{T^{\varepsilon^*}}$, at a time of mean above $\lambda h/(m_{T_0^{-}z}-m_{T^{\varepsilon^*}})$, and at most $(\lambda h+\gamma_{\mathrm{ex}})/(m_{T_0^{-}z}-m_{T^{\varepsilon^*}})$ when γ_{ex} is bounded; and $T^{\varepsilon^*}\to T_{\mathrm{tr}}$ as in (b).*

When these expected switch points increase in this order, the predicted order of the λ-MAPs is $\emptyset\to T_0\to T_0^{-Z}\to T^{\varepsilon^*}\to T_{\mathrm{tr}}$.

Proof. Theorem G.20 with the costs of Proposition G.14 and Lemma G.13: refutation by T_0 has probability p_Δ per datum, independently, and while unrefuted T_0 proves every datum (it is never silent), so $J_\lambda(T_0;D_n)=\lambda\ell_0<n\gamma=J_\lambda(\emptyset;D_n)$ once $n>\lambda\ell_0/\gamma$. T_{tr} has rate 0 and every other form a positive rate ($p_\Delta>0$, $s_\Delta>0$, $\gamma_{\mathrm{si}}\ge\log\mathrm{e}-1>0$), which gives (b) with Theorem G.20(iii) at level $\lambda(h_{\mathrm{tr}}-h)$. The bound in (d) for $T_0^{-Z}\to T^{\varepsilon^*}$ is Theorem G.20(iii) applied to the race between these two forms from $n=0$, at level λh. □

A simulation of the Q/det cost model ($\ell_0 = 200$, $h = 20$, $h_{\mathrm{tr}} = 2000$, $c_{\mathrm{time}} = 22$; 400 runs per setting; code in the accompanying repository) shows the five phases in this order when the predicted switch points are ordered: for a sinusoidal error at $\Delta = 60'$ and $\lambda = 1$ (rates 0.323 and 0.283) the mean switch time to the honest law is 529.8, inside the interval $[495.6, 1251.3]$ given by (d), the mean switch time to the true theory is 7010.1, inside $[6994.9, 7024.9]$, and the mean first refutation is 91.8 against 94.25; when the two rates are close, the patched and the honest law alternate several times before the honest law settles. With rates tuned to be equal, the sign of $J_\lambda(T_0^{-Z}) - J_\lambda(T^{\varepsilon^*})$ changes 148 times on average up to $n = 2 \times 10^5$, and 43% of runs still change sign after $n = 10^5$. Under SAI's pricing with exact theory-machine costs (R/rand, sinusoid, $\varepsilon^* = 1'$, $\Delta = 60'$, $\lambda = 1$; rates 0.1749 and 0.02953), the mean first time the honest law is the λ-MAP is 133.6 against a predicted switch near 138, and the mean first time the true theory is is 67060 against 67055; all 400 runs reach it.

Proposition G.22 (Humility)**.** *Let $\Pi_\lambda(T \mid D) \propto 2^{-J_\lambda(T;D)}$ be the power-prior posterior (Hänni and Claude (Anthropic), 2026d, Definition 3.6) restricted to a finite pool of deterministic forms (code Q) containing $\emptyset$, T_0^{-Z}, T^{ε^*} and possibly T_{tr}, and ask a new question at a fresh time t', with cell width $\Delta' \le 2\varepsilon^*$.*

(a) The pool's three-valued prediction (the analogue, for the pool, of the trichotomy of Proposition 3.1) puts mass $\Pi_\lambda(T^{\varepsilon^*} \mid D) + \Pi_\lambda(\emptyset \mid D)$ on "independent", mass $\Pi_\lambda(T_0^{-Z} \mid D)$ on the cell of $f(t')$, which is wrong with probability $p_{\Delta'}$ over the dither, and mass $\Pi_\lambda(T_{\mathrm{tr}} \mid D)$ on the true cell.

(b) Without T_{tr}, if the least rate at the training cell width Δ is attained by one form only, the posterior concentrates on it almost surely: the inducer ends humble (independent mass tending to 1 on finer questions) if T^{ε^*} or $\emptyset$ is selected, and overconfident (a wrong cell with probability $p_{\Delta'}$) if T_0^{-Z} is; Proposition G.16 at Δ and Proposition G.19 decide which. If two forms tie in rate with a non-degenerate cost difference of finite variance, the posterior oscillates between them forever.

(c) In a randomised pool (code R) a band-confined completion $\mathrm{BC}_{\mathcal{G}}$ is not silent: it puts on each cell its $\mathcal{G}$-mass, a spread over the cells its band meets and never probability one on a wrong cell; the exact law with random exceptions puts probability $1 - \rho$ on the cell of $f(t')$.

Proof. (a) T^{ε^*} is silent on every question with $\Delta' \le 2\varepsilon^*$ (Lemma G.13), and $\emptyset$ is silent. T_0^{-Z} proves the cell of $f(t')$, since t' is fresh and so not a patched time, and T_{tr} proves the true cell. (b) Theorem G.20(i), (ii), with $\Pi_\lambda(X \mid D)/\Pi_\lambda(Y \mid D) = 2^{-(J_\lambda(X;D) - J_\lambda(Y;D))}$, whose exponent diverges linearly when the rates differ. (c) The definition of $\mathrm{BC}_{\mathcal{G}}$, and the withdrawal probability ρ. □

The Kepler residuals. Table 8 summarises the error laws. For each body and window, one fixed Kepler ellipse (seven elements) was fitted four ways (three-dimensional least squares, least squares on longitude and latitude, a robust fit for small mean error, and an L_p continuation for small maximum error), and two series of longitude errors at 8000 uniform times were kept: that of the fit with the least sample maximum (the *minimax fit*) and that of the fit with the least sample mean (the *mean fit*). The ephemerides are plan94 for the planets and moon98 for the Moon. An honest law whose tolerance is the sample maximum is slightly false if the window's true supremum is larger. A local parabola through the three time-sorted samples around the maximum estimates the excess at a median 2×10^{-4} (relative) and at most 1.9% in 33 of the

body	ε^*, minimax fit (12 / 100 / 2000 yr)	pk, minimax fit	pk, mean fit
Mercury	$0.38'$ / $2.02'$ / $37.4'$	2.80–3.19	3.29–4.71
Venus	$0.36'$ / $0.47'$ / $3.6'$	2.27–3.65	2.81–4.07
Earth–Moon barycentre	$0.27'$ / $0.62'$ / $7.3'$	2.34–3.95	2.91–4.03
Mars	$1.44'$ / $3.04'$ / $49.9'$	2.71–3.38	3.33–4.47
Jupiter	$0.70'$ / $6.57'$ / $34.0'$	1.60–2.65	3.00–7.08
Saturn	$0.17'$ / $7.46'$ / $81.9'$	1.66–2.39	3.21–6.91
Moon (1 month / 1 yr / 18.6 yr)	$30.9'$ / $156'$ / $480'$	1.89–2.21	1.89–3.17

Table 8: Longitude errors of one fixed Kepler ellipse per body and window, against the plan94 and moon98 ephemerides (not the sky): the largest error of the minimax fit, and the peakedness of each fit over the windows of 12 to 2000 years (for the Moon, one month to 18.6 years). Computed; code in the accompanying repository.

39 series apart from one, Mercury over 2000 years, where the estimate is unreliable because the sampling interval of about 91 days is comparable to Mercury's 88-day period. The band machines therefore use $\varepsilon = (1 + \kappa_{\mathrm{b}})\varepsilon^*$ with $\kappa_{\mathrm{b}} \in \{0, 0.01, 0.05\}$, and every verdict below is the same for the three values. Within the pool $\{\emptyset, T_0^{-Z}, T^{\varepsilon^*}\}$ under R/rand (and under R/det at $\lambda = 1$ and 4), the uniform band has the least rate in all 429 cells, with exact closed-form costs; in that pool's tolerance family with whole-circle exceptions the optimum is a tolerance of about 0.7–1.0 ε^* with at most 1% of the data as exceptions. With band-confined completions, the histogram machine SH has the least rate in all 429 cells at each κ_{b}, between 0.4% and 25.7% below the uniform band (median 6.6%), and lies within 0.0011 bits of the Gibbs bound of its test half in every coarse-regime pair; the best two-point form beats the uniform band in 379 of the 429 cells at $\kappa_{\mathrm{b}} = 0.01$, and in the coarse regime in 34 of the 39 body–window pairs, with best ρ between 0.52 and 0.997, exactly where the criterion of Proposition 9.3(d) says. The five pairs in which the uniform band beats every two-point form are Jupiter over 12 and 30 years, Saturn over 100 years and the Moon over one month and 18.6 years, whose minimax-fit peakedness is 1.60–1.96. The gain persists at fine precision: for the Earth–Moon barycentre over 100 years at $\Delta = 1'$ the rates of the uniform band, the best two-point form and SH are 0.651, 0.560 and 0.503 bits per datum, and for Mars over 12 years 1.660, 1.558 and 1.496. Under Q/det (naming a time costs $c_{\mathrm{time}} = \log(\text{days in the window}) + 2$ bits, 14.1–21.5 for the planets and 6.8, 10.5 and 14.7 for the Moon's windows; a finite-horizon cost model) the threshold $(\lambda c_{\mathrm{time}} + \gamma)/(2\gamma)$ grows as the cells coarsen. At $\lambda = 1$ the exact law with exceptions wins from cell widths of about ε^* upwards, the honest law only at the coarsest cells (30°, and from 10° for Jupiter over 30 years and Saturn over 2000 years) and not even there for 17 of the 39 pairs, and at finer cells no law pays; the best member of the tolerance family is then a small tolerance (0.05–0.5 ε^*) with exception shares up to tens of per cent. At $\lambda = 2$ the honest law takes over at cell widths from $8.1'$ to $962'$ (median $134'$), and at $\lambda = 4$ from $0.72'$ to $962'$ (median $13.4'$). Under Q/rand the exact law with random exceptions wins almost everywhere.

G.5 Derivation horizons and sound reasoning with approximate models

Work over $B = \mathrm{RCF}$ with constants $x_0, x_1, \ldots$ for the states of a system in $\mathbb{R}^m$ (one constant per coordinate), a rational initial state a, a map Ψ definable in B, and a norm $|\cdot|$ for which B proves that Ψ is L-Lipschitz (for a linear $\Psi(x) = Ax$ with rational A and the sup-norm, $L = \|A\|_\infty$, provably in RCF). The *honest dynamical law* is $T_\Psi^\varepsilon := \{|x_0 - a| \le \varepsilon^{(0)}\} \cup \{|x_{s+1} - \Psi(x_s)| \le \varepsilon :$

$s \geq 0\}$, and the *exact law* is $T^0_\Psi := \{x_0 = a\} \cup \{x_{s+1} = \Psi(x_s) : s \geq 0\}$.

Proposition G.23 (Derivation horizon)**.**

(a) For every $L \geq 0$, $T^\varepsilon_\Psi \vdash |x_r - \Psi^r(a)| \leq \varepsilon^{(r)}$, with $\varepsilon^{(r)} := L^r\varepsilon^{(0)} + \varepsilon(L^r - 1)/(L - 1)$ for $L \neq 1$ and $\varepsilon^{(r)} := \varepsilon^{(0)} + r\varepsilon$ for $L = 1$.

(b) Let $\Psi(x) = Ax$ be linear, with sup-norm tolerances. The values of the i-th coordinate $x_{r,i}$ over the models of $B \cup T^\varepsilon_\Psi$ form exactly the interval $(A^r a)_i \pm h_{r,i}$, where $h_{r,i} := \varepsilon^{(0)}\|(A^r)_{i\cdot}\|_1 + \varepsilon \sum_{s<r}\|(A^s)_{i\cdot}\|_1$ (the projection of a zonotope). So T^ε_Ψ proves "$x_{r,i} \in [v - \delta, v + \delta]$" iff that interval contains $(A^r a)_i \pm h_{r,i}$. In dimension one, $h_r = \varepsilon^{(r)}$ with $L = |A|$, so (a) is tight.

(c) Put $r^*(\delta) := \max\{r : \varepsilon^{(r)} \leq \delta\}$. For $\varepsilon^{(0)} = 0$: $r^*(\delta) = \lfloor \delta/\varepsilon \rfloor$ for $L = 1$; $r^*(\delta) = \lfloor \log_L(1 + \delta(L - 1)/\varepsilon) \rfloor$ for $L > 1$; and for $L < 1$, $r^*(\delta) = \infty$ iff $\delta \geq \varepsilon/(1 - L)$, and $r^*(\delta) = \lfloor \log_L(1 - \delta(1 - L)/\varepsilon) \rfloor$ otherwise. For every L-Lipschitz Ψ, T^ε_Ψ proves a datum at precision δ about x_r, centred at $\Psi^r(a)$, if $r \leq r^*(\delta)$; for linear Ψ in dimension one with $L = |A|$, also only if.

Proof. (a) By induction, $|x_{s+1} - \Psi^{s+1}(a)| \leq |x_{s+1} - \Psi(x_s)| + |\Psi(x_s) - \Psi(\Psi^s(a))| \leq \varepsilon + L\varepsilon^{(s)}$; each step is an instance of the axioms and of the provable Lipschitz inequality, and the recursion $\varepsilon^{(s+1)} = L\varepsilon^{(s)} + \varepsilon$ has the stated solution. (b) $x_r - A^r a = A^r(x_0 - a) + \sum_{s<r} A^{r-1-s}u_s$ with $u_s := x_{s+1} - Ax_s$, and in the real model the deviations $x_0 - a$ and u_s range freely over sup-norm boxes of half-widths $\varepsilon^{(0)}$ and ε. A linear image of a product of boxes is a zonotope, which is convex; its projection on coordinate i is the interval of half-width the sum of the ℓ_1-norms of the corresponding rows, with the extreme values attained by sign choices. So exactly the stated interval of values occurs in $\mathbb{R}$. Since RCF is complete and the claim about $x_{r,i}$ involves only the finitely many axioms with $s < r$, T^ε_Ψ proves an interval statement about $x_{r,i}$ iff it holds in every real model, that is iff the interval contains $(A^r a)_i \pm h_{r,i}$. (c) Solve $\varepsilon^{(r)} \leq \delta$; the "if" is (a), and the "only if" for linear Ψ in dimension one is (b). □

For non-linear Ψ the "only if" fails. Let $\Psi(x) = 0$ on $[-1, 1]$ and $\Psi(x) = 2(x - \operatorname{sign} x)$ outside, so that $L = 2$, with $a = 0$, $\varepsilon^{(0)} = 0$ and $\varepsilon = 1/10$. Then $|x_s| \leq 1/10 \leq 1$ gives $\Psi(x_s) = 0$ and so $|x_{s+1}| \leq 1/10$; by induction T^ε_Ψ proves "$x_r \in [-1/10, 1/10]$" at every depth, while $\varepsilon^{(r)} = (2^r - 1)/10$ gives $r^*(1/10) = 1$. A computation (code in the accompanying repository) confirms the closed form against exact interval propagation for $r \leq 200$ and several L between 0.5 and 2; at $\delta/\varepsilon = 10^4$ the horizons are 10^4, 463, 72 and 13 for $L = 1$, 1.01, 1.1 and 2, and infinite for $L \leq 0.99$. In dimensions 3 and 10, for random matrices of spectral radius 0.95–1.05, propagating boxes by interval arithmetic overestimates the exact tolerance $h_{r,i}$ by a median factor of 5–300 at $r = 10$ and up to 10^9 at $r = 30$, and the ∞-norm Lipschitz bound by up to 10^{13}: a reasoner that propagates boxes has a much shorter effective horizon than the theory.

Proposition G.24 (The horizon of an exact approximation)**.** *Suppose the true trajectory satisfies T^ε_Ψ, but the inducer holds the exact law T^0_Ψ with the same a. Then T^0_Ψ is consistent and proves $x_r = \Psi^r(a)$, which is within $\varepsilon^{(r)}$ of the true x_r. If $r \leq r^*(\delta)$, the depth-r conclusions of T^0_Ψ are δ-correct for every trajectory compatible with the honest law. For linear Ψ in dimension one with $L = |A|$, some trajectory compatible with the honest law attains the deviation $\varepsilon^{(r)}$, so in that case they are δ-correct for all such trajectories only if $r \leq r^*(\delta)$. And T^0_Ψ together with one true datum about x_r whose interval excludes $\Psi^r(a)$ is inconsistent, and so proves every sentence.*

Proof. The exact trajectory is a model of T^0_Ψ, and the true trajectory is a model of T^ε_Ψ. The middle claims are Proposition G.23(a), (b). For the last, the datum and $x_r = \Psi^r(a)$ are jointly refutable in B. □

Proposition G.25 (Sound reasoning with approximate models)**.** *Let $T_i := \{\forall t \in \mathrm{Dom}_i : \ |q(t) - f_i(t)| \le \varepsilon_i\}$, $i \le k$, be honest pointwise models of one quantity q (scalar, or vector-valued with a norm), with q otherwise unconstrained, and let T_i^0 be their exact versions ($\varepsilon_i = 0$).*

(a) Let B be true in the intended structure. If every T_i is true, then $\bigcup_i T_i$ is consistent and every sentence derivable from $B \cup \bigcup_i T_i$ is true; every derived statement about $q(t)$ is a set containing the true value.

For (b)–(d) assume moreover that B decides every sentence built from the f_i, the Dom_i and the ε_i at the times concerned (for example $B = \mathrm{RCF}$, with semialgebraic f_i and Dom_i and rational ε_i).

(b) $\bigcup_i T_i^0$ is inconsistent, and so proves every sentence, iff $f_i(t) \ne f_j(t)$ for some $t \in \mathrm{Dom}_i \cap \mathrm{Dom}_j$.

(c) $\bigcup_i T_i$ is consistent iff for each t the balls $\{v : |v - f_i(t)| \le \varepsilon_i\}$ with $t \in \mathrm{Dom}_i$ have a common point. For scalar q, and for vector q with sup-norm (box) tolerances, this holds iff it holds pairwise, $|f_i(t) - f_j(t)| \le \varepsilon_i + \varepsilon_j$ on every overlap. For Euclidean tolerances in $\mathbb{R}^m$, $(m+1)$-wise intersection is necessary and sufficient, and pairwise intersection is not sufficient.

(d) If B proves that a map Ψ is L_Ψ-Lipschitz, then $T_i \vdash |\Psi(q(t)) - \Psi(f_i(t))| \le L_\Psi \varepsilon_i$.

Proof. (a) The intended structure is a model, and derivations preserve truth. (b) If two exact models disagree at a common time t, together they prove $f_i(t) = f_j(t)$, which B refutes by the hypothesis. If they agree on all overlaps, then in every model of B (in which B's decisions fix the values $f_i(t)$) setting $q(t) := f_i(t)$ for $t \in \mathrm{Dom}_i$, and arbitrarily elsewhere, gives a model. (c) The constants $q(t)$ for distinct t are distinct and otherwise unconstrained, so by compactness $B \cup \bigcup_i T_i$ is consistent iff for each t, B is consistent with "some v lies in all the balls at t". By the hypothesis B decides this sentence, and B is true in the intended structure, so it is consistent with B iff it holds there, that is iff the balls have a common point. Intervals that meet pairwise have $\max_i$(left end) $\le \min_j$(right end), which is a common point (Helly's theorem in dimension one), and boxes are products of intervals. Helly's theorem in $\mathbb{R}^m$ (standard; Helly, 1923) gives the $(m+1)$-wise criterion. Three discs of radius 1.1 centred at the vertices of an equilateral triangle of side 2 meet pairwise but have no common point, since the circumradius is $2/\sqrt{3} \approx 1.155$. (d) is the Lipschitz inequality in B. □

Without the hypothesis on B, (b) and (c) fail: take $B = \emptyset$, $f_i(t)$ and $f_j(t)$ distinct constant symbols and $\mathrm{Dom}_i = \mathrm{Dom}_j$ a single time; then $T_i^0 \cup T_j^0$ is consistent (interpret the two constants as equal), although the values differ in the intended structure. A computation finds pairwise and joint overlap in agreement on 10^5 random families of 2 to 6 intervals and on 2×10^4 random families of boxes in $\mathbb{R}^2$.

G.6 Best systems

Proposition G.26 (Marginal slopes)**.** *Let T^* be a λ-MAP within a family of theories that contains $T^* \setminus \mathcal{A}$ and $T^* \cup \mathcal{B}$ for the modules considered. For every module $\mathcal{A} \subseteq T^*$, $\mathrm{Exc}_D(T^* \setminus \mathcal{A}) - \mathrm{Exc}_D(T^*) \ge \lambda \|\mathcal{A}\|$; and for every $\mathcal{B}$ disjoint from T^*, $\mathrm{Exc}_D(T^*) - \mathrm{Exc}_D(T^* \cup \mathcal{B}) \le \lambda \|\mathcal{B}\|$.*

Proof. $J_\lambda(T^*; D) \le J_\lambda(T^* \setminus \mathcal{A}; D)$ and $J_\lambda(T^*; D) \le J_\lambda(T^* \cup \mathcal{B}; D)$, with $\|T^*\| - \|T^* \setminus \mathcal{A}\| = \|\mathcal{A}\|$ and $\|T^* \cup \mathcal{B}\| - \|T^*\| = \|\mathcal{B}\|$. □

Proof of Proposition 9.5. By the hypotheses (Proposition G.7) the hull lemma (Hänni and Claude (Anthropic), 2026d, Lemma 3.8(v)) applies: for $\lambda \in (\sigma_{i+1}, \sigma_i)$ the λ-MAPs are exactly the theories at v_i, and at $\lambda = \sigma_j$ ($j \geq 1$) they are the theories on the closed edge $[v_{j-1}, v_j]$. (b)⇒(a): $(\tau, \sigma) \subseteq (\sigma_{i+1}, \sigma_i)$. (a)⇒(b): if some edge slope σ_j lay in (τ, σ), the λ-MAP set at $\lambda = \sigma_j$, the theories on $[v_{j-1}, v_j]$, would differ from the set just below σ_j, the theories at v_j only, since $v_{j-1} \neq v_j$. So no edge slope lies in (τ, σ), and (τ, σ) lies in one interval (σ_{i+1}, σ_i), which is (b). (b)⇔(c) is Hänni and Claude (Anthropic) (2026d, Corollary 5.4(c)): a zero-window (σ, τ)-kink with $\tau < \sigma$ is a vertex v_i with $(\tau, \sigma) \subseteq (\sigma_{i+1}, \sigma_i)$, and conversely every vertex v_i is a zero-window (σ_i, σ_{i+1})-kink, hence a (σ, τ)-kink whenever $\sigma \leq \sigma_i$ and $\tau \geq \sigma_{i+1}$. □

A computation on 400 random point families with 10 random intervals each, testing "the λ-MAP set is constant on (τ, σ)" exactly at every crossing point and between consecutive ones, agrees with (b) in all 4000 cases.

Proposition G.27 (The trivialisation is blocked)**.** *Let $\mathcal{L}' \supseteq \mathcal{L}$ be any extension of the language, for example by a predicate* Act *true of exactly the actual things, with the axiom $\forall x\, \mathrm{Act}(x)$; fix a syntax code for $\mathcal{L}'$, and let the data be base-language answers whose provability from the theories considered is decidable. Then every finite theory T in $\mathcal{L}'$ consistent with D has $\|T\| + \mathrm{Exc}_D^{\mathcal{Q}}(T) \geq K(y_D \mid \mathrm{dom}\, D) - K(\mathcal{L}') - c$, where $K(\mathcal{L}')$ is the complexity of a decoder for $\mathcal{L}'$ and its syntax code. Without decidability, add n on the left. If $\mathcal{L}'$ and its code are fixed independently of D, $K(\mathcal{L}')$ is a constant.*

Proof. The weights in the proof of Proposition G.8 search for, or decide, proofs from T of base-language answers. Given a decoder for the syntax of $\mathcal{L}'$ they are computable (respectively lower semicomputable) from T and the questions, and they do not otherwise depend on the language. So the argument of Proposition G.8(b) (respectively (a)) gives $K(y_D \mid \mathrm{dom}\, D) \leq K(\mathcal{L}') + K(T \mid \mathcal{L}') + \mathrm{Exc}_D^{\mathcal{Q}}(T) + c$ (respectively with n added), and $K(T \mid \mathcal{L}') \leq \|T\| + c$. □

Proposition G.28 (Facts used once never enter for $\lambda > 1$)**.**

(a) Let T^* be a λ-MAP with $\lambda > 1$ and χ a sentence. Suppose that, in the context $T^* \setminus \{\chi\}$, adding χ turns exactly the exceptions in a set $\mathcal{D}_\chi \subseteq D$ into proved data and changes nothing else, and that $\sum_{(\mathcal{Q},a)\in\mathcal{D}_\chi} \gamma(\mathcal{Q}) < \lambda\ell(\chi)$. Then $\chi \notin T^*$. In particular this applies to a datum's answer a stated as an axiom whenever $\gamma(\mathcal{Q}) \leq \ell(a)$ and stating it proves no other datum in that context.

*(b) Let provability on the data be decidable in the class, with κ the constant of Proposition G.8, and let T_s be b-*sufficient*: $\|T_s\| + \mathrm{Exc}_D(T_s) \leq K(y_D \mid \mathrm{dom}\, D) + b$. Every theory T has $\mathrm{Exc}_D(T) \geq \mathrm{Exc}_D(T_s) - (\|T\| - \|T_s\|) - (b+\kappa)$; a module $\mathcal{A}$ added to T_s has $\mathrm{EP}_D(\mathcal{A}; T_s \cup \mathcal{A}) \leq 1 + (b+\kappa)/\|\mathcal{A}\|$; and every λ-MAP T with $\lambda > 1$ has $\|T\| \leq \|T_s\| + (b+\kappa)/(\lambda - 1)$.*

Proof. (a) If $\chi \in T^*$, deleting it raises Exc_D by exactly $\sum_{\mathcal{D}_\chi} \gamma(\mathcal{Q}) < \lambda\ell(\chi) = \lambda\|\{\chi\}\|$, contradicting Proposition G.26. For an answer stated as an axiom, $\mathcal{D}_\chi$ is the one datum, and $\gamma(\mathcal{Q}) \leq \ell(a) < \lambda\ell(a)$. (b) Proposition G.8(b) gives $\|T\| + \mathrm{Exc}_D(T) \geq K(y_D \mid \mathrm{dom}\, D) - \kappa \geq \|T_s\| + \mathrm{Exc}_D(T_s) - b - \kappa$, which is the first claim, a right line of slope 1 through the point of T_s. For a module $\mathcal{A}$ disjoint from T_s, $\|T_s \cup \mathcal{A}\| - \|T_s\| = \|\mathcal{A}\|$, which gives the second. For a λ-MAP T, $\lambda\|T\| + \mathrm{Exc}_D(T) \leq \lambda\|T_s\| + \mathrm{Exc}_D(T_s)$; combined with the first claim this gives $(\lambda - 1)(\|T\| - \|T_s\|) \leq b + \kappa$, which is Hänni and Claude (Anthropic) (2026d, Corollary 5.5(a)) in this class. □

So at $\lambda = 1$ a fact that explains only itself is excluded only by the code's per-axiom overhead, while every $\lambda > 1$ excludes every module whose explanatory power is below λ.

The solar-system toy. Bodies $b \le k_{\mathrm{b}}$ have n_b observations each, each of literal cost γ (code Q), and $n_{\mathrm{tot}} := \sum_b n_b$. The modules are the law Λ (L bits), body b's elements El_b (k bits), and Kepler's third law K3 (L_3 bits), which shortens every elements module by s bits, $0 < s < k$, because it derives the period from the semi-major axis. $\Lambda \cup \mathrm{El}_b$ proves all of b's observations, and no other combination proves any observation: laws alone answer no position question, and neither do elements alone. The theories are Λ or not, K3 or not, and a set S of bodies whose elements are axioms.

Proposition G.29 (Laws and initial conditions in the solar-system toy)**.** *Put $k_\Lambda := k$, $k_{\Lambda+\mathrm{K3}} := k - s$ and $S_x(\lambda) := \{b : \gamma n_b > \lambda k_x\}$ for $x \in \{\Lambda, \Lambda + \mathrm{K3}\}$. The optimal values of the three law choices are*

$$J^*_\emptyset = \gamma n_{\mathrm{tot}}, \qquad J^*_\Lambda = \lambda L + \sum_b \min(\gamma n_b, \lambda k), \qquad J^*_{\Lambda+\mathrm{K3}} = \lambda(L + L_3) + \sum_b \min(\gamma n_b, \lambda(k - s)),$$

*attained by $S = S_x(\lambda)$ (bodies with $\gamma n_b = \lambda k_x$ may go either way); the λ-MAP is the best of the three (*K3 *without Λ is useless).*

(a) Λ beats the empty theory iff $\sum_{b \in S_\Lambda(\lambda)} (\gamma n_b - \lambda k) > \lambda L$.

(b) $J^*_{\Lambda+\mathrm{K3}} - J^*_\Lambda = \lambda L_3 - \sum_b \big[\min(\gamma n_b, \lambda k) - \min(\gamma n_b, \lambda(k - s))\big]$. If no body has $\lambda(k - s) < \gamma n_b \le \lambda k$, so that S is the same with and without K3*, then $J^*_{\Lambda+\mathrm{K3}} < J^*_\Lambda$ iff $s|S| > L_3$, whatever λ is.* K3 *is in the λ-MAP iff in addition $J^*_{\Lambda+\mathrm{K3}} < J^*_\emptyset$.*

(c) In the λ-MAP $\Lambda \cup \{\mathrm{El}_b : b \in S\}$ (with or without K3*), $\mathrm{EP}(\Lambda) = \gamma \sum_{b \in S} n_b / L$ and $\mathrm{EP}(\mathrm{El}_b) = \gamma n_b / k_x$.*

Proof. Without Λ no observation is proved, so the best such theory is empty, with $J = \gamma n_{\mathrm{tot}}$. For a law choice x with Λ, J_λ is a constant plus a sum over bodies, where including El_b costs λk_x and excluding it costs γn_b; minimise each term. (a) is the comparison of J^*_Λ with $J^*_\emptyset$. (b) The bracket is λs when $\gamma n_b > \lambda k$, $\gamma n_b - \lambda(k - s)$ when $\lambda(k - s) < \gamma n_b \le \lambda k$, and 0 otherwise; without bodies in the middle range the difference is $\lambda(L_3 - s|S|)$. The λ-MAP is the best of the three law choices, the empty one included. (c) Deleting Λ (respectively El_b) makes exactly the observations of the bodies in S (respectively of b) literal. □

Part (b) compares law choices; it is not membership of the λ-MAP. With two bodies of one observation each, $\gamma = 10$, $L = 1000$, $k = 5$, $s = 4$, $L_3 = 1$ and $\lambda = 1$, S is both bodies with and without K3 and $s|S| = 8 > L_3$, but the λ-MAP is the empty theory: $J^*_\emptyset = 20$ against $J^*_{\Lambda+\mathrm{K3}} = 1003$ and $J^*_\Lambda = 1010$. The middle-range condition of (b) matters too: in the population of Section 9.5 (eight planets with 655 to 3838 observations, sixty asteroids with a median of about five, thirty comets with one or two; $L = 300$, $L_3 = 150$, $k = 60$, $s = 15$, $\gamma = 12.4$), K3 is in the λ-MAP at $\lambda = 6.75$ although $s|S_\Lambda(\lambda)| = 135 < L_3$, because with K3 more bodies pay for their elements. A computation (code in the accompanying repository) confirms the closed form against brute force over all theories in 2100 random cases, and the criterion of (b) in all 13250 random cases with the same S under both law choices.

Proposition G.30 (Laws and initial conditions in the question format)**.** *Let the data be conditional questions "given the elements α_b of body b, is ℓ the case?", with compatibility as in* (E3)*: a datum is compatible with T when $B \cup T \cup \{\alpha_b\}$ is consistent and proves ℓ. In the toy, every theory of the family (the law,* K3*, and elements modules stating the true elements) is consistent with every true antecedent. Then an elements module El_b with the same content as α_b adds nothing, so $\mathrm{EP}(\mathrm{El}_b) = 0$ and no λ-MAP contains elements; and Λ alone answers every question, with $\mathrm{EP}(\Lambda) = \gamma n_{\mathrm{tot}} / L$ however the n_{tot} observations are spread over bodies.*

Proof. Immediate from the definition of compatibility: $T \cup \{\alpha_b\} \cup \mathrm{El}_b$ and $T \cup \{\alpha_b\}$ prove the same sentences, and $\Lambda \cup \{\alpha_b\}$ proves all of b's observations. Adding El_b raises J_λ by $\lambda k > 0$ and lowers no exception cost. □

Without the consistency condition the encoding is degenerate: a theory that refutes the antecedents would prove every datum (Proposition E.3). In the question format the distinction between laws and initial conditions is the distinction between theory and question antecedent.

G.7 Explanatory power, unification and correction

Proposition G.31 (Explanatory power as a likelihood ratio; unification)**.** *Write* $P_T(D) := 2^{-\mathrm{Exc}_D(T)}$ *(code Q or R). For a hypothesis* h*, a background theory* $T_\flat$*, data sets* D_1, D_2 *on disjoint questions and a theory* x*, put* $I(h; D \mid T_\flat) := \log(P_{T_\flat \cup h}(D)/P_{T_\flat}(D))$*,* $I_x(D_1; D_2) := \log \frac{P_x(D_1 \cup D_2)}{P_x(D_1)P_x(D_2)}$ *and* $U(D_1, D_2; h \mid T_\flat) := I(h; D_1 \cup D_2 \mid T_\flat) - I(h; D_1 \mid T_\flat) - I(h; D_2 \mid T_\flat)$*.*

(a) $\|\mathcal{A}\| \,\mathrm{EP}_D(\mathcal{A}; T) = I(\mathcal{A}; D \mid T \setminus \mathcal{A})$*.*

(b) $U(D_1, D_2; h \mid T_\flat) = I_{T_\flat \cup h}(D_1; D_2) - I_{T_\flat}(D_1; D_2)$*. If* P *is a product over questions for both theories, then* $U = 0$ *for every split.*

(c) Let h *be a law family with a* k*-bit parameter* θ *under the uniform prior,* $P_h(D) := 2^{-k} \sum_\theta P_{h \cup \{\theta\}}(D)$*, and* $P_\emptyset$ *a product. If each of* D_1, D_2 *is proved by* $h \cup \{\theta^*\}$ *and has likelihood* 0 *under every other* θ*, then* $U(D_1, D_2; h) = k$*. If* $\theta = (\theta_1, \theta_2)$ *with* D_i *depending only on* θ_i*, then* $U = 0$*. If* $\theta_i = (\theta_{\mathrm{sh}}, \theta_i')$ *share a* k_{sh}*-bit constant* θ_{sh} *and each* D_i *identifies* $(\theta_{\mathrm{sh}}, \theta_i')$*, then* $U = k_{\mathrm{sh}}$*.*

Proof. (a) is the definition of EP. (b) Expand the six logarithms: the first term of the difference is $\log P_{T_\flat \cup h}(D_1 \cup D_2) - \log P_{T_\flat \cup h}(D_1) - \log P_{T_\flat \cup h}(D_2)$, and the second is the same expression for $T_\flat$; a product law makes $I_x(D_1; D_2) = 0$. (c) Use (b) with $I_\emptyset = 0$. In the first case $P_h(D_1) = P_h(D_2) = P_h(D_1 \cup D_2) = 2^{-k}$, so $I_h(D_1; D_2) = k$. In the second, $P_h(D_1 \cup D_2) = P_h(D_1)P_h(D_2)$. In the third, with θ_i' of k' bits, $P_h(D_i) = 2^{-k_{\mathrm{sh}} - k'}$ and $P_h(D_1 \cup D_2) = 2^{-k_{\mathrm{sh}} - 2k'}$, so $I_h(D_1; D_2) = k_{\mathrm{sh}}$. □

A computation with $k = 6$ and observations given by a random table gives $U = 0.00$ for a stated parameter, 6.00 for a marginalised parameter and one body, 0.00 for two bodies with independent parameters, and 3.00 for two bodies sharing a 3-bit constant; the identity (b) holds to 3×10^{-15} on 1000 random joint laws. Myrvold describes unification as the degree to which a hypothesis renders phenomena informationally relevant to one another, a contribution to evidential support (Myrvold, 2003); by (b), U is the change in the mutual information between D_1 and D_2 that adding h brings about, which has that form. We do not claim that his formula coincides with U in every detail.

Proposition G.32 (Unification without better prediction)**.** *Consider three theories with a common part:* Pt_f*, Ptolemy's system with free epicycle phases, which pays for the coincidence of each superior planet* p*'s retrograde motion with its opposition in its fit,* a_p *bits per planet;* Pt_s*, the same with the coincidences stated as axioms,* $\sum_p a_p$ *law bits more than* Pt_f *and with the fit of* Cop*; and* Cop*, Copernicus's system,* H *law bits more than* Pt_f *and with the fit of* Pt_s*.*

(a) Cop *beats* Pt_s*, which makes the same predictions, at every* λ *iff* $H < \sum_p a_p$*.*

(b) Cop *is* Pt_f *plus a module with* $\mathrm{EP} = \sum_p a_p/H$*, so it beats* Pt_f *iff* $\lambda < \sum_p a_p/H$*.*

(c) Pt_s *beats* Pt_f *iff* $\lambda < 1$*.*

Hence if $H < \sum_p a_p$, Cop *is the best of the three exactly for $\lambda < \sum_p a_p/H$, and* Pt_f *above; if $H \geq \sum_p a_p$,* Cop *is never strictly best:* Pt_s *is best for $\lambda < 1$ (tying* Cop *if $H = \sum_p a_p$) and* Pt_f *for $\lambda > 1$.*

Proof. Measure lengths and fits relative to Pt_f and write $a := \sum_p a_p$. Then $J_\lambda(\mathrm{Pt}_f) = 0$, $J_\lambda(\mathrm{Pt}_s) = \lambda a - a$ and $J_\lambda(\mathrm{Cop}) = \lambda H - a$. (a) $J_\lambda(\mathrm{Cop}) - J_\lambda(\mathrm{Pt}_s) = \lambda(H - a)$, negative for every $\lambda > 0$ iff $H < a$. (b) $J_\lambda(\mathrm{Cop}) - J_\lambda(\mathrm{Pt}_f) = \lambda H - a$, and the module of H bits lowers the fit by a. (c) $J_\lambda(\mathrm{Pt}_s) - J_\lambda(\mathrm{Pt}_f) = (\lambda - 1)a$. If $H < a$, then $a/H > 1$: for $\lambda < a/H$, Cop beats both others; for $\lambda > a/H > 1$, Pt_f beats both. If $H \geq a$: for $\lambda < 1$, Pt_s beats Pt_f and is at least as good as Cop, with equality iff $H = a$; for $\lambda > 1 \geq a/H$, Pt_f beats both; at $\lambda = 1$, Pt_s and Pt_f tie and Cop is no better. □

Newton's theory does not entail Kepler's laws: it contradicts them (cf. Duhem, 1954). What it derives, with masses and initial conditions, are corrected statements, and Kepler's laws within tolerances on windows, as in the model of reduction to a corrected reduced theory (cf. Schaffner, 1967). The following toy makes this provable. Over $B = \mathrm{RCF}$ with constants $x_{t,1}, x_{t,2}$ for $t \leq \bar{t}$ (a planar state x_t per step), let Rot be the rotation with cosine $3/5$ and sine $4/5$, $\mathbf{P}$ a rational 2×2 matrix with operator norm $\|\mathbf{P}\|$, $\eta_c := 1/100$, $A := \mathrm{Rot} + \eta_c\mathbf{P}$ and $a := (1, 0)$; $|\cdot|$ is the Euclidean norm, and $|v| \leq \varepsilon$ is stated as $|v|^2 \leq \varepsilon^2$ with ε^2 rational. The *old exact law* is $T_0 := \{x_t = \mathrm{Rot}^t a : t \leq \bar{t}\}$ (uniform circular motion), the *successor* is $T_{\mathrm{new}} := \{x_0 = a\} \cup \{x_{t+1} = Ax_t : t < \bar{t}\}$, the *honest old law* is $T^\varepsilon := \{|x_t - \mathrm{Rot}^t a| \leq \varepsilon : t \leq \bar{t}\}$, and a *time-dependent form* is $T^{(\varepsilon_\cdot)} := \{|x_t - \mathrm{Rot}^t a| \leq \varepsilon_t : t \leq \bar{t}\}$.

Proposition G.33 (Explanation by approximate correction)**.**

(a) $T_{\mathrm{new}} \vdash T^\varepsilon$ *for every rational* $\varepsilon \geq \varepsilon(\bar{t}) := ((1 + \eta_c\|\mathbf{P}\|)^{\bar{t}} - 1)|a|$.

(b) $T_{\mathrm{new}} \vdash T^\varepsilon$ *iff* $\varepsilon^2 \geq \varepsilon_{\min}(\bar{t})^2 := \max_{t \leq \bar{t}} |A^t a - \mathrm{Rot}^t a|^2$*, a rational number. So* $\varepsilon_{\min}(\bar{t}) \leq \varepsilon(\bar{t})$*, and* $T^{\varepsilon_{\min}(\bar{t})}$ *is the strongest uniform-tolerance form of the old law on the window that the successor proves.*

(c) If $\eta_c\mathbf{P}a \neq 0$ *and* $\bar{t} \geq 1$*, then* $T_{\mathrm{new}} \vdash \neg \bigwedge T_0$.

(d) Among uniform-tolerance forms, T^ε *is a common consequence of* T_0 *and* T_{new} *iff* $\varepsilon \geq \varepsilon_{\min}(\bar{t})$*; the strongest common one is* $T^{\varepsilon_{\min}(\bar{t})}$.

(e) The time-dependent form with $\varepsilon_t := |A^t a - \mathrm{Rot}^t a|$ *is a common consequence of both theories, strictly stronger than* $T^{\varepsilon_{\min}(\bar{t})}$ *whenever* $\varepsilon_t < \varepsilon_{\min}(\bar{t})$ *for some* t.

(f) The strongest common consequence of the two finite theories is their disjunction $\bigwedge T_0 \vee \bigwedge T_{\mathrm{new}}$*, which is not a tolerance law; if* $\varepsilon_{\min}(\bar{t}) > 0$*,* $T^{\varepsilon_{\min}(\bar{t})}$ *does not entail it.*

Proof. Write $\varepsilon_t := |A^t a - \mathrm{Rot}^t a|$; ε_t^2 is rational because $A^t a$ and $\mathrm{Rot}^t a$ are rational vectors. In every model of $B \cup T_{\mathrm{new}}$, that is in any real closed field with interpretations of the constants, the recurrence determines $x_t = A^t a$, a rational vector, since $\mathbb{Q}$ embeds uniquely into every ordered field; instantiating the recurrence t times gives $T_{\mathrm{new}} \vdash x_t = A^t a$.

(a) $\|\mathrm{Rot}\| = 1$ and $\|A\| \leq 1 + \eta_c\|\mathbf{P}\|$, so $\varepsilon_{t+1} = |A(A^t a - \mathrm{Rot}^t a) + \eta_c\mathbf{P}\,\mathrm{Rot}^t a| \leq (1 + \eta_c\|\mathbf{P}\|)\varepsilon_t + \eta_c\|\mathbf{P}\||a|$ with $\varepsilon_0 = 0$, and by induction $\varepsilon_t \leq ((1 + \eta_c\|\mathbf{P}\|)^t - 1)|a| \leq \varepsilon(\bar{t})$ for $t \leq \bar{t}$. Since T_{new} proves $|x_t - \mathrm{Rot}^t a|^2 = \varepsilon_t^2$, it proves each conjunct of T^ε for rational $\varepsilon \geq \varepsilon(\bar{t})$.

(b) Over $B \cup T_{\mathrm{new}}$ each conjunct of T^ε is equivalent to the inequality $\varepsilon_t^2 \leq \varepsilon^2$ between rationals, which RCF proves or refutes. So $T_{\mathrm{new}} \vdash T^\varepsilon$ iff all $\bar{t} + 1$ inequalities hold, iff $\varepsilon^2 \geq \max_{t \leq \bar{t}} \varepsilon_t^2 = \varepsilon_{\min}(\bar{t})^2$; and $\varepsilon_{\min}(\bar{t}) \leq \varepsilon(\bar{t})$ by the bound in (a).

(c) $T_{\text{new}} \vdash x_1 = Aa$ and $T_0 \ni (x_1 = \operatorname{Rot} a)$, with $Aa - \operatorname{Rot} a = \eta_{\text{c}}\mathbf{P}a \neq 0$.

(d) T_0 proves $x_t = \operatorname{Rot}^t a$, hence T^ε for every $\varepsilon \geq 0$; combine with (b).

(e) T_{new} proves $|x_t - \operatorname{Rot}^t a|^2 = \varepsilon_t^2$, and T_0 proves $|x_t - \operatorname{Rot}^t a|^2 = 0 \leq \varepsilon_t^2$. If $\varepsilon_t < \varepsilon_{\min}(\bar{t})$ at some t, choose a real point x_t with $\varepsilon_t < |x_t - \operatorname{Rot}^t a| \leq \varepsilon_{\min}(\bar{t})$ and put $x_s := \operatorname{Rot}^s a$ for $s \neq t$: this is a model of $T^{\varepsilon_{\min}(\bar{t})}$ that violates the time-dependent form.

(f) For finite theories, a sentence follows from both iff it follows from their disjunction (proof by cases), so the disjunction is the strongest common consequence. Put $x_t := \operatorname{Rot}^t a + (\varepsilon_{\min}(\bar{t}), 0)$ for every $t \leq \bar{t}$: if $\varepsilon_{\min}(\bar{t}) > 0$, this is a model of $T^{\varepsilon_{\min}(\bar{t})}$ with $x_0 \neq a$, so it is a model of neither theory, and $T^{\varepsilon_{\min}(\bar{t})}$ does not entail the disjunction; in particular the disjunction is not equivalent to a uniform-tolerance law. □

Remark G.34 (Derivations of polynomial size). *Proof sketch.* What is not written out is the size of derivations of arithmetic on binary numerals from the field axioms, which is standard. The natural derivation of T^ε from T_{new} instantiates the recurrence t times, writes each x_t as an exact rational vector and checks $\bar{t} + 1$ numerical inequalities; the numerals it writes have total bit size $\sum_{t \leq \bar{t}} \operatorname{bits}(A^t a) = O(\bar{t}^2 \log \operatorname{ht})$, where ht is the height of the entries of A. If the numeral arithmetic has derivations of size polynomial in the numerals' size, the whole derivation has size polynomial in $\bar{t}$. For $\mathbf{P} = \mathbf{J}\operatorname{Rot}$, with $\mathbf{J}$ the rotation by 90°, the numerals have 1894 bits at $\bar{t} = 10$ and about 1.8×10^5 at $\bar{t} = 100$.

Exact rational arithmetic (code in the accompanying repository) gives, for three perturbations with $\|\mathbf{P}\| = 1$ ($\mathbf{J}$ is the rotation by 90°):

$\mathbf{P}$	$\varepsilon_{\min}(\bar{t})$ at $\bar{t} = 1, 10, 100, 200$	bound $\varepsilon(\bar{t})$
$\mathbf{J}\operatorname{Rot}$ (outward spiral with precession)	0.0100, 0.0999, 0.961, 1.69	0.0100, 0.105, 1.70, 6.32
$\operatorname{diag}(1, -1)\operatorname{Rot}$ (distortion)	0.0100, 0.0127, 0.0186, 0.0243	the same
identity (spiral)	0.0100, 0.103, 1.33, 3.49	the same

In each case T_{new} refutes T_0 at $t = 1$. The bound $\varepsilon(\bar{t})$ is nearly tight for small windows when the perturbation accumulates, and very loose for a bounded distortion (ratio 0.004 at $\bar{t} = 200$). For $\mathbf{P} = \mathbf{J}\operatorname{Rot}$ the time-dependent tolerances have mean 0.0500 at $\bar{t} = 10$ and 0.490 at $\bar{t} = 100$, about half of $\varepsilon_{\min}(\bar{t})$. A circle refitted on the window (radius and angular speed; floating point, for comparison only) absorbs the precession almost entirely in the first case (2.5×10^{-4} and 2.5×10^{-3} at $\bar{t} = 10, 100$) and the growth not at all in the third (3.7×10^{-2} and 0.80): the toy analogue of the fitted ellipses of Table 8.

G.8 Domains of applicability and retention

Let a law Λ with per-system elements prove the data of every system but one, $\star$, whose $n_\star$ data it fits, with $\star$'s elements ($k_\star$ bits), at the per-datum rate $m^\star_{\text{pat}}$ of the exact law with exceptions, or at the rate $m^\star_{\text{hon}}$ of an honest tolerance for $\star$, at a one-off cost of $h_\star$ bits. The *domain-restricted law* $\Lambda_{\text{Dom}} := \forall b\,(\text{Dom}(b) \to \text{law}(b))$, with Dom a definable predicate that B proves of the other systems and refutes of $\star$, costs $\|\text{Dom}\|$ bits more than Λ (a few bits if "planet" is a base predicate, about the logarithm of the number of systems to name $\star$) and leaves $\star$'s data literal, at rate γ. So $\star$'s data have three treatments: restriction (rate γ, constant $\lambda\|\text{Dom}\|$), patched (rate $m^\star_{\text{pat}}$, constant $\lambda k_\star$) and honest (rate $m^\star_{\text{hon}}$, constant $\lambda(k_\star + h_\star)$).

Proposition G.35 (Domain restriction)**.**

(a) $J_\lambda(\Lambda_{\mathrm{Dom}}) - J_\lambda(\Lambda \cup \mathrm{El}_\star$ *with* $\star$ *'s exceptions*$) = \lambda\|\mathrm{Dom}\| - \lambda k_\star + \sum_{\star\text{'s data}}(\gamma - c_{\mathrm{pat}})$*, and likewise against the honest treatment of* $\star$*, with* $\lambda(k_\star + h_\star)$ *and* c_{hon}*, where* $c_{\mathrm{pat}}, c_{\mathrm{hon}}$ *are the per-datum costs.*

(b) *With i.i.d. data and* $n_\star \to \infty$*, the restriction is eventually the* λ*-MAP's treatment of* $\star$ *iff* $\gamma < \min(m^\star_{\mathrm{pat}}, m^\star_{\mathrm{hon}})$*, or* $\gamma = m^\star_{\mathrm{hon}} < m^\star_{\mathrm{pat}}$ *with the honest cost almost surely* γ *and* $\|\mathrm{Dom}\| < k_\star + h_\star$*. If the least rate is shared by two treatments whose costs differ non-degenerately, there is no eventual winner.*

(c) *Under SAI's pricing (code R),* $m^\star_{\mathrm{hon}} < \gamma$ *whenever* $\varepsilon^*_\star < \mathsf{R}/(2\mathrm{e})$ *(for* $\Delta \le 2\varepsilon^*_\star$*) or* $\Delta < \mathsf{R}/\mathrm{e}$ *(for* $\Delta \ge 2\varepsilon^*_\star$*). So a system that the law confines within about* 66° *of longitude (a sufficient bound) is never cut out asymptotically at any cell width below* R/e*: it is kept with its own, wider tolerance. Under code Q with* $\Delta \le 2\varepsilon^*_\star$ *the honest treatment costs exactly* γ *per datum, and the restriction is eventually selected iff* $m^\star_{\mathrm{pat}} > \gamma$ *and* $\|\mathrm{Dom}\| < k_\star + h_\star$*.*

Proof. (a) The two theories differ in the domain clause, in $\star$'s elements, and in the treatment of $\star$'s data; the other systems' data are proved by both. (b) Theorem G.20(i), (ii) applied to the three treatments (code-Q costs are bounded and code-R costs have all moments): a unique least rate wins eventually; a tie with an almost surely zero cost difference is decided by the constants, which is the case of the restriction and the honest treatment under code Q in the fine regime, where the honest law is silent on every datum and costs exactly γ; and a tie with a non-degenerate difference has no eventual winner. (c) In the fine regime Lemma G.18(ii) gives $m^\star_{\mathrm{hon}} \le \log(2\varepsilon^*_\star/\Delta) + \log \mathrm{e}$, which is below $\log(\mathsf{R}/\Delta)$ iff $2\mathrm{e}\varepsilon^*_\star < \mathsf{R}$. In the coarse regime $m^\star_{\mathrm{hon}} = s_\Delta \beta_e \le \log \mathrm{e}$, which is below $\log(\mathsf{R}/\Delta)$ iff $\Delta < \mathsf{R}/\mathrm{e}$. Under code Q with $\Delta \le 2\varepsilon^*_\star$ the honest law is silent on every datum (Lemma G.13), and (b) applies. □

For the Moon over one year, with one fixed ellipse ($\varepsilon^* = 156.3' = 2.6^\circ$, $\mathbb{E}|e| = 61.7'$ for the mean fit, $k_\star = 60$, $h_\star = 20$, $\|\mathrm{Dom}\| = 4$, $c_{\mathrm{time}} = 10.5$), an exact computation (code in the accompanying repository) gives the eventually selected treatment of the Moon's data, with rates in bits per datum ("excluded" is the domain restriction, chosen at an exact tie between the honest and the literal treatment):

Δ	γ	code Q, $\lambda = 1$	code Q, $\lambda = 2$	code R, $\lambda = 1, 2$
$10'$	11.08	excluded (tie; patched 20.52)	excluded (patched 30.51)	honest (4.973)
$60'$	8.49	excluded (tie; patched 13.66)	excluded (patched 21.22)	honest (2.445)
$300'$	6.17	patched (3.430)	patched (5.592)	honest (0.686)
$600'$	5.17	patched (1.612; honest 2.694)	patched (2.6933; honest 2.6939)	honest (0.343)
$1800'$	3.58	patched (0.483; honest 0.623)	honest (0.623; patched 0.843)	honest (0.114)

At the exact ties the constants decide ($\|\mathrm{Dom}\| = 4$ against $k_\star + h_\star = 80$). In code R the honest treatment overtakes the restriction after about $\lambda \cdot 76/(\gamma - m^\star_{\mathrm{hon}})$ of the Moon's data in expectation, about 12 at $\lambda = 1$ and 25 at $\lambda = 2$ at $\Delta = 10'$ and $60'$.

For retention, let a datum proved from T cost in addition $r_{\mathrm{d}}\,\mathrm{dl}_T$ bits, where dl_T is the size of the derivation used, and let $J^{\mathrm{der}}_\lambda(T; D) := \lambda\|T\| + \mathrm{Exc}_D(T) + \sum_{\text{data proved}} r_{\mathrm{d}}\,\mathrm{dl}_T$. Compare an honest old law T_{old} (tolerance $\varepsilon_{\mathrm{old}}$ on a window) with an exact successor T_{new} that proves every datum, with per-datum derivation charges $c_{\mathrm{old}} < c_{\mathrm{new}}$.

Proposition G.36 (Retention of a superseded theory)**.**

(a) T_{old} *beats* T_{new} *on* D *iff*

$$\lambda(\|T_{\mathrm{new}}\| - \|T_{\mathrm{old}}\|) + (c_{\mathrm{new}} - c_{\mathrm{old}})\, N_{\mathrm{p}} + c_{\mathrm{new}}\,(n - N_{\mathrm{p}}) > \mathrm{Exc}_D(T_{\mathrm{old}}),$$

where N_{p} is the number of data that T_{old} proves and $n = |D|$.

(b) With i.i.d. data the rates are $m_{\mathrm{old}} = s_{\mathrm{old}}\gamma_{\mathrm{si}} + (1 - s_{\mathrm{old}})c_{\mathrm{old}}$, with s_{old} the silence probability of T_{old}, and $m_{\mathrm{new}} = c_{\mathrm{new}}$. If $m_{\mathrm{old}} < m_{\mathrm{new}}$ (respectively $>$), then almost surely T_{old} (respectively T_{new}) is eventually the λ-MAP of the two, whatever λ and the lengths are. At $m_{\mathrm{old}} = m_{\mathrm{new}}$ the lengths decide if the cost difference is almost surely constant, and otherwise there is no eventual winner.

(c) Let $s_{\mathrm{old}} = \min(2\varepsilon_{\mathrm{old}}/\Delta, 1)$. Under code Q, m_{old} is non-increasing in Δ on $\{\Delta : \gamma(\Delta) \geq c_{\mathrm{old}}\}$; under code R it is non-increasing in Δ on $(0, \infty)$ if $\beta_e \geq c_{\mathrm{old}}$. In either case the cell widths in that range at which T_{old} is retained form an interval with no upper bound inside the range; call its lower end Δ^*. For $\Delta \geq 2\varepsilon_{\mathrm{old}}$ under code Q, Δ^* solves $2\varepsilon_{\mathrm{old}}(\gamma(\Delta) - c_{\mathrm{old}})/\Delta = c_{\mathrm{new}} - c_{\mathrm{old}}$.

Proof. (a) $J^{\mathrm{der}}_\lambda(T_{\mathrm{new}}) = \lambda\|T_{\mathrm{new}}\| + c_{\mathrm{new}}n$ and $J^{\mathrm{der}}_\lambda(T_{\mathrm{old}}) = \lambda\|T_{\mathrm{old}}\| + \mathrm{Exc}_D(T_{\mathrm{old}}) + c_{\mathrm{old}}N_{\mathrm{p}}$; compare term by term. (b) T_{old} is honest, so it never refutes a datum; a datum costs c_{old} when it proves it and γ_{si} when it is silent. Apply Theorem G.20. (c) *Code Q.* In the coarse regime $m_{\mathrm{old}} = c_{\mathrm{old}} + s_{\mathrm{old}}(\gamma(\Delta) - c_{\mathrm{old}})$, and s_{old} and $\gamma(\Delta) - c_{\mathrm{old}} \geq 0$ are both non-negative and non-increasing in Δ, so their product is too; in the fine regime $m_{\mathrm{old}} = \gamma(\Delta)$, which is decreasing. *Code R.* In the fine regime T_{old} is silent on every datum and $m_{\mathrm{old}} = \mathbb{E}\bar{c}_\Delta(w)$, strictly decreasing by Lemma G.18(iii); in the coarse regime $m_{\mathrm{old}} = c_{\mathrm{old}} + s_{\mathrm{old}}(\beta_e - c_{\mathrm{old}})$, non-increasing when $\beta_e \geq c_{\mathrm{old}}$. The two forms agree at $\Delta = 2\varepsilon_{\mathrm{old}}$. A non-increasing rate stays below c_{new} once it is below it, which gives the interval, and the equation for Δ^* is $m_{\mathrm{old}} = c_{\mathrm{new}}$ in the coarse regime under code Q. □

With derivations from the successor assumed a thousand times larger ($\mathrm{dl}_{T_{\mathrm{new}}}/\mathrm{dl}_{T_{\mathrm{old}}} = 1000$, as for a validated integration against solving Kepler's equation), exact code-R costs and bisection (code in the accompanying repository) give the least retained cell width Δ^* for $c_{\mathrm{new}} = 0.01$, 0.1 and 1 bit per datum:

body, window	ε^*	code Q	code R
Mercury, 100 yr	$2.02'$	$1542'$, $258.5'$, $37.2'$	$229.8'$, $22.97'$, $2.33'$
Mars, 12 yr	$1.44'$	$1202'$, $195.6'$, $27.7'$	$174.2'$, $17.42'$, $1.744'$
Mars, 2000 yr	$49.9'$	$10450'$, $2895'$, $533'$	$5717'$, $572'$, $57.8'$
Jupiter, 100 yr	$6.57'$	$3468'$, $661'$, $101.6'$	$867'$, $86.7'$, $8.61'$
Saturn, 2000 yr	$81.9'$	never, $3994'$, $784'$	$10320'$, $1032'$, $102.9'$
Moon, 1 month	$30.9'$	$8414'$, $2086'$, $364'$	$4473'$, $447'$, $44.4'$

Under random completion the retention region reaches precisions 1.8 to 6.9 times finer than under literal pricing at $c_{\mathrm{new}} = 0.01$, 3.9 to 11 times at 0.1, and 7.6 to 16 times at 1, because the honest law's silence is cheap.

Question G.37. What is the posterior over an explicit pool of domain predicates (base predicates such as "planet", windows, and their conjunctions, with stated syntax costs) on the Kepler residuals? By Proposition G.35 we expect the answer to depend on the pricing: tolerances with error models under SAI's pricing, and sharp domains under literal pricing at fine precision.

H The Kepler computations

This appendix gives the proofs of the propositions of Section 10, the definitions behind each computation, the full tables, and the elements that witness each reported fit. All computations

take the plan94 analytical ephemeris of the planets and the moon98 ephemeris of the Moon, as implemented in the ERFA library, as the data; every arcminute figure is an error against plan94, not against the sky.

H.1 Proofs

The Ramsey sentence $R(T)$ of a theory T is obtained by replacing its theoretical function symbols by second-order variables and quantifying them existentially; its consequences are taken in second-order semantic consequence.

Proposition H.1 (the Ramsey sentence of Kepler's laws)**.**

(a) $R(T)$ has the same $\mathcal{L}_{\mathrm{hel}}$-consequences as T.

(b) For every B true in $\mathbb{R}$, $R(\mathrm{K12} \cup \mathrm{K3}) \cup B$ entails no datum $\mathrm{Near}(b, t, \mathbf{v}, \delta)$ with $\delta < \pi$.

Proof. (a) A model of $R(T)$ is an $\mathcal{L}_{\mathrm{hel}}$-structure that expands to a model of T, so $R(T)$ and T have the same $\mathcal{L}_{\mathrm{hel}}$-reducts of models. (b) Fix b, t, $\mathbf{v}$ and $\delta < \pi$, and a model of $B \cup \mathrm{K12} \cup \mathrm{K3}$ on the standard reals. If b is a planet, keep $a(b)$ and $P(b)$, so that K3 is untouched, and choose $i(b), \Omega(b), \varpi(b)$ and $M_0(b)$ so that the Keplerian direction of b at t is $-\mathbf{v}$: the orbital plane can be any plane through the Sun (the three angles give every rotation), and the phase can place the body at any point of its orbit, which meets every direction of its plane. If b is not a planet, $\mathrm{K12} \cup \mathrm{K3}$ does not constrain its direction at t (for the Sun it requires only $r(\mathrm{Sun}, t) = 0$), and we set that direction to $-\mathbf{v}$. Defining θ, β, r from the positions gives a model of $B \cup \mathrm{K12} \cup \mathrm{K3}$, hence of $R(\mathrm{K12} \cup \mathrm{K3}) \cup B$, in which the direction of b at t makes the angle $\pi > \delta$ with $\mathbf{v}$. □

So the laws alone answer only structural questions; the laws together with the elements answer position questions.

Proof of Proposition 10.2. The position $\mathbf{r}(b, t)$ that K12 gives is $a(b)$ times an expression free of a, so a Kepler direction depends on $(e, i, \Omega, \varpi, M_0, P)$ and not on a. Given a model of $B \cup \mathrm{K12}$, put $a'(b) := (k^2 P(b)^2 / 4\pi^2)^{1/3}$ for a fixed $k > 0$ (cube roots exist in a real closed field) and leave every other element unchanged. Positions and distances scale by $a'(b)/a(b) > 0$, directions are unchanged, and K3 holds because $a'(b)^3 / P(b)^2 = k^2 / 4\pi^2$ is the same for every body. So an $\mathcal{L}_{\mathrm{dir}}$-sentence true in every model of $B \cup \mathrm{K12} \cup \mathrm{K3}$ is true in every model of $B \cup \mathrm{K12}$; the converse is trivial. □

Proof of Proposition 10.3. (b) Let $\mathcal{M}_0$ have the Earth with $a = 1$, $e = 0$, $i = 0$, period P_0 and some phase, and Mars with $a = \rho > 1$, $e = 0$, $i = 0$, the same period P_0 and the same phase. Then $\mathbf{r}(\mathrm{Ma}, t) = \rho\, \mathbf{r}(\mathrm{Ea}, t)$ for all t, so $\mathbf{r}(\mathrm{Ma}, t) - \mathbf{r}(\mathrm{Ea}, t) = (\rho - 1)\mathbf{r}(\mathrm{Ea}, t) \neq 0$ and $\mathrm{dir}_g(\mathrm{Ma}, t) = \mathbf{r}(\mathrm{Ea}, t)/|\mathbf{r}(\mathrm{Ea}, t)| = -\mathrm{dir}_g(\mathrm{Sun}, t)$ for every t. So $\mathcal{M}_0 \models \neg\chi_{\mathrm{opp}}$. $\mathcal{M}_0$ satisfies K12, since each body moves on a Kepler orbit and K12 imposes no relation between bodies, and it violates K3 (equal periods, unequal semi-major axes). It is a model of every B true in $\mathbb{R}$, so $B \cup \mathrm{K12} \nvdash \chi_{\mathrm{opp}}$.

(a), standard models. Let $\mathcal{M}$ be a model of $\mathrm{K12} \cup \mathrm{K3}$ whose real sort is $\mathbb{R}$, and suppose $\mathcal{M} \models \neg\chi_{\mathrm{opp}}$. The Earth moves in the ecliptic ($i(\mathrm{Ea}) = 0$), so $\mathbf{u}(t) := \mathbf{r}(\mathrm{Ea}, t)/|\mathbf{r}(\mathrm{Ea}, t)|$ lies in the ecliptic and differs from $\mathbf{z}$. By $\neg\chi_{\mathrm{opp}}$, $\mathrm{dir}_g(\mathrm{Ma}, t) = \mathbf{u}(t) \neq \mathbf{z}$ for all t, so the bridge gives $\mathbf{r}(\mathrm{Ma}, t) \neq \mathbf{r}(\mathrm{Ea}, t)$ and $\mathbf{r}(\mathrm{Ma}, t) = \mathbf{r}(\mathrm{Ea}, t) + c(t)\mathbf{u}(t) = (r_{\mathrm{E}}(t) + c(t))\, \mathbf{u}(t)$ with $c(t) > 0$, where $r_b(t) := |\mathbf{r}(b, t)|$. So Mars is always on the ray from the Sun through the Earth, beyond the Earth: it moves in the ecliptic, its heliocentric direction equals the Earth's at every t, and

$r_{\mathrm{M}}(t) > r_{\mathrm{E}}(t)$. Take continuous polar angles $\theta_{\mathrm{M}} = \theta_{\mathrm{E}}$ in the ecliptic. Kepler's second law, a consequence of K12, gives $r_b(t)^2\dot\theta_b(t) = h_b$ with a nonzero constant h_b for each body (from $r = a(1 - e\cos\psi)$, $\tan\frac{\nu}{2} = \sqrt{(1+e)/(1-e)}\,\tan\frac{\psi}{2}$ for the true anomaly ν, and $\dot M = 2\pi/P$, one gets $r^2\dot\nu = 2\pi a^2\sqrt{1-e^2}/P$; standard). Hence $r_{\mathrm{M}}^2/r_{\mathrm{E}}^2 = h_{\mathrm{M}}/h_{\mathrm{E}}$ is constant: $r_{\mathrm{M}}(t) = \kappa\, r_{\mathrm{E}}(t)$ with a constant $\kappa > 1$, so $\mathbf{r}(\mathrm{Ma}, t) = \kappa\,\mathbf{r}(\mathrm{Ea}, t)$ for all t. The period of a Kepler orbit is the least period of $t \mapsto \mathbf{r}(b, t)$, and its semi-major axis is half the sum of its largest and smallest distances from the Sun ($r = a(1 - e\cos\psi)$). So $P(\mathrm{Ma}) = P(\mathrm{Ea})$ and $a(\mathrm{Ma}) = \kappa\, a(\mathrm{Ea}) \neq a(\mathrm{Ea})$, contradicting K3.

(a), arbitrary models. Let ζ_{K3} be the following sentence of the language of the real field with sin and cos:

$$\forall\vec\varepsilon_{\mathrm{Ea}}\,\forall\vec\varepsilon_{\mathrm{Ma}}\Big(\mathrm{Adm}(\vec\varepsilon_{\mathrm{Ea}}, \vec\varepsilon_{\mathrm{Ma}}) \to \exists t\,\forall\psi_{\mathrm{Ea}}\,\forall\psi_{\mathrm{Ma}}\big(\mathrm{Kep}_{\mathrm{Ea}}(t, \psi_{\mathrm{Ea}}) \wedge \mathrm{Kep}_{\mathrm{Ma}}(t, \psi_{\mathrm{Ma}}) \to \Delta(t, \psi_{\mathrm{Ea}}, \psi_{\mathrm{Ma}})\big)\Big).$$

Here $\vec\varepsilon_b$ are the seven elements of body b; Adm says that $0 \le e < 1$ and $a, P > 0$ for both bodies, that $i_{\mathrm{Ea}} = 0$, and that K3 holds for the pair; $\mathrm{Kep}_b(t, \psi)$ is Kepler's equation $\psi - e_b\sin\psi = M_{0,b} + 2\pi t/P_b$; and Δ says that the positions $\boldsymbol\xi_b := \mathrm{Rot}_b(a_b(\cos\psi_b - e_b),\, a_b\sqrt{1-e_b^2}\sin\psi_b,\, 0)$ either coincide or satisfy $(\boldsymbol\xi_{\mathrm{Ma}} - \boldsymbol\xi_{\mathrm{Ea}})/|\boldsymbol\xi_{\mathrm{Ma}} - \boldsymbol\xi_{\mathrm{Ea}}| \neq \boldsymbol\xi_{\mathrm{Ea}}/|\boldsymbol\xi_{\mathrm{Ea}}|$. Let ζ_{pos} be the sentence $\forall a\,\forall e\,\forall\psi\,(a > 0 \wedge 0 \le e < 1 \to a(1 - e\cos\psi) > 0)$. Both are true in $\mathbb{R}$: ζ_{pos} obviously, and ζ_{K3} because, for standard element values satisfying Adm, Kepler's equation has exactly one solution for each t (its left side is strictly increasing in ψ when $0 \le e < 1$), so the antecedent of the inner implication fixes the true positions, and the standard case above gives a t at which they coincide or the directions differ.

Now let $\mathcal{M}$ be any model of $B \cup \mathrm{K12} \cup \mathrm{K3}$ with $\zeta_{\mathrm{K3}}, \zeta_{\mathrm{pos}} \in B$; its real sort may be non-standard. Instantiate ζ_{K3} at the elements of the Earth and Mars in $\mathcal{M}$ (Adm holds by K12, K3 and the frame convention) and obtain t. K12 supplies, for this t, values $\psi_{\mathrm{Ea}}, \psi_{\mathrm{Ma}}$ that solve Kepler's equations and give $\mathbf{r}(b, t) = \boldsymbol\xi_b$; the universal quantifiers over $\psi_{\mathrm{Ea}}, \psi_{\mathrm{Ma}}$ apply to them, so Δ holds of the actual positions. By ζ_{pos}, $|\mathbf{r}(\mathrm{Ea}, t)| = a_{\mathrm{Ea}}(1 - e_{\mathrm{Ea}}\cos\psi_{\mathrm{Ea}}) > 0$, so $-\mathrm{dir}_g(\mathrm{Sun}, t) = \mathbf{r}(\mathrm{Ea}, t)/|\mathbf{r}(\mathrm{Ea}, t)|$. If the positions coincide, the bridge gives $\mathrm{dir}_g(\mathrm{Ma}, t) = \mathbf{z}$, which differs from $\mathbf{r}(\mathrm{Ea}, t)/|\mathbf{r}(\mathrm{Ea}, t)|$, whose third coordinate is 0 since $i_{\mathrm{Ea}} = 0$. Otherwise the bridge gives $\mathrm{dir}_g(\mathrm{Ma}, t) \neq \mathbf{r}(\mathrm{Ea}, t)/|\mathbf{r}(\mathrm{Ea}, t)|$. Either way $\mathcal{M} \models \chi_{\mathrm{opp}}$, and by the completeness theorem $B \cup \mathrm{K12} \cup \mathrm{K3} \vdash \chi_{\mathrm{opp}}$. □

The universal quantifier over ψ in ζ_{K3} is what makes the instantiation valid in a model in which Kepler's equation might have several solutions; an existential quantifier would leave that case open. The sentence ζ_{pos} supplies $\mathbf{r}(\mathrm{Ea}, t) \neq 0$ in non-standard models, the only fact about the reals used outside ζ_{K3}.

Remarks on Proposition 10.3. (i) The bridge needs a convention at coincidence: without it, a K3-model in which Mars coincides with the Earth would leave $\mathrm{dir}_g(\mathrm{Ma}, t)$ unconstrained there and could imitate $\mathcal{M}_0$. (ii) For an inner co-moving planet ($\rho < 1$) the analogue fails: a K3-model with the planet on the Earth's orbit, diametrically opposite (behind the Sun), is always in the direction of the Sun; so χ_{opp} is stated for an outer planet. (iii) A computation (code in the accompanying repository) illustrates the difference in the circular coplanar case with $\rho = 1.5237$: $\mathcal{M}_0$ keeps Mars at opposition to rounding error (1.2×10^{-6} degrees in floating point), while the K3-model with the same Earth, the same ρ and the same initial phase leaves opposition by 1.3° within a day, 13° within 10 days and 87° within 100 days. A second, independently written implementation finds, over 300 random $\mathrm{K12} \cup \mathrm{K3}$ models (an outer planet, the Earth in the ecliptic, the planet's plane random), a largest angle from opposition over 20 years of at least 175° in every model, and over 50 random models of the kind of $\mathcal{M}_0$ never more than 6×10^{-14} degrees. Whether finitely many interval data true in $\mathcal{M}_0$ refute every K3-model was not examined.

Proof of Proposition 10.4. (a) K3 with $a(\mathrm{Ea}) = 1$ gives $a(\mathrm{Ma})^3 = P(\mathrm{Ma})^2/P(\mathrm{Ea})^2$, so $\rho = (P(\mathrm{Ma})/P(\mathrm{Ea}))^{2/3}$; then K12 and Num^- determine $\mathbf{r}(\mathrm{Ma}, t)$ and $\mathbf{r}(\mathrm{Ea}, t)$, and the bridge determines the geocentric direction. A datum $\mathrm{Near}_g(\mathrm{Ma}, t, \mathbf{v}, \delta)$ satisfied by the computed direction then follows once B proves the numerical bound that places the computed direction within δ of $\mathbf{v}$.

(b) Fix t with $\mathbf{r}(\mathrm{Ea}, t)$ and $\mathbf{F}(t)$ not parallel. Every $\rho > 0$ gives a model of $\mathrm{K12} \cup \mathrm{Num}^-$ over $\mathbb{R}$, since K12 constrains a only by $a > 0$. The geocentric vector is $\mathbf{g}(\rho) = \rho\mathbf{F}(t) - \mathbf{r}(\mathrm{Ea}, t)$; as ρ runs over $(0, \infty)$ it runs along the open ray from $-\mathbf{r}(\mathrm{Ea}, t)$ in the direction $\mathbf{F}(t)$, inside the plane spanned by the two vectors. The ray does not pass through the origin because the two vectors are not parallel. Seen from the origin, its direction therefore moves continuously and monotonically along the great circle through $-\mathbf{u}(t)$ and $\mathbf{F}(t)/|\mathbf{F}(t)|$, tends to $-\mathbf{u}(t)$ as $\rho \to 0$ and to $\mathbf{F}(t)/|\mathbf{F}(t)|$ as $\rho \to \infty$, and covers the open arc between them, whose length $\alpha(t)$ lies in $(0, \pi)$. If $B \cup \mathrm{K12} \cup \mathrm{Num}^-$ proved $\mathrm{Near}_g(\mathrm{Ma}, t, \mathbf{v}, \delta)$ for a B true in $\mathbb{R}$, the whole arc would lie in the closed cap of angular radius δ about $\mathbf{v}$; its two end points, at angular distance $\alpha(t)$ from each other, would then lie in that cap too, so $\alpha(t) \le 2\delta$. □

A computation over 1580–1600 (daily, plan94) gives the distribution of $\alpha(t)$ quoted in Section 10.2: percentiles $0, 5, 25, 50, 75, 95, 100$ of $0.11°$, $9.52°$, $46.68°$, $90.22°$, $133.91°$, $170.34°$, $179.96°$; no sampled day is within $2'$ of conjunction or opposition. Along the arc the direction moves monotonically as ρ grows on every one of 21 sampled days (a second implementation finds the same on 60 seeded days, with the great-circle identity to 9×10^{-15} radians), and the sensitivity of the geocentric direction to ρ at the true ρ has median $969'$ and maximum $3555'$ per unit of ρ.

H.2 Data, fits and the error budget

Table 9: How wrong Kepler's laws are, against plan94: largest angular separation (arcminutes) between one fixed Kepler ellipse, fitted by least squares, and the plan94 positions, over windows centred on J2000; for the Moon (moon98, geocentric), in degrees. Last column: the fitted third-law constant $q := (2\pi/P)^2 a^3/k^2$ (k Gauss's constant) over the windows of 30 years and more, with the Newtonian two-body value $1 + m/M$ in brackets.

window (years)	12	30	100	300	1000	2000	third-law constant q
Mercury	0.46	0.89	2.17	6.04	19.3	38.6	
Mars	1.58	1.86	3.61	9.17	26.2	51.8	
Jupiter	1.05	4.19	7.64	7.74	31.8	38.5	1.00075–1.00099 (1.00095)
Saturn	1.70	3.32	9.73	11.7	91.3	92.4	1.0013–1.0019 (1.00029)
Moon	1.12° (one month), 3.0° (one year), 9.6° (18.6 years)						

Residuals of one fixed ellipse. For each body and window, centred on J2000, one fixed Kepler ellipse (seven elements) is fitted by least squares to the plan94 heliocentric positions (for the Moon, the moon98 geocentric positions) in the J2000 ecliptic, and the table reports the largest angular separation between the fit and the data. The fitted third-law constant is $q := (2\pi/P)^2 a^3/k^2$, with k Gauss's constant. Over the 12-year window the fits give $q = 1.00062$ (Jupiter) and 1.00487 (Saturn); over the windows of 30 years and more they give the ranges of the table. The Newtonian two-body value is $1 + m/M$; for Saturn, a secular, circular and coplanar estimate that

treats the orbit-averaged pull of each other planet as that of a ring gives $q = 1.00154$, of which Jupiter supplies 1.25×10^{-3}. This is a heuristic estimate, not a perturbation theory. A second computation, with other sample times and longitude errors only, agrees with the table to within about 8% in every planetary cell. The same fits, with a precessing ellipse, give Mercury's line of apsides a rotation of $575.4''$ per century in the fixed J2000 frame (fits over 100–2000 years), and the 2000-year longitude residuals of Saturn and Jupiter carry oscillations of period about 900 and 910 years and amplitude $43.8'$ and $18.3'$.

The third law's own error. With $q - 1 = 9.5 \times 10^{-4}$ (Jupiter, the two-body value) and 1.5×10^{-3} (Saturn, the middle of its fitted range), an exact K3 calibrated on the Earth changes the semi-major axes by the relative amounts 3.170×10^{-4} and 4.985×10^{-4} and misplaces Jupiter and Saturn geocentrically by at most $0.233'$ and $0.193'$ over 1580–1600 (medians $0.150'$ and $0.129'$).

The error budget. The documentation of ERFA's plan94 routine quotes, from comparisons with JPL's DE102 ephemeris over 1800–2050, maximum errors in heliocentric longitude of $17''$ (Mars), $71''$ (Jupiter) and $81''$ (Saturn), an accuracy over 1000–3000 no worse than 1.5 times that, and, against DE200 over 1800–2100, $26''$, $78''$ and $87''$. Hence: (i) over 1580–1600 plan94's longitude error for Mars is at most about $25'' \approx 0.43'$, the size of the gap $1.41' - 1.07' = 0.34'$ between the two best forms of the Mars pool, so the ordering of those two forms is established against plan94, and against the sky it rests on the errors against an exact ellipse in Table 5; (ii) the 12-year entries for Jupiter ($1.05'$) and Saturn ($1.70'$) lie inside 1800–2050, where plan94's own errors reach $1.18'$ and $1.35'$, so they measure the ellipse against plan94; (iii) plan94's Mars distance error (at most 7700 km over 1800–2050, about 3×10^{-5} of Mars's distance) is far below the distance tolerances of Appendix H.6.

The data of the Mars pool. The 7306 daily heliocentric positions of Mars over 1580.0–1600.0 are reduced to the best plane through the Sun (inclination $1.8828°$ and node $50.757°$ to the J2000 ecliptic; the largest angle of a position out of this plane is $0.070'$), and v_i is the longitude in that plane. The geocentric data of Appendix H.5 are the ecliptic longitudes of plan94 Mars minus plan94 Earth–Moon barycentre (J2000 ecliptic): geometric positions, without aberration or light time, and taken from the barycentre rather than from the Earth.

H.3 The Mars pool: forms, law texts and Occam factors

The forms. Write $\mathbf{u}(\varphi) := (\cos\varphi, \sin\varphi)$ and $\Lambda(t) := L + n(t - t_{\mathrm{mid}})$ for the mean longitude, with t_{mid} the middle of the window, L the mean longitude at t_{mid} and n the mean motion; w is the longitude of perihelion and $M(t) := \Lambda(t) - w$ the mean anomaly. Each form gives the heliocentric longitude $\theta(t)$ in the plane of the data; in longitude the forms are scale-free, so the circle has radius 1.

- *Circle about the Sun* (elements L, n): $\theta(t) = \Lambda(t)$.
- *Circle with centre $C = -e_1\mathbf{u}(w)$ and equant point $Q = -(e_1 + e_2)\mathbf{u}(w)$*, both on the line of apsides on the aphelion side of the Sun: the planet moves on the unit circle about C so that its direction seen from Q is $\Lambda(t)$, and $\theta(t)$ is its direction seen from the Sun. The *eccentric circle* is the case $e_2 = 0$ (uniform motion about the centre; elements e, w, L, n with $e = e_1$), the *bisected equant* the case $e_1 = e_2 = e$ (centre halfway between the Sun and the equant point; elements e, w, L, n), and the *vicarious hypothesis* the case of free e_1, e_2 (elements e_1, e_2, w, L, n).

- *Ellipse* (elements e, w, L, n): $\theta(t) = w + \nu$, where ν is the true anomaly that Kepler's equation $\psi - e\sin\psi = M(t)$ gives for eccentricity e.

The element ranges are $e, e_1, e_2 \in [0, 1)$, $w, L \in [0, 2\pi)$ and $n \in [0, 0.1)$ radians per day. Each element is written as a fixed-length 64-bit binary numeral for the midpoint of a cell of its range. The weight of the compatible numerals of a form is then $\mathrm{Vol}_{\mathsf{F}}(\delta)/\prod(\text{ranges})$ up to a grid error of order 2^{-64}, which gives the posterior weight $2^{-(|\mathrm{LAW}_{\mathsf{F}}|_S + \mathrm{Occ}_{\mathsf{F}}(\delta))}$ of Section 10.4.

Minimax errors. Each form is fitted by least squares from several starts, refined by L_p fits with increasing p, and its largest residual is then minimised by sequential linear programming. The following table lists the minimax errors (arcminutes, against plan94) and the elements that witness them (w and L in degrees, L at the middle of the window; n in radians per day); each value is an upper bound on the true minimax error. A second implementation, an exchange algorithm written independently and started elsewhere, finds the same values to four decimals. The circle's value is exact: its minimax problem is a one-dimensional convex minimisation. For the others, global optimality is not proved. Against an exact, unperturbed Kepler ellipse (the ellipse's own least-squares elements on these data, $e = 0.0930$, whose largest deviation from plan94 is $1.25'$), the minimax errors are $22.34'$ (eccentric), $7.43'$ (bisected equant) and $0.62'$ (vicarious hypothesis). The eccentric's error against the exact ellipse agrees with the second-order expansions of the two equations of centre, $2e^2 \sin 2M$ for the eccentric and $\frac{5}{4}e^2 \sin 2M$ for the ellipse, whose difference $\frac{3}{4}e^2$ is about $22.5'$ for Mars.

form	minimax	witnessing elements
circle	640.7908	$L = 309.45$, $n = 0.00914611$
eccentric	23.0508	$e = 0.184516$, $w = 283.471$, $L = 309.448$, $n = 0.0091461$
equant	8.1653	$e = 0.0933767$, $w = 283.465$, $L = 309.446$, $n = 0.00914613$
vicarious	1.4127	$e_1 = 0.115724$, $e_2 = 0.0707907$, $w = 283.462$, $L = 309.446$, $n = 0.00914609$
ellipse	1.0733	$e = 0.0930389$, $w = 283.445$, $L = 309.442$, $n = 0.00914609$

The law texts. A law text is a sentence $\forall b\, \forall t\, (\theta(b, t) = \ldots)$, or with an existential quantifier as below, in which the elements are unary function symbols $L(b), n(b), w(b), e(b), e_1(b), e_2(b)$ and the mean longitude is written out as $L(b) + n(b) \cdot t$, with t measured from the middle of the window, wherever it occurs (below we abbreviate it by Λ, and $\Lambda - w(b)$ by M; the abbreviations are expanded in the counted texts). The syntaxes are: *S1*, with $+, -, \cdot, /, \sin, \cos, \mathrm{atan2}, \sqrt{}$, the numerals $0, 1, 2$, $\forall$, $\exists$, $\wedge$, $=$, $>$; *S2*, S1 with a unit-vector primitive $\mathrm{uc}(\varphi) = \mathbf{u}(\varphi)$, a direction primitive $\mathrm{dir}(\mathbf{x})$ (the polar angle of $\mathbf{x}$) and $\mathrm{rc}(Q, C, \varphi)$, the point where the ray from Q at angle φ meets the unit circle about C; and *S3*, S1 with $\mathrm{kep}(e, M)$, the true anomaly for eccentricity e and mean anomaly M. The texts are (arguments b of the element symbols omitted):

- circle (all syntaxes): $\theta(b, t) = L + n \cdot t$;
- eccentric: in S1 and S3, $\theta(b, t) = \mathrm{atan2}(\sin\Lambda - e \cdot \sin w,\ \cos\Lambda - e \cdot \cos w)$; in S2, $\theta(b, t) = \mathrm{dir}(\mathrm{uc}(\Lambda) + (-e) \cdot \mathrm{uc}(w))$;
- vicarious hypothesis: in S1 and S3, $\exists s\, (s > 0 \wedge s \cdot s - 2 \cdot (e_2 \cdot (s \cdot \cos M)) + e_2 \cdot e_2 = 1 \wedge \theta(b, t) = \mathrm{atan2}(-((e_1 + e_2) \cdot \sin w) + s \cdot \sin\Lambda,\ -((e_1 + e_2) \cdot \cos w) + s \cdot \cos\Lambda))$; in S2, $\theta(b, t) = \mathrm{dir}(\mathrm{rc}(-(e_1 + e_2) \cdot \mathrm{uc}(w),\ (-e_1) \cdot \mathrm{uc}(w),\ \Lambda))$;
- bisected equant: the vicarious texts with e for e_1 and e_2 in S1 and S3, and with $2 \cdot e$ for $e_1 + e_2$ and e for e_1 in S2;

- ellipse: in S1 and S2, $\exists E\,(E - e\cdot\sin E = M \wedge \theta(b,t) = w + \mathrm{atan2}(\sqrt{1-e\cdot e}\cdot\sin E,\ \cos E - e))$; in S3, $\theta(b,t) = w + \mathrm{kep}(e, M)$;
- K3 (all syntaxes; with $P = 2\pi/n$): $\forall b\,\forall c\,((n(b)\cdot n(b))\cdot(a(b)\cdot(a(b)\cdot a(b))) = (n(c)\cdot n(c))\cdot(a(c)\cdot(a(c)\cdot a(c))))$.

A text is serialised with one opening and one closing bracket per compound, and its length is its number of tokens times $\lceil \log_2(\text{alphabet size})\rceil$. The alphabets have 31 (S1), 34 (S2) and 32 (S3) symbols, so a token costs 5, 6 and 5 bits. This gives the text lengths of Table 6: 31, 89, 198, 198, 109 and 75 tokens in S1 for the circle, eccentric, equant, vicarious hypothesis, ellipse and K3; 31, 57, 75, 78, 109, 75 in S2; 31, 89, 198, 198, 52, 75 in S3. Every text was evaluated numerically and compared with the model it is attached to, at 40 random elements and times each: all agree to within 1.4×10^{-14} radians. A second serialisation, Polish notation with fixed arities and a separate symbol for unary minus, gives about half these lengths: in S1 75, 195, 440, 440, 245 and 165 bits, in S2 90, 150, 198, 204, 294, 198 and in S3 75, 195, 440, 440, 120, 165; a compact Polish code for K3 with a power symbol and the numeral 3 has 23 tokens, 115 bits.

Occam factors. $\mathrm{Occ}_{\mathsf{F}}(\delta)$ is estimated by hit-or-miss Monte Carlo with the exact model on all 7306 data: points are drawn uniformly in a parallelepiped built from the whitened linearisation of the residuals at the minimax point, enlarged where hits come close to a face, and capped at the element ranges; about 2000 hits per entry give a standard error of about 0.03 bits. The table below gives every value. Two cross-checks support them where the region is far from linear. Hit-or-miss sampling in an axis-aligned box with the full range of the eccentricities gives, for the vicarious hypothesis, 22.73 ± 0.03, 33.36 ± 0.03, 37.87 ± 0.07 and 43.02 ± 0.24 bits at $720'$, $120'$, $60'$ and $30'$, within about one combined standard error of the table. An independent implementation, with boxes found by its own random-walk explorer, gives 21.16 (ellipse, $720'$), 20.21 (eccentric, $720'$), 21.17 (equant, $720'$), 31.72 (equant, $120'$), 31.69 (ellipse, $120'$), 35.67 (ellipse, $60'$), each ± 0.03 or better, and 20.929 (circle, $720'$, an exact one-dimensional integral): all within 0.06 bits of the table. At $720'$ the compatible regions have a thin part with the apsidal angle opposite the true perihelion and small eccentricity (since the circle fits to $640.8' < 720'$, every apsidal angle is compatible for small e); the parallelepiped at $720'$ spans 6.5 radians in w, and a full-range rerun shows the effect of this part to be within the noise.

δ	$720'$	$120'$	$60'$	$30'$	$20'$	$12'$	$9'$	$6'$	$4'$	$3'$	$2'$	$1.6'$	$1.3'$
circle	20.93	–	–	–	–	–	–	–	–	–	–	–	–
eccentric	20.24	31.17	36.34	45.38	–	–	–	–	–	–	–	–	–
equant	21.20	31.70	35.88	40.35	43.63	49.49	57.34	–	–	–	–	–	–
vicarious	22.77	33.34	37.93	42.83	45.77	49.56	51.78	54.98	58.56	61.53	67.56	74.54	–
ellipse	21.18	31.63	35.68	39.65	42.01	45.01	46.81	49.21	51.86	53.88	57.41	59.93	63.81

Occam factors $\mathrm{Occ}_{\mathsf{F}}(\delta)$ in bits ("–": δ below the form's minimax error).

The MAP form. At these 13 precisions the MAP form is: in S1, the circle at $720'$, the eccentric at $120'$–$30'$ and the ellipse from $20'$ on; in S2, the circle at $720'$, the eccentric at $120'$–$30'$, the bisected equant at $20'$–$9'$, the vicarious hypothesis at $6'$–$1.6'$ and the ellipse at $1.3'$; in S3, the circle at $720'$ and the ellipse from $120'$ on. Below $1.07'$ no form is compatible. In every row the runner-up has at most $2^{-12.4}$ of the MAP's weight. Each K3 variant of a form has exactly $2^{-|\mathrm{K3}|}$ times the form's weight: with one planet K3 is vacuous, and in heliocentric direction data it has no content (Proposition 10.2).

The data without the text prior. With a uniform prior over the five forms, the element prior above and no text prior, the MAP is the eccentric at $720'$ and $120'$ (margins 0.69 and 0.46 bits) and the ellipse at every $\delta \le 60'$, with margins 0.20, 0.70, 1.62, 4.48, 4.97, 5.77, 6.70, 7.65, 10.15 and 14.61 bits at $60'$, $30'$, $20'$, $12'$, $9'$, $6'$, $4'$, $3'$, $2'$ and $1.6'$. These margins are largest just above a competitor's minimax error, where its compatible volume shrinks to zero. They depend on the element ranges: halving the ranges of the eccentricities (the compatible regions lie inside $[0, 0.5)$ at every $\delta \le 120'$) adds one bit of prior density per eccentricity, which moves the vicarious hypothesis by one bit against the four-element forms and the circle by one bit the other way.

Text against data. Form F beats form G iff $|\mathrm{LAW}_{\mathsf{F}}|_S - |\mathrm{LAW}_{\mathsf{G}}|_S < \mathrm{Occ}_{\mathsf{G}}(\delta) - \mathrm{Occ}_{\mathsf{F}}(\delta)$. For the epicyclic syntax's equant and vicarious texts (18 bits apart), the Occam-factor difference moves from 2.14 bits in the equant's favour at $20'$ to 5.56 bits in the vicarious hypothesis's favour at $9'$, so the equant keeps a margin of 12.4 bits at $9'$; text and data contributions are of the same order there. For every other pair of forms the text difference is between 100 and 800 bits, an order of magnitude larger than any Occam-factor difference, except that in S1 and S3 the equant and vicarious texts have equal length; both lose to the ellipse there by more than 400 bits.

Robustness. A variable-length numeral code (an Elias-gamma length prefix per element) adds about 5–11 bits per element; this moves the vicarious hypothesis against the four-element forms by 6–11 bits and the circle against them by 12–20 bits, and changes none of the MAP rows. In Polish notation the text differences shrink by about half and no MAP row changes; the closest case is the epicyclic syntax's equant and vicarious texts, which then differ by 6 bits (198 against 204), against an Occam-factor difference of at most 5.56 bits. A natural form outside the pool, the equation-of-centre series truncated at second or third order (four elements, like the ellipse), has S1 texts of 680 and 1345 bits, against the ellipse's 545, and minimax errors $4.11'$ and $1.31'$, so it changes no S1 row.

H.4 Graded fit

Gaussian noise. Least-squares fits to the 7306 longitudes have root-mean-square residuals $454.4'$ (circle), $15.81'$ (eccentric), $5.26'$ (equant), $0.628'$ (vicarious hypothesis) and $0.446'$ (ellipse). For a sampler with Gaussian errors of standard deviation σ, the expected log-likelihood penalty per independent observation relative to the ellipse is $(\mathrm{RMS}^2_{\mathsf{F}} - \mathrm{RMS}^2_{\mathrm{ell}})/(2\sigma^2 \ln 2)$ bits. A form's text advantage over the ellipse is overcome after (text advantage)/(penalty) observations, which omits the parameter-fitting terms:

form	penalty (bits per observation) $\sigma = 1'$	$\sigma = 2'$	$\sigma = 4'$	observations needed to overcome the text advantage at $\sigma = 1', 2', 4'$
eccentric	180.2	45.05	11.26	S1 (100 bits): 0.6, 2.2, 8.9; S2 (312 bits): 1.7, 6.9, 27.7
equant	19.84	4.96	1.24	S2 (204 bits): 10.3, 41.1, 164.5
vicarious	0.142	0.035	0.0089	S2 (186 bits): 1313, 5253, 21013

In S1 and S3 the ellipse already has the shorter text against the equant and the vicarious hypothesis, by 445 and 730 bits, and in S3 also against the eccentric.

A simulation. For n days drawn at random from 1580–1600 (seeded; 20 runs per cell), the plan94 longitude plus Gaussian noise is fitted by each form by least squares, and the evidence is

computed by the Laplace approximation under the element prior above; the noise model's text is common to all forms and cancels. At $\sigma = 2'$, in S1 and S3 the MAP is the ellipse in 20 of 20 runs for every $n \geq 10$ (at $n = 5$, S1's MAP is the eccentric in all runs); in S2 it is the vicarious hypothesis in 20 of 20 runs for $n = 20$ to 2560, in 10 of 20 at $n = 5120$, and the ellipse in 20 of 20 at $n = 7306$. At $\sigma = 1'$ the S2 switch to the ellipse comes between $n = 640$ (vicarious hypothesis in 20 of 20) and $n = 2560$ (ellipse in 20 of 20). The median evidence ratio of the ellipse over the vicarious hypothesis, without text prior, grows from $2^{2.2}$ at $n = 5$ to 2^{276} at $n = 7306$ ($\sigma = 2'$). The simulation agrees with the expected-penalty estimates.

Honest uniform-tolerance theories. Give each form the honest law with tolerance equal to its minimax error (Proposition G.3). For cells much narrower than the tolerances, the cost per datum against the ellipse is about $\log_2(\varepsilon_{\mathsf{F}}/\varepsilon_{\text{ell}})$: 4.43 bits (eccentric), 2.93 (equant) and 0.40 (vicarious hypothesis). The text advantages are then overtaken after 23 (S1) and 71 (S2) data for the eccentric, 70 for the equant (S2) and about 470 for the vicarious hypothesis (S2). These are estimates from the pricing of Section 9, not a separate posterior computation, and they are conservative: SAI's mixture prefers a law with a calibrated error model to the uniform band (Proposition 9.3).

H.5 What K3 is worth

Setting. The data are the daily geocentric ecliptic longitudes of Mars over 1580.0–1600.0 (Appendix H.2) as interval sentences at a common tolerance δ. The hypotheses are Kepler ellipses for the Earth (in the ecliptic, $a = 1$) and for Mars (its plane fixed at the best heliocentric plane of Appendix H.2), each with elements e, w, L, n, and either $\rho = a(\text{Ma})/a(\text{Ea})$ as a ninth free element in $[0, 4)$ (without K3) or $\rho := (n_{\text{Ea}}/n_{\text{Ma}})^{2/3}$ (with K3, eight elements). As in Appendix H.3, the posterior odds of the hypothesis with K3 against the one without are $2^{s_{\text{K3}}(\delta) - |\text{K3}|}$ with $s_{\text{K3}}(\delta) := \log(4\,\text{Vol}_8(\delta)/\text{Vol}_9(\delta))$, where Vol_8 and Vol_9 are the volumes of the exact compatible regions with and without K3.

Fit. The minimax geocentric error is $0.627'$ with ρ free (fitted $\rho = 1.523676$, against the K3 value 1.523687 from the fitted periods) and $0.632'$ with K3; an independent fit gives $0.6266'$ and $0.6318'$ with the same values of ρ. So K3 costs $0.005'$ of fit and is compatible with the data at every $\delta \geq 0.632'$. These minimax values are upper bounds witnessed by elements. The data are values of a model, so plan94's own error does not affect the comparison.

Method. With $y := \rho - (n_{\text{Ea}}/n_{\text{Ma}})^{2/3}$, a shear of unit Jacobian turns the volume ratio into a density: $\text{Vol}_8/\text{Vol}_9 = p_y(0)$, the density at 0 of y under the uniform distribution on the nine-dimensional compatible region. That distribution is sampled by hit-and-run (Smith, 1984), with stepping-out brackets in the manner of slice sampling (Neal, 2003) and the exact membership test on all data; $p_y(0)$ is estimated as the fraction of samples with $|y| < h$ divided by $2h$, for $h = 0.05$, 0.1 and 0.2 standard deviations of y. Box sampling cannot be used here: the linearised compatible region fills a fraction of 10^{-5} or less of its bounding box.

δ	$8'$	$4'$	$2'$	$1'$
$s_{\text{K3}}(\delta)$ (bits)	11.28 ± 0.04	12.31 ± 0.03	13.39 ± 0.03	14.73 ± 0.04
second implementation	11.27–11.32	12.28–12.32	13.32–13.35	14.69–14.71
standard deviation of y	$6.47 \cdot 10^{-4}$	$3.10 \cdot 10^{-4}$	$1.47 \cdot 10^{-4}$	$5.54 \cdot 10^{-5}$
$\log_2\left(4/(\sqrt{2\pi}\,\text{sd}(y))\right)$	11.27	12.33	13.41	14.81

The fourth row is the value for a Gaussian y, as a check. The errors in the first row are one standard error over the 492–583 independent chains (of 600) that start at exactly compatible points. The three kernel widths agree to within 0.03 bits, and the first and second halves of the kept steps agree to within 0.05 bits except at $8'$ (11.33 against 11.23), where we quote 11.3 ± 0.1. The second implementation, written independently and run with three kernel widths, is a hit-and-run on the linearised compatible polytope with exact chords (60,000 samples per δ); its shares of linearised samples that are exactly compatible, 0.819, 0.901, 0.958, 0.968, match the first implementation's 0.820, 0.895, 0.948, 0.972. The sampler failed to bracket or to accept in 38, 5, 1 and 0 of about 128,000–152,000 chain steps per row; the direct effect is at most about 0.01 bits at $8'$. A cruder estimate from the single most sensitive day (9.8, 10.8, 11.8 bits at $8'$, $4'$, $2'$) is lower, as it should be. s_{K3} includes $\log_2$ of the range of ρ ($[0, 4)$, 2 bits); the range $[0, 2)$ would lower every entry by one bit.

Against the text. |K3| is 375 bits in S1 with brackets counted (450 in S2), 165 bits in Polish notation and 115 bits in the compact Polish code (Appendix H.3). For one pair of planets the posterior odds against K3 at $8'$ are therefore about 2^{-364} (S1) to 2^{-104} (compact code). By the criterion of Proposition G.29(b), when the same bodies' elements are in the theory with and without K3, K3 pays, with the law in place, iff the number $|S|$ of bodies whose size it fixes satisfies $s_{\mathrm{K3}}|S| > |\mathrm{K3}|$; at $s_{\mathrm{K3}} = 11.3$ and 14.7 bits per body this needs $|S| > 33.2$ and 25.5 (375 bits), 14.6 and 11.2 (165 bits), 10.2 and 7.8 (115 bits). The extrapolation to the five planets other than the Earth (about 55–75 bits) assumes that each saves about as much as Mars; the other planets were not computed.

H.6 Distances as data

The data are those of the Mars pool together with, for every day, the sentence "$r(t_i)$ lies within a factor $1 \pm \tau_{\mathrm{d}}$ of r_i", with r_i the plan94 heliocentric distance in the best plane. Each form gets one more element, its scale. An entry of Table 10 is the least longitude tolerance found at which a form, with suitable elements and scale, proves every longitude datum and every distance datum. The search combines sequential linear programming from many starts: earlier optima, perturbations of them, the optimum at the neighbouring τ_{d} in a decreasing and in an increasing sweep, and six seeded random starts; a second, independently written search was run as well, and each entry is the best value found by either. Every entry is witnessed by printed elements and was re-checked with the same residual function; none is proved minimal. The circle about the Sun is not in this computation. The equant and ellipse rows come from a coarser search; their entries at 3%, 1.5% and 0.75% follow by monotonicity (elements feasible at a smaller τ_{d} are feasible at a larger one, and no form beats its longitude minimax), and the independent search confirms them: at its longitude minimax the ellipse needs only a distance tolerance of 0.0153% and the equant 0.2487%.

The witness at $\tau_{\mathrm{d}} = 1\%$. The elements $e_1 = 0.102835781$, $e_2 = 0.0837969992$, $w = -1.33587144$, $L = -0.882335245$ (radians), $n = 0.00914611825$ radians per day and scale 1.52509743 fit every distance of the vicarious hypothesis to within 1.00000% and every longitude to within $5.119'$. The fit is sensitive to rounding: with n rounded to six decimals the longitude error rises to $6.2'$. The vicarious entry at 0.05% ($132.04'$) is the independent search's; the other search found $132.35'$ there.

Table 10: Distances as data: the least longitude tolerance found (arcminutes, against plan94) at which a form proves every longitude datum and every distance datum at relative distance tolerance τ_{d}; "none": no elements meeting τ_{d} were found; r^*: the least distance tolerance found with no longitude constraint. Every entry is an upper bound on the least tolerance, witnessed by elements; minimality is not proved.

form	r^*	5%	3%	2%	1.5%	1%	0.75%	0.5%	0.2%	0.1%	0.05%
eccentric	0.226%	150.73	218.31	252.45	269.56	286.70	295.32	305.33	none	none	none
equant	0.222%	8.17	8.17	8.17	8.17	8.17	8.17	8.17	none	none	none
vicarious	0.0164%	1.41	1.41	2.01	3.54	5.12	5.97	6.95	23.74	94.95	132.04
ellipse	0.0117%	1.07	1.07	1.07	1.07	1.07	1.07	1.07	1.07	1.07	1.07

Reading. Distances order the forms differently from longitudes. The bisected equant fits the distances to within 0.5% without giving up any longitude accuracy, and to 0.22% at best. The eccentric circle fits distances alone to 0.23%, but not with the elements its longitudes need: the distance fit wants a smaller eccentricity ($e = 0.142$ at 5%, against 0.185 for the longitude fit). The vicarious hypothesis fits distances alone to 0.016% (with $e_1 \approx e_2$ it is close to the equant), but not together with its longitude fit. The ellipse fits both at once, at its longitude minimax, for every τ_{d} down to 0.05%. Not computed: the Occam factors with distance data, and hence the MAP form above the thresholds; a lower bound on any least tolerance (it would need, for example, interval arithmetic over a covering of the element space); and the precision of Kepler's own distance determinations, so the table does not say which τ_{d} was historically available.

H.7 Reproducing the computations

The scripts in the accompanying repository are deterministic or seeded (seed 20261015 for the residual fits and the Mars pool, 20261016 for the graded-fit and distance computations) and were run with Python 3.11.15, NumPy 2.4.6, SciPy 1.17.1 and pyerfa 2.0.1.5. Their saved outputs contain every number of Section 10 and of this appendix. Re-running reproduces them apart from timing lines; the Mars-pool posterior with the value of K3 takes about 80 minutes on one core, the distance search about 36 minutes, the residual fits about 2 minutes and the graded-fit computations under a minute. The independently written implementations quoted above are in the same repository.

I Notation

The table lists the symbols used in more than one place, with the place where each is defined. Symbols local to one proof, example or subsection are defined where they occur. Logarithms are to base 2 and losses are in bits; $c, c', \ldots$ are constants that depend only on the fixed universal machines, the coding of syntax and the background B, unless a statement says otherwise.

symbol	meaning	where
Algorithmic information and probability		
V; K, $K(\cdot \mid \cdot)$	the fixed universal prefix machine; prefix complexity	§ 2.1
$\mathbf{m}$	universal discrete semimeasure, $\mathbf{m}(i) = \sum_{V(p)=i} 2^{-\lvert p\rvert}$	§ 2.1

symbol	meaning	where
Ω_V; $\Omega_V^{\geq \ell}$	halting probability of V; halting mass of the V-programs of length at least ℓ	§ 2.1; Thm. 3.18
$K^{\mathrm{pad}}(n)$	$\min_{n' \geq n} K(n')$, nondecreasing and unbounded	§ 4.3
Kt; K^q	Levin's complexity $Kt(x) = \min\{\lvert p\rvert + \log t : V(p) = x \text{ within } t \text{ steps}\}$; the least $\lvert p\rvert$ with $V(p) = x$ within $q(\lvert x\rvert)$ steps, q a polynomial	Thm. 7.3; § 7.7
$\mathbb{U}$; $[w]$; $\omega_{1:n}$	fair-coin measure on tapes $\omega \in \{0,1\}^\infty$; cylinder of a finite string w; first n bits of ω	§ 2.1
$\mathbb{P}$, $\mathbb{E}$; $\mathbf{1}[\cdot]$; KL	law of a data process and its expectation; indicator; Kullback–Leibler divergence in bits	Def. 9.1; Thm. 3.14
$\asymp$	equality within a factor $2^{\pm c}$ on every finite data set	§ 4
Logic		
$\mathcal{L}$, Sent; B; $\mathrm{Cn}(\Gamma)$	language, its sentences; background theory (decidable, consistent); $\{\varphi : B \cup \Gamma \vdash \varphi\}$	Def. 2.1
φ^1, φ^0; "decides"	φ, $\neg\varphi$; Γ decides φ if it proves φ or $\neg\varphi$	Def. 2.1
W_e	the e-th r.e. set of sentences	§ 2.1
$\mathcal{L}_{\mathrm{ar}}$, $\underline{n}$; Q, PA	the language $\{0, S, +, \cdot, <\}$, numerals; Robinson and Peano arithmetic	§ 5
Π_1^*; "B true"	conjunctions of Π_1 sentences and negations of Σ_1 sentences; B has a model with $\mathcal{L}_{\mathrm{ar}}$-reduct $\mathbb{N}$	§ 5
Ind_B	sentences that B decides neither way	§ 4.4
$[T]$; $K_{\equiv}(T)$; $\mathbf{m}_{\equiv}(T)$	indices e with $\mathrm{Cn}(\mathsf{W}_e) = \mathrm{Cn}(T)$; $\min_{e \in [T]} K(e)$; $\sum_{e \in [T]} \mathbf{m}(e)$	Def. 2.9
$\mathrm{Ext}(T)$; $\mathbf{m}(\mathrm{Ext}T)$	indices of the consistent theories that extend $\mathrm{Cn}(T)$; their total $\mathbf{m}$-weight	Def. 2.9
K_{enum}, $\mathbf{m}_{\mathrm{enum}}$, $\mathbb{U}_{\mathrm{enum}}$	enumeration complexity of a set, its universal discrete semimeasure, its probability under the universal theory machine	App. A.6
Data, conventions and questions		
D, $\mathrm{dom}\, D$; Y_3, ι	data set (finite partial function to $\{0,1\}$); three-valued labels $\{1, 0, \iota\}$, ι "independent"	Def. 2.1
$D^{\pm}$; ψ_D; $D + (\varphi, y)$	the literals of D; their conjunction; one more datum	Def. 2.1
D_t, D_∞; $\nu(y \mid D, \varphi)$	prefixes of a reveal sequence, a stream; conditional, with the repeated-input convention	Def. 2.1
fresh atoms; $\mathtt{e}$, $\mathtt{s}_0$, $\mathtt{s}_1$, R, t_w, φ_w	the literal setting and language; $\varphi_w := R(t_w)$	Def. 2.2
D^X, f_X; $\mathfrak{T}$; fit class	literal sequence and labeller of a set of words X; the named structure; the sets X on whose D^X an inducer has bounded loss	Def. 2.2
conventions	semimeasure (primary), normalised and three-valued loss conventions	Def. 2.5
$\mathcal{Q}$; (α, ℓ); (E3)	question (pairwise B-incompatible direct answers); conditional question; its encoding	Def. 2.11
$\mathrm{Un}_D(\mathcal{Q})$	posterior probability that the random theory proves no direct answer	Prop. 6.3
Theory machines and the one-tape law		

symbol	meaning	where
G; A^G_w, A^G_ω	theory machine (c.e. set of pairs (prefix, sentence)); axioms emitted under a prefix, on a tape	Def. 2.3
O^G_χ; Cons^G; E^G_D	tapes on which the theory proves χ; on which it is consistent; consistent and proving ψ_D	Def. 2.3
ν_G, ν^3_G; tri^G_ω; $K(G)$, $K(\nu)$	law and three-valued law of G; trichotomy function; complexities of a machine and a law	Def. 2.3
G_U, A_ω, O_χ, Cons, E_D, E^3_D	the universal theory machine and its events	Def. 2.4
M_A; M^3_A; L_A	SAI, the one-tape proof-requiring law ν_{G_U}; its three-valued law; $-\log M_A$	Def. 2.4
Inc_0	probability that the random theory is inconsistent	Def. 2.4
Bel_D, Dis_D, Indep_D	posterior probabilities that the random theory proves, refutes, decides neither	Def. 2.4
Bel_0; $\widehat{\mathrm{Bel}}_0 = \mathrm{Bel}_\emptyset$; Pl_0, Pl_D	prior probability of provability (unnormalised); its normalisation; plausibilities	Def. 2.4
$\mathbb{U}_A(\mathcal{C}; D)$; $\mathbb{U}_A[T]$; $\mathbb{U}_A(\mathrm{Ext}T)$	class-restricted laws: a class of theories; the class of T; the consistent extensions of T	Def. 2.4
E_∞; P_∞	$\bigcap_n E_{D_n}$; the limit posterior $\mathbb{U}(\cdot \mid E_\infty)$	§ 3.7
$\varepsilon_\emptyset$	mass of the empty theory	Def. 2.10
M^{nc}_A, $\mathrm{Bel}^{\mathrm{nc}}_D$, $\mathrm{Dis}^{\mathrm{nc}}_D$; $M^{1/2}_A$; Ξ	consistency-requiring weight and predictor; graded weight; generative mixture	Def. 2.10
Versions of SAI		
W_A, K_A; W^{dec}_A, K^{dec}_A	coin-free and Kolmogorov versions over enumerators; over deciders	Def. 2.7
K_{fin}, W_{fin}	the same over single sentences (finite axiomatisations)	Def. 2.7
M^{dec}_A	one-tape mixture over random deciders	Def. 4.1
$M^{\mathrm{chk},t}_A$, $M^{\mathrm{gen},t}_A$; $W^{\mathrm{chk},t}_A$, $W^{\mathrm{gen},t}_A$; $W^{\mathrm{Kt},t}_A$	laws with membership-time or generation-time penalties on axioms, one-tape and coin-free; Levin's penalty	§ 4.2, Def. B.4, and Prop. B.16
W^3_A	coin-free three-valued SAI	§ 4.4
(S), ($\mathrm{S}_{\mathrm{tape}}$), (F1)	self-simulation assumptions; closure under renaming of parameters	Ass. 4.5, 4.6 and 4.17
$W_A[t, s]$; W^{a1}_A, W^{a2}_A, W^b_A, W^{b2}_A	charged deduction: derivation size s; readings (a1), (a2), (b), (b2)	§ 4.6
$M^{\mathcal{L}_O}_A$, $W^{\mathcal{L}_O}_A$, $K^{\mathcal{L}_O}_A$	hypotheses restricted to observational axioms	Def. 8.1
The operator-induction side		
F, f^F_ω, ν_F; F_U, f_ω; M_F	function machine, its random labeller and law; universal ones; one-tape SOI	Def. 4.1
Γ_f; consistent, closed	$\{\varphi^{f(\varphi)} : \varphi \in \mathrm{dom}\, f\}$; labellers consistent with B, closed under consequence	Def. 4.1
M^{cons}_F, M^{cl}_F; M^{mw}_F	one-tape consistent and closed SOI; machines consistent on every tape	Def. 4.1; Prop. B.14
W^{cons}, W^{cl}, W^{all}; K^{cons}_F, K^{cl}_F	coin-free consistent, closed and plain SOI on sentences; Kolmogorov versions	Def. 4.1
W, K_F	the companion paper's coin-free and Kolmogorov versions on strings	Def. 4.1
$M^{3,\mathrm{cons}}_F$, $W^{3,\mathrm{cons}}$	three-valued consistent SOI, one-tape and coin-free	§ 4.4

symbol	meaning	where
$W^{\mathrm{cons},\tau}$, $W^{\mathrm{all},\tau}$; $M_F^{\mathrm{cons},\tau}$, $M_F^{\mathrm{all},\tau}$	clocked consistent and plain SOI, coin-free and one-tape	§ 4.5
W^{cert}; M_{prod}	certificate SOI; the companion paper's product mixture	§ 4.6; Cor. 4.9
Random theories and arithmetic		
Win, Lot_k; G_{lot}	lottery relation and sentences; the random lottery theory machine	§ 5.1
$\mathrm{Tick}_{k,j}$, $\mathrm{Lot}_k^{\mathrm{ar}}$	arithmetic tickets and lotteries	Thm. 5.5
G_{thr}	the threshold machine	Thm. 5.7
$\mathrm{Kgt}_m(x)$	the Π_1 sentence "$K(x) > m$"	§ 5.3
$\mathcal{U}$; $P_D(\mathcal{C})$	the Σ_1-unsound theories; posterior probability of a class	§ 5
Bit_k; ξ	sentences for the binary digits of Ω_V; the exponent of the lower bound on true arithmetic	App. C.7; Thm. 5.19
Theory shape		
m; x_i, y_j	number of base variables; base variables, auxiliary variables (intermediate concepts)	Def. 7.1
Φ; $\mathrm{proj}\,\Phi$; $\mathcal{E}_m^{\mathrm{base}}$, $\mathcal{E}_m$	explicit or linear theory; its projection; classes without and with auxiliary variables	Def. 7.1
$\mathcal{S}$; $\mathcal{Q}(\mathcal{S})$	law; its conditional questions	Def. 7.1
$\mathcal{R}$; UP_k, UP, GE, PS_∞; FC_k	reasoner; unit propagation, elimination, proof search; forward chaining	Def. 7.1; Prop. E.20
$K_A^{\mathcal{R}}$, $W_A^{\mathcal{R}}$; $K_{\mathrm{lst}}^{\mathcal{R}}$, $W_{\mathrm{lst}}^{\mathcal{R}}$; $Kt_A^{\mathcal{R}}$	program length; listing (the theory charged as written); time-charged	Def. 7.1
$\mathrm{PI}(\mathcal{S})$; d_*	prime implicates; least weight of a nonzero derived equation	§ 7.2; § 7.3
Par_m^b; $\mathcal{H}_q$; ia_k	parity law; hidden-state law; inverse-Ackermann rows	§§ 7.4, 7.5 and 7.8
Theoretical terms		
$\mathcal{L}_O$, $\mathcal{L}_T$; $T_O(A)$; $\mathcal{C}_\Theta$	observational and theoretical vocabulary; observational content; class of theories with content in Θ	Def. 8.1
$\mathrm{Th}_{\mathfrak{T}}$; about, compatible with the named structure	the theory of $(\mathfrak{T}; \mathsf{e}, \mathsf{s}_0, \mathsf{s}_1)$; two relations of a theory to it	Def. 8.3
C; $\mathcal{L}^+$; Inf_n	binary theoretical relation; three-sorted extension; "there are at least n things"	§ 8.3; Thm. F.28; Thm. F.23
Approximate laws, explanation, the Kepler example		
$(\mathcal{Q}, a)$; $\gamma(\mathcal{Q})$	answered datum; literal cost $\log\lvert\mathcal{Q}\rvert$	Def. 9.1
$\ell(\chi)$, $\lVert\Gamma\rVert$; Exc^Q, Exc^F, Exc^R	code lengths; exception codes	Def. 9.1
λ; J_λ; λ-MAP	steepness; $\lambda\lVert T\rVert + \mathrm{Exc}_D(T)$; its minimisers	Def. 9.1
T_0, T^ε; $e(t)$, ε^*	exact and honest laws; error and its essential supremum	Def. 9.1
δ, δ_0; Δ; $\mathrm{pk}(e)$	data precision, measurement-error bound; dithered cell width; peakedness	Def. 9.1
$\mathrm{BC}_{\mathcal{G}}$, $\mathrm{HY}(\rho)$	band-confined completions	§ 9.3
$\mathrm{EP}_D(\mathcal{A}; T)$	explanatory power of a module	Def. 9.6
μ, η, μ_η	raw world, extractor, induced law on question data	§ 6.3
$\mathcal{L}_{\mathrm{hel}}$, $\mathcal{L}_{\mathrm{dir}}$, $\mathcal{L}_{\mathrm{geo}}$; K12, K3	Kepler data languages; Kepler's first two laws, third law	Def. 10.1

symbol	meaning	where
$\mathrm{Occ}_{\mathsf{F}}(\delta)$; S1, S2, S3	Occam factor of a form of the Mars pool; syntaxes of the law texts	§ 10.4